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(54) **COMPOSITIONS AND METHODS FOR PRODUCTION OF SALIDROSIDE, ICARISIDE D2, AND PRECURSORS OF SALIDROSIDE AND ICARISIDE D2**

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(73) Assignee: **Whitehead Institute for Biomedical Research**, Cambridge, MA (US)

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This patent is subject to a terminal disclaimer.

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(51) **Int. Cl.**

C12N 15/52 (2006.01)

C12N 15/70 (2006.01)

C12N 15/74 (2006.01)

C12N 15/81 (2006.01)

C12N 15/82 (2006.01)

C12P 17/06 (2006.01)

(52) **U.S. Cl.**

CPC **C12N 15/8257** (2013.01); **C12N 15/52** (2013.01); **C12N 15/70** (2013.01); **C12N 15/743** (2013.01); **C12N 15/81** (2013.01); **C12N 15/8243** (2013.01); **C12N 15/8245** (2013.01); **C12P 17/06** (2013.01); **C12Y 114/13068** (2013.01); **C12Y 204/0117** (2013.01); **C12Y 401/01** (2013.01)

(58) **Field of Classification Search**

CPC **C12N 15/8257**; **C12N 15/52**; **C12N 15/70**; **C12N 15/743**; **C12N 15/81**; **C12N 15/8243**; **C12N 15/8245**; **C12P 17/06**; **C12P 7/22**; **C12P 7/24**; **C12P 19/44**; **C12Y 114/13068**; **C12Y 204/0117**; **C12Y 401/01**

See application file for complete search history.

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(57) **ABSTRACT**

Transgenic host cells, vectors useful for making transgenic host cells, and kits useful for making transgenic host cells are described. Also described are transgenic plants. In some embodiments, transgenic host cells express a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, transgenic host cells express a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). The transgenic host cells are useful for biosynthesis of one or more of salidroside, icaraside D2, tyrosol, and 4-hydroxyphenylacetaldehyde.

21 Claims, 43 Drawing Sheets
(33 of 43 Drawing Sheet(s) Filed in Color)
Specification includes a Sequence Listing.

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FIG.1A

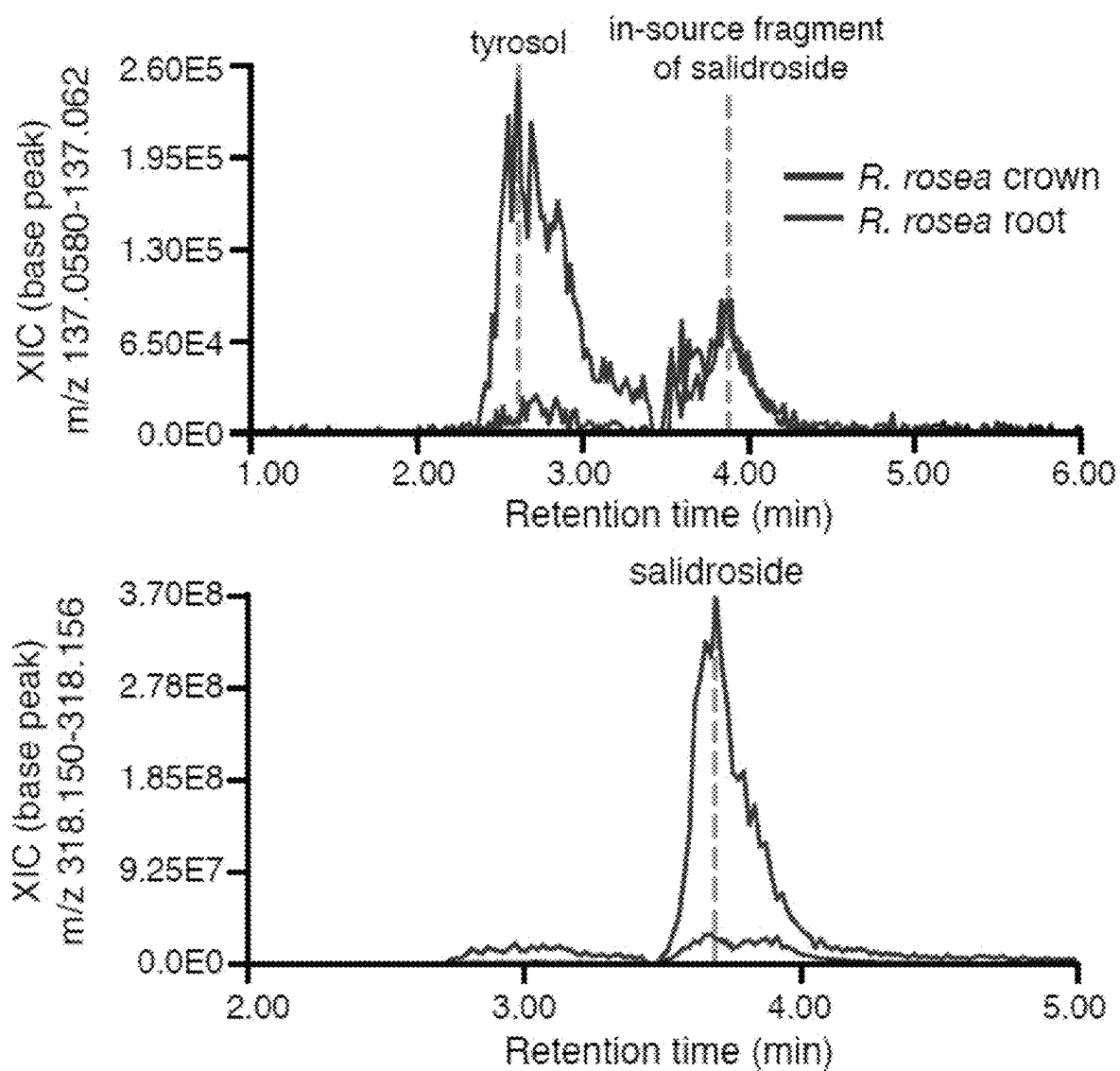


FIG. 1B

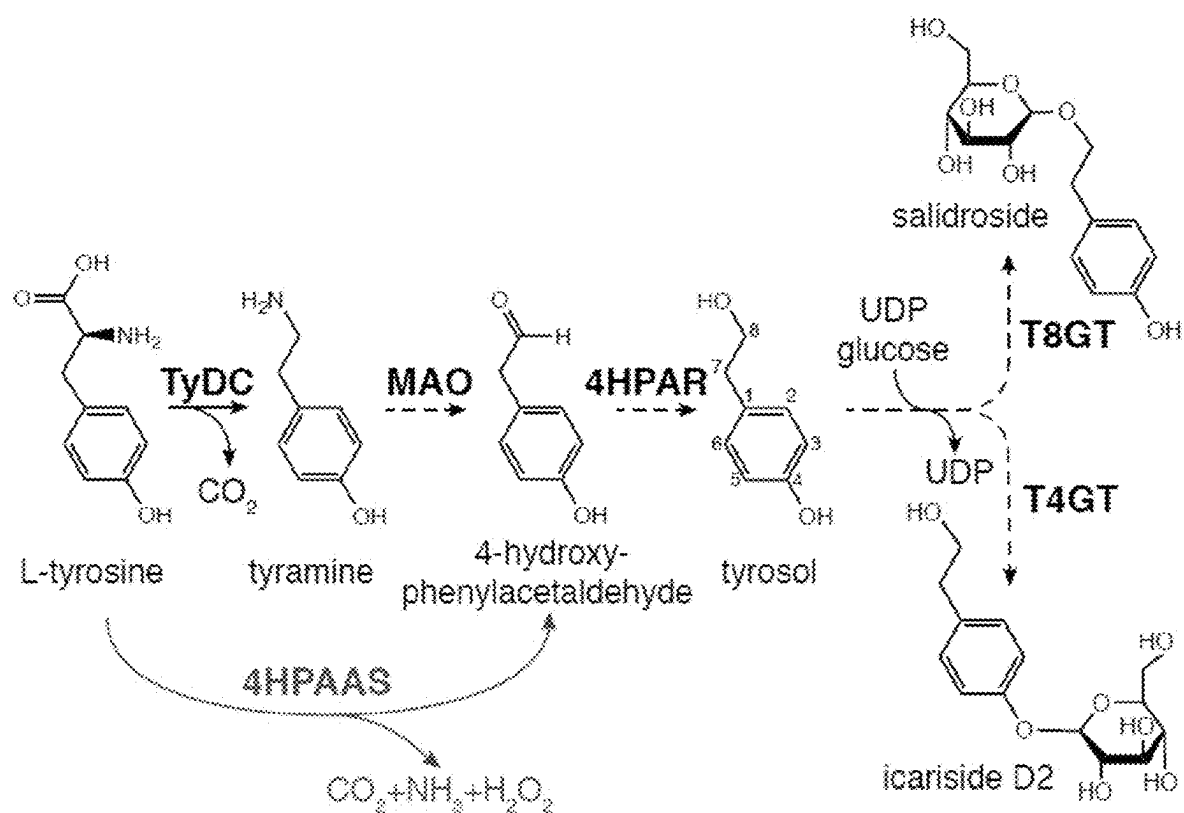


FIG. 1C

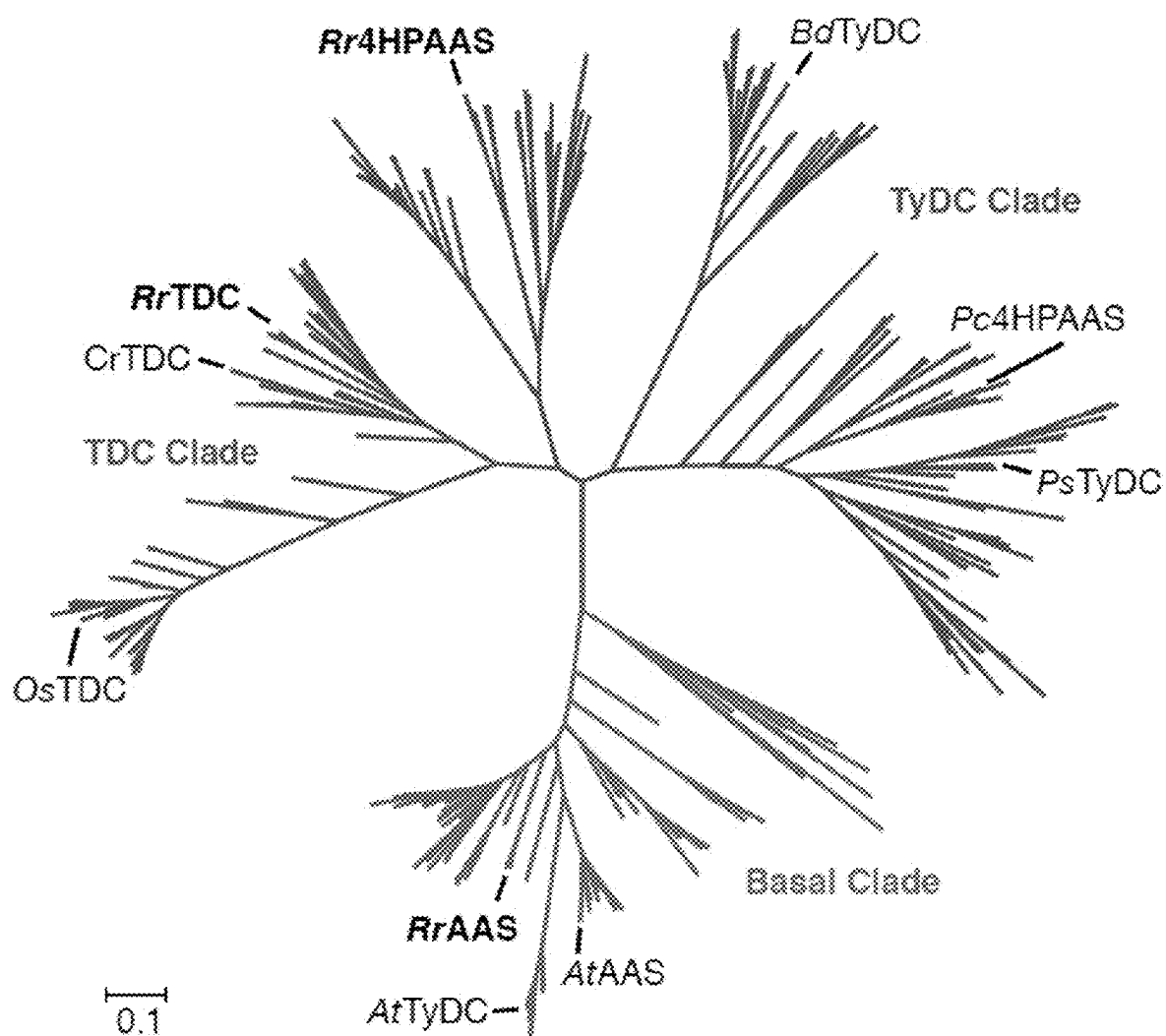


FIG. 2A

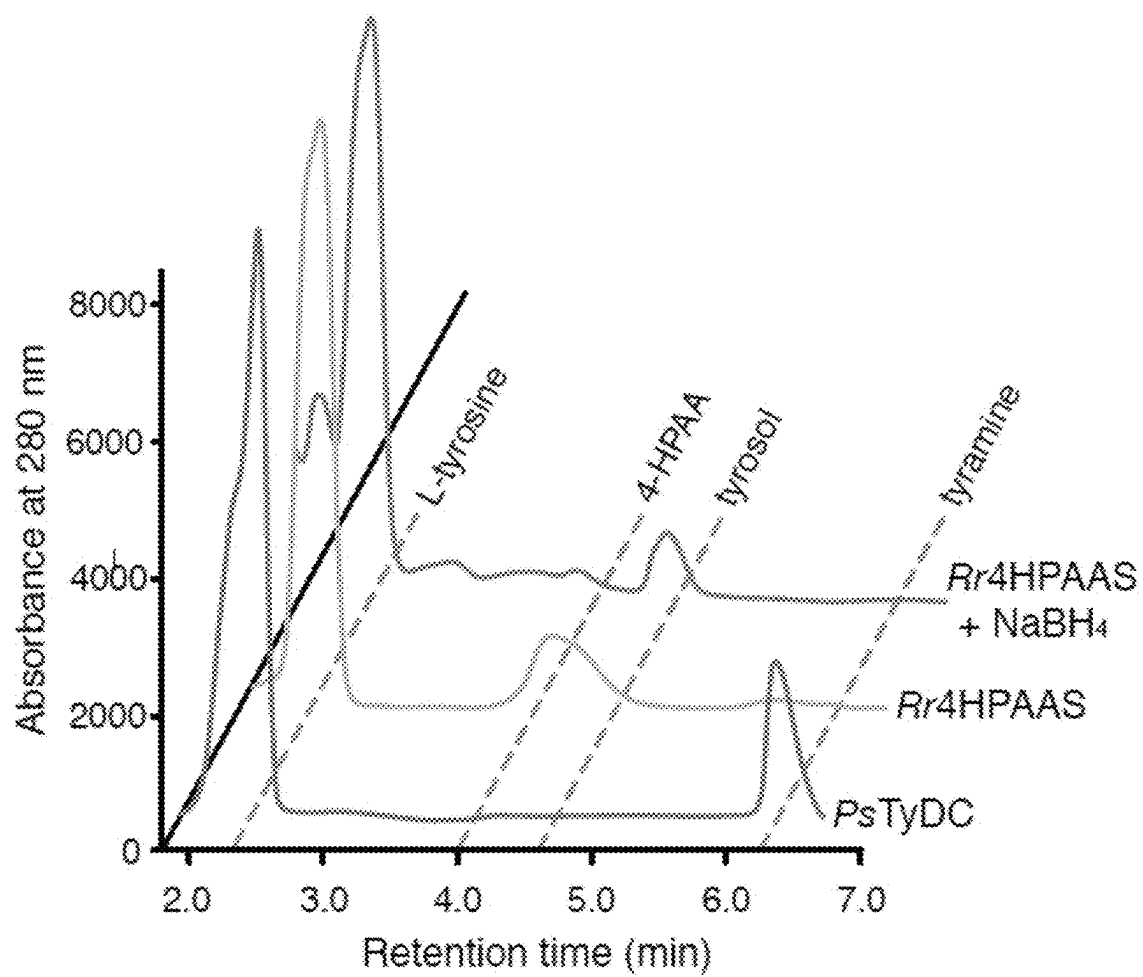


FIG. 2B

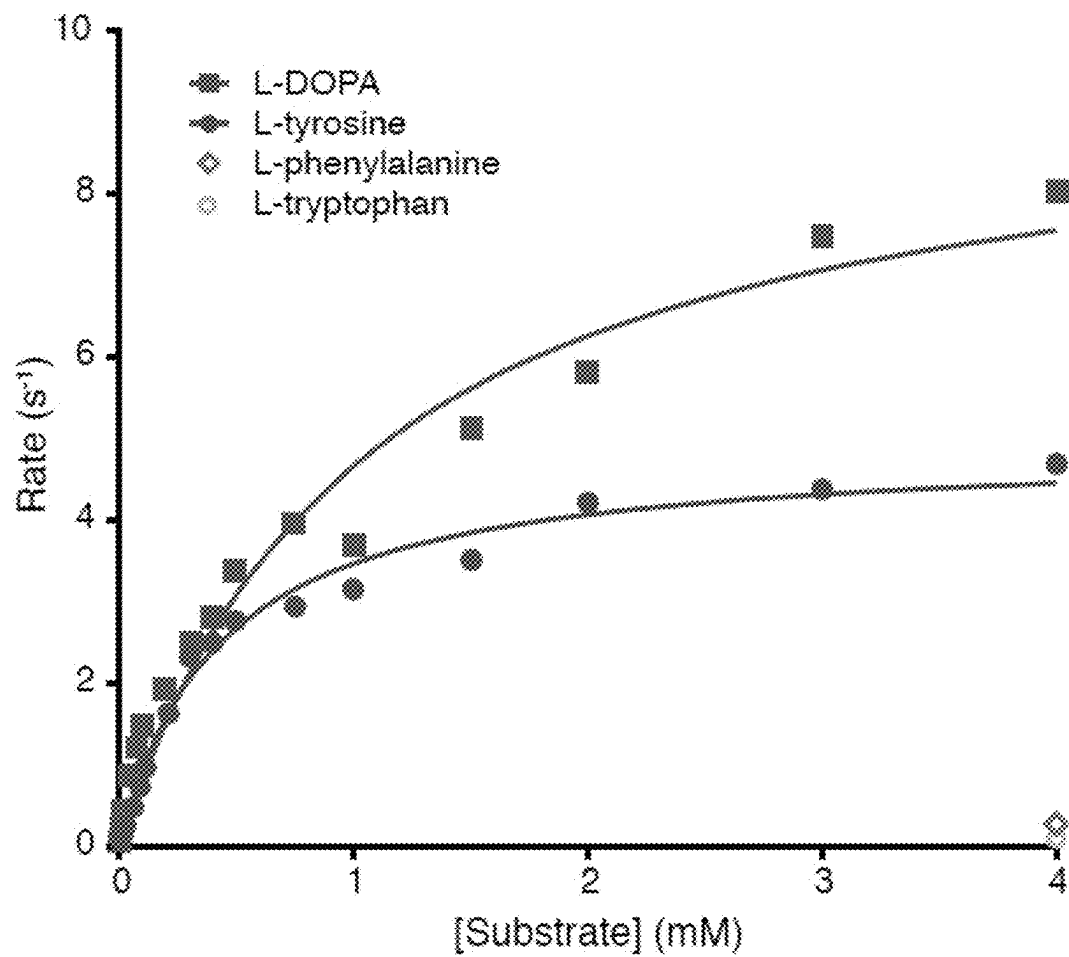


FIG. 2C

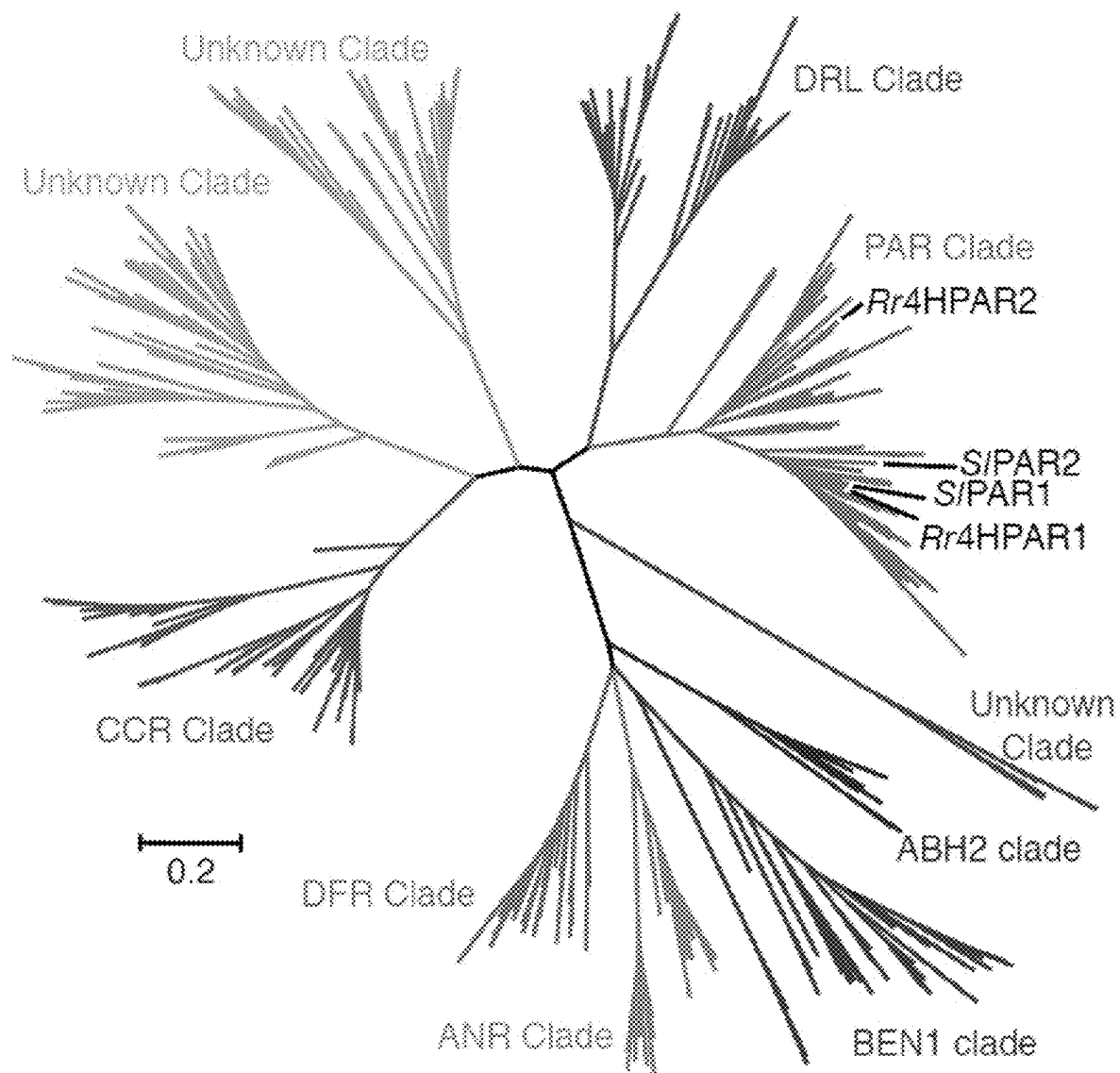


FIG. 3A

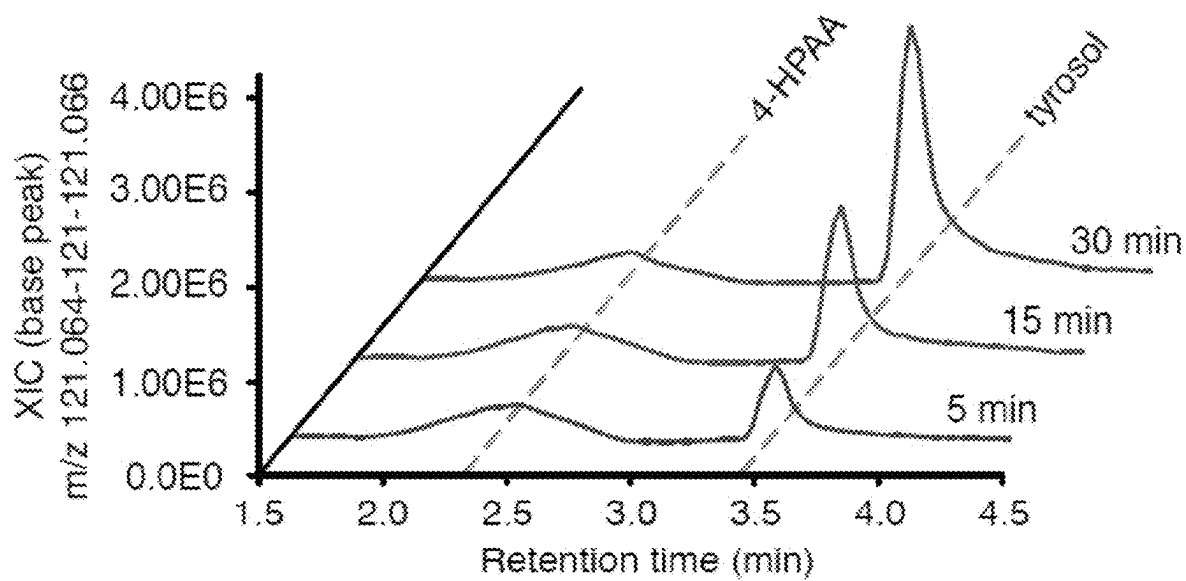


FIG. 3B

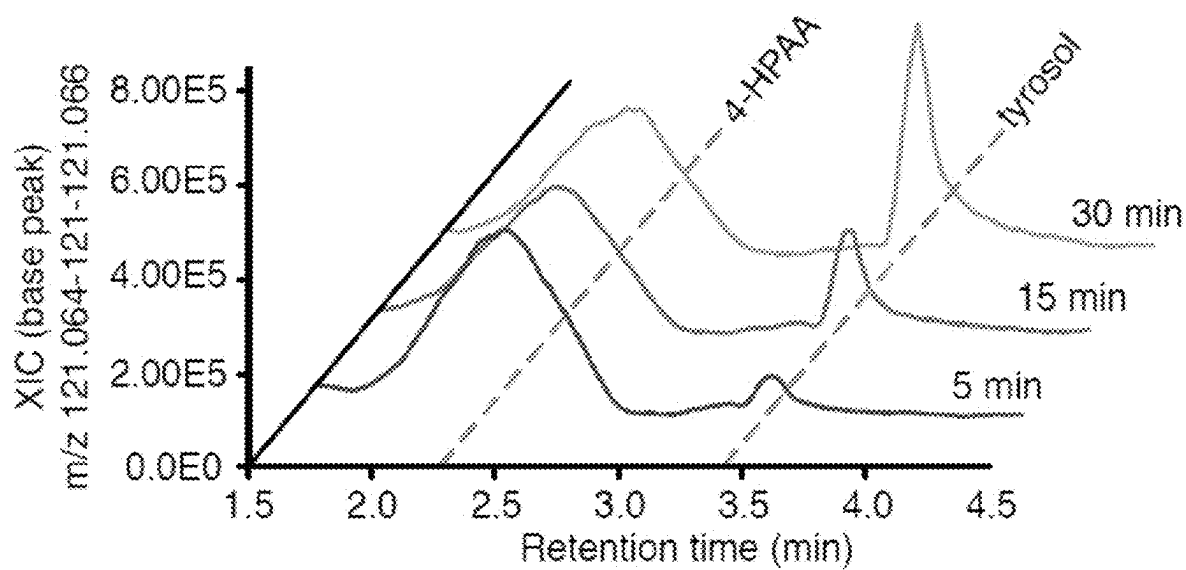


FIG. 3C

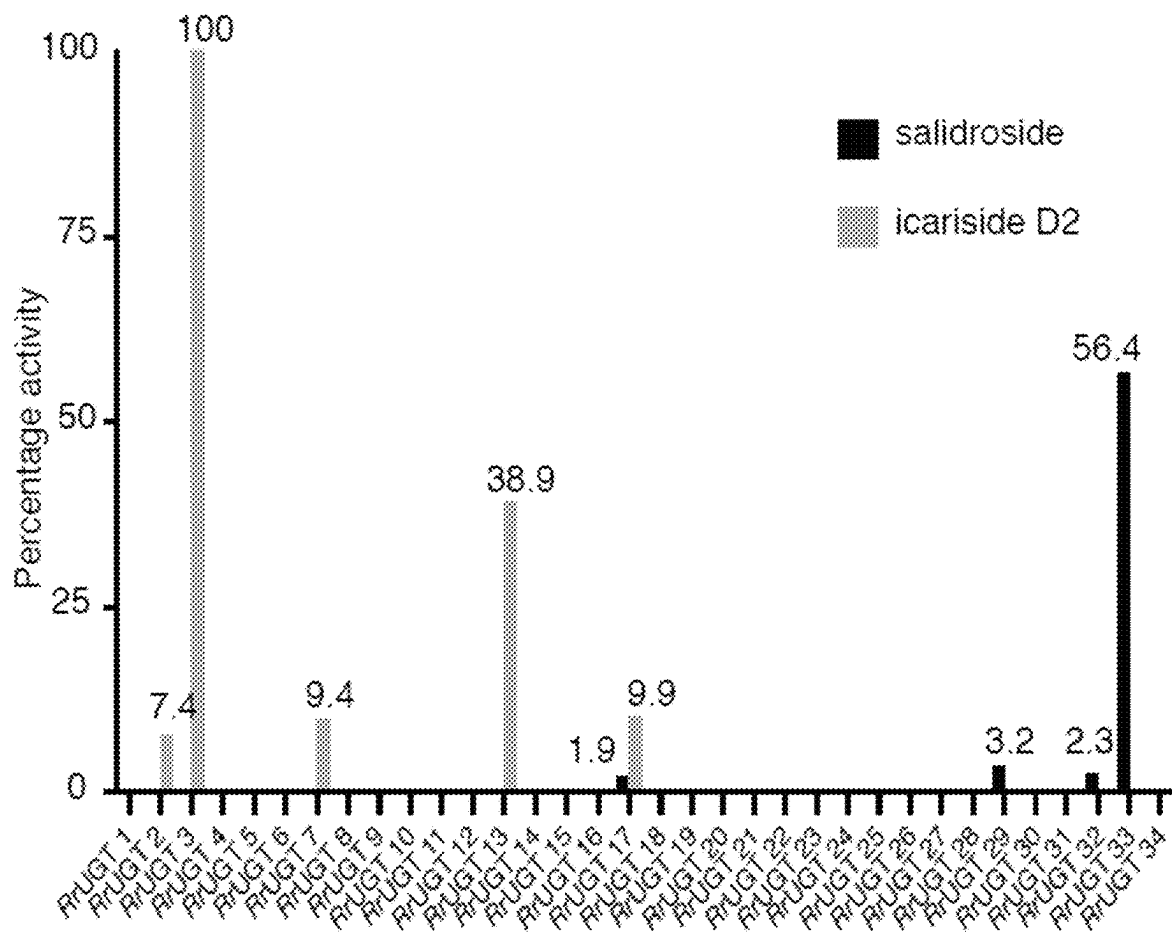


FIG. 4B

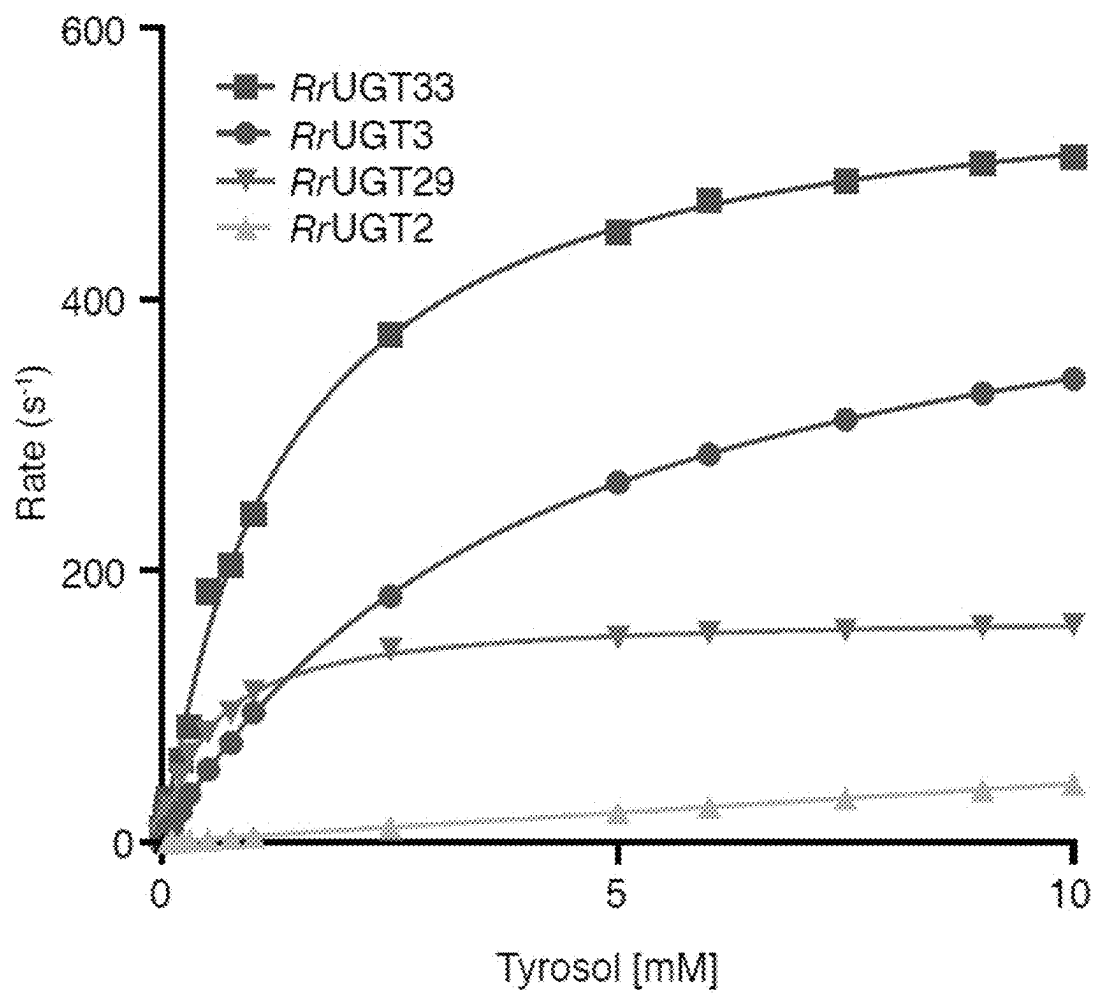


FIG. 4C

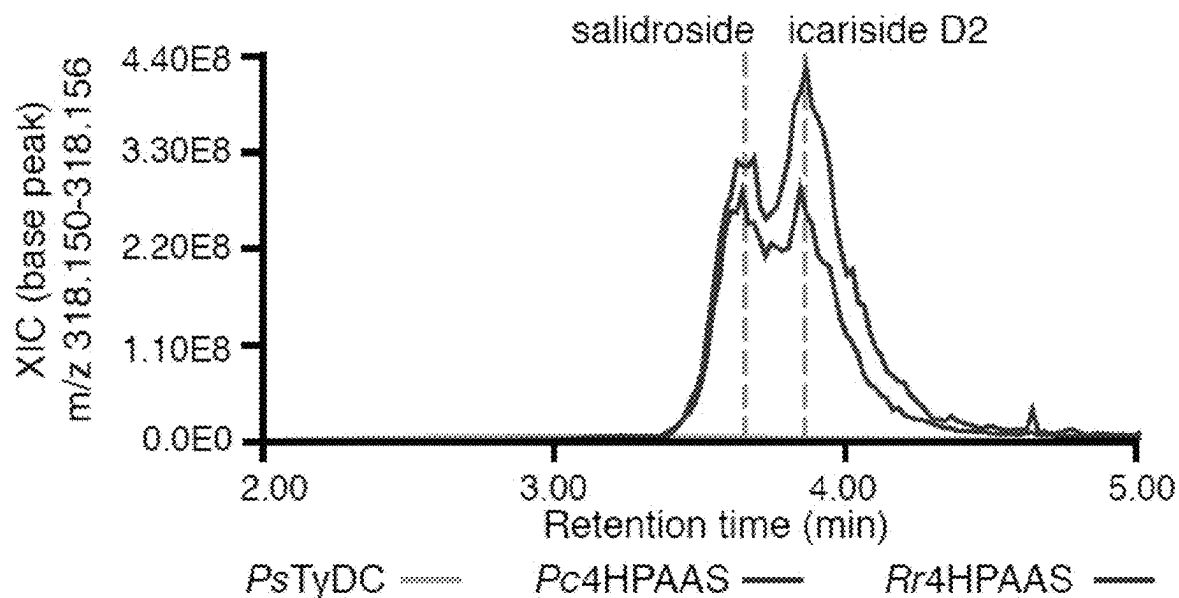


FIG. 5A

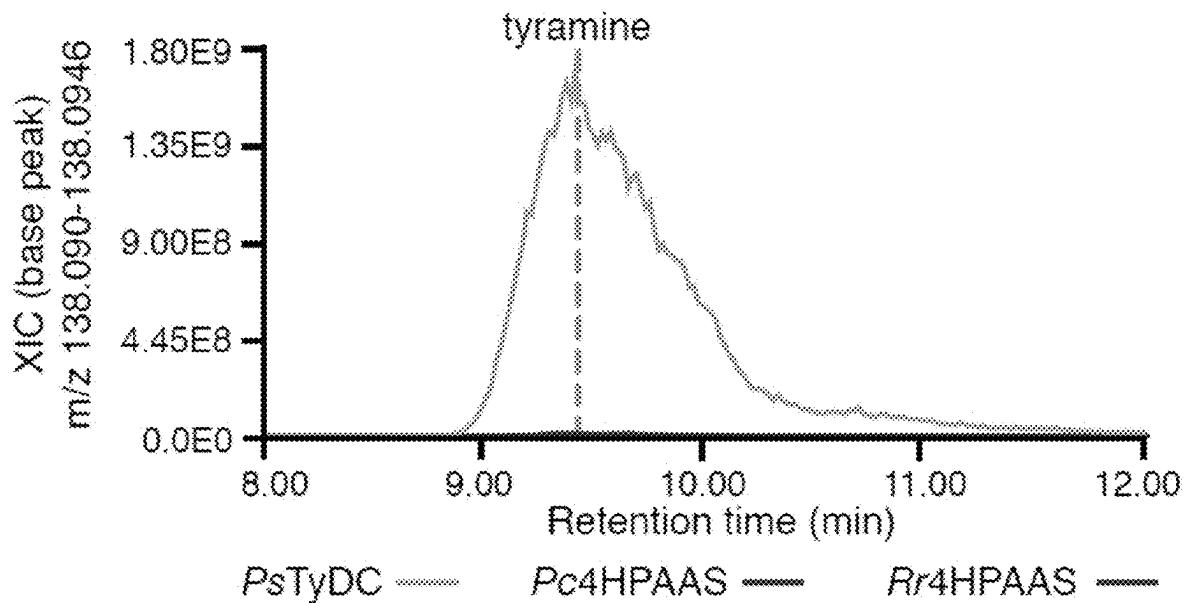


FIG. 5B

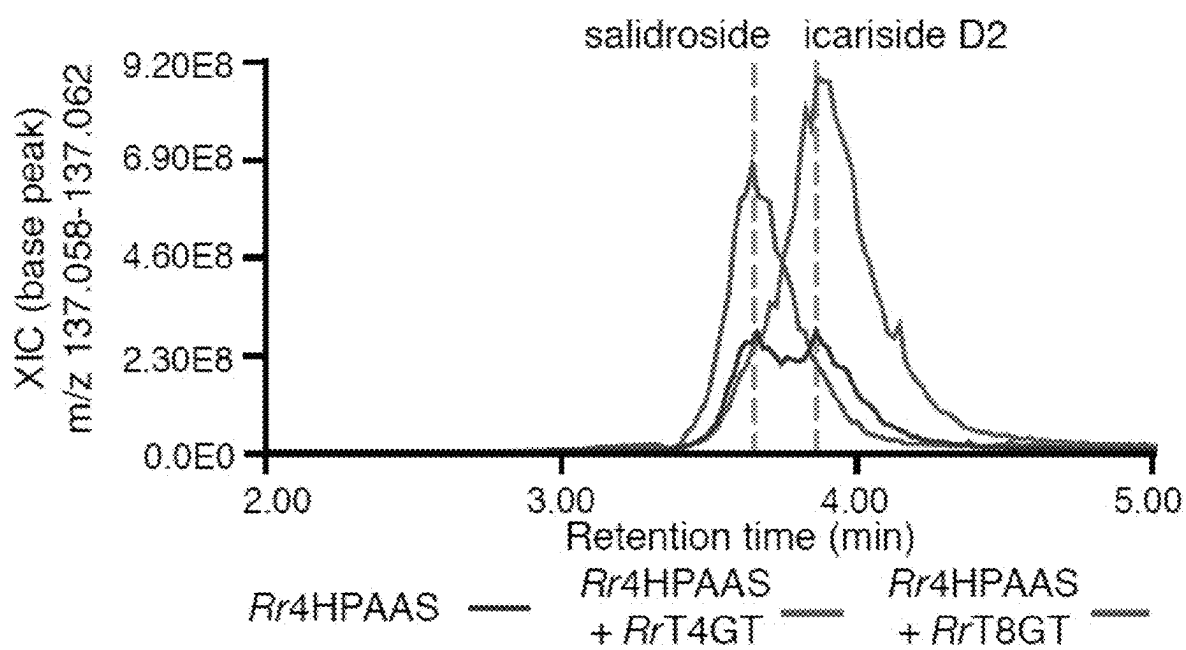


FIG. 5C

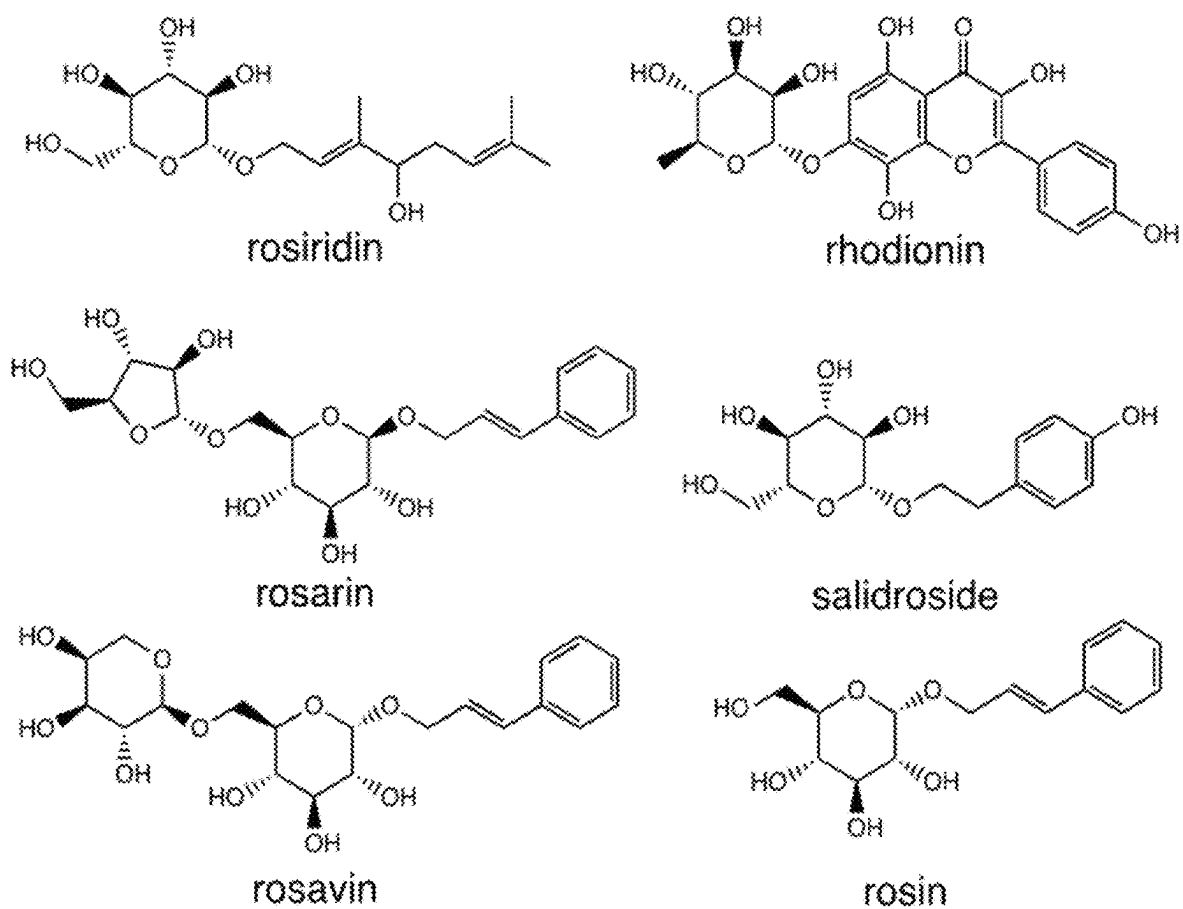


FIG. 6

Substrate selectivity
Gly 370

Indolic Selective	C. roseus	P17770.1	360	VDFKNWQIATGRKRSLSKLL
	C. annuum	XP_016579458	360	VDYKDWQIGTGRRKSLRL
	O. pumila	BAC41515.1	359	VDFKDWQIGTGRRKALRL
	C. acuminata	AAB39708.1	360	VDYKDWQVGTGRRKALRL
	C. acuminata	AAB39709.1	357	VDFKDWQVGTGRRKALRL
	O. sativa	XP_015648701.1	371	TDLKDMQVGVGRRRGLKLM
phenolic Selective	P. hybrid	ABE72475.1	359	VDYKDWQITLSRRRSLSKLL
	P. crispum	Q06086.1	358	VDYKDWQIMLSRRFRALKL
	P. somniferum	AAC61842.1	362	IDYKDWQIALSRRFRSMKLL
	P. somniferum	P54769.1	360	VDYKDWQIALSRRRSLSKLL
	T. flavum	AAG60665.1	359	VDYKDWQIALSRRGRAMKL
	R. hybrid	ABB04522.1	360	VDYKDWQIALSRRRALKLL
	R. crenulata	AFN89854.1	351	VDYKDWQISLSRRRAIKML

FIG. 7

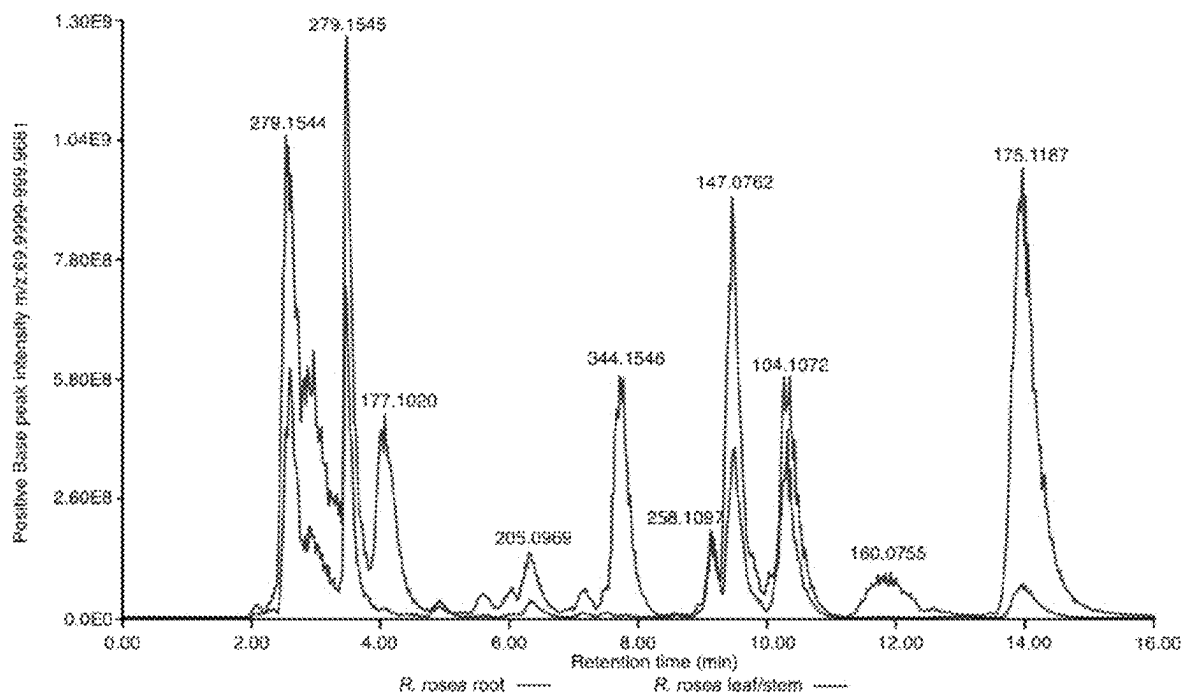


FIG. 8A

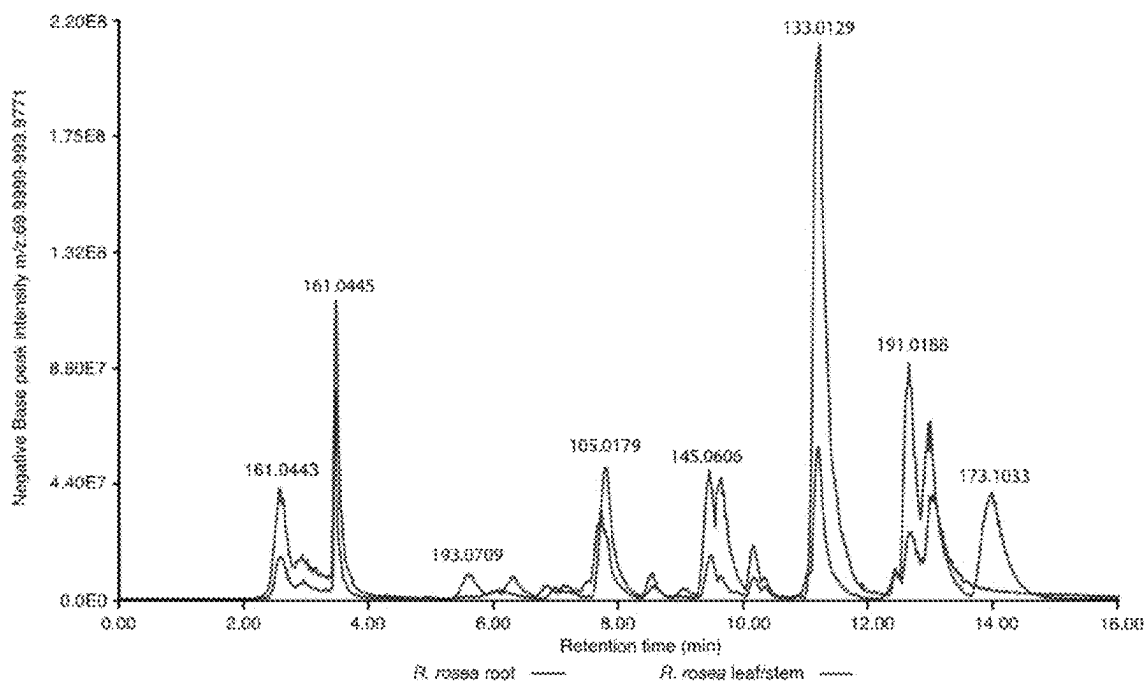


FIG. 8B

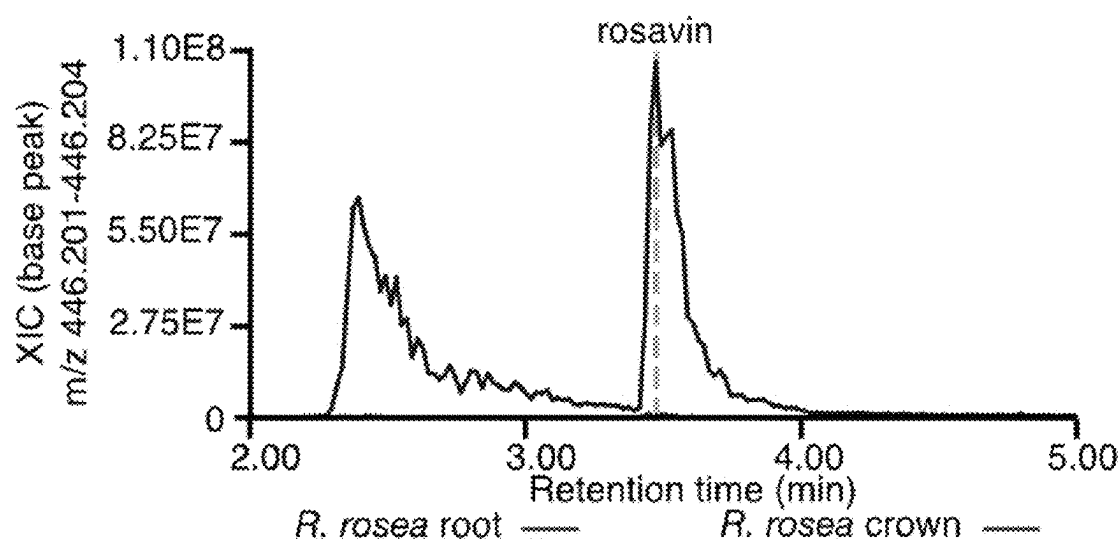


FIG. 9

				Tyr 350
Decarboxylases	P. somniferum	AAC61842.1	340	LVKALSTSAEYLRN.KATESK
	B. distachyon	XP_003569907.1	358	LIAALGTEQEVILKDSASEGH
	O. sativa	XP_015633932.1	358	LVAAAGTEQEVILRDAAAEGH
	T. flavum	AAG60665.1	337	LIKALSTNPEYLRN.KATESH
	P. somniferum	P54769.1	338	LVKALSTNPEYLRN.KATESR
	C. acuminata	AAB39708.1	338	LVKALSTDPEYLRN.QPSESK
	C. acuminata	AAB39709.1	335	LVKALSTDPEYLRN.KPSESN
	O. pumila	BAC41515.1	337	MVKAISTNPEYLRN.KRSEFD
	O. sativa	XP_015648701.1	349	LTGSLSTNPEYLRN.HASDSG
	C. annuum	XP_016541857.1	324	LIQSLSTNPEYLRN.KASQGN
	A. thaliana	NP_194597.1	380	LIDALSTNPEYLRN.KVSKKD
	C. annuum	XP_016541857.1	325	LIQSLSTNPEYLRN.KASQGN
	C. roseus	P17770.1	338	LLRALTTNPEYLRN.KQSDLD
	R. rosea	TDC Candidate	338	LTKALSTNPEYLRN.QQSELN
	R. hybrid	ABB04522.1	338	LASSLSTNPEYLRN.KASDSK
Aldehyde Synthases	P. crispum	Q06086.1	336	LIXSLSTYPEYLRN.NASETN
	A. thaliana	Q8RY79.1	328	LTLLALSTNPEYLRN.KASQAN
	O. europaea	TORX-2002737	323	LIQSLSTNPEYLRN.KASEGN
	R. crenulata	AFN89854.1	333	LIESLAEEANFLK....GNS
	R. rosea	basal clade AAS	324	LIQSLSTYPEYLRN.KASQSN
	R. rosea	TyDC clade AAS	333	LIESLAEEANFLK....GGS

FIG. 10

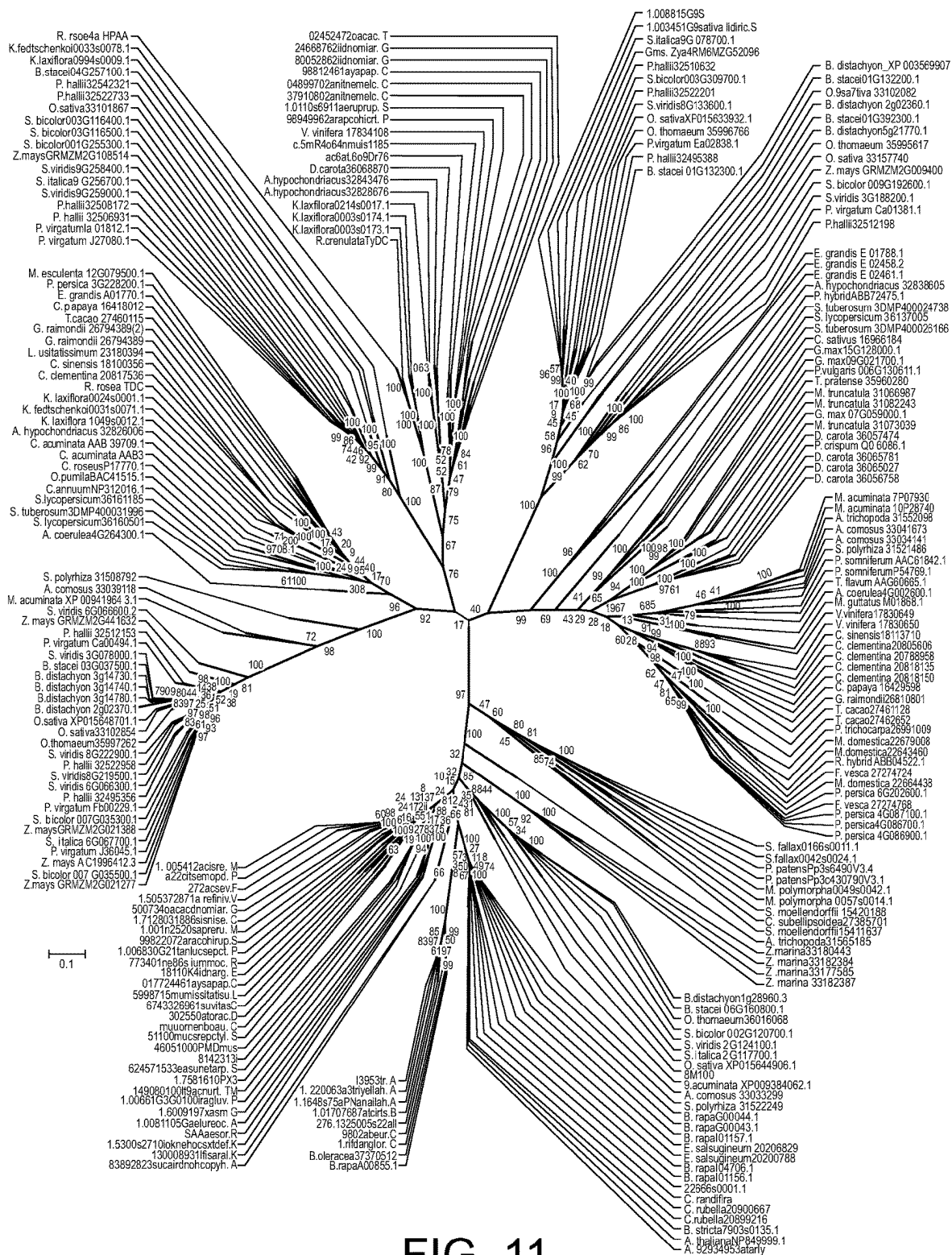


FIG. 11

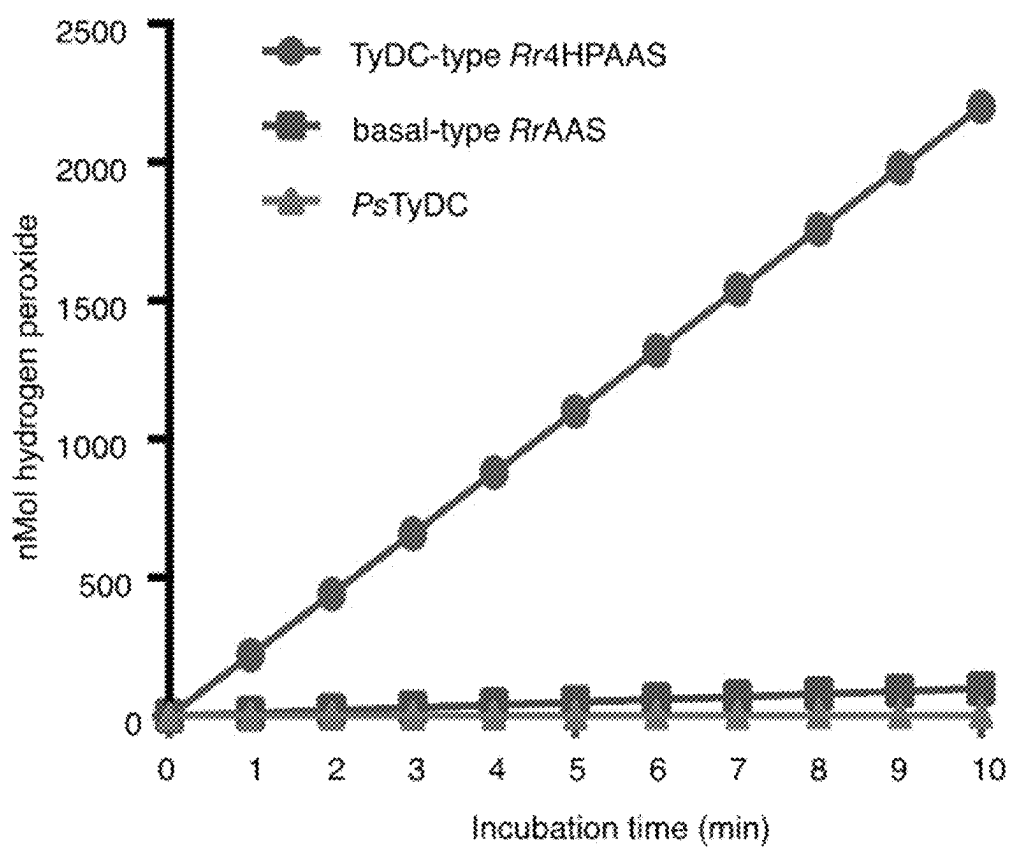


FIG. 12

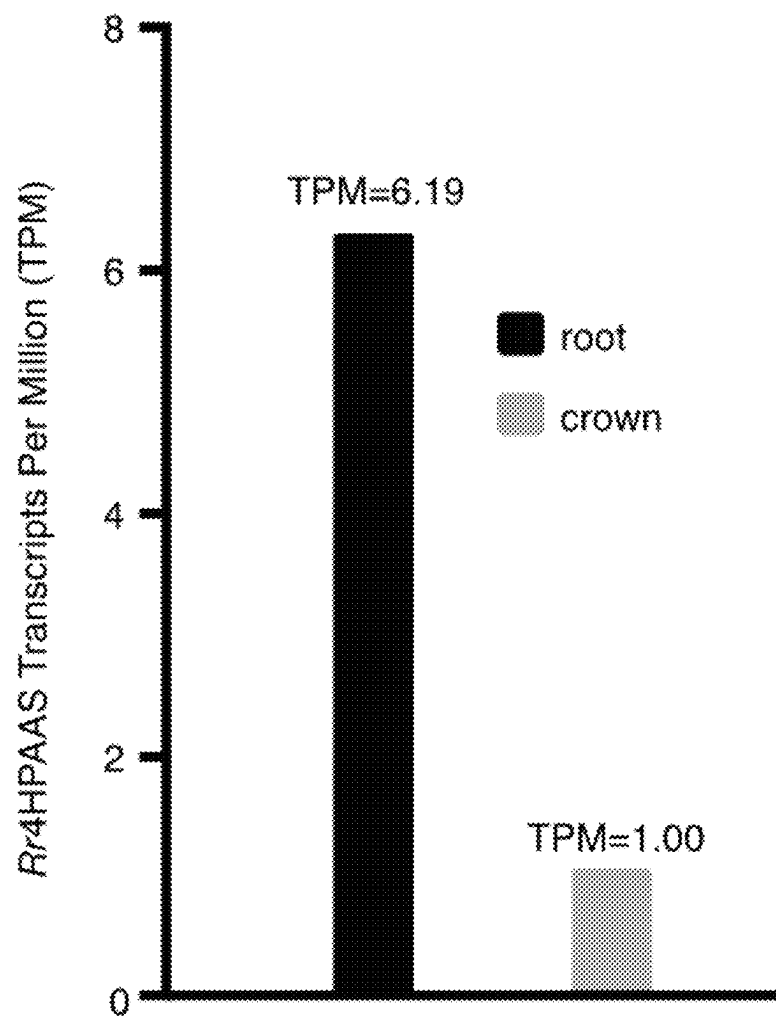
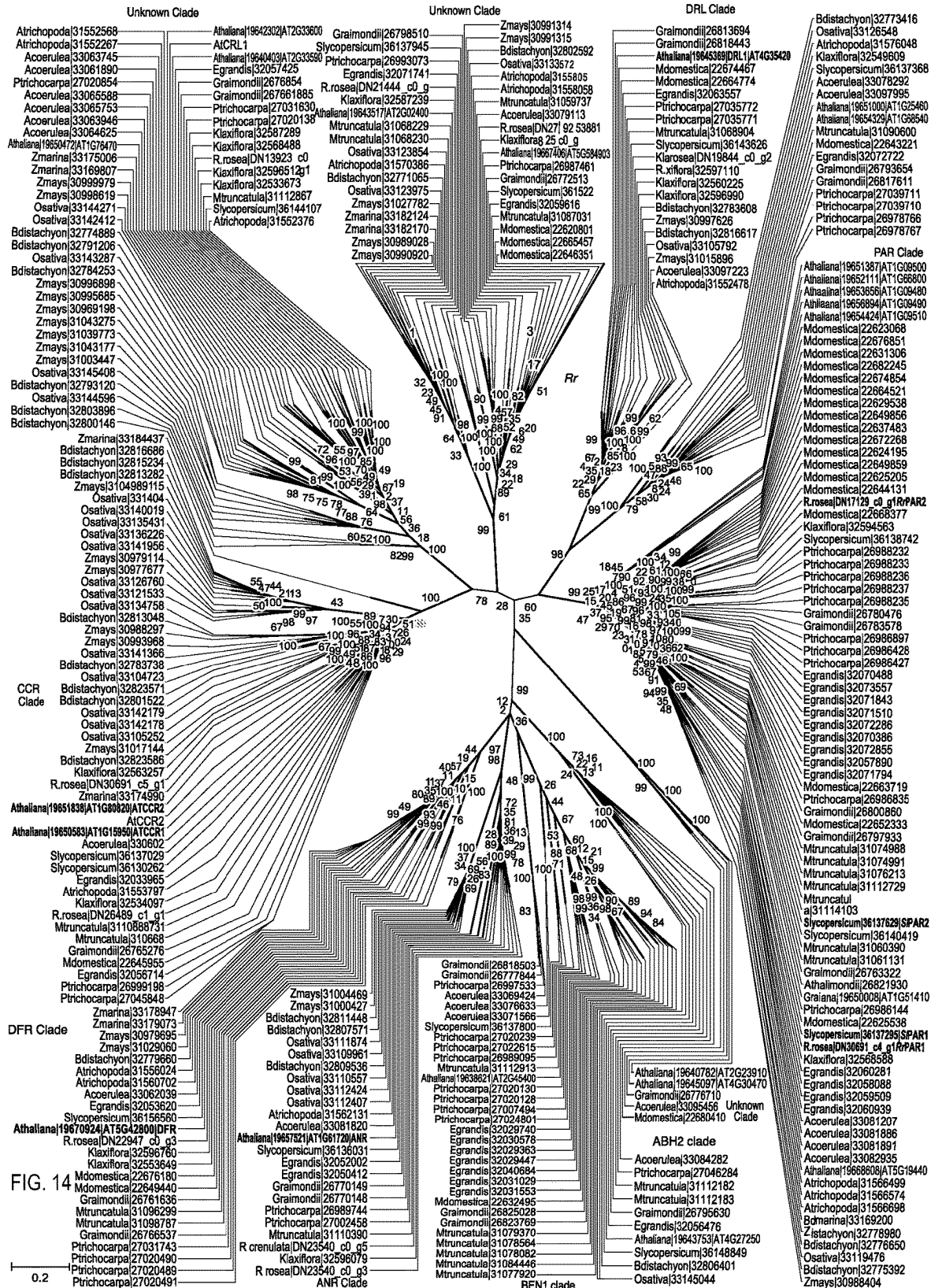


FIG. 13



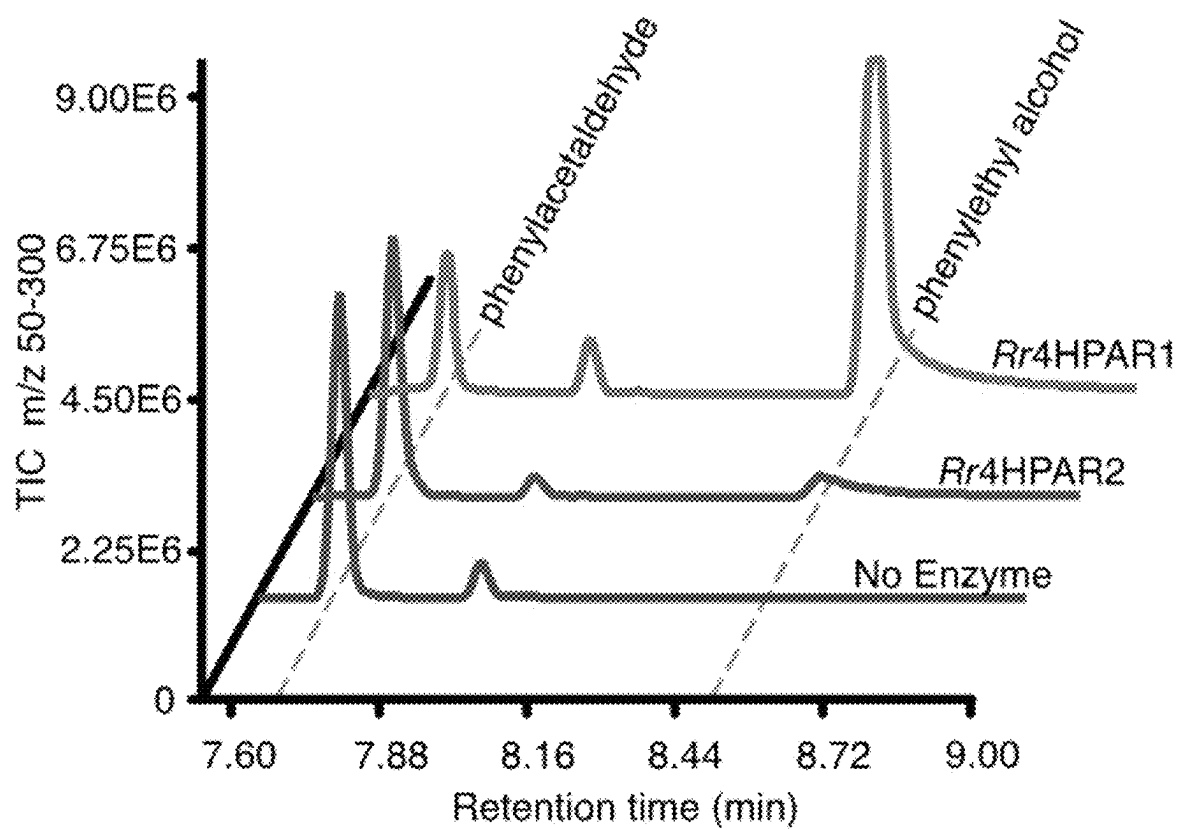


FIG. 15

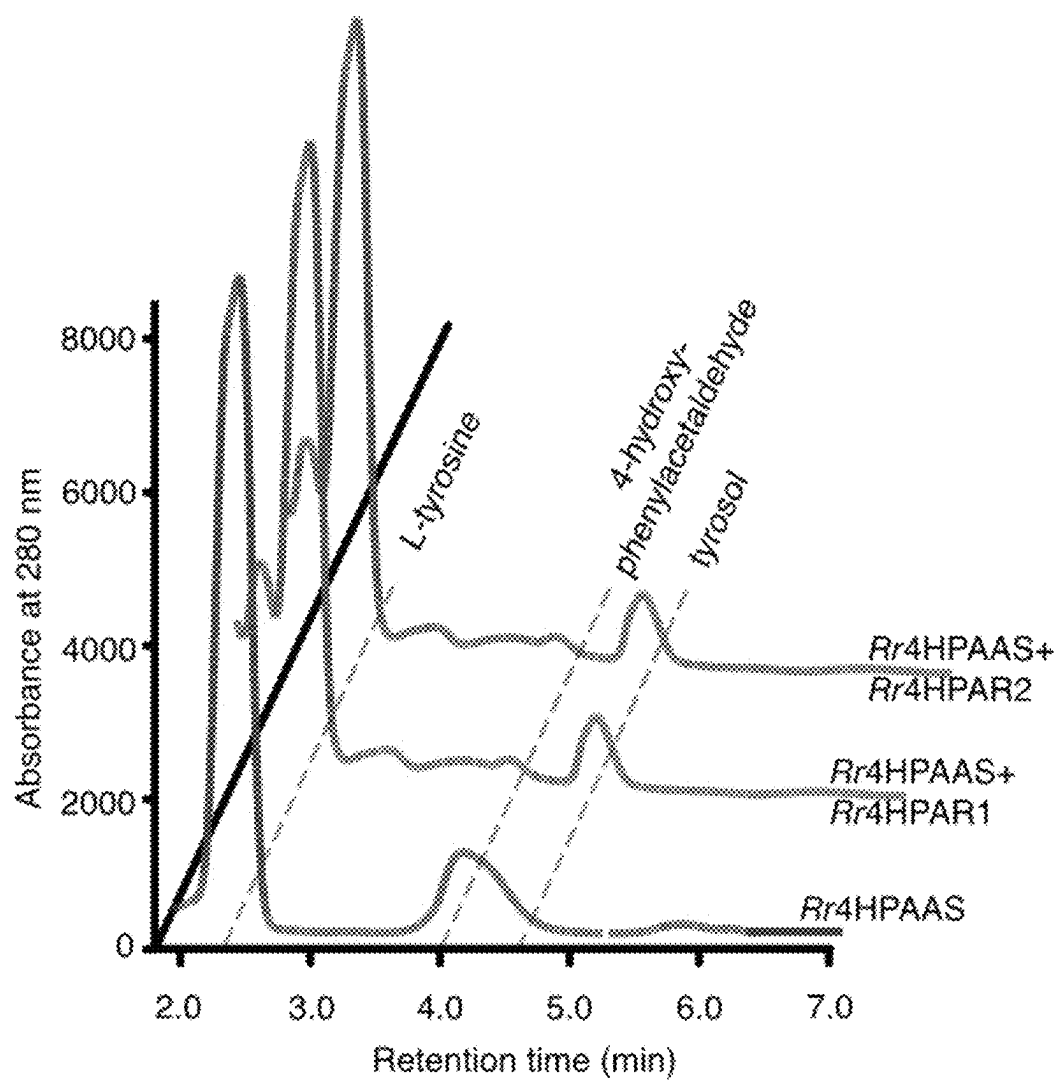


FIG. 16

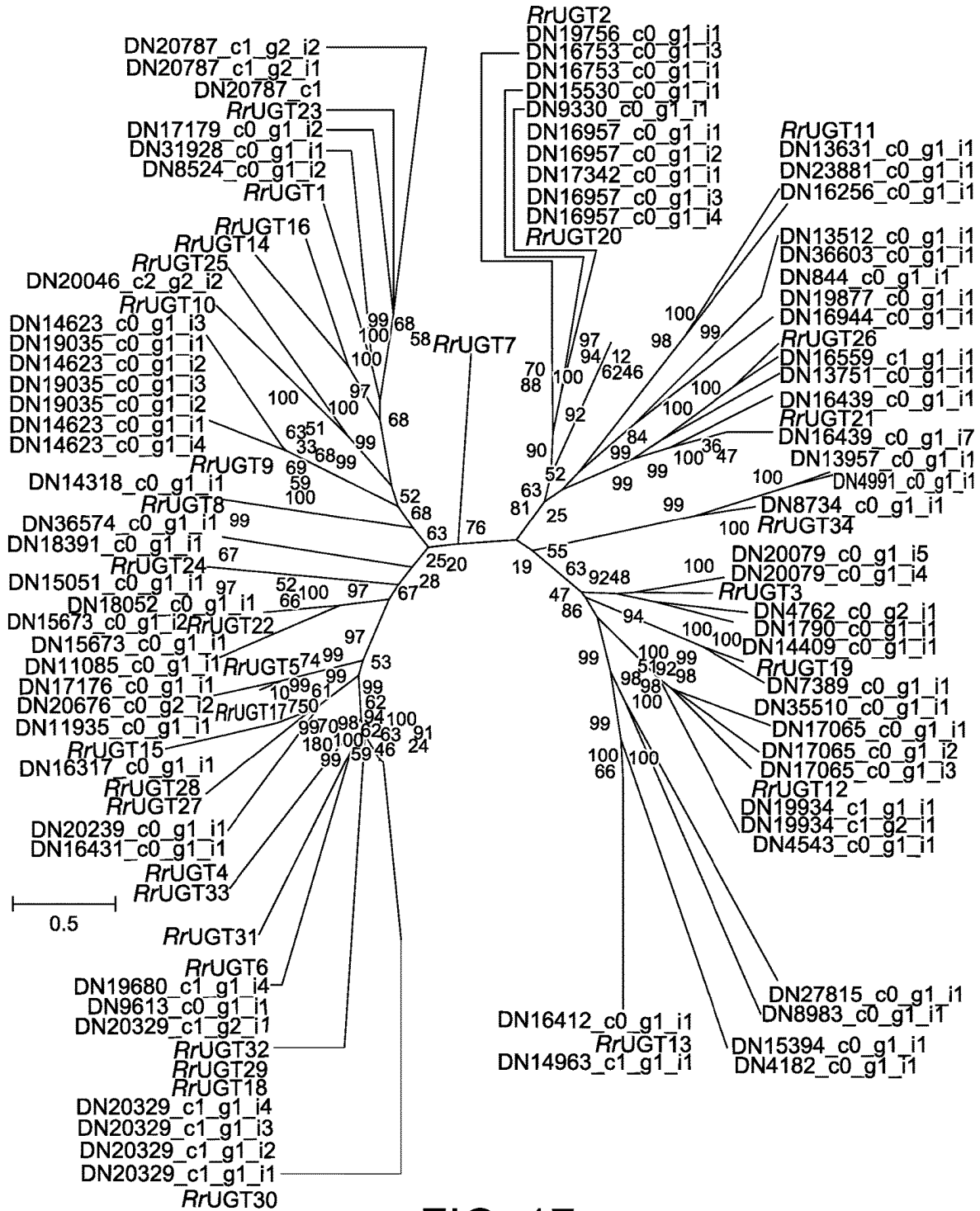


FIG. 17

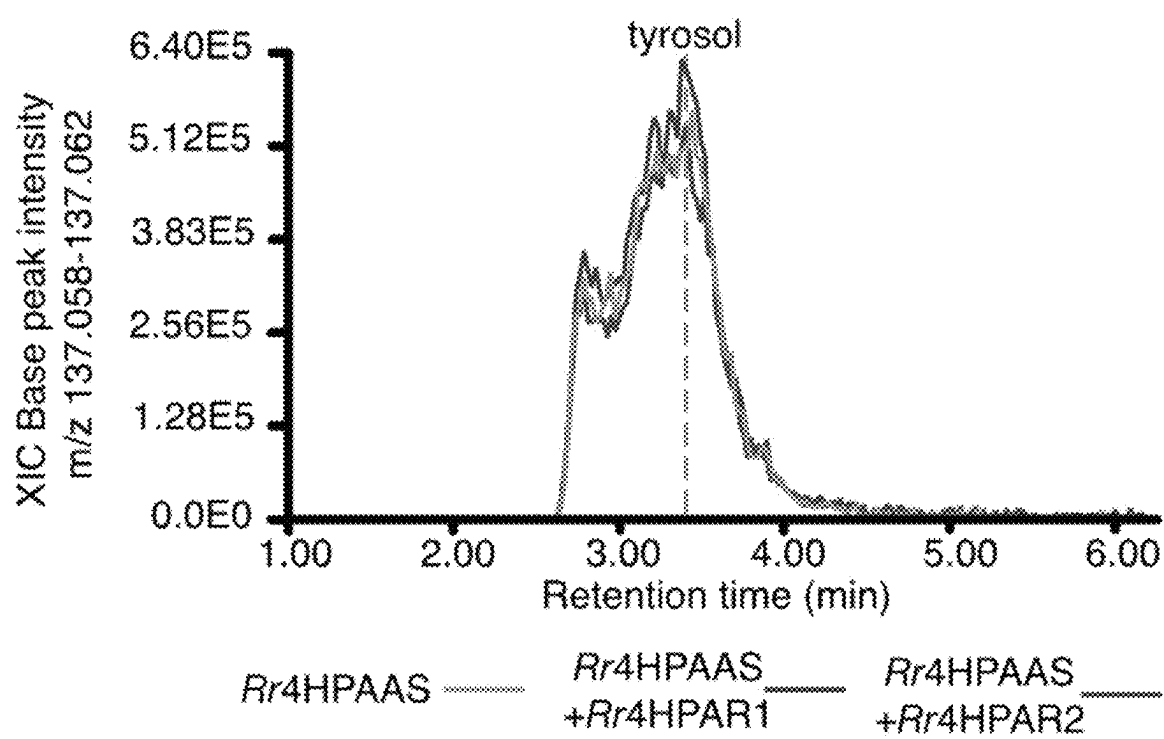


FIG. 18

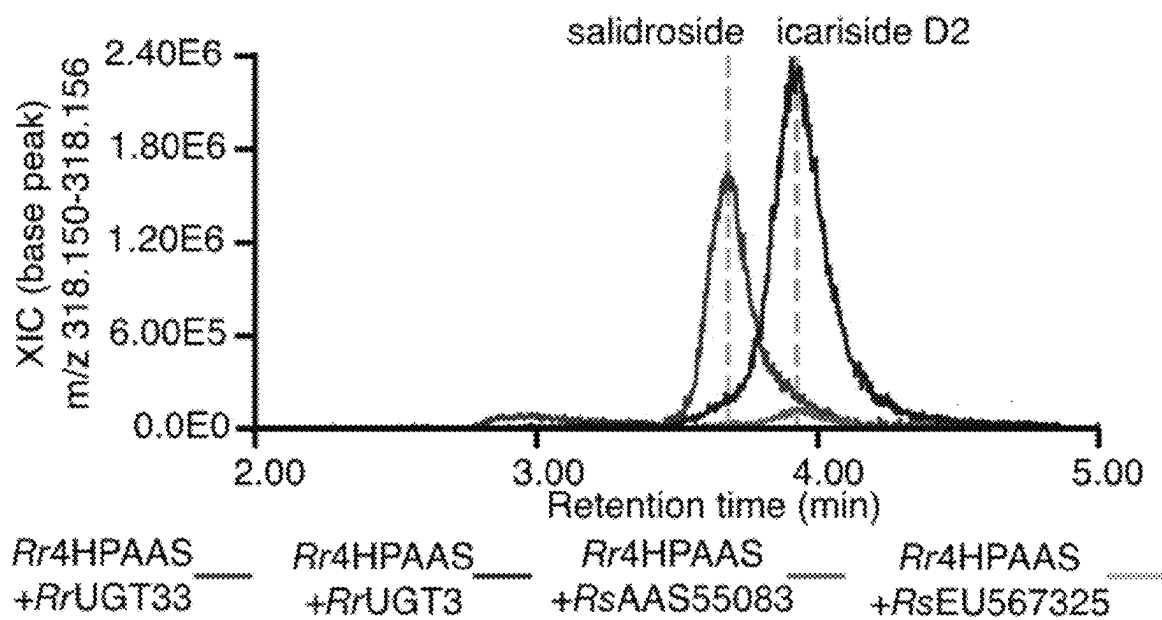


FIG. 19A

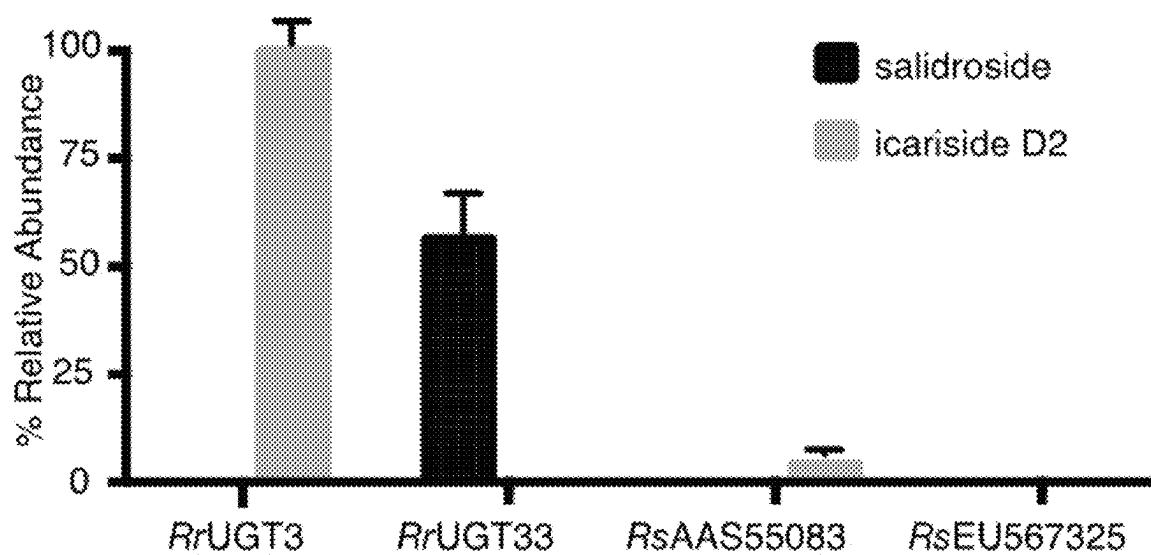


FIG. 19B

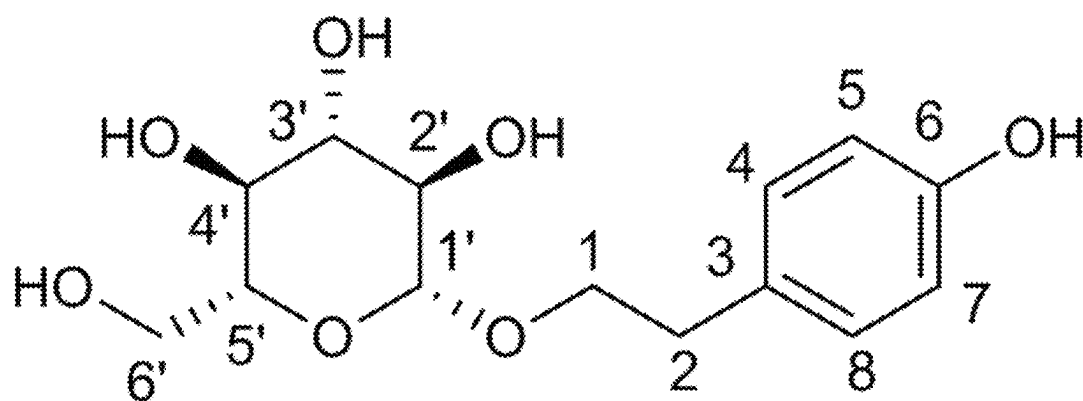


FIG. 20

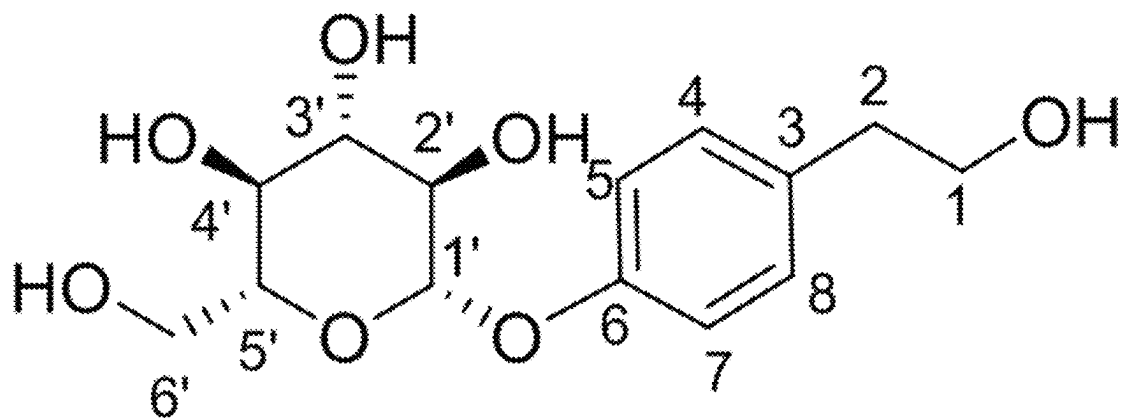


FIG. 21

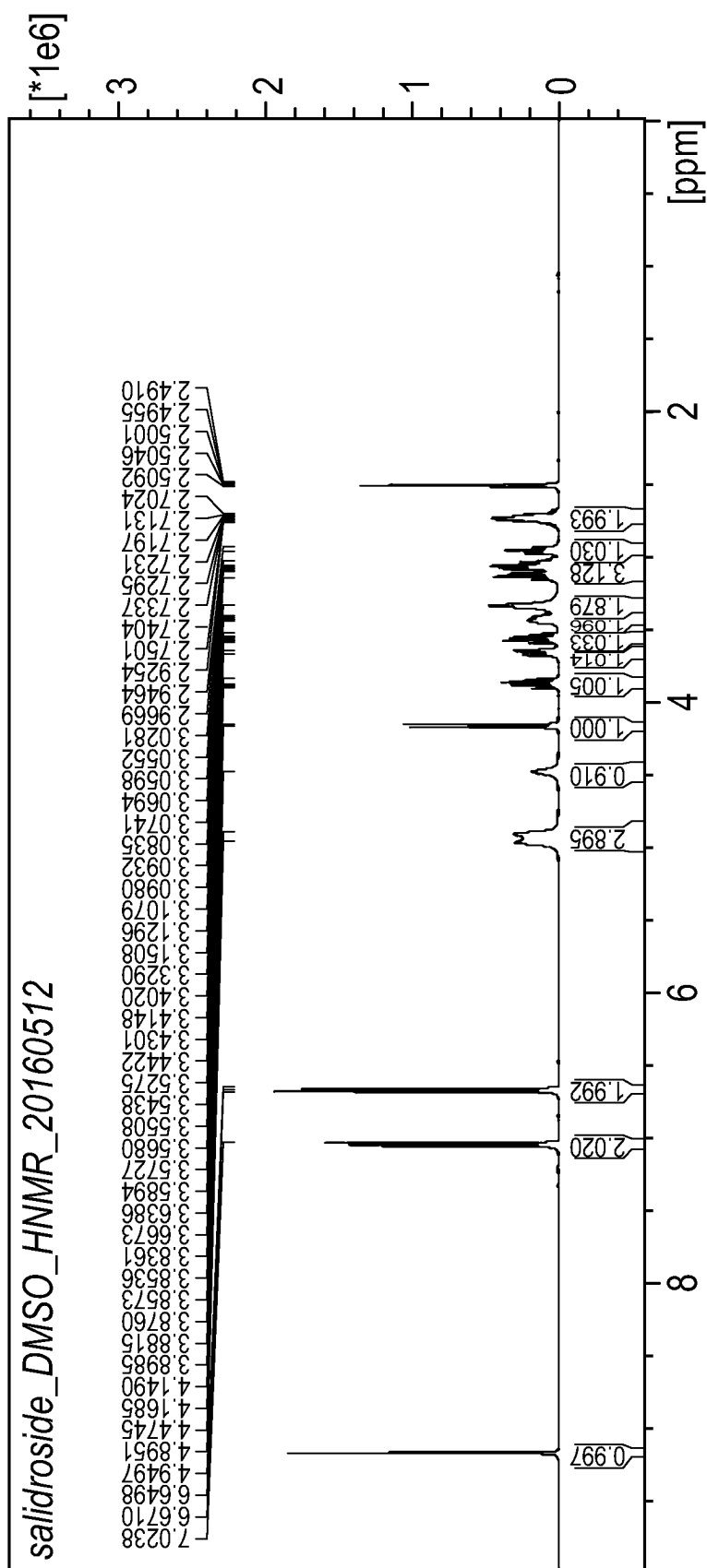


FIG. 22

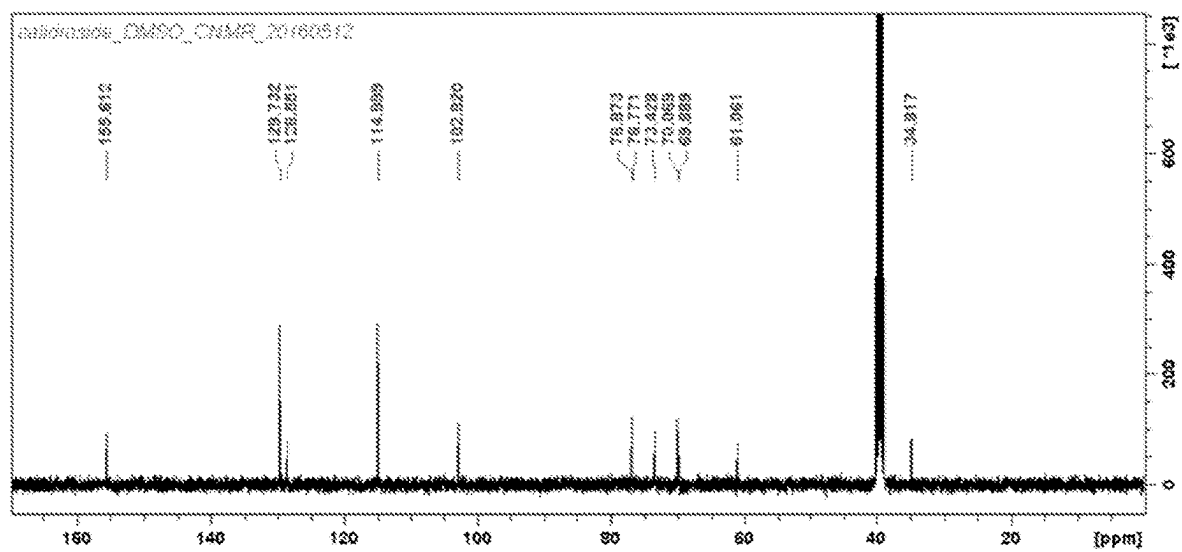


FIG. 23

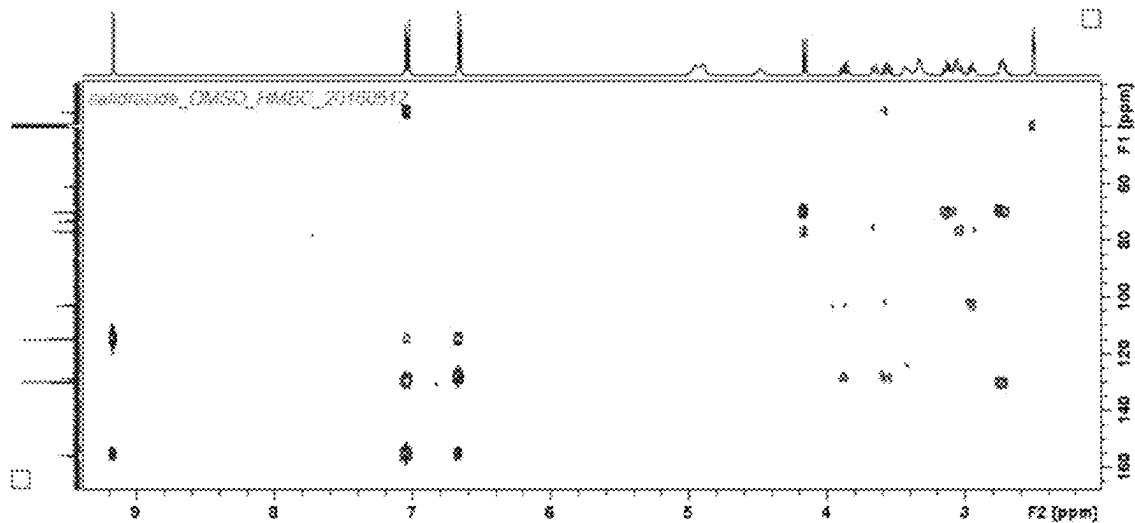


FIG. 24

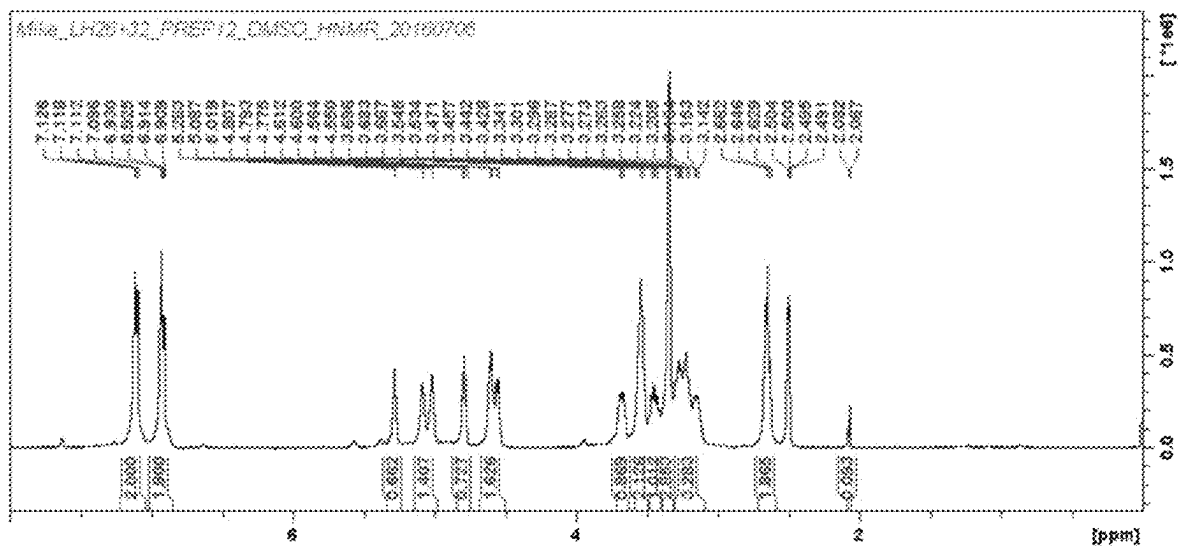


FIG. 25

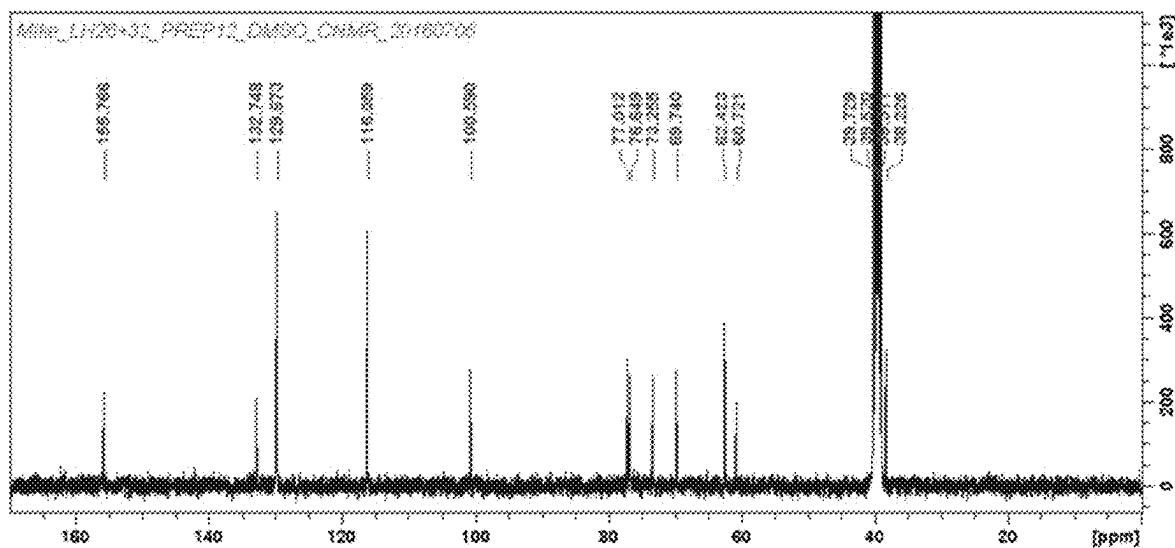


FIG. 26

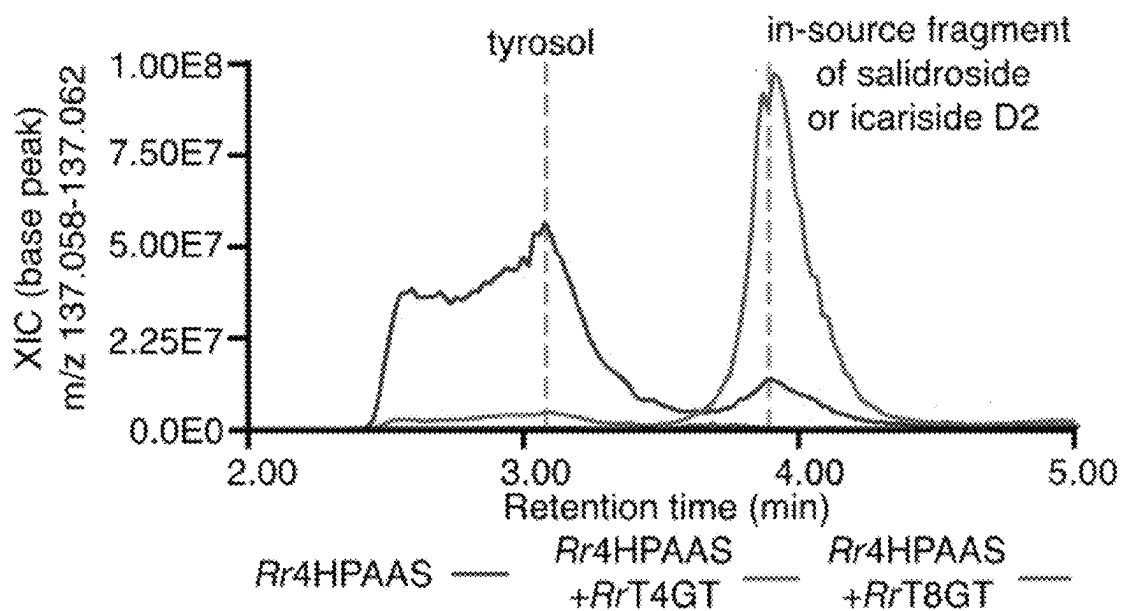


FIG. 27

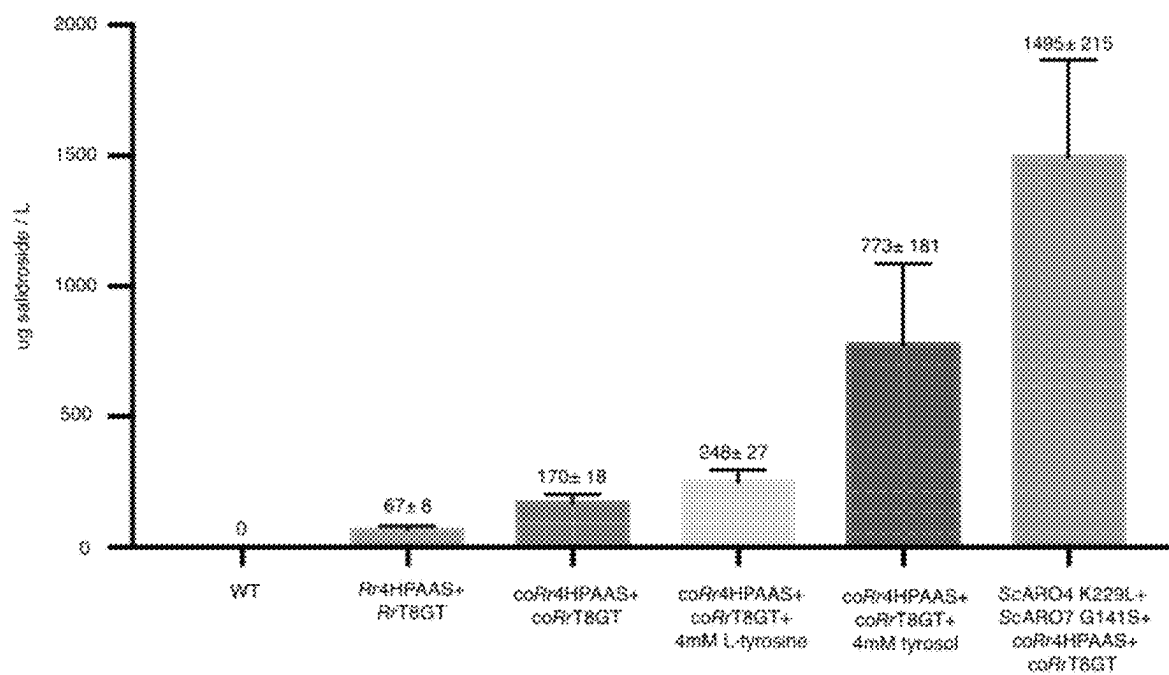


FIG. 28

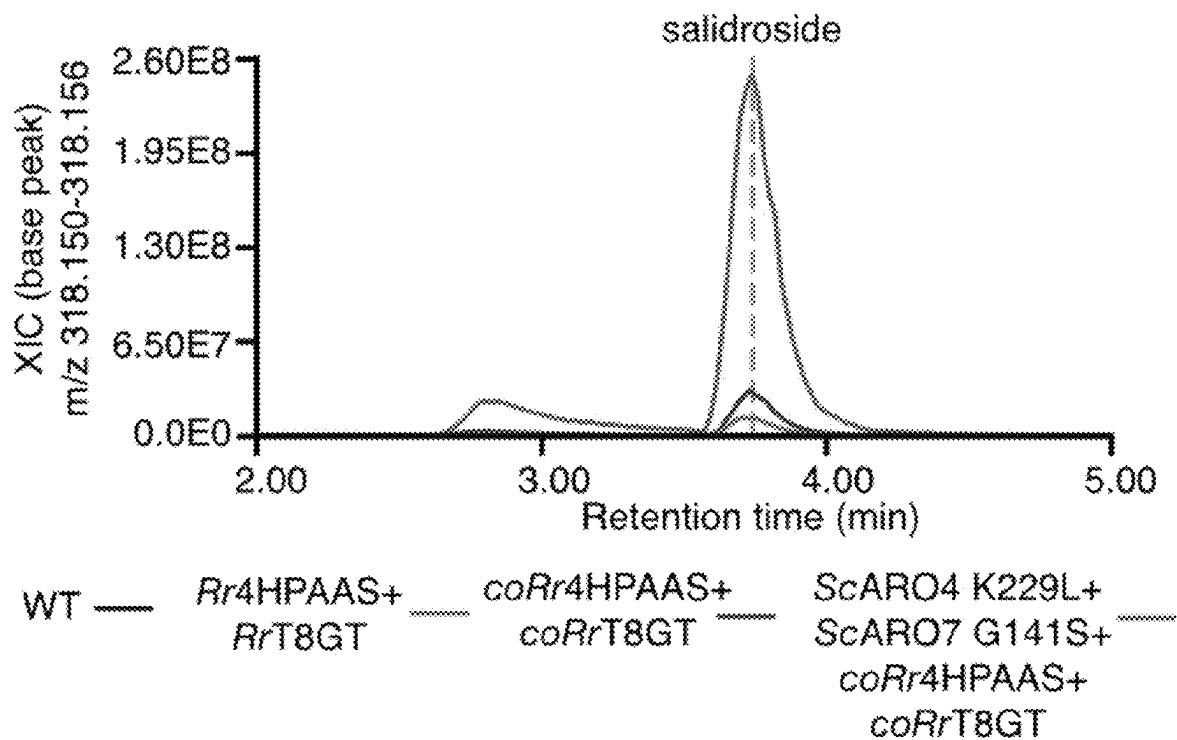


FIG. 29A

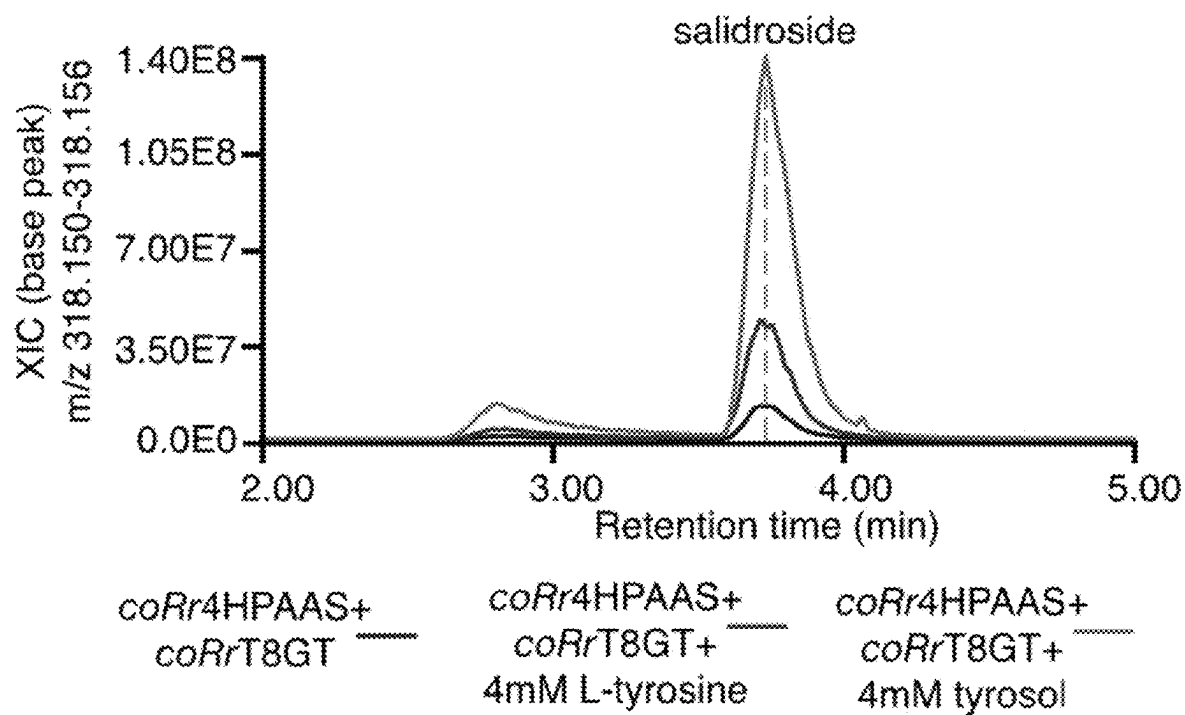


FIG. 29B

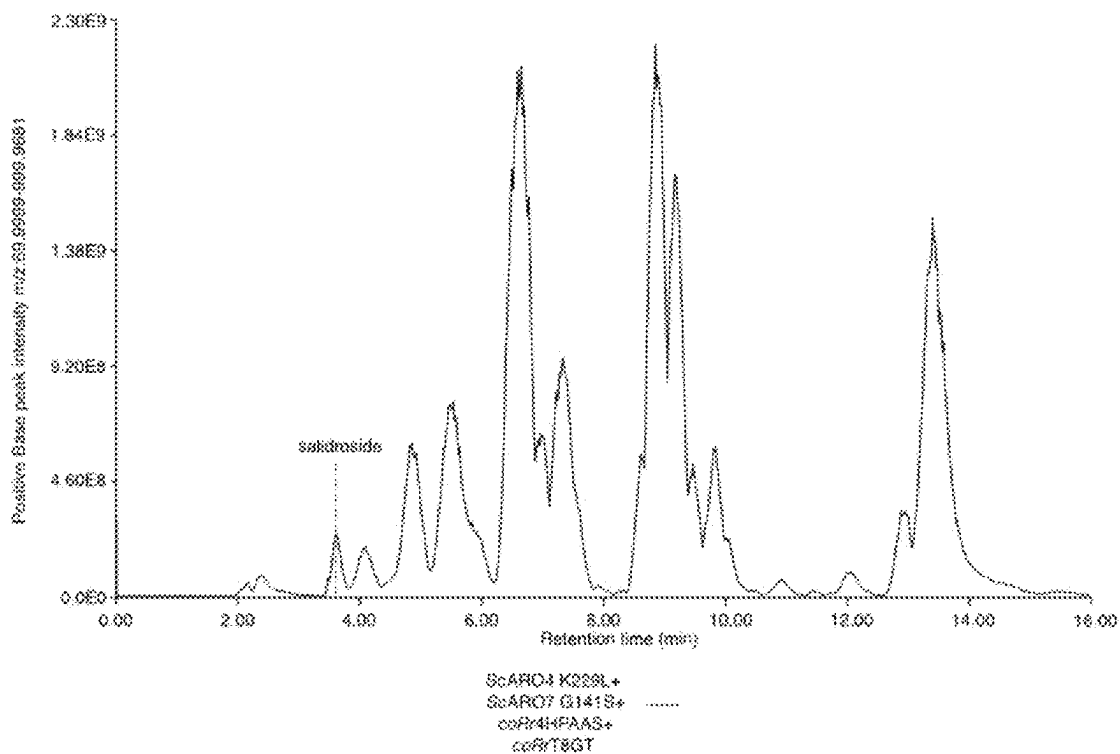


FIG. 30

	190	198	332	343
R. rosea 4HPAAS MF674522	VVLCSDDTHTFTIHGAK	ALTESAAEANFLK...	GGSEM	VV
A. thaliana PAAS NP 849999	VVLCSDDTHTSALQAAQ	SLTLASTNPEFLKN...	KASQANL	VV
A. thaliana TyDC NP 0010784 61	VVLCSDDTHTSFRACL	SLIDAKTNPEYLFK	VXVSKKDT	VV
C. annuum TDC ACN62126	VVLCSDDTHTSSLOAAQ	ALIQSSTNPEYLRN...	KASQGNL	VV
O. sativa TDC AK103253	VVLCSDDTHTSALQAAQ	FLIQSSTNPEFLKN...	KASQANS	VV
O. sativa TDC XP 015648701	AVLAADDTHTSTFFACR	RLTGSSTNPEYLRN...	HASDSGE	VT
C. acuminata TDC AAB39708	VVLCSDDTHTSTYAAACK	LLVKAISTDPEYLRN...	QPESKSV	VV
C. acuminata TDC AAB39709	VVLCSDDTHTSTYAAACN	VLVKAISTDPEYLRN...	KPSESNS	VV
C. annuum TDC NP 001312016	VVLCSDDTHTSMYAAACK	VLVKAISTNPEYLRN...	KRSEHGS	VV
O. pumila TDC BAC41515	VVLCSDDTHTSFFQTCK	LMVKAISTNPEYLRN...	KRSEFDG	VV
C. roseus TDC P17770	VVLCSDDTHTMTFTCK	LLLRATSTNPEYLRN...	XQSOLDK	VV
P. crispum 4HAAS Q06086	VVLCSDDTHTSALQAAK	ALIKSSTYPEFLKN...	NASETNK	VV
R. hybrid PAAS ABB04522	VVLCSDDTHTSTLOAAQ	VLAASSTNPEFLRN...	KASDSKQ	VV
P. hybrida PAAS ABB72475	VVLCSDDTHTSFRQAAV	SLTKASTNPEYLRN...	DATDSEQ	VV
T. flavum TyDC AAG60665	VVLCSDDTHTCALQAAQ	ALIKASTNPEYLRN...	KATESHQ	VV
B. distachyon TyDC XP 00356 9907	VVLCSDDTHTFAFRQAAQ	ELIAAGTEQEQYILK	D.SASEGH	IV
P. somniferum TyDC AAC61842	VVLCASNTHTCALQAAQ	SLVKAISTSAEYLRN...	KATESKQ	VV

FIG. 31

R. rosea_4HPAAS_MF674522	80	90	100	110	120	130
A. thaliana_PAAS_NP_849999	81	91	101	111	121	131
A. thaliana_TyDC_NP_001078461	82	92	102	112	122	132
C. annuum_TDC_ACN62126	83	93	103	113	123	133
O. sativa_TDC_AK103253	84	94	104	114	124	134
O. sativa_TDC_NP_015648701	85	95	105	115	125	135
C. acuminata_TDC_AAB39708	86	96	106	116	126	136
C. annuum_TDC_NP_001312016	87	97	107	117	127	137
O. pumila_TDC_BAC41515	88	98	108	118	128	138
C. roseus_TDC_P17770	89	99	109	119	129	139
P. crispum_4HAAS_Q06086	90	100	110	120	130	140
R. hybrid_PAAS_ABB04522	91	101	111	121	131	141
P. hybrida_PAAS_ABB72475	92	102	112	122	132	142
T. flavum_TyDC_AAG60665	93	103	113	123	133	143
B. distachyon_TyDC_XP_003569907	94	104	114	124	134	144
P. somniferum_TyDC_AAC61842	95	105	115	125	135	145
R. rosea_4HPAAS_MF674522	140	150	160	170	180	190
A. thaliana_PAAS_NP_849999	141	151	161	171	181	191
A. thaliana_TyDC_NP_001078461	142	152	162	172	182	192
C. annuum_TDC_ACN62126	143	153	163	173	183	193
O. sativa_TDC_AK103253	144	154	164	174	184	194
O. sativa_TDC_NP_015648701	145	155	165	175	185	195
C. acuminata_TDC_AAB39708	146	156	166	176	186	196
C. annuum_TDC_NP_001312016	147	157	167	177	187	197
O. pumila_TDC_BAC41515	148	158	168	178	188	198
C. roseus_TDC_P17770	149	159	169	179	189	199
P. crispum_4HAAS_Q06086	150	160	170	180	190	200
R. hybrid_PAAS_ABB04522	151	161	171	181	191	201
P. hybrida_PAAS_ABB72475	152	162	172	182	192	202
T. flavum_TyDC_AAG60665	153	163	173	183	193	203
B. distachyon_TyDC_XP_003569907	154	164	174	184	194	204
P. somniferum_TyDC_AAC61842	155	165	175	185	195	205

FIG. 32 (continued)

R. rosea 4HPAAS MF674522	190	200	210	220	230	240
A. thaliana PAAS NP 849999	N...	KLVLN	CSUQTH	IIH	GA	KI
A. thaliana TyDC NP 001078461	E...	KLVLN	SSUQTH	SALQ	ACQIA	CIHRE
C. annuum TDC ACN62126	P...	QVLVN	GSEUQTH	SSER	NACLI	GC
O. sativa TDC AK103253	S...	KLVLN	CSEUQTH	SLO	KACQIA	COIG
O. sativa TDC XP 015648701	E...	KLVLN	ASEUQTH	SALO	KACQIA	CIHRE
C. acuminata TDC AAB39708	AC	HLRIAN	AAUQTH	SIFER	KACRI	LA
C. acuminata TDC AAB39709	T...	SEVLN	GSEUQTH	SIYA	KACMLA	CIHRE
C. annuum TDC NP 001312016	H...	KLVLN	GSEUQTH	SMYA	KACMLA	CIHRE
O. pumila TDC BAC41515	G...	KLVLN	GSEUQTH	PEQ	TC	KVA
C. roseus TDC P1770	G...	KLVLN	GSEUQTH	MEP	TC	KVA
P. crispum 4HAAS Q06086	G...	KLVLN	GSEUQTH	SALO	KACMLA	CIHRE
R. hybrid PAAS ABB04522	G...	KLVLN	GSEUQTH	SILO	KACMLA	CIHRE
P. hybrida PAAS ABB72475	D...	KLVLN	ASEUQTH	SEFO	KACMLA	CIHRE
T. flavum TyDC AAG60665	C...	KLVLN	GSEUQTH	CALO	KACMLA	CIHRE
B. distachyon TyDC XP 003569907	G...	KLVLN	ASEUQTH	FAER	KACMLA	CIHRE
P. somniferum TyDC AAC61842	N...	KLVLN	ASUQTH	CALO	KACMLA	CIHRE

R. rosea 4HPAAS MF674522	250	260	270	280	290	300
A. thaliana PAAS NP 849999	GLVPE	YV	CGII	CH	ALGV	VP
A. thaliana TyDC NP 001078461	GLVPE	YV	CGII	CH	ALGV	VP
C. annuum TDC ACN62126	GLVPE	YV	CGII	CH	ALGV	VP
O. sativa TDC AK103253	GLVPE	YV	CGII	CH	ALGV	VP
O. sativa TDC XP 015648701	GLVPE	YV	CGII	CH	ALGV	VP
C. acuminata TDC AAB39708	GLVPE	YV	CGII	CH	ALGV	VP
C. acuminata TDC AAB39709	GLVPE	YV	CGII	CH	ALGV	VP
C. annuum TDC NP 001312016	GLVPE	YV	CGII	CH	ALGV	VP
O. pumila TDC BAC41515	GLVPE	YV	CGII	CH	ALGV	VP
C. roseus TDC P1770	GLVPE	YV	CGII	CH	ALGV	VP
P. crispum 4HAAS Q06086	GLVPE	YV	CGII	CH	ALGV	VP
R. hybrid PAAS ABB04522	GLVPE	YV	CGII	CH	ALGV	VP
P. hybrida PAAS ABB72475	GLVPE	YV	CGII	CH	ALGV	VP
T. flavum TyDC AAG60665	GLVPE	YV	CGII	CH	ALGV	VP
B. distachyon TyDC XP 003569907	GLVPE	YV	CGII	CH	ALGV	VP
P. somniferum TyDC AAC61842	GLVPE	YV	CGII	CH	ALGV	VP

FIG. 32 (continued)

R. rosea_4HPAAS_MF674522	LVDS	ISMNA	HKW	FLTNF	CELVLC	SPNA	LIES	AAEA	NFLK	..	..	GGSE	MDV	MDV	MDV
A. thaliana_PAAS_NP_849999	TAOS	FMNNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KASQ	CANL	VVDI	KDWS
A. thaliana_TyDC_NP_001078461	NADS	FMNNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. annuum_TDC_ACN62126	EAOS	FMNNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. sativa_TDC_AK103253	EAOS	FMNNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. sativa_TDC_XP_015648701	RVDS	ISMSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. acuminata_TDC_AAB39708	RACS	LSLSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. acuminata_TDC_AAB39709	RACS	LSLSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. annuum_TDC_NP_001312016	RACS	LSLSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. pumila_TDC_BAC41515	RACS	LSLSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. roseus_TDC_P1770	RACS	LSLSP	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. crispum_4HAAS_Q06086	NADS	FSLNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
R. hybrid_PAAS_ABB04522	GANS	FSFNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. hybrida_PAAS_ABB72475	GANS	FSFNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
T. flavum_TyDC_AAG60665	NADS	FSLNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
B. distachyon_TyDC_XP_003569907	SVDS	FSMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. somniferum_TyDC_AAC61842	DAOS	FSLNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS

R. rosea_4HPAAS_MF674522	360	370	380	390	400	410	420	430	440	450	460	470	480	490	500
A. thaliana_PAAS_NP_849999	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
A. thaliana_TyDC_NP_001078461	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. annuum_TDC_ACN62126	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. sativa_TDC_AK103253	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. sativa_TDC_XP_015648701	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. acuminata_TDC_AAB39708	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. acuminata_TDC_AAB39709	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. annuum_TDC_NP_001312016	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
O. pumila_TDC_BAC41515	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
C. roseus_TDC_P1770	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. crispum_4HAAS_Q06086	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
R. hybrid_PAAS_ABB04522	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. hybrida_PAAS_ABB72475	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
T. flavum_TyDC_AAG60665	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
B. distachyon_TyDC_XP_003569907	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS
P. somniferum_TyDC_AAC61842	ISLSP	ISMNA	HKW	FLTNF	CELVLC	QDQS	LTIA	STNP	EFLK	..	..	KVSK	QANL	VVDI	KDWS

FIG. 32 (continued)

R. rosea_4HPAAS_MF674522	420	430	440	450
A. thaliana_PAAS_NP_849999	GRUV	EKIRPGR.....	DSELRRELMER.....	AYLGGVQMGRIE
A. thaliana_TyDC_NP_001078461	GRUAE	VKDEEKK.....	CNERREKELLAAVNS	EFWSHTALSCKI
C. annuum_TDC_ACN62126	GRULI	VDGDEDO.....	CNERREKELLAAVNS	EFISHTALSCKE
O. sativa_TDC_AK103253	GRULV	PESNEDZ.....	ANKEKRELLDSVNS	EFISHTLLSDRY
O. sativa_TyDC_NP_015648701	GRIRAG	PISDHEH.....	GRKENYDMMEGVNS	EFLSHTVLSCKE
C. acuminata_TDC_AAB39708	GRUNP	ACAAAATEED.....	ADEANREKELMERLNS	AYVAHTVVGGRF
C. annuum_TDC_NP_001312016	GRUNP	ISGSD.....	PCTEALNKKELLEWNS	VYMTHTKVGGIY
O. pumila_TDC_EAC41515	PRENE	FSKSD.....	PCNTEALNKKELLEWNS	VYMTHTKVGGIY
C. roseus_TDC_P1770	PRINP	DKEYE.....	PAYTELNKKELLEWNS	VYMTHTKVGGIY
P. crispum_4HAAS_Q06086	PRINP	GKCYDD.....	EIDKEIKSKELELINS	AYMTHTKAGGIY
R. hybrid_PAAS_ABB04522	PRINP	DVSSL.....	HVEEVNKKELLEMLNS	VYMTHTKVGGIY
P. hybrida_PAAS_ABB72475	PRINP	SAMIGKNDEDE.....	VNEINNKELLEMLNS	IYVSHTVLGGIY
T. flavum_TyDC_AAG60665	PRINP	SALISGNDEDE.....	ICMVNEVNCKELLE	AYMTHAVVGGIY
B. distachyon_TyDC_XP_003569907	PRVGL	LALEKKNFVD.....	ETQVNEFNARLLE	VYMTHTVVEGGY
P. somniferum_TyDC_AAC61842	PRVGL	PRSPLIITKINGYQNGVYH	KDESRANELMERLLE	AYMTHTVVEGGY
	PRVGL	PEKYGGEK.....	TANELNKSLEEVMA	VYMTHTVVEGGY
	PRVGL	AAIFKQKIVDN.....	DYIEDQINENVNARLLE	VYMTHTVVEGGY

R. rosea_4HPAAS_MF674522	460	470	480	490
A. thaliana_PAAS_NP_849999	ELRCVIG	SGLTEER.....	VDMER.....	QETIQOSIVUS
A. thaliana_TyDC_NP_001078461	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
C. annuum_TDC_ACN62126	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
O. sativa_TDC_AK103253	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
O. sativa_TyDC_NP_015648701	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
C. acuminata_TDC_AAB39708	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
C. acuminata_TDC_AAB39709	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
O. pumila_TDC_NP_001312016	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
C. roseus_TDC_P1770	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
P. crispum_4HAAS_Q06086	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
R. hybrid_PAAS_ABB04522	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
P. hybrida_PAAS_ABB72475	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
T. flavum_TyDC_AAG60665	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
B. distachyon_TyDC_XP_003569907	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS
P. somniferum_TyDC_AAC61842	VLRCVIG	CAPLTEER.....	VDMER.....	QETIQOSIVUS

FIG. 32 (continued)

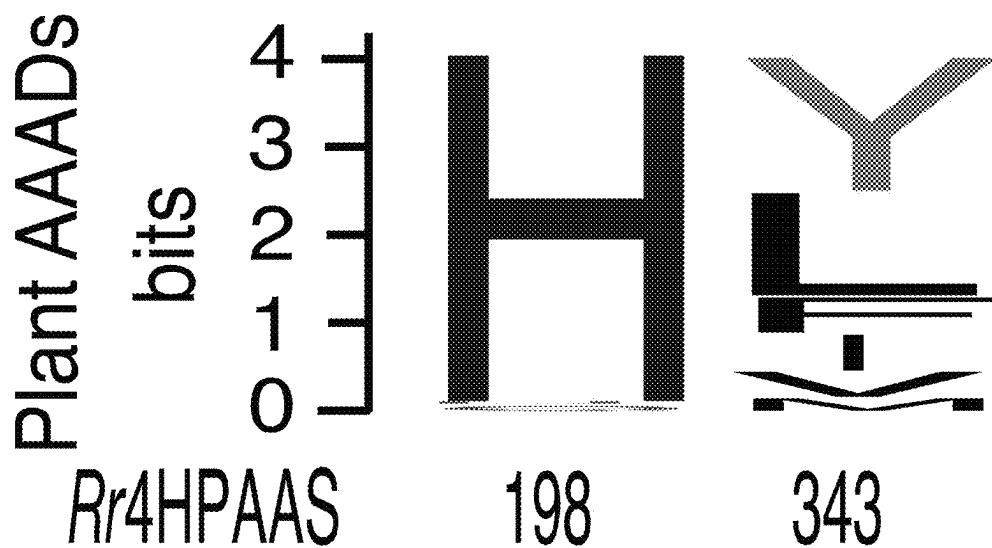


FIG. 33

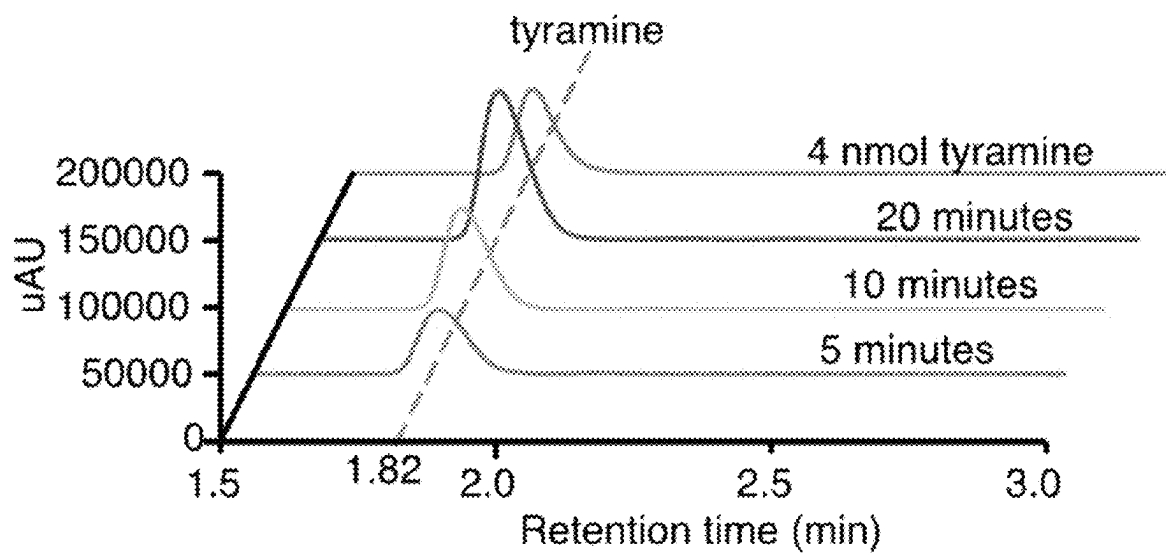


FIG. 34A

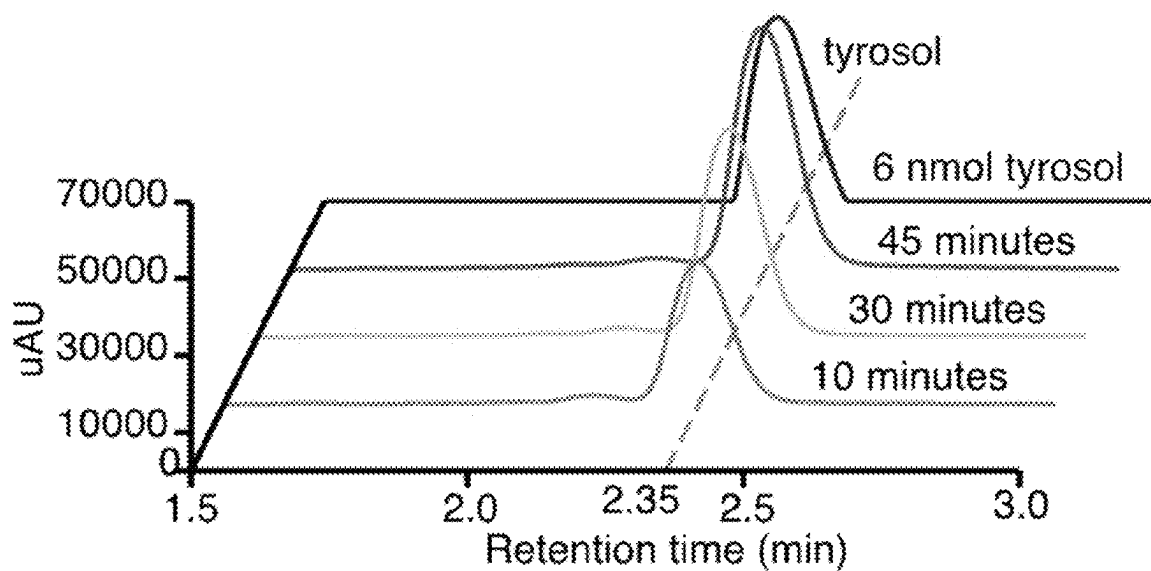


FIG. 34B

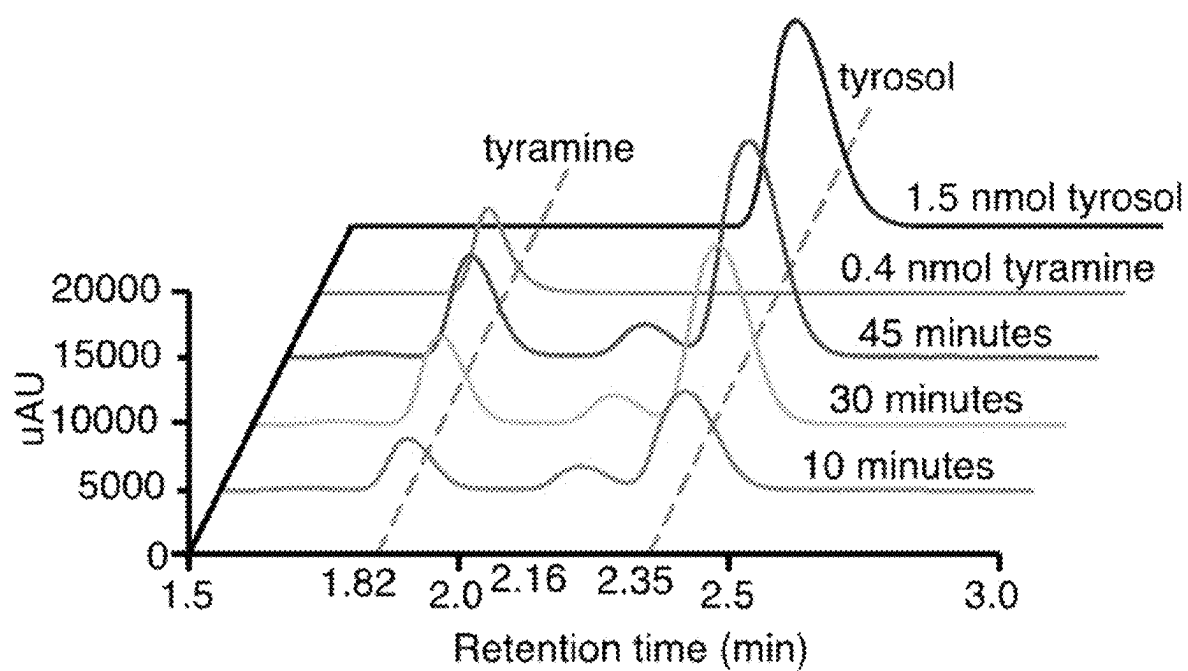


FIG. 34C

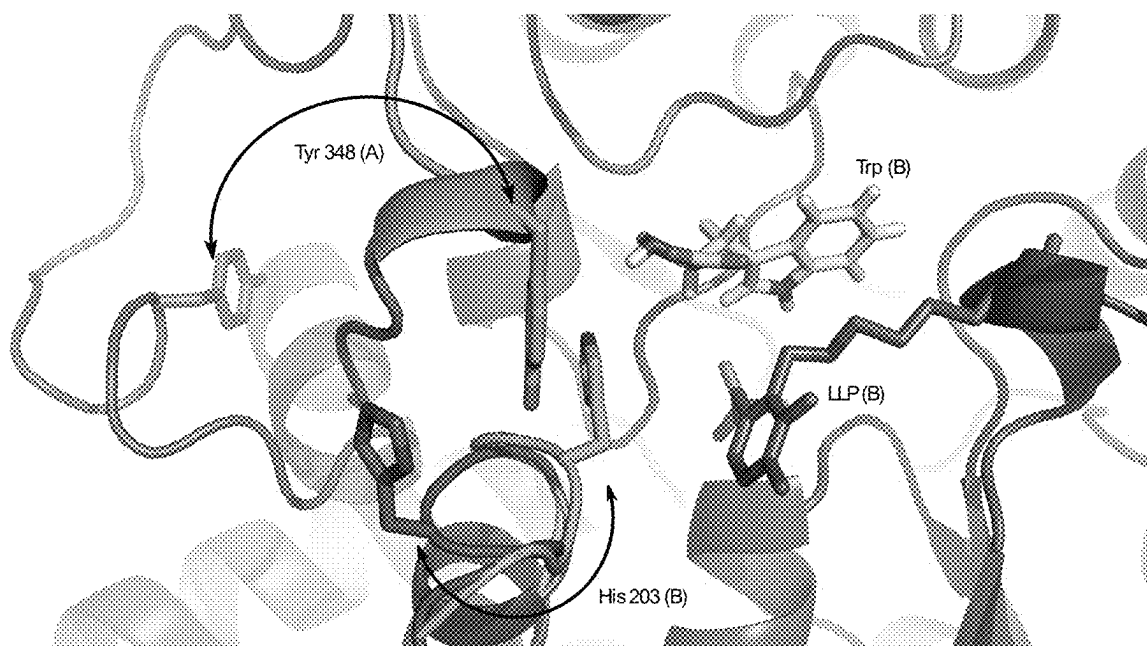


FIG. 35

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COMPOSITIONS AND METHODS FOR PRODUCTION OF SALIDROSIDE, ICARISIDE D2, AND PRECURSORS OF SALIDROSIDE AND ICARISIDE D2

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/224,257 (now U.S. Pat. No. 11,408,009), filed on Dec. 18, 2018, which claims the benefit of U.S. Provisional Application No. 62/607,271, filed on Dec. 18, 2017. The entire teachings of the above applications are incorporated herein by reference.

INCORPORATION BY REFERENCE OF MATERIAL IN ASCII TEXT FILE

This application incorporates by reference the Sequence Listing contained in the following ASCII text file being submitted concurrently herewith:

File name: 03992060002_SL.txt; created May 2, 2025, 506,369 Bytes in size.

BACKGROUND

Salidroside, also known as tyrosol 8-O-glucoside, is naturally produced by plants within the *Rhodiola* genus. Salidroside is of particular interest and value because of its unique reported biological activities (Cifani et al., 2010; Guan et al., 2012; Panossian et al., 2014). However, commercially available salidroside in its pure form is currently obtained through a lengthy purification process from its native plant host, which poses a significant bottleneck hindering further clinical development of salidroside as a potential therapeutic agent. Accordingly, improved methods of making salidroside are needed.

SUMMARY

Salidroside is a bioactive tyrosine-derived phenolic natural product found in medicinal plants under the *Rhodiola* genus. In addition to their anti-fatigue and anti-anoxia roles in traditional medicine, *Rhodiola* total extract and salidroside have also displayed medicinal properties as anti-cardiovascular disease, and anti-cancer, agents. The resulting surge in global demand of *Rhodiola* plants and salidroside has driven some species close to extinction.

Described herein is a *Rhodiola* salidroside biosynthetic pathway that was elucidated utilizing comprehensive transcriptomics and metabolomics datasets for *Rhodiola rosea*. This pathway includes a pyridoxal phosphate (PLP)-dependent 4-hydroxyphenylacetaldehyde synthase (4HPAAS) that directly converts tyrosine to 4-HPAA. Genes encoding the subsequent 4-HPAA reductase (4HPAR) and tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), respectively, were further identified to complete salidroside biosynthesis in *Rhodiola*. As described herein, heterologous production of salidroside can be achieved in yeast *Saccharomyces cerevisiae* as well as in plant *Nicotiana benthamiana* through transgenic expression of *Rhodiola* salidroside biosynthetic genes. Accordingly, the methods and compositions described herein provide useful tools for engineering sustainable production of salidroside in heterologous hosts.

Described herein are vectors and kits that include vectors. Those vectors include a nucleic acid encoding one or more of a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), a

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4-hydroxyphenylacetaldehyde reductase (4HPAR), a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), and a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). Described herein are methods of using the vectors and kits to make a transgenic host cell having a transgene encoding one or more of a 4HPAAS, a 4HPAR, a T8GT, and a T4GT. Described herein are methods of making one or more of 4-hydroxyphenylacetaldehyde (4-HPAA), tyrosol, tyrosol 8-O-glucoside (salidroside), and icaraside D2 in a transgenic host cell. The tyrosol, salidroside, and/or icaraside D2 can subsequently be obtained, e.g., by separation and purification processes. A variety of transgenic host cells can be used, such as yeast cells, plant cells, and bacterial cells. In some embodiments, the tyrosol, tyrosol 8-O-glucoside (salidroside), or icaraside D2 can be obtained in greater quantities than by purification from the native plant host. In some embodiments, the tyrosol, tyrosol 8-O-glucoside (salidroside), or icaraside D2 can be obtained more cost-effectively than by purification from the native plant host.

Certain embodiments provide a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), wherein the 4HPAAS has at least 70% sequence identity to either SEQ ID NO: 2 (*Rhodiola rosea* 4HPAAS), or a biologically active fragment thereof. The 4HPAAS includes: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof.

Certain embodiments provide a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR), wherein the 4HPAR includes at least 70% amino acid sequence identity to SEQ ID NO: 4, or a biologically active fragment thereof.

Certain embodiments provide a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the T8GT comprises a plant secondary product glycosyltransferase (PSPG) motif. In some embodiments, the T8GT comprises at least 70% amino acid sequence identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, or a biologically active fragment thereof.

Certain embodiments provide a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the T4GT comprises a plant secondary product glycosyltransferase (PSPG) motif. In some embodiments, the T4GT comprises at least 70% amino acid sequence identity to one or more of SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, or a biologically active fragment thereof.

Certain embodiments provide a kit that includes: a) a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS); b) a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR); and c) one or more of i) a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) and ii) a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the kit includes both a T8GT and a T4GT.

Some embodiments provide a host cell that includes a transgene encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell further includes a transgene encoding 4-hydroxyphenylac-

etaldehyde reductase (4HPAR). In some embodiments, the host cell further includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a 4HPAAS. In some embodiments, a single transgene encodes multiple genes, such as one or more of the T8GT, the 4HPAR, and the 4HPAAS. In some embodiments, separate transgenes encode one or more of T8GT, 4HPAR, and 4HPAAS.

Some embodiments provide a host cell that includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell further includes a transgene encoding 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further includes a transgene encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell further includes a transgene encoding tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a T8GT. In some embodiments, the host cell further includes a transgene encoding both a 4HPAR and a T4GT. In some embodiments, a single transgene encodes multiple genes, such as one or more of the 4HPAAS, the 4HPAR, the T8GT, and the T4GT. In some embodiments, separate transgenes encode one or more of the 4HPAAS, the 4HPAR, the T8GT, and the T4GT.

Some embodiments provide a method of making a transgenic host cell. The method can include introducing a vector into the host cell, wherein the vector includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

Some embodiments provide a method of making a transgenic host cell. The method can include introducing a vector into the host cell, wherein the vector includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The method can further include introducing into the host cell a vector that includes a nucleic acid encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) or a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT). In some embodiments, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a T8GT and a nucleic acid encoding a T4GT.

Certain embodiments provide a method of making tyrosol 8-O-glucoside (salidroside). In some embodiments, the salidroside is made in a host cell. In certain embodiments, the salidroside is made in a cell-free system or cell lysate. The method can include expressing in a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell includes tyrosol, either produced endogenously or provided to the cell exogenously.

In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, tyrosol is secreted by the host cell into the cell culture media, from which it can be obtained.

Certain embodiments provide a method of making 4-hydroxyphenylacetaldehyde (4-HPAA). In some embodiments, the 4-HPAA is made in a host cell. In some embodiments, the 4-HPAA is made in a cell-free system or lysate. The method can include expressing in the host cell a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell includes L-tyrosine, produced endogenously or provided to the cell exogenously. In some embodiments, the method further includes making tyrosol in the host cell, and the host cell further expresses a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, tyrosol is secreted by the host cell into the cell culture media, from which it can be obtained.

In some embodiments, the host cell is a yeast cell, such as *Saccharomyces cerevisiae*. In some embodiments, the host cell is a plant cell, such as a cell from a *Nicotiana benthamiana* plant. In some embodiments, the host cell is a bacterial cell, such as *Escherichia coli* or *Agrobacterium tumefaciens*.

In some embodiments, nucleic acids encoding two or more of 4HPAAS, 4HPAR, T8GT, and T4GT are included in a single vector. In some embodiments, the transgene encoding an enzyme (e.g., 4HPAAS, 4HPAR, T8GT, and T4GT) can be integrated into the genome of the host transgenic cell.

Certain embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT).

Some embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT).

Some embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

Certain embodiments provide a transgenic plant, such as a *Nicotiana benthamiana* plant, that includes a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR).

Certain embodiments provide an isolated deoxyribonucleic acid (DNA) coding sequence encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the nucleic acid includes SEQ ID NO: 13. In some embodiments, the nucleic acid includes SEQ ID NO: 15. In some embodiments, the nucleic acid includes SEQ ID NO: 17. In some embodiments, the nucleic acid includes SEQ ID NO: 19.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

The foregoing will be apparent from the following more particular description of example embodiments, as illustrated in the accompanying drawings in which like reference characters refer to the same parts throughout the different views. The drawings are not necessarily to scale, emphasis instead being placed upon illustrating embodiments.

FIGS. 1A-C show salidroside biosynthesis in *R. rosea*. FIG. 1A is a photograph of greenhouse-grown *R. rosea*. FIG. 1B is a chromatogram showing metabolic profiling of *R. rosea* root and crown tissues by LC-HRAM-MS. Enrichment of tyrosol and salidroside is observed in the root.

Extracted ion chromatogram (XIC) is shown with mass windows set to display the $[M-H]^-$ ion for tyrosol and the $[M+NH_4]^+$ ion for salidroside. The identity of the metabolites was verified in comparison to authentic standards. FIG. 1C is a schematic showing alternative salidroside biosynthetic pathways in *Rhodiola*.

FIGS. 2A-C show identification and characterization of the Rr4HPAAS. FIG. 2A is a simplified maximum likelihood (ML) phylogenetic tree of land plant AAADs. A fully annotated version of this tree is shown in FIG. 11. The three major groups of the tree have been annotated as the basal (green), TyDC (blue) and TDC (red) clades based on taxonomic distribution, cladding, and conservation of the substrate-specifying active site residue. Representative characterized enzymes are labeled at the tree branches, while the *R. rosea* TDC, AAS and 4HPAAS are displayed in bold. The scale measures evolutionary distances in substitutions per amino acid. FIG. 2B is LC-UV chromatograms of the reaction product of L-tyrosine and Rr4HPAAS enzyme (with and without $NaBH_4$ reduction) in comparison to enzyme assay conducted using PsTyDC as a control. The identity of the products was verified by comparison with authentic standards. FIG. 2C is a graph showing kinetic characterization of Rr4HPAAS against various aromatic amino acid substrates.

FIGS. 3A-C show identification and characterization of two *R. rosea* 4HPARs. FIG. 3A is a simplified ML phylogenetic tree of angiosperm ADHs. A fully annotated version of this tree is shown in FIG. 14. Major clades are annotated based on representative characterized enzymes when possible. The two *R. rosea* 4HPARs and the previously characterized SIPARs are labeled at the tree branches. The scale measures evolutionary distances in substitutions per amino acid. FIG. 3B is LC-MS chromatograms of the reaction product of 4-HPAA and 0.2 μ g recombinant Rr4HPAR1 after incubation for various time points. FIG. 3C is LC-MS chromatograms of the reaction product of 4-HPAA and 15 μ g recombinant Rr4HPAR2 after incubation for various time points. The identity of the tyrosol product was verified by comparison with an authentic standard.

FIGS. 4A-C show identification and characterization of *R. rosea* tyrosol-modifying UGTs. FIG. 4A is a maximum likelihood (ML) phylogenetic tree of 34 *R. rosea* UGTs together with 88 full-length UGTs encoded by the *A. thaliana* genome. UGTs that show T4GT and T8GT activities are denoted by black circles and stars, respectively. Bootstrap values (based on 500 replicates) are indicated at the major nodes. The scale measures evolutionary distances in substitutions per amino acid. FIG. 4B is a chart showing relative in vivo T4GT and T8GT activities of *R. rosea* UGTs as examined in engineered yeast. FIG. 4C is a graph showing Michaelis-Menten kinetic characterization of four *R. rosea* tyrosol-modifying UGTs.

FIGS. 5A-C show heterologous production of tyrosine-derived metabolites in transgenic *N. benthamiana* as detected by LC-HRAM-MS. FIG. 5A is a chromatogram showing that *N. benthamiana* transiently expressing Rr4HPAAS or Pc4HPAAS produces both salidroside and icaricide D2. FIG. 5B is a chromatogram showing that *N. benthamiana* transiently expressing PsTyDC produces tyramine. FIG. 5C is a chromatogram showing that *N. benthamiana* leaves transiently co-expressing Rr4HPAAS and RrT4GT or RrT8GT produce predominantly icaricide D2 or salidroside, respectively. XICs are shown with mass windows set to display the $[M+NH_4]^+$ ion for salidroside and

icaricide D2, and the $[M+H]^+$ ion for tyramine. The identity of the metabolites was verified by comparison with authentic standards.

FIG. 6 shows the chemical structures of a number of *Rhodiola* glycosylated natural products.

FIG. 7 is a multiple sequence alignment highlighting the sequence regions that can influence enzyme substrate selectivity in select plant AAAD family members. Sequences represent various enzymes from the TyDC and TDC clades. The residue framed in black (identified as Gly 370 for *C. roseus*) can impact substrate selectivity (indolic vs. phenolic). Columns framed in blue indicate greater than 70% conservation of residue physico-chemical properties. Identical amino acids are in white font boxed in red, while similar residues are displayed in red font. FIG. 7 includes SEQ ID Nos. 207 through 219, in order from top-to-bottom.

FIGS. 8A-B show total ion count of the root and crown *R. rosea* extractions. FIG. 8A is a chromatogram of positive ion mode metabolites. FIG. 8B is a chromatogram of negative ion mode metabolites.

FIG. 9 is a chromatogram showing relative abundance of rosavin between *R. rosea* tissue types. The differential location of the natural product rosavin $[M+NH_4]^+$ ion between *R. rosea* root and crown. The identity of rosavin was verified by comparison to an authentic standard.

FIG. 10 is a multiple sequence alignment highlighting the residue that dictate decarboxylation and aldehyde synthase chemistry in plant AAADs family members. Sequences represent plant AAADs performing either decarboxylation chemistry or aldehyde synthase chemistry (highlighted in green). The three identified *R. rosea* AAAD sequences are also displayed. Investigation of the activity influencing residue (boxed in black) suggests that the *R. rosea* sequence from the TDC clade likely catalyzes decarboxylation chemistry while the basal and TyDC clade *R. rosea* AAS catalyze aldehyde synthase chemistry. Columns framed in blue indicate greater than 70% conservation of residue physico-chemical properties. Identical amino acids are in white font boxed in red, while similar residues are displayed in red font. FIG. 10 includes SEQ ID Nos. 220 through 240, in order from top-to-bottom.

FIG. 11 is a phylogenetic tree of embryophyte AAADs. This tree is populated with sequences from all Phytozome V12 embryophyte species, the three AAAD like sequences from the *R. rosea* transcriptomes (shown in bold) and all attainable characterized NCBI AAAD sequences (also shown in bold). Green, red and blue branches correspond to the basal, TDC and TyDC clades, respectively. These clades were determined through the application of the indolic substrate selective active site glycine (red clade), the phenolic substrate selective serine (blue clade), their taxonomic distribution (green clade exists in all sampled species and is most closely related to chlorophytes species) and representative characterized sequences.

FIG. 12 is a graph showing relative hydrogen peroxide production for the Rr4HPAAS, the RrAAS and the PsTyDC. 100 μ L reaction mixtures containing 50 μ g of recombinant enzyme, 2 mM L-tyrosine, 50 mM Tris pH 8.0 and 200 μ M PLP were incubated at 30° C. for various time points prior to quenching with 100 μ L of 0.8 M formic acid. Hydrogen peroxide levels of quenched reaction mixtures were subsequently analyzed using Pierce Quantitative Peroxide Assay Kit against a standard curve of hydrogen peroxide.

FIG. 13 is a chart showing relative TPM for the Rr4HPAAS transcript between the root and crown transcriptomes.

FIG. 14 is a phylogenetic tree of angiosperm ADHs. This tree is populated with sequences from various Phytozome V12 angiosperm species, ADLs like sequences from the *R. rosea* transcriptomes and two characterized SIPAR sequences. Characterized *A. thaliana* enzymes, the two SIPARs and the two Rr4HPARs are shown in bold. The different colors have been applied to distinguish between various clades.

FIG. 15 is a chromatogram showing the enzymatic reduction of phenylacetaldehyde to phenylethyl alcohol by Rr4HPAR1 and Rr4HPAR2. Chromatogram of positive ion mode metabolites between 50 and 300 m/z show the depletion of phenylacetaldehyde and production of phenylethyl alcohol when exposed to NADPH and Rr4HPAR1 or Rr4HPAR2. Reactions were carried out in 200 μ L 50 mM Tris, pH 8.0 in the presence of 2 mM phenylacetaldehyde, 5 mM NADPH and 50 μ g of recombinant enzyme. The reactions were incubated at 30° C. for 25 minutes prior to quenching with 200 μ L of 0.8 M formic acid, extracted with 100 μ L of ethyl acetate and analyzed by gas chromatography-mass spectrometry. Phenylethyl alcohol was verified by comparison to an authentic standard.

FIG. 16 is a chromatogram showing the enzymatic reduction of 4-HPAA to tyrosol by Rr4HPAR1 and Rr4HPAR2. LC-UV chromatograms of products generated from coupled enzyme assays conducted using Rr4HPAAS in combination with Rr4HPAR1 or Rr4HPAR2, respectively. Enzyme assay conducted using Rr4HPAAS alone is included as a control. 100 μ L reaction mixtures containing 50 mM Tris pH 8.0, 4 mM tyrosine, 2 μ g catalase and 100 μ g of Rr4HPAAS were incubated at 30 degrees C. for 1 hour. 10 mM NADPH and 10 μ g of Rr4HPAR1 or Rr4HPAR2 was then added and incubated for an additional 15 minutes at which point the reactions were quenched with an equal volume of 0.8 M formic acid and analyzed by LC-UV. The identity of the product was verified by comparing the elution profile and UV spectrum to that of an authentic tyrosol standard.

FIG. 17 is a phylogenetic tree of the 113 curated full-length non redundant *R. rosea* transcriptome UGTs. Sequences profiled for tyrosol glycosylation activity in yeast have been annotated RrUGT1-34.

FIG. 18 is a chromatogram of the tyrosol [M-H]⁻ ion generated in transgenic yeast expressing the Rr4HPAAS, the Rr4HPAAS+Rr4HPAR1 or Rr4HPAAS+Rr4HPAR2. The identity was verified by comparison to commercially purchased tyrosol.

FIGS. 19A-B show *R. rosea* tyrosol UGTs as compared to previously characterized *Rhodiola* tyrosol UGTs. FIG. 19A is a chromatogram of the tyrosol glycoside [M+NH₄]⁺ production using the newly described RrUGT3, RrUGT33 and the previously described *R. sachalinensis* UGTs (GenBank: AAS55083 and EU567325). FIG. 19B is a graph showing relative icaricide D2 and salidroside production from RrT8HGT, RrT4GHT RsAAS55083 and RsEU567325. The identity of the ions was confirmed by comparison to NMR verified standards.

FIG. 20 is the structure of salidroside.

FIG. 21 is the structure of icaricide D2.

FIG. 22 is a ¹H NMR spectrum (400 MHz, CDCl₃) of salidroside isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 9.16 (1H, s, OH), 7.03 (2H, d, J=8.4, 4-H, 8-H), 6.65 (2H, d, J=8.4, 5-H, 7-H), 4.92 (3H, m, Glu-OH), 4.47 (1H, s, Glu-OH), 4.16 (1H, d, J=7.6, 1'-H), 3.87 (1H, m, 1-H), 3.65 (1H, m, 6'-H), 3.56 (1H, m, 1-H), 3.42 (1H, m, 6'-H), 3.12 (1H, m, 3'-H), 3.07 (1H, m, 5'-H), 3.04 (1H, m, 4'-H), 2.95 (1H, m, 2'-H), 2.73 (2H, m, 2-H).

FIG. 23 is a ¹³C NMR spectrum (100 MHz, CDCl₃) of salidroside isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 155.6 (6-C), 129.7 (4, 8-C), 128.6 (3-C), 115.0 (5, 7-C), 102.8 (1'-C), 76.9 (3'-C), 76.8 (5'-C), 73.4 (2'-C), 70.1 (1-C), 69.9 (4'-C), 61.1 (6'-C), 34.8 (2-C).

FIG. 24 is a heteronuclear multiple bond correlation (HMBC) spectrum of salidroside isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes.

FIG. 25 is a ¹H NMR spectrum (400 MHz, CDCl₃) of icaricide D2 isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 7.11 (2H, d, J=8.8, 4-H, 8-H), 6.92 (2H, d, J=8.8, 5-H, 7-H), 5.28 (1H, s, Glu-OH), 5.09 (1H, s, Glu-OH), 5.02 (1H, s, Glu-OH), 4.78 (1H, d, J=7.2, 1'-H), 4.61 (1H, m, 1-H), 4.56 (1H, m, 1-H), 3.68 (1H, s, 6'-H), 3.55 (1H, m, OH), 3.45 (1H, s, 6'-H), 3.14-3.32 (4H, m, 2', 3', 4', 5'-H), 2.66 (2H, m, 2-H).

FIG. 26 is a ¹³C NMR spectrum (100 MHz, CDCl₃) of icaricide D2 isolated from *N. benthamiana* leaves overexpressing *R. rosea* salidroside biosynthetic genes. δ : 155.8 (6-C), 132.7 (3-C), 129.7 (4, 8-C), 116.1 (5, 7-C), 100.6 (1'-C), 77.0 (3'-C), 76.6 (5'-C), 73.3 (2'-C), 69.7 (4'-C), 62.4 (1-C), 60.7 (6'-C), 38.2 (2-C).

FIG. 27 is a chromatogram of the tyrosol [M-H]⁻ ion generated in transgenic *N. benthamiana* expressing the Rr4HPAA. The addition of either the RrT4HGT or the RrT8HGT depletes the tyrosol substrate in the production of icaricide D2 or salidroside. The identity of the ions was verified against authentic standards.

FIG. 28 is a chart showing titer of salidroside producing *S. cerevisiae* strains with and without substrate feeding. The first bar of the bar graph represents salidroside titer in wild type (WT) BY4743 yeast while the second bar demonstrates salidroside titer from the yeast strains expressing the native Rr4HPAAS and RrT8GT genes in separate pTEF 2 μ plasmids. The third bar of the graph illustrates the salidroside titer from the *S. cerevisiae* strain transformed with a pTDH3 promoter 2 μ multi gene plasmid containing coRr4HPAAS and coRrT8GT genes. The fourth and fifth bars show salidroside production from the aforementioned codon optimized multi gene plasmid with the addition of either 4 mM L-tyrosine or 4 mM tyrosol. The final bar shows the salidroside production from a strain containing the multi gene coRr4HPAAS and coRrT8GT plasmid additionally transformed with a second 2 μ pTDH3 ARO4 K229L and ARO7 G141S multi gene plasmid.

FIG. 29A-B are chromatograms of the salidroside [M+NH₄]⁺ ion generated in transgenic *S. cerevisiae*. FIG. 29A is a chromatogram of salidroside production in wild type (WT), native Rr4HPAAS and RrT8GT expressing, coRr4HPAAS and coRrT8GT expressing or ARO4 K229L, ARO7 G141S, coRr4HPAAS and coRrT8GT expressing *S. cerevisiae* strains. FIG. 29B is a chromatogram of salidroside production in *S. cerevisiae* expressing coRr4HPAAS and coRrT8GT with and without the addition of L-tyrosine and tyrosol.

FIG. 30 is a graph showing total ion count of salidroside producing transgenic *S. cerevisiae*. Salidroside, labeled in the chromatogram, appears as one of the principle metabolites.

FIG. 31 is a multiple sequence alignment of key residues within biochemically characterized plant AAADs. The multiple sequence alignment of FIG. 31 shows portions of the full alignment of FIG. 32. FIG. 31 includes SEQ ID Nos. 241 through 257 corresponding to sequences beginning at position 190, in order from top-to-bottom; and SEQ ID Nos.

275-291 corresponding to sequences beginning at position 332, in order from top-to-bottom.

FIG. 32 is a multiple sequence alignment of key residues within biochemically characterized plant AAADs. FIG. 32 includes SEQ ID Nos. 258 through 274, in order from top-to-bottom.

FIG. 33 is a chart showing sequence conservation for plant AAAD activity dictating residues. Multiple sequence alignments of the queried AAAD sequences evaluated for active site conservation using WebLogo. Polar amino acids are green, basic amino acids are blue, acidic amino acids are red and hydrophobic amino acids are black. The y-axis units (bits) display the maximum entropy for the given residue. The representative residues from the Rr4HPAAS MF674522 sequence are listed below with residue numbers.

FIGS. 34A-C show product formation of PsTyDC and mutants. FIG. 34A is a chromatogram showing the reduced enzyme product of tyrosine incubated with wildtype PsTyDC. FIG. 34B is a chromatogram showing the reduced enzyme product of tyrosine incubated with PsTyDC Y350F. FIG. 34C is a chromatogram showing the reduced enzyme product of tyrosine incubated with PsTyDC H204N.

FIG. 35 is a depiction of active site conformations of *Catharanthus roseus* tryptophan decarboxylase. In this homodimer, the A chain is shown in green while the B chain is shown in blue. The active site ligand (tryptophan) is shown in yellow. The active site lysine bound pyridoxal phosphate (LLP) cofactor is visible in the B chain.

DETAILED DESCRIPTION

A description of example embodiments follows.

Rhodiola and Salidroside Biosynthesis

The *Rhodiola* genus consists of approximately 90 species of high-altitude and cold tolerant perennial plants of the Crassulaceae family native to the arctic regions of Eurasia and North America (FIG. 1A). Select species from this genus have a long history in traditional medicine with purported roles in bolstering immunity, memory and learning, while ameliorating depression, altitude sickness and fatigue (Fu, 2009; Lei et al., 2006). Recent studies of *Rhodiola* extract have also demonstrated antioxidant and anti-inflammatory properties with potential applications in the prevention of cardiovascular diseases and cancer (Gauger et al., 2010; Khanum et al., 2005; Skopinska-Rozewska et al., 2008; Tu et al., 2008; Zhang et al., 2007). Extensive phytochemical analysis of *Rhodiola* has identified a number of specialized glycosides, including rosiridin, rhodionin, rosarin, rosin, rosavin and salidroside (FIG. 6) (Du and Xie, 1995; Rohloff, 2002; Yang et al., 2012; Yousef et al., 2006). Salidroside, or tyrosol 8-O-glucoside, is of particular interest and value because of its unique reported biological activities (Cifani et al., 2010; Guan et al., 2012; Panossian et al., 2014). However, commercially available salidroside in its pure form is currently obtained through a lengthy purification process from its native plant host, which poses a significant bottleneck hindering further clinical development of salidroside as a potential therapeutic agent. Moreover, surging global demand of wild *Rhodiola* plants as a herbal supplement has led to overharvesting of these ecologically vulnerable plants from their native habitats with some species now threatened by extinction (Booker et al., 2016; Dorji, 2016).

Metabolic engineering is a promising approach to gain access to high-value plant natural products as an alternative to direct compound isolation from plant hosts (O'Connor, 2015). Previous attempts to engineer salidroside biosynthe-

sis in heterologous hosts have utilized a selection of plant and yeast enzymes to assemble artificial salidroside biosynthetic pathways (Bai et al., 2014; Chung et al., 2017). Although these studies demonstrated the feasibility of engineering salidroside production in bacterial hosts (Bai et al., 2014), an unresolved native salidroside biosynthetic pathway in planta hinders further development and improvement of salidroside biosynthetic strategies in bacteria and other alternative chassis organisms. In postulated salidroside biosynthetic pathway, the salidroside aglycone tyrosol is generated from tyrosine through sequential decarboxylation, oxidative deamination, and aldehyde reduction reactions, catalyzed by three discrete enzymes, tyrosine decarboxylase (TyDC), monoamine oxidase (MAO) and 4HPAR, respectively (FIG. 1C) (Lan et al., 2013). Tyrosol is then glycosylated at its 8-OH group by a regio-specific uridine 5'-diphospho-glucosyltransferase (UGT) to yield salidroside. Although the proposed salidroside pathway seems plausible, to date, only one enzyme of this proposed pathway, *Rhodiola crenulata* TyDC (RcTyDC) (GenBank AFN89854.1), has been previously recombinantly expressed and experimentally examined (Lan et al., 2013). Overexpression of this TyDC-like gene in *R. crenulata* hairy roots culture led to increased accumulation of salidroside (Lan et al., 2013).

TyDCs, together with tryptophan decarboxylases (TDCs) and aromatic acetaldehyde synthases (AASs), encompass a large family of PLP-dependent enzymes broadly referred to as the plant AAAD family (Facchini et al., 2000; Kaminaga et al., 2006). As their respective names imply, TyDCs, TDCs and AAS catalyze discrete decarboxylation or decarboxylation-deamination reactions using specific aromatic amino acids as substrates.

To resolve *Rhodiola* salidroside biosynthesis, tissue-specific transcriptomics and metabolomics datasets were generated for *R. rosea*. Using a combination of differential expression analysis, phylogenetic analysis, biochemical characterization, and heterologous expression, a set of *Rhodiola* genes encoding 4HPAAS, 4HPAR, and T8GT to complete salidroside biosynthesis from tyrosine were identified. In addition, a number of regio-specific T4GTs capable of producing icariside D2 were identified. The newly acquired knowledge about phenolic glycoside biosynthesis in *Rhodiola* allowed reconstitution of salidroside or icariside D2 biosynthesis in yeast *S. cerevisiae* as well as in the plant *N. benthamiana*.

Aromatic Amino Acid Decarboxylases (AAAD) Family of Enzymes

TyDCs, together with tryptophan decarboxylases (TDCs) and aromatic acetaldehyde synthases (AASs), encompass a large family of PLP-dependent enzymes broadly referred to as the plant AAAD family (Facchini et al., 2000; Kaminaga et al., 2006). Thus, the AAAD family encompasses enzymes with aromatic amino acid decarboxylase activity and enzymes with aromatic acetaldehyde synthase activity. (Torrens-Spence et al., 2012; Torrens-Spence et al., 2013). Without wishing to be bound by theory, the catalytic mechanism of the AAAD family of enzymes is contingent on the conformational change of two active site loops, which is illustrated in FIG. 35 with respect to a tryptophan decarboxylase from *Catharanthus roseus*. The large loop from the A chain (342-359) undergoes a dramatic conformational change from a solvent exposed active site "open" conformation to an active site obscured "closed" conformation. Concurrently, a small loop from chain B (201-205) undergoes a crank shaft conformational change to move from a solvent exposed "open" conformation to a pyridoxal phosphate (LLP) associated "closed" conformation. Key residues

in these dynamic loops play important roles in the catalytic mechanism of AAAD enzymes. In the tryptophan decarboxylase from *Catharanthus roseus*, tyrosine 348 (Chain A) functions as a catalytic acid to donate a proton to the carbanion intermediate in the decarboxylation reaction mechanism while histidine 203 (Chain B) functions as a molecular chaperon responsible for coordinating and enabling proton donation of the acid tyrosine 348. Substitution of either residue abolishes the protonation and enables a peroxy-aldimine intermediate through the attack of molecular oxygen which spontaneously decomposes to yield the corresponding aromatic acetaldehyde, peroxide and ammonia aldehyde synthase products. Consequently, substitutions at either location function as a primary sequence means for biochemical functional prediction. One of skill in the art will understand that the precise location within the sequence (here, tyrosine at 348 and histidine at 203) varies among related enzymes within the AAAD family.

Nucleic Acids

As used herein, the term "nucleic acid" refers to a polymer comprising multiple nucleotide monomers (e.g., ribonucleotide monomers or deoxyribonucleotide monomers). "Nucleic acid" includes, for example, DNA (e.g., genomic DNA and cDNA), RNA, and DNA-RNA hybrid molecules. Nucleic acid molecules can be naturally occurring, recombinant, or synthetic. In addition, nucleic acid molecules can be single-stranded, double-stranded or triple-stranded. In certain embodiments, nucleic acid molecules can be modified. In the case of a double-stranded polymer, "nucleic acid" can refer to either or both strands of the molecule.

The terms "nucleotide" and "nucleotide monomer" refer to naturally occurring ribonucleotide or deoxyribonucleotide monomers, as well as non-naturally occurring derivatives and analogs thereof. Accordingly, nucleotides can include, for example, nucleotides comprising naturally occurring bases (e.g., adenosine, thymidine, guanosine, cytidine, uridine, inosine, deoxyadenosine, deoxythymidine, deoxyguanosine, or deoxycytidine) and nucleotides comprising modified bases known in the art.

As used herein, "wildtype" refers to the canonical amino acid sequence as found in nature. As those of skill in the art would appreciate, a nucleic acid sequence can be modified, e.g., for codon optimization in a host cell (e.g., bacteria, yeast, and plant host cells).

As used herein, the term "sequence identity," refers to the extent to which two nucleotide sequences, or two amino acid sequences, have the same residues at the same positions when the sequences are aligned to achieve a maximal level of identity, expressed as a percentage. For sequence alignment and comparison, typically one sequence is designated as a reference sequence, to which a test sequences are compared. The sequence identity between reference and test sequences is expressed as the percentage of positions across the entire length of the reference sequence where the reference and test sequences share the same nucleotide or amino acid upon alignment of the reference and test sequences to achieve a maximal level of identity. As an example, two sequences are considered to have 70% sequence identity when, upon alignment to achieve a maximal level of identity, the test sequence has the same nucleotide or amino acid residue at 70% of the same positions over the entire length of the reference sequence.

Alignment of sequences for comparison to achieve maximal levels of identity can be readily performed by a person of ordinary skill in the art using an appropriate alignment method or algorithm. In some instances, the alignment can

include introduced gaps to provide for the maximal level of identity. Examples include the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), the homology alignment algorithm of Needleman & Wunsch, *J. Mol. Biol.* 48:443 (1970), the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, Wis.), and visual inspection (see generally Ausubel et al., *Current Protocols in Molecular Biology*).

When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequent coordinates are designated, if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters. A commonly used tool for determining percent sequence identity is Protein Basic Local Alignment Search Tool (BLASTP) available through National Center for Biotechnology Information, National Library of Medicine, of the United States National Institutes of Health. (Altschul et al., 1990).

In various embodiments, two nucleotide sequences, or two amino acid sequences, can have at least, e.g., 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more, sequence identity. When ascertaining percent sequence identity to one or more sequences described herein, the sequences described herein are the reference sequences.

Some embodiments of the invention relate to a nucleic acid coding sequence (e.g., dsDNA, cDNA) encoding one or more of the enzymes described herein, including those nucleic acid sequences provided in SEQ ID NO: 1, SEQ ID NO: 3, SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, and SEQ ID NO: 19.

Enzymes

As used herein, the term 4-hydroxyphenylacetaldehyde synthase (4HPAAS) refers to an enzyme that catalyzes conversion of L-tyrosine to 4-hydroxyphenylacetaldehyde. Methods and assays for determining whether an enzyme catalyzes conversion of L-tyrosine to 4-hydroxyphenylacetaldehyde are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. An example of a 4HPAAS is SEQ ID NO: 2, which is the amino acid sequence of a 4HPAAS identified in *Rhodiola rosea* (Rr4HPAAS). In some embodiments, a 4HPAAS has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 2, or a biologically active fragment thereof. In some embodiments, a 4HPAAS has: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. Typically, a 4HPAAS has at least 70% sequence identity to SEQ ID NO: 2, or a biologically active fragment thereof, and also: a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position

343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. An example of a nucleic acid coding sequence that encodes a 4HPAAS is SEQ ID NO: 1, which encodes an amino acid having SEQ ID NO: 2. Many different nucleic acids can encode the 4HPAAS of SEQ ID NO: 2 due to the degeneracy of the genetic code. Nucleic acids can also differ from SEQ ID NO: 1, for example, as a result of one or more substitutions (e.g., silent substitutions).

In some embodiments, modified enzymes can be used in the methods and host cells described herein to provide 4HPAAS activity in those host cells and methods. Typically, those modified enzymes have a) an amino acid residue selected from the group consisting of F, L, I, M and V at a position corresponding to the F residue at position 343 in SEQ ID NO: 2; b) an amino acid residue selected from the group consisting of N and D at a position corresponding to the H residue at position 198 in SEQ ID NO: 2; or c) a combination thereof. In certain embodiments, modified *Papaver somniferum* tyrosine decarboxylase (PsTyDC) enzymes comprising a substitution of the active site histidine (e.g., with N or D) at the position corresponding to the H residue at position 198 in SEQ ID NO: 2, and/or the active site tyrosine (e.g., with F, L, I, M or V) corresponding to the F residue at position 343 in SEQ ID NO: 2, can be used in the methods and host cells described herein to provide 4HPAAS activity in those host cells and methods. In some embodiments, modified nucleic acids encoding the modified enzymes can be used in the vectors, kits, and methods described herein. In some embodiments, those nucleic acids may be codon optimized for expression in a host cell.

As used herein, the term 4-hydroxyphenylacetaldehyde reductase (4HPAR) refers to an enzyme that catalyzes conversion of 4-hydroxyphenylacetaldehyde to tyrosol. Methods and assays for determining whether an enzyme catalyzes conversion of 4-hydroxyphenylacetaldehyde to tyrosol are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. An example of a 4HPAR is SEQ ID NO: 4, which is the amino acid sequence of a 4HPAR identified in *Rhodiola rosea* (Rr4HPAR). In some embodiments, a 4HPAR has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to SEQ ID NO: 4, or a biologically active fragment thereof. An example of a nucleic acid that encodes a 4HPAR is SEQ ID NO: 3, which encodes an amino acid having SEQ ID NO: 4. Many different nucleic acids can encode the 4HPAR of SEQ ID NO: 4 due to the degeneracy of the genetic code. Nucleic acids can also differ from SEQ ID NO: 3, for example, as a result of one or more substitutions (e.g., conservative substitutions, non-conservative substitutions), deletions, or insertions, or a combination thereof, with respect to the wild-type Rr4HPAR sequence (SEQ ID NO: 3).

As used herein, the term tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) refers to an enzyme that catalyzes conversion of tyrosol to tyrosol 8-O-glucoside (salidroside). Methods and assays for determining whether an enzyme catalyzes conversion of tyrosol to tyrosol 8-O-glucoside (salidroside) are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic reso-

nance (NMR) or liquid chromatography-mass spectrometry. Examples of T8GTs are SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, which are the amino acid sequences of T8GTs identified in *Rhodiola rosea* (RrT8GTs). In some embodiments, a T8GT has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, or biologically active fragments thereof. Examples of nucleic acids that encode T8GTs are SEQ ID NO: 13, SEQ ID NO: 15, SEQ ID NO: 17, and SEQ ID NO: 19, which encode amino acids having SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20, respectively. Many different nucleic acids can encode the T8GTs due to the degeneracy of the genetic code. Nucleic acids can also differ, for example, as a result of one or more substitutions (e.g., silent substitutions), with respect to any of the wild-type RrT8GT nucleic acid sequences.

As used herein, the term tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT) refers to an enzyme that catalyzes conversion of tyrosol to tyrosol 4-O-glucoside (icariside D2). Methods and assays for determining whether an enzyme catalyzes conversion of tyrosol to tyrosol 4-O-glucoside (icariside D2) are known in the art, and include enzyme activity assays and liquid chromatography to assess retention time of metabolites, as described herein. Chemical structure can also be assessed by nuclear magnetic resonance (NMR) or liquid chromatography-mass spectrometry. Examples of T4GTs are SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, which are the amino acid sequences of T4GTs identified in *Rhodiola rosea* (RrT4GTs). In some embodiments, a T4GT has at least about 70%, 75%, 80%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or more sequence identity to one or more of SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, or biologically active fragments thereof. Examples of nucleic acids that encode T8GTs are SEQ ID NO: 5, SEQ ID NO: 7, SEQ ID NO: 9, SEQ ID NO: 11, and SEQ ID NO: 13, which encode amino acids having SEQ ID NO: 6, SEQ ID NO: 8, SEQ ID NO: 10, SEQ ID NO: 12, and SEQ ID NO: 14, respectively. Many different nucleic acids can encode the T4GTs due to the degeneracy of the genetic code. Nucleic acids can also differ, for example, as a result of one or more substitutions (e.g., silent substitutions) with respect to any of the wild-type RrT4GT nucleic acid sequences.

Vectors

The terms “vector”, “vector construct” and “expression vector” mean the vehicle by which a DNA or RNA sequence (e.g. a foreign gene) can be introduced into a host cell, so as to transform the host and promote expression (e.g. transcription and translation) of the introduced sequence. Vectors typically comprise the DNA of a transmissible agent, into which foreign DNA encoding a protein is inserted by restriction enzyme technology. A common type of vector is a “plasmid”, which generally is a self-contained molecule of double-stranded DNA that can readily accept additional (foreign) DNA and which can readily introduced into a suitable host cell. A large number of vectors, including plasmid and fungal vectors, have been described for replication and/or expression in a variety of eukaryotic and prokaryotic hosts.

The terms “express” and “expression” mean allowing or causing the information in a gene or DNA sequence to become manifest, for example producing a protein by activating the cellular functions involved in transcription and

translation of a corresponding gene or DNA sequence. A DNA sequence is expressed in or by a cell to form an "expression product" such as a protein. The expression product itself, e.g. the resulting protein, may also be said to be "expressed" by the cell. A polynucleotide or polypeptide is expressed recombinantly, for example, when it is expressed or produced in a foreign host cell under the control of a foreign or native promoter, or in a native host cell under the control of a foreign promoter. Gene delivery vectors generally include a transgene (e.g., nucleic acid encoding an enzyme) operably linked to a promoter and other nucleic acid elements required for expression of the transgene in the host cells into which the vector is introduced. Suitable promoters for gene expression and delivery constructs are known in the art. For bacterial host cells, suitable promoters, include, but are not limited to promoters obtained from the *E. coli* lac operon, *Streptomyces coelicolor* agarase gene (*dagA*), *Bacillus subtilis* levansucrase gene (*sacB*), *Bacillus licheniformis* alpha-amylase gene (*amyL*), *Bacillus stearothermophilus* maltogenic amylase gene (*amyM*), *Bacillus amyloliquefaciens* alpha-amylase gene (*amyQ*), *Bacillus licheniformis* penicillinase gene (*penP*), *Bacillus subtilis* *xy1A* and *xy1B* genes, and prokaryotic beta-lactamase gene (See e.g., Villa-Kamaroff et al., *Proc. Natl. Acad. Sci. USA* 75: 3727-3731, 1978), as well as the tac promoter (See e.g., DeBoer et al., *Proc. Natl. Acad. Sci. USA* 80: 21-25, 1983). Examples of promoters for filamentous fungal host cells, include, but are not limited to promoters obtained from the genes for *Aspergillus oryzae* TAKA amylase, *Rhizomucor miehei* aspartic proteinase, *Aspergillus niger* neutral alpha-amylase, *Aspergillus niger* acid stable alpha-amylase, *Aspergillus niger* or *Aspergillus awamori* glucoamylase (*glaA*), *Rhizomucor miehei* lipase, *Aspergillus oryzae* alkaline protease, *Aspergillus oryzae* triose phosphate isomerase, *Aspergillus nidulans* acetamidase, and *Fusarium oxysporum* trypsin-like protease (See e.g., WO 96/00787), as well as the NA2-tpi promoter (a hybrid of the promoters from the genes for *Aspergillus niger* neutral alpha-amylase and *Aspergillus oryzae* triose phosphate isomerase), and mutant, truncated, and hybrid promoters thereof. Examples of yeast cell promoters can be from the genes for *Saccharomyces cerevisiae* enolase (*ENO-1*), *Saccharomyces cerevisiae* galactokinase (*GAL1*), *Saccharomyces cerevisiae* alcohol dehydrogenase/glyceraldehyde-3-phosphate dehydrogenase (*ADH2/GAP*), and *Saccharomyces cerevisiae* 3-phosphoglycerate kinase. Other useful promoters for yeast host cells are known in the art (See e.g., Romanos et al., *Yeast* 8:423-488, 1992). The selection of a suitable promoter is within the skill in the art. The recombinant plasmids can also comprise inducible, or regulatable, promoters for expression of an enzyme in cells.

Various gene delivery vehicles are known in the art and include both viral and non-viral (e.g., naked DNA, plasmid) vectors. Viral vectors suitable for gene delivery are known to those skilled in the art. Such viral vectors include, e.g., vector derived from the herpes virus, baculovirus vector, lentiviral vector, retroviral vector, adenoviral vector and adeno-associated viral vector (AAV). Vectors derived from plant viruses can also be used, such as the viral backbones of the RNA viruses Tobacco mosaic virus (TMV), Potato virus X (PVX) and Cowpea mosaic virus (CPMV), and the DNA geminivirus Bean yellow dwarf virus. The viral vector can be replicating or non-replicating.

Non-viral vectors include naked DNA and plasmids, among others. Non-limiting examples include pKK plasmids (Clontech), pUC plasmids, pET plasmids (Novagen, Inc., Madison, Wis.), pRSET or pREP plasmids (Invitrogen,

San Diego, Calif.), or pMAL plasmids (New England Biolabs, Beverly, Mass.), and such vectors may be introduced into many appropriate host cells, using methods disclosed or cited herein or otherwise known to those skilled in the relevant art.

In certain embodiments, the vector comprises a transgene operably linked to a promoter. The transgene encodes a biologically active molecule, such as an enzyme described herein.

To facilitate the introduction of the gene delivery vector into host cells, the vector can be combined with different chemical means such as colloidal dispersion systems (macromolecular complex, nanocapsules, microspheres, beads) or lipid-based systems (oil-in-water emulsions, micelles, liposomes).

Some embodiments relate to a vector comprising a nucleic acid encoding any enzyme described herein. In certain embodiments, the vector is a plasmid, and includes any one or more plasmid sequences such as, e.g., a promoter sequence, a selection marker sequence, or a locus-targeting sequence. Suitable plasmid vectors include p423TEF 2μ, p425TEF 2μ, and p426TEF 2μ. Another suitable vector is pHis8-4 (Whitehead Institute, Cambridge, Massachusetts, United States of America), which is identified as SEQ ID NO: 94. Another suitable vector is pEAQ-HT, which is identified as SEQ ID NO: 95. Another suitable vector is pJKW 1410, which is identified as SEQ ID NO: 96. pJKW 1410 is a backbone vector used to construct the multi gene yeast expression vector used for salidroside production in the work described in the Examples.

Although the genetic code is degenerate in that most amino acids are represented by multiple codons (called "synonyms" or "synonymous" codons), it is understood in the art that codon usage by particular organisms is nonrandom and biased towards particular codon triplets. Accordingly, in some embodiments, the vector includes a nucleotide sequence that has been optimized for expression in a particular type of host cell (e.g., through codon optimization). Codon optimization refers to a process in which a polynucleotide encoding a protein of interest is modified to replace particular codons in that polynucleotide with codons that encode the same amino acid(s), but are more commonly used/recognized in the host cell in which the nucleic acid is being expressed. In some aspects, the polynucleotides described herein are codon optimized for expression in a bacterial cell, e.g., *E. coli*. In some aspects, the polynucleotides described herein are codon optimized for expression in a yeast cell, e.g., *S. cerevisiae*.

Host Cells

A wide variety of host cells can be used, including fungal cells, bacterial cells, plant cells, insect cells, and mammalian cells.

In some embodiments, the host cell is a fungal cell, such as a yeast cell and an *Aspergillus* spp cell. A wide variety of yeast cells are suitable, such as cells of the genus *Pichia*, including *Pichia pastoris* and *Pichia stipitis*; cells of the genus *Saccharomyces*, including *Saccharomyces cerevisiae*; cells of the genus *Schizosaccharomyces*, including *Schizosaccharomyces pombe*; and cells of the genus *Candida*, including *Candida albicans*.

In some embodiments, the host cell is a bacterial cell. A wide variety of bacterial cells are suitable, such as cells of the genus *Escherichia*, including *Escherichia coli*; cells of the genus *Bacillus*, including *Bacillus subtilis*; cells of the genus *Pseudomonas*, including *Pseudomonas aeruginosa*; and cells of the genus *Streptomyces*, including *Streptomyces griseus*.

In some embodiments, the host cell is a plant cell. A wide variety of cells from a plant are suitable, including cells from a *Nicotiana benthamiana* plant. In other embodiments, the plant belongs to a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.

In some embodiments, the host cell is an insect cell, such as a *Spodoptera frugiperda* cell, such as *Spodoptera frugiperda* Sf9 cell line and *Spodoptera frugiperda* Sf21

In some embodiments, the host cell is a mammalian cell.

In some embodiments, the host cell is an *Escherichia coli* cell, and the vector is pHis8-4. In some embodiments, the host cell is a *Nicotiana benthamiana* cell, and the vector is pEAQ-HT. In some embodiments, the cell is a *Saccharomyces cerevisiae* cell, and the vector is a p423TEF 2μ plasmid, a p425TEF 2μ plasmid, or a p426TEF 2μ plasmid.

As used herein, the term “host cell” encompasses cells in cell culture and also cells within an organism (e.g., a plant).

Some embodiments relate to a host cell comprising a vector as described herein. In certain embodiments, the host cell is an *Escherichia coli* cell, a *Nicotiana benthamiana* cell, or a *Saccharomyces cerevisiae* cell.

In some embodiments, the hosts cells are cultured in a cell culture medium, such as a standard cell culture medium known in the art to be suitable for the particular host cell. In some embodiments, the culture medium is supplemented with one or more of L-tyrosine, 4-hydroxyphenylacetaldehyde (4-HPAA), and tyrosol. In some embodiments, the culture medium is supplemented with tyrosine, for example, between 0.1 mM and 100 mM L-tyrosine. In some embodiments, the culture medium is supplemented with 4-HPAA, for example, between 0.1 mM and 100 mM of 4-HPAA. In some embodiments, the culture medium is supplemented with tyrosol, for example, between 0.1 mM and 100 mM of tyrosol.

Methods of Making Transgenic Host Cells

Described herein are methods of making a transgenic host cell. The transgenic host cells can be made, for example, by introducing one or more of the vector embodiments described herein into the host cell.

In one embodiment, the method comprises introducing into a host cell a vector that includes a nucleic acid encoding a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In another embodiment, the method can also include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR) in addition to introducing a nucleic acid encoding a T8GT. In another embodiment, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS) in addition to introducing one or more nucleic acids encoding one or more of T8GT and 4HPAR.

In another embodiment, the method comprises introducing a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In another embodiment, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR) in addition to introducing a nucleic acid encoding a 4HPAAS. In another embodiment, the method can further include introducing into the host cell a vector that includes one or more of a nucleic acid encoding tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) and a tyrosol:UDP-glucose 4-O-glucosyltransferase (T4GT), in addition to introducing a nucleic acid encoding one or more of a 4HPAAS and a

4HPAR. In some embodiments, the method can further include introducing into the host cell a vector that includes a nucleic acid encoding a T8GT and a nucleic acid encoding a T4GT, in addition to introducing a nucleic acid encoding one or more of a 4HPAAS and a 4HPAR.

In some embodiments, nucleic acids encoding two or more of 4HPAAS, 4HPAR, T8GT, and T4GT are included in a single vector, such that a single vector encoding one or more enzymes is introduced into a host cell.

In some embodiments, one or more of the nucleic acids are integrated into the genome of the host cell. In some embodiments, the nucleic acids to be integrated into a host genome can be introduced into the host cell using any of a variety of suitable methodologies known in the art, including, for example, CRISPR-based systems (e.g., CRISPR/Cas9; CRISPR/Cpf1), TALEN systems and *Agrobacterium*-mediated transformation. However, as those skilled in the art would recognize, transient transformation techniques can be used that do not require integration into the genome of the host cell. In some embodiments, nucleic acid (e.g., plasmids) can be introduced that are maintained as episomes, which need not be integrated into the host cell genome.

In certain embodiments, the nucleic acid is introduced into a tissue, cell, or seed of a plant cell. Various methods of introducing nucleic acid into the tissue, cell, or seed of plants are known to one of ordinary skill in the art, such as protoplast transformation. The particular method can be selected based on several considerations, such as, e.g., the type of plant used. For example, the floral dip method, as described herein, is a suitable method for introducing genetic material into a plant. In certain embodiments, the nucleic acid can be delivered into the plant by an *Agrobacterium*.

In some embodiments, a host cell is selected or engineered to have increased activity of the synthesis pathway for one or more of L-tyrosine, 4-hydroxyphenylacetaldehyde (4-HPAA) and tyrosol. In some embodiments, a host cell is selected or engineered to have increased activity of the synthesis pathway for L-tyrosine. In some embodiments a host cell may be selected or engineered to have reduced feedback inhibition of one or more enzymes in the L-tyrosine synthesis pathway. In some embodiments, the host cell is engineered to increase uptake of a precursor, such as L-tyrosine, 4-HPAA, or tyrosol, from the medium.

Methods of Making Salidroside, Icariside D2, and Salidroside Precursors

Described herein are methods of making salidroside, icariside D2, and salidroside precursors. Salidroside, icariside D2, and salidroside precursors can be produced by expressing one or more of the enzymes described herein in a host cell.

Some embodiments provide a method of making tyrosol 8-O-glucoside (salidroside) in a host cell. The method can include expressing in a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT). In some embodiments, the host cell synthesizes tyrosol and includes, either endogenously or transgenically, enzymes to synthesize tyrosol. In some embodiments, tyrosol is provided in the culture media. In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). In some embodiments, the host cell further expresses a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS).

Some embodiments provide a method of making tyrosol 8-O-glucoside (salidroside) in a host cell. The method can include expressing in a host cell a transgene that encodes a

4-hydroxyphenylacetaldehyde synthase (4HPAAS) and a transgene that encodes a 4-hydroxyphenylacetaldehyde reductase (4HPAR). The host cell expresses, either endogenously or transgenically, one or more enzymes that catalyze conversion of tyrosol to tyrosol 8-O-glucoside (salidroside).

Certain embodiments provide a method of making 4-hydroxyphenylacetaldehyde (4HPAA) in a host cell. The method can include expressing in the host cell a transgene that encodes a 4-hydroxyphenylacetaldehyde synthase (4HPAAS). In some embodiments, the host cell includes L-tyrosine, produced endogenously or provided to the cell exogenously. In some embodiments, L-tyrosine is provided in the cell culture medium. In some embodiments, the method further includes making tyrosol in the host cell, and the host cell further expresses a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR).

In some embodiments, particularly those optimized for producing salidroside, the host cell can have low or absent T4GT activity in order to reduce competition from T4GT for the substrate tyrosol. In some embodiments, the host cell is engineered to reduce or eliminate expression of T4GT.

In some embodiments, particularly those optimized for producing icarisiide D2, the host cell can have low or absent T8GT activity in order to reduce competition from T8GT for the substrate tyrosol. In some embodiments, the host cell is engineered to reduce or eliminate expression of T8GT.

In some embodiments, a host cell (e.g., a bacterial host cell) endogenously expresses enzymes that catalyze the production of salidroside or icarisiide D2 from tyrosol. For example, some bacteria express UGTs that exhibit T8GT and/or T4GT activity (Fan et al., 2017). In some embodiments, nucleic acids encoding the bacterial-derived T8GTs can be used in vectors and methods described herein. In some embodiments, host cells and methods can express a T8GT that is a bacterial-derived T8GT.

In some embodiments, one or more copies of one or more of the nucleic acids are integrated into the genome of the host cell. However, as those skilled in the art would recognize, transient transformation techniques can be used that do not require integration into the genome of the host cell.

Methods of obtaining, or extracting, salidroside, icarisiide D2, and precursors of salidroside and icarisiide D2 are described herein and are well known to one of ordinary skill in the art. For example, as described herein, salidroside, icarisiide D2, and/or precursors of salidroside and icarisiide D2 can be separated by liquid chromatography. Larger scale separation can be obtained by, e.g., simulated moving bed (SMB) chromatography and/or ion exchange chromatography. Any of the methods described herein can further include isolating salidroside, icarisiide D2, and/or a salidroside precursor from a host cell. Any of the methods described herein can include harvesting tissue (e.g., leaves, roots) of a transgenic plant described herein and processing the harvested tissue to obtain salidroside, icarisiide D2, and/or a precursor of salidroside and icarisiide D2 therefrom.

Values and Ranges

Unless otherwise indicated or otherwise evident from the context and understanding of one of ordinary skill in the art, values that are expressed as ranges can assume any specific value or subrange within the stated ranges in various embodiments, unless the context clearly dictates otherwise. "About" in reference to a numerical value generally refers to a range of values that fall within $\pm 8\%$, in some embodiments $\pm 6\%$, in some embodiments $\pm 4\%$, in some embodiments $\pm 2\%$, in some embodiments $\pm 1\%$, in some embodiments $\pm 0.5\%$ of the value unless otherwise stated or otherwise evident from the context.

EXEMPLIFICATION

Example #1: Results

5 Generating Metabolomics and Transcriptomics Resources for *R. rosea*

To survey the metabolic profile of *Rhodiola* cultivated under lab conditions, crown tissue (aerial tissue including leaves and stems) and root tissue were collected separately from a three-month old greenhouse-grown *R. rosea* plant (FIG. 1A). The fresh tissues were extracted by 50% methanol, and analyzed by untargeted liquid chromatography high-resolution accurate-mass mass spectrometry (LC-HRAM-MS). This analysis confirmed the presence of tyrosol, rosavin and salidroside in greenhouse-grown *R. rosea*, all of which accumulate at much higher levels in the root compared to the crown (FIGS. 1, 8, and 9).

The higher accumulation of these metabolites in the root suggests that the requisite biosynthetic genes may also obey a similar tissue-specific expression pattern. An RNA-Seq experiment was then performed using total RNAs prepared from the two tissues. This experiment yielded about 30 million paired-end sequencing reads (100×100 bp) per sample. While 84,645 and 105,132 unique transcripts were assembled de novo from the crown and root tissues separately, a total of 128,623 unique transcripts were assembled combining all raw sequencing reads from both tissues. The combined transcriptome was evaluated as 90.3% complete by the metric of Benchmarking Universal Single-Copy Orthologs (BUSCO) (Simao et al., 2015). The Transcripts Per Million (TPM) value of unique transcripts in each tissue type was calculated to infer the relative expression level of the corresponding genes (Li et al., 2010). The identification and prioritization of candidate salidroside biosynthetic genes from the *R. rosea* transcriptome were based upon our hypothetical salidroside biosynthetic model, subsequent large-scale phylogenetic analyses, and the relative expression level of plausible candidate genes in the two examined tissue types. The biochemical function of selected candidate genes was further investigated both in vitro and in vivo.

R. rosea Contains a Neofunctionalized 4HPAAS

A BLAST search using PsTyDC as the query against the *R. rosea* transcriptome identified three AAAD homologs. Using the sequence motifs correlating to AAAD substrate specificity and catalytic mechanism (Torrens-Spence et al., 2014; Torrens-Spence et al., 2013), two of the three *R. rosea* AAAD homologs were predicted to possibly function as AASs, and the other is likely to catalyze decarboxylation chemistry (FIG. 10). A phylogenetic analysis including the three *R. rosea* AAAD homologs together with other AAAD sequences from taxonomically diverse plant species was conducted (FIGS. 2B and 11). Whereas the predicted *R. rosea* decarboxylase candidate clusters within the TDC clade (red) containing largely previously known TDCs, the two *R. rosea* AAS candidates fall into two distinct clades, designated as the basal clade (green) and the TyDC clade (blue), respectively (FIGS. 2B and 11). It is noted that the TyDC-type AAS candidate isolated in this study is likely orthologous to the RcTyDC previously reported by Bai et al. (Bai et al., 2014), sharing 96% sequence identity at the protein level.

To experimentally assess the biochemical activities of the two *R. rosea* AAS candidates, full-length open reading frame corresponding to both the basal and TyDC-type AAS candidate genes from *R. rosea* cDNA were cloned. Their encoded proteins were recombinantly expressed in *E. coli*, purified to homogeneity, and tested for enzymatic activity

using L-tyrosine as the substrate. Both enzymes readily yield hydrogen peroxide, a co-product of AAS as opposed to canonical TyDC (Kaminaga et al., 2006), while the TyDC-type AAS candidate exhibits much higher activity than the basal AAS candidate (FIG. 12). To confirm the chemical identity of the AAS reaction products, the enzyme assays were analyzed by LC coupled with a UV detector (FIG. 2B). Incubation of L-tyrosine with both AAS candidate enzymes led to the production of 4-HPAA, which is distinct from the tyramine product yielded by PsTyDC as a control (FIG. 2B). The identity of the 4-HPAA product was further confirmed by sodium borohydride reduction of 4-HPAA to yield tyrosol (FIG. 2B). Notably, the transcript corresponding to the TyDC-type AAS candidate is highly enriched in the root versus the crown (FIG. 13), whereas such pattern was not observed for the basal AAS candidate. In light of these results, the TyDC-type AAS candidate is likely the primary AAS involved in salidroside biosynthesis in *R. rosea* root. The TyDC-type AAS candidate is referred to as Rr4HPAAS hereafter.

The Michaelis-Menten kinetics of Rr4HPAAS was measured against four aromatic amino acids, namely L-tyrosine, L-3,4-dihydroxyphenylalanine (L-DOPA), L-phenylalanine, and L-tryptophan (FIG. 2C and Table 1). Rr4HPAAS demonstrates the highest catalytic efficiency toward L-tyrosine ($k_{cat}/K_m=11.7 \text{ s}^{-1} \text{ mM}^{-1}$) followed by L-DOPA ($k_{cat}/K_m=9.1 \text{ s}^{-1} \text{ mM}^{-1}$), whereas L-phenylalanine and L-tryptophan are much less preferred substrates (FIG. 2C and Table 1). L-DOPA and any potential phenolic compound derived from it were not detected in the *R. rosea* metabolomics datasets, and thus the kinetic characteristics of Rr4HPAAS is consistent with its role in salidroside biosynthesis. These results also suggest that the previously reported RcTyDC was likely functionally mischaracterized (Bai et al., 2014). Identification and Biochemical Characterization of *Rhodiola* Phenolic Aldehyde Reductases

To identify *R. rosea* enzymes involved in the next step of salidroside biosynthesis, a BLAST search was conducted using the previously characterized *Solanum lycopersicum* PAR (SIPAR, GenBank: ABR15768.1) as a query (Tiemann et al., 2007) against our *R. rosea* transcriptome. A phylogenetic analysis was performed using the returned *R. rosea* hits together with other homologous ADHs from select plant species (FIGS. 3A and 14). This analysis revealed two *R. rosea* ADH homologs, referred to as RrPAR-like1 and RrPAR-like2, that cluster phylogenetically with SIPAR (Tiemann et al., 2007), and share 76% and 58% protein sequence identity to SIPAR, respectively. Both genes were cloned from *R. rosea* cDNA as candidate genes encoding 4HPAR.

To examine the biochemical activity of the two 4HPAR candidates, recombinant enzymes were expressed in *E. coli*, purified to homogeneity, and assayed against 4-HPAA or phenylacetaldehyde in the presence of NADPH as the co-substrate. Both enzymes are capable of reducing phenylacetaldehyde to phenylethanol with RrPAR-like1 displaying higher activity (FIG. 15). Likewise, RrPAR-like1 exhibited orders of magnitude higher specific activity towards 4-HPAA ($6.9 \mu\text{mol min}^{-1} \text{ mg}^{-1}$) than RrPAR-like2 ($8.4 \text{ nmol min}^{-1} \text{ mg}^{-1}$) (FIGS. 3B, 3C, and 16). Nonetheless, RrPAR-like1 and RrPAR-like2 were renamed as Rr4HPAR1 and Rr4HPAR2, respectively, as both enzymes displayed 4-HPAA reductase activity.

Identification of Regio-Specific Tyrosol-Modifying UGTs from *R. rosea*

To complete the salidroside biosynthetic pathway, candidate UGT genes encoding T8GT were identified. The UGT superfamily is one of the largest enzyme families in the plant

kingdom (Li et al., 2001). The plant secondary product glycosyltransferase (PSPG) motif is described in Gachon et al., 2005, particularly at FIG. 2A and associated text. The transcriptome was queried using a UGT superfamily signature motif (Li et al., 2001), and conducted an unbiased phylogenetic analysis using 113 curated non-redundant full-length UGT homologs retrieved from the *R. rosea* transcriptome (FIG. 17). Thirty-four candidate UGT genes were then prioritized for further functional analysis according to a combination of criteria including phylogenetic distribution pattern and transcript levels in the root and crown transcriptome datasets.

To facilitate functional assessment of a large number of UGT candidates, an in vivo tyrosol glycosylation assay in the yeast *S. cerevisiae* was devised. Initial iterations of the tyrosol-producing yeast strains were generated by transforming wild type *S. cerevisiae* BY4743 with separate 2-micron TEF-promoter expression plasmids containing Rr4HPAAS and Rr4HPAR1, respectively. It was later observed that yeast contains endogenous ADH activity sufficient to reduce 4-hydroxyphenylacetaldehyde produced by Rr4HPAAS to tyrosol. Therefore, the Rr4HPAR1-containing plasmid was omitted in the final tyrosol-producing strain (FIG. 18). Each of the 34 *R. rosea* UGT candidate genes, carried on the yeast 2-micron TEF-promoter expression plasmids, was transformed into the background strain expressing Rr4HPAAS. After auxotrophic selection, colonies were cultured, harvested and subjected to metabolic profiling by LC-HRAM-MS. From this screen, we identified three UGTs (RrUGT 29, 32, and 33) with regio-specific T8GT activity, four UGTs (RrUGT 2, 3, 7, and 13) with regio-specific T4GT activity, and RrUGT17 with both T8GT and T4GT activities (FIG. 4B). Further phylogenetic analysis of the 34 cloned *R. rosea* UGTs against the 88 unique and complete *A. thaliana* UGTs suggests a correlation between the cladding of the UGTs and their respective biochemical activities (FIG. 4A) (Li et al., 2001). The UGTs that contain T4GT activity appear to be phylogenetically diverse with representative enzymes falling into the D, G, E and K groups, while all of identified T8GTs cluster within the G group (FIG. 4A). These results also show that RrUGT 3 and RrUGT33, the most active T4GT and T8GT, respectively, display significantly higher regio-specific tyrosol glycoside-producing activities than the two previously reported UGTs from *R. sachalinensis* (GenBank: AAS55083 and EU567325) (FIG. 19) (Ma et al., 2007; Yu et al., 2011).

Using recombinant enzymes produced and purified from *E. coli*, the kinetic parameters for the salidroside-producing RrUGT29 and RrUGT33 and the icaricide D2-producing RrUGT2 and RrUGT3 were measured (FIG. 4C and Table 1). RrUGT33 exhibits the highest T8GT catalytic efficiency with a k_{cat}/K_m value of $420.6 \text{ s}^{-1} \text{ mM}^{-1}$ and was subsequently referred to as RrT8GT (Table 1). In contrast, RrUGT3 exhibits the greatest T4GT catalytic efficiency with a k_{cat}/K_m value of $117.2 \text{ s}^{-1} \text{ mM}^{-1}$ and was subsequently referred to as RrT4GT (Table 1).

Heterologous Production of Salidroside and Icaricide D2 in *N. benthamiana*

To further evaluate the biochemical function of *R. rosea* tyrosol glycoside biosynthetic genes in planta, these genes were expressed in *N. benthamiana* leaves using the *Agrobacterium tumefaciens*-mediated transient protein production technique (Sainsbury et al., 2009) followed by LC-HRAM-MS-based metabolic profiling. To first demonstrate the biochemical function of Rr4HPAAS in planta, Rr4HPAAS alone was transiently expressed in *N. benthamiana* leaves. PsTyDC and the previously reported Petrose-

linum crispum 4HPAAS (Pc4HPAAS, GenBank: AAA33861) (Torrens-Spence et al., 2012) were also tested in parallel as controls. Interestingly, expression of Rr4HPAAS or Pc4HPAAS in *N. benthamiana* led to significant accumulation of both salidroside and icariside D2 in *N. benthamiana* leaves (FIG. 5A). The chemical identity of these compounds was confirmed by both LC-HRAM-MS and nuclear magnetic resonance (NMR) analyses (FIGS. 20-26). This result suggests that 4-HPAA produced by transgenic 4HPAAS can be readily metabolized by endogenous *N. benthamiana* reductase and glycosyltransferase enzymes to yield both salidroside and icariside D2. In contrast, the expression of PsTyDC yielded tyramine in high abundance in *N. benthamiana* leaves with no measurable production of tyrosol glycosides (FIG. 5B). Next, the in planta regio-specificity of RrT8GT and RrT4GT in tyrosol glycosylation was evaluated. Co-expression of either RrT8GT or RrT4GT with Rr4HPAAS led to regio-specific glycosylation of tyrosol and accumulation of salidroside or icariside D2, respectively (FIG. 5C). Meanwhile, the accumulation of free tyrosol was reduced in these plants compared to those with Rr4HPAAS expression alone (FIG. 27). Notably, the paired expression of Rr4HPAAS and one of the two regio-specific *R. rosea* tyrosol glycosyltransferases yielded up to 2% dry weight for salidroside or icariside D2 production in *N. benthamiana* leaves. This set of in planta experiments demonstrate that Rr4HPAAS and regio-specific RrT8GT are specialized metabolic enzymes underpinning salidroside biosynthesis in *Rhodiola*. Although icariside D2 does not naturally accumulate in *Rhodiola*, the identification of the regio-specific RrT4GT adds to the tool box for metabolic engineering of valuable tyrosol-derived glycosides.

Optimization of Salidroside Production in *S. cerevisiae*

The complete elucidation of salidroside biosynthesis in *Rhodiola* provides new opportunities for bioengineering of sustainable salidroside production in heterologous hosts. Although *N. benthamiana* has been used for the commercial production of high value natural products and recombinant proteins, its scalability currently does not match to industrial yeast fermentation. To increase the salidroside titer in yeast, the Rr4HPAAS and RrT8GT genes were optimized according to *S. cerevisiae* codons, and assembled in a custom 2 μ plasmid for constitutive expression driven by pTDH3 promoter in yeast (FIGS. 28 and 29A). The increased promoter strength and codon optimization of Rr4HPAAS (coRr4HPAAS) and RrT8GT (coRrT8GT) resulted in a 2.5-fold increase in salidroside titer as compared to the initial strain. To probe the potential bottlenecks in salidroside biosynthesis in yeast, we next fed the culture containing the codon optimized construct with either L-tyrosine or tyrosol. Both feeding experiments demonstrated significant increase in salidroside titer, suggesting that improved tyrosine flux may further improve salidroside titer (FIGS. 28 and 29B). Thus, a yeast strain was engineered to include the previously described feedback-insensitive mutants of the yeast L-tyrosine pathway enzymes ARO4 and ARO7 (Gold et al., 2015). Incorporation of both ARO4 K229L and ARO7 G141S into the prior best engineered yeast strain produced salidroside as one of the most abundant metabolites with a titer of 1.5 mg L⁻¹, when grown for 48 h in 4% glucose 2x yeast nitrogen base in shake flasks (FIGS. 28, 29A, and 30). In summary, this preliminary metabolic engineering exercise in yeast yielded a prototype salidroside-producing strain, which can be improved through additional rounds of targeted and untargeted genetic modifications to further increase titer.

Example #1: Discussion

As described herein, the *R. rosea* ortholog of the previously reported RcTyDC is a 4HPAAS, which catalyzes the direct conversion of tyrosine to 4-HPAA. This discovery therefore corrected a major long-standing misconception about the biosynthetic route towards tyrosol, an important precursor for many important phenolic natural products in plants (Chapple et al., 1986; Wyk, 2010).

UGTs play important roles in plant specialized metabolism as they alter the solubility, reactivity, bioactivity, intercellular and subcellular transport of a wide array of plant metabolites by glycosylation (Jones and Vogt, 2001). The resulting glycosides also have profound impact on human health with diverse pharmacological and nutraceutical indications (Jones and Vogt, 2001). Since natural product glycosides often contain distinct pharmacokinetic properties as compared to their aglycones, chemical derivatization via glycosylation has received considerable attention in pharmaceutical research (Gantt et al., 2011). Several *R. rosea* UGTs capable of producing salidroside and icariside D2 from the aglycone tyrosol in a regio-specific manner were identified. In this gene-mining process, a library was established containing phylogenetically diverse UGTs from *R. rosea*, which likely contain enzymes responsible for the biosynthesis of other phenolic glycosides from *Rhodiola*, such as rosiridin, rhodionin, rosarin, rosin, and rosavin (FIG. 6).

Unlike bacterial natural product biosynthetic pathways, which are encoded by operons ubiquitously present in bacterial genomes, enzyme-encoding genes of a given plant specialized metabolic pathway often scatter randomly across the plant genome, making metabolic pathway elucidation unattainable simply by genome mining. Plants, like many other multicellular eukaryotes, contain rich tissue types where specific natural products accumulate under developmental and environmental regulations. In recent years, this feature of plant specialized metabolism has been exploited for pathway and enzyme discovery in medicinal plants that lack classical genetic tools (Torrens-Spence et al., 2016). Through mining transcriptomics and metabolomics datasets generated separately from the root and crown tissues of *R. rosea*, candidate salidroside biosynthetic genes were prioritized based on correlation between transcript and metabolite abundances in these two tissues. Extended phylogenomics analyses of the involved enzyme families further provided additional information that facilitates salidroside biosynthetic gene discovery. The biochemical functions of the identified candidate enzymes were then examined in vitro using recombinantly expressed proteins, and in vivo through expression of the candidate enzymes in heterologous hosts, e.g. yeast and *N. benthamiana* in this case. Collectively, this work describes a rare de novo elucidation of the complete biosynthetic pathway of a given plant natural product. The workflow adopted in this study is generally applicable for future investigation of other largely unexplored specialized metabolic pathways in non-model plants, and will ultimately contribute to a capability of synthesizing structurally diverse plant natural products through the means of metabolic engineering.

Materials and Methods

Reagents

Salidroside, tyrosine, tyramine, tyrosol, phenylacetaldehyde, phenylethyl alcohol, sodium borohydride, NADPH,

UDP-glucose, and PLP were purchased from Sigma-Aldrich. 4-HPAA was purchased from Santa Cruz Biotechnology, Inc.

Plant Materials

R. rosea seeds were purchased from Horizon Herbs. Seeds were stratified at 4° C. for three days, and germinated in potting soil. *R. rosea*, *P. crispum*, and *N. benthamiana* plants were grown under a 16-h-light/8-h-dark photoperiod at 23° C. in a local greenhouse.

RNA Isolation, Library Preparation, Transcriptome Assembly, cDNA Production and Molecular Cloning

Tissue of seventy-day-old *R. rosea* plants were harvested for total RNA extraction using the Qiagen's RNeasy Mini Kit (Qiagen). RNA quality was assessed by Bioanalyzer (Agilent Technologies). For the RNAseq experiment, strand-specific mRNA libraries were prepared using total RNA prepared separately from the root and crown tissue using the TruSeq Stranded mRNA Library Prep Kit (Illumina), and sequenced on a HiSeq2000 sequencer (Illumina) in paired-end mode (PE100). Sequence FASTQ files were trimmed for sequencing adaptors using Trimmomatic (Bolger et al., 2014) and assembled into de novo transcriptomes using Trinity in strand-specific mode (Grabherr et al., 2011). Gene expression statistics (TPM values) were determined by RSEM (Li and Dewey, 2011). Completeness of the combined *R. rosea* root and crown transcriptome was evaluated using the BUSCO tool, with 'embryophyta_odb9' set as lineage and 'Arabidopsis' set as model species (Simao et al., 2015). Putative coding regions were predicted using Transdecoder (Haas et al., 2013). Transcripts and predicted protein sequences were annotated with TPM values and closest BLAST hits using in-house scripts. Transcriptome mining was performed on a local BLAST server (Anurag Priyam, 2015). First-strand cDNAs were synthesized by RT-PCR using total RNA sample as template and the Invitrogen SuperScript™ III kit (Invitrogen) with the oligo(dT)20 primer. The coding sequences (CDS) of candidate genes were amplified from cDNAs by PCR using gene-specific primers (Table 2). Select *R. rosea* and *R. sachalinensis* genes were also synthesized as gBlocks (IDT) with yeast codon optimization. Gibson assembly was used to ligate PCR amplicons or gBlocks into several base vectors. These include pHis8-4, a bacterial expression vector containing an N-terminal 8xHis tag followed by a tobacco etch virus (TEV) cleavage site for recombinant protein production in *E. coli*; pEAQ-HT, a binary vector designed for transient expression of heterologous proteins in *N. benthamiana* (Peyret and Lomonosoff, 2013); p423TEF, p425TEF and p426TEF 2μ plasmids (Mumberg et al., 1995) with various auxotrophic growth markers for constitutive expression in *S. cerevisiae*; and a custom plasmid containing 2μ, pTDH3, tTDH1, HIS3 for constitutive multi gene expression in *S. cerevisiae* (Lee et al., 2015).

Sequence Alignment and Phylogenetic Analysis

The protein multiple sequence alignments were generated using ClustalW2 with default settings (Thompson et al., 2002). ESPript 3.0 (Gouet et al., 2003) was used to display the multiple sequence alignments. The phylogeny was inferred using the Maximum Likelihood method based on the Poisson correction model (L., 1965). The bootstrap consensus unrooted trees were inferred from 500 replicates to represent the phylogeny of the analyzed enzyme families (Sanderson and Wojciechowski, 2000). The phylogenetic analysis of the AAAD family includes 242 sequences from the Phytozome V12 embryophyte species with fully sequenced genome (*A. thaliana*, *G. raimondii*, *P. trichocarpa*, *M. domestica*, *M. truncatula*, *E. grandis*, *K.*

laxiflora, *S. lycopersicum*, *A. coerulea*, *Z. mays*, *B. distachyon*, *O. sativa*, *Z. marina*, and *A. trichopoda*), the *R. rosea* transcriptome, and previously characterized AAAD proteins. The phylogenetic analysis of ADHs includes 346 PAR homologs from the Phytozome V12 embryophyte species and *R. rosea* transcriptome. The phylogenetic analysis of UGTs contains 113 non-redundant full-length UGT homologs from the *R. rosea* transcriptome. A second UGT tree was also generated using the 34 cloned *R. rosea* UGTs in addition to the 88 full length and unique UGTs from *Arabidopsis thaliana* (Li et al., 2001). Initial trees for the heuristic search were obtained automatically by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using a JTT model, and then selecting the topology with superior log likelihood value. All phylogenetic analyses were conducted in MEGA7 (Kumar et al., 2016).

Agrobacterium-Mediated Transient Expression of Heterologous Proteins in *N. benthamiana*

A. tumefaciens (LBA4404) containing the transgene construct was grown to optical density (OD) 600 of 1.6 in 50 mL of YM medium (0.4 g/L yeast extract, 10 g/L mannitol, 0.1 g/L NaCl, 0.2 g/L MgSO₄·7H₂O, 0.5 g/L K₂HPO₄·3H₂O), washed with washing buffer (10 mM MES (2-(N-morpholino)ethanesulfonic acid), pH 5.6), and resuspended in MMA buffer (10 mM MES, pH 5.6, 10 mM MgCl₂, 100 μM acetosyringone) to OD 600 of 0.8. For co-expressing multiple genes, individual *A. tumefaciens* cultures containing the unique transgene constructs were grown, pelleted, and washed separately. The cultures were then resuspended together at a higher optical density so that each individual culture was present at a concentration equivalent to OD 600 of 0.8. 1 mL of culture was used to infiltrate the underside of six-week-old *N. benthamiana* leaves.

Metabolomic Profiling by LC-HRAM-MS

Crown tissue and root tissue of a three-month-old *R. rosea* plant was harvested and stored at -80° C. before subsequent metabolomic analysis. Various transgene-carrying *S. cerevisiae* BY4743 strains and transiently transformed *N. benthamiana* plants were generated to test the activity of candidate genes involved in the tyrosol glycoside biosynthesis. 3 mL of saturated *S. cerevisiae* culture was used to inoculate 50 mL of synthetic minimal media (SD) in a shake flask. After 24 hours of shaking at 30° C., the culture was pelleted by centrifugation, washed with water, and stored at -80° C. before further processing. *N. benthamiana* leaf tissue was harvested 5 days after *Agrobacterium* infiltration and was stored at -80° C. before further processing. Frozen yeast or plant tissue was disrupted with a TissueLyser (Qiagen) using acid-washed metal beads in 50% methanol (500 μL per 100 mg fresh weight). The extracts were then analyzed by LC-HRAM-MS. Metabolite profiling was conducted on a QExactive benchtop orbitrap mass spectrometer equipped with an Ion Max source and a HESI II probe, which was coupled to a Dionex UltiMate 3000 UPLC system (Thermo Fisher Scientific). 2 μL of each sample was injected onto a 150×2.1 mm ZIC-pHILIC column (5 μm particle size, EMD Millipore). Solvent A was 20 mM ammonium carbonate, 0.1% ammonium hydroxide; solvent B was acetonitrile. The column oven and autosampler tray were held at 25° C. and 4° C., respectively. The chromatographic gradient was run at a flow rate of 0.15 mL/min as follows: 0-20 min, linear gradient from 80% to 20% solvent B; 20-20.5 min, linear gradient from 20% to 80% solvent B; 20.5-28 min, hold at 80% solvent B. The mass spectrometer was operated in full-scan, polarity-switching mode with the spray voltage set to 3.0 kV, the heated capillary held at 275°

C., and the HESI probe held at 350° C. The sheath gas flow was set to 40 units, the auxiliary gas flow was set to 15 units, and the sweep gas flow was set to 1 unit. The MS data acquisition was performed in a range of 70-1000 m/z, with the resolution set at 70,000, the AGC target at 10 e6, and the maximum injection time at 20 msec. The raw data was converted to mzML format using MSConvert (Chambers et al., 2012), and analyzed using MetaboAnalyst (Xia and Wishart, 2016) and MZmine2 (Pluskal et al., 2010).

Small Molecule Isolation and NMR

For large-scale compound isolation from *Agrobacterium*-transformed *N. benthamiana* leaves, 15 g (dry weight) of *N. benthamiana* leaves (harvested 5 days post infection) were extracted with 70% EtOH. The solvent was evaporated from the extracts under reduced pressure using a rotary evaporator (Buchi). The residue was suspended in 100 mL of water, and extracted successively with hexane, chloroform and butanol. The water-soluble portion was separated by Sephadex LH20 using a H₂O/MeOH gradient of 0-100% MeOH. Fractions 26-32 and 36-44 were combined separately for further purification by a preparative HPLC (Shimadzu) equipped with a SPD-20A UV-VIS detector and a 150×21.2 mm 100 Å Kinetex 5 µC₁₈ column (Phenomenex). 7 mg of salidroside and 13 mg of icariside D2 were purified using water (solvent A) and a 60-minute gradient of 5-80% acetonitrile (solvent B) at a flow rate of 10 mL/min. The samples were dried by lyophilization and subjected to NMR analysis in DMSO-d₆. The solution NMR spectra were recorded on a Bruker AVANCE-400 NMR spectrometer with a Spectro Spin superconducting magnet.

Recombinant Protein Production and Purification

BL21(DE3) *E. coli* containing appropriate constructs were grown at 37° C. in terrific broth (TB) to OD 600 of 0.9, induced with 0.15 mM isopropyl-β-D-thiogalactoside (IPTG), and allowed to grow for an additional 20 h at 18° C. Cells were harvested by centrifugation, washed with phosphate buffered saline (PBS) (137 mM NaCl, 2.7 mM KCl, 10 mM Na₂HPO₄ and 1.8 mM KH₂PO₄), resuspended in 150 mL of lysis buffer (50 mM Tris pH 8.0, 0.5 M NaCl, 20 mM imidazole, and 0.5 mM dithiothreitol (DTT)), and lysed with five passes through a M-110L microfluidizer (Microfluidics). The resulting crude protein lysate was clarified by centrifugation prior to Qiagen Ni-NTA gravity flow chromatographic purification. After loading the clarified lysate, His-tagged recombinant protein-bound Ni-NTA resin was washed with 20 column volumes of lysis buffer, and eluted with 1 column volume of elution buffer (50 mM Tris pH 8.0, 0.5 M NaCl, 250 mM imidazole and 0.5 mM DTT). 1 mg of His-tagged TEV protease was added to the eluted protein, followed by dialysis at 4° C. for 16 h in dialysis buffer (50 mM Tris pH 8.0, 0.1 M NaCl, 20 mM imidazole and 2 mM DTT). After dialysis, protein solution was then passed through Ni-NTA resin to remove uncleaved protein and His-tagged TEV. The recombinant protein was further purified by gel filtration on a fast protein liquid chromatography (FPLC) system (GE Healthcare Life Sciences). The principle peaks were collected, verified for molecular weight by SDS-PAGE, and stored in storage buffer (20 mM Tris pH 8.0, 25 mM NaCl, and 0.5 mM DTT) at a protein concentration of 10 mg/mL. The purity of the recombinant protein was evaluated by ImageJ densitometric analysis using bovine serum albumin as the standard (Schneider et al., 2012). 200 µM PLP was added to all buffers during the purification of all AAAD family enzymes.

Enzyme Assays

The AAS enzyme assays were performed in 100 µL of reaction buffer (50 mM Tris, pH 8.0) containing 50 µg of

recombinant enzyme, 200 µM PLP. Kinetic reactions were incubated with a range of amino acid substrate concentrations (1 µM-4 mM) at 30° C. for 30 minutes prior to quenching with 100 µL of 0.8 M formic acid. The reaction mixture was centrifuged, and the supernatant was analyzed by Pierce Quantitative Peroxide Assay Kit (Pierce) against a standard curve of hydrogen peroxide to demonstrate AAS activity or determine AAS kinetic parameters. Rr4HPAAS reactions were also analyzed by LC-MS-UV. 50 µL of reaction mixture was analyzed by an Ultimate 3000 liquid chromatography system (Dionex), equipped with a 150 mm C18 Column (Kinetex 2.6 µm silica core shell C18 100 Å pore, Phenomenex) and coupled to an UltiMate 3000 diode-array detector (DAD) in-line UV-Vis spectrophotometer (Dionex) and a TSQ Quantum Access MAX triple-quadrupole mass spectrometer (Thermo-Scientific). To resolve chromatographically L-tyrosine, tyrosol and 4-HPAAA, compounds were separated through the use of an isocratic mobile phase containing 50 mM monopotassium phosphate pH 4.6, 15% (v/v) acetonitrile and 0.5 mM octyl sulfate. Rr4HPAAS product formation was quantified using the UV absorbance at 280 nm and compared to analytical standards using the diode array detector wavelength at a wavelength range of 200-500 nm and chromatographic retention time. The reduction of aldehyde products was achieved by addition of saturated sodium borohydride in ethanol or by addition of 10 mM NADPH and 10 g of Rr4HPPAR1 or Rr4HPPAR2.

The phenylacetaldehyde reductase activity assays using Rr4HPPAR1 and Rr4HPPAR2 were carried out in 200 µL reaction buffer (50 mM Tris, pH 8.0) at the presence of 2 mM phenylacetaldehyde, 5 mM NADPH and 5 µg of recombinant enzyme. The reactions were incubated at 30° C. for various time points, quenched with an equal volume of 0.8 M formic acid, and extracted by 100 µL of ethyl acetate. The organic phase was then analyzed by gas chromatography-mass spectrometry (GC-MS) using an 5% Phenyl Methyl Silox column (30 m×250 m×0.25 m, Agilent) with a temperature gradient as follows: 0-1 min 45° C., 4-13.33 min 45-185° C. The quadrupole MS was set to EI mode, electron energy at 70 eV, MS-source temperature at 230° C., MS-quad temperature at 150° C., scan mass range at 50-300 m/z and SIM for 120 m/z and 122 m/z. EI-MS spectra were compared against analytical standards. The 4-HPAA reductase activity assays were carried out in 100 µL of 50 mM Tris pH 8.0 with the addition of 5 mM 4-HPAA, 10 mM NADPH and 0.2 µg of Rr4HPPAR1 or 15 ag of Rr4HPPAR2. The reactions were incubated at 30° C. and then quenched at various time points with the addition of 100 µL methanol. The reaction mixture was then centrifuged and analyzed by LC-MS. Compounds were separated by reversed-phase chromatography with a ramp gradient of solvent A (0.1% formic acid in H₂O) and solvent B (0.1% formic acid in acetonitrile): 10% solvent B for 0.5 min, 5-40% solvent B over 8.5 min, 95% solvent B for 1.8 min followed by a final equilibration of 10% solvent B for 1 min with a flow rate at 0.7 mL/min. Product formation was measured using select ion monitoring in positive mode for a centroid center mass of 121.065 with a scan width of 0.002. The specific activity was determined at a five-minute reaction time point, and quantified against a standard curve of tyrosol.

Kinetic characterization of UGTs was conducted in 200 µL reaction buffer (50 mM Tris, pH 8.0) containing 10 mM UDP-glucose and various concentrations of tyrosol (0.01-5.0 mM). Reactions were started with addition of recombinant enzyme, incubated at 30° C. for 10 minutes, and quenched by addition of 200 µL of methanol. The reaction

mixed was then analyzed by LC-HRMS as described above. Compounds were separated by reversed-phase chromatography with a ramp gradient of solvent A (0.1% formic acid in H₂O) and solvent B (0.1% formic acid in acetonitrile): 5% solvent B for 0.5 min, 5-55% solvent B over 6 min, 55-5% solvent B over 1.0 min and a final equilibration of 5% solvent B for 1 min with a flow rate at 0.6 mL/min. Product formation was measured using select ion monitoring in positive mode for a centroid center mass of 318.15. Product mass was calculated by comparison to a standard curve of the NMR verified plant purified salidroside and icariside D2 samples.

Kinetic constants such as K_m and V_{max} were determined by fitting raw data to the Michaelis-Menten equation using the nonlinear regression function in Prism (version 7.0).

Accession Codes

The sequences of *R. rosea* genes reported in this article are deposited into NCBI GenBank under the following accession numbers: Rr4HPAAS (MF674522), RrAAS (MF674523), Rr4HPAR1-2 (MF674524-MF674525) and RrUDP1-34 (MF674526-MF674558, MG385659). Raw RNA-Seq reads have been submitted to NCBI SRA (SRR5936536 and SRR5936537). The de novo transcriptomes assembled from the raw reads have been submitted to NCBI TSA (GFVD000000000 for merged transcriptome, GFVE000000000 for crown transcriptome, and GFVF000000000 for root transcriptome). Raw and mzTab format feature called metabolomic data from the *R. rosea* crown and root have been uploaded to the EBI MetaboLights database (MTBLS566).

TABLE 1

Kinetic parameters of characterized enzymes.					
	Enzyme				
	RrT4GT	RrT8GT	RrUGT29	Rr4HPAAS	Rr4HPAAS
	Substrate				
	tyrosol	tyrosol	tyrosol	L-tyrosine	L-DOPA
k_{cat} (sec)	481.60 $\pm$ 3.91	576.20 $\pm$ 5.68	167.5 $\pm$ 0.82	4.92 $\pm$ 0.08	9.52 $\pm$ 0.37
K_m (mM)	4.11 $\pm$ 0.08	1.37 $\pm$ 0.05	0.53 $\pm$ 0.01	0.42 $\pm$ 0.02	1.04 $\pm$ 0.10
k_{cat}/K_m (sec ⁻¹ mM ⁻¹)	117.18	420.58	316.04	11.71	9.15

TABLE 2

Cloning primers.			
Gene	Vector/ direction	SEQ ID NO:	Sequence
Rr4HPAAS	pHis8-4 Forward	97	GAAACTTGTACTTCCAGGCCCATGGCATGGGC AGCTTGCCTTCTCCTAATG
Rr4HPAAS	pHis8-4 Reverse	98	CTCGAATTCGGATCCGCCATGGCTAAGACACGA TGCTTTGAGCTGTTTCTTG
Rr4HPAAS	pEAQ-HT Forward	99	GTATATTCTGCCAAATTCGCGACCGGTATGGGC AGCTTGCCTTCTCCTAATG
Rr4HPAAS	pEAQ-HT Reverse	100	GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGCTAAGACACGATGCTTTGAGCTGTTTCTTG
Rr4HPAAS	p423TEF Forward	101	GCATAGCAATCTAATCTAAGTTTCTAGAACTAG TATGGGCAGCTTGCTTCTCC
Rr4HPAAS	p423TEF Reverse	102	CAGCCCGGGGATCCACTAGTCTAAGACACGAT GCTTTGAGCTGTTTCTTG
RrAAS	pHis8-4 Forward	103	GAAACTTGTACTTCCAGGCCCATGGCATGGAG GAGGAGTTGAAGCCG
RrAAS	pHis8-4 Reverse	104	CTCGAATTCGGATCCGCCATGGTCATGCATTAT ATGCTTTTGTAGCAGTGAAGTG
RrPAR1	pHis8-4 Forward	105	GAAACTTGTACTTCCAGGCCCATGGCATGAGTT TAAGCGGAGCGGGG
RrPAR1	pHis8-4 Reverse	106	CTCGAATTCGGATCCGCCATGGTCAGAGTTTGGC GAAACCCTTTCC
RrPAR1	p425TEF Forward	107	GCATAGCAATCTAATCTAAGTTTCTAGAACTAG TATGAGTTTAAGCGGAGCGGGG
RrPAR1	p425TEF Reverse	108	CAGCCCGGGGATCCACTAGTTCAGAGTTTGGC GAAACCCTTTCC
RrPAR2	pHis8-4 Forward	109	GAAACTTGTACTTCCAGGCCCATGGCATGGGTT TATCTGAAGAGAAGAAGTTAG

TABLE 2-continued

Cloning primers.			
Gene	Vector/ direction	SEQ ID NO:	Sequence
RrPAR2	pHis8-4 Reverse	110	CTCGAATTCGGATCCGCCATGGTCATTGTCTTT CAAACCTTCGACAGTGTCTC
RrUGT1	p426TEF Forward	111	CAATCTAATCTAAGTTTCTAGAACTAGTATGGT GACGAAAAAACTCACATTCTTATCC
RrUGT1	p426TEF Reverse	112	CAGCCCGGGGATCCACTAGTTCAGGTAAGACC AGACACAACTTGAC
RrUGT2	p426TEF Forward	113	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG TTCTGATTCACGGCCTC
RrUGT2	p426TEF Reverse	114	CAGCCCGGGGATCCACTAGTCTAGGACAAAGT CTCTCTTCTCAACTTCAATTC
RrUGT2	pHis8-4 Forward	115	GAAAACCTTGACTTCCAGGCCCATGGCATGGGT CTGATTCACGGCCTC
RrUGT2	pHis8-4 Reverse	116	CTCGAATTCGGATCCGCCATGGCTAGGACAAAG TCTCTTCTCAACTTCAATTC
RrUGT2	pEAQ-HT Forward	117	GTATATTCTGCCCAAATTCGCGACCGGTATGGGT TCTGATTCACGGCCTC
RrUGT2	pEAQ-HT Reverse	118	GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AG CTAGGACAAAGTCTCTTCTCAACTTC
RrUGT3	p426TEF Forward	119	CAATCTAATCTAAGTTTCTAGAACTAGTATGTC AGGCACACCACACATCG
RrUGT3	p426TEF Reverse	120	CAGCCCGGGGATCCACTAGTTCATGCTTCATC GAATCCGCC
RrUGT3	pHis8-4 Forward	121	GAAAACCTTGACTTCCAGGCCCATGGCATGTCAG GCACACCACACATCG
RrUGT3	pHis8-4 Reverse	122	CTCGAATTCGGATCCGCCATGGTCAATGCTTCAT CGAACTCCGCC
RrUGT3	pEAQ-HT Forward	123	GTATATTCTGCCCAAATTCGCGACCGGTATGTCA GGCACACCACACATCG
RrUGT3	pEAQ-HT Reverse	124	GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGTCAATGCTTCATCGAACTCCGCC
RrUGT4	p426TEF Forward	125	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG TTCACAAGCCTCTCCAAAACC
RrUGT4	p426TEF Reverse	126	CAGCCCGGGGATCCACTAGTTCATTCCTTGAAC TGGAGAATATCTTTCACAAGCC
RrUGT5	p426TEF Forward	127	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA ACCGAGACCTCACGCAG
RrUGT5	p426TEF Reverse	128	CAGCCCGGGGATCCACTAGTTTAATTAGTGTCA CCAAGATGAGTTTCTTTAGTAAG
RrUGT6	p426TEF Forward	129	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA ATCTGTACAAGGTGTTCAAGAAAAGC
RrUGT6	p426TEF Reverse	130	CAGCCCGGGGATCCACTAGTTCAGTTTGAATTC CTCGACAGGAGCAC
RrUGT7	p426TEF Forward	131	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC TGAAAACACTCATGCTCATGC
RrUGT7	p426TEF Reverse	132	CAGCCCGGGGATCCACTAGTTCATTTCTTGAAG ATTGTAGGTCGTGGATG
RrUGT8	p426TEF Forward	133	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC TTCCTCCTCTTAGCTTGTGATTC
RrUGT8	p426TEF Reverse	134	CAGCCCGGGGATCCACTAGTTTATTAACTGTT TCTTGTTTTTGACAGGACAGAATGAATG

TABLE 2-continued

Cloning primers.			
Gene	Vector/ direction	SEQ ID NO:	Sequence
RrUGT9	p426TEF Forward	135	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG GTCTGAGCCACTAGTCC
RrUGT9	p426TEF Reverse	136	CAGCCCGGGGGATCCACTAGTTTATGCTGAAATT GCATCCTTAGCAACTGG
RrUGT10	p426TEF Forward	137	CAATCTAATCTAAGTTTCTAGAACTAGTATGAC GAGGCGCCACCAC
RrUGT10	p426TEF Reverse	138	CAGCCCGGGGGATCCACTAGTTCATCCAAGGCC ATTGACAAAACGAC
RrUGT11	p426TEF Forward	139	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC AGGCGAGATTCTAATACTTCCG
RrUGT11	p426TEF Reverse	140	CAGCCCGGGGGATCCACTAGTTCACTTGTGGGA GATAATGAAGTCCCTG
RrUGT12	p426TEF Forward	141	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA GGAGGCGGCCAG
RrUGT12	p426TEF Reverse	142	CAGCCCGGGGGATCCACTAGTTTAAACACAGAGT CCAAATGTCCAGCAAC
RrUGT13	p426TEF Forward	143	CAATCTAATCTAAGTTTCTAGAACTAGTATGCT ACCTCTCTTACATGTTACACTAAC
RrUGT13	p426TEF Reverse	144	CAGCCCGGGGGATCCACTAGTTTACAAGCCAAT GTTGGTCCTGAGATCAC
RrUGT14	p426TEF Forward	145	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA CACCACCGCCGC
RrUGT14	p426TEF Reverse	146	CAGCCCGGGGGATCCACTAGTTTATCCCTTCCA AGTTGAGTCAACGAC
RrUGT15	p426TEF Forward	147	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC TGATGCTGCTCAACATGTC
RrUGT15	p426TEF Reverse	148	CAGCCCGGGGGATCCACTAGTTTATTGAACTTGG TGAAATTGAAGATGACTCAAAAGG
RrUGT16	p426TEF Forward	149	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC AGAGGAAAACAGAACACG
RrUGT16	p426TEF Reverse	150	CAGCCCGGGGGATCCACTAGTTCATACAGCTGA AGATATTTTGGATATGAATTGGTC
RrUGT17	p426TEF Forward	151	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG CTCACTTCCTTCCAC
RrUGT17	p426TEF Reverse	152	CAGCCCGGGGGATCCACTAGTTCAGACGCTAAA CTGGACCACTTTTTC
RrUGT18	p426TEF Forward	153	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG CTCCCGAGGAAAGCCACATG
RrUGT18	p426TEF Reverse	154	CAGCCCGGGGGATCCACTAGTTCATTTGGGGA ATTAGACAGCAGG
RrUGT19	p426TEF Forward	155	CAATCTAATCTAAGTTTCTAGAACTAGTATGAC GTCATCAACACCTCCTCCTC
RrUGT19	p426TEF Reverse	156	CAGCCCGGGGGATCCACTAGTCTAAAAAATGC TTTAACATAGCTAGCGTCCG
RrUGT20	p426TEF Forward	157	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG TTCACGACGTCGTC
RrUGT20	p426TEF Reverse	158	CAGCCCGGGGGATCCACTAGTTCATTTCAATAA GCTTCATCAATCAACTCGG
RrUGT21	p426TEF Forward	159	CAATCTAATCTAAGTTTCTAGAACTAGTATGAA GTCCAACACTCATCTATTCCTC

TABLE 2-continued

Cloning primers.			
Gene	Vector/ direction	SEQ ID NO:	Sequence
RrUGT21	p426TEF Reverse	160	CAGCCCGGGGGATCCACTAGTTCATACAACCGG CTCCAGTTGAC
RrUGT22	p426TEF Forward	161	CAATCTAATCTAAGTTTCTAGAACTAGTATGAA AACTCCTCAAAATCCACACGTAG
RrUGT22	p426TEF Reverse	162	CAGCCCGGGGGATCCACTAGTTCATCCTGATA AATCTTTGAACTCATCTTGCTC
RrUGT23	p426TEF Forward	163	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA AAGGCAGAGTGATCACCAG
RrUGT23	p426TEF Reverse	164	CAGCCCGGGGGATCCACTAGTTCATTGGTGGAT ATCACATCTCTAACAACCTG
RrUGT24	p426TEF Forward	165	CAATCTAATCTAAGTTTCTAGAACTAGTATGAG CAACGCCGCCG
RrUGT24	p426TEF Reverse	166	CAGCCCGGGGGATCCACTAGTTTAGTTTATGACT TCATTCACTTGCTCCAACAAC
RrUGT25	p426TEF Forward	167	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC GCGCCACCACCTTG
RrUGT25	p426TEF Reverse	168	CAGCCCGGGGGATCCACTAGTTTAGCAGGTAAC AAGGTTATTAACCAATCCTTGAG
RrUGT26	p426TEF Forward	169	CAATCTAATCTAAGTTTCTAGAACTAGTATGTC ATCAGATTCGGCCACATTATCC
RrUGT26	p426TEF Reverse	170	CAGCCCGGGGGATCCACTAGTCTATATTATTTT CTTAATGCCATGACTTGTCGGACC
RrUGT27	p426TEF Forward	171	CAATCTAATCTAAGTTTCTAGAACTAGTATGAG TTCAGTCAATGCTCAAAAGCC
RrUGT27	p426TEF Reverse	172	CAGCCCGGGGGATCCACTAGTTCAAAAGTGCAT TAGTAGTCCTCCACAATC
RrUGT28	p426TEF Forward	173	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA CTCGGTTGATCTGAACAAG
RrUGT28	p426TEF Reverse	174	CAGCCCGGGGGATCCACTAGTCTAGTTGGCACTT GGCAACACAATCG
RrUGT29	p426TEF Forward	175	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG ATCTCTAGGAAAGAAGATTCAAC
RrUGT29	p426TEF Reverse	176	CAGCCCGGGGGATCCACTAGTTTAGGTTGTAAC ACAATTTTTTTTTTGGAC
RrUGT29	pHis8-4 Forward	177	GAAAACCTTGTAAGTCCAGGCCATGGCATGGGA TCTCTAGGAAAGAAGATTCAAC
RrUGT29	pHis8-4 Reverse	178	CTCGAATTCGGATCCGCCATGGTTAGGTTGTAAC TACAATTTTTTTTTTGGAC
RrUGT29	pEAQ-HT Forward	179	GTATATCTGCCCAAATTCGCGACCGGTATGGGA TCTCTAGGAAAGAAGATTCAAC
RrUGT29	pEAQ-HT Reverse	180	GAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGTTAGGTTGTAACATAATTTTTTTTTTGGAC
RrUGT30	p426TEF Forward	181	CAATCTAATCTAAGTTTCTAGAACTAGTATGGG CTCCCGAGGAAAGCCACATG
RrUGT30	p426TEF Reverse	182	CAGCCCGGGGGATCCACTAGTTCATTTGGGGA ATTAGACAGCAGG
RrUGT31	p426TEF Forward	183	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA ATCTGTACAAGGTGTTCAAGAAAAG
RrUGT31	p426TEF Reverse	184	CAGCCCGGGGGATCCACTAGTTCAGTTGAATTC CTCGACAGGAGCAC

TABLE 2-continued

Cloning primers.			
Gene	Vector/ direction	SEQ ID NO:	Sequence
RrUGT32	p426TEF Forward	185	CAATCTAATCTAAGTTTCTAGAACTAGTATGGA CTCGGTTGATCTGAACAAGAAACC
RrUGT32	p426TEF Reverse	186	CAGCCCGGGGATCCACTAGTCTACAATTTTTT TTGGACAGAAGTACGTCATTATAAGTC
RrUGT33	p426TEF Forward	187	CAATCTAATCTAAGTTTCTAGAACTAGTATGAG CTTAATTGAAAAACCACTCACG
RrUGT33	p426TEF Reverse	188	CAGCCCGGGGATCCACTAGTCTAACGGATATG TTTTGTTTTTGAGAGCAGGAC
RrUGT33	pHis8-4 Forward	189	GAAAACTTGACTTCCAGGCCCATGGCATGAGCT TAATTGAAAAACCACTCACG
RrUGT33	pHis8-4 Reverse	190	CTCGAATTCGGATCCGCCATGGCTAACGGATATG TTTTGTTTTTGAGAGCAGGAC
RrUGT33	pEAQ-HT Forward	191	GTATATCTGCCCCAAATTCGCGACCGGTATGAGC TTAATTGAAAAACCACTCACG
RrUGT33	pEAQ-HT Reverse	192	GAAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGCTAACGGATATGTTTTGTTTTTGAGAGCAGGA C
RrUGT34	p426TEF Forward	193	GCATAGCAATCTAATCTAAGTTTCTAGAACTAG TTGGACCTGACGACAGCGTTTTG
RrUGT34	p426TEF Reverse	194	CAGCCCGGGGATCCACTAGTTTAGTTTTGTTC TCGTACAATAATGCACAACTCATC
Pc4HPAAS	pHis8-4 Forward	195	GAAAACTTGACTTCCAGGCCCATGGCATGGGCT CCATCGATAATC
Pc4HPAAS	pHis8-4 Reverse	196	CTCGAATTCGGATCCGCCATGGTTAGGATAAAAT ATTACGATCTTCT
Pc4HPAAS	pEAQ-HT Forward	197	GTATATCTGCCCCAAATTCGCGACCGGTATGGGC TCCATCGATAATC
Pc4HPAAS	pEAQ-HT Reverse	198	GAAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGTTAGGATAAAATATTCAGATCTTC
PsTyDC	pHis8-4 Forward	199	GAAAACTTGACTTCCAGGCCCATGGCATGGGA AGCCTCCGACTAATAACCTTG
PsTyDC	pHis8-4 Reverse	200	CTCGAATTCGGATCCGCCATGGCTAGGCACCAA GTATGGCATCTGTATG
PsTyDC	pEAQ-HT Forward	201	GTATATCTGCCCCAAATTCGCGACCGGTATGGGA AGCCTCCGACTAATAACCTTG
PsTyDC	pEAQ-HT Reverse	202	GAAAAATTTAATGAAACCAGAGTTAAAGGCCTCG AGCTAGGCACCAAGTATGGCATCTGTATG
AAS55083	p426TEF Forward	203	CAATCTAATCTAAGTTTCTAGAACTAGTATGGC AGGCAGTGGGACTG
AAS55083	p426TEF Reverse	204	CAGCCCGGGGATCCACTAGTTCAGTGTTAACT GAGGATCTCCACTTTTAGC
EU567325	p426TEF Forward	205	GCATAGCAATCTAATCTAAGTTTCTAGAACTAG TATGGGTTCTGAAACTCGGCCTTG
EU567325	p426TEF Reverse	206	CAGCCCGGGGATCCACTAGTCTAGACTTTCTTT AACTTGAGTTCCTGAAGCAG

Enzymes of the plant aromatic amino acid decarboxylases (AAAD) family that can be used in the production of one or more of tyrosol, salidroside, and icaride D2 were identified. These plant AAAD-family enzymes contain substitutions in one of two active site residues responsible for influencing aldehyde synthase chemistry. These activity-influencing residues are boxed in the multiple sequence alignment of biochemically characterized plant AAADs show in FIG. 26.

Plant AAAD enzymes that contain an active site histidine to asparagine or aspartic acid substitution have an aldehyde synthase activity. This active site substitution is represented by the Rr4HPAAS MF674522 histidine 198 in FIGS. 31 and 32.

Plant AAAD enzymes that contain an active site tyrosine to leucine, isoleucine, phenylalanine, methionine or valine substitution have an aldehyde synthase activity. This active site substitution is represented by the Rr4HPAAS MF674522 phenylalanine 343 in FIGS. 31 and 32.

These active site substitutions at positions 198 and 343 were selected from natural variation, shown in FIG. 33, found within all plant AAAD sequences available on Phytozome V12.1. We have curated 226 plant AAAD sequences from Phytozome 12. The list was queried for sequences containing substitution in one of the two activity dictating residues to make a list of 73 enzymes that likely have some 4HPAAS activity. These 73 AAS enzymes are identified as SEQ ID NOS: 21-93. In some embodiments, any of the enzymes of SEQ ID NOS: 21-93 can provide 4HPAAS activity in a host cell or method described herein.

To demonstrate the roles of these residues in aldehyde synthase chemistry, the biochemical activity of wild type and mutant *Papaver somniferum* tyrosine decarboxylase (PsTyDC) enzymes were characterized. The substitution of the active site histidine (Rr4HPAAS MF674522 histidine 198) or the active site tyrosine (Rr4 PAAS MF674522 phenylalanine 343) within a *Papaver somniferum* tyrosine decarboxylase (PsTyDC) results in aldehyde synthase chemistry. FIGS. 34A-C are chromatograms showing product formation of PsTyDC and mutants.

SEQUENCES

TABLE 2

Summary of Sequences.	
SEQ ID NO.:	Description
1	Rr4HPAAS DNA
2	Rr4HPAAS GenBank accession MF674522
3	Rr4HPAR1 DNA
4	Rr4HPAR1 amino acid GenBank accession MF674524
5	RrUGT2 DNA
6	RrUGT2 amino acid GenBank accession MF674527
7	RrUGT3 DNA
8	RrUGT3 amino acid GenBank accession MF674528
9	RrUGT7 DNA
10	RrUGT7 amino acid GenBank accession MF674532
11	RrUGT13 DNA
12	RrUGT13 amino acid GenBank accession MF674538
13	RrUGT17DNA
14	RrUGT17 amino acid GenBank accession MF674542
15	RrUGT29 DNA
16	RrUGT29 amino acid GenBank accession MF674554

TABLE 2-continued

Summary of Sequences.	
SEQ ID NO.:	Description
17	RrUGT32 DNA
18	RrUGT32 amino acid GenBank accession MF674557
19	RrUGT33 DNA
20	RrUGT33 amino acid GenBank accession MF674558
21	<i>Arabidopsis thaliana</i> AT2G20340.1
22	<i>Brachypodium distachyon</i> 1g28960.3
23	<i>Carica papaya</i> 16427710
24	<i>Ricinus communis</i> 16804377
25	<i>Cucumis sativus</i> 16963476
26	<i>Vitis vinifera</i> 17835588
27	<i>Citrus sinensis</i> 18113817
28	<i>Capsella rubella</i> 20900667
29	<i>Malus domestica</i> 22636618
30	<i>Linum usitatissimum</i> 23178995
31	<i>Eutrema salsugineum</i> 20200788
32	<i>Populus trichocarpa</i> 27022899
33	<i>Brachypodium stacei</i> 06G160800.1
34	<i>Physcomitrella patens</i> Pp3c4_30790V3.1
35	<i>Ananas comosus</i> 33033299
36	<i>Zostera marina</i> 33182387
37	<i>Daucus carota</i> subsp. <i>sativus</i> 36055203
38	<i>Trifolium pratense</i> 35974269
39	<i>Arabidopsis lyrata</i> 35943929
40	<i>Sorghum bicolor</i> 002G120700.1
41	<i>Sphagnum fallax</i> 0166s0011.1
42	<i>Kalanchoe laxiflora</i> 1398s0003.1
43	<i>Manihot esculenta</i> 12G038600.1
44	<i>Prunus persica</i> 8G214500.1
45	<i>Eucalyptus grandis</i> K01418.1
46	<i>Amborella trichopoda</i> 31565185
47	<i>Salix purpurea</i> 0252s0200.1
48	<i>Medicago truncatula</i> 31080941
49	<i>Brassica rapa</i> I01156.1
50	<i>Brassica rapa</i> I04706.1
51	<i>Brassica rapa</i> G00043.1
52	<i>Glycine max</i> 03G167900.1
53	<i>Fragaria vesca</i> 27261550
54	<i>Kalanchoe fedtschenkoi</i> 0172s0035.1
55	<i>Capsella grandiflora</i> 22666s0001.1
56	<i>Selaginella moellendorffii</i> 15420188
57	<i>Setaria italica</i> 3G188200.1
58	<i>Kalanchoe fedtschenkoi</i> 0033s0078.1
59	<i>Daucus carota</i> subsp. <i>sativus</i> 36068870
60	<i>Daucus carota</i> subsp. <i>sativus</i> 36056758
61	<i>Solanum tuberosum</i> 3DMP400026166
62	<i>Solanum tuberosum</i> 3DMP400024738
63	<i>Solanum lycopersicum</i> 36137005
64	<i>Daucus carota</i> subsp. <i>sativus</i> 36065781
65	<i>Oropetium thomaeum</i> 35995617
66	<i>Oryza sativa</i> 33157740
67	<i>Brachypodium stacei</i> 01G392300.1
68	<i>Amaranthus hypochondriacus</i> 32828676
69	<i>Brachypodium distachyon</i> 5g21770.1
70	<i>Brachypodium distachyon</i> 2g02360.1
71	<i>Sorghum bicolor</i> 009G192600.1
72	<i>Kalanchoe laxiflora</i> 0994s0009.1
73	<i>Kalanchoe laxiflora</i> 0003s0173.1
74	<i>Panicum hallii</i> 32512198
75	<i>Prunus persica</i> 6G202600.1
76	<i>Prunus persica</i> 4G086700.1
77	<i>Prunus persica</i> 4G087100.1
78	<i>Medicago truncatula</i> 31073039
79	<i>Zea mays</i> GRMZM2G009400
80	<i>Glycine max</i> 07G059000.1
81	<i>Panicum virgatum</i> Ca01381.1
82	<i>Theobroma cacao</i> 27425420
83	<i>Fragaria vesca</i> 27274768
84	<i>Gossypium raimondii</i> 26786642
85	<i>Populus trichocarpa</i> 26994989
86	<i>Malus domestica</i> 22679008
87	<i>Citrus Clementina</i> 20801973
88	<i>Citrus Clementina</i> 20818150
89	<i>Vitis vinifera</i> 17834108
90	<i>Petunia hybrida</i> ABB72475.1

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TABLE 2-continued

Summary of Sequences.	
SEQ ID NO.:	Description
91	<i>Carica papaya</i> 16421889
92	<i>Sphagnum fallax</i> 0042s0024.1
93	<i>Eucalyptus grandis</i> E01788.1
94	pHis8-4

42

TABLE 2-continued

Summary of Sequences.	
SEQ ID NO.:	Description
95	pEAQ-HT
96	pJKW 1410

Rr4HPAAS DNA (SEQ ID NO: 1):

ATGGGCAGCTTGCCCTTCCTAATGATCCATCAAACACCTTCAACCCCATGGACCTC

ACCGAGTTATCCACCGAGTCGAACTCGTCGTAGATTTTCATAACTCAGTACTACCAA

ACCCCTAGAGACCCGACCCGTCAGCCACGGGTCAAGCCAGGTTTCTTAACGGGCCA

GCTTCCAGATAAAGCACCTTTTCATGGTGAATCAATGGAAGTAATATTGTCTGATGT

AAATGAGAAGATTGTCCCTGGCCTCACTCATTGGCAAAGCCCTAATTTCCATGCATA

CTTCCAGCCAGTTCAGCAACGCAGGGCTGTTGGGAGAGTTACTATGCTCCGGACT

CAGTGTCTATTGGGTTACATGGAGCTCCTCCCTGCCGCGACGGAGCTTGAGAATGT

CGTGGTTGACTGGATGGCCAAGATGCTTAACCTTCCATCCTCTTTCTGCTTCTCCGGC

GGAGCGGTGGCGTCTTGCAAGCAAACTTGCAGGGCTGTGTTGTGCACTTTAGCC

GCTGCGAGGGACAAGGCTCTTAACCGGGTGGGAGATGATCAGATCAATAAATCGGT

CCTCTACTGCTCCGACCAAACACATTTCAATCCACAAGGGCGCAAAGTTGATAGG

AATCCGATCAAAGAACATAAAATCAATCACTACTAAGAAAGAGAACGAGTTTAAAC

TCTGTCTTAACGACCTACGCGACGCGATAAGGAGTGATCTGGAAGCAGGACTAGTT

CCGTTTTACGTATGCGGAACGATTGGAACGACCGCGTTAGGAGTTGTGGATCCGATT

AAAGAGCTGGTAAGGTGGCAAGAGAGTTTGATTTGTGGTTACATGTTGATGGAGC

TTATGGTGGCAGTGATGCATATGCCCTGAGTTTCAGCATTACCTTGATGGAGTTGA

CCTTGTGACTCGATCAGCATGAATGCACATAAATGGCTTTTATCCAATCTAGATTG

CTGCTTCCTGTGGCTTCAATCTCCTAACGCCCTAATCGAATCCCTGGCGCAGAAGC

TAACTTTCTGAAAGGTGGTAGTGAGATGGTGGATTACAAGGACTGGCAGATATCGTT

GAGTCGTCGATTTAGAGCGATCAAGATGTGGATGGTGATAAGGCGATACGGTGTGA

GTAATCTCATTGAGCATATTCGATCCGACGTGAGCATGGCGGTGAGATTCGAAGAG

ATGGTGGCGGGCGACGACCGGTTTGAATCGTGTTTCTAGAAAGTTTGCGCTTGTT

TGCTTCAAGCTTAGTAGCGAGAAGACACCACCGGCCGCGACTCGGAGTTAACTCG

TGAGCTGATGGAGAGAGTCAACTCGAGTGGGAAGGCTTACTTGAGTGGAGTTCAAA

TGGGTCGGATCTTCTTCATCAGGTGTGTATCGGGTCGAGTTTGACTGAGGAGAGAC

ACGTCGATAATCTGTGGAGGCTCATTCAAGAAACAGCTCAAAGCATCGTGTCTTAG

Rr4HPAAS GenBank accession MF674522 (SEQ ID NO: 2):

MGSLPSPNDPSNTFNPMDLTELSTESKLVDFITQYYQLETRPVQPRVKPGFLTGQLPD

KAPPHGESMEVILSDVNEKIVPGLTHWQSPNFHAYFPASSNAGLLGELLCSGLSVIGFT

WSSSPAATELENNVVDWMKMLNLPSSFCFSGGGGVQLQANTCEAVLCTLAAARDKA

LNRVGDDQINKLVLYCSDQTHFTIHKGAKLIGIRSKNIKSITTKKENEFKLCPNDRDAIR

SDLEAGLVFPFYVCGTIGTTALGVVDP IKELGKVAREFDLWLHVDGAYGGSACICPEFQH

YLDGVDLVDSISMNAHKWLLSNLDCFLWLQSPNALIESLAAEANFLKGGSEMVDYKD

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WQISLSRRFRAIKMMWVIRRYGVSNLIEHIRSDVSMVRFEEEMVAADDRFEIVFPRKFA
 LVCFKLSSEKTPPGRDSELTRELMERNSSGKAYLSGVQMGRIFPIRCFIGSSLTEERHVD
 NLWRLIQETAQSIVS

Rr4HPAR1 DNA (SEQ ID NO: 3):
 ATGAGTTTAAGCGGAGCGGGGAAGGTGGTTTTCGTTACCGCGCGTCTGGCTACAT
 AGCGTCCTGGCTCGTCAAGCTTCTTCTCCAGCGCGTTATACCGTCAAGGCCTCCGT
 TCGCGATCCTAATGATCCGAAAAGACTCAGCACTTGACGGCACTTGATGGAGCTA
 AGGAGAGGCTGCAGTTGTACAAAGCCAATTTGCTTGAACAAGGCTCGTTTGATCCCA
 TAGTTGAAGGATGTGAAGGTGTTTTCCACACCGCGTCTCCCTTTTATCATGCAGTGG
 ATGATCCGCAGGCCGAGTTAATTGACCCTGCTGTCAAGGGAACACTCAATGTTCTTT
 CTTTCATGTGCTAAAGTTGCGTCTCTTAAAGAGTAGTCTGACTTCTTCGATTGCTGC
 TGTTCATATAAATGGGAAACCCGTA CTCCGAGGTTGTAGTTGACGAGACTTGGT
 TTCTAACCAGATGTTTGTAGGAGATGAAGCTTTGGTATGTCATATCCAAGACACT
 CGCTGAAGAAGCAGCATGGAAGTTTGTGAAAGAGAAAGGAATAGACATGGTTACCA
 TAAATCCGGCCATGGTGATTGGTCCCCCTTCTGCAACCAACTCAATACCAGTGCTG
 CTGCTATTCTGAACCTTGATCAATGGATCGGAGACATACCCAATGCTTCTTTTGAT
 GGGTCAATGTGAAAGATGTTGCAGAAGCACAGTCTTTCATTTGAGGTTCTTCAG
 CTAATGGTAGATACTGCTTGGTGAAAGAGTTGCCACAGTCTGAAGTGGTGAACA
 TGCTCCATGAGCTCTACCTGATATCAAACCTCCCGCAAGTGTGCAGATGACAAAC
 CATTGTGCCAATTTATCAAGTTTCAAAGAAAAGGCACATACCTTAGGGGTAAAT
 TCATTCTTTAGAGGTAAGCCTCAAGGAAACAGTTGAAAGCTTGAAAGAAAAGGT
 TTCGCCAAACTCTGA

Rr4HPAR1 amino acid GenBank accession MF674524 (SEQ ID NO: 4):
 MSLSGAGKVVCVTGASGYIASWLVLKLLQRGYTVKASVRDPNDPKKTQHLTALDGAK
 ERLQLYKANLLEQGSFDPIVEGCEGVFHTASPFYHAVDDPQAEIDPAVKGTNLNVLSSC
 AKVASLKRVLTLSSIAAVAYNGKPRTPPEVVVDETWFSPNDVCKEMKLWYVISKTLAEE
 AAWKFVKEKGIDMVTINPAMVIGPLLQPTLNTSAAAILNLINGSETYPNASFGWVNVKD
 VAEAHVLAFEPVSANGRYCLVERVAHSSEVVNMLHELYPDIKLPKACDDKPFVPIYQ
 VSKEKAHTLGVKFIPLEVSLKETVESLKEKGFACL

RrUGT2 DNA (SEQ ID NO: 5):
 ATGGGTTCGATTACGGCCTCTACGCGTCTTCTTCTTCCCTTCATGGCTCACGGCC
 ATCTGATTCGATGTCGACATCGCCAGACTCTTCTCTTCTCAAGGAGTCCACTCCA
 CCATCATCACCACCCCACTAAACGCCAATTACATCTCCAAACGACGCTCTATCCA
 TCAAAACGATACCGTTTCTGCTGCGGAAGTTGGGCTTCCGACGGCTGCGAGAATA
 TCGACATGCTTCTTCGCCGATCTCTTCTTCAAATTTTCCAAGCCGCCAATTTACT
 CCAAGCGCGTTCGAGAACCTTCTAGAACTCGAAAGGCCGATTGCTTAATCTCCGA
 CATCTTCTTCCCTGGTCAGTCGACTCCGCCGAGAAATTCAACATCCCAGACTCGT
 TTTCCACGGCACGAGCTTCTTCGCCATGTGCCCATGGAGAGCTTGAAGACCCACAA
 GCCCTATAAATCGGTAAGCACCGACTCTGAACCGTCTTAATCCCGAATCTCCCTGA
 TGAAATCAAAATGACTAAAAGTCAGTTCACGGTTGACGCTTGGGAAGACACCGAAA
 AGGGCCTTGGGAAGCTGTTGGCTGATGCGAGAGCTTACGGGCTGAGGAGCTTCGGC
 ATGATCGTAAACAGCTTCCACGAGCTCGAACC GGCTTACGCGATTATTACAAGAAT

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GTGTTGAACATGAAAGCGTGGTGTGTCGGGCCTGTTTCGTTATATAACCGAAACGAT
 GACGAGAAAAATTGCAAGAGGGAAGAAATCAGCAATCGATGATCATGAGTGTTTAAA
 ATGGCTGGAGGAAAGCAGCCAGACTCCGTCGTGTACGTTTGTTCGGGAGCAGCG
 CGAGCTTCCCTGATGAGCAGTTGCGCGATATCGCATTGGGGCTGGAAGAATCTGGA
 GTAAATTTTCATCTGGGTGATCAGGAGAAGTTCGAGTCAGGATCAGAAGATTACTTG
 CCGGAGGGGTTTGAAGACCGGTGAAGGACAGAGGGCTCGTGATCCGAGGTTGGGC
 GCCACAGGTACTGATTTTGGACCATCCGTCGGTTGGGGGATTTGTGACTCACTGCGG
 ATGGAATTCGGCATTTGAGGGGATTTTCAGCTGGCTTGCCGATGGTGACTTGGCCACT
 GTTCGAGAGCAGTTTTTCAACCAGAAATTGATTACGGATGTGTTGAAAGTTGGGGT
 TGAGGTTGGAGTGCAGAAATGGTCTCGGAACGGGGAGGATCGCGTGACGAAGGAG
 AAGGTTGAGAAGCGGTGAGGGCTGTTATGGTTGGGGAGGACGCTGAGGAGAGGC
 GTGGCAGAGCTCGTCAGCTTGGGAAATTGGCAAAGAAAGCTGTGGCGAAAGATGGG
 TCTTCGTACATTGATCTCCACAATTTGCTTGATGAATTGAAGTTGAGAAGAGAGACT
 TTGTCCTAG

RrUGT2 amino acid GenBank accession MF674527 (SEQ ID NO: 6):
 MGSDSRPLRVFFPFMAHGLIPMVDIARLFSSQGVHSTIIITPLNANYISKTTSLSIKTIPF

PAAEVGLPDGCENIDMLSPDLFFKFFQAANLLQAPFENLLELERPDCLISDIFFPWSVDS
 AEKFNIPRLVPHGTSFFAMCAMESLKHKPYKSVSTDSEPFLIPNLPEIKMTKSQFTVD
 AWEDTEKGLGKLLADARASGLRSFGMIVNSFHELEPAYADYYKNVLNMKAWCVGPVS
 LYNRNDDEKIARGKKSADDHECLKWLEKQPDVVVYVCFGSSASFPDEQLRDIALGLE
 ESGVNFIIWIRRSSESGSEDYLPFGFEDRVKDRGLVIRGWAPQVLILDHPSVGGFVTHCG
 WNSALEGISAGLPMVTWPLFAEQFFNQKLITDVLKVGVEVGVQKWSRNGEDRVTEK
 VEKAVRAVMVGEDAERRGRARQLGKLAKKAVAKDGSSYIDLHNLDELKLRRETLS

RrUGT3 DNA (SEQ ID NO: 7):
 ATGTCAGGCACACCACACATCGCCATCCTCCCAGCCCCGGCATGGGCCACCTCATC
 CCCATGGCCGAGTTCGCCAAGCGCCTAGTCCACCACCACAACCTTCAGTATCACCTTC
 GTCATCCCTACCGACGGCCACCTTCCTCCGCCTACCAACAAGTCTCACCTCCCTCC
 CATCTTCCATAGATCACATCTTCCTTCCACAAGTCGACTTAACCGACGTCGTATCAC
 AATCACCGACTCATCCAGAATCGAAACCTAATCTCCCTCACCGTCGCTCGCTCCC
 TCTCCTCCCTCCGACCACCTTATCCTCTCTCCAATCGTCTAAAACCTCGTCTCGCT
 CGTCGTTGATCTTTTCGGCACTGATGCATTGACCCGGCCATCGAGCTCGGCATCTC
 GCCCTACATTTTCTTCCCTTCCACAGCCATGACGCTCTCGCTCTTCTATACATGCCT
 CAGCTTGACAAATCAGTCACGTGCGAATTTGTCACATGACGGATTTGTTTGAATT
 CCTGGATGCGTTTCTGTCCGTGGATCGGATTTATTCGACCCGGTTCAAGACAGGACC
 GACGAGGCTTATAAATGGGTCATACATCACTCCAACAGGTACCTATGGCGGAGGG
 TGTATAGAGAATAGCTTCATGGAGTTGGAACATGGTGCCTTAAAGTATTTGCAAAAC
 GGTTCAATCGGGTAAGCCGCTGTCTACGCGTCGGACCGTTGATTAAATGGATTAA
 TGATGTTGACGATTCGGGTCGAAGATAATCGAGTGGCTCGATGATCAACCGTTGG
 TTCGGTTTTATTGTTTCGTTTGAAGCGGCGGAACGCTCTCGTATGAGCAAATGAC
 CGAGCTGGCTCACGGTTTGAATCGAGCCAGCAACGGTTCTTATGGGTGGTTTCGGAG
 TCCGAATCAAATCCCAACAGCACGTATTTTCAGTGTACAAAGCCAAAAAGACCCGT
 TGGCTTACTTGCCAGAAGGATTTTAAACCGAACCGAGGGTAGGGGTCTGGTCGTAT

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CGAATTGGGCCCCACAGGCTCAAATTTTGAGTCACGGTTCGACCGGTGGGTTCATGA
GCCACTGTGGTTGGAATTCGATTTTGAGAGTGTGGTGCACGGCGTCCCGATCATAG
CGTGGCCGTTGTACGCCGAGCAGAAGATGAATTCGATAATCGTGGTGGAGGACGTT
AAGGTGGCGCTGAGGCCGCGGGGTAGGGGAGAGGGTGGTGGAGAGGTGCGAGA
TAACCGCAGTGGTGAAGGCGTTGATGGAGGGTGAAGAGGGGAAGAAGGTAAGGAA
TAGGATGAAGGAACCAAGGAAGCGCGGCACGTGCGGTTAGTGATGACGGTGCCT
CGACCATAGCGATTGCGGACTTGGCGCAAAAATGGCGGAGTTCGATGAAGCATTGA
RrUGT3 amino acid GenBank accession MF674528 (SEQ ID NO: 8):
MSGTPHIAILPSPGMGHLIPMAEFKRLVHHNFSITFVIPTDGPSSAYQQVLTSLPSSID
HIFLPQVDLTDVVSQSPAHPRIETLISLTVARSLSSLRTTLLSSLQSSKNLVSLVVDLFGTDA
FDPAILGISPIYFFPSTAMTSLSLFLYMPQLDKSVTCEFRHMTDLVRIPGCVPRGSDLPD
PVQDRDTEAYKWVIHHSNRYPMAEGVIENSFMELEHGALKYLQTVQSGKPPVYAVGPL
IKMDYDVDDSGSKII EWLDDQPVGSVLFVSPGSGGTLSYEQMTLAHGLESSQQRFLWV
VRSPNQIPNSTYFSVQSKDPLAYLPEGFLNRTEGRGLVVSNWAPQAQILSHGSTGGFM
SHCGWNSILESUVVHGPPIIAWPLYAEQKMNSIIIVEDVKVALRPAGVGERVVERSEITAV
VKALMEGEEGKKVRNRMKELKEAAARAVSDDGASTIAIADLAQKWRSSMKH-
RrUGT7 DNA (SEQ ID NO: 9):
ATGGCTGAAAACACTCATGCTCATGCCATAGTGGTACCATTTCAGTTC AAGGACAC
ATAAAGCCCTCGCTGAATCTAGCCCTCAAGCTAGCATCTCAAGGCTTCACCATCACT
TTTGTCAACCACTCATTTCACCCACCAGCAAATCTCCCAAGCTCACAAAACAGTACA
AATACAAACCATGACATGTTTTTCCAGGCACGAACTCCAGTCTCGATATCCGCCAT
GTAACGGTGACAGACACTTTTCCTTTGGGATTCGATCGCGCAGGGAATCAGGATCAG
TTTTGGGAGGGCATGCTTCACGTATTCCTGCACATGTTGATGAAGTGGTGGATCAG
TTAATGAATCTTCGAAGCCGAGACCACTTGTTTGATTCTGGATACATTTTATAACT
GGGTTTCCAAAATTGCTAACAAGTTTAATTTAGTGCATATTTTCATTTTGGACTCAGTC
TGCTCTTTCTTTCACTTTGTTTTACCATTGGGAACTTTTAAAGAAAAATGGTCACTTT
GGCTCTCCAGATAATCGCACGGATGTCATCGATTATATTCCTGGTGTGCAAGAGATC
AAGCCCGCAGACTTAATATCCTACCTTCAGATGAGTGATACAATACTGTGGCTCAC
AGGACTTGTTTTCACAGCATTTGAAGATGTCAGGAAGGCAGATTTTCATCCTGGCTAAT
ACAATCCAAGAATTTGAAACTGATACAATTTCTTCTATCCGATTTACCAGCCATTTT
TCTACCCAATTGGACCTGTTTTTTTAAACAAGTCTGAACAACAAGCTAGCTCAGCTTT
GTGGTCTGAGTCAGACTGTGAGCAGTGGCTAAGTACAAAACAAAAGGTCTGTTC
TCTATGCCTCATTTGGGAGCTATGCTCGTGAAC TAGGCATGATATCGCAGAGATAG
CCTACGATTGATGCAAAGTGAGGTGAATTTTATTTGGGTGATTCGCGACGATATTG
TGGGTGCACACGAGACTGATTTTTTACCAACAGAATTCATAAATGGAATCAAACCTCA
AAGATCAGGGACTACTAGTTTCTGGTGCTCTCAAACGAAGTTTGTCCAATGCGG
CGATTGGAGGATTTCTGACTCATTGTGGATGGAAC TCGATACTCGAAAGCGTATGGT
GTGAAGTTCCATTATTGTGTTTTCCAATAATGACTGATCAGCCTAGTAACAGGAAAC
TGGTGGTGGATGACTGGAGGATCGGCGTCAACCTATCTGCGGCGGAGGAGGTCAGT
AGAGAAGAAGTGTC AATGAAGTCAGGAAC TTGATTCTGGAGAATTGGGGAATGA
GTTGAGAGTGCAGATTCAAAAGTACAAAAGTTGATGGAGAATGGTATAATGGAAG

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GTGGATCATCACATTCCAATTGGAACAAGTTCATCCACGACCTACAAATCTTCAAGA

AATGA

RrUGT7 amino acid GenBank accession MF674532 (SEQ ID NO: 10):
MAENTHAHAIVVPPVQGHKPSLNLALKLASQGFTITFVTTHTHQQISQAHKNSTNTN

HDMFFQARNSSLDIRHVTVDTFPLGFDRAGNQDQFWEGLMHVFPAHVDELVDQLMN

SSKPRPTCLILDTFYNWGSKIANKFNLVHISFWTQSALSFTLFYHWELLKKNHFGSPDN

RTDVIDIYIPGVQEIKPADLISYLMQSDTTTVAHRTCFTAFEDVRKADFILANTIQEFETDTI

SSIRFHQPPFFYPPIGVFLTKSEQQASSALWSESDCEQWLSTKPKGSVLYASFSGSYARVTR

HDIAEIAYGLMQSEVNFIIWIRDDIVGAHETDFLPTEFINGIKLKDQGLLVSWCSQTEVLS

NAAIGGFLTHCGWNSILESVCVPLLCFPIMTDQPSNRKLVDDWRIGVNLSAEEVS

REEVSMKVRNLISGELGNELRVQIQYKKLMENGIMEGSSHSNWNKFIHDLQIFKK-

RrUGT13 DNA (SEQ ID NO: 11):

ATGGCAGAAATAAGTCTCATCTTCATCCCTTTTCCCGTAATCAGCCATCTCACTCCA

CAATCGAAATCGCCAAAATCCTCCTCAGCAGAGACCACCGCCTTTCATCACCTTCC

TCGTATCATGACATCCCCCAACGAGACGCCTCACTCGCCTCCCTCACCACCTCCATCA

TCTCCGATCGCCTCCACTTCCTCGATGTCGTAATTCCTCCCAACCACTCCCAATC

ATCCAAGCCATCAGGCATCGCGGCTATCGAGTCCGCCAAACCCGAGTCAAGAAAA

CGATCAGCGATCTTGTGTACGATCTCAGTCCGCCGATCTGGTCCGCGGATAGCTG

GCTTCGTGTGGACATGTTCTGCACGGCCATGATCGACATCGCAACTGAGTTTAAC

TTCCTTCGTATATTTACTACACTTGCAGGCTCTTCGTTTCTTTCAATCGTGCTCCAGTC

CAGAAGCTCTGCGATGACGACGCTCTCGATATCGCCGATTTCAAAAACCTCGAGTGTG

GAGTTTTCGTTACCTGAGTTTTTCAAACTTGATTCCGGCTAGGCTGCTTCCATCCATGG

CGCTCGATAAGGACTTCTCGGCTTCATTCGTCGGCAAAGCTAGAGCGTTCAAGGAAGA

CGAAGGGCATTGTTGTCAACTCGCTGTAGAGTTGGAGCCTCACGCAATCGAGTCGA

TGAAATTAGACCGGTCTGTTCTCCGATTTACTCGGTCGGACAGTGCTCAACATGA

ATAGCAACACTGCATTTATCAGACAGGAGCAGGAGAAGGAGATCATGGAGTGGCTG

GACCAACAGCCTCCAGCATCTGTAGTTTCTTGTGTTTGGCAGCAGGGAGCGTTC

AAGCCGGACAGGTGAAGGAAATCGCACGGGGTTGGAGTTCGAGCGGCTGCCGTT

CCTCTGGGCGCTTCGGCAGCCTTCATCAAGCAATGTGAGGTTTTACCTCCTACAGA

TTATGAAGATTTCTCTGAGGTTCTGCCTGAAGGTTTTTGACGCGGACATATGGTGT

GGGAAAGTGATTGGTTGGGCACCCAGACAGCTGTTTAGACCACCTTCGGTGGGT

GGATTCGTATCGCATTGCGGTTGGAACCTGATACTGGAATCTCTTTGGTTTGGTGTGC

CGATTGCGACTTGGCCTCTGTATGCTGAGCAGCAGATGAATGCGTTTGAGGTTGTGA

AGGAGATGAAGATTGGAGTGGAGATAAGTTTGGATTATCGGCTTGAAATGGGCGGT

AAACAAGCAGAAGGTTCTGGGATTATAAGTGGTGAACAGATTGAGAGAGGGATTAG

AGATGTGATGCAGGAGGATAGTGAAGTGAAGAAGGTGAAGCTGATGATGGAA

AAGAGTAGAGAGGCAGTTGTGGAGGGAGGCTCCTCTATAATTATATCCAAAATTC

ATCAGTGATCTCAGGACCAACATTGGCTTGTA

RrUGT13 amino acid GenBank accession MF674538 (SEQ ID NO: 12):

MAEISLIFIPFPVISHLTPTIEIAKILLSRDHRLSITFLVIDIPQRDASLASLTTSIISDRLHFLD

VVLPPNQHSQSSKPSGIAIESAKPAVKKTIISDLVVRSSQSAAGPRIAGFVLDMFCTAMID

IATEFNLPYSYIYTCGSSFLSIVLHVQKLCDDALDIADFNSSVEFSLPEFSNLIPARLLPS

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MALDKDFSASVFGKARAFRKTGILVNSLVELEPHAIESMKLDRSVPPIYSVGPVLNMN
 SNTAFIRQEQEKEIMEWLDQQPPASVVFLCFGSRGAFKPDQVKEIARGLESSGCRFLWAL
 RQPSSSNVRFSPPTDYEDFSEVLPEGFLQRTYGVGKVI GWAPQTAVLDHPSVGGFVSHC
 GWNSILES LWFVP IATWPLYAEQQMNAFEVVKEMKIGVEISLDYRLEMGGKQAE GSGI
 ISGEQIERGIRDVMQEDSEVRKKVKLMMEKSREAVVEGGSSYNYIQNFISDLRTNIGL-

RrUGT17 DNA (SEQ ID NO: 13):

ATGGGCTCACTTCCTCCACAAAATCCCATGCAGTCCTCGTCCCATACCTGCCCAA
 GGCCACATCAACCTTTCATGCAACTGCCAAGCTCCTACACTCAAAGGTTTCCAC
 ATAACCTTCGTCAACAATGACCACAACCATCGCCGTTTGCTCAGAACAAAAGGGCA
 TGATTTTGTTCAAGGGTTGGAAGGTTTAAGGTTTGAAGCTGTGCCGATGGCTTACC
 TCCATCTGACCGTGATGCCACTCAGGATGTCCCTAAGCTGACTGAATCTATTACAA
 TAAGAGCATGAACCAACCGTTCAGTGATCTGCTTCAGAGGCTAAACTCAACGCCCG
 GTTCCCCCTCCGGTCACTTGTGTCATATCCGATGTTGCCATGTTTTTGTCTGGGACGT
 GGCGGATGAGCTTGGCATCCCTAATGTTTCAGTTTTGGACAGCTTCAGCTTGTGGCT
 TTTGGGATACTTACAGTATGATGAGCTCCTAAGAAGAGCCATAGTCCCATTCAAAGA
 TGAAAATTTTCATGACGGATGTTTCGTTGGAGGCTTGATTGACTGGATTCTGGCAT
 GCCTAACATGAGGCTGAAGGACTTGCCAAGCTTCATGCGGACCACAAGCCCTGACG
 ACGTGTGTTC AATTACTTGC GTACAATAACCACGAAAGCTCTAAATCCTCGGCCT
 TGTTGCTGAACACATTTGATGATTTGAACATGAAGTAGTTGAAGAGATGAAGAAA
 ATGCAACCAACATATTCTTAGGAGGTCCACTCAACATGCTTCTCAGGCACACATCA
 AAAACTGAAATCACATCCTTAACAACAAGTTTATGGAAAGAGGACACTCATTGTTTA
 GAATGGCTGGACAAGCAAGAACCGGAGTCAGTGGTATACATCAATTACGGATCGGT
 GACGATAATGCTGATCACCATTAAATGAGTTTGCTTGGGGTTTGGCTAACAGCAA
 GCACCCTTTTTTGTGGATCGTGAGGCCGGATGTTGTGAGGGGCGAGTCGGGGACTTT
 GCCAAGGAGTTTTATGATGAGATCAAGGACAGGGATTGATAACGAGCTGGTGTC
 CGCAACCAGAGGTGCTTAACATCCATCCGTAGGTGTATACTTGACGCATTGTGGTT
 GGAACTCTATCACGGAGAGTGTGGCCGGAGGAGTGCCATTGATGTGCTGGCCGTTTT
 TCGCTGAGCAACAGACGAATAGCCGATTCGCGTGTACGGTGTGGGGCACTGGAGTG
 GAGGTGAATGCGGATGTGAAGAGGGAGGAGCTAGCGGAACAAGTGATGGAGATGT
 TGAAGGAAAGAGGGGGCAAGAGTTGAGGAAAAATGCTAAGGAGTGGAGGAGGAA
 GGCGGAGGAGGCGACGACATTGGCGGTTCTGCCTATGCTGATTTGATAGGTTTAT
 GGAAAAAGTGGTCCAGTTTAGCGTCTGA

RrUGT17 amino acid GenBank accession MF674542 (SEQ ID NO: 14):

MGSLPSTKSHAVLVYPYPAQGHINPFMQLAKLLHSKGFHITFVNNDHNHRRLLRTKGHDF
 VQGLEGLRFEAVPDGLPPSDRDATQDVPKLTESIYNKSMNQPFSDLLQRLNSTPGSPPVVT
 CVISDVAMFFAWDVADELGIPNVQFWTASACGLLGYLQYDELLRRAIVPFKDENFMTD
 GSLEALIDWIPGPMNMRKLDLPSFMRTTSPDDVLFNYLRTITTKALKSSALLLNTFDDFE
 HEVVEEMKKMQPNIFLGGPLNMLLRHTSKTEITSLTTS LWKEDTHCLEWLDKQEPESVV
 YINYSVTIMSDHHLNEFAWGLANSKHPFLWIVRPDVVRGESGTLPKIFYDEIKDRGLIT
 SWCPQPEVLKHP SVGVYLTHCGWNSITESVAGGVPLMCWPPFAEQQTNSRFACTVWGT
 GVEVNADV KREELAEQVMEMLEGKRGQELRKNAKEWRRKAE EATDIGGSAYADFDRF
 MEKVVQFSV-

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RrUGT29 DNA (SEQ ID NO: 15):
 ATGGGATCTCTAGGAAAGAAGATTCAACAAAAGCCACATGCAATATGCACCCCATATA
 CCCAGCACAAAGCCATATTAAATCCCATGCTTAACTAGCCAAGCTCCTACACCACTC
 AGGCTTCTACATAACCTTTGTTACACAACTACAATCGCCTTCTCAAGAC
 CCACGGGTCTGATTCCCTAAGTGGTCTACCAGATTCCAATTTGAGACCATCCCTGAT
 GGACTACCACCATCAGATGCAGCTGATGTCACACAAGACATCCCTGCCTTGTGTAA
 TCAACCACCGAAACCTGCTTAGTCCCATTCAAAGAGCTCCTGGCTAAGCTGCATAAC
 AAGTCAATGGCGTCACCGGAGGAAGTTCCCTCAGTGACATGATAGTTTCTGATGGT
 TGCATGTCATTTACTGTGGATGCTGCAGAAGAGGCAGGGGTTCTAATGTGCTTCTT
 TGGACTACCAGTGCATGCGGATTTTGTAGGATATGCTAATTACCCGAAACTTATTGAC
 AGAGGCATAATCCACTCAAAGATGAGAGCTACTTTACGAATGGGTACCTAGACAA
 GACAGTAGATGGAATACCTGGAATGAAAGGCATACGGCTACGAGACTTCCCAAAT
 TTGTATGCACCACAAACCCAGATGAGTTTATGGTGAAATATGCAATTCAAGAGATCA
 CTAGAGCTGCCAGAGCAGATGCTGTTATTTGAACACCTTTGACGCTTTGGAACATG
 ATTTCTTAGATGGCTATCAAACATATACCCAAAGGTCCTCCCTATTGGCCCGCTCC
 AGCTTCGCTCAACCAAATCCAGAGAGCTCACCTCTACATTCAATCTGTTCTAGTC
 TCTGGAAGATGAACCACAGTGCATTACCTGGTTAACTCCCAAAACCAAATCA
 GTCGTTTATGTTAACTACGGAAGTATCACAGTTATGACTCCGCAACAAATGGTGGAG
 TTCGCATGGGGACTGGCTAATACAAAATACCCCTTTTCTGTGGATTATTAGACCTGAT
 TTGGTTGCTGGTGAGACAGCTGTCTACCTCCAGATTTTGTGAAGTGACAAAAGGA
 AGGAGCTGCTTGGCTAGTTGGTGCCACAGGAACAAGTTCTTAGTCACACATCCATA
 GGAGGGTTCTTAACCCATTGTGGGTGGAACCTAATGCTAGAAAGCGTGGTCGAAGG
 AGTTCCAATGGTATGCTGGCCGTTTTTGTGAGCAACAGACTAATTGCTGGGCTGC
 TCGGACAAAATGGGTATAGGTATGGAATGACAAATGATGTTAAGAGGGATAAGG
 TTCAGAAAATGGTGACAGAGCTTATGGAGGGCGAAAAGGGAAGGAGATGAAGAG
 GAAGGGCGGAGAATGGAAGAAGCTTGGGGCAGAAGCTGCCGCTCCTAATGGCTCAG
 CTACCTTAAACTTCAGCAGACTTATAAATGACGTACTTCTGTCCAAAAAAAATTG
 TAGTTACAACCTAA

RrUGT29 amino acid GenBank accession MF674554 (SEQ ID NO: 16):
 MGS LGK K I Q Q K P H A I C T P Y P A Q G H I N P M L K L A K L L H S G F Y I T F V H T T Y N Y N R L L K T H G
 S D S L S G L P D F Q F E T I P D G L P P S D A A D V T Q D I P A L C K S T T E T C L V P F K E L L A K L H N K S M A S P
 E E V P P V T C I V S D G C M S F T V D A A E E A G V P N V L L W T T S A C G F L G Y A N Y P K L I D R G I I P L K D E
 S Y F T N G Y L D K T V D G I P G M K G I R L R D F P N F V C T T N P D E F M V K Y A I Q E I T R A A R A D A V I L N T
 F D A L E H D F L D G L S N I Y P K V L P I G P L Q L P L N Q I P E S S P L H S I C S S L W K D E P Q C I T W L N S Q K P K
 S V V Y V N Y G S I T V M T P Q Q M V E F A W G L A N T K Y P F L W I I R P D L V A G E T A V L P P D F L E V T K G R
 S C L A S W C P Q E Q V L S H T S I G G F L T H C G W N S M L E S V V E G V P M V C W P F F A E Q Q T N C W A A R
 T K W G I G M E I D N D V K R D K V Q K M V T E L M E G E K G K E M K R K G G E W K K L G A E A A G P N G S A T
 L N F S R L I N D V L L S K K I V V T T

RrUGT32 DNA (SEQ ID NO: 17):
 ATGGGATCTCTAGGAAAGAAGATTCAACAAAAGCCACATGCAATATGCACCCCATATA
 CCCAGCACAAAGCCATATTAAATCCCATGCTTAACTAGCCAAGCTCCTACACCACTC
 AGGCTTCTACATAACCTTTGTTACACAACTACAATCGCCTTCTCAAGAC

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CCACGGGTCTGATTCTTAAGTGGTCTACCAGATTCCAATTGAGACCATCCCTGAT
 GGACTACCACCATCAGATGCAGCTGATGTCACACAAGACATCCCTGCCTTGTGTAAA
 TCAACCACCGAAACCTGCTTAGTCCCATTCAAAGAGCTCCTGGCTAAGCTGCATAAC
 AAGTCAATGGCGTCACCGGAGGAAGTTCCTCCAGTGACATGCATAGTTTCTGATGGT
 TGCATGTCATTTACTGTGGATGCTGCAGAAGAGGAGGGGTTCTAATGTGCTTCTT
 TGGACTACCAGTGCATGCGGATTTTAGGATATGCTAATTACCCGAAACTTATTGAC
 AGAGGCATAATTCCACTCAAAGATGAGAGCTACTTACGAATGGGTACCTAGACAA
 GACAGTAGATGGAATACCTGGAATGAAAGGCATACGGCTACGAGACTTCCCAA
 TTGTATGCACCACAAACCCAGATGAGTTTATGGTGAAATATGCAATCAAGAGATCA
 CTAGAGCTGCCAGAGCAGATGCTGTTATTTGAACACCTTTGACGCTTTGGAACATG
 ATTTCTTAGATGGCCTATCAAACATATACCCAAAGGTCCTCCCTATTGGCCCGCTCC
 AGCTTCCGCTCAACCAAATCCAGAGAGCTCACCTCTACATTCAATCTGTTCTAGTC
 TCTGGAAGATGAACCACAGTGCATTACCTGGTTAACTCCCAAAACCAAATCA
 GTCGTTTATGTTAACTACGGAAGTATCACAGTTATGACTCCGCAACAAATGGTGGAG
 TTCGATGGGACTGGCTAATACAAATACCCCTTTCTGTGGATTATTAGACCTGAT
 TTGGTTGCTGGTGAGACAGCTGTCTACCTCCAGATTTTGGAAAGTGACAAAAGGA
 AGGAGCTGCTTGGCTAGTTGGTGCCACAGGAACAAGTCTTAGTCACACATCCATA
 GGAGGGTTCTTAACCCATTGTGGGTGGAACCAATGCTAGAAAGCGTGGTCGAAGG
 AGTTCCAATGGTATGCTGGCCGTTTTTGTGAGCAACAGACTAATTGCTGGGCTGC
 TCGGACAAAATGGGTATAGGTATGGAATGACAAATGATGTTAAGAGGGATAAGG
 TTCAGAAAATGGTGACAGAGCTTATGGAGGGCGAAAGGGAAGGAGATGAAGAG
 GAAGGGCGGAGAATGGAAGAAGCTTGGGGCAGAAGCTGCCGGTCTAATGGCTCAG
 CTACCTTAAACTTCAGCAGACTTATAAATGACGTACTTCTGTCCAAAAAAAATTGT
 AG

RrUGT32 amino acid GenBank accession MF674557 (SEQ ID NO: 18):
 MGSLGKKIQQKPHAICTPYPAQGHINPMLKLAKLLHHSGFYITFVHTTNYNRLKTHG

SDSLGSLPDFQFETIPDLPPSDAADVTQDIPALCKSTTETCLVPFKELLAKLHNKSMASP
 EEVPPVTCIVSDGCMSTVDAAEEAGVPMVLLWTTTACGFLGYANYPKLIDRGIIPLKDE
 SYFTNGYLDKTDVGIPGMKGI RLDFPNFVCTTNPDEFMVKYAIQEI TRARADAVILNT
 FDALEHDFLDGLSNIYPKVLPIGPLQLPLNQIPSSPLHSICSSLWKDEPQCITWLNSQKPK
 SVVYVNYGSITVMT PQQMVFAWGLANTKYPFLWII RPD LVAGETAVLPDFLEVTKGR
 SCLASWCPQEQLSHTSIGGFLTHCGWNSMLESVVEGVP MCV PPFABEQQTNCWAAR
 TKWIGIMEIDNDVKRDKVQKMVTELMERGEKGKEMKRKGGEWKKLGAEAAAGPNSAT
 LNFSRLINDVLLSKKKL-

RrUGT33 DNA (SEQ ID NO: 19):
 ATGAGCTTAATTGAAAAACCACTCACGGCCATAGAGACTCGTGAAAAACACACGC
 TGTGTGCATCCCATACCCAGCTCAAGGCCATATCAATCCCATGATGCAACTTGCAA
 GCTCTCCACCACTCTGGTTTCCACATAACGTTTGTCCACACTGAGTATAATTATGAC
 CGTCTAGTGAAGTCTCAAGGTTTCTAGCTTGTGTGGCTGGTTTACCGATTTCGCTTTG
 AAGCCATCCAGATGGCTTGCCCTCGACGAATGGTGATGTTACTCAAGACATTCCTC
 TGTGAGTAGCTCTACTTCTAAAACCTGCTTGAAGCCGTTTAAAGAGTTATTGAAGA

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GGTTGCAGGACAAATGCAAAGAGTTACCTGATGATGTTCCGCCTCTGTCGTGCATCG
 TGTCTGATGCAGCCATGTCGTTTACGATCGATGCATCTGAGGAGTTTGGAGTGCCCA
 TAGCGCTTCTTTGGACTGCAAGTGCCTGCGGGTTCTTGGGTTACACGCATTACCCAT
 ATCTAATTGACAGAGGTGTCATCCATTGAAAGATGAGAGCCAATTAACAAACGGA
 TACCTAGATATGAGCATAGATGGCATACCTTGTATGGAAGGTATCCGCTTACGAGAC
 CTCCCAAGCTTTCTACGCACAACCTGATTTAGATGATATGATGTTTAGTTATATACTGC
 ACGAAATAAAACAAGTTTCAAGAGGCAGTGCTATCATTCTGAACACCTTTGAAGCTT
 TGGACCATGATGTCCTTGGATAGTCTCTCCAAAATTACCAAATGTCATCCTGCCAG
 TTGGCCCTCTACATGTCTCGCTCAACAAGATCCCAAAACACTACCCACTTCAATCTTT
 AAGCTCGAATTTATGGAAGATGACACAGACTGCATTCCCTGGCTGAGCTCTAAGGC
 TTCAAAATCAGTTATATACGTTAACTTTGGGAGCATCACGACGGTATACCAAAACA
 AATTGTGGAGTTTGCCTGGGGATTGGCTAACAGCAAACACCTTTCTCTTGGATAAT
 CAGACCGGACTTGGTGGCAGGTGAGGCATCCATCATTCCGCAGGACTTCATGGATG
 AAACAAAAGGAAGAGGTTTGTGGCTGGTTGGTGTGACCAAGAGCTTGTCTCAACC
 ATCCATCCATTGGAGGGTTTCTTACGCACTGTGGCTGGAACCAATTATGAAAGCA
 TTAGCGCAGGAGTCCCTACGGTCTGCTGGCCATTTTTTGTCTGAGCAGCAACAAATT
 GTTGGTTTGTCTGCAAAAATGGTGCATTGGGATGGAGATGCATACTGATGTAAAGA
 GGGATGAGGTTGACAAGCTGTTGAGAGAGCTAATGGAAGGTGACAAAGGGGAGGA
 GTTGAAGAGGAAGGCAACCACTGGAAGAGGCTGGCAGAAGAAGCTGTTTCCTCCA
 CTGGCTTATCAACCTTAAACTTCAGGACGTAGTGAATCAAGTCTGCTCTCAAAA
 CAAAACATATCCGTTAG

RrUGT33 amino acid GenBank accession MF674558 (SEQ ID NO: 20):
 MSLIEKPLTAIETREKPHAVCIPYPAQGHINPMMQLAKLLHHSGFHITFVHTEYNYDRLV
 KSQGSACVAGLPDRFEAIPDGLPSTNGDVTQDIPLSSSTSKTCLKPFKELLKRLQDKCK
 ELPDDVPPLSCIVSDAAMSFTIDASEEFGVPIALLWTASACGFLGYTHYPYLIDRGVPLK
 DESQLTNGYLDMSIDGIPCMEGIRLRDLPSFLRTTDLDDMMFSYILHEIKQVSRGSAIILN
 TFEALDHDVLDLSKIYQNVILPVGPLHVSLNKIPKHYPLQSLSSNLWKDDTDCIPWLSS
 KASKSVIYVNFSGITTVSPKQIVEFAWGLANSKHPFLWIIIRPDLVAGEASII PQDFMDETK
 GRGLLAGWCDQELVLNHPISIGFLTHCGWNSIIESISAGVPTVCWPPFAEQQTNCWFAC
 KKWCIGMEMHTDVKRDEVDKLLRELMEGDKGEELKRKATNWKRLAEEAVSSTGLSTL
 NFRTLNVQVLLSKTKHIR

Arabidopsis thaliana AT2G20340.1 (SEQ ID NO: 21):
 MENGSGKVLKPMDSQLREYGHLMVDFIADYYKTIEDFPVLSQVQPGYLHKLLPDSAP
 DHPETLDQVLDDVRAKILPGVTHWQSPSFFAYYPSNSSVAGFLGEMLSAGLGIVGFSWV
 TSPAATELEMIVLDWVAKLLNLPEQFMSKNGGGGVIQGSASEAVLVVLI AARDKVLRSV
 GKNAL EKL VVYS DQTHSALQKACQIAGIHPENCRVLTDSSTNYALRPESLQEA VSRD
 LEAGLIPFLCANVGTSTSTAVDPLAALGKIANSNGIWFHVDAAYAGSACI CPEYRQYID
 GVETADSFNMNAHKWFLTNFDCSLLWVKDQDSLTLALSTNPEFLKNKASQANLVVDY
 KDWQIPLGRRFRSLKLMVLRLYGSETLKS YIRNHIKLAKEFEQLVSQDPNFEIVTPRIF
 ALVCFRLVPVKDEEKKCNRRNRELLDAVNSSGKLFMSHTALSGKIVLRCAIGAPLTEEK
 HVKEAWKIIQEEASYLLHK

Brachypodium distachyon 1g28960.3 (SEQ ID NO: 22):

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MDGSTTSNGGGWMRPMDEEQLECGHRMVDFIADYYKSIETYPVLSQVQPGYLKEL
 LPDSAPNQPDTLDALFDDIREKIVPGVTHWQSPNYFAYYPSNSSTAGFLGEMLSAAFNIV
 GFSWITSPAATELEVIVLDWVAKMLKLPSEFLSAALGGGVIQGTASEAILVVLLSARDRT
 LRKHGKKSLEKIVVYASDQTHSALKKACQIAGIFPENIRIVKADCSMNYAVTPGAVSEAI
 SIDLSAGLIPFFICATVGTSTSSAVDPLHELQIAQAHDMWFHIDAAYAGSACICEYRKY
 LNGVEEADSFNMNAHKWFLTNFDCSLLWVKDRNYLIQALSTNPEFLKNKASQENSVID
 FKDWQIPLGRRFRSLKLMVLRLYGVENLQSYIRKHIQLAQHFEQLVISDPRFEVVTPR
 NFSLVCFCLVPPTCEVDNGHKLNYDLMSANS SGKIFISHTVLSGKFVLR FVVGAPLTEE
 QHVDAAWKLLQDEATKLLGNVVQ

Carica papaya 16427710 (SEQ ID NO: 23):
 MDAEQLENGHKMVDFIADYYKTIENFPVLSQVEPGYLRLIPDSAPNSPESFQQLLDD
 VRTKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATELEMIVL
 DWLAKLLKLPEDFHSTGNGGGVIQGTASEAILVVLLAARDKVLKRVGKNALEKLVVYT
 SDQTHSAFQKACQIGGIHPENCRVLKTDSSSTNYALSPDLLKEAISCDVAAGLIPFFFCATV
 GTTSSTAVDPLMALGKIATSNEIWFHVDAAYAGSACICEYRPHYIDGVEEADSFNMNAH
 KWFLTNFDCSVLWVKDYSLIQSLSTNPEFLKNKASQADMVVDYKDWQIPLGRRFRSL
 KLMVLRLYGVENLKS YIRNHILAKHFEELVTQDPRFEVVTPRIFSLVCFRLLPPGND E
 NHGKNLQDLLETVNSTGKLFISHTVLSGKYILRFAVGAPLTEERHVNEAWKILQDEAS
 TLENP

Ricinus communis 16804377 (SEQ ID NO: 24):
 MPREGELRPMDAEQLEHGHKMVDFIADYYKTIENFPVLSQVEPGYLRLKLLPDSAPNQ
 ESLQNVLDVQAKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSP
 AATELEMIVLDWLKMLKLPEEFLSTGQGGGVIQGTASEAVLVALVAARDKVLRRVGK
 DALRKLVVYGSQTHSALQKACQIGGIHPVNCRLLETDSSTNYALAPDLLSRAISEDISL
 GLIPFFLCATVGTSTSTAVDPLALGKIAKSNMWFHVDAAYAGSACVCPEYRCYMDG
 VEEADSFNMNAHKWFLTNFDCSALWVKDRNALIQSLSTSPEFLQNKPSQTNVTVDYKD
 WQIPLGRRFRSLKLMVLRLYGVEKLQCYIRNHINLAKYFEGLIAEDTRFEVVSPPIFAL
 VCFRLLPPDNNVDHGNKLSHDLDAVNSTGKIFISHTVLSGKYILRFAVGAPLTEERHVT
 AAWKVLQDEACALLETSRIS

Cucumis sativus 16963476 (SEQ ID NO: 25):
 MDNELKPMDAEQLEHGHKMVDFIADYYKNIEDFPVLSQVEPGYLQNLPPESAPLN PES
 LQSVLDDVQKKIPPGVTHWQSPNYFAYYPSNSSIAGFLGEMLSAAFNVIGFSWITSPAA
 TELEMIVLDWLAKLLKLPDDFLSSGNGGGVIQGTASEAVLVLLAARDRALRRFGKDY
 LKKLVVYASDQTHSALQKACQIGGIHPENCRWLKADISTNYALSPDVLSEELSRDTARG
 LIPFFLCATVGTSTSTAVDPLPELGTIAKRHEMWFHVDAAYAGSACVCPEYRQYIDGVE
 EADSFNMNLHKWFLTNFDCSALWIKDRHALIRSLSTNPEFLKNKASEAELVVDYKDWQI
 PLGRRFRSLKVMVLRLYGTENLQKYIRNHISLAERFEALVREDPRFEIVTPRIFSLVCFR
 LLPSRKNEGGNRLNQSLDAVNASGNIFISHTVLSGKYILRFAVGAPLTEEKHINS AWK
 LLQDVASTLLAI

Vitis vinifera 17835588 (SEQ ID NO: 26):
 MDAEQLENGHKMVDFIADYYKSIENFPVLSQVEPGYLRELLPDSAPNQPSLQQVFFD
 LQAKILPGVTHWQSPNFFAYYPSNSSTAGFLGEMLSAGLNIVGFSWITSPAATELEMIVL
 DWLAKLLNLPDDFLSAGNGGGVIQGTASEAVLVLLAARDVLRVTGKTALEKLVVY

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GSDQTHSALQKACQIGGIHPENCCLKKADSSSTGYALSPDLLSEAVSHDITNGLIPFFLCAN
 VGTTSSTAVDPLLELGKVKTSNGIWFHVDAAYAGSACVCPEYRHYIDGVEEADSFNMN
 AHKWFLTNFDCSVLWVKDRNALVQALSTNPVFLKNKASDANMVVDYKDWQVPLGRR
 FRSLKLWMLRLYGVENLQRYIRNHIKLAKQFEELVAQDPRFEIVAPRKFAVLCFRLLPP
 HRNEDFSNKLHNHLLDTVNSTGKVYISHTALSGKYTLRLAVGAPLTEERHVNAAWKVI
 QEKASVLLSEFGMNGLFDNINLKFILNHQIDISILLNYN

Citrus sinensis 18113817 (SEQ ID NO: 27):
 MDAEQLRENAHKMVDFIADYYKSIENFPVLSQVQPGYLHNLIIPDSAPHHPELQNVLDG
 YIDIQEKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATELEMI
 VLDWLAKLLKLPEDFLSSGGGGVIQGTASEAVLVVLLAARDKALKRVGKNSLEKLVV
 YASDQTHSALQKACQIGGIHPQNFRVLKTDSSSTNYSLSPDSLAEAISRDLTIGLIPFFLCAT
 VGTTSSTAVDPLLALGNIAGSNGMWFHVDAAYAGSACI CPEYRQYIDGVEEADSFNMN
 AHKWFLTNFDCSALWVKDRNTLIQSLSTNPEFLKNKASQANMVVDYKDWQIPLGRRFR
 SLKLWMLRLYGLLENLQGYIRNHIQLAKHFEGLVAQDLRFVVTTPRIFSLVCFRLLPPHN
 DEDHGNKLNHKLDDINSTGKIFISHTVLSGKYILRFVAVGAPLTEWRHVNAAWVEMQD
 KASALLARLSIE

Capsella rubella 20900667 (SEQ ID NO: 28):
 MGFCQIELLRHINKHNMQNGSGKNVLKPMDSQLREYGHMVDFIADYYKTIEDFPVL
 SQVQPGYLHQLLPDSAPDHPETLDQVLDDVRAKILPGVTHWQSPGFFAYYPSNSSVAGF
 LGEMLSAGLGIVGFSWVTSAPAATELEMIIVLDWLAKLLNLPKEFLSKNGGGGVIQGSASE
 AVLVLIAARDKVLRSAGKNALGKLVVYSSDQTHSALQKACQIAGIHPENCRVLETDAS
 TNYALRPELLQEAVSKDLKAGLIPFFLCANVGTTSSTAVDPLAALGKIANSNEIWFHVDA
 AYAGSACI CPEYRKYIDGVETADSFNMNAHKWFLTNFDCSLWVKEQDSLTEALSTNP
 EFLKNKASQANLVVDYKDWQIPLGRRFRSLKLWMLRLYGAETLKSYYIRNHIKLAKYF
 EKLVSQDPNFEIVTPRIFSLVCFRLVPKNDDEKKNQNRKLEAANS SGKLFMSHTALS
 GKIVLRCAIGAPLTEEHMKEAWKVIQDEASFLHK

Malus domestica 22636618 (SEQ ID NO: 29):
 MSGLKPMDAEQLENAHKMVDFIADYYKTIEDFPVLSQVQPGYLRDLLPDSAPTHPEL
 QQVFDDIQAKILPGVTHWQSPNFFGYPSNSSVAGFLGEMLSAGLNIVGFSWITSPAATE
 LEMIVLDWFAKMLKLPPEFLSAGQGGGGVIQGTASEAVLVVLLAARDRI LRAEGKKALE
 KLVVYASDQTHSALQKACQIGGIHPENCRVLSSTDSTNYALSPNVLEAISNDIASGLVP
 FFLCATVGTTSSTAVDPLLELGKITKSNMWFHVDAAYAGSACI CPEYRHHIDGVEEAD
 SFNMNAHKWFLTNFDCSLWIKDRNALVQALSTNPEFLKNKASQANLVVDYKDWQIPL
 GRRFRSLKLWMLRLYGLLENLQSYIRNHIDLAKCFEDLVAQDSRFEIVTPRIFSLVCFRL
 LPPHNDETYATKLNHDLDDTVNSTGKIFVSHTVLSGKYVLRFAVGAPLTEERHVLA
 KLLQEEASALLAPL

Linum usitatissimum 23178995 (SEQ ID NO: 30):
 MGGYRSLNLI FIFISFVADIRDLGYNTKEGDDGGGALKPMDAEQLRQNAHQMVDFIADY
 YKNIETYPVLSQVEPGYLRELLPDSAPNRPESLQSVLDDVQSKIMPGVTHWQSPNYFAY
 YPSNSSVAGFLGEMLSAGINMVGFSWITSPAATELEMIIVLDWLKLLKLPPEFLSSGHG
 GGVIIQGTASEAILVVLLAARDKMLRKFGKSALEKLVVYASDQTHSALQKACQIGGIYPE
 NCRLLKTDSSVNYSLTPELVSEAVSQDISAGLIPFFLCGTGTSSATVDPLGLGKIKN

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NDMWFHVDAAAYAGSACICPEYRQYLDGVVEADSFNMNAHKWFLTNFDCSTLWVKDK
 SALIQALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRSLKLMVLRLYGVENLQQY
 LRNHIELARHFEECVNHDPRFEALSGKYTLRVAIGAPLTEKRHVAAALKVLQDEATSL
 VATSPLENGNSS
Eutrema salsugineum 20200788 (SEQ ID NO: 31):
 MENGKNKVLKPMDSQLREYGHMVDFIADYYKTIEDFPVLSQVQPGYLHNLLPDSAP
 DQPETLEEVLDVVKGKILPGVTHWQSPSPFAYYPSNSSVAGFLGEMLSAGLGIVGFSWIT
 SPAATELEMIVLDWLAKLLNLPQFLSRNGGGVIQGSASEAELVVLLAARDKVLRSVG
 KKALEKLVVYSSDQTHSALQKACQIAGIHPENCRVLKADYSTNYALRPETLQEA VSKDL
 EAGLIPFFLCANVGTTSSSTAVDPLAALGEIAKSNEMWFHVDAAAYAGSACICPEYRQYID
 GVTADSFNMNAHKWFLTNFDCSLLWVKDQYALTEARSTNPEFLKNKASQANLVVDY
 KDWQIPLGRRFRSLKLMVLRLYGSENLSYIRNHIKAKDFEQLVSEDPNFEIVTPRIFS
 LVCFRIVPAENDEKKCNNQNRNLLDAVNSSGKFLSHTALSGKIVLRCAIGAPLTEEKHV
 KEAWKVIQEEASYLLRK

Populus trichocarpa 27022899 (SEQ ID NO: 32):
 MESKGLQPMDSQLRENHAKMVDFIADYYKSIENFPVLSQVEPGYLRELLPDSAPNQPE
 TLQNVLDVQAKILPGVTHWQSPSYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSPA
 ATELEMIVLDWLGKLLKLPEDFLSTGQGGGVIQGTASEAVLVLLAARDVLRKLGKN
 ALEKLVVYASDQTHSALQKACQIGGIHPENCKLLKTGSSTNYALSPDLLGKAISDDISTG
 LVPFFLCATVGTSSSTAVDPLLSLGKIAKNNGIWFHVDAAAYAGSACICPEYRCYIDGVVE
 ADSFNMNAHKWFLTNFDCSALWVKDRNALIQSLSTNPEFLKNKASQANMVVDYKDW
 QIPLGRRFRSLKLMVLRLYGLENLQCYIRNHINLAKYFEGLVAAADSRFEVVTPRIFSLV
 CFRLLPPNNNEDHGNNLNHDLDAVNSTGKIFISHTVLSGKYILRFVAGAPLTEERHVTA
 AWKVLQDEASALLGSL

Brachypodium stacei 06G160800.1 (SEQ ID NO: 33):
 MDGSTTSNGDGGGWMRPMDEEQLRECGHRMVDFIADYYKSIETYPVLSQVQPGYLK
 ELLPDSAPNQPTLDALFDDIQEKIVPGVTHWQSPNYFAYYPSNSSTAGFLGEMLSAAFN
 IVGFSWITSPAATELEVIVLDWVAKMLKLPQSQFLSAGLGGGVIQGTASEAILVVLLSARD
 RTLRKHGKKSLEKLVVYASDQTHSALQKACQIAGIFSDNIRIVKADCSMNAYVTPGSVS
 EAISIDLSSGLIPFFICATLGTSSSAVDPLHELQIAQAHDMWFHIDAAYAGSACICPEYQ
 QYLVNGVEADSFNMNAHKWFLTNFDCSLLWVKDRNYLIQALSTNPEFLKNKASQENSV
 IDFKDWQIPLGRRFRSLKLMVLRLYGVENLQSYIRKHIQLAQRFELVISDSRFEVVT
 RNFSLVCFCLVPPTSEVDNGHKLNYDLMDSVNSSGKIFISHTVLSGKFVLRFAVGAPLTE
 EQHVNAAWKLLQDEATKLLGSVVV

Physcomitrella patens Pp3c4_30790V3.1 (SEQ ID NO: 34):
 MGSEAGSRSSLTKPFDPPEFRKHAHRMVDFIADYHRDIENFPVQSQVEPGYLQKLLPEN
 APDEPESLDDILADVQSKIVPGVTHWQSPNFYGYPSNGSTAGFLGEMLSGGFNIIIGFSW
 ITSPAATELEIIVMDWLKLLKLPNEFLSSGKGGVIQGTASEAVLVMLAARKRAVEK
 LTKEQGISEFEALAKLVAYTSDQAHSCVNKASQIAGISIENLRLLIPTDVSTNYAMSSKVL
 A
 NTLANDVKAGLVPPFLCGVIGSTSSAAVDPLSELGDLAQEYGMWFHVDGAYAGNACIC
 PEFRPYLVNGVEKADSFDMNPKWLLTNFDCSTLWVKNPSSLVDALSTNPVFLRNKQSD
 NNLVVDYKDWQIPLGRRFRSLKLMVLRMYGSNGLRSYITNHCNLAHFEE LLRTDSR
 FEVVAPRVFSLVCFRLKSPANDADNSCSLSAKLVDALNSDGNILITNTVLGGRYTIRFTV

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GASRTEL RHVDAAWKVIQQ LASKLLKECSS

Ananas comosus 33033299 (SEQ ID NO: 35):
MESELKPM DSEQ LREYAHKMVDFIADYYKMIESFPVLSQVKPGYLKELLPSAPCKPEN

LEDVFDDIRQKIIPGITHWQSPDYFAYYPSNSSTAGFLGEMLSAGFNII GFSWIASPAATEL
EMIVLDWFAKMLKLP EQFLSTGQGGGVIQGTASEAVLVVLLAARDKILKAGRKSLEKL
VVYCS DQTHSAMQKACQIAGIFPENFRVLKTDSSSNYALLPEVLS EAI SKDLSFGLIPFFL
CATVGTTSAAVDPLLLKLGNI SKVHDMWFHVDAAHAGSACICPEYRHHIDGVEEADSF
CMNAHKWFLTNFDCSLLWVKDRSALIQLSTNPEFLKNKASQENS VVDFKDWQIPLGR
RFRSLKLMVLRLYGLENLQSYIREHIKLA EQFEQLISSDSRFEIVAPRTFSLVCFRLLPPL
YDQDDGYKLNYNLLDAVNRSKGFMSHTVLSGKFVLRFAIGAPLTEERHVVA AWKVLQ
DEATILLRGS

Zostera marina 33182387 (SEQ ID NO: 36):
MLNGNMGENEPFKPM DSEQ LREYGHKMVDFIADYKSI EKFPVLSQVQPYLKDLLPD
AAPDQPEKFDVLD DITKKIIPGVTHWQSPNFFGYYPGNSSIAGFLGEMICSGLNVIGFS
WITSPASTELEVIVLDWLAKLLNLPDQFLSSGHGGGVIQGTASEAILVVLLAARDKILGRI
GRNSLDKLVVYSSDQVHAAPKKACQIAGIYTENFRVLKTDASSGYGIDPKKFDQAIHDD
MEAGLIPFFLCSTVGTTSASVDPLVEIGQITEENDMWFHVDAAYAGSACICPEYRHYLD
GVEYADSF CMNAHKWLLTNFDCSALWVKDSSALVNSLSTNPEFLKNKMSEQKVVDF
KDWQIPLGRRFRSLKLMVLRLYGAENLREYIRNHIKLANLFEQLVRSDSRFEIVCPTLF
SLVCFRFLPSNDNDNGYELNSMLLD AVNSTGQLFFHTHTIISDKYILRFAVGAALTEERHV
RESWKVIQNQATII SRQHILSKTNMKS KCEGMIANE

Daucus carota subsp. *sativus* 36055203 (SEQ ID NO: 37):
MDGVLKPM DAEQLRENAHKMVDFIADYYKNIETFPVLSQVEPGYLRDLLPHSAPDQPE
SLQNILDDIQAKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSPAA
TELEMIVLDWLAKLLKLPDHFLSTGQGGGVIQGTASEAVLVVLLAARDKVL RITGKDAL
GKLVVYCS DQTHSALQKACQIAGIHPGNCRVLKTESCNDYSLSPETFEQAISTDVASGLI
PLLLCATVGTTSSTAVDPLLELGKITKMKGIWLHVDAAYAGSACVCPEFRHYIDGVEEA
DSFNMNAHKWFLTNFDCSALWVKDRSALIHSLSSTNPEFLKNKASQENLVVDYKDWQIP
LGRRFRSLKLMVLRLYGLENLQSYIRNHIQLAATFESFVTEDPRFEVVAPRK FALVCFR
LLPPSHKDEDCSNQLNRDLLDAVNATGKAFVSHTALSGRYVVRFAIGAPLTEESHI IEAW
KIFQEVATVLLKSLKMNHTRPLN

Trifolium pratense 35974269 (SEQ ID NO: 38):
MVDFIADYYKTIENFPVLSQVEPGYLGKLLPDSAPTYP TLEHVLNDVQHKILPGVTHW
QSPNYFAYYPSNSSIAGFLGEMLSAGINIVGFSWITSPAATELESIVLDWLAKALFLPQDF
LSNGKGGGVIQGTASEAVLVVLLAARDKILRTVGRSALPKLVTYASDHVHSSLLKACQI
GGLDPELCRLLLKTDSSTNFALSPDVLSEAINDIASGLIPFFLCANVGTTSSTAVDPLPALA
KVTKTNNIWLHVDAAYAGSACICPEYRHFIDGVEEADSFNMNAHKWFLTNFDCSLLWV
KDRSALIQLSTNPEFLKNKASEGNMVIDYKDWQIPLGRRFRSLKLMVLRLYGLEGLR
SHIRNHIALAASFEEVLVQDARFKVVTPTFTSLVCFRLLPPPNS EDNGNKLNHDLDDL VN
STGSVFITHTVLSGEYILRLAVGAPLTEVRHVNA AWQILQE KATALLENL

Arabidopsis lyrata 35943929 (SEQ ID NO: 39):
MDSEQ LREYGH RMVDFIADYYKTIEDFPVLSQVQPGYLHKL LPSADHPETLDQVLDD
VRAKILPGVTHWQSPSFAYYPSNSSVAGFLGEMLSAGLGIVGFSWVTS PAATELEMIVL

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DWLAKLLNLPEQFM SKNGGGV IQGSASEAVLVV LIAARDKVLRSV GKNALQKLVVYS
 SDQTHSALQKACQIAGIHPENCRVLKTDSSSTNYALRPELLQEAVSQDL DAGLIPFFLCAN
 VGTTSSTAVDPLAALGKIANRNEMWFHVDAAYAGSACICPEYRQYIDGVETADSFNMN
 AHKWFLTNFDCSLLWVKDQDSLTLALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFR
 SLKLWMVLRLYGSETLKS YIRNHIKLAKEFEQLVSQDPNFEIVTPRIFSLVCFRLVPVKNE
 EKKCNNRRELDAVNSSGKLFISHTVSDFSFFLLFLLDNVNLNRNLCRGKSYCVA
 Q

Sorghum bicolor 002G120700.1 (SEQ ID NO: 40):
 MDGSGSSGGTNGSGGDGAGWLRPMDAEQLRECGHRMVDVFADYYKSIETFPVLSQV
 QPGYLKELLPDTAPNKPDTLEALFDDIREKIVPGVTHWQSPNYFAYYPSNSSTAGFLGE
 MLSAAFNIVGFSWITSPAATELEVIVLDWFAKMLRLPSQFLSTALGGGVIQGTASEAVLV
 VLLAARDRTL RKHGKTSLEKLVVYASDQTHSALQKACQIAGIFPENVR LVKADCNRNY
 AVAPLAISDAIATDLSSGLIPFFICATVGTSSSAVDPLPELGQIAKANDMWLHIDAAYAG
 SACICPEYRHHNLNGVEEADSFNMNAHKWFLTNFDCSLLWVKDRSYLIQSLSTNPEFLKN
 KASEANSVDFPKDWQIPLGRRFRSLKLWMVLRLYGVENLQSYIRKHIELAKEFEQLVIS
 DSRFEVVTPTFSLVCFRLVPLASDQDNGRKLNYDLMDAANSSGKIFISHTVLSGKFVLR
 FAVGAPLTEGQHIFSAWKILQDLATKQLLESS

Sphagnum fallax 0166s0011.1 (SEQ ID NO: 41):
 MGSEAGEGSR LSKPLDVEEFRKHAHQMVDFVADYHRDIESFPVRSQVKPGYLRPLLPDS
 APAEPETVEDVFADLWSKILPGLTHWQSPKFFGYPCNVSTAGMLGEMLCGGLNVNGF
 SWITSPAATELETIVLDWLGKLLHLP EEFSTSGKGGGVIQGTASEAVLVVMLAARKRA
 LKQVSSAAQGMSEAEALS KLVVYSSDQTHSCVIKACQVASIATENFRPLPTDASTNFALS
 PAVVRKAIATDVEAGLIPFFLCGTLGTTSSAAVDPLEELGDIKEYGMWYHIDAAYAGN
 ACICPEFRHYLNGVEKADSYNMNPHKWLLTNFDCSTLWMKDSEFLAALS NKPVFLRN
 EATDNNLVVDYKDWQIPLGRRFRALKLWMVMRLYGTSGLQSFIRSHVSSAKHFESLVR
 ADSRFEVMAPMTFSLVCFRLRTLPGSQDNSNSLNSKLVDA LNRKGNILVTHTELSGIYTV
 RFAVGATHTEHQHVQA AWEVIQAEASHLLNGKQ

Kalanchoe laxiflora 1398s0003.1 (SEQ ID NO: 42):
 MILSIHPFFPTLSARFSGAAAANILSKASCWLRLRSMEGELKPMDAEQ LREYGHMVD
 FVADYYKTIEDHPVLSQVEPGYLRKLLPDSAPDKPESFENVLSDVKTKIIPGVTHWQSPN
 YFAYFPSNSSTAGFLGEMLSACFNIVGFSWITSPAATELEMIVLDWFAKMLKLPDFFLST
 GQGGGVIQGTASEAVLVVLLAARDIFLRKLKGFLEKLVVYASDQTHSALQKACQIAGI
 HPENVRALKTDSSSTNYGLSPDLLSKEICHDIANGLVPPFACASVGTTSSTAVDPILELANV
 TKS YNIWLHVDSAYAGSACVCPEYRHHIDGVEEVDSFNMNAHKWFLTNFDCSLLWVK
 DRNALIQSLSTNPEFLKNKASQSN SVLDYKDWQIPLGRRFRSLKLWLVRLYGVENLQA
 YIRNHIELALNFEELVSQDMRFEIVAPRTFALVCFRLLPCGFEDHTNDVNSDLLQAVNS
 TGKIFISHTVLSGTYVLRFAVGAPLTEERHIDA AAWKLIQDQASSLLEKL

Manihot esculenta 12G038600.1 (SEQ ID NO: 43):
 MEGELRPMDAEQLREYGHQMVDFIADYYKT IENFPVLSQVEPGYLHKL LPDSAPNQPE
 ALQNVLD D VVKILPGVTHWQSPNYFAYYPSNSSVAGFLGEMLSAGINMIGFSWITSPA
 ATELEMIVLDWLGKMLKLP EEFSSGQGGGVIQGTASEAVLVVLLAARDKVLTRVGKD
 SLKKLVVYGSDQTHSALQKACQIAGVHLDNCRLLKTDSSKNYALSPDILCDAISQDMSN

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GLIPFFLCATVGTGTTSSATVDPLALGKIACKYGMWFHVDAAAYAGSACICPEYRCYIDGV
 EADSFNMAHKWFLTNFDCSALWVKDRNALIQSLSTNPEFL1<NKASQANMVVDYKD
 WQIPLGRRFRSLKLMVLRLYGVANLQSYIRNHINLAKYFEGLVAGDSRFEVVAAPRLFS
 LVCFRLLPPDNDENHGKLNHDLDAANSTGKIFISHTVLSGKYILRFAVGAPLTEERHV
 TAAWKVLQDEASALLGSL

Prunus persica 8G214500.1 (SEQ ID NO: 44):
 MESGLKPMDAEQLENNAHKMVDFIADYYKTIEFPVLSQVQPGYLRELLPDSAPTHPEP
 LQHIFFDDIQAKILPGVTHWQSPNFFGYPSNSSIAGFLGEMMSAGLNIVGFSWITSPAATE
 LEMIVLDWFGKMLKLPPEEFLSAGKGGGVIQGTASEAVLVVLLAARDKILRRVGKNSLE
 KLVVYASDQTHSALQKACQIGGIHPENCRLRLTDSSSTNYALSPNVLEAISNDVTSGLIP
 FFLCATVGTTSSTAVDPLLELGIKSNMWFHVDAAAYAGSACICPEYRHYIDGVEEAD
 SFNTNAHKWFLTNFDCSVLWIKDRNALIQALSTNPEFLKNKASQANLVVDYKDWQIPL
 GRRFRSLKLMVLRLYGLENLQSYIRNHINLAKHFKELVAQDPRFEIVTPRLFSLVCFRL
 LPPHNDETCAKLNHGLLDAVNATGKIFISHTVLSGKYLLRLAVGAPLTEERHVNAAWK
 LLQDEASALLATL

Eucalyptus grandis K01418.1 (SEQ ID NO: 45):
 MEERLKPMDAEQLRSAHRMVDFIADYYKIESFPVLSQVEPGYLRKLLPDSAPDHPESL
 QQVLEDVQAKILPGVTHWQSPNYFAYYPSNSSIAGFMGEMLSAGLNIVGFSWITSPAAT
 ELEIIVLDWLAKLLNLPDDFLSTGPGGGVIQGTASEAVLVVLLAARDKFLSRIGKSSLDK
 LVVYSSDQTHSALQKACQIGGIYPENCRVLKTDASTNYALSPDLLNEVISQDISTGLVPFL
 LCATVGTTSSTAVDPLPALATVAKRNGMWFHIDAAYAGSACICPEYRPYIDGVEEADSF
 NMNAHKWFLTNFDCSALWIKDRKALIQALSTNPEFLKNKASQANMVVDYRDWQIPLG
 RRRFRSLKLMVLRLYGVQNLQQYIRNHIELARQFEDLVIQDPRFEVVTPRIFSLVCFRLL
 SPDNDGDKGNKLNRLDLDTVNSTGKIFISHTVLSGTYLRLFAVGAPLTEERHVNAAWKV
 LQDEASKLLATIQNN

Amborella trichopoda 31565185 (SEQ ID NO: 46):
 MDAEELREHGHMVDFIADYYKEIESYPVRSQVQPGYLRNLIPDSAPDMPESEFIEIDIR
 HKIIPGVTHWQSPKYFAYYPSNSSSTAGFLGEMLSAGFNIVGFSWVTSPAATELEVIVLDW
 LAKVLKLPBQFLSTGKGGGVIQGTASEAMLVALLAARDKALRRVGQNLLENLVVYGS
 QTHSALIKACKIAGINPMNCRLLQATFMTNYALSPVASESISNDIAAGLLPIFLCATVGT
 TSSTAVDPLAALGRKAKANDMWFHIDAAYAGSACICPEYRHYIDGVEEADSFNMNPHK
 WLLTNFDCSTLWVKSSNLIQSLSTNPEFLRNKASEEDLVVDYKDWQIPLGRRFRSLKL
 WMVLRMYGVANLQNHIRTHINLAKHFEELIATDTRFEIIVPRVFALVCFALKPMPNGQD
 DASKNLKLEAVNNSGAMFLTHTVLSGRFVLRVVGAPLTEERHVNTAWKVLQDHA
 NLILGTV

Salix purpurea 0252s0200.1 (SEQ ID NO: 47):
 MESKGLKPMDSQLENNAHKMVDFIADYYKSIENFPVLSQVEPGYLRRELLPDSAPNQPE
 TLQNVLDDVQAKILPGVTHWQSPSYFAYYPSNSSVAGFLGEMLSAGINMVGFSWITSPA
 ATELEMIVLEWLKLLKLPEDFLSTGQGGGVIQGTASESVLVVLLAARDRVLTCLGKNA
 LEKLVVYASDQTHSALQKACKIGGIHPENCCLKLTDSSSTNYALSPDLLSKAISDDISTGLI
 PFFLCATVGTTSSTAVDPLHALGKIKNNGIWFHVDAAAYAGSACICPEYRCYIDGVEEA
 DSFNMAHKWLLTNFDCSALWVKDRNALIQALSTNPEFLKNKASQANMVVDYKDWQI
 PLGRRFRSLKLMVLRLYGLENLQCYIRNHINLAKYFEGLVAAADSRFEVVTPRIFSLVCF

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RLPPSNEDHGNNLRDLDAVNSSGKIFISHTVLSGKYILRFVAGAPLTEERHVIAAW
KVLQDESTSLGSL

Medicago truncatula 31080941 (SEQ ID NO: 48):
MVLQIWCLTHDSKKLGGGYLLFPVIKVAYTVHTLTWCCVTEEGGSELKAMDAEQ
LREQGHMMVDFIADYYKTIENTFPVLSQVQPGYLGKLLPDSAPTHPESLQHVLNDVQEKI
LPGVTHWQSPNYFAYPPSNSSIAGFLGEMLSAGLSIVGFSWISSPAATELETIVLDWLAK
ALLLPHDFFSTQGGGVIQGTASEAVLVVLAARDKILRTVGRSALPKLVTYASDQTHS
SLQKACQIAGLNPELCRLKTDSSSTNFALSPDVLSEAISNDIASGLTPFFLCATVGTTSSTA
VDPLPALAKVTKPNNIWLHVDAAYAGSACICEYRHFIDGVEEADSFNMNAHKWFLTN
FDCSVLWVKDRSALIQSLSTNPEFLKNKASQENTVIDYKDWQIPLGRRFRSLKLWMVM
RLYGLEGLRTHIRSHIALAVYFEELVVQDTRFKVAPRTFSLVCFRLPQPQNSDNGNKL
NHDLLDAVNSTGDVFITHTVLSGEYILRLAVGAPLTEVRHVHAAWQILQEKATALLES

Brassica rapa I01156.1 (SEQ ID NO: 49):
MQIRAKIPVFGRENGSRHVLKPMDEQLREYGHMVDFIADYYKTIESFPVLSQVQPGY
LHNLLPDSAPDHPETVEQVLDDVKTILPGVTHWQSPNFFAYYPSNSSVAGFLGEMLSA
GVGIVGFSWVTSAPATELEMIIVLDWLAKLLNPEHFLSKGNGGGVIQGSASEAILVVM
AARDKVLRSAGKNALGKLVVYSSDQTHSALQKACQIAGIHPENCRVLKADSSSTNYALR
PELLQEAVSRDLEAGLIPFFLCGNVGTSSAAVDPLAALGKIAXSNEIWFHVDAAYAGS
ACICEYRQYIDGVETADSFNMNAHKWFLTNFDCSLLWVKDQHALTEALSTNPEFLKN
KASQANLVVDYKDWQIPLGRRFRSLKLWMVLRLYGAEALKNYIRNHIKLAKDLEQLVS
QDPNFEVITPRIFSLVCFRIVPTDNDKCKNSRNLELLEAVNSSGKLFISHTALSGKIVLRC
AIGAPLTEEKHVKETWKVIEKVSYLLRK

Brassica rapa I04706.1 (SEQ ID NO: 50):
MDSEQLREYGHMVDFIADYYKTIEFTFPVLSQVQPGYLNLLPDSAPDQPETVEQVLDD
VKTILPGITHWQSPTFYAYYPSNSSVAGFLGEMLSAGLGIVGFSWVTSAPATELEMIIV
DWLAKLLNPEQFLSKGNGGGVIQGSASEAILVVMIGAREKVLRRVGKNALGKLVVYS
SDQTHSALQKACQIAGIHPENCRVLKADSSSTNYALRPELLQEAVSKDIEAGLIPFFLCGN
VGTTSSTAVDPLAALGKIAXSNEIWFHVDAAYAGSACICEYRQYIDGVETADSFNMNA
HKWFLTNFDCSLLWVKDQYVLTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRS
LKLWMVLRLYGAETLKSIRNHIKLAKDLEQLVSQDPNFEVVTPRIFSLVCFRILPVDND
EKECNRRNRNLLDAVNSSGKLFISHTALSGKIVLRCAIGAPLTEERHVKETWKVIEEAS
RLLGK

Brassica rapa G00043.1 (SEQ ID NO: 51):
MDSEQLREYGHMVDFIADYYKTIEFTFPVLSQVQPGYLNLLPDSAPDQPETLEQVLDD
VKEKILPGVTHWQSPSFYAYYPANSSVAGFLGEMLSAALNIVGFSWVSSPAATELEMIIV
DWFALLLNPEQFLSRGNGGGVIQGTASEAILVVMIAARDKVLRSLGKKALEKLVVYSS
DQTHSSLLKACQIAGIHLENCRMLKTDSSSTNYALRPESLQEAVSGDLEAGLIPFFLCGTV
GTTSTAVDPLAELGKIAXSNEMWFHVDAAYAGSACICEYRQYIDGVETADSFNMNA
HKWFLTNFDCSLLWVKDRYALTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFRS
LKLWMVLRLYGAETLKSIRNHIKLAKDLEQLVSQDPNFEVVTPRIFSLVCFRIVPVDND
EKTCCNNLRNLLDAVNSSGKLFISHTTSLSGKFVLRRLAIGAPLTEEKHVMDAWKVIQEEAS
FLLASQVK

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Glycine max 03G167900.1 (SEQ ID NO: 52):
MEEESALRPMDAEQLEQAHKMVDFIADYYKTIEDFPVLSQVQPGYLGKLLPDSAPDSP

ESLQNVLDVQEKILPGVTHWQSPNYFAYFPSNSSIAGFLGEMLSAGLNIVGFSWITSPA
ATELETIVLDWLAKAFQLPDYFYSSGKGGGVIQGTASEAVLVVLLAARDKILRRVGRNA
LPKLVMYASDQTHSALLKACQIAGINPELCRLKTDSSSTNYALSPDVLSEISNDIAGGL
VPFFLCATVGTTSSTAVDPLPALGKIAKTNKLWFHVDAAAYGSACVCPEYRHCIDGVVEE
ADSFNMNAHKWFLTNFDCSLLWVKDRSSLIQSLSTNPEFLKNKASQGNMVIDYKDWQI
PLGRRFRSLKLWMVRLRYGLDGLRSHIRNHIELAANFEELVRQDTRFKVVPARTFSLVC
FRLPHPNASDHGNKLNLDLSDVNSTGNAFITHTVLSGEYILRFVAVGAPLTERRHVNM
AWQILQDKATALLES

Fragaria vesca 27261550 (SEQ ID NO: 53):
MDAEQLRENAHKMVDIADYYKTIEDFPVLSQVQPGYLRELLPDSAPTQPELQHFDDI

QAKILPGVTHWQSPNFFAYFPSNSSIAGFLGEMLSAGLNIVGFSWVTSPAATELEMIVLD
WLAKLIKLPDEFLSAGQGGGVIQGTASEAILVVMLAARDKILRRVGKNALEKLVVYASD
QTHSALQKACQIAGIHPENCRIILSTNSTNYALSPSVGTTSSTAVDPLGELGKIAKNEM
WFHVDAAAYGSACICPEYRHYIDGVEKADSFNMNAHKWFLTNFDCSVLWIKDRNALV
QSLSTNPEFLKNKASQANMVVDYKDWQVPLGRRFRSLKLWMVRLRYGLENLQSYIRT
HINLAKHFEELVAQDPRFEIVTPRLYSLVCFRLLPPHGNEACASKLNHDLDAVNSTGKI
YISHTVLSGAYILRFVAVGAPLTEKHVTAAWKKLSVIRDVLALANSFVSI TFSHMYREA
NFLTDALASVGHSLSSSMCWFDGIPPQAQMALMDSSCIGHLRGSSL

Kalanchoe fedtschenkoi 0172s0035.1 (SEQ ID NO: 54):
MEGELKPMDAEQLEQREYGHMVDVADYYKTIEDHPVLSQVEPGYLRKLLPDSAPDKPE

SPENVLSDVTKIIPGVTHWQSPNYFAYFPSNSSIAGFLGEMLSACFNIVGFSWITSPAAT
ELEMIVLDWFAKMLKLPDFFLSTGQGGGVIQGTASEAVLVVLLAARDIFLRKLKGKGFLE
KLVVYASDQTHSALQKACQIAGIHPENVKALKTDSSSTNYGLSPDLSKEICHDIANGLVP
FFACASVGTTSSTAIIDPILELANVTKSYNIWLHVDSAYAGSACVCPEYRHHIDGVVEEDS
FNMNAHKWFLTNFDCSLLWVKDRNALIQSLSTNPEFLKNKASQSKSVLDYKDWQIPLG
RRFRSLKLWLRLRYGVENLQAYIRNHIELAIHFEELVSQDMRFEIVAPRTFALVCFRLL
LPCGFEDRTNDVNGDLLQAVNSTGKIFISHTVLSGTYVMRFVAVGAPLTEERHIDAANKL
IQDQASSLLEKL

Capsella grandiflora 22666s0001.1 (SEQ ID NO: 55):
MDSEQLREYGHMVDIADYYKTIEDFPVLSQVQPGYLHKLLPDSAPDQPETLDQVLDD

VRAKILPGVTHWQSPGFFAYFPSNSSVAGFLGEMLSAGLGIVGFSWVTSPAATELEMIV
LDWLAKLLNLPKEFLSKNGGGVIQGSASEAVLVVLLIARDKVLRSAKNALGKLVVY
SSDQTHSALQKACQIAGIHPENCRLVETDASTNYALRPELLQEAHSVSKDLKAGLIPFFLCA
NVGTTSSTAVDPLAALGKIANSNEIWFHVDAAAYGSACICPEYRKYIDGVETADSFNMN
AHKWFLTNFDCSLLWVKEQDSLTEALSTNPEFLKNKASQANLVVDYKDWQIPLGRRFR
SLKLWMVRLRYGAETLSYIRNHILAKYIEKLVSDPNFEIVTPRIFSLVCFRLVPKNE
DEKKCNNQNRKLEAANSSGKLFSHTALSGKIVLRCAIGAPLTEEHMKEAWKVIQD
EASFLHKL

Selaginella moellendorffii 15420188 (SEQ ID NO: 56):
MGEANIGPKPIDAEFRKHAHEMVDIADYYRDI ESFPVRSQVSQPGYLKTLPPAAPED
PEALEEVFADIQSKIIPGVTHWQSPNFFGYFPSNSSIAGLLGEMLSAGLNIVGFSWITSPA

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ATELEIIVLDWLAKLLKLPDEFLFGNGGGVIGGTASEAVSVVLLAARTRAISENKRKGL
 SEAEILSKLAVYTSQDQTHSCLOKGCIAIGIPLNLVIVPTDSSTNYAVSPAAMRQALEDG
 VKQGLLPFFLCGTGTSSAVDPLSALGDIKDFGMWFHVDAAYAGSACICEPRHHL
 DGVEKADSFNMNAHKWLLTNFDCSALWVKESHLVSALSTTPEFLRNKASDLNQVVD
 YKDWQIPLGRRFRSLKLVFVMRMNGASGLRSYIRNHVRLAKRFEGFVREDPRFQLLVP
 RTFGLICFRLKPESDDPDNGRTLNLSTLEAVNSSGRMFITHTVLSGVYTLRMAIGGPLTQ
 DKHVDAAWKLIQEEATTLVKGPSHILANNLRLSPILANNLRLSPILANNRI

Setaria italica 3G188200.1 (SEQ ID NO: 57):
 MDILNHADTTTANGTSPAAAAAAVAPATPSSSLVTPPLDADEFRQGRLVVDFIADYY
 TRINEYPVRPAVAPGFLARQLPETAPARPERDALAAALRDVLDLILPGVTHWQSPRHFA
 HFAATASNVGALGEALAAGLNINPFTWAASPAATELEVVDWLGKALHLPERLLFSG
 GGGGTLTGTSCEAMLCITIVAARDRKLAEIGEERIGDLVVYFSDQTHFSFQKAARIAGIRR
 GNCREIPTSRESGFTLSPKALRAAVRADEASGRVPLFLCATVGTTPATAIDPLRELCAAV
 SGHGVMVHVDAAYAGAACVCEPRHAIAGAEAVDSFSTNPHKWLLANMDCCALWVT
 RPAALVAALGTDHVDILKDPASAAQDGHVVDYKDWQVALSRRFRALKLWLVLR
 HGVEGLRGPVRAHVRMAAAFEAMVRADTRFEVPVPRQFALVCFRLRPASAGEKRTRG
 GEVVEPNELNRRLLLEAVNATGRAYISSAVVGGVYVLRCAIGNSLTEERHVRSAWVSVQ
 EQANVVLAATATCPDERAVHRARCVETDAADAPASVPPVQMRFPSPAQS

Kalanchoe fedtschenkoi 0033s0078.1 (SEQ ID NO: 58):
 MGSLSPPHDPNSAFNPMDDVAELSIESRLVMDFITQYYQTLTRPVQPRVKPGFLTGLPLPE
 KAPFHAESMEEILSDVSEKIVPGLTHWQSPNFHAYFPASSNAGLLGEMLCGLSVIGFT
 WNSSPAATELENVVDWLDMLNLPPSFRFSGGGGGGVLQSNCEAVLCTLAAARD
 KVLERIGDDKINKLVAYCSDQTHFTLHKGAKLIGIRANIKSIGTRENGFGLCPNDLRN
 AITGDLEAGLVPPYLCGTIGTTALGAVDPIKELGKVAREFDLWFHIDAAYGGSACICEP
 RHYLDGVELVDSISMNAHKWLLSNLDCFLWLQNPCKLIQCLAAEAFLKSGSEMVDY
 KDWQISLSRRFRAIKMMVFRRYGVSNLMEHIRSDVSMAARFEEMVSADDRFEIVFPR
 KFALVCFKLNTKGSVQHGEDDGEDGLDGSVLTRELMGRVNSSGKAYLSGVEMGRIF
 IRCVIGSSLTEERHVDNLWNLIQEKTQSIMPCRA

Daucus carota subsp. *sativus* 36068870 (SEQ ID NO: 59):
 MGSLSSTQKFNPLNLDFFSSESNKVEFITAYYKNVEKYPVRSQVEPGFLNMYPKKAPSQ
 PVSLDTILQELEADIIIPGMTHWQSPNFYAYFRTTTSNAAFQGEMLCNALNVAGFNWICSP
 AATELEMIVMDWLGKMLSLPQSFLFAGNGGGVLQGSTSEALICVLSAARDALKQYGE
 DSITKLVVYASDQTHFVVKAAKLVGIPTKNFRVPTSATCFALKPNDIKMAIERDLESG
 LVPLFVCATVGATPSGSDPVEGLGLLAKNYGLWLHIEAAYAGSAFICPELTHYLRGIEH
 AHSISINLHKWLLTNMDCSCLWVKSPLVLESLSMTDEILRNEASESKKVDFMDWQIA
 TSKLFRALKLVFVLRRYGVDNLMAHIRSDIELAKHFEALVNSDKRFEVVVPVNFSLVCF
 RLKPNEEGEESLKVLMNWNLMNAVNSSGRAYMTHAVLGDIFVIRCAIGTSLTEERHVNE
 LWKLILEKTEVILKRDQ

Daucus carota subsp. *sativus* 36056758 (SEQ ID NO: 60):
 MNTFTDTEFRKQAHLLIIDFLADYYQNIKFPVRSQVSPGYLGEILPDSAPHDPEPIEKILED
 VRSNIIIPGITHWQSPNFYAYFPSCGSTAGFLGEMLANGFNVVGFNWISSPAATELETIVM
 DWLGKMLQLPEAFLFSGGGGGVLQGTTCAMLCITLVAARDRTLREQGMENFDKLLCP
 VQLELEILSDVQNGLIPLFLCVTIGTTPSTAVDPLATLSEVAKKYKLWVHVDAAYAGSA

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CICPEFRHFLDGLNVNSFSMNAHKWFLTTLDCCCLWVNDPSALIKSLSTYPEFLRNHAS
 ESNKVVDYKDWQIMLSRRFRALKLWFLVLSYGVVEKLREFIRVHVEMAKYFEGLVAMD
 QRFEVVVPRLFAMVCFRVVCCGENDVNEINEKLLSVNQSGRIYVSHAVLDGVYVIRFA
 IGATLTDYSHVSAAEVVEVQEHADALLA

Solanum tuberosum 3DMP400026166 (SEQ ID NO: 61):
 MGTNLNINHELDDQIFNTINPLDPEEFRRQGHKIVNFLADYYQNIQYVPVCSQVNPGLYQK
 IVPNSAPNNSESLEKILKDVERDIIIPGLTHWQSPNFFAYFPSSGSTAGFLGEMLSVGFNVV
 GFNWISSPAATELESIVMDWFGKMLNLPNCFLFASGGGGVLQGTTCEAMLC TIVAARD
 QMLRKISRENFGKL VVYASDQTHFSLKKAHIAAGIDPGNFRVIPTIKANEYTLCPKSLRL
 AILNDLKEGNVPLFLCATIGTTATTSTVDPLRLLCIEAKEFGIWHVDAAYAGSACICPEFQ
 VPLDGVENANSFSLNAHKWFFSTLDCCCLWVKDPSALTNALSTNPECLRNKATELNQVI
 DYKDWQIALSKRFRALKLWLVLSYGVNTLRNLIRSHVNMMAKHFEGLVATDKRFEIFV
 PRKFAMVCFRISPLVLSQVSTKFDDEKEVNMFNTKLVESINSCGKLYLTHGVVGGTYIIR
 FAIGASLTHYRHVDVAWKVIQDHANALLNQGYV

Solanum tuberosum 3DMP400024738 (SEQ ID NO: 62):
 MGMTKINPEHEFDGQFSINTSSRLDPPEEFRRQGHMMVDFLADYFQNIQYVPRVSQVE
 PGYLKLLPDSAPYKPEPIAKILEDVERDIFPGLTHWQSPNFFAYFPCTSSTAGILGEMLS
 AGLNVVGFSLIASPAATELESIVMDWLGKMISLPKTYLFSGHGGGGVIQGTTCEAMLC
 TIVAAREQMLEKVGREKVDKL VVYASDQTHFSFEKAVKISGIKLENFRVIPTTKDTEFAL
 DPKLSRTIEQDIKSGFIPLFCATIGTTSTTVVDPLKLLCEITKDYGIWHVDAAYAGGA
 CICPEFQHFLDGIENANSFSFNAHKWLFSLNLDCCCLWVKDPSALTNALSTRPECLRNKAT
 DTKQVVDYKDWQLSLRRFRALKLWLVLSYGIDNLRNFIIRSHVMAKHFEQLVMSD
 ERFEIVAPRNFMSVCFRVSPALGNKQVNKFMELLESINSCGNIHMTALVGGVYMIR
 FAIAAPLTEYKHIDMAWEVICNHANAMLDVN

Solanum lycopersicum 36137005 (SEQ ID NO: 63):
 MGTNLNINHELDDQIFNTINPLDPEEFRRQGHKIVNFLADYYQNIQYVPVCSQVNPGLYQN
 IVPNSAPNNPESLDKILKDQNDIIIPGLTHWQSPNFFAYFPSSGSTVGFVGEMLSVGFNV
 VGFNWISSPAATELESIVMDWFGKMLNLPNCFLFASGGGGVLQGTTCAILCTIVAARD
 QMLRKISRENFGKL VVYASGQTHFSLKKSAAHIAAGIDPGNFRVIPTIKAEYTLCPKSLRLA
 ILNDLKEGNVPLFLCATIGTTSTTSVDPLRLLCDISKEFGIWHVDAAYVGSACICPEFQV
 FLDGVENANSFSLNDPSALTNALSTNLEFLRNKATELNQVIDYKDWQIALSRRFRALKL
 WLVLSYGVNTLRNLIRSHVNMTHFEGLIAMDKRFEIFVPRKFAMVCFRISPLVLSQVS
 IKFDDEKEVNMFNTKLLESINSCSKLYLTHGIVGGTYIIRFAIGASLTHYRHVDIA

Daucus carota subsp. *sativus* 36065781 (SEQ ID NO: 64):
 MCKPKSSPASHINWQSPNFFAYFPSSGSTAGFLGEMLS TGFNVVGFHWMASPAATELEN
 VVTDWFGKMLQLPKSFLFSGGGGGVLQGTTCEAMLC TLVAARDKNLRQHGMENIGKL
 VVYCSQDTHSAMQKAAKIAAGIDPKNFRTVETSRSANFQLCPRRLESAILTDIQNGLIPLYL
 CATVGTTSSTAVDPLPALTEVAKKYDLWVHVDAAYAGSACICPELRQYLVNGVENADSF
 SLNAHKWFLTTLDCCCLWVKNPALIKSLSTYPEFLRNNASETNKKVVDYKDWQIMLSR
 RFRALKLWFLVLSYGVGQLREFIRGHVDMAKYFEGLVGDKRFEVVVPRLFSMVCIRV
 RPSAMTGKSCGNDVNELNRKLLLES LNESGRIYVSHTVLDGIYIIRFAIGATLTDINHVSAA
 WKVVQDHATALDDTNFLAKKVADIILS

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Oropetium thomaeum 35995617 (SEQ ID NO: 65):
 MAILNHADDASPANDDNPATAPAMAPATNPRPLDADEFRRQGRLVVDIADYYARVEE

YPVRPSVTPGFLSRKLPETAPEQPEPGHGDAFASALRDVRDLILPGITHWQSPNHFAHFA
 ATASNVGALGEALAAGLNINPFTWAASSAATELEVVTDLGKALHLPQELLFSGGGG
 GTLLGTSCEAMLCTVVAARDRLGEIGEHRIGDLVVYCSQTHFSFRKAARVAGIRAN
 CREIPTSLESDFALSPSALLAAVRADEAAGLVPLVLCVTVGTTPTAAVDPVRELCAAVA
 GRGVVHVDAAYAGAARVCEPLL RHAGAIVDGVSFSTNPHKWLLANMDCALWVQ
 QPDALVAALGTDHDVILKDPAAAAAGDVVDYKDWQVALSRRFRALKLWLLLRCHG
 VEGRLAHVRDGLRMAEAFEAMVRADARFEVPVRRQLSLVCFRLRPTAVIREKQQQORG
 RRRDHDDDTAAANELNRRLLEAVNATGRITYMSCAVVGGVYMLRCAIGNSLTEDRHVE
 EAWN VVQE QASAILDAAMVVRAD ECTVCTAAHCVMGMVDDILAASFPTGNEVTIR

Oryza sativa 33157740 (SEQ ID NO: 66):
 MAILNHSDAAPPVAATTPLLGRRPLDAGEFRQGRQVVDIADYYAGINDYVVRPAVAP
 GFLAGKL PATAPSTPEPDALTAGLRDVRELMPLGLTHWQSPRHFAHFSATASNVGALGE
 ALAAGLNVNPFTEWASPAATELEVVTDLGKALHLPERLLFAGGGGGTLLGTSCEAM
 LCTIVAARDEKLAIEIGERIGDLVVYCSQTHFSFQKAARIAGIRRGNCREIPTCRESGFV
 LTATALQA AVADEAAGRVPLFLCATVGTTPTAAVDPLRELCAAVEGRGVVHVDA
 YAGAACVCEPFRHAIAGAEAVDSFSTNPHKWLLANMDCALWVARPAALVAALGTDD
 DVILKDAAAAARPARGDHHHHAADVYKDWQVALSRRFRALKLWLVLRCHGVDGLRA
 VVRSHVRMAAALERMVRADARFEVPVPRQFALVCFRLRGGGAAQLVGGDELTASNE
 LNRRLLEAVNATGRAYMSSAVVGGMYVLRCVGNLSLTEEHHVREAWSVVQGQAAV
 LATAGAAADTARTKDHAAGDDHGADQPHAMTTTTMTGCRSGPWEL

Brachypodium stacei 01G392300.1 (SEQ ID NO: 67):
 MAPASSTRQVITDHTQKENS SCTVINHLDADEFRRQGHKVIDFIADYYSGIADYPVHP
 SVTPGFLNQLPADPPDPDTFASALQDVDRDLILPGMTHWQSPRHLAHFPASSSVTGALG
 EALAAGINAVPFMWSASPAATELEMVAVDWLGKALHLPKTLFSGAGGGTLLGTSYRK
 LAETGAGRIGDLVVYGSQTHFALRKAARIAGIRHGRCRELRTCIADMFALSPAALSAA
 MDADAGAGLVPLFLCATVGTTQTKAVDPICALCAEAAPHGVVHVDAAYGGSALVCP
 ELARDAIDGVEAVDSFSMNAHKWLLVNTDCCALWVKRPALLVSALGTQDEDEVILRD
 AAAQGHADVVDYKDWAVTLTRFRALKLWLVLR CYGVEGLREHIRGHVRMAALFEGM
 VNADPRFEVVTERRFALVCFRLRPDQLPDEGNKKKTMAAANELNRRLLQEVNAAALGP
 YMSAANVGGIYVLRCAVGSTLTEKRHRVQAWVEVQEKATSI LRA

Amaranthus hypochondriacus 32828676 (SEQ ID NO: 68):
 SLHDETLQGIKYVTQYYKNVEKYPVSVKWKGYLRQILPENAPSLPESIDQILEDVTKI
 VPGLTHWQSPNFFAYFPATASNAAMLGDIVCSGLNVVGFSSWISSPAATELEAIVMDWM
 AKLLMLPPTFLFSGGGGGVIHGSTCEAIVCTQAAARDVALNIHGEKITKLVVYASDQT
 HISFQKAAKLIGIPPRNFRVLPTSSATDFALSPTTLRASIEVDLSQGLVPFYICATIGATPSG
 AVDPIDGLGQIARDYGAWLHVDAAFAGNACICEYRHYLDGVELADSISMNPHKWLLT
 NMECSCLWLKNPKLMVDSLSTKPEILNNKATQSGDVIDYKDWQIALSRRFRALKLWIVI
 RRYGSTYLMNHVRSDIELAKYFESLIKQDERFELVVPKFSLVCFRMKLVGREDVETLT
 NQKLEDVNSSGKAYMTHAVIGGKFVIRCAIGGTLTEKRHIDSLWKLIIEKVPLTTCEL

Brachypodium distachyon 5g21770.1 (SEQ ID NO: 69):
 MSSNSCPAAAAATFTTPGAHPLPLDADAFRRQGRQVADFIADYYDRIEDYVVRPNVSP

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GFLAAQLPDAAPSWPEEPDALASALRDVRLILPGLTHWQSPRHFAHFAATASNAGAL
 GEFLAAGLVNPNFTWAASPAAAELEVVDWLGLQALGLPEKLLFRGGSGGGTLLGTS
 CEAMLCITIVAARDQKLLKIGEDRIGDLVVCSDQTHFSFKKAARVAGIRRGNCRVIPTRF
 EDGFALSPAALAAAVRDDVARGKVPLFLCATVGTATGAVDPVRELCAAVGAGHGSG
 VVHVHDAAYAGGACVCPEFRHVAAGAEEDSFSTNPHKLLANMDCALWIRPGL
 LVAALGAGEDEDAILNKAPPAARGMQADLMVDYKDWQVPLSRRFRALKLWLVLRCH
 GVEGLRGVVRGHVRMAAAFEAMVRADPRFEVPPAPAFALVCFRLRPLAAHPGSSSGID
 EVNGRLLEAVNGTGRAYMSGAVVGGAYVLRCAVGNSLTEDRHVREAWSVVQEQA
 ILAPSDDEDRCCCTDQIQTEMELQRRPLGAAADVFA

Brachypodium distachyon 2g02360.1 (SEQ ID NO: 70):
 MAPASSKLHAITDDKTQQNSSCPAASNGAIEPSNAKCAASSNHLDDADEFRRQGHKVI
 DFIADYYAGIADYPVHPSVTPGFLNLQLPADPPSRPEDHPDGAFGPALQDVRDVILPGMT
 HWQSPRHFAHFPASSSVAGVLGEALAAGINAVPFTWAASPAAAELEMVAVDWLKAL
 HLPESLLFSGAGGGTLLGTSCEAILCALVAARDRKLADIGTRIGDLVVGSDQTHFALR
 KAARIAGIRHDCRELQTCCLADMFAISPAALSAAMDADAGGLVPLFLCATVGTQTT
 AVDQVGALCAAAPHGVVHVDAAYAGSALVCPELARDAIDGIEVDSFSMNAHW
 LLANTDCCALWVKQPKLLVVS LGTQNEELILRDAAAEGHDVVDYKDWAITLRRFRAL
 KLWLVFRCYGVEGLREHRAHVRMAALFEGLVKDDPRFEVVTERRFALVCFRLRAPDQ
 LMDEGNEKKKTTAAANELNRLLREVNGVALGPYMSAAVVGGIYILCAVGSTLTEER
 HVRQAWEVVQERATSILRG

Sorghum bicolor 009G192600.1 (SEQ ID NO: 71):
 MGVAVTAEVVHARCKGTPPVGAAASVMVWDGAGQGYSCQPVGTTTANGGTPPAAP
 VAIAMPSPHPLLDADDEFRRQGRLLVVDIFIADYYARIDEYPVRPAVAPGFLARQLPETAPA
 RPEPDALAAALRDVRLILPGVTHWQSPRHFAHFAATASNVGALGEALAAGLNINPFT
 WAASPAATELEVVDWLGLKALHLPESLLFSGGGGGTLLGTSCEAMLCITIVAARDKL
 AEVGEERMGLDVVYCSQTHFSFQKAARIAGIRRGNCREIPTSMAGFTLSPKALAAV
 RADEAAGRVPLFLCATVGTTPTAADVDPVRELCAAVAGRGVHVHDAAYAGAASVCE
 LRHAVAGVERVDSFSTNPHKLLANMDCALVRRPAALTAALGTDHVDVILKDPSAQ
 AAQEGGAVVDYKDWQVALSRRFRALKLWLVLRCHGVVEGLRGLVRAHVRMAAAFEA
 MVRTDARFEVPPVPRQFALVCFRLRAAAVLVVGKRRARDGDEVVTAGNELNRLLLEA
 VNATGRVYMSSAVVGTYILRCAIGNSLTEERHVREAWSVVQEQAAILAAARRPTAR
 TNRRTVRRHAAL

Kalanchoe laxiflora 0994s0009.1 (SEQ ID NO: 72):
 MGSLSQSPHPDNAPNMDVAELSIESRLVMDFITQYYQTLETRPVQPRVKPGFLTQGLPE
 KPPFHAESMEEILSDVSEKIVPGLTHWQSPNFHAYFPASSNAGLLGEMLCGLSVIGFT
 WNSSPAATELENVVDWLADMLNLPPSFRFSGGGGVLQSNCEAVLCTLAAARDKV
 LERIGDDKINKLVVYCSQTHFTLHKGAKLIGIRRANIKSISTRENGFLCPNDLRNAIK
 SDLEAGLVPPFYLCTGTGTALGAVDPIKELGKVAREFDLWFHIDAAYGGSACICPEFRHY
 LDGVELVDSISMAHKWLLSNLDCCFLWLQNPCKLIQCLAAEGEFLKSGEMVDYKD
 WQISLSRRFRAIKMWMVFRYGVSNLMEHIRSDVSMAARFEEMVAADDRFEIVFPRKF
 ALVCFKLNKTSVQHGVEDGEDGLDGSVLTRMLMGRVNSSGKAYLSGVEMGRIFIR
 CVIGSSSLTEERHVDNLWNLIQEKTSIMPRRA

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Kalanchoe laxiflora 0003s0173.1 (SEQ ID NO: 73):
MGSLSPPRDLTKPFNPLDPTELAVESSLVTDFAEYYRTVEQRPVQPHVTPGFLTSQLPSA

APFASESVESILQDVYDKILPGLVQWQSPNFHAYYPATCSNAGLLGEMLCSGLN NVGFT
WSASPAAAEELQVVVDWMGKMMGLPQSFLFSGGGGGVLQGSTCEAVVCTLAAADR
ALERVGD DMFNKLVVYCS DQTHFTLKKGSKLVGIRPANVKAIKTTKNNEYGLCPTDLR
NLVASDVKAGFIPIYLCGTIGTTAFGAVDPIRELKGVAREFNMWFHVDAAAYAGSAFICPE
FRHYMDGVELADSFSTNPHKWLLSNMDCCVLWLKFPKRVIKSLAAEGVFLEGGSETMV
DYKD WQIALSRRFRAIKLWMVIKRYGLKNLISHIRSDVSMARFEELLSDRRFEVVFPFR
KFSLVCFKLDVMKNVPEVVD EDDGELSHDSKLTRELMASVNVGTGKAPLTGVRLGRIFFI
RCAIGSTLTEDRHIQDLWKLIQEKAKHICANHDLKFRV

Panicum hallii 32512198 (SEQ ID NO: 74):
MAILNHGDTTANGSSPADAAAVAPAMPSLVQPPLDADEFRRQGRLVVD FIA DYYTRID
EHPVRPAVAPGFLARQLPDTAPARPEPGDDALAAALRDVRDLILPGVTHWQSPRFAHF
AATASN VGALGEALTAGLNINPFTWAASPAATELEV VVTDWL GKALHLPESLLFSGGG
GATLLGTSCEAMLCTLVAARDRKLA EIGEERIGDLV VYCS DQTHFSFQKAARIAGIRRG
NYREIPTSRESGFTLSPKVLRAAVRADEAAGRVPLFLCATVGTTPTAAVDPLRELCATVA
GHGVVHVDAAYAGAACVCPEFRHAIAGAEAVDSFSTNPHKWLLANMDCCALWVRR
PEALTAALGTDHDVILKDPSSERDCGRGVVDYKD WQVALSRFRALKLWLVL RCHGV
EGLRGFVRAHVMAAAFE DMVRADARFEVVPVRQFALVCFRLRSAAAGEKRARDGDD
AEPNELNRRLLEAVNATGRAYMSSAVVGGIYVLRCAIGNSLTEERHVR EAWCVVQEQA
TVVLA AACTEERAVHSARCADAPAAVPPVQNEGYGEPTSIAAKIFGTSIARCSIKSEAS
TYHSWSTLWRTL MFKLLTWIISRL

Prunus persica 6G202600.1 (SEQ ID NO: 75):
MTSALDPVEFRQGHMMVD FIA DYYQNIDKYPVLSQVDPGYLRKRLPESAPDNPEPIETI
LQDVQEHIVPGLTHWQSPSFAYFASNVS IAGFLGEMLSTGFNVVGFNVWSSPAATELE
SIVMDWLG NLLSLPKSFLFSGNGGGVIHGSTCEAIVCTMAASRDQMLSRIGDNIGKLV
VYGS DQTHSALQKASQIVGINPKNFRAIEATRSTTFALSPESLKLAISSDIEAGLVPLFLCA
TVGT TATTAVDPLGPLCDVAKHHGMVHVDAAYAGSACICPEFRHFIDGIEGVDSFSFN
AHKWFFTGLDCCCLWVKNP GALISSLSANPEFLRNKPTDSKQVVDYKD WQIALSRFR
AMKLWLVLRSYG VVNLRNFLRSHVKMAKLFEGLVAMDQRFEIVVPRNFSMVPPTTPTS
NSFHQNGIEINVEKCTNEVNCKLLEAINASGRVFMTHAMVGGMYVIRCAVGVTQTEEK
HIAMAWKVQEHADVILKNNGDDGDANLKLPLLDKIA

Prunus persica 4G086700.1 (SEQ ID NO: 76):
MGSLNFDHPQENNSAHMSGPLDLVELRRQGHMIIDFITDYYQNI EKHPVLSQVQPGYLK
QRLPESAPYNPEPIETILRDVQDHIVPGLTHWQSPNHFA YFPATISTAGFLGEMLTTCFNV
VGFNWMASPAATELETIVMDWLG DMLKLPNSFLFSGTGGGV LHGSTHESVVCTMAAA
RDQILSRIGEENIGKLVVYGS DQTHSVIQKVSQIVGIPSKNFRAIETTISSTLSPETLRLT
VCSDMEAGLVPPFYLCATVGTTATTAVDPLGPLCDVAKDYGMMVHVDAAYAGSACICP
EFRQYIDGIEGANSFSFNAQWFFTALDCCCLWVKNP SALT KSMSTDLEVL RNKASESK
RVVDFKD WQIALTRRFRAIKLWLVLRSYG VANLRNFLRSHVKMAKRFEGLVRTDERFE
VVVPRIFALVCFRISPSAISKANPTPSDEKCVNEVNCKLLEAINGSGWVYMTHAVVGGM
YVLRCAIGASLTKEKHVAMAWKVQEHVDAILPLTMY

Prunus persica 4G087100.1 (SEQ ID NO: 77):

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MMGSVEFEHPQENNSAHMTTSPLDPEEFRRQGHMVIDFIADYYKTIEKYPVLSQVQPGY
 LKKRLPESAPYDPEPIETILQDVQDHLVPGLTHWLSPNHFGYFPAAISTAAFLGEMLTG
 FNVVGFNMWASPAATELENIVMDWLGDMLKLPKSLFSGNGGGVLQGTTCBAIVCTM
 AAARDQMLRQIGRENIGKLVVYGSQDTHSALQKASQIVGIHPKNFRAIETTTSTSFALSP
 EVLKSTICSDIEAGLVPLFLCATVGTTAITAVDPLGPLECAKEHDMVHVDAAYAGSA
 FICPEFYFIDGVEGADSFSLNAHKWFFTTLDCCCLWVKNPSALVSSSLTNPEFLRNKAT
 DSKQVVYDKDWQIALSRRFKAIKLWLVLSYGVGNLNRNFLRSHVKMAKIFEGLVGMD
 KRFEIVAPRHFSLVCFRVSPSAISKANPSLSDHDNGKLKAHNYELNGVKCVVNEVNSK
 LLEAINGSGLVMSHAVVGGMYVLRCAIGASLTEEKHVAMAWKVQEHADAILGTKII
 VDQT

Medicago truncatula 31073039 (SEQ ID NO: 78):

MNTSSNPQSDPQKTMNPLDLEEFKRQGYMMIDFLTDYYKNIENYPVLSKVEPGYLAK
 ILPSSAPFPQESIESILEDVQQHIIPIGITHWMSPNYYAYFPSSGSIAGFIGEMLSTGFNVVGF
 NWLSSPAATELETIVMNWLGKLLNLPKSFIFSSNIKGGGEIKKLSQIGKDNIGKLVVYCS
 QTHSALQKATQIVGIHSENFRVIKTKGSNLFALSPDSSLSTILLVDNGLIPYFLCATIGTT
 STNAVDPIKLLCNVTKEYDIWVHVDAAYAGSVCIPEPRHCIDGIEELNSFSFNAHKWFL
 TNLACCCLWVKDHNAITLSTLNPEFLRNKKSDEVIDYKDWQIPLSRKFNAKLWIV
 LRSYGVENLKNFLRNHVEMAKIFEGLVRKDERFEIVVPSKFSLVCFRISPFASIANDEG
 YVVGKMMNDAYLVNEMNHKLLDLINSSGKAYMSHGEVEGSFVIRCAIGATLTEEHVT
 MTWKLQVQIASFLLGTPLN

Zea mays GRMZM2G009400 (SEQ ID NO: 79):

MAILNRADTSHTTASNGSATPAAPVAIAMPPLDDEFRQGRLLVDFIADYYA
 RIDGYVPRPAVAPGFLIRQLPEAAPARPEPDALAAALRDVRDLILPGVTHWQSPRHFAHF
 AATASNVALGALGEALAGLVNPNPTWAASPAATELEVVDWLKALHLPESLFSGGG
 GGTLTGTSCEAMLCITVAARDRKLAEVGEERIGDLVYCSQDTHFSFQKAARIAGIRRG
 NCREIPTSRSGFTLSPKALAAAVRADEAAGRVPLFLCATVGTTPTAADVPLRELCAAV
 AGHDVWVHVDAAYAGAACVCPEFSHVAVGEAESEFSTNPHKWL LANMDCCALWV
 RRPAALTAALGTDHDVILKDPAAAQAQQAQQQCSGQGVVDYKDWQVALSRRFRALKL
 WLVLRCCHGVEGLRGLVRAHVRMAAFVAMVRGDARFEVHVPRQFALVCFRLRAVAV
 AVAGEKRAGDYDGAAGNELNRRLLEAVNATGRVYMSSAVVGAYILCAIGNSLTE
 ERHVREAWSVVQEQATAILSAATATARTNGLTVRRARCDADVSDVPTPQQPLPLG

Glycine max 07G059000.1 (SEQ ID NO: 80):

MEMKNTMNRNPQSDAPIIKPLDPEEFKRQGYMMVDFLADYIRNVSHYPVLSKVEPGYL
 KQRLPTSAPCGPEPIESILKDVQDHIIPGLTHWQSPNFYGYFPSSGSIAGFMGEMLSAGLN
 VVGFNVVSSPSATELESIVMDWLGGVNLNPKSLFCGDHGGGVVLGTTCEAILCTLVAA
 REKKLSQVGKENIGKLVVYGSQDTHSALQKAAQIAGIHPANFRVIKTKRSNSFALSPDSL
 LSTILLDVERGLIPCFLCATVGTTAIATIDPIGPLCNVAKDYGIWVHVDAAYAGSACICPE
 FRHCIDGVEEVNSFSLNAHKWFLTNLTCCCLWVKDHIALTKSLTVNPQFLRNKASESKR
 VIDYKDWQIPLSRKFNAKLWLVLSYGVENIRNFLRNHVQMAKTFEGLVRLDKRFEIV
 VPPKFSLVCFRIAPSAIIANGLSKGVEACYNGKLVNDEYMVNEVNRKLLDSVNSSGDAF
 MTHGEVEGAFMIRCAIGGTLTEHHVIMAWKLVQEHANSLGL

Panicum virgatum Ca01381.1 (SEQ ID NO: 81):

MAILNHGDTTAASGTSPAAAANVAPPMHSLVQPVLDADAFRRQGRLLVDFIADYYTR

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IDEYPVRPAVAPGFLARQLPEAAPARPEPGDALAAALRDVRLILPGVTHWQSPRHFA
 HFATTGSNVGALGEALAAGLNINPFTWAASPAATELEVVTDWLGKALHLPERLLFSG
 GGGTLLGTSCEAMLCTLVAAARDKLAETIGEERMGLVVCSDQTHFSFRKAARIAGIR
 RGNCREIPTRESGFALQFRTLLAAVRADAEAGRVPMFLCATVGTPTTAAVDPLRELCA
 AVAGRGVWHVDAAYAGAACVCEFRGATAGAEAVDSFSTNPHKWL LANMDCCAL
 WVRPEALTAAAGTDHVDVILKDPSSERGGGVVDYKDWQVALSRRFRALKLWLVL RCH
 GVEGLRGLVRADARFEVPVPRQFALVCFRLRAAAAAAVGEKGRDRDND AEPNELNR
 RLLEAVNATGRAYMSSAVVGGIYVLRCAIGNSLTEERHVR EAWRVVQEQTAVLAAA
 ACTEERAVRSAR

Theobroma cacao 27425420 (SEQ ID NO: 82):
 MSSASRKTFLPLEPTSFTNESKAVIDFIADYYKNI EEPVQSGVEPGYLSAKLPDSAPYCP
 ESLEDILKDVNDCCIIPGLTHWQSPNFFAYFQANASTAGFLGEMLCSGFN VVGFNWISSPA
 ATELESIVLDWMGKLLKLPSSFLFSGTGGGVLHGSTCEAAVCTLAAARDKALKELGGW
 ENITKLMVYASDQTHFTFQKAAKLVGIPPSNFRFIETSLS TGFSSSDQVRLAIEHDIKSG
 LVPLFLCATIGTTACGAIDPIAELGQVAREYKWLHLHIDAAYAGSACI CPELRHFLDGVEL
 ANSVSMNPHKWFLTNMDCCLWITEPRLLVDSLSTDPEILRNKASEFKAVLDYKDWQV
 ALSRRFRALKLWIVIRRHGLANLVYHIRSDISMAERFEAFVAKDDRFDIVVPRKFALVCF
 RLKPKQELEGLELNSRLLEAINS SGRAPMTHAVVGGIYVIRCAIGTTMTEERHVDALWK
 LIQEKAQGLLME

Fragaria vesca 27274768 (SEQ ID NO: 83):
 MGS�DFHHVPEKTNSDPPMANPMDPEEFRRQGHIMIDFIADYYKNI EKYPVLSQVQPGY
 LKLLLPESAPYNPEPIETILQDVQDHIVPGITHWQSPSYFAYFPSSGSIAGFLGEMLSTGFN
 VVGFNWMSSPAATELERTTCEAIVCTMAAARDQMLSRIGKDNIGKLVVYGS DQTHSAL
 KKASQIVGIHPNNFRAIKTTKSTEFALSPELLRSTICSDIDKGLVPLFLCATMGTTATTSDV
 PLRGLCDVAKDYDLWVHVDAAYAGSICICEPRHFIEGVDGANSFSFNAHKWF FTTLDLC
 CCLWVKNPALINSLS TNPEFLRNKASDSKQVVDYKDWQVALSRRFRALKLWLVL RSY
 GVANLRSFLRSHVKMAEVFEKLVRENKWFVVPRNFAMVCFRISPSAIRKAPTDD DGI
 DVVINEVNSKLL EAMNTSGSVYMTHAVVGGMYVLRCAIGATMTEEKHVLMAWKCS
 ALERKDVAANETLSFNFQRRFDRRARQRRGHVGFRLAITMLDLKT SERDGARRWSIGA
 YANQITTISQANSSVAWTFMEFHS CFIFFCGSIKLDTQVPNDDFVLSARWPPSPFVSGWSTI
 NFHETIKIYVGSLSLDSWTMEFHSCTFFCGS

Gossypium raimondii 26786642 (SEQ ID NO: 84):
 MVSASRKTFLPLDPVTF SNESKAVIDFIADYYENVEKYPVQSTVEPGYLSAMLPESAPYC
 PEPLQDILEDVSNCCIIPGLTHWQSPNFFAYFPHANASTAGFFGEMLCSGFN VVGFNWISSP
 AATELESIVLDWMGKMLKLPSSFLFSGTGGGVLHGSSCEAAVCVLA AARDKALKELGG
 WENITKLVVYASDQAHFTFQKAAKLVGIPPSNFR LIETSFSSTGFSLSPENLRFVIEDNIRSG
 LVPLFLCATIGTTPSGAVDPIAELGKVAMEFKWLHLHIDAAYAGSGCICPELRHYLDGVEL
 ANSISMNPHKWFLTNMDCCLWIKPKLLVDSLSTDPEILRNNA SKKAVVDCKDWQI
 ALSRRFRALKLWVIVIRRHGLANLMCHIRSDIAMAKRFEALVGEDERFEIVVPRKFALVC
 FRLKPKVEEEDLNCKLVEAINSSGRAFM SHAVLSGIYVIRCAIGTTLTQOHHVDALWKLI
 QDKAQSLLM

Populus trichocarpa 26994989 (SEQ ID NO: 85):

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MGSLSTNTFSPLDPNNGFTNDSKMVIDFIADYYKNIENNFVQSQVKPGYLLTQLPDTAPYC
 EESLEDVLKDVDSIIPGLTHWQSPNFFAYFQANASTAGFVGEMLCTGLNVVGFNWIAS
 PAATELESIVMDWMGKMLKLPSTFLFSGNGGGVLHGSTCEAIVCTLVAAARDETLRMIGA
 ENITKLVVYASDQTHSTLLKGVKLVGIPSSNFRCLSTSFSSEFSLSPQALEDAIENDIKAGL
 VPLFLCATVGTACGAVDPVMDLGEIARKYNLWFHIDAAYAGSACICPEFRHYLDGVEL
 ADSLSMNPBKLLTNMDCCCLWVKQPRLLIESLSSDAEFLRNNAESEDVVDYKDWQI
 ALSRRFRALKLWIVIRRHGLANLMCHIRSDVNLAKRFESLVAKDSRFEVVVRRRFSLVC
 FRLKHNDQCQGLELNRKLLAAVNESGRAFMTHAVVGGFLFIIRCAIGSTLTEERHVDDLW
 KLIQEKAADLLSKKQVLLDN

Malus domestica 22679008 (SEQ ID NO: 86):
 MSLLAIFYNSNGERSKRVHLSASTYGNSTPNSYISLPYALFSSATQLINIHSNSNFQMGSLI
 SQENNSPNVPTNPLDPEEFRRQGHVDFIADYYKSIEKHPVLSQVQPGYLLKRLPDTAP
 YNPEPLETILQDVQDHIVPGITHWQSPNYFAYFPSSGSVAGFLGEMLSSGFNVVGFNWM
 SSPAATELESTVRDWFNMLKLPKSFLFSGNGGDDVIQGTTCALVCAMVAARDQKLSK
 FGRHNIGKLVVYGSQTHSALQKASQIVGIHPENFRSIETTRSTSFALSPESLKVIIYSDIEA
 GLVPLFLCATVGTTAIATVDPLGPLCGVAGDYGMMVHVDAAYAGSACICPSFDISLMA
 SRVQIHSVSTRTNGSSPLSTVVAFLRIPTRWNKATELKQVVDYKDWQIALSRRFRSMK
 LWLVLSYGVANLNRNFLRSHVKMAKIFEGLVAMDKRFEIVAPRNFSLVCFRVSPSSISN
 KASSDQNGKTDYCCDANGDENSVIINEVNRKLESINVSGHVMTGHVVGGLYMLRFA
 VGATLTEHHIALAWKVQEHADQILTKY

Citrus Clementina 20801973 (SEQ ID NO: 87):
 MRAGEASIIKMGSPGLSANNITHGSSFSADLEPKSFSDESKAVIDFIADYYKNIIEKYPVQS
 KVEPGYLSARLPDTAPHSPESLDDILKDVTDICILPGLTHWQSPNFFGYFQANASTAGFLG
 EMLCSGFNVVGFNWLASPVATELESIVMDWMGKMLKLPSSFLFSGTGGGVHLHGSTCES
 LVCTLAAARDKALEKGGGFDNITKLAVYASDQTHFALQKSAKIGIPPANFRPLRTSFS
 TEFSLSPDTVRAAIEDDIKSGHVPLYLCATVGTGAGAVDPIEELGKIANEYKLWLHIDA
 AYAGSACICPEYRHYLNGVELADSI SLNPHKWFLTNMDCCCLWVKHPSFLVDSLSTESD
 IMRNRSPASNTSTNAAPVIDYKDWQIALSRRFKALKLWTVIRKHGYSGLMYHIRSDVSM
 AKRFAAMVAKDERFEIVVPRKFALVCFRLKPKRESEGSSELNRELVDALNGSGRAFLTQA
 MLGGVYVIRCISIGTTLTQDRHVDDLWKLIQEKADRLLSLQEPEHASR

Citrus Clementina 20818150 (SEQ ID NO: 88):
 MGSLNSDHELKTNASAFNNPMDSEEFRRQGHMIIIDFIADYYRDVEKYPVLSQVEPGYLQ
 KRLPESAPYNPEPIETILQDVQDHIVPGITHWQSPYFAYFPSSGSIAGFLGEMLSSGFNV
 VGFNWMSSPAATELENIVMDWLGEMLKLPKSFLFSGTGGGVIQGTTCAILCTLAAARD
 QILNEIGRENISRLLVYGSQTHSALQKAAQIAGIDPKNFRAIKTKSSSFTLTPESLQAAI
 DLDIQSGLIPLFLCATVGTTAITVDPLGPLCDIAKRYSIWIHVDAAYAGSACICPEFRHFI
 DGIESADSFSLNAHKWFFTTLDCCCMWVKNPNALIKALSTNPEFLRNKASDSKQVVDY
 KDWQISLSRRFRALKLWLVLSFGVANLNRNFLRSHVGMAQLFQELVGGDNRFIVAPR
 NFAVVCFRVLPASGLNGKANEGANELNRKLESINASGQLYVSHGMVAGIYFIRFAV
 GATLTEDRHVIAAWKVQEKLDGILATS

Vitis vinifera 17834108 (SEQ ID NO: 89):
 MGSLSFNTFSPLDPQSFSESKMVVDIFIADYYKNVEKYPVQSQVDPGYLMHHCPDTAPY
 CPEPLETILKDVSDGIIIPGLTHWQSPNFFGYFQANASTAGFLGEMLCTGLNVVGFNWIAS

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PAATELESIAIICSLAAARDKVLKKLGHHKITKLVVYGSQDQTHSTLQKASKLVGIPASNFR
SLPTSFSNYFALCPDDVRTAMEEDIGAGLVPLFLCATVGTSSGAVDPLEALGHVAKDF
KVHHLNGVELAHSISMNPHKWLLTNMCCCLWIKPKLFVDSLSTAPEFLRNNAESK
KVIDYKDWQIALSRRFRAIKVWAVVPRRFALVCFRLRPREEGESTELNSRLLMVANGSG
AAFMTHAVVGGIYIIRCAIGSTLTETRHVDSLWKLIQEKAQLVLQEPGLALEEDYIDPCIG
VSATSLHAVVRWYCNYSSSEINAHLVFIAFFVVCKENRENYVLGVNGPPN

Petunia hybrida ABB72475.1 (SEQ ID NO: 90):
MDTIKINPEFDGQFCKTTSLLDPEEFRRNGHMMVDFLADYFHNIEKYPVRSQVEPGYLE
RLLPDSAPIQPEPIEKILKDVRSDIFPGLTHWQSPNFFAYFPCSSTAGILGEMLSAGLNVV
GFSWIASPAATELESIVMDWLKGLINLPKTYLFSGGGGVMQGTTCVLMCTIVAARDK
MLEKFGRENIDKLVVYASDQTHFSQKAVKISGIKPNFRAIPTTKATEFSLNPESLRRAI
QEDKKAGLIPLFLCTSIGTTSTTAVDPLKPLCEIAEEYGIWVHVDAAAYAGSACICEFHQHF
LDGVEHANSFSFNAHKWLFITLDCCCLWLKDPSSLTKALSTNPEVLRNDATDSEQVVD
YKDWQITLSRRFRSLKWLVLKSYGVANLRFIRSHIEMAKHFEELVAMDERFEIMAPR
NFSLVCFRVSLLALEKKNFVDETQVNEFNALLESIISSGNVYMTHTVVEGVYMRFAV
GAPLTDYPHIDMAWNVVRNATMMLNA

Carica papaya 16421889 (SEQ ID NO: 91):
MSSLSRDLNASPLEPENFRVESKRVIDFIADYKNIETYPVQSRVKPGYLAGRLPSSAPFS
PESLETILQDIAENISPGLTHWQSPNFFGYFQANASTAGFHGEMLCSGLNVVGFNWISSP
AATELESIVMDWMGNMLKLPSFLFSGSGGVLHGSTCEAVVCTLAAARDKTLNQLG
GNYQNI TKFVVYASDQTHFTLQKAAKLGIPPSNFRSLTTSFPSPGFSLSPEKLQSAIKDDIK
SGYVPLYVCATVGTTAAGAVDPIELGKVAQEYNLWFHIDAAYAGSACICTEFRHYLN
GVELADSI STNPHKWLLTNMECSCLWVKSPSSLVDSLSTKSEIMRNAATDSNQVIDYKD
WQIALSRRFRALKLWIVIRRHGLSGLTSHIHKDIKMAELFESLVAKDKRFEIVVPRKFAL
VCFRFPKEKENQDSELNSKLLNAVNSSGCAFMTHAVLEGVYTIRCAIGTTLTEEHVHV
NLWKLIQEKAQSLIINEY

Sphagnum fallax 0042s0024.1 (SEQ ID NO: 92):
MSSKVAPWRLSKPLDVEEFRTHAHRMVDFIADYHHNIQSPFVHSQKPGYLRPLLPDT
APTEPEVVEDVFADVWNKILPGITHWQSPKFFGYYPFNVSTAGILGEILSGGVNVTGFSW
ITSPVVTELEIIVDLWLGLLHLPPEEFLSSGKGGVIGQTSSEAVVCTSQHMEAEALTKL
VVTSDQAQSCVLRACQIAGIATANFRPLPTDASSHFSLSPAVLIKAAATDVAAGLFPFF
LCGKVGTTSSSAVDPLLELGDIAKRYGMWYHIDAAYAGSACICEFRHYLNGVEKADS
YMNPHDWMLTNFD CSTLWVKNSSELLVAALSNKPVYLQNEATDNNLVDCSHIRNHISI
AKHFESLVRADFRFEMIVPTNFSLVCFRLRTPAGSKDNSRTLNSKLVEALNRKGDILVTH
TELSGRYTTLRFAVGGTHMELHHVQAAWNRLQRQVF

Eucalyptus grandis E01788.1 (SEQ ID NO: 93):
MNPLDPGEFRRQGHMVVDFLAKYYENIEKYPVLSQVEPGYLSKRLPSSAPQDEEPMET
LDDVHQHIFPGLTHWQSPNFFAYYQNTSTAAILGEMLCAGFNVAGFNWVSSPAATELE
SLVMDWLKMLDLPRPFLPFGNGGVI EGN TSEAI ICTLTAARDVLRKLGHNSIAKL
VYGSQDQTNCSFQKAARVVGIDPRNFRALKMTRSTLFLGLSPDSLEKAI RLDINAGLIPLYL
CATVGTTS CAAVDPLEPLCKVASKFSMWIHVDAAYAGASCICEYRK FINGVEFADSF
FNAHKWLLTPLDCCCLWVKDPNALVKSLSLTDPEYLNKNEATESKQVIDYADWQLSLSR

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FRALKLWLVLRSHGVLNLRSHIKNHCRLAKLFEEELVEEDPQFEVVFPRNFALVCFRIHPS

GVAGMLNAQLLHAINASGRVMSHTTVGGVYVLRFAVGATLVTEKHVIMAWKVQQE

HANSLLSMPASEQHS

pHis8-4 (SEQ ID NO: 94):

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CGCGCAGCGTGACCGCTACACTTGCCAGCGCCCTAGCGCCCGCTCCTTTTCGCTTTCTT

CCCTTCCTTTCTCGCCACGTTCCCGGCTTTCCCGTCAAGCTCTAAATCGGGGGCTC

CCTTTAGGTTCCGATTTAGTGCTTTACGGCACCTCGACCCCAAAAACCTTGATTAG

GGTGATGGTTCACGTAGTGGGCCATCGCCCTGATAGACGGTTTTTCGCCCTTTGACG

TTGGAGTCCACGTTCTTTAATAGTGGACTCTTGTTCCAACTGGAACAACACTCAAC

CCTATCTCGGTCTATTCTTTTGATTATAAGGGATTTTGCCGATTTTCGGCCTATTGGTT

AAAAAATGAGCTGATTTAACAAAAATTTAACGCGAATTTTAACAAAATATTAACGTT

TACAATTTCAGGTGGCACTTTTCGGGAAATGTGCGCGGAACCCCTATTGTTTATTT

TTCTAAATACATTCAAATATGTATCCGCTCATGAATTAATCTTAGAAAACTCATC

GAGCATCAAATGAACTGCAATTTATTCATATCAGGATTATCAATACCATATTTTGTG

AAAAAGCCGTTTCTGTAATGAAGGAGAAAACACCGAGGCAGTTCATAGGATGG

CAAGATCCTGGTATCGGTCTGCGATTCCGACTCGTCCAAATCAATACAACCTATTA

ATTTCCCTCGTCAAAAATAAGGTTATCAAGTGAGAAATCACCATGAGTGACGACTG

AATCCGGTGAGAATGGCAAAAGTTTATGCATTTCTTTCCAGACTTGTTCAACAGGCC

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TGAATCAGGATATTCTTCTAATACCTGGAATGCTGTTTCCCGGGATCGCAGTGGT

GAGTAACCATGCATCATCAGGAGTACGGATAAAATGCTTGATGGTCGGAAGAGGCA

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TACGGTTCTCTGGCCTTTTGCTGGCCTTTTGCTCACAATGTTCTTTCTCGCTTATCCCCT
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CGGGCTTGTCTGCTCCCGGCATCCGCTTACAGACAAGCTGTGACCGTCTCCGGGAGC
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CTTGAGCGAGGGCGTGCAAGATTCCGAATACCGCAAGCGACAGGCCGATCATCGTC
GCGCTCCAGCGAAAGCGGTCTCGCCGAAAATGACCCAGAGCGCTGCCGGCACCTG
TCCTACGAGTTGCATGATAAAGAAGACAGTCATAAGTGCGGCGACGATAGTCATGC
CCCCGCCCCACCGGAAGGAGCTGACTGGGTTGAAGGCTCTCAAGGCATCGGTGCA
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GCTTTCAGTCGGGAAACCTGTCGTGCCAGCTGCATTATGAATCGGCCAACGCGCG
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GGGCAACAGCTGATTGCCCTTACCGCCTGGCCCTGAGAGAGTTGAGCAAGCGGT
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CAGTCGCGTACCGTCTTCATGGGAGAAAATAATACTGTTGATGGGTGCTGGTCAGA
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CAGTTGTTGTGCCACGCGGTTGGGAATGTAATTCAGCTCCGCCATCGCCGCTTCCAC
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AAAACCTTGTACTTCCAGGCCCATGGCGGATCCGAATTCGAGCTCCGTGACAAGCTT
GCGGCCGCACTCGAGCACCACCACCACCACCACCTGAGATCCGGCTGCTAACAAAGC
CCGAAAGGAAGCTGAGTTGGCTGCTGCCACCGCTGAGCAATAACTAGCATAACCCC
TTGGGGCCTCTAAACGGGTCTTGAGGGGTTTTTTGCTGAAAGGAGGAACATATACCG
GAT

pEAQ-HT (SEQ ID NO: 95):

CCTGTGGTTGGCATGCACATACAAATGGACGAACGGATAAACCTTTTCACGCCCTTT
TAAATATCCGATTATTTCTAATAAACGCTCTTTTCTCTTAGGTTTACCCGCCAATATAT
CCTGTCAAACACTGATAGTTTGTGAACCATCACCCAAATCAAGTTTTTTGGGGTCGA
GGTGCCGTAAAGCACTAAATCGGAACCCCTAAAGGGAGCCCCGATTTAGAGCTTGA
CGGGGAAAGCCGGCGAACGTGGCGAGAAAGGAAGGAAGAAAGCGAAAGGAGCG
GGCGCCATTAGGCTGCGCAACTGTTGGGAAGGGCGATCGGTGCGGGCCTCTTCGCT
ATTACGCCAGCTGGCGAAAGGGGGATGTGCTGCAAGGCGATTAAGTTGGGTAACGC
CAGGGTTTTTCCAGTCACGACGTTGTAAAACGACGGCCAGTGAATTGTTAATTAAGA
ATTGAGCTCCACCGCGAAACCTCCTCGGATTCCATTGCCAGCTATCTGTCACTTT
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GTGGATTGATGTGATATCTCCACTGACGTAAGGGATGACGCACAATCCCCTATCCT
TCGCAAGACCCTTCCTCTATATAAGGAAGTTCAATTCATTTGGAGAGGTATTAAAT
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CCTATTCTGTGCGTGTGGTCTTGGGAAAAGAAAGCTTGCTGGAGGCTGCTGTTTCA
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TCACCATCATCCCGGCATCACCATCACCATCACTAGCTCGAGGCCTTTAACTCTGG
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CGAGGAGCATCGTGGAAAAGAAGACGTTCCAACCACGCTTCAAAGCAAGTGGAT
TGATGTGACATCTCCACTGACGTAAGGGATGACGCACAATCCCACTATCCTTCGCAA
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TCGACCAGATCGGATGTACCTATAGTATTCGGTTTCGAGGAGTTAGTATCACCGTTT
CTGGAGGGTCTCGAACTCTTCAGCATCTCTGTGAGATGGCAATTCGGTCTAAGCAAG
AACTGCTACAGCTTGCCCCAATCGAAGTGGAAGTAATGTATCAAGAGGATGCCCT
GAAGGTACTGAGACCTTCGAAAAAGAAAGCGAGTAAGGGGAGCTCGAATTCGCTGA
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GCATATAAGAAACCTTAGTATGTATTTGTATTTGTAAAATACTTCTATCAATAAAA
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INCORPORATION BY REFERENCE AND EQUIVALENTS

The teachings of all patents, published applications and references cited herein are incorporated by reference in their entirety.

While example embodiments have been particularly shown and described, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the embodiments encompassed by the appended claims.

SEQUENCE LISTING

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Ala	Met	Glu	Ser	Leu	Lys	Thr	His	Lys	Pro	Tyr	Lys	Ser	Val	Ser	Thr
145					150					155					160
Asp	Ser	Glu	Pro	Phe	Leu	Ile	Pro	Asn	Leu	Pro	Asp	Glu	Ile	Lys	Met
				165					170					175	
Thr	Lys	Ser	Gln	Phe	Thr	Val	Asp	Ala	Trp	Glu	Asp	Thr	Glu	Lys	Gly
			180					185					190		
Leu	Gly	Lys	Leu	Leu	Ala	Asp	Ala	Arg	Ala	Ser	Gly	Leu	Arg	Ser	Phe
		195					200					205			
Gly	Met	Ile	Val	Asn	Ser	Phe	His	Glu	Leu	Glu	Pro	Ala	Tyr	Ala	Asp
		210				215					220				
Tyr	Tyr	Lys	Asn	Val	Leu	Asn	Met	Lys	Ala	Trp	Cys	Val	Gly	Pro	Val
225					230					235					240
Ser	Leu	Tyr	Asn	Arg	Asn	Asp	Asp	Glu	Lys	Ile	Ala	Arg	Gly	Lys	Lys
			245						250					255	
Ser	Ala	Ile	Asp	Asp	His	Glu	Cys	Leu	Lys	Trp	Leu	Glu	Gly	Lys	Gln
			260				265						270		
Pro	Asp	Ser	Val	Val	Tyr	Val	Cys	Phe	Gly	Ser	Ser	Ala	Ser	Phe	Pro
		275					280					285			
Asp	Glu	Gln	Leu	Arg	Asp	Ile	Ala	Leu	Gly	Leu	Glu	Glu	Ser	Gly	Val
		290				295					300				
Asn	Phe	Ile	Trp	Val	Ile	Arg	Arg	Ser	Ser	Glu	Ser	Gly	Ser	Glu	Asp
305					310					315					320
Tyr	Leu	Pro	Glu	Gly	Phe	Glu	Asp	Arg	Val	Lys	Asp	Arg	Gly	Leu	Val
			325						330					335	
Ile	Arg	Gly	Trp	Ala	Pro	Gln	Val	Leu	Ile	Leu	Asp	His	Pro	Ser	Val
			340				345						350		
Gly	Gly	Phe	Val	Thr	His	Cys	Gly	Trp	Asn	Ser	Ala	Leu	Glu	Gly	Ile
		355					360					365			
Ser	Ala	Gly	Leu	Pro	Met	Val	Thr	Trp	Pro	Leu	Phe	Ala	Glu	Gln	Phe
		370				375					380				
Phe	Asn	Gln	Lys	Leu											

-continued

Val Ala Lys Asp Gly Ser Ser Tyr Ile Asp Leu His Asn Leu Leu Asp
450 455 460

Glu Leu Lys Leu Arg Arg Glu Thr Leu Ser
465 470

<210> SEQ ID NO 7
<211> LENGTH: 1419
<212> TYPE: DNA
<213> ORGANISM: *Rhodiola rosea*

<400> SEQUENCE: 7

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atgtcaggca caccacacat cgccatcctc cccagccccg gcatgggcca cctcatcccc    60
atggccgagt tcgccaagcg cctagtccac caccacaact tcagtatcac cttcgtcacc    120
cctaccgacg gccacacctc ctcgccttac caacaagtcc tcacctccct cccatcttcc    180
atagatcaca tcttccctcc acaagtgcac ttaaccgacg tcgtatcaca atcaccagct    240
catcccagaa tcgaaacctc aatctccctc accgtcgtc gctccctctc ctcctccgcg    300
accaccttat cctctctcca atcgtctaaa aacctcgtct cgctcgtcgt tgatcttttc    360
ggcactgatg cattcgaccc ggccatcgag ctcggcatct cgccctacat tttcttcctc    420
tccacagcca tgacgtctc gctcttctta tacatgcctc agcttgacaa atcagtcacg    480
tgcgaaatttc gtcacatgac ggatttggtt cgaattcctg gatgcgttcc tgtccgtgga    540
tcggatttat tcgacccggt tcaagacagg accgacgagg cttataaatg ggtcatacat    600
cactccaaca ggtacctat ggccggagggt gttatagaga atagcttcat ggagtggaa    660
catggtgcgt taaagtattt gcaaacggtt caatcgggta agccgcctgt ctacgcggtc    720
ggaccgttga ttaaaatgga ttatgatgtt gacgattccg ggtcgaagat aatcgagtgg    780
ctcgatgac aaccggttgg ttcggtttta tttgtttcgt ttggaagcgg cggaacgctc    840
tcgtatgagc aaatgaccga gctggctcac ggtttggaat cgagccagca acggttctta    900
tgggtgggtc ggagtcggaa tcaaatcccc aacagcacgt atttcagtg acaaagccaa    960
aaagaccctg tggttactt gccagaagga tttttaaac gaaccgagg taggggtctg    1020
gtcgtatcga attgggcccc acaggctcaa attttgagtc acggttcgac cgttgggttc    1080
atgagccact gtggttgga ttcgattttg gagagtgtgg tgcacggcgt gccgatcata    1140
gcgtggccgt tgtacgccga gcagaagatg aattcgataa tcgtggtgga ggacgttaag    1200
gtggcgctga ggccggcggg ggtaggggag aggggtggtg agaggtcgga gataaccgca    1260
gtggtgaagg cgttgatgga gggtagaggag gggaagaagg taaggaaatag gatgaaggaa    1320
ctcaagggaag cggcggcacg tgcggttagt gatgacggtg cgtcgaccat agcgattgcg    1380
gacttggcgc aaaaatggcg gagttcgatg aagcattga                                1419

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<210> SEQ ID NO 8
<211> LENGTH: 472
<212> TYPE: PRT
<213> ORGANISM: *Rhodiola rosea*
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: GenBank MF674528

<400> SEQUENCE: 8

Met Ser Gly Thr Pro His Ile Ala Ile Leu Pro Ser Pro Gly Met Gly
1 5 10 15
His Leu Ile Pro Met Ala Glu Phe Ala Lys Arg Leu Val His His
20 25 30
Asn Phe Ser Ile Thr Phe Val Ile Pro Thr Asp Gly Pro Pro Ser Ser

-continued

35	40	45
Ala Tyr Gln Gln Val Leu Thr Ser Leu Pro Ser Ser Ile Asp His Ile		
50	55	60
Phe Leu Pro Gln Val Asp Leu Thr Asp Val Val Ser Gln Ser Pro Ala		
65	70	75 80
His Pro Arg Ile Glu Thr Leu Ile Ser Leu Thr Val Ala Arg Ser Leu		
	85	90 95
Ser Ser Leu Arg Thr Thr Leu Ser Ser Leu Gln Ser Ser Lys Asn Leu		
	100	105 110
Val Ser Leu Val Val Asp Leu Phe Gly Thr Asp Ala Phe Asp Pro Ala		
	115	120 125
Ile Glu Leu Gly Ile Ser Pro Tyr Ile Phe Phe Pro Ser Thr Ala Met		
	130	135 140
Thr Leu Ser Leu Phe Leu Tyr Met Pro Gln Leu Asp Lys Ser Val Thr		
145	150	155 160
Cys Glu Phe Arg His Met Thr Asp Leu Val Arg Ile Pro Gly Cys Val		
	165	170 175
Pro Val Arg Gly Ser Asp Leu Phe Asp Pro Val Gln Asp Arg Thr Asp		
	180	185 190
Glu Ala Tyr Lys Trp Val Ile His His Ser Asn Arg Tyr Pro Met Ala		
	195	200 205
Glu Gly Val Ile Glu Asn Ser Phe Met Glu Leu Glu His Gly Ala Leu		
	210	215 220
Lys Tyr Leu Gln Thr Val Gln Ser Gly Lys Pro Pro Val Tyr Ala Val		
225	230	235 240
Gly Pro Leu Ile Lys Met Asp Tyr Asp Val Asp Asp Ser Gly Ser Lys		
	245	250 255
Ile Ile Glu Trp Leu Asp Asp Gln Pro Val Gly Ser Val Leu Phe Val		
	260	265 270
Ser Phe Gly Ser Gly Gly Thr Leu Ser Tyr Glu Gln Met Thr Glu Leu		
	275	280 285
Ala His Gly Leu Glu Ser Ser Gln Gln Arg Phe Leu Trp Val Val Arg		
	290	295 300
Ser Pro Asn Gln Ile Pro Asn Ser Thr Tyr Phe Ser Val Gln Ser Gln		
305	310	315 320
Lys Asp Pro Leu Ala Tyr Leu Pro Glu Gly Phe Leu Asn Arg Thr Glu		
	325	330 335
Gly Arg Gly Leu Val Val Ser Asn Trp Ala Pro Gln Ala Gln Ile Leu		
	340	345 350
Ser His Gly Ser Thr Gly Gly Phe Met Ser His Cys Gly Trp Asn Ser		
	355	360 365
Ile Leu Glu Ser Val Val His Gly Val Pro Ile Ile Ala Trp Pro Leu		
	370	375 380
Tyr Ala Glu Gln Lys Met Asn Ser Ile Ile Val Val Glu Asp Val Lys		
385	390	395 400
Val Ala Leu Arg Pro Ala Gly Val Gly Glu Arg Val Val Glu Arg Ser		
	405	410 415
Glu Ile Thr Ala Val Val Lys Ala Leu Met Glu Gly Glu Glu Gly Lys		
	420	425 430
Lys Val Arg Asn Arg Met Lys Glu Leu Lys Glu Ala Ala Ala Arg Ala		
	435	440 445
Val Ser Asp Asp Gly Ala Ser Thr Ile Ala Ile Ala Asp Leu Ala Gln		
	450	455 460

-continued

Lys Trp Arg Ser Ser Met Lys His
465 470

<210> SEQ ID NO 9
<211> LENGTH: 1431
<212> TYPE: DNA
<213> ORGANISM: *Rhodiola rosea*

<400> SEQUENCE: 9

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atggctgaaa acactcatgc tcatgccata gtggtaccat ttccagttca aggacacata    60
aagccctcgc tgaatctagc cctcaagcta gcatctcaag gcttcaccat cacttttgtc    120
accactcatt tcaccacca gcaaatctcc caagctcaca aaaacagtac aaatacaaac    180
catgacatgt ttttcaggc acgaaactcc agtctcgata tccgccatgt aacggtgaca    240
gacacttttc ctttgggatt cgatcgcgca gggaaatcagg atcagttttg ggagggcatg    300
cttcacgtat tcctgcaca tgttgatgaa ctggtggatc agttaatgaa ttcttcgaag    360
ccgagaccaa cttgtttgat tctggataca ttttataact ggggttccaa aattgctaac    420
aagtttaatt tagtgcatat ttcatTTTgg actcagctctg ctctttcttt cactttgttt    480
taccattggg aacttttaaa gaaaaatggt cactttggct ctccagataa tcgcacggat    540
gtcatcgatt atattcccg tgtgcaagag atcaagcccg cagacttaat atcctacctt    600
cagatgagtg atacaactac tgtggctcac aggacttgtt tcacagcatt tgaagatgtc    660
aggaaggcag atttcactct ggctaataca atccaagaat ttgaaactga tacaatttct    720
tctatccgat ttcaccagcc atttttctac ccaattggac ctgttttttt aacaaagtct    780
gaacaacaag ctagctcagc tttgtggtct gagtcagact gtgagcagtg gctaagtaca    840
aaacaaaag ggtctgttct ctatgcctca tttgggagct atgctcgtgt aactaggcat    900
gatatcgag agatagccta cggattgatg caaagtgagg tgaattttat ttgggtgatt    960
cgcgacgata ttgtgggtgc acacgagact gattttttac caacagaatt cataaatgga   1020
atcaaaactca aagatcaggg actactagtt tcttgggtgt ctcaaactga agttttgtcc   1080
aatgcggcga ttggaggatt tctgactcat tgtggatgga actcgatact cgaaagcgta   1140
tgggtgtgaag ttccattatt gtgttttcca ataatgactg atcagcctag taacaggaaa   1200
ctggtggtgg atgactggag gatcggcgtc aacctatctg cggcggagga ggtcagtaga   1260
gaagaagtg caatgaaggt caggaacttg atttctggag aattggggaa tgagttgaga   1320
gtgcagattc aaaagtacaa aaagttgatg gagaatggt taatggaagg tggatcatca   1380
cattccaatt ggaacaagtt catccacgac ctacaaatct tcaagaaatg a           1431

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<210> SEQ ID NO 10
<211> LENGTH: 476
<212> TYPE: PRT
<213> ORGANISM: *Rhodiola rosea*
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: GenBank MF674532

<400> SEQUENCE: 10

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Met Ala Glu Asn Thr His Ala His Ala Ile Val Val Pro Phe Pro Val
1           5           10          15

Gln Gly His Ile Lys Pro Ser Leu Asn Leu Ala Leu Lys Leu Ala Ser
          20          25          30

Gln Gly Phe Thr Ile Thr Phe Val Thr Thr His Phe Thr His Gln Gln
          35          40          45

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Ile 50	Ser	Gln	Ala	His	Lys	Asn 55	Ser	Thr	Asn	Thr	Asn 60	His	Asp	Met	Phe
Phe 65	Gln	Ala	Arg	Asn	Ser 70	Ser	Leu	Asp	Ile	Arg 75	His	Val	Thr	Val	Thr 80
Asp	Thr	Phe	Pro	Leu 85	Gly	Phe	Asp	Arg	Ala 90	Gly	Asn	Gln	Asp	Gln 95	Phe
Trp	Glu	Gly	Met 100	Leu	His	Val	Phe	Pro 105	Ala	His	Val	Asp	Glu 110	Leu	Val
Asp	Gln	Leu 115	Met	Asn	Ser	Ser	Lys 120	Pro	Arg	Pro	Thr	Cys 125	Leu	Ile	Leu
Asp	Thr 130	Phe	Tyr	Asn	Trp	Gly 135	Ser	Lys	Ile	Ala	Asn 140	Lys	Phe	Asn	Leu
Val 145	His	Ile	Ser	Phe	Trp 150	Thr	Gln	Ser	Ala	Leu 155	Ser	Phe	Thr	Leu	Phe 160
Tyr	His	Trp	Glu	Leu 165	Leu	Lys	Lys	Asn	Gly 170	His	Phe	Gly	Ser	Pro 175	Asp
Asn	Arg	Thr	Asp 180	Val	Ile	Asp	Tyr	Ile	Pro	Gly	Val	Gln	Glu 190	Ile	Lys
Pro	Ala 195	Asp	Leu	Ile	Ser	Tyr	Leu 200	Gln	Met	Ser	Asp	Thr 205	Thr	Thr	Val
Ala 210	His	Arg	Thr	Cys	Phe	Thr 215	Ala	Phe	Glu	Asp	Val 220	Arg	Lys	Ala	Asp
Phe 225	Ile	Leu	Ala	Asn 230	Thr	Ile	Gln	Glu	Phe	Glu 235	Thr	Asp	Thr	Ile	Ser 240
Ser	Ile	Arg	Phe	His 245	Gln	Pro	Phe	Phe	Tyr 250	Pro	Ile	Gly	Pro	Val 255	Phe
Leu	Thr	Lys	Ser 260	Glu	Gln	Gln	Ala	Ser 265	Ser	Ala	Leu	Trp 270	Ser	Glu	Ser
Asp	Cys 275	Glu	Gln	Trp	Leu	Ser	Thr 280	Lys	Pro	Lys	Gly	Ser 285	Val	Leu	Tyr
Ala 290	Ser	Phe	Gly	Ser	Tyr	Ala 295	Arg	Val	Thr	Arg	His 300	Asp	Ile	Ala	Glu
Ile 305	Ala	Tyr	Gly	Leu 310	Met	Gln	Ser	Glu	Val	Asn 315	Phe	Ile	Trp	Val	Ile 320
Arg	Asp	Asp	Ile	Val 325	Gly	Ala	His	Glu	Thr 330	Asp	Phe	Leu	Pro	Thr 335	Glu
Phe	Ile	Asn	Gly 340	Ile	Lys	Leu	Lys	Asp 345	Gln	Gly	Leu	Leu 350	Val	Ser	Trp
Cys 355	Ser	Gln	Thr	Glu	Val	Leu	Ser 360	Asn	Ala	Ala	Ile	Gly 365	Gly	Phe	Leu
Thr 370	His	Cys	Gly	Trp	Asn	Ser 375	Ile	Leu	Glu	Ser	Val 380	Trp	Cys	Glu	Val
Pro 385	Leu	Leu	Cys	Phe 390	Pro	Ile	Met	Thr	Asp	Gln 395	Pro	Ser	Asn	Arg	Lys 400
Leu	Val	Val	Asp	Asp 405	Trp	Arg	Ile	Gly	Val 410	Asn	Leu	Ser	Ala	Ala	Glu
Glu	Val	Ser	Arg	Glu 420	Glu	Val	Ser	Met	Lys 425	Val	Arg	Asn	Leu	Ile	Ser
Gly	Glu 435	Leu	Gly	Asn	Glu	Leu	Arg 440	Val	Gln	Ile	Gln	Lys 445	Tyr	Lys	Lys
Leu 450	Met	Glu	Asn	Gly	Ile	Met 455	Glu	Gly	Gly	Ser	Ser 460	His	Ser	Asn	Trp
Asn	Lys	Phe	Ile	His	Asp	Leu	Gln	Ile	Phe	Lys	Lys				

-continued

465	470	475	
<210> SEQ ID NO 11			
<211> LENGTH: 1458			
<212> TYPE: DNA			
<213> ORGANISM: <i>Rhodiola rosea</i>			
<400> SEQUENCE: 11			
atggcagaaa taagtctcat cttcatccct tttcccgtaa tcagccatct cactcccaca			60
atcgaaatcg ccaaaatcct cctcagcaga gaccaccgcc tttccatcac cttectcgtc			120
atcgacatcc cccaacgaga cgctcactc gcctccctca ccacctccat catctccgat			180
cgctccact tctcgatgt cgtacttctt cccaaccaac actcccaatc atccaagcca			240
tcaggcatcg cggctatcga gtcgcgcaaa cccgcagtca agaaaacgat cagcgatctt			300
gttgtacgat ctcagtcgcg cgcacttgtt ccgcggatag ctggcttcgt gctggacatg			360
ttctgcacgg ccatgatcga catcgcaact gagtttaacc ttcttctgta tattttactac			420
acttgcggtt ctctgtttct ttcaatcgtg ctccacgtcc agaagctctg cgatgaacgac			480
gctctcgata tcgccgatct caaaaactcg agtgtggagt tttcgttacc tgagttttca			540
aaattgattc cggctaggtt gcttccatcc atggcgctcg ataaggactt ctcggtttca			600
ttcgtcggca aagctagagc gttcaggaag acgaagggca ttttgggtcaa ctcgcttgta			660
gagttggagc ctcacgcaat cgagtcgatg aaattagacc ggtctgttcc tccgatttac			720
tcggtcggac cagtgtctaa catgaatagc aacactgcat ttatcagaca ggagcaggag			780
aaggagatca tggagtggct ggaccaacag cctccagcat ctgtagtttt cttgtgtttt			840
ggcagcaggg gagcgttcaa gccggaccag gtgaaggaaa tcgcacgggg gttggagtcg			900
agcggctgcc ggttctcttg ggcgcttcgg cagccttcat caagcaatgt gaggttttca			960
cctcctacag attatgaaga tttctctgag gttctgcctg aagggttttt gcagcggaca			1020
tatggtgttg ggaaagtgat tggttgggca cccagacag ctgtttttaga ccacccttcg			1080
gtgggtggat tcgtatcgca ttgcggttg aactcgatac tggaatctct ttggtttggt			1140
gtgccgattg cgacttggtc tctgtatgct gagcagcaga tgaatgcgtt tgaggttgtg			1200
aaggagatga agattggagt ggagataagt ttggattatc ggcttgaaat gggcggtaaa			1260
caagcagaag gttctgggat tataagtgtt gaacagattg agagagggat tagagatgtg			1320
atgcaggagg atagtgaagt gaggaagaag gtgaagctga tgatggaaaa gagtagagag			1380
gcagttgtgg agggaggctc ctcttataat tatatccaaa acttcacag tgatctcagg			1440
accaacattg gcttgtaa			1458

<210> SEQ ID NO 12
 <211> LENGTH: 485
 <212> TYPE: PRT
 <213> ORGANISM: *Rhodiola rosea*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: GenBank MF674538

<400> SEQUENCE: 12

Met Ala Glu Ile Ser Leu Ile Phe Ile Pro Phe Pro Val Ile Ser His
 1 5 10 15

Leu Thr Pro Thr Ile Glu Ile Ala Lys Ile Leu Leu Ser Arg Asp His
 20 25 30

Arg Leu Ser Ile Thr Phe Leu Val Ile Asp Ile Pro Gln Arg Asp Ala
 35 40 45

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Ser	Leu	Ala	Ser	Leu	Thr	Thr	Ser	Ile	Ile	Ser	Asp	Arg	Leu	His	Phe
50						55				60					
Leu	Asp	Val	Val	Leu	Pro	Pro	Asn	Gln	His	Ser	Gln	Ser	Ser	Lys	Pro
65					70					75					80
Ser	Gly	Ile	Ala	Ala	Ile	Glu	Ser	Ala	Lys	Pro	Ala	Val	Lys	Lys	Thr
			85						90					95	
Ile	Ser	Asp	Leu	Val	Val	Arg	Ser	Gln	Ser	Ala	Ala	Ser	Gly	Pro	Arg
			100					105					110		
Ile	Ala	Gly	Phe	Val	Leu	Asp	Met	Phe	Cys	Thr	Ala	Met	Ile	Asp	Ile
		115					120					125			
Ala	Thr	Glu	Phe	Asn	Leu	Pro	Ser	Tyr	Ile	Tyr	Tyr	Thr	Cys	Gly	Ser
	130					135					140				
Ser	Phe	Leu	Ser	Ile	Val	Leu	His	Val	Gln	Lys	Leu	Cys	Asp	Asp	Asp
145					150					155					160
Ala	Leu	Asp	Ile	Ala	Asp	Phe	Lys	Asn	Ser	Ser	Val	Glu	Phe	Ser	Leu
			165						170					175	
Pro	Glu	Phe	Ser	Asn	Leu	Ile	Pro	Ala	Arg	Leu	Leu	Pro	Ser	Met	Ala
			180					185					190		
Leu	Asp	Lys	Asp	Phe	Ser	Ala	Ser	Phe	Val	Gly	Lys	Ala	Arg	Ala	Phe
		195					200					205			
Arg	Lys	Thr	Lys	Gly	Ile	Leu	Val	Asn	Ser	Leu	Val	Glu	Leu	Glu	Pro
	210					215					220				
His	Ala	Ile	Glu	Ser	Met	Lys	Leu	Asp	Arg	Ser	Val	Pro	Pro	Ile	Tyr
225					230					235					240
Ser	Val	Gly	Pro	Val	Leu	Asn	Met	Asn	Ser	Asn	Thr	Ala	Phe	Ile	Arg
			245						250					255	
Gln	Glu	Gln	Glu	Lys	Glu	Ile	Met	Glu	Trp	Leu	Asp	Gln	Gln	Pro	Pro
			260					265					270		
Ala	Ser	Val	Val	Phe	Leu	Cys	Phe	Gly	Ser	Arg	Gly	Ala	Phe	Lys	Pro
		275					280					285			
Asp	Gln	Val	Lys	Glu	Ile	Ala	Arg	Gly	Leu	Glu	Ser	Ser	Gly	Cys	Arg
	290					295					300				
Phe	Leu	Trp	Ala	Leu	Arg	Gln	Pro	Ser	Ser	Ser	Asn	Val	Arg	Phe	Ser
305					310					315					320
Pro	Pro	Thr	Asp	Tyr	Glu	Asp	Phe	Ser	Glu	Val	Leu	Pro	Glu	Gly	Phe
			325						330					335	
Leu	Gln	Arg	Thr	Tyr	Gly	Val	Gly	Lys	Val	Ile	Gly	Trp	Ala	Pro	Gln
			340					345					350		
Thr	Ala	Val	Leu	Asp	His	Pro	Ser	Val	Gly	Gly	Phe	Val	Ser	His	Cys
		355					360					365			
Gly	Trp	Asn	Ser	Ile	Leu	Glu	Ser	Leu	Trp	Phe	Gly	Val	Pro	Ile	Ala
	370					375					380				
Thr	Trp	Pro	Leu	Tyr	Ala	Glu	Gln	Gln	Met	Asn	Ala	Phe	Glu	Val	Val
385					390					395					400
Lys	Glu	Met	Lys	Ile	Gly	Val	Glu	Ile	Ser	Leu	Asp	Tyr	Arg	Leu	Glu
			405						410					415	
Met	Gly	Gly	Lys	Gln	Ala	Glu	Gly	Ser	Gly	Ile	Ile	Ser	Gly	Glu	Gln
			420					425					430		
Ile	Glu	Arg	Gly	Ile	Arg	Asp	Val	Met	Gln	Glu	Asp	Ser	Glu	Val	Arg
		435					440					445			
Lys	Lys	Val	Lys	Leu	Met	Met	Glu	Lys	Ser	Arg	Glu	Ala	Val	Val	Glu
	450					455					460				
Gly	Gly	Ser	Ser	Tyr	Asn	Tyr	Ile	Gln	Asn	Phe	Ile	Ser	Asp	Leu	Arg

Met Gly Ser Leu Pro Ser Thr Lys Ser His Ala Val Leu Val Pro Tyr
1 5 10 15

Pro Ala Gln Gly His Ile Asn Pro Phe Met Gln Leu Ala Lys Leu Leu
20 25 30

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His	Ser	Lys	Gly	Phe	His	Ile	Thr	Phe	Val	Asn	Asn	Asp	His	Asn	His
		35					40					45			
Arg	Arg	Leu	Leu	Arg	Thr	Lys	Gly	His	Asp	Phe	Val	Gln	Gly	Leu	Glu
	50					55					60				
Gly	Leu	Arg	Phe	Glu	Ala	Val	Pro	Asp	Gly	Leu	Pro	Pro	Ser	Asp	Arg
65					70					75					80
Asp	Ala	Thr	Gln	Asp	Val	Pro	Lys	Leu	Thr	Glu	Ser	Ile	Tyr	Asn	Lys
			85						90					95	
Ser	Met	Asn	Gln	Pro	Phe	Ser	Asp	Leu	Leu	Gln	Arg	Leu	Asn	Ser	Thr
			100					105					110		
Pro	Gly	Ser	Pro	Pro	Val	Thr	Cys	Val	Ile	Ser	Asp	Val	Ala	Met	Phe
		115					120					125			
Phe	Ala	Trp	Asp	Val	Ala	Asp	Glu	Leu	Gly	Ile	Pro	Asn	Val	Gln	Phe
	130					135					140				
Trp	Thr	Ala	Ser	Ala	Cys	Gly	Leu	Leu	Gly	Tyr	Leu	Gln	Tyr	Asp	Glu
145					150					155					160
Leu	Leu	Arg	Arg	Ala	Ile	Val	Pro	Phe	Lys	Asp	Glu	Asn	Phe	Met	Thr
				165					170					175	
Asp	Gly	Ser	Leu	Glu	Ala	Leu	Ile	Asp	Trp	Ile	Pro	Gly	Met	Pro	Asn
			180					185					190		
Met	Arg	Leu	Lys	Asp	Leu	Pro	Ser	Phe	Met	Arg	Thr	Thr	Ser	Pro	Asp
		195					200					205			
Asp	Val	Leu	Phe	Asn	Tyr	Leu	Arg	Thr	Ile	Thr	Thr	Lys	Ala	Leu	Lys
	210					215					220				
Ser	Ser	Ala	Leu	Leu	Leu	Asn	Thr	Phe	Asp	Asp	Phe	Glu	His	Glu	Val
225					230					235					240
Val	Glu	Glu	Met	Lys	Lys	Met	Gln	Pro	Asn	Ile	Phe	Leu	Gly	Gly	Pro
			245						250					255	
Leu	Asn	Met	Leu	Leu	Arg	His	Thr	Ser	Lys	Thr	Glu	Ile	Thr	Ser	Leu
		260					265						270		
Thr	Thr	Ser	Leu	Trp	Lys	Glu	Asp	Thr	His	Cys	Leu	Glu	Trp	Leu	Asp
		275				280						285			
Lys	Gln	Glu	Pro	Glu	Ser	Val	Val	Tyr	Ile	Asn	Tyr	Gly	Ser	Val	Thr
	290					295					300				
Ile	Met	Ser	Asp	His	His	Leu	Asn	Glu	Phe	Ala	Trp	Gly	Leu	Ala	Asn
305					310					315					320
Ser	Lys	His	Pro	Phe	Leu	Trp	Ile	Val	Arg	Pro	Asp	Val	Val	Arg	Gly
			325						330					335	
Glu	Ser	Gly	Thr	Leu	Pro	Lys	Glu	Phe	Tyr	Asp	Glu	Ile	Lys	Asp	Arg
			340					345					350		
Gly	Leu	Ile	Thr	Ser	Trp	Cys	Pro	Gln	Pro	Glu	Val	Leu	Lys	His	Pro
		355					360					365			
Ser	Val	Gly	Val	Tyr	Leu	Thr	His	Cys	Gly	Trp	Asn	Ser	Ile	Thr	Glu
	370					375					380				
Ser	Val	Ala	Gly	Gly	Val	Pro	Leu	Met	Cys	Trp	Pro	Phe	Phe	Ala	Glu
385					390					395					400
Gln	Gln	Thr	Asn	Ser	Arg	Phe	Ala	Cys	Thr	Val	Trp	Gly	Thr	Gly	Val
			405						410					415	
Glu	Val	Asn	Ala	Asp	Val	Lys	Arg	Glu	Glu	Leu	Ala	Glu	Gln	Val	Met
			420					425					430		
Glu	Met	Leu	Glu	Gly	Lys	Arg	Gly	Gln	Glu	Leu	Arg	Lys	Asn	Ala	Lys
		435					440					445			
Glu	Trp	Arg	Arg	Lys	Ala	Glu	Glu	Ala	Thr	Asp	Ile	Gly	Gly	Ser	Ala

[illegible]

<400> SEQUENCE: 15

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tacataacct	ttgttcacac	aacctacaac	tacaatcgcc	ttctcaagac	ccacgggtct	180
gatttcctta	gtgggtctacc	agattttcaa	tttgagacca	tccttgatgg	actaccacca	240
tcagatgcag	ctgatgtcac	acaagacatc	cctgccttgt	gtaaatcaac	caccgaaacc	300
tgtttagtc	cattcaaaga	gtcctcggtc	aagctgcata	acaagtcaat	ggcgtcaccc	360
gaggaagttc	ctccagtgac	atgcatagtt	tctgatggtt	gcattgtcatt	tactgtggat	420
gctgcagaag	aggcaggggt	tcctaattgt	cttctttgga	ctaccagtgc	atgcggattt	480
ttaggatatg	ctaattaccc	gaaacttatt	gacagaggca	taattccact	caaagatgag	540
agctacttta	cgaattggga	cctagacaag	acagtagatg	gaatacctgg	aatgaaaggc	600
atacggctac	gagacttccc	aaactttgta	tgcaccacaa	accagatga	gtttatggtg	660
aaatatgcaa	ttcaagagat	cactagagct	gccagagcag	atgctgttat	tttgaacacc	720
tttgacgctt	tggaacatga	tttcttagat	ggcctatcaa	acatatatccc	aaaggtcctc	780
cctattggcc	cgctccagct	tccgtcaac	caaatcccag	agagctcacc	tctacattca	840
atctgttcta	gtctctggaa	agatgaacca	cagtgcatta	cctgggttaa	ctcccaaaaa	900
ccaaaatcag	tcgtttatgt	taactacgga	agtatcacag	ttatgactcc	gcaacaatat	960
gtggagttcg	catggggact	ggctaataca	aaataccctt	ttctgtggat	tatttagacct	1020
gatttggttg	ctggtgagac	agctgtccta	cctccagatt	ttttggaagt	gacaaaagga	1080
aggagctgct	tggttagttg	gtgccacag	gaacaagttc	ttagtcacac	atccatagga	1140
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tggggtatag	gtatggaaat	tgacaatgat	gttaagaggg	ataaggttca	gaaaatggtg	1320
acagagctta	tggaggggca	aaagggaaag	gagatgaaga	ggaagggcgg	agaatggaag	1380
aagcttgggg	cagaagctgc	cggctccta	ggctcagcta	ccttaaaact	cagcagactt	1440
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<400> SEQUENCE: 16

Met Gly Ser Leu Gly Lys Lys Ile Gln Gln Lys Pro His Ala Ile Cys
1 5 10 15

Thr Pro Tyr Pro Ala Gln Gly His Ile Asn Pro Met Leu Lys Leu Ala
20 25 30

Lys	Leu	Leu	His	His	Ser	Gly	Phe	Tyr	Ile	Thr	Phe	Val	His	Thr	Thr
35							40					45			
Tyr	Asn	Tyr	Asn	Arg	Leu	Leu	Lys	Thr	His	Gly	Ser	Asp	Ser	Leu	Ser
50						55					60				
Gly	Leu	Pro	Asp	Phe	Gln	Phe	Glu	Thr	Ile	Pro	Asp	Gly	Leu	Pro	Pro
65					70					75					80
Ser	Asp	Ala	Ala	Asp	Val	Thr	Gln	Asp	Ile	Pro	Ala	Leu	Cys	Lys	Ser
				85					90				95		
Thr	Thr	Glu	Thr	Cys	Leu	Val	Pro	Phe	Lys	Glu	Leu	Leu	Ala	Lys	Leu
			100					105					110		
His	Asn	Lys	Ser	Met	Ala	Ser	Pro	Glu	Glu	Val	Pro	Pro	Val	Thr	Cys
115							120					125			
Ile	Val	Ser	Asp	Gly	Cys	Met	Ser	Phe	Thr	Val	Asp	Ala	Ala	Glu	Glu
130						135					140				
Ala	Gly	Val	Pro	Asn	Val	Leu	Leu	Trp	Thr	Thr	Ser	Ala	Cys	Gly	Phe
145					150					155					160
Leu	Gly	Tyr	Ala	Asn	Tyr	Pro	Lys	Leu	Ile	Asp	Arg	Gly	Ile	Ile	Pro
				165					170				175		
Leu	Lys	Asp	Glu	Ser	Tyr	Phe	Thr	Asn	Gly	Tyr	Leu	Asp	Lys	Thr	Val
			180					185					190		
Asp	Gly	Ile	Pro	Gly	Met	Lys	Gly	Ile	Arg	Leu	Arg	Asp	Phe	Pro	Asn
195						200						205			
Phe	Val	Cys	Thr	Thr	Asn	Pro	Asp	Glu	Phe	Met	Val	Lys	Tyr	Ala	Ile
210						215					220				
Gln	Glu	Ile	Thr	Arg	Ala	Ala	Arg	Ala	Asp	Ala	Val	Ile	Leu	Asn	Thr
225					230					235					240
Phe	Asp	Ala	Leu	Glu	His	Asp	Phe	Leu	Asp	Gly	Leu	Ser	Asn	Ile	Tyr
				245					250				255		
Pro	Lys	Val	Leu	Pro	Ile	Gly	Pro	Leu	Gln	Leu	Pro	Leu	Asn	Gln	Ile
			260					265					270		
Pro	Glu	Ser	Ser	Pro	Leu	His	Ser	Ile	Cys	Ser	Ser	Leu	Trp	Lys	Asp
275							280					285			
Glu	Pro	Gln	Cys	Ile	Thr	Trp	Leu	Asn	Ser	Gln	Lys	Pro	Lys	Ser	Val
290						295					300				
Val	Tyr	Val	Asn	Tyr	Gly	Ser	Ile	Thr	Val	Met	Thr	Pro	Gln	Gln	Met
305					310					315					320
Val	Glu	Phe	Ala	Trp	Gly	Leu	Ala	Asn	Thr	Lys	Tyr	Pro	Phe	Leu	Trp
				325					330					335	
Ile	Ile	Arg	Pro	Asp	Leu	Val	Ala	Gly	Glu	Thr	Ala	Val	Leu	Pro	Pro
			340					345					350		
Asp	Phe	Leu	Glu	Val	Thr	Lys	Gly	Arg	Ser	Cys	Leu	Ala	Ser	Trp	Cys
355							360					365			
Pro	Gln	Glu	Gln	Val	Leu	Ser	His	Thr	Ser	Ile	Gly	Gly	Phe	Leu	Thr
370						375					380				
His	Cys	Gly	Trp	Asn	Ser	Met	Leu	Glu	Ser	Val	Val	Glu	Gly	Val	Pro

450
 465
 475
 485
 490
 495

Glu Ala Ala Gly Pro Asn Gly Ser Ala Thr Leu Asn Phe Ser Arg Leu
 Ile Asn Asp Val Leu Leu Ser Lys Lys Lys Ile Val Val Thr Thr

<210> SEQ ID NO 17
 <211> LENGTH: 1476
 <212> TYPE: DNA
 <213> ORGANISM: *Rhodiola rosea*
 <400> SEQUENCE: 17

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 gcacaaggcc atattaatcc catgcttaaa ctagccaagc tcctacacca ctcaggcttc 120
 tacataacct ttgttcacac aacctacaac tacaatcgcc ttctcaagac ccacgggtct 180
 gattccttaa gtggcttacc agatttccaa ttgagacca tccctgatgg actaccacca 240
 tcagatgcag ctgatgtcac acaagacatc cctgccttgt gtaaatcaac caccgaaacc 300
 tgcttagtcc cattcaaaga gctcctggct aagctgcata acaagtcaat ggcgctaccg 360
 gaggaagttc tcccagtgc atgcatagtt tctgatgggt gcattgtcatt tactgtggat 420
 gctgcagaag aggcaggggg tcctaattgtg cttctttgga ctaccagtgc atgcggattt 480
 ttaggatatg ctaattaccg gaaacttatt gacagaggca taattccact caaagatgag 540
 agctacttta cgaatgggta cctagacaag acagtagatg gaatacctgg aatgaaaggc 600
 atacggctac gagacttccc aaactttgta tgcaccacaa acccagatga gtttatgggtg 660
 aaatatgcaa ttcaagagat cactagagct gccagagcag atgctgttat ttgaaacacc 720
 ttgacgctt tggaacatga tttcttagat ggccatcaaa acatataccc aaaggctctc 780
 cctattggcc cgctccagct tccgctcaac caaatcccag agagctcacc tctacattca 840
 atctgttcta gtctctggaa agatgaacca cagtgcatta cctgggttaa cteccaaaaa 900
 ccaaaatcag tcgtttatgt taactacgga agtatcacag ttatgactcc gcaacaaatg 960
 gtggagttcg catggggact ggctaataca aaataccctt ttctgtggat tatttagacct 1020
 gatttggttg ctgggtgagc agctgtccta cctccagatt ttttggaagt gacaaaagga 1080
 aggagctgct tggctagtgt gtgccacag gaacaagttc ttagtcacac atccatagga 1140
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 atgggtatgct ggcggttttt tgctgagcaa cagactaatt gctgggctgc tcggacaaaa 1260
 tggggtatag gtatggaaat tgacaatgat gttaagaggg ataagggtca gaaaatgggtg 1320
 acagagctta tggaggggca aaaggggaaag gagatgaaga ggaagggcgg agaatggaag 1380
 aagcttgggg cagaagctgc cggctccta at ggctcagcta ccttaaaact cagcagactt 1440
 ataaatgacg tacttctgtc caaaaaaaaa ttgtag 1476

<210> SEQ ID NO 18
 <211> LENGTH: 491
 <212> TYPE: PRT
 <213> ORGANISM: *Rhodiola rosea*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: GenBank MF674557
 <400> SEQUENCE: 18

Met Gly Ser Leu Gly Lys Lys Ile Gln Gln Lys Pro His Ala Ile Cys
 1 5 10 15

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Thr	Pro	Tyr	Pro	Ala	Gln	Gly	His	Ile	Asn	Pro	Met	Leu	Lys	Leu	Ala
			20					25					30		
Lys	Leu	Leu	His	His	Ser	Gly	Phe	Tyr	Ile	Thr	Phe	Val	His	Thr	Thr
	35					40					45				
Tyr	Asn	Tyr	Asn	Arg	Leu	Leu	Lys	Thr	His	Gly	Ser	Asp	Ser	Leu	Ser
	50				55					60					
Gly	Leu	Pro	Asp	Phe	Gln	Phe	Glu	Thr	Ile	Pro	Asp	Gly	Leu	Pro	Pro
65					70					75				80	
Ser	Asp	Ala	Ala	Asp	Val	Thr	Gln	Asp	Ile	Pro	Ala	Leu	Cys	Lys	Ser
				85				90					95		
Thr	Thr	Glu	Thr	Cys	Leu	Val	Pro	Phe	Lys	Glu	Leu	Leu	Ala	Lys	Leu
		100						105					110		
His	Asn	Lys	Ser	Met	Ala	Ser	Pro	Glu	Glu	Val	Pro	Pro	Val	Thr	Cys
		115					120					125			
Ile	Val	Ser	Asp	Gly	Cys	Met	Ser	Phe	Thr	Val	Asp	Ala	Ala	Glu	Glu
	130					135					140				
Ala	Gly	Val	Pro	Asn	Val	Leu	Leu	Trp	Thr	Thr	Ser	Ala	Cys	Gly	Phe
145					150					155					160
Leu	Gly	Tyr	Ala	Asn	Tyr	Pro	Lys	Leu	Ile	Asp	Arg	Gly	Ile	Ile	Pro
				165				170						175	
Leu	Lys	Asp	Glu	Ser	Tyr	Phe	Thr	Asn	Gly	Tyr	Leu	Asp	Lys	Thr	Val
		180						185					190		
Asp	Gly	Ile	Pro	Gly	Met	Lys	Gly	Ile	Arg	Leu	Arg	Asp	Phe	Pro	Asn
		195					200					205			
Phe	Val	Cys	Thr	Thr	Asn	Pro	Asp	Glu	Phe	Met	Val	Lys	Tyr	Ala	Ile
	210					215					220				
Gln	Glu	Ile	Thr	Arg	Ala	Ala	Arg	Ala	Asp	Ala	Val	Ile	Leu	Asn	Thr
225					230					235					240
Phe	Asp	Ala	Leu	Glu	His	Asp	Phe	Leu	Asp	Gly	Leu	Ser	Asn	Ile	Tyr
			245					250						255	
Pro	Lys	Val	Leu	Pro	Ile	Gly	Pro	Leu	Gln	Leu	Pro	Leu	Asn	Gln	Ile
		260						265					270		
Pro	Glu	Ser	Ser	Pro	Leu	His	Ser	Ile	Cys	Ser	Ser	Leu	Trp	Lys	Asp
		275					280					285			
Glu	Pro	Gln	Cys	Ile	Thr	Trp	Leu	Asn	Ser	Gln	Lys	Pro	Lys	Ser	Val
	290					295					300				
Val	Tyr	Val	Asn	Tyr	Gly	Ser	Ile	Thr	Val	Met	Thr	Pro	Gln	Gln	Met
305					310					315					320
Val	Glu	Phe	Ala	Trp	Gly	Leu	Ala	Asn	Thr	Lys	Tyr	Pro	Phe	Leu	Trp
			325						330					335	
Ile	Ile	Arg	Pro	Asp	Leu	Val	Ala	Gly	Glu	Thr	Ala	Val	Leu	Pro	Pro
			340					345					350		
Asp	Phe	Leu	Glu	Val	Thr	Lys	Gly	Arg	Ser	Cys	Leu	Ala	Ser	Trp	Cys
		355					360					365			
Pro	Gln	Glu	Gln	Val	Leu	Ser	His	Thr	Ser	Ile	Gly	Gly	Phe	Leu	Thr
	370					375					380				
His	Cys	Gly	Trp	Asn	Ser	Met	Leu	Glu	Ser	Val	Val	Glu	Gly	Val	Pro
385				390						395					400
Met	Val	Cys	Trp	Pro	Phe	Phe	Ala	Glu	Gln	Gln	Thr	Asn	Cys	Trp	Ala
			405					410						415	
Ala	Arg	Thr	Lys	Trp	Gly	Ile	Gly	Met	Glu	Ile	Asp	Asn	Asp	Val	Lys
			420					425					430		
Arg	Asp	Lys	Val	Gln	Lys	Met	Val	Thr	Glu	Leu	Met	Glu	Gly	Glu	Lys

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435	440	445	
Gly Lys Glu Met Lys Arg	Lys Gly Gly Glu Trp	Lys Lys Leu Gly Ala	
450	455	460	
Glu Ala Ala Gly Pro Asn Gly Ser Ala Thr Leu	Asn Phe Ser Arg Leu		
465	470	475	480
Ile Asn Asp Val Leu Leu Ser Lys Lys Leu			
	485	490	

<210> SEQ ID NO 19
 <211> LENGTH: 1497
 <212> TYPE: DNA
 <213> ORGANISM: *Rhodiola rosea*
 <400> SEQUENCE: 19

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tgcattcccat acccagctca aggccatata aatcccatga tgcaacttgc aaagctctctc    120
caccactctg gtttcccat aacgtttgtc cacactgagt ataattatga ccgtctagtgt      180
aagtctcaag gttcagcttg tgtggctggt ttaccggatt tccgcttga agccatccca      240
gatggcttgc cctcgacgaa tggatgatgt actcaagaca ttctctgtgt gagtagctct      300
acttctaaaa cctgcttgaa gccgtttaag gagttattga agagggttga ggacaaatgc      360
aaagagttac ctgatgatgt tccgctctg tcgtgcatcg tgtctgatgc agccatgtcg      420
tttacgatcg atgcatttga ggagtttga gtgcccatac cgcttctttg gactgcaagt      480
gcctgcgggt tcttgggtta cagcattac ccatatctaa ttgacagagg tgtcatccca      540
ttgaaagatg agagccaatt aacaaacgga tacctagata tgagcataga tggcatacct      600
tgtatggaag gtatccgctt acgagacctc ccaagcttct tacgcacaac tgatttagat      660
gatatgatgt ttatgttatat actgcacgaa ataaaaaag tttcaagagg cagtgtctatc      720
attctgaaca cctttgaagc tttggaccat gatgtcttgg atagtctctc caaaatttac      780
caaaatgtca tcttgccagt tggccctcta catgtctcgc tcaacaagat cccaaaacac      840
taccacttcc aatctttaag ctggaattta tggaaagatg acacagactg cattccctgg      900
ctgagctcta aggttcaaa atcagttata tacgttaact ttgggagcat cagcaggtta      960
tcacaaaaac aaattgtgga gtttgcgtgg ggattggcta acagcaaaac ccctttcctt    1020
tggataatca gaccggactt ggtggcaggt gaggcattca tcattccgca ggacttcatt    1080
gatgaaacaa aaggaagagg tttgttggtt ggttggtgtg accaagagct tgttctcaac    1140
catccatcca ttggagggtt tcttacgcac tgtggttgga actcaattat tgaaagcatt    1200
agcgcaggag tccctacggt ctgctggcca ttttttctg agcagcaaac aaattgttgg    1260
tttgccttga aaaaatgggt cattgggatg gagatgcata ctgatgtaaa gagggatgag    1320
gttgacaagc tgttgagaga gctaattgaa ggtgacaaaag gggaggagtt gaagaggaag    1380
gcaaccaact ggaagaggct ggcagaagaa gctgtttcct cacttggtt atcaacctta    1440
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<210> SEQ ID NO 20
 <211> LENGTH: 498
 <212> TYPE: PRT
 <213> ORGANISM: *Rhodiola rosea*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: GenBank MF674558
 <400> SEQUENCE: 20

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Met	Ser	Leu	Ile	Glu	Lys	Pro	Leu	Thr	Ala	Ile	Glu	Thr	Arg	Glu	Lys	1	5	10	15
Pro	His	Ala	Val	Cys	Ile	Pro	Tyr	Pro	Ala	Gln	Gly	His	Ile	Asn	Pro	20	25	30	
Met	Met	Gln	Leu	Ala	Lys	Leu	Leu	His	His	Ser	Gly	Phe	His	Ile	Thr	35	40	45	
Phe	Val	His	Thr	Glu	Tyr	Asn	Tyr	Asp	Arg	Leu	Val	Lys	Ser	Gln	Gly	50	55	60	
Ser	Ala	Cys	Val	Ala	Gly	Leu	Pro	Asp	Phe	Arg	Phe	Glu	Ala	Ile	Pro	65	70	75	80
Asp	Gly	Leu	Pro	Ser	Thr	Asn	Gly	Asp	Val	Thr	Gln	Asp	Ile	Pro	Leu	85	90	95	
Leu	Ser	Ser	Ser	Thr	Ser	Lys	Thr	Cys	Leu	Lys	Pro	Phe	Lys	Glu	Leu	100	105	110	
Leu	Lys	Arg	Leu	Gln	Asp	Lys	Cys	Lys	Glu	Leu	Pro	Asp	Asp	Val	Pro	115	120	125	
Pro	Leu	Ser	Cys	Ile	Val	Ser	Asp	Ala	Ala	Met	Ser	Phe	Thr	Ile	Asp	130	135	140	
Ala	Ser	Glu	Glu	Phe	Gly	Val	Pro	Ile	Ala	Leu	Leu	Trp	Thr	Ala	Ser	145	150	155	160
Ala	Cys	Gly	Phe	Leu	Gly	Tyr	Thr	His	Tyr	Pro	Tyr	Leu	Ile	Asp	Arg	165	170	175	
Gly	Val	Ile	Pro	Leu	Lys	Asp	Glu	Ser	Gln	Leu	Thr	Asn	Gly	Tyr	Leu	180	185	190	
Asp	Met	Ser	Ile	Asp	Gly	Ile	Pro	Cys	Met	Glu	Gly	Ile	Arg	Leu	Arg	195	200	205	
Asp	Leu	Pro	Ser	Phe	Leu	Arg	Thr	Thr	Asp	Leu	Asp	Asp	Met	Met	Phe	210	215	220	
Ser	Tyr	Ile	Leu	His	Glu	Ile	Lys	Gln	Val	Ser	Arg	Gly	Ser	Ala	Ile	225	230	235	240
Ile	Leu	Asn	Thr	Phe	Glu	Ala	Leu	Asp	His	Asp	Val	Leu	Asp	Ser	Leu	245	250	255	
Ser	Lys	Ile	Tyr	Gln	Asn	Val	Ile	Leu	Pro	Val	Gly	Pro	Leu	His	Val	260	265	270	
Ser	Leu	Asn	Lys	Ile	Pro	Lys	His	Tyr	Pro	Leu	Gln	Ser	Leu	Ser	Ser	275	280	285	
Asn	Leu	Trp	Lys	Asp	Asp	Thr	Asp	Cys	Ile	Pro	Trp	Leu	Ser	Ser	Lys	290	295	300	
Ala	Ser	Lys	Ser	Val	Ile	Tyr	Val	Asn	Phe	Gly	Ser	Ile	Thr	Thr	Val	305	310	315	320
Ser	Pro	Lys	Gln	Ile	Val	Glu	Phe	Ala	Trp	Gly	Leu	Ala	Asn	Ser	Lys	325	330	335	
His	Pro	Phe	Leu	Trp	Ile	Ile	Arg	Pro	Asp	Leu	Val	Ala	Gly	Glu	Ala	340	345	350	
Ser	Ile	Ile	Pro	Gln	Asp	Phe	Met	Asp	Glu	Thr	Lys	Gly	Arg	Gly	Leu	355	360	365	
Leu	Ala	Gly	Trp	Cys	Asp	Gln	Glu	Leu	Val	Leu	Asn	His	Pro	Ser	Ile	370	375	380	
Gly	Gly	Phe	Leu	Thr	His	Cys	Gly	Trp	Asn	Ser	Ile	Ile	Glu	Ser	Ile	385	390	395	400
Ser	Ala	Gly	Val	Pro	Thr	Val	Cys	Trp	Pro	Phe	Phe	Ala	Glu	Gln	Gln	405	410	415	
Thr	Asn	Cys	Trp	Phe	Ala	Cys	Lys	Lys	Trp	Cys	Ile	Gly	Met	Glu	Met				

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420					425					430				
His Thr Asp Val Lys Arg Asp Glu Val Asp Lys Leu Leu Arg Glu Leu														
435					440					445				
Met Glu Gly Asp Lys Gly Glu Glu Leu Lys Arg Lys Ala Thr Asn Trp														
450					455					460				
Lys Arg Leu Ala Glu Glu Ala Val Ser Ser Thr Gly Leu Ser Thr Leu														
465					470					475			480	
Asn Phe Arg Thr Leu Val Asn Gln Val Leu Leu Ser Lys Thr Lys His														
				485				490					495	
Ile Arg														
<210> SEQ ID NO 21														
<211> LENGTH: 490														
<212> TYPE: PRT														
<213> ORGANISM: Arabidopsis thaliana														
<300> PUBLICATION INFORMATION:														
<308> DATABASE ACCESSION NUMBER: AT2G20340.1														
<400> SEQUENCE: 21														
Met Glu Asn Gly Ser Gly Lys Val Leu Lys Pro Met Asp Ser Glu Gln														
1				5				10				15		
Leu Arg Glu Tyr Gly His Leu Met Val Asp Phe Ile Ala Asp Tyr Tyr														
			20					25				30		
Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr														
			35					40				45		
Leu His Lys Leu Leu Pro Asp Ser Ala Pro Asp His Pro Glu Thr Leu														
			50					55				60		
Asp Gln Val Leu Asp Asp Val Arg Ala Lys Ile Leu Pro Gly Val Thr														
65								70				75	80	
His Trp Gln Ser Pro Ser Phe Phe Ala Tyr Tyr Pro Ser Asn Ser Ser														
								85				90	95	
Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala Gly Leu Gly Ile Val														
			100					105				110		
Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile														
			115					120				125		
Val Leu Asp Trp Val Ala Lys Leu Leu Asn Leu Pro Glu Gln Phe Met														
			130					135				140		
Ser Lys Gly Asn Gly Gly Gly Val Ile Gln Gly Ser Ala Ser Glu Ala														
145								150				155	160	
Val Leu Val Val Leu Ile Ala Ala Arg Asp Lys Val Leu Arg Ser Val														
								165				170	175	
Gly Lys Asn Ala Leu Glu Lys Leu Val Val Tyr Ser Ser Asp Gln Thr														
			180					185				190		
His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala Gly Ile His Pro Glu														
			195					200				205		
Asn Cys Arg Val Leu Thr Thr Asp Ser Ser Thr Asn Tyr Ala Leu Arg														
			210					215				220		
Pro Glu Ser Leu Gln Glu Ala Val Ser Arg Asp Leu Glu Ala Gly Leu														
225								230				235	240	
Ile Pro Phe Phe Leu Cys Ala Asn Val Gly Thr Thr Ser Ser Thr Ala														
								245				250	255	
Val Asp Pro Leu Ala Ala Leu Gly Lys Ile Ala Asn Ser Asn Gly Ile														
			260					265				270		
Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro														
			275					280				285		

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Glu Tyr Arg Gln Tyr Ile Asp Gly Val Glu Thr Ala Asp Ser Phe Asn
 290 295 300
 Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu Leu
 305 310 315 320
 Trp Val Lys Asp Gln Asp Ser Leu Thr Leu Ala Leu Ser Thr Asn Pro
 325 330 335
 Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn Leu Val Val Asp Tyr
 340 345 350
 Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu
 355 360 365
 Trp Met Val Leu Arg Leu Tyr Gly Ser Glu Thr Leu Lys Ser Tyr Ile
 370 375 380
 Arg Asn His Ile Lys Leu Ala Lys Glu Phe Glu Gln Leu Val Ser Gln
 385 390 395 400
 Asp Pro Asn Phe Glu Ile Val Thr Pro Arg Ile Phe Ala Leu Val Cys
 405 410 415
 Phe Arg Leu Val Pro Val Lys Asp Glu Glu Lys Lys Cys Asn Asn Arg
 420 425 430
 Asn Arg Glu Leu Leu Asp Ala Val Asn Ser Ser Gly Lys Leu Phe Met
 435 440 445
 Ser His Thr Ala Leu Ser Gly Lys Ile Val Leu Arg Cys Ala Ile Gly
 450 455 460
 Ala Pro Leu Thr Glu Glu Lys His Val Lys Glu Ala Trp Lys Ile Ile
 465 470 475 480
 Gln Glu Glu Ala Ser Tyr Leu Leu His Lys
 485 490

<210> SEQ ID NO 22
 <211> LENGTH: 498
 <212> TYPE: PRT
 <213> ORGANISM: Brachypodium distachyon
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 1g28960.3

<400> SEQUENCE: 22

Met Asp Gly Ser Thr Thr Ser Asn Gly Gly Gly Trp Met Arg Pro
 1 5 10 15
 Met Asp Glu Glu Gln Leu Arg Glu Cys Gly His Arg Met Val Asp Phe
 20 25 30
 Ile Ala Asp Tyr Tyr Lys Ser Ile Glu Thr Tyr Pro Val Leu Ser Gln
 35 40 45
 Val Gln Pro Gly Tyr Leu Lys Glu Leu Leu Pro Asp Ser Ala Pro Asn
 50 55 60
 Gln Pro Asp Thr Leu Asp Ala Leu Phe Asp Asp Ile Arg Glu Lys Ile
 65 70 75 80
 Val Pro Gly Val Thr His Trp Gln Ser Pro Asn Tyr Phe Ala Tyr Tyr
 85 90 95
 Pro Ser Asn Ser Ser Thr Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
 100 105 110
 Ala Phe Asn Ile Val Gly Phe Ser Trp Ile Thr Ser Pro Ala Ala Thr
 115 120 125
 Glu Leu Glu Val Ile Val Leu Asp Trp Val Ala Lys Met Leu Lys Leu
 130 135 140
 Pro Ser Glu Phe Leu Ser Ala Ala Leu Gly Gly Gly Val Ile Gln Gly
 145 150 155 160

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Thr Ala Ser Glu Ala Ile Leu Val Val Leu Leu Ser Ala Arg Asp Arg
 165 170 175
 Thr Leu Arg Lys His Gly Lys Lys Ser Leu Glu Lys Ile Val Val Tyr
 180 185 190
 Ala Ser Asp Gln Thr His Ser Ala Leu Lys Lys Ala Cys Gln Ile Ala
 195 200 205
 Gly Ile Phe Pro Glu Asn Ile Arg Ile Val Lys Ala Asp Cys Ser Met
 210 215 220
 Asn Tyr Ala Val Thr Pro Gly Ala Val Ser Glu Ala Ile Ser Ile Asp
 225 230 235 240
 Leu Ser Ala Gly Leu Ile Pro Phe Phe Ile Cys Ala Thr Val Gly Thr
 245 250 255
 Thr Ser Ser Ser Ala Val Asp Pro Leu His Glu Leu Gly Gln Ile Ala
 260 265 270
 Gln Ala His Asp Met Trp Phe His Ile Asp Ala Ala Tyr Ala Gly Ser
 275 280 285
 Ala Cys Ile Cys Pro Glu Tyr Arg Lys Tyr Leu Asn Gly Val Glu Glu
 290 295 300
 Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe
 305 310 315 320
 Asp Cys Ser Leu Leu Trp Val Lys Asp Arg Asn Tyr Leu Ile Gln Ala
 325 330 335
 Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Glu Asn
 340 345 350
 Ser Val Ile Asp Phe Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe
 355 360 365
 Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Val Glu Asn
 370 375 380
 Leu Gln Ser Tyr Ile Arg Lys His Ile Gln Leu Ala Gln His Phe Glu
 385 390 395 400
 Gln Leu Val Ile Ser Asp Pro Arg Phe Glu Val Val Thr Pro Arg Asn
 405 410 415
 Phe Ser Leu Val Cys Phe Cys Leu Val Pro Pro Thr Cys Glu Val Asp
 420 425 430
 Asn Gly His Lys Leu Asn Tyr Asp Leu Met Asp Ser Ala Asn Ser Ser
 435 440 445
 Gly Lys Ile Phe Ile Ser His Thr Val Leu Ser Gly Lys Phe Val Leu
 450 455 460
 Arg Phe Val Val Gly Ala Pro Leu Thr Glu Glu Gln His Val Asp Ala
 465 470 475 480
 Ala Trp Lys Leu Leu Gln Asp Glu Ala Thr Lys Leu Leu Gly Asn Val
 485 490 495
 Val Gln

<210> SEQ ID NO 23
 <211> LENGTH: 480
 <212> TYPE: PRT
 <213> ORGANISM: Carica papaya
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 16427710

<400> SEQUENCE: 23

Met Asp Ala Glu Gln Leu Arg Glu Asn Gly His Lys Met Val Asp Phe
 1 5 10 15

Ile	Ala	Asp	Tyr	Tyr	Lys	Thr	Ile	Glu	Asn	Phe	Pro	Val	Leu	Ser	Gln
			20				25						30		
Val	Glu	Pro	Gly	Tyr	Leu	Arg	Asp	Leu	Ile	Pro	Asp	Ser	Ala	Pro	Asn
			35				40				45				
Ser	Pro	Glu	Ser	Phe	Gln	Gln	Leu	Leu	Asp	Asp	Val	Arg	Thr	Lys	Ile
			50				55				60				
Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser	Pro	Asn	Tyr	Phe	Ala	Tyr	Tyr
			65				70				75				
Pro	Ser	Asn	Ser	Ser	Val	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu	Ser	Ala
			85				90				95				
Gly	Leu	Asn	Ile	Val	Gly	Phe	Ser	Trp	Ile	Thr	Ser	Pro	Ala	Ala	Thr
			100				105				110				
Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp	Leu	Ala	Lys	Leu	Leu	Lys	Leu
			115				120				125				
Pro	Glu	Asp	Phe	His	Ser	Thr	Gly	Asn	Gly	Gly	Gly	Val	Ile	Gln	Gly
			130				135				140				
Thr	Ala	Ser	Glu	Ala	Ile	Leu	Val	Val	Leu	Leu	Ala	Ala	Arg	Asp	Lys
			145				150				155				
Val	Leu	Lys	Arg	Val	Gly	Lys	Asn	Ala	Leu	Glu	Lys	Leu	Val	Val	Tyr
			165				170				175				
Thr	Ser	Asp	Gln	Thr	His	Ser	Ala	Phe	Gln	Lys	Ala	Cys	Gln	Ile	Gly
			180				185				190				
Gly	Ile	His	Pro	Glu	Asn	Cys	Arg	Val	Leu	Lys	Thr	Asp	Ser	Ser	Thr
			195				200				205				
Asn	Tyr	Ala	Leu	Ser	Pro	Asp	Leu	Leu	Lys	Glu	Ala	Ile	Ser	Cys	Asp
			210				215				220				
Val	Ala	Ala	Gly	Leu	Ile	Pro	Phe	Phe	Phe	Cys	Ala	Thr	Val	Gly	Thr
			225				230				235				
Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Met	Ala	Leu	Gly	Lys	Ile	Ala
			245				250				255				
Thr	Ser	Asn	Glu	Ile	Trp	Phe	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser
			260				265				270				
Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	Pro	Tyr	Ile	Asp	Gly	Val	Glu	Glu
			275				280				285				
Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe
			290				295				300				
Asp	Cys	Ser	Val	Leu	Trp	Val	Lys	Asp	Lys	Tyr	Ser	Leu	Ile	Gln	Ser
			305				310				315				
Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asp
			325				330				335				
Met	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe
			340				345				350				
Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Val	Glu	Asn
			355				360				365				
Leu	Lys	Ser	Tyr	Ile	Arg	Asn	His	Ile	Lys	Leu	Ala	Lys	His	Phe	Glu
			370				375				380				
Glu	Leu	Val	Thr	Gln	Asp	Pro	Arg	Phe	Glu	Val	Val	Thr	Pro	Arg	Ile
			385				390				395				
Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu	Pro	Pro	Gly	Asn	Asp	Glu	Asn
			405				410				415				
His	Gly	Asn	Lys	Leu	Asn	Gln	Asp	Leu	Leu	Glu	Thr				

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435	440	445
Arg Phe Ala Val Gly Ala	Pro Leu Thr Glu Glu	Arg His Val Asn Glu
450	455	460
Ala Trp Lys Ile Leu Gln Asp	Glu Ala Ser Thr	Leu Leu Glu Asn Pro
465	470	475 480

<210> SEQ ID NO 24
 <211> LENGTH: 492
 <212> TYPE: PRT
 <213> ORGANISM: Ricinus communis
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 16804377

<400> SEQUENCE: 24

Met Phe Arg Glu Gly Glu Leu Arg	Pro Met Asp Ala Glu Gln Leu Arg
1	5 10 15
Glu His Gly His Lys Met Val Asp	Phe Ile Ala Asp Tyr Tyr Lys Thr
20	25 30
Ile Glu Asn Phe Pro Val Leu Ser	Gln Val Glu Pro Gly Tyr Leu Arg
35	40 45
Lys Leu Leu Pro Asp Ser Ala Pro	Asn Gln Pro Glu Ser Leu Gln Asn
50	55 60
Val Leu Asp Asp Val Gln Ala Lys	Ile Leu Pro Gly Val Thr His Trp
65	70 75 80
Gln Ser Pro Asn Tyr Phe Ala Tyr	Tyr Pro Ser Asn Ser Ser Val Ala
85	90 95
Gly Phe Leu Gly Glu Met Leu Ser	Ala Gly Ile Asn Met Val Gly Phe
100	105 110
Ser Trp Ile Thr Ser Pro Ala Ala	Thr Glu Leu Glu Met Ile Val Leu
115	120 125
Asp Trp Leu Gly Lys Met Leu Lys	Leu Pro Glu Glu Phe Leu Ser Thr
130	135 140
Gly Gln Gly Gly Gly Val Ile Gln	Gly Thr Ala Ser Glu Ala Val Leu
145	150 155 160
Val Ala Leu Val Ala Ala Arg Asp	Lys Val Leu Arg Arg Val Gly Lys
165	170 175
Asp Ala Leu Arg Lys Leu Val Val	Tyr Gly Ser Asp Gln Thr His Ser
180	185 190
Ala Leu Gln Lys Ala Cys Gln Ile	Gly Gly Ile His Pro Val Asn Cys
195	200 205
Arg Leu Leu Glu Thr Asp Ser Ser	Thr Asn Tyr Ala Leu Ala Pro Asp
210	215 220
Leu Leu Ser Arg Ala Ile Ser Glu	Asp Ile Ser Leu Gly Leu Ile Pro
225	230 235 240
Phe Phe Leu Cys Ala Thr Val Gly	Thr Thr Ser Ser Thr Ala Val Asp
245	250 255
Pro Leu Leu Ala Leu Gly Lys Ile	Ala Lys Ser Asn Gly Met Trp Phe
260	265 270
His Val Asp Ala Ala Tyr Ala Gly	Ser Ala Cys Val Cys Pro Glu Tyr
275	280 285
Arg Cys Tyr Met Asp Gly Val Glu	Glu Ala Asp Ser Phe Asn Met Asn
290	295 300
Ala His Lys Trp Phe Leu Thr Asn	Phe Asp Cys Ser Ala Leu Trp Val
305	310 315 320
Lys Asp Arg Asn Ala Leu Ile Gln	Ser Leu Ser Thr Ser Pro Glu Phe

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325	330	335
Leu Gln Asn Lys Pro Ser Gln Thr Asn Thr Val Val Asp Tyr Lys Asp		
340	345	350
Trp Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met		
355	360	365
Val Leu Arg Leu Tyr Gly Val Glu Lys Leu Gln Cys Tyr Ile Arg Asn		
370	375	380
His Ile Asn Leu Ala Lys Tyr Phe Glu Gly Leu Ile Ala Glu Asp Thr		
385	390	395
Arg Phe Glu Val Val Ser Pro Pro Ile Phe Ala Leu Val Cys Phe Arg		
405	410	415
Leu Leu Pro Pro Asp Asn Asn Val Asp His Gly Asn Lys Leu Ser His		
420	425	430
Asp Leu Leu Asp Ala Val Asn Ser Thr Gly Lys Ile Phe Ile Ser His		
435	440	445
Thr Val Leu Ser Gly Lys Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro		
450	455	460
Leu Thr Glu Glu Arg His Val Thr Ala Ala Trp Lys Val Leu Gln Asp		
465	470	475
Glu Ala Cys Ala Leu Leu Glu Thr Ser Arg Ile Ser		
485	490	

<210> SEQ ID NO 25
 <211> LENGTH: 486
 <212> TYPE: PRT
 <213> ORGANISM: Cucumis sativus
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 16963476
 <400> SEQUENCE: 25

Met Asp Asn Glu Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu His		
1	5	10
Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu		
20	25	30
Asp Phe Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Gln Asn Leu		
35	40	45
Leu Pro Glu Ser Ala Pro Leu Asn Pro Glu Ser Leu Gln Ser Val Leu		
50	55	60
Asp Asp Val Gln Lys Lys Ile Phe Pro Gly Val Thr His Trp Gln Ser		
65	70	75
Pro Asn Tyr Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Ile Ala Gly Phe		
85	90	95
Leu Gly Glu Met Leu Ser Ala Ala Phe Asn Val Ile Gly Phe Ser Trp		
100	105	110
Val Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp		
115	120	125
Leu Ala Lys Leu Leu Lys Leu Pro Asp Asp Phe Leu Ser Ser Gly Asn		
130	135	140
Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val		
145	150	155
Leu Leu Ala Ala Arg Asp Arg Ala Leu Arg Arg Phe Gly Lys Asp Tyr		
165	170	175
Leu Lys Lys Leu Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu		
180	185	190
Gln Lys Ala Cys Gln Ile Gly Gly Ile His Pro Glu Asn Cys Arg Trp		

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195	200	205
Leu Lys Ala Asp Ile Ser Thr Asn Tyr Ala Leu Ser Pro Asp Val Leu		
210	215	220
Ser Glu Glu Leu Ser Arg Asp Thr Ala Arg Gly Leu Ile Pro Phe Phe		
225	230	235 240
Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Thr Ala Val Asp Pro Leu		
	245	250 255
Pro Glu Leu Gly Thr Ile Ala Lys Arg His Glu Met Trp Phe His Val		
	260	265 270
Asp Ala Ala Tyr Ala Gly Ser Ala Cys Val Cys Pro Glu Tyr Arg Gln		
	275	280 285
Tyr Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn Leu His		
	290	295 300
Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Ala Leu Trp Ile Lys Asp		
305	310	315 320
Arg His Ala Leu Ile Arg Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys		
	325	330 335
Asn Lys Ala Ser Glu Ala Glu Leu Val Val Asp Tyr Lys Asp Trp Gln		
	340	345 350
Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Val Trp Met Val Leu		
	355	360 365
Arg Leu Tyr Gly Thr Glu Asn Leu Gln Lys Tyr Ile Arg Asn His Ile		
	370	375 380
Ser Leu Ala Glu Arg Phe Glu Ala Leu Val Arg Glu Asp Pro Arg Phe		
385	390	395 400
Glu Ile Val Thr Pro Arg Ile Phe Ser Leu Val Cys Phe Arg Leu Leu		
	405	410 415
Pro Ser Arg Lys Asn Glu Asp Gly Gly Asn Arg Leu Asn Gln Ser Leu		
	420	425 430
Leu Asp Ala Val Asn Ala Ser Gly Asn Ile Phe Ile Ser His Thr Val		
	435	440 445
Leu Ser Gly Lys Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro Leu Thr		
	450	455 460
Glu Glu Lys His Ile Asn Ser Ala Trp Lys Leu Leu Gln Asp Val Ala		
465	470	475 480
Ser Thr Leu Leu Ala Ile		
	485	

<210> SEQ ID NO 26

<211> LENGTH: 508

<212> TYPE: PRT

<213> ORGANISM: Vitis vinifera

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 17835588

<400> SEQUENCE: 26

Met Asp Ala Glu Gln Leu Arg Glu Asn Gly His Lys Met Val Asp Phe		
1	5	10 15
Ile Ala Asp Tyr Tyr Lys Ser Ile Glu Asn Phe Pro Val Leu Ser Gln		
	20	25 30
Val Glu Pro Gly Tyr Leu Arg Glu Leu Leu Pro Asp Ser Ala Pro Asn		
	35	40 45
Gln Pro Glu Ser Leu Gln Gln Val Phe Asp Asp Leu Gln Ala Lys Ile		
	50	55 60
Leu Pro Gly Val Thr His Trp Gln Ser Pro Asn Phe Phe Ala Tyr Tyr		

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65				70				75				80				
Pro	Ser	Asn	Ser	Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu	Ser	Ala	
				85					90					95		
Gly	Leu	Asn	Ile	Val	Gly	Phe	Ser	Trp	Ile	Thr	Ser	Pro	Ala	Ala	Thr	
				100					105					110		
Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp	Leu	Ala	Lys	Leu	Leu	Asn	Leu	
				115					120					125		
Pro	Asp	Asp	Phe	Leu	Ser	Ala	Gly	Asn	Gly	Gly	Gly	Val	Ile	Gln	Gly	
				130					135					140		
Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val	Leu	Leu	Ala	Ala	Arg	Asp	Arg	
				145					150					155		
Val	Leu	Arg	Thr	Val	Gly	Lys	Thr	Ala	Leu	Glu	Lys	Leu	Val	Val	Tyr	
				165					170					175		
Gly	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu	Gln	Lys	Ala	Cys	Gln	Ile	Gly	
				180					185					190		
Gly	Ile	His	Pro	Glu	Asn	Cys	Lys	Leu	Leu	Lys	Ala	Asp	Ser	Ser	Thr	
				195					200					205		
Gly	Tyr	Ala	Leu	Ser	Pro	Asp	Leu	Leu	Ser	Glu	Ala	Val	Ser	His	Asp	
				210					215					220		
Ile	Thr	Asn	Gly	Leu	Ile	Pro	Phe	Phe	Leu	Cys	Ala	Asn	Val	Gly	Thr	
				225					230					235		
Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Leu	Glu	Leu	Gly	Lys	Val	Thr	
				245					250					255		
Lys	Ser	Asn	Gly	Ile	Trp	Phe	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	
				260					265					270		
Ala	Cys	Val	Cys	Pro	Glu	Tyr	Arg	His	Tyr	Ile	Asp	Gly	Val	Glu	Glu	
				275					280					285		
Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	
				290					295					300		
Asp	Cys	Ser	Val	Leu	Trp	Val	Lys	Asp	Arg	Asn	Ala	Leu	Val	Gln	Ala	
				305					310					315		
Leu	Ser	Thr	Asn	Pro	Val	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Asp	Ala	Asn	
				325					330					335		
Met	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Val	Pro	Leu	Gly	Arg	Arg	Phe	
				340					345					350		
Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Val	Glu	Asn	
				355					360					365		
Leu	Gln	Arg	Tyr	Ile	Arg	Asn	His	Ile	Lys	Leu	Ala	Lys	Gln	Phe	Glu	
				370					375					380		
Glu	Leu	Val	Ala	Gln	Asp	Pro	Arg	Phe	Glu	Ile	Val	Ala	Pro	Arg	Lys	
				385					390					395		
Phe	Ala	Leu	Val	Cys	Phe	Arg	Leu	Leu	Pro	Pro	His	Arg	Asn	Glu	Asp	
				405					410					415		
Phe	Ser	Asn	Lys	Leu	Asn	His	Asn	Leu	Leu	Asp	Thr	Val	Asn	Ser	Thr	
				420					425					430		
Gly	Lys	Val	Tyr	Ile	Ser	His	Thr	Ala	Leu	Ser	Gly	Lys	Tyr	Thr	Leu	
				435					440					445		
Arg	Leu	Ala	Val	Gly	Ala	Pro	Leu	Thr	Glu	Glu	Arg	His	Val	Asn	Ala	
				450					455					460		
Ala	Trp	Lys	Val	Ile	Gln	Glu	Lys	Ala	Ser	Val	Leu	Leu	Ser	Glu	Phe	
				465					470					475		
Gly	Met	Asn	Gly	Leu	Phe	Asp	Asn	Ile	Asn	Leu	Lys	Phe	Ile	Leu	Asn	
				485					490					495		

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His Gln Ile Asp Ile Ser Ile Leu Leu Asn Tyr Asn
500 505

<210> SEQ ID NO 27
<211> LENGTH: 486
<212> TYPE: PRT
<213> ORGANISM: Citrus sinensis
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 18113817

<400> SEQUENCE: 27

Met Asp Ala Glu Gln Leu Arg Glu Asn Ala His Lys Met Val Asp Phe
1 5 10 15
Ile Ala Asp Tyr Tyr Lys Ser Ile Glu Asn Phe Pro Val Leu Ser Gln
20 25 30
Val Gln Pro Gly Tyr Leu His Asn Leu Ile Pro Asp Ser Ala Pro His
35 40 45
His Pro Glu Ser Leu Gln Asn Val Leu Asp Gly Tyr Ile Asp Ile Gln
50 55 60
Glu Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser Pro Asn Tyr Phe
65 70 75 80
Ala Tyr Tyr Pro Ser Asn Ser Ser Val Ala Gly Phe Leu Gly Glu Met
85 90 95
Leu Ser Ala Gly Leu Asn Ile Val Gly Phe Ser Trp Ile Thr Ser Pro
100 105 110
Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp Leu Ala Lys Leu
115 120 125
Leu Lys Leu Pro Glu Asp Phe Leu Ser Ser Gly Gln Gly Gly Gly Val
130 135 140
Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val Leu Leu Ala Ala
145 150 155 160
Arg Asp Lys Ala Leu Lys Arg Val Gly Lys Asn Ser Leu Glu Lys Leu
165 170 175
Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys
180 185 190
Gln Ile Gly Gly Ile His Pro Gln Asn Phe Arg Val Leu Lys Thr Asp
195 200 205
Ser Ser Thr Asn Tyr Ser Leu Ser Pro Asp Ser Leu Ala Glu Ala Ile
210 215 220
Ser Arg Asp Leu Thr Ile Gly Leu Ile Pro Phe Phe Leu Cys Ala Thr
225 230 235 240
Val Gly Thr Thr Ser Ser Thr Ala Val Asp Pro Leu Leu Ala Leu Gly
245 250 255
Asn Ile Ala Lys Ser Asn Gly Met Trp Phe His Val Asp Ala Ala Tyr
260 265 270
Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg Gln Tyr Ile Asp Gly
275 280 285
Val Glu Glu Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu
290 295 300
Thr Asn Phe Asp Cys Ser Ala Leu Trp Val Lys Asp Arg Asn Thr Leu
305 310 315 320
Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser
325 330 335
Gln Ala Asn Met Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly
340 345 350

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Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly
   355                               360               365

Leu Glu Asn Leu Gln Gly Tyr Ile Arg Asn His Ile Gln Leu Ala Lys
   370                               375               380

His Phe Glu Gly Leu Val Ala Gln Asp Leu Arg Phe Glu Val Val Thr
  385                               390               395               400

Pro Arg Ile Phe Ser Leu Val Cys Phe Arg Leu Leu Pro Pro His Asn
   405                               410               415

Asp Glu Asp His Gly Asn Lys Leu Asn His Lys Leu Leu Asp Asp Ile
   420                               425               430

Asn Ser Thr Gly Lys Ile Phe Ile Ser His Thr Val Leu Ser Gly Lys
   435                               440               445

Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro Leu Thr Glu Trp Arg His
   450                               455               460

Val Asn Ala Ala Trp Glu Val Met Gln Asp Lys Ala Ser Ala Leu Leu
  465                               470               475               480

Ala Arg Leu Ser Ile Glu
   485

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<210> SEQ ID NO 28
<211> LENGTH: 507
<212> TYPE: PRT
<213> ORGANISM: Capsella rubella
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 20900667

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<400> SEQUENCE: 28

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Met Gly Phe Cys Gln Ile Glu Leu Leu Arg His Ile Asn Lys His Asn
 1           5           10           15

Met Gln Asn Gly Ser Gly Lys Asn Val Leu Lys Pro Met Asp Ser Glu
 20           25           30

Gln Leu Arg Glu Tyr Gly His Arg Met Val Asp Phe Ile Ala Asp Tyr
 35           40           45

Tyr Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln Val Gln Pro Gly
 50           55           60

Tyr Leu His Gln Leu Leu Pro Asp Ser Ala Pro Asp His Pro Glu Thr
 65           70           75           80

Leu Asp Gln Val Leu Asp Asp Val Arg Ala Lys Ile Leu Pro Gly Val
 85           90           95

Thr His Trp Gln Ser Pro Gly Phe Phe Ala Tyr Tyr Pro Ser Asn Ser
100          105          110

Ser Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala Gly Leu Gly Ile
115          120          125

Val Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr Glu Leu Glu Met
130          135          140

Ile Val Leu Asp Trp Leu Ala Lys Leu Leu Asn Leu Pro Lys Glu Phe
145          150          155          160

Leu Ser Lys Gly Asn Gly Gly Gly Val Ile Gln Gly Ser Ala Ser Glu
165          170          175

Ala Val Leu Val Val Leu Ile Ala Ala Arg Asp Lys Val Leu Arg Ser
180          185          190

Ala Gly Lys Asn Ala Leu Gly Lys Leu Val Val Tyr Ser Ser Asp Gln
195          200          205

Thr His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala Gly Ile His Pro
210          215          220

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Glu Asn Cys Arg Val Leu Glu Thr Asp Ala Ser Thr Asn Tyr Ala Leu
 225 230 235 240
 Arg Pro Glu Leu Leu Gln Glu Ala Val Ser Lys Asp Leu Lys Ala Gly
 245 250 255
 Leu Ile Pro Phe Phe Leu Cys Ala Asn Val Gly Thr Thr Ser Ser Thr
 260 265 270
 Ala Val Asp Pro Leu Ala Ala Leu Gly Lys Ile Ala Asn Ser Asn Glu
 275 280 285
 Ile Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys
 290 295 300
 Pro Glu Tyr Arg Lys Tyr Ile Asp Gly Val Glu Thr Ala Asp Ser Phe
 305 310 315 320
 Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu
 325 330 335
 Leu Trp Val Lys Glu Gln Asp Ser Leu Thr Glu Ala Leu Ser Thr Asn
 340 345 350
 Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn Leu Val Val Asp
 355 360 365
 Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys
 370 375 380
 Leu Trp Met Val Leu Arg Leu Tyr Gly Ala Glu Thr Leu Lys Ser Tyr
 385 390 395 400
 Ile Arg Asn His Ile Lys Leu Ala Lys Tyr Phe Glu Lys Leu Val Ser
 405 410 415
 Gln Asp Pro Asn Phe Glu Ile Val Thr Pro Arg Ile Phe Ser Leu Val
 420 425 430
 Cys Phe Arg Leu Val Pro Lys Asn Asp Asp Glu Lys Lys Cys Asn Asn
 435 440 445
 Gln Asn Arg Lys Leu Leu Glu Ala Ala Asn Ser Ser Gly Lys Leu Phe
 450 455 460
 Met Ser His Thr Ala Leu Ser Gly Lys Ile Val Leu Arg Cys Ala Ile
 465 470 475 480
 Gly Ala Pro Leu Thr Glu Glu Lys His Met Lys Glu Ala Trp Lys Val
 485 490 495
 Ile Gln Asp Glu Ala Ser Phe Leu Leu His Lys
 500 505

<210> SEQ ID NO 29

<211> LENGTH: 486

<212> TYPE: PRT

<213> ORGANISM: Malus domestica

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 22636618

<400> SEQUENCE: 29

Met Ser Gly Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu Asn Ala
 1 5 10 15
 His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asp
 20 25 30
 Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Arg Asp Leu Leu
 35 40 45
 Pro Asp Ser Ala Pro Thr His Pro Glu Ser Leu Gln Gln Val Phe Asp
 50 55 60
 Asp Ile Gln Ala Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser Pro
 65 70 75 80

Asn	Phe	Phe		Gly	Tyr	Tyr	Pro	Ser	Asn	Ser	Ser	Val	Ala	Gly	Phe	Leu
				85						90					95	
Gly	Glu	Met	Leu	Ser	Ala	Gly	Leu	Asn	Ile	Val	Gly	Phe	Ser	Trp	Ile	
			100					105					110			
Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp	Phe	
		115					120					125				
Ala	Lys	Met	Leu	Lys	Leu	Pro	Glu	Glu	Phe	Leu	Ser	Ala	Gly	Gln	Gly	
		130				135					140					
Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val	Leu	
145					150					155					160	
Leu	Ala	Ala	Arg	Asp	Arg	Ile	Leu	Arg	Ala	Glu	Gly	Lys	Lys	Ala	Leu	
			165						170					175		
Glu	Lys	Leu	Val	Val	Tyr	Ala	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu	Gln	
			180					185					190			
Lys	Ala	Cys	Gln	Ile	Gly	Gly	Ile	His	Pro	Glu	Asn	Cys	Arg	Val	Leu	
		195				200						205				
Ser	Thr	Asp	Ser	Ser	Thr	Asn	Tyr	Ala	Leu	Ser	Pro	Asn	Val	Leu	Asn	
		210				215					220					
Glu	Ala	Ile	Ser	Asn	Asp	Ile	Ala	Ser	Gly	Leu	Val	Pro	Phe	Phe	Leu	
225					230					235					240	
Cys	Ala	Thr	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Leu	
			245						250					255		
Glu	Leu	Gly	Lys	Ile	Thr	Lys	Ser	Asn	Gly	Met	Trp	Phe	His	Val	Asp	
			260					265					270			
Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	His	His	
		275					280					285				
Ile	Asp	Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	
		290				295					300					
Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu	Leu	Trp	Ile	Lys	Asp	Arg	
305				310						315					320	
Asn	Ala	Leu	Val	Gln	Ala	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys	Asn	
				325					330					335		
Lys	Ala	Ser	Gln	Ala	Asn	Leu	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	
			340				345						350			
Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	
		355					360					365				
Leu	Tyr	Gly	Leu	Glu	Asn	Leu	Gln	Ser	Tyr	Ile	Arg	Asn	His	Ile	Asp	
		370				375					380					
Leu	Ala	Lys	Cys	Phe	Glu	Asp	Leu	Val	Ala	Gln	Asp	Ser	Arg	Phe	Glu	
385				390						395					400	
Ile	Val	Thr	Pro	Arg	Ile	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu	Pro	
			405						410					415		
Pro	His	Asn														

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<210> SEQ ID NO 30
<211> LENGTH: 483
<212> TYPE: PRT
<213> ORGANISM: Linum usitatissimum
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 23178995

<400> SEQUENCE: 30

Met Gly Gly Tyr Arg Ser Leu Asn Leu Ile Phe Ile Phe Ile Ser Phe
 1             5             10             15

Val Ala Asp Ile Arg Asp Leu Gly Tyr Asn Thr Lys Glu Gly Asp Asp
 20             25             30

Gly Gly Gly Ala Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Gln Asn
 35             40             45

Ala His Gln Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu
 50             55             60

Thr Tyr Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Glu Leu
 65             70             75             80

Leu Pro Asp Ser Ala Pro Asn Arg Pro Glu Ser Leu Gln Ser Val Leu
 85             90             95

Asp Asp Val Gln Ser Lys Ile Met Pro Gly Val Thr His Trp Gln Ser
 100            105            110

Pro Asn Tyr Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Val Ala Gly Phe
 115            120            125

Leu Gly Glu Met Leu Ser Ala Gly Ile Asn Met Val Gly Phe Ser Trp
 130            135            140

Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp
 145            150            155            160

Leu Gly Lys Leu Leu Lys Leu Pro Glu Glu Phe Leu Ser Ser Gly His
 165            170            175

Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Ile Leu Val Val
 180            185            190

Leu Leu Ala Ala Arg Asp Lys Met Leu Arg Lys Phe Gly Lys Ser Ala
 195            200            205

Leu Glu Lys Leu Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu
 210            215            220

Gln Lys Ala Cys Gln Ile Gly Gly Ile Tyr Pro Glu Asn Cys Arg Leu
 225            230            235            240

Leu Lys Thr Asp Ser Ser Val Asn Tyr Ser Leu Thr Pro Glu Leu Val
 245            250            255

Ser Glu Ala Val Ser Gln Asp Ile Ser Ala Gly Leu Ile Pro Phe Phe
 260            265            270

Leu Cys Gly Thr Val Gly Thr Thr Ser Ser Ala Thr Val Asp Pro Leu
 275            280            285

Gly Thr Leu Gly Lys Ile Ala Lys Asn Asn Asp Met Trp Phe His Val
 290            295            300

Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg Gln
 305            310            315            320

Tyr Leu Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn Ala His
 325            330            335

Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Thr Leu Trp Val Lys Asp
 340            345            350

Lys Ser Ala Leu Ile Gln Ala Leu Ser Thr Asn Pro Glu Phe Leu Lys
 355            360            365

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Asn Lys Ala Ser Gln Ala Asn Leu Val Val Asp Tyr Lys Asp Trp Gln
 370 375 380
 Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu
 385 390 395 400
 Arg Leu Tyr Gly Val Glu Asn Leu Gln Gln Tyr Leu Arg Asn His Ile
 405 410 415
 Glu Leu Ala Arg His Phe Glu Glu Cys Val Asn His Asp Pro Arg Phe
 420 425 430
 Glu Ala Leu Ser Gly Lys Tyr Thr Leu Arg Val Ala Ile Gly Ala Pro
 435 440 445
 Leu Thr Glu Lys Arg His Val Ala Ala Ala Leu Lys Val Leu Gln Asp
 450 455 460
 Glu Ala Thr Ser Leu Leu Val Ala Thr Ser Pro Leu Leu Glu Asn Gly
 465 470 475 480
 Asn Ser Ser

<210> SEQ ID NO 31
 <211> LENGTH: 490
 <212> TYPE: PRT
 <213> ORGANISM: Eutrema salsugineum
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 20200788

<400> SEQUENCE: 31

Met Glu Asn Gly Asn Lys Asn Val Leu Lys Pro Met Asp Ser Glu Gln
 1 5 10 15
 Leu Arg Glu Tyr Gly His Arg Met Val Asp Phe Ile Ala Asp Tyr Tyr
 20 25 30
 Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr
 35 40 45
 Leu His Asn Leu Leu Pro Asp Ser Ala Pro Asp Gln Pro Glu Thr Leu
 50 55 60
 Glu Glu Val Leu Asp Asp Val Lys Gly Lys Ile Leu Pro Gly Val Thr
 65 70 75 80
 His Trp Gln Ser Pro Ser Phe Phe Ala Tyr Tyr Pro Ser Asn Ser Ser
 85 90 95
 Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala Gly Leu Gly Ile Val
 100 105 110
 Gly Phe Ser Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile
 115 120 125
 Val Leu Asp Trp Leu Ala Lys Leu Leu Asn Leu Pro Glu Gln Phe Leu
 130 135 140
 Ser Arg Gly Asn Gly Gly Gly Val Ile Gln Gly Ser Ala Ser Glu Ala
 145 150 155 160
 Glu Leu Val Val Leu Ile Ala Ala Arg Asp Lys Val Leu Arg Ser Val
 165 170 175
 Gly Lys Lys Ala Leu Glu Lys Leu Val Val Tyr Ser Ser Asp Gln Thr
 180 185 190
 His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala Gly Ile His Pro Glu
 195 200 205
 Asn Cys Arg Val Leu Lys Ala Asp Tyr Ser Thr Asn Tyr Ala Leu Arg
 210 215 220
 Pro Glu Thr Leu Gln Glu Ala Val Ser Lys Asp Leu Glu Ala Gly Leu
 225 230 235 240
 Ile Pro Phe Phe Leu Cys Ala Asn Val Gly Thr Thr Ser Ser Thr Ala

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245				250				255							
Val	Asp	Pro	Leu	Ala	Ala	Leu	Gly	Glu	Ile	Ala	Lys	Ser	Asn	Glu	Met
			260												270
Trp	Phe	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro
			275												285
Glu	Tyr	Arg	Gln	Tyr	Ile	Asp	Gly	Val	Glu	Thr	Ala	Asp	Ser	Phe	Asn
			290												300
Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu	Leu
															320
Trp	Val	Lys	Asp	Gln	Tyr	Ala	Leu	Thr	Glu	Ala	Arg	Ser	Thr	Asn	Pro
															335
Glu	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asn	Leu	Val	Val	Asp	Tyr
															350
Lys	Asp	Trp	Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu
															365
Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Ser	Glu	Asn	Leu	Lys	Ser	Tyr	Ile
															380
Arg	Asn	His	Ile	Lys	Leu	Ala	Lys	Asp	Phe	Glu	Gln	Leu	Val	Ser	Glu
															400
Asp	Pro	Asn	Phe	Glu	Ile	Val	Thr	Pro	Arg	Ile	Phe	Ser	Leu	Val	Cys
															415
Phe	Arg	Ile	Val	Pro	Ala	Glu	Asn	Asp	Glu	Lys	Lys	Cys	Asn	Asn	Gln
															430
Asn	Arg	Asn	Leu	Leu	Asp	Ala	Val	Asn	Ser	Ser	Gly	Lys	Leu	Phe	Leu
															445
Ser	His	Thr	Ala	Leu	Ser	Gly	Lys	Ile	Val	Leu	Arg	Cys	Ala	Ile	Gly
															460
Ala	Pro	Leu	Thr	Glu	Glu	Lys	His	Val	Lys	Glu	Ala	Trp	Lys	Val	Ile
															480
Gln	Glu	Glu	Ala	Ser	Tyr	Leu	Leu	Arg	Lys						
															490

<210> SEQ ID NO 32

<211> LENGTH: 488

<212> TYPE: PRT

<213> ORGANISM: Populus trichocarpa

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 27022899

<400> SEQUENCE: 32

Met	Glu	Ser	Lys	Gly	Leu	Gln	Pro	Met	Asp	Ser	Glu	Gln	Leu	Arg	Glu
1				5					10					15	
Asn	Ala	His	Lys	Met	Val	Asp	Phe	Ile	Ala	Asp	Tyr	Tyr	Lys	Ser	Ile
				20				25					30		
Glu	Asn	Phe	Pro	Val	Leu	Ser	Gln	Val	Glu	Pro	Gly	Tyr	Leu	Arg	Glu
				35			40					45			
Leu	Leu	Pro	Asp	Ser	Ala	Pro	Asn	Gln	Pro	Glu	Thr	Leu	Gln	Asn	Val
				50			55				60				
Leu	Asp	Asp	Val	Gln	Ala	Lys	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln
				65		70				75				80	
Ser	Pro	Ser	Tyr	Phe	Ala	Tyr	Tyr	Pro	Ser	Asn	Ser	Ser	Val	Ala	Gly
				85				90						95	
Phe	Leu	Gly	Glu	Met	Leu	Ser	Ala	Gly	Ile	Asn	Met	Val	Gly	Phe	Ser
				100				105					110		
Trp	Ile	Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Met	Ile	Val	Leu	Asp

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115					120					125					
Trp	Leu	Gly	Lys	Leu	Leu	Lys	Leu	Pro	Glu	Asp	Phe	Leu	Ser	Thr	Gly
130					135					140					
Gln	Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val
145					150					155					160
Val	Leu	Leu	Ala	Ala	Arg	Asp	Arg	Val	Leu	Arg	Lys	Leu	Gly	Lys	Asn
				165					170					175	
Ala	Leu	Glu	Lys	Leu	Val	Tyr	Ala	Ser	Asp	Gln	Thr	His	Ser	Ala	
			180				185					190			
Leu	Gln	Lys	Ala	Cys	Gln	Ile	Gly	Gly	Ile	His	Pro	Glu	Asn	Cys	Lys
		195					200					205			
Leu	Leu	Lys	Thr	Gly	Ser	Ser	Thr	Asn	Tyr	Ala	Leu	Ser	Pro	Asp	Leu
		210					215					220			
Leu	Gly	Lys	Ala	Ile	Ser	Asp	Asp	Ile	Ser	Thr	Gly	Leu	Val	Pro	Phe
225						230					235				240
Phe	Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro
				245					250					255	
Leu	Leu	Ser	Leu	Gly	Lys	Ile	Ala	Lys	Asn	Asn	Gly	Ile	Trp	Phe	His
			260					265					270		
Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg
		275					280					285			
Cys	Tyr	Ile	Asp	Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala
		290				295					300				
His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Ala	Leu	Trp	Val	Lys
305					310					315					320
Asp	Arg	Asn	Ala	Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu
				325					330					335	
Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asn	Met	Val	Val	Asp	Tyr	Lys	Asp	Trp
			340					345					350		
Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val
		355					360					365			
Leu	Arg	Leu	Tyr	Gly	Leu	Glu	Asn	Leu	Gln	Cys	Tyr	Ile	Arg	Asn	His
		370				375					380				
Ile	Asn	Leu	Ala	Lys	Tyr	Phe	Glu	Gly	Leu	Val	Ala	Ala	Asp	Ser	Arg
385					390					395					400
Phe	Glu	Val	Val	Thr	Pro	Arg	Ile	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu
				405				410						415	
Leu	Pro	Pro	Asn	Asn	Asn	Glu	Asp	His	Gly	Asn	Asn	Leu	Asn	His	Asp
			420					425					430		
Leu	Leu	Asp	Ala	Val	Asn	Ser	Thr	Gly	Lys	Ile	Phe	Ile	Ser	His	Thr
		435					440					445			
Val	Leu	Ser	Gly	Lys	Tyr	Ile	Leu	Arg	Phe	Ala	Val	Gly	Ala	Pro	Leu
		450				455					460				
Thr	Glu	Glu	Arg	His	Val	Thr	Ala	Ala	Trp	Lys	Val	Leu	Gln	Asp	Glu
465					470					475					480
Ala	Ser	Ala	Leu	Leu	Gly	Ser	Leu								
				485											

<210> SEQ ID NO 33

<211> LENGTH: 500

<212> TYPE: PRT

<213> ORGANISM: Brachypodium stacei

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 06G160800.1

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<400> SEQUENCE: 33

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Met Asp Gly Ser Thr Thr Ser Asn Gly Asp Gly Gly Gly Gly Trp Met
1      5      10      15

Arg Pro Met Asp Glu Glu Gln Leu Arg Glu Cys Gly His Arg Met Val
20      25      30

Asp Phe Ile Ala Asp Tyr Tyr Lys Ser Ile Glu Thr Tyr Pro Val Leu
35      40      45

Ser Gln Val Gln Pro Gly Tyr Leu Lys Glu Leu Leu Pro Asp Ser Ala
50      55      60

Pro Asn Gln Pro Asp Thr Leu Asp Ala Leu Phe Asp Asp Ile Gln Glu
65      70      75      80

Lys Ile Val Pro Gly Val Thr His Trp Gln Ser Pro Asn Tyr Phe Ala
85      90      95

Tyr Tyr Pro Ser Asn Ser Ser Thr Ala Gly Phe Leu Gly Glu Met Leu
100     105     110

Ser Ala Ala Phe Asn Ile Val Gly Phe Ser Trp Ile Thr Ser Pro Ala
115     120     125

Ala Thr Glu Leu Glu Val Ile Val Leu Asp Trp Val Ala Lys Met Leu
130     135     140

Lys Leu Pro Ser Gln Phe Leu Ser Ala Gly Leu Gly Gly Gly Val Ile
145     150     155     160

Gln Gly Thr Ala Ser Glu Ala Ile Leu Val Val Leu Leu Ser Ala Arg
165     170     175

Asp Arg Thr Leu Arg Lys His Gly Lys Lys Ser Leu Glu Lys Leu Val
180     185     190

Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys Gln
195     200     205

Ile Ala Gly Ile Phe Ser Asp Asn Ile Arg Ile Val Lys Ala Asp Cys
210     215     220

Ser Met Asn Tyr Ala Val Thr Pro Gly Ser Val Ser Glu Ala Ile Ser
225     230     235     240

Ile Asp Leu Ser Ser Gly Leu Ile Pro Phe Phe Ile Cys Ala Thr Leu
245     250     255

Gly Thr Thr Ser Ser Ser Ala Val Asp Pro Leu His Glu Leu Gly Gln
260     265     270

Ile Ala Gln Ala His Asp Met Trp Phe His Ile Asp Ala Ala Tyr Ala
275     280     285

Gly Ser Ala Cys Ile Cys Pro Glu Tyr Gln Gln Tyr Leu Asn Gly Val
290     295     300

Glu Glu Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr
305     310     315     320

Asn Phe Asp Cys Ser Leu Leu Trp Val Lys Asp Arg Asn Tyr Leu Ile
325     330     335

Gln Ala Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln
340     345     350

Glu Asn Ser Val Ile Asp Phe Lys Asp Trp Gln Ile Pro Leu Gly Arg
355     360     365

Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Val
370     375     380

Glu Asn Leu Gln Ser Tyr Ile Arg Lys His Ile Gln Leu Ala Gln Arg
385     390     395     400

Phe Glu Gln Leu Val Ile Ser Asp Ser Arg Phe Glu Val Val Thr Pro
405     410     415

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Arg Asn Phe Ser Leu Val Cys Phe Cys Leu Val Pro Pro Thr Ser Glu
420 425 430

Val Asp Asn Gly His Lys Leu Asn Tyr Asp Leu Met Asp Ser Val Asn
435 440 445

Ser Ser Gly Lys Ile Phe Ile Ser His Thr Val Leu Ser Gly Lys Phe
450 455 460

Val Leu Arg Phe Ala Val Gly Ala Pro Leu Thr Glu Glu Gln His Val
465 470 475 480

Asn Ala Ala Trp Lys Leu Leu Gln Asp Glu Ala Thr Lys Leu Leu Gly
485 490 495

Ser Val Val Val
500

<210> SEQ ID NO 34
<211> LENGTH: 503
<212> TYPE: PRT
<213> ORGANISM: Physcomitrella patens
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: Pp3c4_30790V3.1

<400> SEQUENCE: 34

Met Gly Ser Glu Ala Gly Ser Arg Ser Ser Leu Thr Lys Pro Phe Asp
1 5 10 15

Pro Glu Glu Phe Arg Lys His Ala His Arg Met Val Asp Phe Ile Ala
20 25 30

Asp Tyr His Arg Asp Ile Glu Asn Phe Pro Val Gln Ser Gln Val Glu
35 40 45

Pro Gly Tyr Leu Gln Lys Leu Leu Pro Glu Asn Ala Pro Asp Glu Pro
50 55 60

Glu Ser Leu Asp Asp Ile Leu Ala Asp Val Gln Ser Lys Ile Val Pro
65 70 75 80

Gly Val Thr His Trp Gln Ser Pro Asn Phe Tyr Gly Tyr Tyr Pro Ser
85 90 95

Asn Gly Ser Thr Ala Gly Phe Leu Gly Glu Met Leu Ser Gly Gly Phe
100 105 110

Asn Ile Ile Gly Phe Ser Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu
115 120 125

Glu Ile Ile Val Met Asp Trp Leu Gly Lys Leu Leu Lys Leu Pro Asn
130 135 140

Glu Phe Leu Ser Ser Gly Lys Gly Gly Val Ile Gln Gly Thr Ala
145 150 155 160

Ser Glu Ala Val Leu Val Val Met Leu Ala Ala Arg Lys Arg Ala Val
165 170 175

Glu Lys Leu Thr Lys Glu Gln Gly Ile Ser Glu Phe Glu Ala Leu Ala
180 185 190

Lys Leu Val Ala Tyr Thr Ser Asp Gln Ala His Ser Cys Val Asn Lys
195 200 205

Ala Ser Gln Ile Ala Gly Ile Ser Ile Glu Asn Leu Arg Leu Ile Pro
210 215 220

Thr Asp Val Ser Thr Asn Tyr Ala Met Ser Ser Lys Val Leu Ala Asn
225 230 235 240

Thr Leu Ala Asn Asp Val Lys Ala Gly Leu Val Pro Phe Phe Leu Cys
245 250 255

Gly Val Ile Gly Ser Thr Ser Ser Ala Ala Val Asp Pro Leu Ser Glu
260 265 270

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Leu Gly Asp Leu Ala Gln Glu Tyr Gly Met Trp Phe His Val Asp Gly
 275 280 285
 Ala Tyr Ala Gly Asn Ala Cys Ile Cys Pro Glu Phe Arg Pro Tyr Leu
 290 295 300
 Asn Gly Val Glu Lys Ala Asp Ser Phe Asp Met Asn Pro His Lys Trp
 305 310 315 320
 Leu Leu Thr Asn Phe Asp Cys Ser Thr Leu Trp Val Lys Asn Pro Ser
 325 330 335
 Leu Leu Val Asp Ala Leu Ser Thr Asn Pro Val Phe Leu Arg Asn Lys
 340 345 350
 Gln Ser Asp Asn Asn Leu Val Val Asp Tyr Lys Asp Trp Gln Ile Pro
 355 360 365
 Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu Arg Met
 370 375 380
 Tyr Gly Ser Asn Gly Leu Arg Ser Tyr Ile Thr Asn His Cys Asn Leu
 385 390 395 400
 Ala Lys His Phe Glu Glu Leu Leu Arg Thr Asp Ser Arg Phe Glu Val
 405 410 415
 Val Ala Pro Arg Val Phe Ser Leu Val Cys Phe Arg Leu Lys Ser Pro
 420 425 430
 Ala Asn Asp Ala Asp Asn Ser Cys Ser Leu Ser Ala Lys Leu Val Asp
 435 440 445
 Ala Leu Asn Ser Asp Gly Asn Ile Leu Ile Thr Asn Thr Val Leu Gly
 450 455 460
 Gly Arg Tyr Thr Ile Arg Phe Thr Val Gly Ala Ser Arg Thr Glu Leu
 465 470 475 480
 Arg His Val Asp Ala Ala Trp Lys Val Ile Gln Gln Leu Ala Ser Lys
 485 490 495
 Leu Leu Lys Glu Cys Ser Ser
 500

<210> SEQ ID NO 35
 <211> LENGTH: 487
 <212> TYPE: PRT
 <213> ORGANISM: Ananas comosus
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 33033299

<400> SEQUENCE: 35

Met Glu Ser Glu Leu Lys Pro Met Asp Ser Glu Gln Leu Arg Glu Tyr
 1 5 10 15
 Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Met Ile Glu
 20 25 30
 Ser Phe Pro Val Leu Ser Gln Val Lys Pro Gly Tyr Leu Lys Glu Leu
 35 40 45
 Leu Pro Asp Ser Ala Pro Cys Lys Pro Glu Asn Leu Glu Asp Val Phe
 50 55 60
 Asp Asp Ile Arg Gln Lys Ile Ile Pro Gly Ile Thr His Trp Gln Ser
 65 70 75 80
 Pro Asp Tyr Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Thr Ala Gly Phe
 85 90 95
 Leu Gly Glu Met Leu Ser Ala Gly Phe Asn Ile Ile Gly Phe Ser Trp
 100 105 110
 Ile Ala Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp
 115 120 125

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Phe Ala Lys Met Leu Lys Leu Pro Glu Gln Phe Leu Ser Thr Gly Gln
 130 135 140
 Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val
 145 150 155 160
 Leu Leu Ala Ala Arg Asp Lys Ile Leu Leu Lys Ala Gly Arg Lys Ser
 165 170 175
 Leu Glu Lys Leu Val Val Tyr Cys Ser Asp Gln Thr His Ser Ala Met
 180 185 190
 Gln Lys Ala Cys Gln Ile Ala Gly Ile Phe Pro Glu Asn Phe Arg Val
 195 200 205
 Leu Lys Thr Asp Ser Ser Ser Asn Tyr Ala Leu Leu Pro Glu Val Leu
 210 215 220
 Ser Glu Ala Ile Ser Lys Asp Leu Ser Phe Gly Leu Ile Pro Phe Phe
 225 230 235 240
 Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Ala Ala Val Asp Pro Leu
 245 250 255
 Leu Lys Leu Gly Asn Ile Ser Lys Val His Asp Met Trp Phe His Val
 260 265 270
 Asp Ala Ala His Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg His
 275 280 285
 His Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Cys Met Asn Ala His
 290 295 300
 Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu Leu Trp Val Lys Asp
 305 310 315 320
 Arg Ser Ala Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys
 325 330 335
 Asn Lys Ala Ser Gln Glu Asn Ser Val Val Asp Phe Lys Asp Trp Gln
 340 345 350
 Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu
 355 360 365
 Arg Leu Tyr Gly Leu Glu Asn Leu Gln Ser Tyr Ile Arg Glu His Ile
 370 375 380
 Lys Leu Ala Glu Gln Phe Glu Gln Leu Ile Ser Ser Asp Ser Arg Phe
 385 390 395 400
 Glu Ile Val Ala Pro Arg Thr Phe Ser Leu Val Cys Phe Arg Leu Leu
 405 410 415
 Pro Pro Leu Tyr Asp Gln Asp Asp Gly Tyr Lys Leu Asn Tyr Asn Leu
 420 425 430
 Leu Asp Ala Val Asn Arg Ser Gly Lys Ile Phe Met Ser His Thr Val
 435 440 445
 Leu Ser Gly Lys Phe Val Leu Arg Phe Ala Ile Gly Ala Pro Leu Thr
 450 455 460
 Glu Glu Arg His Val Val Ala Ala Trp Lys Val Leu Gln Asp Glu Ala
 465 470 475 480
 Thr Ile Leu Leu Arg Gly Ser
 485

<210> SEQ ID NO 36
 <211> LENGTH: 511
 <212> TYPE: PRT
 <213> ORGANISM: *Zostera marina*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 33182387
 <400> SEQUENCE: 36

Met 1	Leu	Asn	Gly	Asn 5	Met	Gly	Glu	Asn	Glu 10	Pro	Phe	Lys	Pro	Met 15	Asp
Ser	Glu	Gln	Leu 20	Arg	Glu	Tyr	Gly	His 25	Lys	Met	Val	Asp	Phe 30	Ile	Ala
Asp	Tyr	Tyr	Lys 35	Ser	Ile	Glu	Lys 40	Phe	Pro	Val	Leu	Ser 45	Gln	Val	Gln
Pro	Tyr	Tyr	Leu 50	Lys	Asp	Leu 55	Leu	Pro	Asp	Ala	Ala 60	Pro	Asp	Gln	Pro
Glu 65	Lys	Phe	Gln	Asp	Val 70	Leu	Asp	Asp	Ile	Thr 75	Lys	Lys	Ile	Ile	Pro 80
Gly	Val	Thr	His 85	Trp	Gln	Ser	Pro	Asn	Phe 90	Phe	Gly	Tyr	Tyr	Pro 95	Gly
Asn	Ser	Ser	Ile 100	Ala	Gly	Phe	Leu	Gly 105	Glu	Met	Ile	Cys	Ser 110	Gly	Leu
Asn	Val	Ile	Gly 115	Phe	Ser	Trp	Ile 120	Thr	Ser	Pro	Ala	Ser 125	Thr	Glu	Leu
Glu 130	Val	Ile	Val	Leu	Asp	Trp 135	Leu	Ala	Lys	Leu	Leu 140	Asn	Leu	Pro	Asp
Gln 145	Phe	Leu	Ser	Ser	Gly 150	His	Gly	Gly	Gly	Val 155	Ile	Gln	Gly	Thr	Ala 160
Ser	Glu	Ala	Ile 165	Leu	Val	Val	Leu	Leu	Ala 170	Ala	Arg	Asp	Lys	Ile	Leu 175
Gly	Arg	Ile	Gly 180	Arg	Asn	Ser	Leu	Asp 185	Lys	Leu	Val	Val	Tyr 190	Ser	Ser
Asp	Gln	Val	His 195	Ala	Ala	Phe	Lys 200	Lys	Ala	Cys	Gln	Ile 205	Ala	Gly	Ile
Tyr 210	Thr	Glu	Asn	Phe	Arg	Val 215	Leu	Lys	Thr	Asp	Ala 220	Ser	Ser	Gly	Tyr
Gly 225	Ile	Asp	Pro	Lys 230	Lys	Phe	Asp	Gln	Ala	Ile 235	His	Asp	Asp	Met	Glu 240
Ala	Gly	Leu	Ile 245	Pro	Phe	Phe	Leu	Cys	Ser 250	Thr	Val	Gly	Thr	Thr	Ser 255
Ser	Ala	Ser	Val 260	Asp	Pro	Leu	Val	Glu 265	Ile	Gly	Gln	Ile	Thr	Glu	Glu
Asn	Asp	Met	Trp 275	Phe	His	Val	Asp 280	Ala	Ala	Tyr	Ala	Gly 285	Ser	Ala	Cys
Ile 290	Cys	Pro	Glu	Tyr	Arg	His 295	Tyr	Leu	Asp	Gly	Val 300	Glu	Tyr	Ala	Asp
Ser 305	Phe	Cys	Met	Asn 310	Ala	His	Lys	Trp	Leu	Leu	Thr 315	Asn	Phe	Asp	Cys 320
Ser	Ala	Leu	Trp 325	Val	Lys	Asp	Ser	Ser	Ala 330	Leu	Val	Asn	Ser	Leu	Ser 335
Thr	Asn	Pro	Glu 340	Phe	Leu	Lys	Asn	Lys	Met 345	Ser	Glu	Gln	Lys 350	Lys	Val
Val	Asp	Phe 355	Lys	Asp	Trp	Gln 360	Ile	Pro	Leu	Gly	Arg	Arg 365	Phe	Arg	Ser
Leu 370	Lys	Leu	Trp	Met	Val	Leu 375	Arg	Leu	Tyr	Gly	Ala 380	Glu	Asn	Leu	Arg
Glu 385	Tyr	Ile	Arg	Asn 390	His	Ile	Lys	Leu	Ala	Asn 395	Leu	Phe	Glu	Gln	Leu 400
Val	Arg	Ser	Asp 405	Ser	Arg	Phe	Glu	Ile	Val 410	Cys	Pro	Thr	Leu	Phe	Ser 415

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Leu Val Cys Phe Arg Phe Leu Pro Ser Asn Asp Asp Asn Asp Gly Tyr
420 425 430

Glu Leu Asn Ser Met Leu Leu Asp Ala Val Asn Ser Thr Gly Gln Leu
435 440 445

Phe Phe Thr His Thr Ile Ile Ser Asp Lys Tyr Ile Leu Arg Phe Ala
450 455 460

Val Gly Ala Ala Leu Thr Glu Glu Arg His Val Arg Glu Ser Trp Lys
465 470 475 480

Val Ile Gln Asn Gln Ala Thr Ile Ile Ser Arg Gln His Ile Leu Ser
485 490 495

Lys Thr Asn Met Lys Ser Lys Cys Glu Gly Met Ile Ala Asn Glu
500 505 510

<210> SEQ ID NO 37
<211> LENGTH: 497
<212> TYPE: PRT
<213> ORGANISM: *Daucus carota* subsp. *sativus*
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 36055203

<400> SEQUENCE: 37

Met Asp Gly Val Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu Asn
1 5 10 15

Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu
20 25 30

Thr Phe Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Asp Leu
35 40 45

Leu Pro His Ser Ala Pro Asp Gln Pro Glu Ser Leu Gln Asn Ile Leu
50 55 60

Asp Asp Ile Gln Ala Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser
65 70 75 80

Pro Asn Tyr Phe Ala Tyr Phe Pro Ser Asn Ser Ser Val Ala Gly Phe
85 90 95

Leu Gly Glu Met Leu Ser Ala Gly Ile Asn Met Val Gly Phe Ser Trp
100 105 110

Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp
115 120 125

Leu Ala Lys Leu Leu Lys Leu Pro Asp His Phe Leu Ser Thr Gly Gln
130 135 140

Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val
145 150 155 160

Leu Leu Ala Ala Arg Asp Lys Val Leu Arg Ile Thr Gly Lys Asp Ala
165 170 175

Leu Gly Lys Leu Val Val Tyr Cys Ser Asp Gln Thr His Ser Ala Leu
180 185 190

Gln Lys Ala Cys Gln Ile Ala Gly Ile His Pro Gly Asn Cys Arg Val
195 200 205

Leu Lys Thr Glu Ser Cys Asn Asp Tyr Ser Leu Ser Pro Glu Thr Phe
210 215 220

Glu Gln Ala Ile Ser Thr Asp Val Ala Ser Gly Leu Ile Pro Leu Leu
225 230 235 240

Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Thr Ala Val Asp Pro Leu
245 250 255

Leu Glu Leu Gly Lys Ile Thr Lys Met Lys Gly Ile Trp Leu His Val
260 265 270

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Asp Ala Ala Tyr Ala Gly Ser Ala Cys Val Cys Pro Glu Phe Arg His
 275 280 285
 Tyr Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn Ala His
 290 295 300
 Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Ala Leu Trp Val Lys Asp
 305 310 315 320
 Arg Ser Ala Leu Ile His Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys
 325 330 335
 Asn Lys Ala Ser Gln Glu Asn Leu Val Val Asp Tyr Lys Asp Trp Gln
 340 345 350
 Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu
 355 360 365
 Arg Leu Tyr Gly Leu Glu Asn Leu Gln Ser Tyr Ile Arg Asn His Ile
 370 375 380
 Gln Leu Ala Ala Thr Phe Glu Ser Phe Val Thr Glu Asp Pro Arg Phe
 385 390 395 400
 Glu Val Val Ala Pro Arg Lys Phe Ala Leu Val Cys Phe Arg Leu Leu
 405 410 415
 Pro Pro Ser His Lys Asp Glu Asp Cys Ser Asn Gln Leu Asn Arg Asp
 420 425 430
 Leu Leu Asp Ala Val Asn Ala Thr Gly Lys Ala Phe Val Ser His Thr
 435 440 445
 Ala Leu Ser Gly Arg Tyr Val Val Arg Phe Ala Ile Gly Ala Pro Leu
 450 455 460
 Thr Glu Glu Ser His Ile Ile Glu Ala Trp Lys Ile Phe Gln Glu Val
 465 470 475 480
 Ala Thr Val Leu Leu Lys Ser Leu Lys Met Asn His Thr Arg Pro Leu
 485 490 495

Asn

<210> SEQ ID NO 38
 <211> LENGTH: 468
 <212> TYPE: PRT
 <213> ORGANISM: Trifolium pratense
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 35974269

<400> SEQUENCE: 38

Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asn Phe Pro
 1 5 10 15
 Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Gly Lys Leu Leu Pro Asp
 20 25 30
 Ser Ala Pro Thr Tyr Pro Thr Thr Leu Glu His Val Leu Asn Asp Val
 35 40 45
 Gln His Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser Pro Asn Tyr
 50 55 60
 Phe Ala Tyr Phe Pro Ser Asn Ser Ser Ile Ala Gly Phe Leu Gly Glu
 65 70 75 80
 Met Leu Ser Ala Gly Ile Asn Ile Val Gly Phe Ser Trp Ile Thr Ser
 85 90 95
 Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Leu Asp Trp Leu Ala Lys
 100 105 110
 Ala Leu Phe Leu Pro Gln Asp Phe Leu Ser Asn Gly Lys Gly Gly Gly
 115 120 125
 Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val Leu Leu Ala

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130					135					140					
Ala	Arg	Asp	Lys	Ile	Leu	Arg	Thr	Val	Gly	Arg	Ser	Ala	Leu	Pro	Lys
145					150					155					160
Leu	Val	Thr	Tyr	Ala	Ser	Asp	His	Val	His	Ser	Ser	Leu	Leu	Lys	Ala
				165					170					175	
Cys	Gln	Ile	Gly	Gly	Leu	Asp	Pro	Glu	Leu	Cys	Arg	Leu	Leu	Lys	Thr
			180					185					190		
Asp	Ser	Ser	Thr	Asn	Phe	Ala	Leu	Ser	Pro	Asp	Val	Leu	Ser	Glu	Ala
			195				200					205			
Ile	Ser	Asn	Asp	Ile	Ala	Ser	Gly	Leu	Ile	Pro	Phe	Phe	Leu	Cys	Ala
			210				215					220			
Asn	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Pro	Ala	Leu
					225		230				235				240
Ala	Lys	Val	Thr	Lys	Thr	Asn	Asn	Ile	Trp	Leu	His	Val	Asp	Ala	Ala
				245					250					255	
Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	His	Phe	Ile	Asp
			260					265					270		
Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Phe
			275					280					285		
Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu	Leu	Trp	Val	Lys	Asp	Arg	Ser	Ala
			290				295				300				
Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys	Asn	Lys	Ala
			305				310				315			320	
Ser	Glu	Gly	Asn	Met	Val	Ile	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Pro	Leu
			325					330					335		
Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	Leu	Tyr
			340					345					350		
Gly	Leu	Glu	Gly	Leu	Arg	Ser	His	Ile	Arg	Asn	His	Ile	Ala	Leu	Ala
			355				360					365			
Ala	Ser	Phe	Glu	Glu	Leu	Val	Val	Gln	Asp	Ala	Arg	Phe	Lys	Val	Val
			370				375					380			
Thr	Pro	Arg	Thr	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu	Pro	Pro	Pro
			385				390				395			400	
Asn	Ser	Glu	Asp	Asn	Gly	Asn	Lys	Leu	Asn	His	Asp	Leu	Leu	Asp	Leu
			405					410					415		
Val	Asn	Ser	Thr	Gly	Ser	Val	Phe	Ile	Thr	His	Thr	Val	Leu	Ser	Gly
			420					425					430		
Glu	Tyr	Ile	Leu	Arg	Leu	Ala	Val	Gly	Ala	Pro	Leu	Thr	Glu	Val	Arg
			435				440					445			
His	Val	Asn	Ala	Ala	Trp	Gln	Ile	Leu	Gln	Glu	Lys	Ala	Thr	Ala	Leu
			450				455				460				
Leu	Glu	Asn	Leu												
465															

<210> SEQ ID NO 39

<211> LENGTH: 475

<212> TYPE: PRT

<213> ORGANISM: Arabidopsis lyrata

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 35943929

<400> SEQUENCE: 39

Met	Asp	Ser	Glu	Gln	Leu	Arg	Glu	Tyr	Gly	His	Arg	Met	Val	Asp	Phe
1				5					10					15	

Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln

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20							25					30				
Val	Gln	Pro	Gly	Tyr	Leu	His	Lys	Leu	Leu	Pro	Asp	Ser	Ala	Pro	Asp	
		35					40					45				
His	Pro	Glu	Thr	Leu	Asp	Gln	Val	Leu	Asp	Asp	Val	Arg	Ala	Lys	Ile	
	50					55					60					
Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser	Pro	Ser	Phe	Phe	Ala	Tyr	Tyr	
65					70					75				80		
Pro	Ser	Asn	Ser	Ser	Val	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu	Ser	Ala	
			85						90					95		
Gly	Leu	Gly	Ile	Val	Gly	Phe	Ser	Trp	Val	Thr	Ser	Pro	Ala	Ala	Thr	
			100					105					110			
Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp	Leu	Ala	Lys	Leu	Leu	Asn	Leu	
		115					120					125				
Pro	Glu	Gln	Phe	Met	Ser	Lys	Gly	Asn	Gly	Gly	Gly	Val	Ile	Gln	Gly	
	130					135					140					
Ser	Ala	Ser	Glu	Ala	Val	Leu	Val	Val	Leu	Ile	Ala	Ala	Arg	Asp	Lys	
145					150					155					160	
Val	Leu	Arg	Ser	Val	Gly	Lys	Asn	Ala	Leu	Gln	Lys	Leu	Val	Val	Tyr	
			165						170					175		
Ser	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu	Gln	Lys	Ala	Cys	Gln	Ile	Ala	
		180						185					190			
Gly	Ile	His	Pro	Glu	Asn	Cys	Arg	Val	Leu	Lys	Thr	Asp	Ser	Ser	Thr	
	195						200					205				
Asn	Tyr	Ala	Leu	Arg	Pro	Glu	Leu	Leu	Gln	Glu	Ala	Val	Ser	Gln	Asp	
	210					215					220					
Leu	Asp	Ala	Gly	Leu	Ile	Pro	Phe	Phe	Leu	Cys	Ala	Asn	Val	Gly	Thr	
225					230					235					240	
Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Ala	Ala	Leu	Gly	Lys	Ile	Ala	
			245						250					255		
Asn	Arg	Asn	Glu	Met	Trp	Phe	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	
		260						265					270			
Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	Gln	Tyr	Ile	Asp	Gly	Val	Glu	Thr	
	275						280					285				
Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	
	290					295					300					
Asp	Cys	Ser	Leu	Leu	Trp	Val	Lys	Asp	Gln	Asp	Ser	Leu	Thr	Leu	Ala	
305					310					315					320	
Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asn	
			325						330					335		
Leu	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe	
		340						345					350			
Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Ser	Glu	Thr	
	355						360					365				
Leu	Lys	Ser	Tyr	Ile	Arg	Asn	His	Ile	Lys	Leu	Ala	Lys	Glu	Phe	Glu	
	370					375					380					
Gln	Leu	Val	Ser	Gln	Asp	Pro	Asn	Phe	Glu	Ile	Val	Thr	Pro	Arg	Ile	
385					390					395					400	
Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Val	Pro	Val	Lys	Asn	Glu	Glu	Lys	
			405						410					415		
Lys	Cys	Asn	Asn	Arg	Asn	Arg	Glu	Leu	Leu	Asp	Ala	Val	Asn	Ser	Ser	
		420						425					430			
Gly	Lys	Leu	Phe	Ile	Ser	His	Thr	Val	Ser	Asp	Phe	Ser	Ser	Phe	Phe	
	435						440						445			

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Leu Leu Phe Phe Leu Leu Asp Asn Val Leu Asn Leu Arg Gly Asn Arg
 450 455 460

Leu Cys Arg Gly Lys Ser Tyr Cys Val Ala Gln
 465 470 475

<210> SEQ ID NO 40
 <211> LENGTH: 505
 <212> TYPE: PRT
 <213> ORGANISM: Sorghum bicolor
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 002G120700.1

<400> SEQUENCE: 40

Met Asp Gly Ser Gly Ser Ser Gly Gly Thr Asn Gly Gly Ser Gly Gly
 1 5 10 15

Asp Gly Ala Gly Trp Leu Arg Pro Met Asp Ala Glu Gln Leu Arg Glu
 20 25 30

Cys Gly His Arg Met Val Asp Phe Val Ala Asp Tyr Tyr Lys Ser Ile
 35 40 45

Glu Thr Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Lys Glu
 50 55 60

Leu Leu Pro Asp Thr Ala Pro Asn Lys Pro Asp Thr Leu Glu Ala Leu
 65 70 75 80

Phe Asp Asp Ile Arg Glu Lys Ile Val Pro Gly Val Thr His Trp Gln
 85 90 95

Ser Pro Asn Tyr Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Thr Ala Gly
 100 105 110

Phe Leu Gly Glu Met Leu Ser Ala Ala Phe Asn Ile Val Gly Phe Ser
 115 120 125

Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Val Ile Val Leu Asp
 130 135 140

Trp Phe Ala Lys Met Leu Arg Leu Pro Ser Gln Phe Leu Ser Thr Ala
 145 150 155 160

Leu Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val
 165 170 175

Val Leu Leu Ala Ala Arg Asp Arg Thr Leu Arg Lys His Gly Lys Thr
 180 185 190

Ser Leu Glu Lys Leu Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala
 195 200 205

Leu Gln Lys Ala Cys Gln Ile Ala Gly Ile Phe Pro Glu Asn Val Arg
 210 215 220

Leu Val Lys Ala Asp Cys Asn Arg Asn Tyr Ala Val Ala Pro Leu Ala
 225 230 235 240

Ile Ser Asp Ala Ile Ala Thr Asp Leu Ser Ser Gly Leu Ile Pro Phe
 245 250 255

Phe Ile Cys Ala Thr Val Gly Thr Thr Ser Ser Ser Ala Val Asp Pro
 260 265 270

Leu Pro Glu Leu Gly Gln Ile Ala Lys Ala Asn Asp Met Trp Leu His
 275 280 285

Ile Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg
 290 295 300

His His Leu Asn Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn Ala
 305 310 315 320

His Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu Leu Trp Val Lys
 325 330 335

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Asp Arg Ser Tyr Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu
 340 345 350
 Lys Asn Lys Ala Ser Glu Ala Asn Ser Val Phe Asp Phe Lys Asp Trp
 355 360 365
 Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val
 370 375 380
 Leu Arg Leu Tyr Gly Val Glu Asn Leu Gln Ser Tyr Ile Arg Lys His
 385 390 395 400
 Ile Glu Leu Ala Lys Glu Phe Glu Gln Leu Val Ile Ser Asp Ser Arg
 405 410 415
 Phe Glu Val Val Thr Pro Arg Thr Phe Ser Leu Val Cys Phe Arg Leu
 420 425 430
 Val Pro Leu Ala Ser Asp Gln Asp Asn Gly Arg Lys Leu Asn Tyr Asp
 435 440 445
 Leu Met Asp Ala Ala Asn Ser Ser Gly Lys Ile Phe Ile Ser His Thr
 450 455 460
 Val Leu Ser Gly Lys Phe Val Leu Arg Phe Ala Val Gly Ala Pro Leu
 465 470 475 480
 Thr Glu Gly Gln His Ile Phe Ser Ala Trp Lys Ile Leu Gln Asp Leu
 485 490 495
 Ala Thr Lys Gln Leu Leu Glu Ser Ser
 500 505

<210> SEQ ID NO 41
 <211> LENGTH: 504
 <212> TYPE: PRT
 <213> ORGANISM: Sphagnum fallax
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 0166s0011.1

<400> SEQUENCE: 41

Met Gly Ser Glu Ala Gly Glu Gly Ser Arg Leu Ser Lys Pro Leu Asp
 1 5 10 15
 Val Glu Glu Phe Arg Lys His Ala His Gln Met Val Asp Phe Val Ala
 20 25 30
 Asp Tyr His Arg Asp Ile Glu Ser Phe Pro Val Arg Ser Gln Val Lys
 35 40 45
 Pro Gly Tyr Leu Arg Pro Leu Leu Pro Asp Ser Ala Pro Ala Glu Pro
 50 55 60
 Glu Thr Val Glu Asp Val Phe Ala Asp Leu Trp Ser Lys Ile Leu Pro
 65 70 75 80
 Gly Leu Thr His Trp Gln Ser Pro Lys Phe Phe Gly Tyr Tyr Pro Cys
 85 90 95
 Asn Val Ser Thr Ala Gly Met Leu Gly Glu Met Leu Cys Gly Gly Leu
 100 105 110
 Asn Val Asn Gly Phe Ser Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu
 115 120 125
 Glu Thr Ile Val Leu Asp Trp Leu Gly Lys Leu Leu His Leu Pro Glu
 130 135 140
 Glu Phe Leu Ser Thr Ser Gly Lys Gly Gly Gly Val Ile Gln Gly Thr
 145 150 155 160
 Ala Ser Glu Ala Val Leu Val Val Met Leu Ala Ala Arg Lys Arg Ala
 165 170 175
 Leu Lys Gln Val Ser Ser Ala Ala Gln Gly Met Ser Glu Ala Glu Ala
 180 185 190

Leu 195	Lys	Leu	Val	Val	Tyr	Ser	Asp	Gln	Thr	His	Ser	Cys	Val		
Ile 210	Lys	Ala	Cys	Gln	Val	Ala	Ser	Ile	Ala	Thr	Glu	Asn	Phe	Arg	Pro
Leu 225	Pro	Thr	Asp	Ala	Ser	Thr	Asn	Phe	Ala	Leu	Ser	Pro	Ala	Val	Val
Arg	Lys	Ala	Ile	Ala	Thr	Asp	Val	Glu	Ala	Gly	Leu	Ile	Pro	Phe	Phe
Leu	Cys	Gly	Thr	Leu	Gly	Thr	Thr	Ser	Ser	Ala	Ala	Val	Asp	Pro	Leu
Glu 275	Glu	Leu	Gly	Asp	Ile	Ala	Lys	Glu	Tyr	Gly	Met	Trp	Tyr	His	Ile
Asp 290	Ala	Ala	Tyr	Ala	Gly	Asn	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	His
Tyr 305	Leu	Asn	Gly	Val	Glu	Lys	Ala	Asp	Ser	Tyr	Asn	Met	Asn	Pro	His
Lys	Trp	Leu	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Thr	Leu	Trp	Met	Lys	Asp
Ser	Glu	Phe	Leu	Leu	Ala	Ala	Leu	Ser	Asn	Lys	Pro	Val	Phe	Leu	Arg
Asn 355	Glu	Ala	Thr	Asp	Asn	Asn	Leu	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln
Ile 370	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp	Met	Val	Met
Arg 385	Leu	Tyr	Gly	Thr	Ser	Gly	Leu	Gln	Ser	Phe	Ile	Arg	Ser	His	Val
Ser	Ser	Ala	Lys	His	Phe	Glu	Ser	Leu	Val	Arg	Ala	Asp	Ser	Arg	Phe
Glu	Val	Met	Ala	Pro	Met	Thr	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Arg
Thr 435	Leu	Pro	Gly	Ser	Gln	Asp	Asn	Ser	Asn	Ser	Leu	Asn	Ser	Lys	Leu
Val 450	Asp	Ala	Leu	Asn	Arg	Lys	Gly	Asn	Ile	Leu	Val	Thr	His	Thr	Glu
Leu 465	Ser	Gly	Ile	Tyr	Thr	Val	Arg	Phe	Ala	Val	Gly	Ala	Thr	His	Thr
Glu	Leu	Gln	His	Val	Gln	Ala	Ala	Trp	Glu	Val	Ile	Gln	Ala	Glu	Ala
Ser	His	Leu	Leu	Asn	Gly	Lys	Gln								

<400> SEQUENCE: 42

Met	Ile	Leu	Ser	Ile	His	Pro	Phe	Pro	Phe	Thr	Leu	Ser	Ala	Arg	Phe
1				5					10					15	
Ser	Gly	Ala	Ala	Ala	Ala	Asn	Ile	Leu	Ser	Lys	Ala	Ser	Cys	Trp	Leu
			20					25					30		
Arg	Cys	Leu	Arg	Ser	Met	Glu	Gly	Glu	Leu	Lys	Pro	Met	Asp	Ala	Glu
		35					40					45			

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Gln	Leu	Arg	Glu	Tyr	Gly	His	Arg	Met	Val	Asp	Phe	Val	Ala	Asp	Tyr
50						55					60				
Tyr	Lys	Thr	Ile	Glu	Asp	His	Pro	Val	Leu	Ser	Gln	Val	Glu	Pro	Gly
65				70					75						80
Tyr	Leu	Arg	Lys	Leu	Leu	Pro	Asp	Ser	Ala	Pro	Asp	Lys	Pro	Glu	Ser
			85						90					95	
Phe	Glu	Asn	Val	Leu	Ser	Asp	Val	Lys	Thr	Lys	Ile	Ile	Pro	Gly	Val
			100					105					110		
Thr	His	Trp	Gln	Ser	Pro	Asn	Tyr	Phe	Ala	Tyr	Phe	Pro	Ser	Asn	Ser
		115					120					125			
Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu	Ser	Ala	Cys	Phe	Asn	Ile
		130				135					140				
Val	Gly	Phe	Ser	Trp	Ile	Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Met
145					150					155					160
Ile	Val	Leu	Asp	Trp	Phe	Ala	Lys	Met	Leu	Lys	Leu	Pro	Asp	Phe	Phe
			165						170					175	
Leu	Ser	Thr	Gly	Gln	Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu
			180					185					190		
Ala	Val	Leu	Val	Val	Leu	Leu	Ala	Ala	Arg	Asp	Ile	Phe	Leu	Arg	Lys
		195					200					205			
Leu	Gly	Lys	Gly	Phe	Leu	Glu	Lys	Leu	Val	Val	Tyr	Ala	Ser	Asp	Gln
	210					215					220				
Thr	His	Ser	Ala	Leu	Gln	Lys	Ala	Cys	Gln	Ile	Ala	Gly	Ile	His	Pro
225					230					235					240
Glu	Asn	Val	Arg	Ala	Leu	Lys	Thr	Asp	Ser	Ser	Thr	Asn	Tyr	Gly	Leu
			245					250						255	
Ser	Pro	Asp	Leu	Leu	Ser	Lys	Glu	Ile	Cys	His	Asp	Ile	Ala	Asn	Gly
			260					265					270		
Leu	Val	Pro	Phe	Phe	Ala	Cys	Ala	Ser	Val	Gly	Thr	Thr	Ser	Ser	Thr
		275					280					285			
Ala	Val	Asp	Pro	Ile	Leu	Glu	Leu	Ala	Asn	Val	Thr	Lys	Ser	Tyr	Asn
		290				295					300				
Ile	Trp	Leu	His	Val	Asp	Ser	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Val	Cys
305					310					315					320
Pro	Glu	Tyr	Arg	His	His	Ile	Asp	Gly	Val	Glu	Glu	Val	Asp	Ser	Phe
			325					330						335	
Asn	Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu
			340					345					350		
Leu	Trp	Val	Lys	Asp	Arg	Asn	Ala	Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn
		355					360					365			
Pro	Glu	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ser	Asn	Ser	Val	Leu	Asp
		370				375					380				
Tyr	Lys	Asp	Trp	Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys
385				390					395						400
Leu	Trp	Leu	Val	Leu	Arg	Leu	Tyr	Gly	Val	Glu	Asn	Leu	Gln	Ala	Tyr
			405					410					415		
Ile	Arg	Asn	His	Ile	Glu	Leu	Ala	Leu	Asn	Phe	Glu	Glu	Leu	Val	Ser
			420					425					430		
Gln	Asp	Met	Arg	Phe	Glu	Ile	Val	Ala	Pro	Arg	Thr	Phe	Ala	Leu	Val
		435					440					445			
Cys	Phe	Arg	Leu	Leu	Leu	Pro	Cys	Gly	Phe	Glu	Asp	His	Thr	Asn	Asp
		450				455					460				

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Val	Asn	Ser	Asp	Leu	Leu	Gln	Ala	Val	Asn	Ser	Thr	Gly	Lys	Ile	Phe
465					470					475					480
Ile	Ser	His	Thr	Val	Leu	Ser	Gly	Thr	Tyr	Val	Leu	Arg	Phe	Ala	Val
				485					490					495	
Gly	Ala	Pro	Leu	Thr	Glu	Glu	Arg	His	Ile	Asp	Ala	Ala	Trp	Lys	Leu
			500					505					510		
Ile	Gln	Asp	Gln	Ala	Ser	Ser	Leu	Leu	Glu	Lys	Leu				
		515					520								

<210> SEQ ID NO 43
 <211> LENGTH: 487
 <212> TYPE: PRT
 <213> ORGANISM: Manihot esculenta
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 12G038600.1

<400> SEQUENCE: 43

Met	Glu	Gly	Glu	Leu	Arg	Pro	Met	Asp	Ala	Glu	Gln	Leu	Arg	Glu	Tyr
1				5					10					15	
Gly	His	Gln	Met	Val	Asp	Phe	Ile	Ala	Asp	Tyr	Tyr	Lys	Thr	Ile	Glu
		20						25					30		
Asn	Phe	Pro	Val	Leu	Ser	Gln	Val	Glu	Pro	Gly	Tyr	Leu	His	Lys	Leu
		35					40					45			
Leu	Pro	Asp	Ser	Ala	Pro	Asn	Gln	Pro	Glu	Ala	Leu	Gln	Asn	Val	Leu
		50				55					60				
Asp	Asp	Val	Arg	Val	Lys	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser
65					70					75					80
Pro	Asn	Tyr	Phe	Ala	Tyr	Tyr	Pro	Ser	Asn	Ser	Ser	Val	Ala	Gly	Phe
			85						90					95	
Leu	Gly	Glu	Met	Leu	Ser	Ala	Gly	Ile	Asn	Met	Ile	Gly	Phe	Ser	Trp
		100						105					110		
Ile	Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp
		115					120					125			
Leu	Gly	Lys	Met	Leu	Lys	Leu	Pro	Glu	Glu	Phe	Leu	Ser	Ser	Gly	Gln
	130					135					140				
Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val
145					150					155					160
Leu	Leu	Ala	Ala	Arg	Asp	Lys	Val	Leu	Thr	Arg	Val	Gly	Lys	Asp	Ser
			165					170						175	
Leu	Lys	Lys	Leu	Val	Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu
		180						185					190		
Gln	Lys	Ala	Cys	Gln	Ile	Ala	Gly	Val	His	Leu	Asp	Asn	Cys	Arg	Leu
		195					200					205			
Leu	Lys	Thr	Asp	Ser	Ser	Lys	Asn	Tyr	Ala	Leu	Ser	Pro	Asp	Ile	Leu
	210					215					220				
Cys	Asp	Ala	Ile	Ser	Gln	Asp	Met	Ser	Asn	Gly	Leu	Ile	Pro	Phe	Phe
225					230					235					240
Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Ser	Ser	Ala	Thr	Val	Asp	Pro	Leu
			245						250					255	
Leu	Ala	Leu	Gly	Lys	Ile	Ala	Lys	Lys	Tyr	Gly	Met	Trp	Phe	His	Val
		260						265					270		
Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	Cys
	275						280					285			
Tyr	Ile	Asp	Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His
	290						295				300				

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Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Ala Leu Trp Val Lys Asp
 305 310 315 320
 Arg Asn Ala Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys
 325 330 335
 Asn Lys Ala Ser Gln Ala Asn Met Val Val Asp Tyr Lys Asp Trp Gln
 340 345 350
 Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu
 355 360 365
 Arg Leu Tyr Gly Val Ala Asn Leu Gln Ser Tyr Ile Arg Asn His Ile
 370 375 380
 Asn Leu Ala Lys Tyr Phe Glu Gly Leu Val Ala Gly Asp Ser Arg Phe
 385 390 395 400
 Glu Val Val Ala Pro Arg Leu Phe Ser Leu Val Cys Phe Arg Leu Leu
 405 410 415
 Pro Pro Asp Asn Asp Glu Asn His Gly Asn Lys Leu Asn His Asp Leu
 420 425 430
 Leu Asp Ala Ala Asn Ser Thr Gly Lys Ile Phe Ile Ser His Thr Val
 435 440 445
 Leu Ser Gly Lys Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro Leu Thr
 450 455 460
 Glu Glu Arg His Val Thr Ala Ala Trp Lys Val Leu Gln Asp Glu Ala
 465 470 475 480
 Ser Ala Leu Leu Gly Ser Leu
 485

<210> SEQ ID NO 44
 <211> LENGTH: 487
 <212> TYPE: PRT
 <213> ORGANISM: Prunus persica
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 8G214500.1

<400> SEQUENCE: 44

Met Glu Ser Gly Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu Asn
 1 5 10 15
 Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu
 20 25 30
 Asn Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Arg Glu Leu
 35 40 45
 Leu Pro Asp Ser Ala Pro Thr His Pro Glu Pro Leu Gln His Ile Phe
 50 55 60
 Asp Asp Ile Gln Ala Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser
 65 70 75 80
 Pro Asn Phe Phe Gly Tyr Tyr Pro Ser Asn Ser Ser Ile Ala Gly Phe
 85 90 95
 Leu Gly Glu Met Met Ser Ala Gly Leu Asn Ile Val Gly Phe Ser Trp
 100 105 110
 Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp
 115 120 125
 Phe Gly Lys Met Leu Lys Leu Pro Glu Glu Phe Leu Ser Ala Gly Lys
 130 135 140
 Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val
 145 150 155 160
 Leu Leu Ala Ala Arg Asp Lys Ile Leu Arg Arg Val Gly Lys Asn Ser
 165 170 175

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Leu Glu Lys Leu Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu
 180 185 190
 Gln Lys Ala Cys Gln Ile Gly Gly Ile His Pro Glu Asn Cys Arg Leu
 195 200 205
 Leu Arg Thr Asp Ser Ser Thr Asn Tyr Ala Leu Ser Pro Asn Val Leu
 210 215 220
 Asn Glu Ala Ile Ser Asn Asp Val Thr Ser Gly Leu Ile Pro Phe Phe
 225 230 235 240
 Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Thr Ala Val Asp Pro Leu
 245 250 255
 Leu Glu Leu Gly Lys Ile Ala Lys Ser Asn Asp Met Trp Phe His Val
 260 265 270
 Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg His
 275 280 285
 Tyr Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Thr Asn Ala His
 290 295 300
 Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Val Leu Trp Ile Lys Asp
 305 310 315 320
 Arg Asn Ala Leu Ile Gln Ala Leu Ser Thr Asn Pro Glu Phe Leu Lys
 325 330 335
 Asn Lys Ala Ser Gln Ala Asn Leu Val Val Asp Tyr Lys Asp Trp Gln
 340 345 350
 Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu
 355 360 365
 Arg Leu Tyr Gly Leu Glu Asn Leu Gln Ser Tyr Ile Arg Asn His Ile
 370 375 380
 Asn Leu Ala Lys His Phe Lys Glu Leu Val Ala Gln Asp Pro Arg Phe
 385 390 395 400
 Glu Ile Val Thr Pro Arg Leu Phe Ser Leu Val Cys Phe Arg Leu Leu
 405 410 415
 Pro Pro His Asn Asp Glu Thr Cys Ala Thr Lys Leu Asn His Gly Leu
 420 425 430
 Leu Asp Ala Val Asn Ala Thr Gly Lys Ile Phe Ile Ser His Thr Val
 435 440 445
 Leu Ser Gly Lys Tyr Leu Leu Arg Leu Ala Val Gly Ala Pro Leu Thr
 450 455 460
 Glu Glu Arg His Val Asn Ala Ala Trp Lys Leu Leu Gln Asp Glu Ala
 465 470 475 480
 Ser Ala Leu Leu Ala Thr Leu
 485

<210> SEQ ID NO 45
 <211> LENGTH: 490
 <212> TYPE: PRT
 <213> ORGANISM: Eucalyptus grandis
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: K01418.1

<400> SEQUENCE: 45

Met Glu Glu Arg Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu Ser
 1 5 10 15
 Ala His Arg Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Ser Ile Glu
 20 25 30
 Ser Phe Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Lys Leu
 35 40 45

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Leu	Pro	Asp	Ser	Ala	Pro	Asp	His	Pro	Glu	Ser	Leu	Gln	Gln	Val	Leu
50						55					60				
Glu	Asp	Val	Gln	Ala	Lys	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser
65					70					75					80
Pro	Asn	Tyr	Phe	Ala	Tyr	Tyr	Pro	Ser	Asn	Ser	Ser	Ile	Ala	Gly	Phe
			85						90					95	
Met	Gly	Glu	Met	Leu	Ser	Ala	Gly	Leu	Asn	Ile	Val	Gly	Phe	Ser	Trp
			100					105					110		
Ile	Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Ile	Ile	Val	Leu	Asp	Trp
		115					120					125			
Leu	Ala	Lys	Leu	Leu	Asn	Leu	Pro	Asp	Asp	Phe	Leu	Ser	Thr	Gly	Pro
	130					135					140				
Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val
145					150					155					160
Leu	Leu	Ala	Ala	Arg	Asp	Lys	Phe	Leu	Ser	Arg	Ile	Gly	Lys	Ser	Ser
				165					170					175	
Leu	Asp	Lys	Leu	Val	Val	Tyr	Ser	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu
			180					185					190		
Gln	Lys	Ala	Cys	Gln	Ile	Gly	Gly	Ile	Tyr	Pro	Glu	Asn	Cys	Arg	Val
		195					200					205			
Leu	Lys	Thr	Asp	Ala	Ser	Thr	Asn	Tyr	Ala	Leu	Ser	Pro	Asp	Leu	Leu
	210					215					220				
Asn	Glu	Val	Ile	Ser	Gln	Asp	Ile	Ser	Thr	Gly	Leu	Val	Pro	Phe	Leu
225					230					235					240
Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu
				245					250					255	
Pro	Ala	Leu	Ala	Thr	Val	Ala	Lys	Arg	Asn	Gly	Met	Trp	Phe	His	Ile
			260					265					270		
Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	Pro
		275					280					285			
Tyr	Ile	Asp	Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His
	290					295					300				
Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Ala	Leu	Trp	Ile	Lys	Asp
305					310					315					320
Arg	Lys	Ala	Leu	Ile	Gln	Ala	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys
				325					330					335	
Asn	Lys	Ala	Ser	Gln	Ala	Asn	Met	Val	Val	Asp	Tyr	Arg	Asp	Trp	Gln
			340					345					350		
Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu
		355					360					365			
Arg	Leu	Tyr	Gly	Val	Gln	Asn	Leu	Gln	Gln	Tyr	Ile	Arg	Asn	His	Ile
	370					375					380				
Glu	Leu	Ala	Arg	Gln	Phe	Glu	Asp	Leu	Val	Ile	Gln	Asp	Pro	Arg	Phe
385				390						395					400
Glu	Val	Val	Thr	Pro	Arg	Ile	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu
				405					410					415	
Ser	Pro	Asp	Asn	Asp	Gly	Asp	Lys	Gly	Asn	Lys	Leu	Asn	Arg	Asp	Leu
			420					425					430		
Leu	Asp	Thr	Val	Asn	Ser	Thr	Gly	Lys	Ile	Phe	Ile	Ser	His	Thr	Val
		435					440					445			
Leu	Ser	Gly	Thr	Tyr	Ile	Leu	Arg	Phe	Ala	Val	Gly	Ala	Pro	Leu	Thr
	450					455					460				
Glu	Glu	Arg	His	Val	Asn	Glu	Ala	Trp	Lys	Val	Leu	Gln	Asp	Glu	Ala

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465 470 475 480
 Ser Lys Leu Leu Ala Thr Ile Gln Asn Asn
 485 490

<210> SEQ ID NO 46
 <211> LENGTH: 480
 <212> TYPE: PRT
 <213> ORGANISM: Amborella trichopoda
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 31565185

<400> SEQUENCE: 46

Met Asp Ala Glu Glu Leu Arg Glu His Gly His Arg Met Val Asp Phe
 1 5 10 15

Ile Ser Asp Tyr Tyr Lys Glu Ile Glu Ser Tyr Pro Val Arg Ser Gln
 20 25 30

Val Gln Pro Gly Tyr Leu Arg Asn Leu Ile Pro Asp Ser Ala Pro Asp
 35 40 45

Met Pro Glu Ser Phe Glu Ser Ile Leu Glu Asp Ile Arg His Lys Ile
 50 55 60

Ile Pro Gly Val Thr His Trp Gln Ser Pro Lys Tyr Phe Ala Tyr Tyr
 65 70 75 80

Pro Ser Asn Ser Ser Thr Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
 85 90 95

Gly Phe Asn Ile Val Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr
 100 105 110

Glu Leu Glu Val Ile Val Leu Asp Trp Leu Ala Lys Val Leu Lys Leu
 115 120 125

Pro Glu Gln Phe Leu Ser Thr Gly Lys Gly Gly Gly Val Ile Gln Gly
 130 135 140

Thr Ala Ser Glu Ala Met Leu Val Ala Leu Leu Ala Ala Arg Asp Lys
 145 150 155 160

Ala Leu Arg Arg Val Gly Gln Asn Leu Leu Glu Asn Leu Val Val Tyr
 165 170 175

Gly Ser Asp Gln Thr His Ser Ala Leu Ile Lys Ala Cys Lys Ile Ala
 180 185 190

Gly Ile Asn Pro Met Asn Cys Arg Leu Leu Gln Ala Thr Phe Met Thr
 195 200 205

Asn Tyr Ala Leu Ser Pro Glu Val Ala Ser Glu Ser Ile Ser Asn Asp
 210 215 220

Ile Ala Ala Gly Leu Leu Pro Ile Phe Leu Cys Ala Thr Val Gly Thr
 225 230 235 240

Thr Ser Ser Thr Ala Val Asp Pro Leu Ala Ala Leu Gly Arg Leu Ala
 245 250 255

Lys Ala Asn Asp Met Trp Phe His Ile Asp Ala Ala Tyr Ala Gly Ser
 260 265 270

Ala Cys Ile Cys Pro Glu Tyr Arg His Tyr Ile Asp Gly Val Glu Glu
 275 280 285

Ala Asp Ser Phe Asn Met Asn Pro His Lys Trp Leu Leu Thr Asn Phe
 290 295 300

Asp Cys Ser Thr Leu Trp Val Lys Asp Ser Ser Asn Leu Ile Gln Ser
 305 310 315 320

Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys Ala Ser Glu Glu Asp
 325 330 335

Leu Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe

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340	345	350
Arg Ser Leu Lys Leu Trp Met Val Leu Arg Met Tyr Gly Val Ala Asn		
355	360	365
Leu Gln Asn His Ile Arg Thr His Ile Asn Leu Ala Lys His Phe Glu		
370	375	380
Glu Leu Ile Ala Thr Asp Thr Arg Phe Glu Ile Ile Val Pro Arg Val		
385	390	395
Phe Ala Leu Val Cys Phe Ala Leu Lys Pro Met Pro Asn Gly Gln Asp		
405	410	415
Asp Ala Ser Lys Leu Asn Leu Lys Leu Leu Glu Ala Val Asn Asn Ser		
420	425	430
Gly Ala Met Phe Leu Thr His Thr Val Leu Ser Gly Arg Phe Val Leu		
435	440	445
Arg Phe Val Val Gly Ala Pro Leu Thr Glu Glu Arg His Val Asn Thr		
450	455	460
Ala Trp Lys Val Leu Gln Asp His Ala Asn Leu Ile Leu Gly Thr Val		
465	470	475
		480

<210> SEQ ID NO 47

<211> LENGTH: 488

<212> TYPE: PRT

<213> ORGANISM: Salix purpurea

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 0252s0200.1

<400> SEQUENCE: 47

Met Glu Ser Lys Gly Leu Lys Pro Met Asp Ser Glu Gln Leu Arg Glu		
1	5	10
Asn Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Ser Ile		
20	25	30
Glu Asn Phe Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Glu		
35	40	45
Leu Leu Pro Asp Ser Ala Pro Asn Gln Pro Glu Thr Leu Gln Asn Val		
50	55	60
Leu Asp Asp Val Gln Ala Lys Ile Leu Pro Gly Val Thr His Trp Gln		
65	70	75
Ser Pro Ser Tyr Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Val Ala Gly		
85	90	95
Phe Leu Gly Glu Met Leu Ser Ala Gly Ile Asn Met Val Gly Phe Ser		
100	105	110
Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Glu		
115	120	125
Trp Leu Gly Lys Leu Leu Lys Leu Pro Glu Asp Phe Leu Ser Thr Gly		
130	135	140
Gln Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ser Val Leu Val		
145	150	155
Val Leu Leu Ala Ala Arg Asp Arg Val Leu Thr Lys Leu Gly Lys Asn		
165	170	175
Ala Leu Glu Lys Leu Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala		
180	185	190
Leu Gln Lys Ala Cys Lys Ile Gly Gly Ile His Pro Glu Asn Cys Lys		
195	200	205
Leu Leu Lys Thr Asp Ser Ser Thr Asn Tyr Ala Leu Ser Pro Asp Leu		
210	215	220
Leu Ser Lys Ala Ile Ser Asp Asp Ile Ser Thr Gly Leu Ile Pro Phe		

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225	230	235	240
Phe Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Thr Ala Val Asp Pro	245	250	255
Leu His Ala Leu Gly Lys Ile Ala Lys Asn Asn Gly Ile Trp Phe His	260	265	270
Val Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg	275	280	285
Cys Tyr Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn Ala	290	295	300
His Lys Trp Leu Leu Thr Asn Phe Asp Cys Ser Ala Leu Trp Val Lys	305	310	315
Asp Arg Asn Ala Leu Ile Gln Ala Leu Ser Thr Asn Pro Glu Phe Leu	325	330	335
Lys Asn Lys Ala Ser Gln Ala Asn Met Val Val Asp Tyr Lys Asp Trp	340	345	350
Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val	355	360	365
Leu Arg Leu Tyr Gly Leu Glu Asn Leu Gln Cys Tyr Ile Arg Asn His	370	375	380
Ile Asn Leu Ala Lys Tyr Phe Glu Gly Leu Val Ala Ala Asp Ser Arg	385	390	395
Phe Glu Val Val Thr Pro Arg Ile Phe Ser Leu Val Cys Phe Arg Leu	405	410	415
Leu Pro Pro Ser Asn Asn Glu Asp His Gly Asn Asn Leu Asn Arg Asp	420	425	430
Leu Leu Asp Ala Val Asn Ser Ser Gly Lys Ile Phe Ile Ser His Thr	435	440	445
Val Leu Ser Gly Lys Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro Leu	450	455	460
Thr Glu Glu Arg His Val Ile Ala Ala Trp Lys Val Leu Gln Asp Glu	465	470	475
Ser Thr Ser Leu Leu Gly Ser Leu	485		

<210> SEQ ID NO 48

<211> LENGTH: 532

<212> TYPE: PRT

<213> ORGANISM: Medicago truncatula

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 31080941

<400> SEQUENCE: 48

Met Val Leu Gln Ile Trp Cys Leu Thr His Asp Ser Asp Lys Lys Leu	1	5	10	15
Gly Gly Gly Tyr Leu Leu Phe Pro Val Ile Lys Val Ala Tyr Thr Val	20	25	30	
His Thr Leu Thr Glu Trp Cys Cys Val Thr Glu Glu Gly Gly Gly Ser	35	40	45	
Glu Leu Lys Ala Met Asp Ala Glu Gln Leu Arg Glu Gln Gly His Met	50	55	60	
Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asn Phe Pro	65	70	75	80
Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Gly Lys Leu Leu Pro Asp	85	90	95	
Ser Ala Pro Thr His Pro Glu Ser Leu Gln His Val Leu Asn Asp Val				

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100						105						110					
Gln	Glu	Lys	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser	Pro	Asn	Tyr		
		115					120					125					
Phe	Ala	Tyr	Phe	Pro	Ser	Asn	Ser	Ser	Ile	Ala	Gly	Phe	Leu	Gly	Glu		
	130					135					140						
Met	Leu	Ser	Ala	Gly	Leu	Ser	Ile	Val	Gly	Phe	Ser	Trp	Ile	Ser	Ser		
145					150					155					160		
Pro	Ala	Ala	Thr	Glu	Leu	Glu	Thr	Ile	Val	Leu	Asp	Trp	Leu	Ala	Lys		
				165					170					175			
Ala	Leu	Leu	Leu	Pro	His	Asp	Phe	Phe	Ser	Thr	Gly	Gln	Gly	Gly	Gly		
			180					185					190				
Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val	Leu	Val	Ala		
		195					200					205					
Ala	Arg	Asp	Lys	Ile	Leu	Arg	Thr	Val	Gly	Arg	Ser	Ala	Leu	Pro	Lys		
	210					215					220						
Leu	Val	Thr	Tyr	Ala	Ser	Asp	Gln	Thr	His	Ser	Ser	Leu	Gln	Lys	Ala		
225					230					235					240		
Cys	Gln	Ile	Ala	Gly	Leu	Asn	Pro	Glu	Leu	Cys	Arg	Leu	Leu	Lys	Thr		
			245						250					255			
Asp	Ser	Ser	Thr	Asn	Phe	Ala	Leu	Ser	Pro	Asp	Val	Leu	Ser	Glu	Ala		
			260						265				270				
Ile	Ser	Asn	Asp	Ile	Ala	Ser	Gly	Leu	Thr	Pro	Phe	Phe	Leu	Cys	Ala		
		275					280					285					
Thr	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Leu	Pro	Ala	Leu		
	290					295					300						
Ala	Lys	Val	Thr	Lys	Pro	Asn	Asn	Ile	Trp	Leu	His	Val	Asp	Ala	Ala		
305					310					315					320		
Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Tyr	Arg	His	Phe	Ile	Asp		
			325						330					335			
Gly	Val	Glu	Glu	Ala	Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Phe		
		340						345					350				
Leu	Thr	Asn	Phe	Asp	Cys	Ser	Val	Leu	Trp	Val	Lys	Asp	Arg	Ser	Ala		
		355					360					365					
Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys	Asn	Lys	Ala		
	370					375						380					
Ser	Gln	Glu	Asn	Thr	Val	Ile	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Pro	Leu		
385				390						395					400		
Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Met	Arg	Leu	Tyr		
			405						410					415			
Gly	Leu	Glu	Gly	Leu	Arg	Thr	His	Ile	Arg	Ser	His	Ile	Ala	Leu	Ala		
		420						425					430				
Val	Tyr	Phe	Glu	Glu	Leu	Val	Val	Gln	Asp	Thr	Arg	Phe	Lys	Val	Val		
	435						440					445					
Ala	Pro	Arg	Thr	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu	Pro	Pro	Gln		
	450					455					460						
Asn	Ser	Glu	Asp	Asn	Gly	Asn	Lys	Leu	Asn	His	Asp	Leu	Leu	Asp	Ala		
465					470					475					480		
Val	Asn	Ser	Thr	Gly	Asp	Val	Phe	Ile	Thr	His	Thr	Val	Leu	Ser	Gly		
			485						490					495			
Glu	Tyr	Ile	Leu	Arg	Leu	Ala	Val	Gly	Ala	Pro	Leu	Thr	Glu	Val	Arg		
			500					505					510				
His	Val	His	Ala	Ala	Trp	Gln	Ile	Leu	Gln	Glu	Lys	Ala	Thr	Ala	Leu		
	515						520					525					

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Leu Glu Ser Leu
530

<210> SEQ ID NO 49
<211> LENGTH: 501
<212> TYPE: PRT
<213> ORGANISM: Brassica rapa
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: I01156.1

<400> SEQUENCE: 49

Met Gln Ile Arg Ala Lys Ile Pro Val Phe Gly Arg Glu Asn Gly Ser
1 5 10 15
Arg His Val Leu Lys Pro Met Asp Ser Glu Gln Leu Arg Glu Tyr Gly
20 25 30
His Arg Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Ser
35 40 45
Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu His Asn Leu Leu
50 55 60
Pro Asp Ser Ala Pro Asp His Pro Glu Thr Val Glu Gln Val Leu Asp
65 70 75 80
Asp Val Lys Thr Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser Pro
85 90 95
Asn Phe Phe Ala Tyr Tyr Pro Ser Asn Ser Ser Val Ala Gly Phe Leu
100 105 110
Gly Glu Met Leu Ser Ala Gly Val Gly Ile Val Gly Phe Ser Trp Val
115 120 125
Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp Leu
130 135 140
Ala Lys Leu Leu Asn Leu Pro Glu His Phe Leu Ser Lys Gly Asn Gly
145 150 155 160
Gly Gly Val Ile Gln Gly Ser Ala Ser Glu Ala Ile Leu Val Val Met
165 170 175
Ile Ala Ala Arg Asp Lys Val Leu Arg Ser Ala Gly Lys Asn Ala Leu
180 185 190
Gly Lys Leu Val Val Tyr Ser Ser Asp Gln Thr His Ser Ala Leu Gln
195 200 205
Lys Ala Cys Gln Ile Ala Gly Ile His Pro Glu Asn Cys Arg Val Leu
210 215 220
Lys Ala Asp Ser Ser Thr Asn Tyr Ala Leu Arg Pro Glu Leu Leu Gln
225 230 235 240
Glu Ala Val Ser Arg Asp Leu Glu Ala Gly Leu Ile Pro Phe Phe Leu
245 250 255
Cys Gly Asn Val Gly Thr Thr Ser Ser Ala Ala Val Asp Pro Leu Ala
260 265 270
Ala Leu Gly Lys Ile Ala Lys Ser Asn Glu Ile Trp Phe His Val Asp
275 280 285
Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg Gln Tyr
290 295 300
Ile Asp Gly Val Glu Thr Ala Asp Ser Phe Asn Met Asn Ala His Lys
305 310 315 320
Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu Leu Trp Val Lys Asp Gln
325 330 335
His Ala Leu Thr Glu Ala Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn
340 345 350

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Lys Ala Ser Gln Ala Asn Leu Val Val Asp Tyr Lys Asp Trp Gln Ile
 355 360 365
 Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met Val Leu Arg
 370 375 380
 Leu Tyr Gly Ala Glu Ala Leu Lys Asn Tyr Ile Arg Asn His Ile Lys
 385 390 395 400
 Leu Ala Lys Asp Leu Glu Gln Leu Val Ser Gln Asp Pro Asn Phe Glu
 405 410 415
 Val Ile Thr Pro Arg Ile Phe Ser Leu Val Cys Phe Arg Ile Val Pro
 420 425 430
 Thr Asp Asn Asp Glu Lys Lys Cys Asn Ser Arg Asn Leu Glu Leu Leu
 435 440 445
 Glu Ala Val Asn Ser Ser Gly Lys Leu Phe Ile Ser His Thr Ala Leu
 450 455 460
 Ser Gly Lys Ile Val Leu Arg Cys Ala Ile Gly Ala Pro Leu Thr Glu
 465 470 475 480
 Glu Lys His Val Lys Glu Thr Trp Lys Val Ile Gln Glu Lys Val Ser
 485 490 495
 Tyr Leu Leu Arg Lys
 500

<210> SEQ ID NO 50
 <211> LENGTH: 479
 <212> TYPE: PRT
 <213> ORGANISM: Brassica rapa
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: I04706.1

<400> SEQUENCE: 50

Met Asp Ser Glu Gln Leu Arg Glu Tyr Gly His Arg Met Val Asp Phe
 1 5 10 15
 Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Thr Phe Pro Val Leu Ser Gln
 20 25 30
 Val Gln Pro Gly Tyr Leu His Asn Leu Leu Pro Asp Ser Ala Pro Asp
 35 40 45
 Gln Pro Glu Thr Val Glu Gln Val Leu Asp Asp Val Lys Thr Lys Ile
 50 55 60
 Leu Pro Gly Ile Thr His Trp Gln Ser Pro Thr Phe Tyr Ala Tyr Tyr
 65 70 75 80
 Pro Ser Asn Ser Ser Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
 85 90 95
 Gly Leu Gly Ile Val Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr
 100 105 110
 Glu Leu Glu Met Ile Val Leu Asp Trp Leu Ala Lys Leu Leu Asn Leu
 115 120 125
 Pro Glu Gln Phe Leu Ser Lys Gly Asn Gly Gly Gly Val Ile Gln Gly
 130 135 140
 Ser Ala Ser Glu Ala Ile Leu Val Val Met Ile Gly Ala Arg Glu Lys
 145 150 155 160
 Val Leu Arg Arg Val Gly Lys Asn Ala Leu Gly Lys Leu Val Val Tyr
 165 170 175
 Ser Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala
 180 185 190
 Gly Ile His Pro Glu Asn Cys Arg Val Leu Lys Ala Asp Ser Ser Thr
 195 200 205

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Asn Tyr Ala Leu Arg Pro Glu Leu Leu Gln Glu Ala Val Ser Lys Asp
 210                215                220

Ile Glu Ala Gly Leu Ile Pro Phe Phe Leu Cys Gly Asn Val Gly Thr
 225                230                235                240

Thr Ser Ser Thr Ala Val Asp Pro Leu Ala Ala Leu Gly Lys Ile Ala
                245                250                255

Lys Ser Asn Glu Ile Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser
                260                265                270

Ala Cys Ile Cys Pro Glu Tyr Arg Gln Tyr Ile Asp Gly Val Glu Thr
                275                280                285

Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe
 290                295                300

Asp Cys Ser Leu Leu Trp Val Lys Asp Gln Tyr Val Leu Thr Glu Ala
 305                310                315                320

Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn
                325                330                335

Leu Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe
 340                345                350

Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Ala Glu Thr
 355                360                365

Leu Lys Ser Tyr Ile Arg Asn His Ile Lys Leu Ala Lys Asp Leu Glu
 370                375                380

Gln Leu Val Ser Gln Asp Pro Asn Phe Glu Val Val Thr Pro Arg Ile
 385                390                395                400

Phe Ser Leu Val Cys Phe Arg Ile Leu Pro Val Asp Asn Asp Glu Lys
                405                410                415

Glu Cys Asn Asn Arg Asn Arg Asn Leu Leu Asp Ala Val Asn Ser Ser
                420                425                430

Gly Lys Leu Phe Leu Ser His Thr Ala Leu Ser Gly Lys Ile Val Leu
 435                440                445

Arg Cys Ala Ile Gly Ala Pro Leu Thr Glu Glu Arg His Val Lys Glu
 450                455                460

Thr Trp Lys Val Ile Gln Glu Glu Ala Ser Arg Leu Leu Gly Lys
 465                470                475

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<210> SEQ ID NO 51
<211> LENGTH: 482
<212> TYPE: PRT
<213> ORGANISM: Brassica rapa
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: G00043.1

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<400> SEQUENCE: 51

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Met Asp Ser Glu Gln Leu Arg Glu Tyr Gly His Arg Met Val Asp Phe
 1          5          10          15

Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Thr Phe Pro Val Leu Ser Gln
 20          25          30

Val Gln Pro Gly Tyr Leu His Asn Leu Leu Pro Asp Ser Ala Pro Asp
 35          40          45

Gln Pro Glu Thr Leu Glu Gln Val Leu Asp Asp Val Lys Glu Lys Ile
 50          55          60

Leu Pro Gly Val Thr His Trp Gln Ser Pro Ser Phe Phe Ala Tyr Tyr
 65          70          75          80

Pro Ala Asn Ser Ser Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
 85          90          95

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Ala Leu Asn Ile Val Gly Phe Ser Trp Val Ser Ser Pro Ala Ala Thr
 100 105 110
 Glu Leu Glu Met Ile Val Leu Asp Trp Phe Ala Lys Leu Leu Asn Leu
 115 120 125
 Pro Glu Gln Phe Leu Ser Arg Gly Asn Gly Gly Gly Val Ile Gln Gly
 130 135 140
 Thr Ala Ser Glu Ala Ile Leu Val Val Met Ile Ala Ala Arg Asp Lys
 145 150 155 160
 Val Leu Arg Ser Leu Gly Lys Lys Ala Leu Glu Lys Leu Val Val Tyr
 165 170 175
 Ser Ser Asp Gln Thr His Ser Ser Leu Leu Lys Ala Cys Gln Ile Ala
 180 185 190
 Gly Ile His Leu Glu Asn Cys Arg Met Leu Lys Thr Asp Ser Ser Thr
 195 200 205
 Asn Tyr Ala Leu Arg Pro Glu Ser Leu Gln Glu Ala Val Ser Gly Asp
 210 215 220
 Leu Glu Ala Gly Leu Ile Pro Phe Phe Leu Cys Gly Thr Val Gly Thr
 225 230 235 240
 Thr Ser Ser Thr Ala Val Asp Pro Leu Ala Glu Leu Gly Lys Ile Ala
 245 250 255
 Lys Ser Asn Glu Met Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser
 260 265 270
 Ala Cys Ile Cys Pro Glu Tyr Arg Gln Tyr Ile Asp Gly Val Glu Thr
 275 280 285
 Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe
 290 295 300
 Asp Cys Ser Leu Leu Trp Val Lys Asp Arg Tyr Ala Leu Thr Glu Ala
 305 310 315 320
 Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn
 325 330 335
 Leu Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe
 340 345 350
 Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Ala Glu Thr
 355 360 365
 Leu Lys Ser Tyr Ile Lys Asn His Ile Lys Leu Ala Lys Asp Leu Glu
 370 375 380
 Gln Leu Val Ser Gln Asp Pro Asn Phe Glu Val Val Thr Pro Arg Ile
 385 390 395 400
 Phe Ser Leu Val Cys Phe Arg Ile Val Pro Val Asp Asn Asp Glu Lys
 405 410 415
 Thr Cys Asn Asn Leu Asn Arg Ser Leu Leu Asp Ala Val Asn Ser Ser
 420 425 430
 Gly Lys Leu Phe Ile Ser His Thr Thr Leu Ser Gly Lys Phe Val Leu
 435 440 445
 Arg Leu Ala Ile Gly Ala Pro Leu Thr Glu Glu Lys His Val Met Asp
 450 455 460
 Ala Trp Lys Val Ile Gln Glu Glu Ala Ser Phe Leu Leu Ala Ser Gln
 465 470 475 480
 Val Lys

<210> SEQ ID NO 52

<211> LENGTH: 489

<212> TYPE: PRT

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<213> ORGANISM: Glycine max
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 03G167900.1

<400> SEQUENCE: 52

Met Glu Glu Glu Ser Ala Leu Arg Pro Met Asp Ala Glu Gln Leu Arg
 1          5          10          15

Glu Gln Ala His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Thr
      20          25          30

Ile Glu Asp Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Gly
      35          40          45

Lys Leu Leu Pro Asp Ser Ala Pro Asp Ser Pro Glu Ser Leu Gln Asn
      50          55          60

Val Leu Asp Asp Val Gln Glu Lys Ile Leu Pro Gly Val Thr His Trp
      65          70          75          80

Gln Ser Pro Asn Tyr Phe Ala Tyr Phe Pro Ser Asn Ser Ser Ile Ala
      85          90          95

Gly Phe Leu Gly Glu Met Leu Ser Ala Gly Leu Asn Ile Val Gly Phe
      100          105          110

Ser Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Thr Ile Val Leu
      115          120          125

Asp Trp Leu Ala Lys Ala Phe Gln Leu Pro Asp Tyr Phe Tyr Ser Ser
      130          135          140

Gly Lys Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu
      145          150          155          160

Val Val Leu Leu Ala Ala Arg Asp Lys Ile Leu Arg Arg Val Gly Arg
      165          170          175

Asn Ala Leu Pro Lys Leu Val Met Tyr Ala Ser Asp Gln Thr His Ser
      180          185          190

Ala Leu Leu Lys Ala Cys Gln Ile Ala Gly Ile Asn Pro Glu Leu Cys
      195          200          205

Arg Leu Leu Lys Thr Asp Ser Ser Thr Asn Tyr Ala Leu Ser Pro Asp
      210          215          220

Val Leu Ser Glu Ala Ile Ser Asn Asp Ile Ala Gly Gly Leu Val Pro
      225          230          235          240

Phe Phe Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Thr Ala Val Asp
      245          250          255

Pro Leu Pro Ala Leu Gly Lys Ile Ala Lys Thr Asn Lys Leu Trp Phe
      260          265          270

His Val Asp Ala Ala Tyr Ala Gly Ser Ala Cys Val Cys Pro Glu Tyr
      275          280          285

Arg His Cys Ile Asp Gly Val Glu Glu Ala Asp Ser Phe Asn Met Asn
      290          295          300

Ala His Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Leu Leu Trp Val
      305          310          315          320

Lys Asp Arg Ser Ser Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe
      325          330          335

Leu Lys Asn Lys Ala Ser Gln Gly Asn Met Val Ile Asp Tyr Lys Asp
      340          345          350

Trp Gln Ile Pro Leu Gly Arg Arg Phe Arg Ser Leu Lys Leu Trp Met
      355          360          365

Val Leu Arg Leu Tyr Gly Leu Asp Gly Leu Arg Ser His Ile Arg Asn
      370          375          380

His Ile Glu Leu Ala Ala Asn Phe Glu Glu Leu Val Arg Gln Asp Thr

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385              390              395              400
Arg Phe Lys Val Val Ala Pro Arg Thr Phe Ser Leu Val Cys Phe Arg
              405              410              415
Leu Leu Pro His Pro Asn Ser Ala Asp His Gly Asn Lys Leu Asn Ser
              420              425              430
Asp Leu Leu Asp Ser Val Asn Ser Thr Gly Asn Ala Phe Ile Thr His
              435              440              445
Thr Val Leu Ser Gly Glu Tyr Ile Leu Arg Phe Ala Val Gly Ala Pro
              450              455              460
Leu Thr Glu Arg Arg His Val Asn Met Ala Trp Gln Ile Leu Gln Asp
465              470              475              480
Lys Ala Thr Ala Leu Leu Glu Ser Leu
              485

<210> SEQ ID NO 53
<211> LENGTH: 520
<212> TYPE: PRT
<213> ORGANISM: Fragaria vesca
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 27261550

<400> SEQUENCE: 53
Met Asp Ala Glu Gln Leu Arg Glu Asn Ala His Lys Met Val Asp Phe
1              5              10              15
Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln
              20              25              30
Val Gln Pro Gly Tyr Leu Arg Glu Leu Leu Pro Asp Ser Ala Pro Thr
              35              40              45
Gln Pro Glu Ser Leu Gln His Ile Phe Asp Asp Ile Gln Ala Lys Ile
50              55              60
Leu Pro Gly Val Thr His Trp Gln Ser Pro Asn Phe Phe Ala Tyr Tyr
65              70              75              80
Pro Ser Asn Ser Ser Ile Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
              85              90              95
Gly Leu Asn Ile Val Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr
              100             105             110
Glu Leu Glu Met Ile Val Leu Asp Trp Leu Ala Lys Leu Ile Lys Leu
              115             120             125
Pro Asp Glu Phe Leu Ser Ala Gly Gln Gly Gly Gly Val Ile Gln Gly
              130             135             140
Thr Ala Ser Glu Ala Ile Leu Val Val Met Leu Ala Ala Arg Asp Lys
145             150             155             160
Ile Leu Arg Arg Val Gly Lys Asn Ala Leu Glu Lys Leu Val Val Tyr
              165             170             175
Ala Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala
              180             185             190
Gly Ile His Pro Glu Asn Cys Arg Ile Leu Ser Thr Asn Ser Thr Thr
              195             200             205
Asn Tyr Ala Leu Ser Pro Ser Val Gly Thr Thr Ser Ser Thr Ala Val
              210             215             220
Asp Pro Leu Gly Glu Leu Gly Lys Ile Ala Lys Asn Asn Glu Met Trp
225             230             235             240
Phe His Val Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu
              245             250             255
Tyr Arg His Tyr Ile Asp Gly Val Glu Lys Ala Asp Ser Phe Asn Met

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260					265					270					
Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Val	Leu	Trp
	275						280					285			
Ile	Lys	Asp	Arg	Asn	Ala	Leu	Val	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu
	290					295					300				
Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asn	Met	Val	Val	Asp	Tyr	Lys
	305					310					315				320
Asp	Trp	Gln	Val	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp
			325						330					335	
Met	Val	Leu	Arg	Leu	Tyr	Gly	Leu	Glu	Asn	Leu	Gln	Ser	Tyr	Ile	Arg
			340					345					350		
Thr	His	Ile	Asn	Leu	Ala	Lys	His	Phe	Glu	Glu	Leu	Val	Ala	Gln	Asp
		355					360					365			
Pro	Arg	Phe	Glu	Ile	Val	Thr	Pro	Arg	Leu	Tyr	Ser	Leu	Val	Cys	Phe
	370					375					380				
Arg	Leu	Leu	Pro	Pro	His	Gly	Asn	Glu	Ala	Cys	Ala	Ser	Lys	Leu	Asn
	385					390					395				400
His	Asp	Leu	Leu	Asp	Ala	Val	Asn	Ser	Thr	Gly	Lys	Ile	Tyr	Ile	Ser
			405						410					415	
His	Thr	Val	Leu	Ser	Gly	Ala	Tyr	Ile	Leu	Arg	Phe	Ala	Val	Gly	Ala
			420					425					430		
Pro	Leu	Thr	Glu	Glu	Lys	His	Val	Thr	Ala	Ala	Trp	Lys	Lys	Leu	Lys
		435					440					445			
Ser	Val	Ile	Arg	Asp	Val	Leu	Ala	Leu	Ala	Asn	Ser	Phe	Val	Ser	Ile
	450					455					460				
Thr	Phe	Ser	His	Met	Tyr	Arg	Glu	Ala	Asn	Phe	Leu	Thr	Asp	Ala	Leu
	465					470					475				480
Ala	Ser	Val	Gly	His	Ser	Leu	Ser	Ser	Ser	Met	Cys	Trp	Phe	Asp	Gly
			485						490					495	
Ile	Pro	Pro	Gln	Ala	Gln	Met	Ala	Leu	Leu	Met	Asp	Ser	Ser	Cys	Ile
			500				505						510		
Gly	His	Leu	Arg	Gly	Ser	Ser	Leu								
		515					520								

<210> SEQ ID NO 54

<211> LENGTH: 487

<212> TYPE: PRT

<213> ORGANISM: Kalanchoe fedtschenkoi

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 0172s0035.1

<400> SEQUENCE: 54

Met	Glu	Gly	Glu	Leu	Lys	Pro	Met	Asp	Ala	Glu	Gln	Leu	Arg	Glu	Tyr
1				5					10					15	
Gly	His	Arg	Met	Val	Asp	Phe	Val	Ala	Asp	Tyr	Tyr	Lys	Thr	Ile	Glu
		20						25					30		
Asp	His	Pro	Val	Leu	Ser	Gln	Val	Glu	Pro	Gly	Tyr	Leu	Arg	Lys	Leu
		35					40					45			
Leu	Pro	Asp	Ser	Ala	Pro	Asp	Lys	Pro	Glu	Ser	Phe	Glu	Asn	Val	Leu
	50					55					60				
Ser	Asp	Val	Lys	Thr	Lys	Ile	Ile	Pro	Gly	Val	Thr	His	Trp	Gln	Ser
	65				70				75					80	
Pro	Asn	Tyr	Phe	Ala	Tyr	Phe	Pro	Ser	Asn	Ser	Ser	Thr	Ala	Gly	Phe
			85					90						95	
Leu	Gly	Glu	Met	Leu	Ser	Ala	Cys	Phe	Asn	Ile	Val	Gly	Phe	Ser	Trp

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100					105					110					
Ile	Thr	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Met	Ile	Val	Leu	Asp	Trp
		115					120					125			
Phe	Ala	Lys	Met	Leu	Lys	Leu	Pro	Asp	Phe	Phe	Leu	Ser	Thr	Gly	Gln
	130					135					140				
Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	Ala	Ser	Glu	Ala	Val	Leu	Val	Val
	145					150					155				160
Leu	Leu	Ala	Ala	Arg	Asp	Ile	Phe	Leu	Arg	Lys	Leu	Gly	Lys	Gly	Phe
				165					170					175	
Leu	Glu	Lys	Leu	Val	Val	Tyr	Ala	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu
			180						185					190	
Gln	Lys	Ala	Cys	Gln	Ile	Ala	Gly	Ile	His	Pro	Glu	Asn	Val	Lys	Ala
		195					200					205			
Leu	Lys	Thr	Asp	Ser	Ser	Thr	Asn	Tyr	Gly	Leu	Ser	Pro	Asp	Leu	Leu
		210					215					220			
Ser	Lys	Glu	Ile	Cys	His	Asp	Ile	Ala	Asn	Gly	Leu	Val	Pro	Phe	Phe
		225				230					235				240
Ala	Cys	Ala	Ser	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Ile	Asp	Pro	Ile
				245					250					255	
Leu	Glu	Leu	Ala	Asn	Val	Thr	Lys	Ser	Tyr	Asn	Ile	Trp	Leu	His	Val
			260						265					270	
Asp	Ser	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Val	Cys	Pro	Glu	Tyr	Arg	His
		275					280					285			
His	Ile	Asp	Gly	Val	Glu	Glu	Val	Asp	Ser	Phe	Asn	Met	Asn	Ala	His
		290					295					300			
Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu	Leu	Trp	Val	Lys	Asp
		305				310					315				320
Arg	Asn	Ala	Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu	Phe	Leu	Lys
				325					330					335	
Asn	Lys	Ala	Ser	Gln	Ser	Lys	Ser	Val	Leu	Asp	Tyr	Lys	Asp	Trp	Gln
		340						345					350		
Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Leu	Val	Leu
		355					360					365			
Arg	Leu	Tyr	Gly	Val	Glu	Asn	Leu	Gln	Ala	Tyr	Ile	Arg	Asn	His	Ile
		370					375					380			
Glu	Leu	Ala	Ile	His	Phe	Glu	Glu	Leu	Val	Ser	Gln	Asp	Met	Arg	Phe
		385				390					395				400
Glu	Ile	Val	Ala	Pro	Arg	Thr	Phe	Ala	Leu	Val	Cys	Phe	Arg	Leu	Leu
			405					410						415	
Leu	Pro	Cys	Gly	Phe	Glu	Asp	Arg	Thr	Asn	Asp	Val	Asn	Gly	Asp	Leu
			420				425						430		
Leu	Gln	Ala	Val	Asn	Ser	Thr	Gly	Lys	Ile	Phe	Ile	Ser	His	Thr	Val
		435					440						445		
Leu	Ser	Gly	Thr	Tyr	Val	Met	Arg	Phe	Ala	Val	Gly	Ala	Pro	Leu	Thr
		450					455					460			
Glu	Glu	Arg	His	Ile	Asp	Ala	Ala	Trp	Lys	Leu	Ile	Gln	Asp	Gln	Ala
		465				470					475				480
Ser	Ser	Leu	Leu	Glu	Lys	Leu									
				485											

<210> SEQ ID NO 55

<211> LENGTH: 479

<212> TYPE: PRT

<213> ORGANISM: Capsella grandiflora

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<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 22666s0001.1

<400> SEQUENCE: 55

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Met Asp Ser Glu Gln Leu Arg Glu Tyr Gly His Arg Met Val Asp Phe
1          5          10          15

Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln
20          25          30

Val Gln Pro Gly Tyr Leu His Lys Leu Leu Pro Asp Ser Ala Pro Asp
35          40          45

Gln Pro Glu Thr Leu Asp Gln Val Leu Asp Asp Val Arg Ala Lys Ile
50          55          60

Leu Pro Gly Val Thr His Trp Gln Ser Pro Gly Phe Phe Ala Tyr Tyr
65          70          75          80

Pro Ser Asn Ser Ser Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala
85          90          95

Gly Leu Gly Ile Val Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr
100         105         110

Glu Leu Glu Met Ile Val Leu Asp Trp Leu Ala Lys Leu Leu Asn Leu
115         120         125

Pro Lys Glu Phe Leu Ser Lys Gly Asn Gly Gly Gly Val Ile Gln Gly
130         135         140

Ser Ala Ser Glu Ala Val Leu Val Val Leu Ile Ala Ala Arg Asp Lys
145         150         155         160

Val Leu Arg Ser Ala Gly Lys Asn Ala Leu Gly Lys Leu Val Val Tyr
165         170         175

Ser Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala
180         185         190

Gly Ile His Pro Glu Asn Cys Arg Val Leu Glu Thr Asp Ala Ser Thr
195         200         205

Asn Tyr Ala Leu Arg Pro Glu Leu Leu Gln Glu Ala Val Ser Lys Asp
210         215         220

Leu Lys Ala Gly Leu Ile Pro Phe Phe Leu Cys Ala Asn Val Gly Thr
225         230         235         240

Thr Ser Ser Thr Ala Val Asp Pro Leu Ala Ala Leu Gly Lys Ile Ala
245         250         255

Asn Ser Asn Glu Ile Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser
260         265         270

Ala Cys Ile Cys Pro Glu Tyr Arg Lys Tyr Ile Asp Gly Val Glu Thr
275         280         285

Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe
290         295         300

Asp Cys Ser Leu Leu Trp Val Lys Glu Gln Asp Ser Leu Thr Glu Ala
305         310         315         320

Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn
325         330         335

Leu Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe
340         345         350

Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Ala Glu Thr
355         360         365

Leu Lys Ser Tyr Ile Arg Asn His Ile Lys Leu Ala Lys Tyr Tyr Glu
370         375         380

Lys Leu Val Ser Gln Asp Pro Asn Phe Glu Ile Val Thr Pro Arg Ile
385         390         395         400

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Phe Ser Leu Val Cys Phe Arg Leu Val Pro Lys Asn Glu Asp Glu Lys
405 410 415

Lys Cys Asn Asn Gln Asn Arg Lys Leu Leu Glu Ala Ala Asn Ser Ser
420 425 430

Gly Lys Leu Phe Met Ser His Thr Ala Leu Ser Gly Lys Ile Val Leu
435 440 445

Arg Cys Ala Ile Gly Ala Pro Leu Thr Glu Glu Lys His Met Lys Glu
450 455 460

Ala Trp Lys Val Ile Gln Asp Glu Ala Ser Phe Leu Leu His Lys
465 470 475

<210> SEQ ID NO 56
<211> LENGTH: 527
<212> TYPE: PRT
<213> ORGANISM: Selaginella moellendorffii
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 15420188

<400> SEQUENCE: 56

Met Gly Glu Ala Asn Ile Gly Pro Lys Pro Ile Asp Ala Glu Glu Phe
1 5 10 15

Arg Lys His Ala His Glu Met Val Asp Phe Ile Ala Asp Tyr Tyr Arg
20 25 30

Asp Ile Glu Ser Phe Pro Val Arg Ser Gln Val Ser Gln Pro Gly Tyr
35 40 45

Leu Lys Thr Leu Leu Pro Pro Ala Ala Pro Glu Asp Pro Glu Ala Leu
50 55 60

Glu Glu Val Phe Ala Asp Ile Gln Ser Lys Ile Ile Pro Gly Val Thr
65 70 75 80

His Trp Gln Ser Pro Asn Phe Phe Gly Tyr Tyr Pro Ser Asn Ser Ser
85 90 95

Thr Ala Gly Leu Leu Gly Glu Met Leu Ser Ala Gly Leu Asn Ile Val
100 105 110

Gly Phe Ser Trp Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Ile Ile
115 120 125

Val Leu Asp Trp Leu Ala Lys Leu Leu Lys Leu Pro Asp Glu Phe Leu
130 135 140

Phe Gly Gly Asn Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala
145 150 155 160

Val Ser Val Val Leu Leu Ala Ala Arg Thr Arg Ala Ile Ser Glu Asn
165 170 175

Lys Arg Lys Gly Leu Ser Glu Ala Glu Ile Leu Ser Lys Leu Ala Val
180 185 190

Tyr Thr Ser Asp Gln Thr His Ser Cys Leu Gln Lys Gly Cys Ala Ile
195 200 205

Ala Gly Ile Pro Leu Glu Asn Leu Val Ile Val Pro Thr Asp Ser Ser
210 215 220

Thr Asn Tyr Ala Val Ser Pro Ala Ala Met Arg Gln Ala Leu Glu Asp
225 230 235 240

Gly Val Lys Gln Gly Leu Leu Pro Phe Phe Leu Cys Gly Thr Val Gly
245 250 255

Thr Thr Ser Ser Ser Ala Val Asp Pro Leu Ser Ala Leu Gly Asp Ile
260 265 270

Ala Lys Asp Phe Gly Met Trp Phe His Val Asp Ala Ala Tyr Ala Gly
275 280 285

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Ser Ala Cys Ile Cys Pro Glu Phe Arg His His Leu Asp Gly Val Glu
 290 295 300
 Lys Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Leu Leu Thr Asn
 305 310 315 320
 Phe Asp Cys Ser Ala Leu Trp Val Lys Glu Ser Ser His Leu Val Ser
 325 330 335
 Ala Leu Ser Thr Thr Pro Glu Phe Leu Arg Asn Lys Ala Ser Asp Leu
 340 345 350
 Asn Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg
 355 360 365
 Phe Arg Ser Leu Lys Leu Trp Phe Val Met Arg Met Asn Gly Ala Ser
 370 375 380
 Gly Leu Arg Ser Tyr Ile Arg Asn His Val Arg Leu Ala Lys Arg Phe
 385 390 395 400
 Glu Gly Phe Val Arg Glu Asp Pro Arg Phe Gln Leu Leu Val Pro Arg
 405 410 415
 Thr Phe Gly Leu Ile Cys Phe Arg Leu Lys Pro Glu Ser Asp Asp Pro
 420 425 430
 Asp Asn Gly Arg Thr Leu Asn Ser Thr Leu Leu Glu Ala Val Asn Ser
 435 440 445
 Ser Gly Arg Met Phe Ile Thr His Thr Val Leu Ser Gly Val Tyr Thr
 450 455 460
 Leu Arg Met Ala Ile Gly Gly Pro Leu Thr Gln Asp Lys His Val Asp
 465 470 475 480
 Ala Ala Trp Lys Leu Ile Gln Glu Glu Ala Thr Thr Leu Leu Val Lys
 485 490 495
 Gly Pro Ser His Ile Leu Ala Asn Asn Leu Arg Leu Ser Pro Ile Leu
 500 505 510
 Ala Asn Asn Leu Arg Leu Ser Pro Ile Leu Ala Asn Asn Arg Ile
 515 520 525

<210> SEQ ID NO 57
 <211> LENGTH: 572
 <212> TYPE: PRT
 <213> ORGANISM: *Setaria italica*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 3G188200.1

<400> SEQUENCE: 57

Met Asp Ile Leu Asn His Ala Asp Thr Thr Thr Ala Asn Gly Thr Ser
 1 5 10 15
 Pro Ala Ala Ala Ala Ala Ala Val Val Ala Pro Ala Thr Pro Ser
 20 25 30
 Ser Leu Val Thr Pro Pro Leu Asp Ala Asp Glu Phe Arg Arg Gln Gly
 35 40 45
 Arg Leu Val Val Asp Phe Ile Ala Asp Tyr Tyr Thr Arg Ile Asn Glu
 50 55 60
 Tyr Pro Val Arg Pro Ala Val Ala Pro Gly Phe Leu Ala Arg Gln Leu
 65 70 75 80
 Pro Glu Thr Ala Pro Ala Arg Pro Glu Arg Asp Ala Leu Ala Ala Ala
 85 90 95
 Leu Arg Asp Val Arg Asp Leu Ile Leu Pro Gly Val Thr His Trp Gln
 100 105 110
 Ser Pro Arg His Phe Ala His Phe Ala Ala Thr Ala Ser Asn Val Gly
 115 120 125

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Ala	Leu	Gly	Glu	Ala	Leu	Ala	Ala	Gly	Leu	Asn	Ile	Asn	Pro	Phe	Thr
130						135					140				
Trp	Ala	Ala	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Val	Val	Val	Thr	Asp
145					150					155					160
Trp	Leu	Gly	Lys	Ala	Leu	His	Leu	Pro	Glu	Arg	Leu	Leu	Phe	Ser	Gly
				165					170					175	
Gly	Gly	Gly	Gly	Thr	Leu	Leu	Gly	Thr	Ser	Cys	Glu	Ala	Met	Leu	Cys
				180				185					190		
Thr	Ile	Val	Ala	Ala	Arg	Asp	Arg	Lys	Leu	Ala	Glu	Ile	Gly	Glu	Glu
		195					200					205			
Arg	Ile	Gly	Asp	Leu	Val	Val	Tyr	Phe	Ser	Asp	Gln	Thr	His	Phe	Ser
	210					215					220				
Phe	Gln	Lys	Ala	Ala	Arg	Ile	Ala	Gly	Ile	Arg	Arg	Gly	Asn	Cys	Arg
225					230					235					240
Glu	Ile	Pro	Thr	Ser	Arg	Glu	Ser	Gly	Phe	Thr	Leu	Ser	Pro	Lys	Ala
				245					250					255	
Leu	Arg	Ala	Ala	Val	Arg	Ala	Asp	Glu	Ala	Ser	Gly	Arg	Val	Pro	Leu
		260						265					270		
Phe	Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Pro	Thr	Ala	Ala	Ile	Asp	Pro
	275						280					285			
Leu	Arg	Glu	Leu	Cys	Ala	Ala	Val	Ser	Gly	His	Gly	Val	Trp	Val	His
	290				295						300				
Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ala	Ala	Cys	Val	Cys	Pro	Glu	Phe	Arg
305					310					315					320
His	Ala	Ile	Ala	Gly	Ala	Glu	Ala	Val	Asp	Ser	Phe	Ser	Thr	Asn	Pro
			325						330					335	
His	Lys	Trp	Leu	Leu	Ala	Asn	Met	Asp	Cys	Cys	Ala	Leu	Trp	Val	Thr
		340						345					350		
Arg	Pro	Ala	Ala	Leu	Val	Ala	Ala	Leu	Gly	Thr	Asp	His	Asp	Val	Ile
		355				360						365			
Leu	Lys	Asp	Pro	Ser	Ala	Ala	Ala	Gln	Asp	Gly	His	Asp	Val	Val	Val
	370				375						380				
Asp	Tyr	Lys	Asp	Trp	Gln	Val	Ala	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu
385					390					395					400
Lys	Leu	Trp	Leu	Val	Leu	Arg	Cys	His	Gly	Val	Glu	Gly	Leu	Arg	Gly
			405						410				415		
Phe	Val	Arg	Ala	His	Val	Arg	Met	Ala	Ala	Ala	Phe	Glu	Ala	Met	Val
		420					425					430			
Arg	Ala	Asp	Thr	Arg	Phe	Glu	Val	Pro	Val	Pro	Arg	Gln	Phe	Ala	Leu
	435					440					445				
Val	Cys	Phe	Arg	Leu	Arg	Pro	Ala	Ser	Ala	Gly	Glu	Lys	Arg	Thr	Arg
	450				455						460				
Gly	Gly	Glu	Val	Val	Glu	Pro	Asn	Glu	Leu	Asn	Arg	Arg	Leu	Leu	Glu
465					470					475					480
Ala	Val	Asn	Ala	Thr	Gly	Arg	Ala	Tyr	Ile	Ser	Ser	Ala	Val	Val	Gly
			485						490				495		
Gly	Val	Tyr	Val	Leu	Arg	Cys	Ala	Ile	Gly	Asn	Ser	Leu	Thr	Glu	Glu
			500					505					510		
Arg	His	Val	Arg	Glu	Ala	Trp	Ser	Val	Val	Gln	Glu	Gln	Ala	Asn	Val
	515					520						525			
Val	Leu	Ala	Ala	Ala	Thr	Ala	Thr	Cys	Pro	Asp	Glu	Arg	Ala	Val	His
	530					535					540				

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Arg Ala Arg Cys Val Glu Thr Asp Ala Ala Asp Ala Pro Ala Ser Val
545 550 555 560

Pro Pro Val Gln Met Arg Phe Pro Ser Ala Gln Ser
565 570

<210> SEQ ID NO 58
<211> LENGTH: 505
<212> TYPE: PRT
<213> ORGANISM: Kalanchoe fedtschenkoi
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 0033s0078.1

<400> SEQUENCE: 58

Met Gly Ser Leu Pro Ser Pro His Asp Pro Ser Asn Ala Phe Asn Pro
1 5 10 15

Met Asp Val Ala Glu Leu Ser Ile Glu Ser Arg Leu Val Met Asp Phe
20 25 30

Ile Thr Gln Tyr Tyr Gln Thr Leu Glu Thr Arg Pro Val Gln Pro Arg
35 40 45

Val Lys Pro Gly Phe Leu Thr Gly Gln Leu Pro Glu Lys Ala Pro Phe
50 55 60

His Ala Glu Ser Met Glu Glu Ile Leu Ser Asp Val Ser Glu Lys Ile
65 70 75 80

Val Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe His Ala Tyr Phe
85 90 95

Pro Ala Ser Ser Ser Asn Ala Gly Leu Leu Gly Glu Met Leu Cys Ser
100 105 110

Gly Leu Ser Val Ile Gly Phe Thr Trp Asn Ser Ser Pro Ala Ala Thr
115 120 125

Glu Leu Glu Asn Val Val Val Asp Trp Leu Ala Asp Met Leu Asn Leu
130 135 140

Pro Pro Ser Phe Arg Phe Ser Gly Gly Gly Gly Gly Gly Val Leu
145 150 155 160

Gln Ser Asn Thr Cys Glu Ala Val Leu Cys Thr Leu Ala Ala Ala Arg
165 170 175

Asp Lys Val Leu Glu Arg Ile Gly Asp Asp Lys Ile Asn Lys Leu Val
180 185 190

Ala Tyr Cys Ser Asp Gln Thr His Phe Thr Leu His Lys Gly Ala Lys
195 200 205

Leu Ile Gly Ile Arg Arg Ala Asn Ile Lys Ser Ile Gly Thr Arg Arg
210 215 220

Glu Asn Gly Phe Gly Leu Cys Pro Asn Asp Leu Arg Asn Ala Ile Thr
225 230 235 240

Gly Asp Leu Glu Ala Gly Leu Val Pro Phe Tyr Leu Cys Gly Thr Ile
245 250 255

Gly Thr Thr Ala Leu Gly Ala Val Asp Pro Ile Lys Glu Leu Gly Lys
260 265 270

Val Ala Arg Glu Phe Asp Leu Trp Phe His Ile Asp Ala Ala Tyr Gly
275 280 285

Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Tyr Leu Asp Gly Val
290 295 300

Glu Leu Val Asp Ser Ile Ser Met Asn Ala His Lys Trp Leu Leu Ser
305 310 315 320

Asn Leu Asp Cys Cys Phe Leu Trp Leu Gln Asn Pro Lys Cys Leu Ile
325 330 335

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Gln Cys Leu Ala Ala Glu Ala Glu Phe Leu Lys Gly Ser Gly Glu Met
 340 345 350

Val Asp Tyr Lys Asp Trp Gln Ile Ser Leu Ser Arg Arg Phe Arg Ala
 355 360 365

Ile Lys Met Trp Met Val Phe Arg Arg Tyr Gly Val Ser Asn Leu Met
 370 375 380

Glu His Ile Arg Ser Asp Val Ser Met Ala Ala Arg Phe Glu Glu Met
 385 390 395 400

Val Ser Ala Asp Asp Arg Phe Glu Ile Val Phe Pro Arg Lys Phe Ala
 405 410 415

Leu Val Cys Phe Lys Leu Asn Thr Lys Gly Ser Val Gln His Gly Glu
 420 425 430

Asp Asp Gly Glu Asp Gly Leu Asp Gly Asp Ser Val Leu Thr Arg Glu
 435 440 445

Leu Met Gly Arg Val Asn Ser Ser Gly Lys Ala Tyr Leu Ser Gly Val
 450 455 460

Glu Met Gly Arg Ile Phe Phe Ile Arg Cys Val Ile Gly Ser Ser Leu
 465 470 475 480

Thr Glu Glu Arg His Val Asp Asn Leu Trp Asn Leu Ile Gln Glu Lys
 485 490 495

Thr Gln Ser Ile Met Pro Cys Arg Ala
 500 505

<210> SEQ ID NO 59
 <211> LENGTH: 493
 <212> TYPE: PRT
 <213> ORGANISM: Daucus carota subsp. sativus
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 36068870

<400> SEQUENCE: 59

Met Gly Ser Leu Ser Thr Gln Lys Phe Asn Pro Leu Asn Leu Asp Phe
 1 5 10 15

Phe Ser Ser Glu Ser Asn Lys Val Ile Glu Phe Ile Thr Ala Tyr Tyr
 20 25 30

Lys Asn Val Glu Lys Tyr Pro Val Arg Ser Gln Val Glu Pro Gly Phe
 35 40 45

Leu Leu Asn Met Tyr Pro Lys Lys Ala Pro Ser Gln Pro Val Ser Leu
 50 55 60

Asp Thr Ile Leu Gln Glu Leu Glu Ala Asp Ile Ile Pro Gly Met Thr
 65 70 75 80

His Trp Gln Ser Pro Asn Phe Tyr Ala Tyr Phe Arg Thr Thr Thr Ser
 85 90 95

Asn Ala Ala Phe Gln Gly Glu Met Leu Cys Asn Ala Leu Asn Val Ala
 100 105 110

Gly Phe Asn Trp Ile Cys Ser Pro Ala Ala Thr Glu Leu Glu Met Ile
 115 120 125

Val Met Asp Trp Leu Gly Lys Met Leu Ser Leu Pro Gln Ser Phe Leu
 130 135 140

Phe Ala Gly Asn Gly Gly Gly Val Leu Gln Gly Ser Thr Ser Glu Ala
 145 150 155 160

Leu Ile Cys Val Leu Ser Ala Ala Arg Asp Arg Ala Leu Lys Gln Tyr
 165 170 175

Gly Glu Asp Ser Ile Thr Lys Leu Val Val Tyr Ala Ser Asp Gln Thr
 180 185 190

His	Phe	Val	Val	Lys	Lys	Ala	Ala	Lys	Leu	Val	Gly	Ile	Pro	Thr	Lys
195						200				205					
Asn	Phe	Arg	Val	Ile	Pro	Thr	Ser	Ile	Ala	Thr	Cys	Phe	Ala	Leu	Lys
210						215				220					
Pro	Asn	Asp	Ile	Lys	Met	Ala	Ile	Glu	Arg	Asp	Leu	Glu	Ser	Gly	Leu
225				230				235				240			
Val	Pro	Leu	Phe	Val	Cys	Ala	Thr	Val	Gly	Ala	Thr	Pro	Ser	Gly	Ser
				245				250						255	
Val	Asp	Pro	Val	Glu	Gly	Leu	Gly	Leu	Leu	Ala	Lys	Asn	Tyr	Gly	Leu
		260				265				270					
Trp	Leu	His	Ile	Glu	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Phe	Ile	Cys	Pro
275						280				285					
Glu	Leu	Thr	His	Tyr	Leu	Arg	Gly	Ile	Glu	His	Ala	His	Ser	Ile	Ser
290						295				300					
Ile	Asn	Leu	His	Lys	Trp	Leu	Leu	Thr	Asn	Met	Asp	Cys	Ser	Cys	Leu
305				310				315				320			
Trp	Val	Lys	Ser	Pro	Asp	Val	Leu	Leu	Glu	Ser	Leu	Ser	Met	Thr	Asp
				325				330						335	
Glu	Ile	Leu	Arg	Asn	Glu	Ala	Ser	Glu	Ser	Lys	Lys	Val	Val	Asp	Phe
		340				345				350					
Met	Asp	Trp	Gln	Ile	Ala	Thr	Ser	Lys	Leu	Phe	Arg	Ala	Leu	Lys	Leu
355						360				365					
Trp	Phe	Val	Leu	Arg	Arg	Tyr	Gly	Val	Asp	Asn	Leu	Met	Ala	His	Ile
370						375				380					
Arg	Ser	Asp	Ile	Glu	Leu	Ala	Lys	His	Phe	Glu	Ala	Leu	Val	Asn	Ser
385				390				395				400			
Asp	Lys	Arg	Phe	Glu	Val	Val	Val	Pro	Val	Asn	Phe	Ser	Leu	Val	Cys
		405				410				415					
Phe	Arg	Leu	Lys	Pro	Asn	Glu	Glu	Gly	Glu	Glu	Ser	Leu	Lys	Val	Leu
		420				425				430					
Met	Asn	Trp	Asn	Leu	Met	Glu	Ala	Val	Asn	Ser	Ser	Gly	Arg	Ala	Tyr
435						440				445					
Met	Thr	His	Ala	Val	Leu	Gly	Asp	Ile	Phe	Val	Ile	Arg	Cys	Ala	Ile
450						455				460					
Gly	Thr	Ser	Leu	Thr	Glu	Glu	Arg	His	Val	Asn	Glu	Leu	Trp	Lys	Leu
465				470				475				480			
Ile	Leu	Glu	Lys	Thr	Glu	Val	Ile	Leu	Lys	Arg	Asp	Gln			
		485				490									

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<210> SEQ ID NO 60
<211> LENGTH: 441
<212> TYPE: PRT
<213> ORGANISM: Daucus carota subsp. sativus
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 36056758
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<400> SEQUENCE: 60

Met	Asn	Thr	Phe	Asp	Thr	Glu	Asp	Phe	Arg	Lys	Gln	Ala	His	Leu	Ile
1				5					10					15	
Ile	Asp	Phe	Leu	Ala	Asp	Tyr	Tyr	Gln	Asn	Ile	Glu	Lys	Phe	Pro	Val
			20					25					30		
Arg	Ser	Gln	Val	Ser	Pro	Gly	Tyr	Leu	Gly	Glu	Ile	Leu	Pro	Asp	Ser
		35				40						45			
Ala	Pro	His	Asp	Pro	Glu	Pro	Ile	Glu	Lys	Ile	Leu	Glu	Asp	Val	Arg
	50					55					60				

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Ser	Asn	Ile	Ile	Pro	Gly	Ile	Thr	His	Trp	Gln	Ser	Pro	Asn	Phe	Phe	65	70	75	80
Ala	Tyr	Phe	Pro	Ser	Cys	Gly	Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu	Met	85	90	95	
Leu	Ala	Asn	Gly	Phe	Asn	Val	Val	Gly	Phe	Asn	Trp	Ile	Ser	Ser	Pro	100	105	110	
Ala	Ala	Thr	Glu	Leu	Glu	Thr	Ile	Val	Met	Asp	Trp	Leu	Gly	Lys	Met	115	120	125	
Leu	Gln	Leu	Pro	Glu	Ala	Phe	Leu	Phe	Ser	Gly	Gly	Gly	Gly	Gly	Val	130	135	140	
Leu	Gln	Gly	Thr	Thr	Cys	Glu	Ala	Met	Leu	Cys	Thr	Leu	Val	Ala	Ala	145	150	155	160
Arg	Asp	Arg	Thr	Leu	Arg	Glu	Gln	Gly	Met	Glu	Asn	Phe	Asp	Lys	Leu	165	170	175	
Leu	Cys	Pro	Val	Gln	Leu	Glu	Leu	Glu	Ile	Leu	Ser	Asp	Val	Gln	Asn	180	185	190	
Gly	Leu	Ile	Pro	Leu	Phe	Leu	Cys	Val	Thr	Ile	Gly	Thr	Thr	Pro	Ser	195	200	205	
Thr	Ala	Val	Asp	Pro	Leu	Ala	Thr	Leu	Ser	Glu	Val	Ala	Lys	Lys	Tyr	210	215	220	
Lys	Leu	Trp	Val	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	225	230	235	240
Cys	Pro	Glu	Phe	Arg	His	Phe	Leu	Asp	Gly	Leu	Glu	Asn	Val	Asn	Ser	245	250	255	
Phe	Ser	Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Thr	Leu	Asp	Cys	Cys	260	265	270	
Cys	Leu	Trp	Val	Asn	Asp	Pro	Ser	Ala	Leu	Ile	Lys	Ser	Leu	Ser	Thr	275	280	285	
Tyr	Pro	Glu	Phe	Leu	Arg	Asn	His	Ala	Ser	Glu	Ser	Asn	Lys	Val	Val	290	295	300	
Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Met	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	305	310	315	320
Lys	Leu	Trp	Phe	Val	Leu	Arg	Ser	Tyr	Gly	Val	Glu	Lys	Leu	Arg	Glu	325	330	335	
Phe	Ile	Arg	Val	His	Val	Glu	Met	Ala	Lys	Tyr	Phe	Glu	Gly	Leu	Val	340	345	350	
Ala	Met	Asp	Gln	Arg	Phe	Glu	Val	Val	Val	Pro	Arg	Leu	Phe	Ala	Met	355	360	365	
Val	Cys	Phe	Arg	Val	Val	Cys	Cys	Gly	Glu	Asn	Asp	Val	Asn	Glu	Ile	370	375	380	
Asn	Glu	Lys	Leu	Leu	Glu	Ser	Val	Asn	Gln	Ser	Gly	Arg	Ile	Tyr	Val	385	390	395	400
Ser	His	Ala	Val	Leu	Asp	Gly	Val	Tyr	Val	Ile	Arg	Phe	Ala	Ile	Gly	405	410	415	
Ala	Thr	Leu	Thr	Asp	Tyr	Ser	His	Val	Ser	Ala	Ala	Trp	Glu	Val	Val	420	425	430	
Gln	Glu	His	Ala	Asp	Ala	Leu	Leu	Ala	435	440									

<210> SEQ ID NO 61

<211> LENGTH: 510

<212> TYPE: PRT

<213> ORGANISM: Solanum tuberosum

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 3DMP400026166

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<400> SEQUENCE: 61

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Met Gly Thr Leu Asn Ile Asn His Glu Leu Asp Asp Gln Ile Phe Asn
1      5      10      15

Thr Ile Asn Pro Leu Asp Pro Glu Glu Phe Arg Arg Gln Gly His Lys
20      25      30

Ile Val Asn Phe Leu Ala Asp Tyr Tyr Gln Asn Ile Glu Gln Tyr Pro
35      40      45

Val Cys Ser Gln Val Asn Pro Gly Tyr Leu Gln Lys Ile Val Pro Asn
50      55      60

Ser Ala Pro Asn Asn Ser Glu Ser Leu Glu Lys Ile Leu Lys Asp Val
65      70      75      80

Glu Arg Asp Ile Ile Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe
85      90      95

Phe Ala Tyr Phe Pro Ser Ser Gly Ser Thr Ala Gly Phe Leu Gly Glu
100     105     110

Met Leu Ser Val Gly Phe Asn Val Val Gly Phe Asn Trp Ile Ser Ser
115     120     125

Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Met Asp Trp Phe Gly Lys
130     135     140

Met Leu Asn Leu Pro Asn Cys Phe Leu Phe Ala Ser Gly Gly Gly Gly
145     150     155     160

Val Leu Gln Gly Thr Thr Cys Glu Ala Met Leu Cys Thr Ile Val Ala
165     170     175

Ala Arg Asp Gln Met Leu Arg Lys Ile Ser Arg Glu Asn Phe Gly Lys
180     185     190

Leu Val Val Tyr Ala Ser Asp Gln Thr His Phe Ser Leu Lys Lys Ala
195     200     205

Ala His Ile Ala Gly Ile Asp Pro Gly Asn Phe Arg Val Ile Pro Thr
210     215     220

Ile Lys Ala Asn Glu Tyr Thr Leu Cys Pro Lys Ser Leu Arg Leu Ala
225     230     235     240

Ile Leu Asn Asp Leu Lys Glu Gly Asn Val Pro Leu Phe Leu Cys Ala
245     250     255

Thr Ile Gly Thr Thr Ala Thr Thr Ser Val Asp Pro Leu Arg Leu Leu
260     265     270

Cys Glu Ile Ala Lys Glu Phe Gly Ile Trp Val His Val Asp Ala Ala
275     280     285

Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Gln Val Phe Leu Asp
290     295     300

Gly Val Glu Asn Ala Asn Ser Phe Ser Leu Asn Ala His Lys Trp Phe
305     310     315     320

Phe Ser Thr Leu Asp Cys Cys Cys Leu Trp Val Lys Asp Pro Ser Ala
325     330     335

Leu Thr Asn Ala Leu Ser Thr Asn Pro Glu Cys Leu Arg Asn Lys Ala
340     345     350

Thr Glu Leu Asn Gln Val Ile Asp Tyr Lys Asp Trp Gln Ile Ala Leu
355     360     365

Ser Lys Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Leu Arg Ser Tyr
370     375     380

Gly Val Thr Asn Leu Arg Asn Leu Ile Arg Ser His Val Asn Met Ala
385     390     395     400

Lys His Phe Glu Gly Leu Val Ala Thr Asp Lys Arg Phe Glu Ile Phe

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405	410	415
Val Pro Arg Lys Phe Ala Met Val Cys Phe Arg Ile Ser Pro Leu Val		
420	425	430
Leu Ser Gln Val Ser Thr Lys Phe Asp Asp Glu Lys Glu Val Asn Met		
435	440	445
Phe Asn Thr Lys Leu Val Glu Ser Ile Asn Ser Cys Gly Lys Leu Tyr		
450	455	460
Leu Thr His Gly Val Val Gly Gly Thr Tyr Ile Ile Arg Phe Ala Ile		
465	470	475
Gly Ala Ser Leu Thr His Tyr Arg His Val Asp Val Ala Trp Lys Val		
485	490	495
Ile Gln Asp His Ala Asn Ala Leu Leu Asn Gln Gly Tyr Val		
500	505	510
 <210> SEQ ID NO 62		
<211> LENGTH: 507		
<212> TYPE: PRT		
<213> ORGANISM: Solanum tuberosum		
<300> PUBLICATION INFORMATION:		
<308> DATABASE ACCESSION NUMBER: 3DMP400024738		
 <400> SEQUENCE: 62		
Met Gly Thr Met Lys Ile Asn Pro Glu His Glu Phe Asp Gly Gln Phe		
1	5	10
Ser Ile Asn Thr Ser Ser Ser Arg Leu Leu Asp Pro Glu Glu Phe Arg		
20	25	30
Arg Gln Gly His Met Met Val Asp Phe Leu Ala Asp Tyr Phe Gln Asn		
35	40	45
Ile Glu Lys Tyr Pro Val Arg Ser Gln Val Glu Pro Gly Tyr Leu Lys		
50	55	60
Lys Leu Leu Pro Asp Ser Ala Pro Tyr Lys Pro Glu Pro Ile Ala Lys		
65	70	75
Ile Leu Glu Asp Val Glu Arg Asp Ile Phe Pro Gly Leu Thr His Trp		
85	90	95
Gln Ser Pro Asn Phe Phe Ala Tyr Phe Pro Cys Thr Ser Ser Thr Ala		
100	105	110
Gly Ile Leu Gly Glu Met Leu Ser Ala Gly Leu Asn Val Val Gly Phe		
115	120	125
Ser Leu Ile Ala Ser Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Met		
130	135	140
Asp Trp Leu Gly Lys Met Ile Ser Leu Pro Lys Thr Tyr Leu Phe Ser		
145	150	155
Gly Gly His Gly Gly Gly Gly Val Ile Gln Gly Thr Thr Cys Glu Ala		
165	170	175
Met Leu Cys Thr Ile Val Ala Ala Arg Glu Gln Met Leu Glu Lys Val		
180	185	190
Gly Arg Glu Lys Val Asp Lys Leu Val Val Tyr Ala Ser Asp Gln Thr		
195	200	205
His Phe Ser Phe Glu Lys Ala Val Lys Ile Ser Gly Ile Lys Leu Glu		
210	215	220
Asn Phe Arg Val Ile Pro Thr Thr Lys Asp Thr Glu Phe Ala Leu Asp		
225	230	235
Pro Lys Ser Leu Ser Arg Thr Ile Glu Gln Asp Ile Lys Ser Gly Phe		
245	250	255
Ile Pro Leu Phe Met Cys Ala Thr Ile Gly Thr Thr Ser Thr Thr Val		

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260	265	270
Val Asp Pro Leu Lys Leu Leu Cys Glu Ile Thr Lys Asp Tyr Gly Ile 275 280 285		
Trp Val His Val Asp Ala Ala Tyr Ala Gly Gly Ala Cys Ile Cys Pro 290 295 300		
Glu Phe Gln His Phe Leu Asp Gly Ile Glu Asn Ala Asn Ser Phe Ser 305 310 315 320		
Phe Asn Ala His Lys Trp Leu Phe Ser Asn Leu Asp Cys Cys Cys Leu 325 330 335		
Trp Val Lys Asp Pro Ser Ala Leu Thr Asn Ala Leu Ser Thr Arg Pro 340 345 350		
Glu Cys Leu Arg Asn Lys Ala Thr Asp Thr Lys Gln Val Val Asp Tyr 355 360 365		
Lys Asp Trp Gln Leu Ser Leu Ser Arg Arg Phe Arg Ala Leu Lys Leu 370 375 380		
Trp Leu Val Leu Arg Ser Tyr Gly Ile Asp Asn Leu Arg Asn Phe Ile 385 390 395 400		
Arg Ser His Val Lys Met Ala Lys His Phe Glu Gln Leu Val Ser Met 405 410 415		
Asp Glu Arg Phe Glu Ile Val Ala Pro Arg Asn Phe Ser Met Val Cys 420 425 430		
Phe Arg Val Ser Pro Leu Ala Leu Gly Asn Lys Gln Val Asn Lys Phe 435 440 445		
Asn Met Glu Leu Leu Glu Ser Ile Asn Ser Cys Gly Asn Ile His Met 450 455 460		
Thr His Ala Leu Val Gly Gly Val Tyr Met Ile Arg Phe Ala Ile Ala 465 470 475 480		
Ala Pro Leu Thr Glu Tyr Lys His Ile Asp Met Ala Trp Glu Val Ile 485 490 495		
Cys Asn His Ala Asn Ala Met Leu Asp Val Asn 500 505		
<210> SEQ ID NO 63		
<211> LENGTH: 476		
<212> TYPE: PRT		
<213> ORGANISM: Solanum lycopersicum		
<300> PUBLICATION INFORMATION:		
<308> DATABASE ACCESSION NUMBER: 36137005		
<400> SEQUENCE: 63		
Met Gly Thr Leu Asn Ile Asn His Glu Leu Asp Asp Gln Ile Phe Asn 1 5 10 15		
Thr Ile Asn Pro Leu Asp Pro Glu Glu Phe Arg Arg Gln Gly His Lys 20 25 30		
Ile Val Asn Phe Leu Ala Asp Tyr Tyr Gln Asn Ile Glu Gln Tyr Pro 35 40 45		
Val Cys Ser Gln Val Asn Pro Gly Tyr Leu Gln Asn Ile Val Pro Asn 50 55 60		
Ser Ala Pro Asn Asn Pro Glu Ser Leu Asp Lys Ile Leu Lys Asp Val 65 70 75 80		
Gln Asn Asp Ile Ile Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe 85 90 95		
Phe Ala Tyr Phe Pro Ser Ser Gly Ser Thr Val Gly Phe Val Gly Glu 100 105 110		
Met Leu Ser Val Gly Phe Asn Val Val Gly Phe Asn Trp Ile Ser Ser		

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115	120	125
Pro Ala Ala Thr Glu Leu	Glu Ser Ile Val Met	Asp Trp Phe Gly Lys
130	135	140
Met Leu Asn Leu Pro Asn Cys Phe Leu Phe Ala Ser Gly Gly Gly Gly		
145	150	155 160
Val Leu Gln Gly Thr Thr Cys Glu Ala Ile Leu Cys Thr Ile Val Ala		
165	170	175
Ala Arg Asp Gln Met Leu Arg Lys Ile Ser Arg Glu Asn Phe Gly Lys		
180	185	190
Leu Val Val Tyr Ala Ser Gly Gln Thr His Phe Ser Leu Lys Lys Ser		
195	200	205
Ala His Ile Ala Gly Ile Asp Pro Gly Asn Phe Arg Val Ile Pro Thr		
210	215	220
Ile Lys Ala Lys Glu Tyr Thr Leu Cys Pro Lys Ser Leu Arg Leu Ala		
225	230	235 240
Ile Leu Asn Asp Leu Lys Glu Gly Asn Val Pro Leu Phe Leu Cys Ala		
245	250	255
Thr Ile Gly Thr Thr Ser Thr Thr Ser Val Asp Pro Leu Arg Leu Leu		
260	265	270
Cys Asp Ile Ser Lys Glu Phe Gly Ile Trp Val His Val Asp Ala Ala		
275	280	285
Tyr Val Gly Ser Ala Cys Ile Cys Pro Glu Phe Gln Val Phe Leu Asp		
290	295	300
Gly Val Glu Asn Ala Asn Ser Phe Ser Leu Asn Asp Pro Ser Ala Leu		
305	310	315 320
Thr Asn Ala Leu Ser Thr Asn Leu Glu Phe Leu Arg Asn Lys Ala Thr		
325	330	335
Glu Leu Asn Gln Val Ile Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser		
340	345	350
Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Leu Arg Ser Tyr Gly		
355	360	365
Val Thr Asn Leu Arg Asn Leu Ile Arg Ser His Val Asn Met Thr Lys		
370	375	380
His Phe Glu Gly Leu Ile Ala Met Asp Lys Arg Phe Glu Ile Phe Val		
385	390	395 400
Pro Arg Lys Phe Ala Met Val Cys Phe Arg Ile Ser Pro Leu Val Leu		
405	410	415
Ser Gln Val Ser Ile Lys Phe Asp Asp Glu Lys Glu Val Asn Met Phe		
420	425	430
Asn Thr Lys Leu Leu Glu Ser Ile Asn Ser Cys Ser Lys Leu Tyr Leu		
435	440	445
Thr His Gly Ile Val Gly Gly Thr Tyr Ile Ile Arg Phe Ala Ile Gly		
450	455	460
Ala Ser Leu Thr His Tyr Arg His Val Asp Ile Ala		
465	470	475

<210> SEQ ID NO 64

<211> LENGTH: 440

<212> TYPE: PRT

<213> ORGANISM: *Daucus carota* subsp. *sativus*

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 36065781

<400> SEQUENCE: 64

Met Cys Lys Pro Lys Ser Ser Pro Ala Ser His Ile Asn Trp Gln Ser

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1	5	10	15
Pro Asn Phe	Phe Ala Tyr	Phe Pro Ser Ser Gly	Ser Thr Ala Gly Phe
	20	25	30
Leu Gly Glu	Met Leu Ser Thr	Gly Phe Asn Val Val	Gly Phe His Trp
	35	40	45
Met Ala Ser	Pro Ala Ala Thr	Glu Leu Glu Asn Val	Val Thr Asp Trp
	50	55	60
Phe Gly Lys	Met Leu Gln Leu	Pro Lys Ser Phe	Leu Phe Ser Gly Gly
	65	70	75
Gly Gly Gly	Val Leu Gln Gly	Thr Thr Cys Glu	Ala Met Leu Cys Thr
	85	90	95
Leu Val Ala	Ala Arg Asp Lys	Asn Leu Arg Gln	His Gly Met Glu Asn
	100	105	110
Ile Gly Lys	Leu Val Val Tyr	Cys Ser Asp Gln	Thr His Ser Ala Met
	115	120	125
Gln Lys Ala	Ala Lys Ile Ala	Gly Ile Asp Pro	Lys Asn Phe Arg Thr
	130	135	140
Val Glu Thr	Ser Arg Ala Ser	Asn Phe Gln Leu	Cys Pro Arg Arg Leu
	145	150	155
Glu Ser Ala	Ile Leu Thr Asp	Ile Gln Asn Gly	Leu Ile Pro Leu Tyr
	165	170	175
Leu Cys Ala	Thr Val Gly Thr	Thr Ser Ser Thr	Ala Val Asp Pro Leu
	180	185	190
Pro Ala Leu	Thr Glu Val Ala	Lys Lys Tyr Asp	Leu Trp Val His Val
	195	200	205
Asp Ala Ala	Tyr Ala Gly Ser	Ala Cys Ile Cys	Pro Glu Leu Arg Gln
	210	215	220
Tyr Leu Asn	Gly Val Glu Asn	Ala Asp Ser Phe	Ser Leu Asn Ala His
	225	230	235
Lys Trp Phe	Leu Thr Thr Leu	Asp Cys Cys Cys	Leu Trp Val Lys Asn
	245	250	255
Pro Ser Ala	Leu Ile Lys Ser	Leu Ser Thr Tyr	Pro Glu Phe Leu Arg
	260	265	270
Asn Asn Ala	Ser Glu Thr Asn	Lys Val Val Asp	Tyr Lys Asp Trp Gln
	275	280	285
Ile Met Leu	Ser Arg Arg Phe	Arg Ala Leu Lys	Leu Trp Phe Val Leu
	290	295	300
Arg Ser Tyr	Gly Val Gly Gln	Leu Arg Glu Phe	Ile Arg Gly His Val
	305	310	315
Asp Met Ala	Lys Tyr Phe Glu	Gly Leu Val Gly	Lys Asp Lys Arg Phe
	325	330	335
Glu Val Val	Val Pro Arg Leu	Phe Ser Met Val	Cys Ile Arg Val Arg
	340	345	350
Pro Ser Ala	Met Thr Gly Lys	Ser Cys Gly Asn	Asp Val Asn Glu Leu
	355	360	365
Asn Arg Lys	Leu Leu Glu Ser	Leu Asn Glu Ser	Gly Arg Ile Tyr Val
	370	375	380
Ser His Thr	Val Leu Asp Gly	Ile Tyr Ile Ile	Arg Phe Ala Ile Gly
	385	390	395
Ala Thr Leu	Thr Asp Ile Asn	His Val Ser Ala	Ala Trp Lys Val Val
	405	410	415
Gln Asp His	Ala Thr Ala Leu	Leu Asp Asp Thr	Asn Phe Leu Ala Lys
	420	425	430

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Lys Val Ala Asp Ile Ile Leu Ser
435 440

<210> SEQ ID NO 65
<211> LENGTH: 577
<212> TYPE: PRT
<213> ORGANISM: Oropetium thomaeum
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 35995617

<400> SEQUENCE: 65

Met Ala Ile Leu Asn His Ala Asp Asp Ala Ser Pro Ala Asn Asp Asp
1 5 10 15
Asn Pro Ala Thr Ala Pro Ala Met Ala Pro Ala Thr Asn Pro Arg Pro
20 25 30
Leu Asp Ala Asp Glu Phe Arg Arg Gln Gly Arg Leu Val Val Asp Phe
35 40 45
Ile Ala Asp Tyr Tyr Ala Arg Val Glu Glu Tyr Pro Val Arg Pro Ser
50 55 60
Val Thr Pro Gly Phe Leu Ser Arg Lys Leu Pro Glu Thr Ala Pro Glu
65 70 75 80
Gln Pro Glu Pro Gly His Gly Asp Ala Phe Ala Ser Ala Leu Arg Asp
85 90 95
Val Arg Asp Leu Ile Leu Pro Gly Ile Thr His Trp Gln Ser Pro Asn
100 105 110
His Phe Ala His Phe Ala Ala Thr Ala Ser Asn Val Gly Ala Leu Gly
115 120 125
Glu Ala Leu Ala Ala Gly Leu Asn Ile Asn Pro Phe Thr Trp Ala Ala
130 135 140
Ser Ser Ala Ala Thr Glu Leu Glu Val Val Val Thr Asp Trp Leu Gly
145 150 155 160
Lys Ala Leu His Leu Pro Gln Glu Leu Leu Phe Ser Gly Gly Gly Gly
165 170 175
Gly Thr Leu Leu Gly Thr Ser Cys Glu Ala Met Leu Cys Thr Val Val
180 185 190
Ala Ala Arg Asp Arg Lys Leu Gly Glu Ile Gly Glu His Arg Ile Gly
195 200 205
Asp Leu Val Val Tyr Cys Ser Asp Gln Thr His Phe Ser Phe Arg Lys
210 215 220
Ala Ala Arg Val Ala Gly Ile Arg Arg Ala Asn Cys Arg Glu Ile Pro
225 230 235 240
Thr Ser Leu Glu Ser Asp Phe Ala Leu Ser Pro Ser Ala Leu Leu Ala
245 250 255
Ala Val Arg Ala Asp Glu Ala Ala Gly Leu Val Pro Leu Tyr Leu Cys
260 265 270
Val Thr Val Gly Thr Thr Pro Thr Ala Ala Val Asp Pro Val Arg Glu
275 280 285
Leu Cys Ala Ala Val Ala Gly Arg Gly Val Trp Val His Val Asp Ala
290 295 300
Ala Tyr Ala Gly Ala Ala Arg Val Cys Pro Glu Leu Leu Arg His Ala
305 310 315 320
Gly Ala Ile Val Asp Gly Val Asp Ser Phe Ser Thr Asn Pro His Lys
325 330 335
Trp Leu Leu Ala Asn Met Asp Cys Cys Ala Leu Trp Val Gln Gln Pro
340 345 350

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Asp Ala Leu Val Ala Ala Leu Gly Thr Asp His Asp Val Ile Leu Lys
 355 360 365
 Asp Pro Ala Ala Ala Ala Ala Gly Asp Val Val Val Asp Tyr Lys Asp
 370 375 380
 Trp Gln Val Ala Leu Ser Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu
 385 390 395 400
 Leu Leu Arg Cys His Gly Val Glu Gly Leu Arg Ala His Val Arg Asp
 405 410 415
 Gly Leu Arg Met Ala Glu Ala Phe Glu Ala Met Val Arg Ala Asp Ala
 420 425 430
 Arg Phe Glu Val Pro Val Arg Arg Gln Leu Ser Leu Val Cys Phe Arg
 435 440 445
 Leu Arg Pro Thr Ala Val Ile Arg Glu Lys Gln Gln Gln Gln Arg Gly
 450 455 460
 Arg Arg Arg Asp His Asp Asp Asp Thr Ala Ala Ala Asn Glu Leu Asn
 465 470 475 480
 Arg Arg Leu Leu Glu Ala Val Asn Ala Thr Gly Arg Thr Tyr Met Ser
 485 490 495
 Cys Ala Val Val Gly Gly Val Tyr Met Leu Arg Cys Ala Ile Gly Asn
 500 505 510
 Ser Leu Thr Glu Asp Arg His Val Glu Glu Ala Trp Asn Val Val Gln
 515 520 525
 Glu Gln Ala Ser Ala Ile Leu Asp Ala Ala Met Val Val Arg Ala Asp
 530 535 540
 Glu Cys Thr Val Cys Thr Ala Ala His Cys Val Gln Met Gly Met Val
 545 550 555 560
 Asp Asp Ile Leu Ala Ala Ser Phe Pro Thr Gly Asn Glu Val Thr Ile
 565 570 575

Arg

<210> SEQ ID NO 66
 <211> LENGTH: 565
 <212> TYPE: PRT
 <213> ORGANISM: Oryza sativa
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 33157740

<400> SEQUENCE: 66

Met Ala Ile Leu Asn His Ser Asp Ala Ala Phe Pro Val Ala Ala Thr
 1 5 10 15
 Thr Pro Leu Leu Gly Arg Arg Pro Leu Asp Ala Gly Glu Phe Arg Arg
 20 25 30
 Gln Gly Arg Gln Val Val Asp Phe Ile Ala Asp Tyr Tyr Ala Gly Ile
 35 40 45
 Asn Asp Tyr Pro Val Arg Pro Ala Val Ala Pro Gly Phe Leu Ala Gly
 50 55 60
 Lys Leu Pro Ala Thr Ala Pro Ser Thr Pro Glu Pro Asp Ala Leu Thr
 65 70 75 80
 Ala Gly Leu Arg Asp Val Arg Glu Leu Met Leu Pro Gly Leu Thr His
 85 90 95
 Trp Gln Ser Pro Arg His Phe Ala His Phe Ser Ala Thr Ala Ser Asn
 100 105 110
 Val Gly Ala Leu Gly Glu Ala Leu Ala Ala Gly Leu Asn Val Asn Pro
 115 120 125

Phe 130	Thr	Trp	Glu	Ala	Ser	Pro 135	Ala	Ala	Thr	Glu	Leu 140	Glu	Val	Val	Val
Thr 145	Asp	Trp	Leu	Gly	Lys 150	Ala	Leu	His	Leu	Pro 155	Arg	Leu	Leu	Phe 160	
Ala	Gly	Gly	Gly	Gly 165	Gly	Thr	Leu	Leu	Gly 170	Thr	Ser	Cys	Glu	Ala 175	Met
Leu	Cys	Thr	Ile 180	Val	Ala	Ala	Arg	Asp 185	Glu	Lys	Leu	Ala	Glu	Ile	Gly
Glu	Glu	Arg 195	Ile	Gly	Asp	Leu	Val 200	Val	Tyr	Cys	Ser	Asp 205	Gln	Thr	His
Phe 210	Ser	Phe	Gln	Lys	Ala	Ala 215	Arg	Ile	Ala	Gly	Ile	Arg	Arg	Gly	Asn
Cys 225	Arg	Glu	Ile	Pro	Thr 230	Cys	Arg	Glu	Ser	Gly 235	Phe	Val	Leu	Thr	Ala
Thr	Ala	Leu	Gln	Ala 245	Ala	Val	Ala	Ala	Asp 250	Glu	Ala	Ala	Gly	Arg 255	Val
Pro	Leu	Phe 260	Leu	Cys	Ala	Thr	Val	Gly 265	Thr	Thr	Pro	Thr	Ala	Ala	Val
Asp	Pro 275	Leu	Arg	Glu	Leu	Cys	Ala 280	Ala	Val	Glu	Gly	Arg 285	Gly	Val	Trp
Val 290	His	Val	Asp	Ala	Ala	Tyr 295	Ala	Gly	Ala	Ala	Cys 300	Val	Cys	Pro	Glu
Phe 305	Arg	His	Ala	Ile 310	Ala	Gly	Ala	Glu	Ala	Val 315	Asp	Ser	Phe	Ser	Thr
Asn	Pro	His	Lys	Trp 325	Leu	Leu	Ala	Asn	Met 330	Asp	Cys	Cys	Ala	Leu	Trp
Val	Ala	Arg 340	Pro	Ala	Ala	Leu	Val	Ala 345	Ala	Leu	Gly	Thr	Asp 350	Asp	Asp
Val	Ile 355	Leu	Lys	Asp	Ala	Ala	Ala 360	Ala	Ala	Arg	Pro	Ala 365	Arg	Gly	Asp
His 370	His	His	His	Ala	Ala	Val 375	Asp	Tyr	Lys	Asp 380	Trp	Gln	Val	Ala	Leu
Ser 385	Arg	Arg	Phe	Arg	Ala 390	Leu	Lys	Leu	Trp	Leu 395	Val	Leu	Arg	Cys	His
Gly	Val	Asp	Gly	Leu 405	Arg	Ala	Val	Val	Arg 410	Ser	His	Val	Arg	Met	Ala
Ala	Ala	Leu 420	Glu	Arg	Met	Val	Arg	Ala 425	Asp	Ala	Arg	Phe	Glu	Val	Pro
Val	Pro 435	Arg	Gln	Phe	Ala	Leu	Val	Cys 440	Phe	Arg	Leu	Arg	Gly	Gly	Gly
Ala 450	Ala	Ala	Gln	Leu	Val	Gly 455	Gly	Asp	Glu	Leu	Thr	Ala	Ser	Asn	Glu
Leu 465	Asn	Arg	Arg	Leu	Leu 470	Glu	Ala	Val	Asn	Ala 475	Thr	Gly	Arg	Ala	Tyr
Met	Ser	Ser	Ala	Val 485	Val	Gly	Gly	Met	Tyr 490	Val	Leu	Arg	Cys	Ala	Val
Gly	Asn	Ser	Leu 500	Thr	Glu	Glu	His	His 505	Val	Arg	Glu	Ala	Trp	Ser	Val
Val	Gln 515	Gly	Gln	Ala	Ala	Ala	Val	Leu 520	Ala	Thr	Ala	Gly	Ala	Ala	Ala
Asp 530	Thr	Ala	Arg	Thr	Lys	Asp 535	His	Ala	Ala	Gly	Asp 540	Asp	His	Gly	Ala
Asp	Gln	Pro	His	Ala	Met	Thr	Thr	Thr	Thr	Thr	Met	Gly	Cys	Arg	Ser

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545 550 555 560

Gly Pro Trp Glu Leu
565

<210> SEQ ID NO 67
 <211> LENGTH: 509
 <212> TYPE: PRT
 <213> ORGANISM: *Brachypodium stacei*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 01G392300.1

<400> SEQUENCE: 67

Met Ala Pro Ala Ser Ser Thr Arg Gln Val Ile Thr Asp His Lys Thr
 1 5 10 15

Gln Lys Glu Asn Ser Ser Cys Thr Val Ile Asn His Leu Leu Asp Ala
 20 25 30

Asp Glu Phe Arg Arg Gln Gly His Lys Val Ile Asp Phe Ile Ala Asp
 35 40 45

Tyr Tyr Ser Gly Ile Ala Asp Tyr Pro Val His Pro Ser Val Thr Pro
 50 55 60

Gly Phe Leu Leu Asn Gln Leu Pro Ala Asp Pro Pro Glu Asp Pro Asp
 65 70 75 80

Thr Phe Ala Ser Ala Leu Gln Asp Val Arg Asp Leu Ile Leu Pro Gly
 85 90 95

Met Thr His Trp Gln Ser Pro Arg His Leu Ala His Phe Pro Ala Ser
 100 105 110

Ser Ser Val Thr Gly Ala Leu Gly Glu Ala Leu Ala Ala Gly Ile Asn
 115 120 125

Ala Val Pro Phe Met Trp Ser Ala Ser Pro Ala Ala Thr Glu Leu Glu
 130 135 140

Met Val Ala Val Asp Trp Leu Gly Lys Ala Leu His Leu Pro Lys Thr
 145 150 155 160

Leu Leu Phe Ser Gly Ala Gly Gly Gly Thr Leu Leu Gly Thr Ser Tyr
 165 170 175

Arg Lys Leu Ala Glu Thr Gly Ala Gly Arg Ile Gly Asp Leu Val Val
 180 185 190

Tyr Gly Ser Asp Gln Thr His Phe Ala Leu Arg Lys Ala Ala Arg Ile
 195 200 205

Ala Gly Ile Arg His Gly Arg Cys Arg Glu Leu Arg Thr Cys Ile Ala
 210 215 220

Asp Met Phe Ala Leu Ser Pro Ala Ala Leu Ser Ala Ala Met Asp Ala
 225 230 235 240

Asp Ala Gly Ala Gly Leu Val Pro Leu Phe Leu Cys Ala Thr Val Gly
 245 250 255

Thr Thr Gln Thr Lys Ala Val Asp Pro Ile Gly Ala Leu Cys Ala Glu
 260 265 270

Ala Ala Pro His Gly Val Trp Val His Val Asp Ala Ala Tyr Gly Gly
 275 280 285

Ser Ala Leu Val Cys Pro Glu Leu Ala Arg Asp Ala Ile Asp Gly Val
 290 295 300

Glu Ala Val Asp Ser Phe Ser Met Asn Ala His Lys Trp Leu Leu Val
 305 310 315 320

Asn Thr Asp Cys Cys Ala Leu Trp Val Lys Arg Pro Ala Leu Leu Val
 325 330 335

Ser Ala Leu Gly Thr Gln Asp Glu Asp Glu Val Ile Leu Arg Asp Ala

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340	345	350
Ala Ala Gln Gly His Asp Val	Val Asp Tyr Lys Asp Trp	Ala Val Thr
355	360	365
Leu Thr Arg Arg Phe Arg Ala	Leu Lys Leu Trp Leu	Val Leu Arg Cys
370	375	380
Tyr Gly Val Glu Gly Leu Arg	Glu His Ile Arg Gly	His Val Arg Met
385	390	400
Ala Ala Leu Phe Glu Gly Met	Val Asn Ala Asp Pro	Arg Phe Glu Val
405	410	415
Val Thr Glu Arg Arg Phe Ala	Leu Val Cys Phe Arg	Leu Arg Pro Asp
420	425	430
Gln Leu Pro Asp Glu Gly Asn	Lys Lys Lys Thr Met	Ala Ala Ala Asn
435	440	445
Glu Leu Asn Arg Arg Leu Leu	Gln Glu Val Asn Ala	Ala Ala Leu Gly
450	455	460
Pro Tyr Met Ser Ala Ala Asn	Val Gly Gly Ile Tyr	Val Leu Arg Cys
465	470	475
Ala Val Gly Ser Thr Leu Thr	Glu Lys Arg His Val	Arg Gln Ala Trp
485	490	495
Glu Val Val Gln Glu Lys Ala	Thr Ser Ile Leu Arg Ala	
500	505	

<210> SEQ ID NO 68
 <211> LENGTH: 475
 <212> TYPE: PRT
 <213> ORGANISM: *Amaranthus hypochondriacus*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 32828676
 <400> SEQUENCE: 68

Ser Leu His Asp Glu Thr Leu	Gln Gly Ile Lys Tyr Val Thr	Gln Tyr
1	5	10
Tyr Lys Asn Val Glu Lys Tyr	Pro Val Val Ser Lys Val	Lys Trp Gly
20	25	30
Tyr Leu Arg Gln Ile Leu Pro	Glu Asn Ala Pro Ser	Leu Pro Glu Ser
35	40	45
Ile Asp Gln Ile Leu Glu Asp	Val Asp Thr Lys Ile	Val Pro Gly Leu
50	55	60
Thr His Trp Gln Ser Pro Asn	Phe Phe Ala Tyr Phe	Pro Ala Thr Ala
65	70	75
Ser Asn Ala Ala Met Leu Gly	Asp Ile Val Cys Ser Gly	Leu Asn Val
85	90	95
Val Gly Phe Ser Trp Ile Ser	Ser Pro Ala Ala Thr	Glu Leu Glu Ala
100	105	110
Ile Val Met Asp Trp Met Ala	Lys Leu Leu Met Leu	Pro Pro Thr Phe
115	120	125
Leu Phe Ser Gly Gly Gly Gly	Gly Val Ile His Gly	Ser Thr Cys Glu
130	135	140
Ala Ile Val Cys Thr Gln Ala	Ala Ala Arg Asp Val	Ala Leu Asn Ile
145	150	155
His Gly Glu Glu Lys Ile Thr	Lys Leu Val Val Tyr	Ala Ser Asp Gln
165	170	175
Thr His Ile Ser Phe Gln Lys	Ala Ala Lys Leu Ile	Gly Ile Pro Pro
180	185	190
Arg Asn Phe Arg Val Leu Pro	Thr Ser Ser Ala Thr	Asp Phe Ala Leu

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195	200	205
Ser Pro Thr Thr Leu Arg Ala	Ser Ile Glu Val Asp	Leu Ser Gln Gly
210	215	220
Leu Val Pro Phe Tyr Ile Cys Ala Thr Ile Gly Ala Thr Pro Ser Gly		
225	230	235 240
Ala Val Asp Pro Ile Asp Gly Leu Gly Gln Ile Ala Arg Asp Tyr Gly		
	245	250 255
Ala Trp Leu His Val Asp Ala Ala Phe Ala Gly Asn Ala Cys Ile Cys		
	260	265 270
Pro Glu Tyr Arg His Tyr Leu Asp Gly Val Glu Leu Ala Asp Ser Ile		
	275	280 285
Ser Met Asn Pro His Lys Trp Leu Leu Thr Asn Met Glu Cys Ser Cys		
	290	295 300
Leu Trp Leu Lys Asn Pro Lys Leu Met Val Asp Ser Leu Ser Thr Lys		
305	310	315 320
Pro Glu Ile Leu Asn Asn Lys Ala Thr Gln Ser Gly Asp Val Ile Asp		
	325	330 335
Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala Leu Lys		
	340	345 350
Leu Trp Ile Val Ile Arg Arg Tyr Gly Ser Thr Tyr Leu Met Asn His		
	355	360 365
Val Arg Ser Asp Ile Glu Leu Ala Lys Tyr Phe Glu Ser Leu Ile Lys		
	370	375 380
Gln Asp Glu Arg Phe Glu Leu Val Val Pro Arg Lys Phe Ser Leu Val		
385	390	395 400
Cys Phe Arg Met Lys Leu Val Gly Arg Glu Asp Val Glu Thr Leu Thr		
	405	410 415
Asn Gln Lys Leu Leu Glu Asp Val Asn Ser Ser Gly Lys Ala Tyr Met		
	420	425 430
Thr His Ala Val Ile Gly Gly Lys Phe Val Ile Arg Cys Ala Ile Gly		
	435	440 445
Gly Thr Leu Thr Glu Lys Arg His Ile Asp Ser Leu Trp Lys Leu Ile		
	450	455 460
Ile Glu Lys Val Pro Leu Thr Thr Cys Glu Leu		
465	470	475

<210> SEQ ID NO 69

<211> LENGTH: 553

<212> TYPE: PRT

<213> ORGANISM: Brachypodium distachyon

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 5g21770.1

<400> SEQUENCE: 69

Met Ser Ser Asn Ser Cys Pro Ala Ala Ala Ala Thr Phe Thr Thr
1 5 10 15
Pro Pro Gly Ala His Pro Leu Pro Leu Asp Ala Asp Ala Phe Arg Arg
20 25 30
Gln Gly Arg Gln Val Ala Asp Phe Ile Ala Asp Tyr Tyr Asp Arg Ile
35 40 45
Glu Asp Tyr Pro Val Arg Pro Asn Val Ser Pro Gly Phe Leu Ala Ala
50 55 60
Gln Leu Pro Asp Ala Ala Pro Ser Trp Pro Glu Glu Pro Asp Ala Leu
65 70 75 80
Ala Ser Ala Leu Arg Asp Val Arg Asp Leu Ile Leu Pro Gly Leu Thr

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85							90					95				
His	Trp	Gln	Ser	Pro	Arg	His	Phe	Ala	His	Phe	Ala	Ala	Thr	Ala	Ser	
			100					105					110			
Asn	Ala	Gly	Ala	Leu	Gly	Glu	Phe	Leu	Ala	Ala	Gly	Leu	Asn	Val	Asn	
			115					120					125			
Pro	Phe	Thr	Trp	Ala	Ala	Ser	Pro	Ala	Ala	Ala	Glu	Leu	Glu	Val	Val	
			130					135					140			
Val	Thr	Asp	Trp	Leu	Gly	Gln	Ala	Leu	Gly	Leu	Pro	Glu	Lys	Leu	Leu	
			145					150					155			
Phe	Arg	Gly	Gly	Ser	Gly	Gly	Gly	Gly	Thr	Leu	Leu	Gly	Thr	Ser	Cys	
			165					170					175			
Glu	Ala	Met	Leu	Cys	Thr	Ile	Val	Ala	Ala	Arg	Asp	Gln	Lys	Leu	Leu	
			180					185					190			
Lys	Ile	Gly	Glu	Asp	Arg	Ile	Gly	Asp	Leu	Val	Val	Tyr	Cys	Ser	Asp	
			195					200					205			
Gln	Thr	His	Phe	Ser	Phe	Lys	Lys	Ala	Ala	Arg	Val	Ala	Gly	Ile	Arg	
			210					215					220			
Arg	Gly	Asn	Cys	Arg	Val	Ile	Pro	Thr	Arg	Phe	Glu	Asp	Gly	Phe	Ala	
			225					230					235			
Leu	Ser	Pro	Ala	Ala	Leu	Ala	Ala	Ala	Val	Arg	Asp	Asp	Val	Ala	Arg	
			245					250					255			
Gly	Lys	Val	Pro	Leu	Phe	Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Ala	Thr	
			260					265					270			
Gly	Ala	Val	Asp	Pro	Val	Arg	Glu	Leu	Cys	Ala	Ala	Val	Gly	Ala	Gly	
			275					280					285			
His	Gly	Ser	Gly	Val	Trp	Val	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Gly	
			290					295					300			
Ala	Cys	Val	Cys	Pro	Glu	Phe	Arg	His	Val	Ala	Ala	Gly	Ala	Glu	Glu	
			305					310					315			
Ala	Asp	Ser	Phe	Ser	Thr	Asn	Pro	His	Lys	Trp	Leu	Leu	Ala	Asn	Met	
			325					330					335			
Asp	Cys	Cys	Ala	Leu	Trp	Ile	Arg	Arg	Pro	Gly	Leu	Leu	Val	Ala	Ala	
			340					345					350			
Leu	Gly	Ala	Gly	Glu	Asp	Glu	Asp	Ala	Ile	Leu	Asn	Lys	Ala	Pro	Pro	
			355					360					365			
Ala	Ala	Arg	Gly	Met	Gln	Ala	Asp	Leu	Met	Val	Asp	Tyr	Lys	Asp	Trp	
			370					375					380			
Gln	Val	Pro	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp	Leu	Val	
			385					390					395			
Leu	Arg	Cys	His	Gly	Val	Glu	Gly	Leu	Arg	Gly	Val	Val	Arg	Gly	His	
			405					410					415			
Val	Arg	Met	Ala	Ala	Ala	Phe	Glu	Ala	Met	Val	Arg	Ala	Asp	Pro	Arg	
			420					425					430			
Phe	Glu	Val	Pro	Val	Pro	Pro	Ala	Phe	Ala	Leu	Val	Cys	Phe	Arg	Leu	
			435					440					445			
Arg	Pro	Leu	Ala	Ala	His	Pro	Gly	Ser	Ser	Ser	Gly	Ile	Asp	Glu	Val	
			450					455					460			
Asn	Gly	Arg	Leu	Leu	Glu	Ala	Val	Asn	Gly	Thr	Gly	Arg	Ala	Tyr	Met	
			465					470					475			
Ser	Gly	Ala	Val	Val	Gly	Gly	Ala	Tyr	Val	Leu	Arg	Cys	Ala	Val	Gly	
			485					490					495			
Asn	Ser	Leu	Thr	Glu	Asp	Arg	His	Val	Arg	Glu	Ala	Trp	Ser	Val	Val	
			500					505					510			

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Gln Glu Gln Ala Asp Ala Ile Leu Ala Pro Ser Asp Asp Glu Asp Arg
 515 520 525

Cys Cys Thr Asp Gln Ile Gln Thr Glu Met Glu Leu Gln Arg Arg Pro
 530 535 540

Leu Gly Ala Ala Ala Asp Val Phe Ala
 545 550

<210> SEQ ID NO 70
 <211> LENGTH: 543
 <212> TYPE: PRT
 <213> ORGANISM: Brachypodium distachyon
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 2g02360.1

<400> SEQUENCE: 70

Met Ala Pro Ala Ser Ser Lys Leu His Ala Ile Thr Asp Asp Lys Thr
 1 5 10 15

Gln Gln Gln Asn Ser Ser Cys Pro Ala Ala Ser Asn Gly Ala Ile Glu
 20 25 30

Pro Ser Asn Ala Lys Cys Ala Ala Ser Ser Asn His Leu Leu Asp Ala
 35 40 45

Asp Glu Phe Arg Arg Gln Gly His Lys Val Ile Asp Phe Ile Ala Asp
 50 55 60

Tyr Tyr Ala Gly Ile Ala Asp Tyr Pro Val His Pro Ser Val Thr Pro
 65 70 75 80

Gly Phe Leu Leu Asn Gln Leu Pro Ala Asp Pro Pro Ser Arg Pro Glu
 85 90 95

Asp His Pro Asp Gly Ala Phe Gly Pro Ala Leu Gln Asp Val Arg Asp
 100 105 110

Val Ile Leu Pro Gly Met Thr His Trp Gln Ser Pro Arg His Phe Ala
 115 120 125

His Phe Pro Ala Ser Ser Ser Val Ala Gly Val Leu Gly Glu Ala Leu
 130 135 140

Ala Ala Gly Ile Asn Ala Val Pro Phe Thr Trp Ala Ala Ser Pro Ala
 145 150 155 160

Ala Ala Glu Leu Glu Met Val Ala Val Asp Trp Leu Gly Lys Ala Leu
 165 170 175

His Leu Pro Glu Ser Leu Leu Phe Ser Gly Ala Gly Gly Gly Thr Leu
 180 185 190

Leu Gly Thr Ser Cys Glu Ala Ile Leu Cys Ala Leu Val Ala Ala Arg
 195 200 205

Asp Arg Lys Leu Ala Asp Ile Gly Thr Asp Arg Ile Gly Asp Leu Val
 210 215 220

Val Tyr Gly Ser Asp Gln Thr His Phe Ala Leu Arg Lys Ala Ala Arg
 225 230 235 240

Ile Ala Gly Ile Arg His Asp Arg Cys Arg Glu Leu Gln Thr Cys Leu
 245 250 255

Ala Asp Met Phe Ala Leu Ser Pro Ala Ala Leu Ser Ala Ala Met Asp
 260 265 270

Ala Asp Ala Gly Ala Gly Leu Val Pro Leu Phe Leu Cys Ala Thr Val
 275 280 285

Gly Thr Thr Gln Thr Thr Ala Val Asp Gln Val Gly Ala Leu Cys Ala
 290 295 300

Ala Ala Ala Pro His Gly Val Trp Val His Val Asp Ala Ala Tyr Ala
 305 310 315 320

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Gly Ser Ala Leu Val Cys Pro Glu Leu Ala Arg Asp Ala Ile Asp Gly
 325 330 335
 Ile Glu Val Val Asp Ser Phe Ser Met Asn Ala His Lys Trp Leu Leu
 340 345 350
 Ala Asn Thr Asp Cys Cys Ala Leu Trp Val Lys Gln Pro Lys Leu Leu
 355 360 365
 Val Val Ser Leu Gly Thr Gln Asn Glu Glu Leu Ile Leu Arg Asp Ala
 370 375 380
 Ala Ala Glu Gly His Asp Val Val Asp Tyr Lys Asp Trp Ala Ile Thr
 385 390 395 400
 Leu Thr Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Phe Arg Cys
 405 410 415
 Tyr Gly Val Glu Gly Leu Arg Glu His Ile Arg Ala His Val Arg Met
 420 425 430
 Ala Ala Leu Phe Glu Gly Leu Val Lys Asp Asp Pro Arg Phe Glu Val
 435 440 445
 Val Thr Glu Arg Arg Phe Ala Leu Val Cys Phe Arg Leu Arg Ala Pro
 450 455 460
 Asp Gln Leu Met Asp Glu Gly Asn Glu Lys Lys Lys Thr Thr Ala Ala
 465 470 475 480
 Ala Asn Glu Leu Asn Arg Arg Leu Leu Arg Glu Val Asn Gly Val Ala
 485 490 495
 Leu Gly Pro Tyr Met Ser Ala Ala Val Val Gly Gly Ile Tyr Ile Leu
 500 505 510
 Arg Cys Ala Val Gly Ser Thr Leu Thr Glu Glu Arg His Val Arg Gln
 515 520 525
 Ala Trp Glu Val Val Gln Glu Arg Ala Thr Ser Ile Leu Arg Gly
 530 535 540

<210> SEQ ID NO 71
 <211> LENGTH: 588
 <212> TYPE: PRT
 <213> ORGANISM: Sorghum bicolor
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 009G192600.1

<400> SEQUENCE: 71

Met Gly Val Ala Val Thr Ala Glu Val Val His Ala Arg Ser Cys Lys
 1 5 10 15
 Gly Thr Pro Pro Val Gly Ala Ala Ala Ser Val Met Val Trp Asp Gly
 20 25 30
 Ala Gly Gln Gly Tyr Ser Cys Gln Pro Val Gly Thr Thr Thr Ala Asn
 35 40 45
 Gly Gly Thr Thr Pro Ala Ala Pro Val Ala Ile Ala Met Pro Ser Leu
 50 55 60
 Pro His Pro Leu Leu Asp Ala Asp Glu Phe Arg Arg Gln Gly Arg Leu
 65 70 75 80
 Val Val Asp Phe Ile Ala Asp Tyr Tyr Ala Arg Ile Asp Glu Tyr Pro
 85 90 95
 Val Arg Pro Ala Val Ala Pro Gly Phe Leu Ala Arg Gln Leu Pro Glu
 100 105 110
 Thr Ala Pro Ala Arg Pro Glu Pro Asp Ala Leu Ala Ala Ala Leu Arg
 115 120 125
 Asp Val Arg Asp Leu Ile Leu Pro Gly Val Thr His Trp Gln Ser Pro
 130 135 140

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Arg	His	Phe	Ala	His	Phe	Ala	Ala	Thr	Ala	Ser	Asn	Val	Gly	Ala	Leu	145	150	155	160
Gly	Glu	Ala	Leu	Ala	Ala	Gly	Leu	Asn	Ile	Asn	Pro	Phe	Thr	Trp	Ala	165	170	175	
Ala	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Val	Val	Val	Thr	Asp	Trp	Leu	180	185	190	
Gly	Lys	Ala	Leu	His	Leu	Pro	Glu	Ser	Leu	Leu	Phe	Ser	Gly	Gly	Gly	195	200	205	
Gly	Gly	Thr	Leu	Leu	Gly	Thr	Ser	Cys	Glu	Ala	Met	Leu	Cys	Thr	Ile	210	215	220	
Val	Ala	Ala	Arg	Asp	Arg	Lys	Leu	Ala	Glu	Val	Gly	Glu	Glu	Arg	Met	225	230	235	240
Gly	Asp	Leu	Val	Val	Tyr	Cys	Ser	Asp	Gln	Thr	His	Phe	Ser	Phe	Gln	245	250	255	
Lys	Ala	Ala	Arg	Ile	Ala	Gly	Ile	Arg	Arg	Gly	Asn	Cys	Arg	Glu	Ile	260	265	270	
Pro	Thr	Ser	Met	Glu	Ala	Gly	Phe	Thr	Leu	Ser	Pro	Lys	Ala	Leu	Ala	275	280	285	
Ala	Ala	Val	Arg	Ala	Asp	Glu	Ala	Ala	Gly	Arg	Val	Pro	Leu	Phe	Leu	290	295	300	
Cys	Ala	Thr	Val	Gly	Thr	Thr	Pro	Thr	Ala	Ala	Val	Asp	Pro	Val	Arg	305	310	315	320
Glu	Leu	Cys	Ala	Ala	Val	Ala	Gly	Arg	Gly	Val	Trp	Val	His	Val	Asp	325	330	335	
Ala	Ala	Tyr	Ala	Gly	Ala	Ala	Ser	Val	Cys	Pro	Glu	Leu	Arg	His	Ala	340	345	350	
Val	Ala	Gly	Val	Glu	Arg	Val	Asp	Ser	Phe	Ser	Thr	Asn	Pro	His	Lys	355	360	365	
Trp	Leu	Leu	Ala	Asn	Met	Asp	Cys	Cys	Ala	Leu	Trp	Val	Arg	Arg	Pro	370	375	380	
Ala	Ala	Leu	Thr	Ala	Ala	Leu	Gly	Thr	Asp	His	Asp	Val	Ile	Leu	Lys	385	390	395	400
Asp	Pro	Ser	Ala	Gln	Ala	Ala	Gln	Glu	Gly	Gly	Ala	Val	Val	Asp	Tyr	405	410	415	
Lys	Asp	Trp	Gln	Val	Ala	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	420	425	430	
Trp	Leu	Val	Leu	Arg	Cys	His	Gly	Val	Glu	Gly	Leu	Arg	Gly	Leu	Val	435	440	445	
Arg	Ala	His	Val	Arg	Met	Ala	Ala	Ala	Phe	Glu	Ala	Met	Val	Arg	Thr	450	455	460	
Asp	Ala	Arg	Phe	Glu	Val	Pro	Val	Pro	Arg	Gln	Phe	Ala	Leu	Val	Cys	465	470	475	480
Phe	Arg	Leu	Arg	Ala	Ala	Ala	Val	Leu	Val	Val	Gly	Glu	Lys	Arg	Ala	485	490	495	
Arg	Asp	Gly	Asp	Asp	Glu	Val	Val	Thr	Ala	Gly	Asn	Glu	Leu	Asn	Arg	500	505	510	
Arg	Leu	Leu	Glu	Ala	Val	Asn	Ala	Thr	Gly	Arg	Val	Tyr	Met	Ser	Ser	515	520	525	
Ala	Val	Val	Gly	Gly	Thr	Tyr	Ile	Leu	Arg	Cys	Ala	Ile	Gly	Asn	Ser	530	535	540	
Leu	Thr	Glu	Glu	Arg	His	Val	Arg	Glu	Ala	Trp	Ser	Val	Val	Gln	Glu	545	550	555	560

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Gln Ala Thr Ala Ile Leu Ala Ala Ala Arg Arg Pro Thr Ala Arg Thr
565 570 575

Asn Arg Arg Thr Val Arg Arg Ala His Ala Ala Leu
580 585

<210> SEQ ID NO 72

<211> LENGTH: 502

<212> TYPE: PRT

<213> ORGANISM: Kalanchoe laxiflora

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 0994s0009.1

<400> SEQUENCE: 72

Met Gly Ser Leu Gln Ser Pro His Asp Pro Asn Ala Phe Asn Pro Met
1 5 10 15

Asp Val Ala Glu Leu Ser Ile Glu Ser Arg Leu Val Met Asp Phe Ile
20 25 30

Thr Gln Tyr Tyr Gln Thr Leu Glu Thr Arg Pro Val Gln Pro Arg Val
35 40 45

Lys Pro Gly Phe Leu Thr Gly Gln Leu Pro Glu Lys Pro Pro Phe His
50 55 60

Ala Glu Ser Met Glu Glu Ile Leu Ser Asp Val Ser Glu Lys Ile Val
65 70 75 80

Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe His Ala Tyr Phe Pro
85 90 95

Ala Ser Ser Ser Asn Ala Gly Leu Leu Gly Glu Met Leu Cys Ser Gly
100 105 110

Leu Ser Val Ile Gly Phe Thr Trp Asn Ser Ser Pro Ala Ala Thr Glu
115 120 125

Leu Glu Asn Val Val Val Asp Trp Leu Ala Asp Met Leu Asn Leu Pro
130 135 140

Pro Ser Phe Arg Phe Ser Gly Gly Gly Gly Gly Val Leu Gln Ser Asn
145 150 155 160

Thr Cys Glu Ala Val Leu Cys Thr Leu Ala Ala Ala Arg Asp Lys Val
165 170 175

Leu Glu Arg Ile Gly Asp Asp Lys Ile Asn Lys Leu Val Val Tyr Cys
180 185 190

Ser Asp Gln Thr His Phe Thr Leu His Lys Gly Ala Lys Leu Ile Gly
195 200 205

Ile Arg Arg Ala Asn Ile Lys Ser Ile Ser Thr Arg Arg Glu Asn Gly
210 215 220

Phe Gly Leu Cys Pro Asn Asp Leu Arg Asn Ala Ile Lys Ser Asp Leu
225 230 235 240

Glu Ala Gly Leu Val Pro Phe Tyr Leu Cys Gly Thr Ile Gly Thr Thr
245 250 255

Ala Leu Gly Ala Val Asp Pro Ile Lys Glu Leu Gly Lys Val Ala Arg
260 265 270

Glu Phe Asp Leu Trp Phe His Ile Asp Ala Ala Tyr Gly Gly Ser Ala
275 280 285

Cys Ile Cys Pro Glu Phe Arg His Tyr Leu Asp Gly Val Glu Leu Val
290 295 300

Asp Ser Ile Ser Met Asn Ala His Lys Trp Leu Leu Ser Asn Leu Asp
305 310 315 320

Cys Cys Phe Leu Trp Leu Gln Asn Pro Lys Cys Leu Ile Gln Cys Leu
325 330 335

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Ala Ala Glu Gly Glu Phe Leu Lys Gly Ser Gly Glu Met Val Asp Tyr
 340 345 350

Lys Asp Trp Gln Ile Ser Leu Ser Arg Arg Phe Arg Ala Ile Lys Met
 355 360 365

Trp Met Val Phe Arg Arg Tyr Gly Val Ser Asn Leu Met Glu His Ile
 370 375 380

Arg Ser Asp Val Ser Met Ala Ala Arg Phe Glu Glu Met Val Ala Ala
 385 390 395 400

Asp Asp Arg Phe Glu Ile Val Phe Pro Arg Lys Phe Ala Leu Val Cys
 405 410 415

Phe Lys Leu Asn Thr Lys Gly Ser Val Gln His Gly Glu Val Asp Gly
 420 425 430

Glu Asp Gly Leu Asp Gly Asp Ser Val Leu Thr Arg Glu Leu Met Gly
 435 440 445

Arg Val Asn Ser Ser Gly Lys Ala Tyr Leu Ser Gly Val Glu Met Gly
 450 455 460

Arg Ile Phe Phe Ile Arg Cys Val Ile Gly Ser Ser Leu Thr Glu Glu
 465 470 475 480

Arg His Val Asp Asn Leu Trp Asn Leu Ile Gln Glu Lys Thr Gln Ser
 485 490 495

Ile Met Pro Arg Arg Ala
 500

<210> SEQ ID NO 73
 <211> LENGTH: 509
 <212> TYPE: PRT
 <213> ORGANISM: Kalanchoe laxiflora
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 0003s0173.1

<400> SEQUENCE: 73

Met Gly Ser Leu Ser Ser Pro Arg Asp Leu Thr Lys Pro Phe Asn Pro
 1 5 10 15

Leu Asp Pro Thr Glu Leu Ala Val Glu Ser Ser Leu Val Thr Asp Phe
 20 25 30

Ile Ala Glu Tyr Tyr Arg Thr Val Glu Gln Arg Pro Val Gln Pro His
 35 40 45

Val Thr Pro Gly Phe Leu Thr Ser Gln Leu Pro Ser Ala Ala Pro Phe
 50 55 60

Ala Ser Glu Ser Val Glu Ser Ile Leu Gln Asp Val Tyr Asp Lys Ile
 65 70 75 80

Leu Pro Gly Leu Val Gln Trp Gln Ser Pro Asn Phe His Ala Tyr Tyr
 85 90 95

Pro Ala Thr Cys Ser Asn Ala Gly Leu Leu Gly Glu Met Leu Cys Ser
 100 105 110

Gly Leu Asn Val Val Gly Phe Thr Trp Ser Ala Ser Pro Ala Ala Ala
 115 120 125

Glu Leu Glu Gln Val Val Val Asp Trp Met Gly Lys Met Met Gly Leu
 130 135 140

Pro Gln Ser Phe Leu Phe Ser Gly Gly Gly Gly Gly Val Leu Gln Gly
 145 150 155 160

Ser Thr Cys Glu Ala Val Val Cys Thr Leu Ala Ala Ala Arg Asp Arg
 165 170 175

Ala Leu Glu Arg Val Gly Asp Asp Met Phe Asn Lys Leu Val Val Tyr
 180 185 190

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Cys Ser Asp Gln Thr His Phe Thr Leu Lys Lys Gly Ser Lys Leu Val
 195 200 205
 Gly Ile Arg Pro Ala Asn Val Lys Ala Ile Lys Thr Thr Lys Asn Asn
 210 215 220
 Glu Tyr Gly Leu Cys Pro Thr Asp Leu Arg Asn Leu Val Ala Ser Asp
 225 230 235 240
 Val Lys Ala Gly Phe Ile Pro Ile Tyr Leu Cys Gly Thr Ile Gly Thr
 245 250 255
 Thr Ala Phe Gly Ala Val Asp Pro Ile Arg Glu Leu Gly Lys Val Ala
 260 265 270
 Arg Glu Phe Asn Met Trp Phe His Val Asp Ala Ala Tyr Ala Gly Ser
 275 280 285
 Ala Phe Ile Cys Pro Glu Phe Arg His Tyr Met Asp Gly Val Glu Leu
 290 295 300
 Ala Asp Ser Phe Ser Thr Asn Pro His Lys Trp Leu Leu Ser Asn Met
 305 310 315 320
 Asp Cys Cys Val Leu Trp Leu Lys Phe Pro Lys Arg Val Ile Lys Ser
 325 330 335
 Leu Ala Ala Glu Gly Val Phe Leu Glu Gly Gly Ser Glu Thr Met Val
 340 345 350
 Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala Ile
 355 360 365
 Lys Leu Trp Met Val Ile Lys Arg Tyr Gly Leu Lys Asn Leu Ile Ser
 370 375 380
 His Ile Arg Ser Asp Val Ser Met Ala Lys Arg Phe Glu Glu Leu Leu
 385 390 395 400
 Leu Ser Asp Arg Arg Phe Glu Val Val Phe Pro Arg Lys Phe Ser Leu
 405 410 415
 Val Cys Phe Lys Leu Asp Val Met Lys Asn Val Pro Glu Val Val Asp
 420 425 430
 Glu Asp Asp Gly Glu Leu Ser His Asp Ser Lys Leu Thr Arg Glu Leu
 435 440 445
 Met Ala Ser Val Asn Val Thr Gly Lys Ala Phe Leu Thr Gly Val Arg
 450 455 460
 Leu Gly Arg Ile Phe Phe Ile Arg Cys Ala Ile Gly Ser Thr Leu Thr
 465 470 475 480
 Glu Asp Arg His Ile Gln Asp Leu Trp Lys Leu Ile Gln Glu Lys Ala
 485 490 495
 His Lys Ile Cys Ala Asn His Asp Leu Lys Phe Arg Val
 500 505

<210> SEQ ID NO 74
 <211> LENGTH: 607
 <212> TYPE: PRT
 <213> ORGANISM: Panicum hallii
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 32512198

<400> SEQUENCE: 74

Met Ala Ile Leu Asn His Gly Asp Thr Thr Thr Ala Asn Gly Ser Ser
 1 5 10 15
 Pro Ala Asp Ala Ala Ala Val Ala Pro Ala Met Pro Ser Leu Val Gln
 20 25 30
 Pro Pro Leu Asp Ala Asp Glu Phe Arg Arg Gln Gly Arg Leu Val Val
 35 40 45

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Asp	Phe	Ile	Ala	Asp	Tyr	Tyr	Thr	Arg	Ile	Asp	Glu	His	Pro	Val	Arg	50	55	60
Pro	Ala	Val	Ala	Pro	Gly	Phe	Leu	Ala	Arg	Gln	Leu	Pro	Asp	Thr	Ala	65	70	75
Pro	Ala	Arg	Pro	Glu	Pro	Gly	Asp	Asp	Ala	Leu	Ala	Ala	Ala	Leu	Arg	85	90	95
Asp	Val	Arg	Asp	Leu	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser	Pro	100	105	110
Arg	His	Phe	Ala	His	Phe	Ala	Ala	Thr	Ala	Ser	Asn	Val	Gly	Ala	Leu	115	120	125
Gly	Glu	Ala	Leu	Thr	Ala	Gly	Leu	Asn	Ile	Asn	Pro	Phe	Thr	Trp	Ala	130	135	140
Ala	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Val	Val	Val	Thr	Asp	Trp	Leu	145	150	155
Gly	Lys	Ala	Leu	His	Leu	Pro	Glu	Ser	Leu	Leu	Phe	Ser	Gly	Gly	Gly	165	170	175
Gly	Ala	Thr	Leu	Leu	Gly	Thr	Ser	Cys	Glu	Ala	Met	Leu	Cys	Thr	Leu	180	185	190
Val	Ala	Ala	Arg	Asp	Arg	Lys	Leu	Ala	Glu	Ile	Gly	Glu	Glu	Arg	Ile	195	200	205
Gly	Asp	Leu	Val	Val	Tyr	Cys	Ser	Asp	Gln	Thr	His	Phe	Ser	Phe	Gln	210	215	220
Lys	Ala	Ala	Arg	Ile	Ala	Gly	Ile	Arg	Arg	Gly	Asn	Tyr	Arg	Glu	Ile	225	230	235
Pro	Thr	Ser	Arg	Glu	Ser	Gly	Phe	Thr	Leu	Ser	Pro	Lys	Val	Leu	Arg	245	250	255
Ala	Ala	Val	Arg	Ala	Asp	Glu	Ala	Ala	Gly	Arg	Val	Pro	Leu	Phe	Leu	260	265	270
Cys	Ala	Thr	Val	Gly	Thr	Thr	Pro	Thr	Ala	Ala	Val	Asp	Pro	Leu	Arg	275	280	285
Glu	Leu	Cys	Ala	Thr	Val	Ala	Gly	His	Gly	Val	Trp	Val	His	Val	Asp	290	295	300
Ala	Ala	Tyr	Ala	Gly	Ala	Ala	Cys	Val	Cys	Pro	Glu	Phe	Arg	His	Ala	305	310	315
Ile	Ala	Gly	Ala	Glu	Ala	Val	Asp	Ser	Phe	Ser	Thr	Asn	Pro	His	Lys	325	330	335
Trp	Leu	Leu	Ala	Asn	Met	Asp	Cys	Cys	Ala	Leu	Trp	Val	Arg	Arg	Pro	340	345	350
Glu	Ala	Leu	Thr	Ala	Ala	Leu	Gly	Thr	Asp	His	Asp	Val	Ile	Leu	Lys	355	360	365
Asp	Pro	Ser	Ser	Glu	Arg	Asp	Cys	Gly	Arg	Gly	Val	Val	Asp	Tyr	Lys	370	375	380
Asp	Trp	Gln	Val	Ala	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp	385	390	395
Leu	Val	Leu	Arg	Cys	His	Gly	Val	Glu	Gly	Leu	Arg	Gly	Phe	Val	Arg	405	410	415
Ala	His	Val	Arg	Met	Ala	Ala	Ala	Phe	Glu	Asp	Met	Val	Arg	Ala	Asp	420	425	430
Ala	Arg	Phe	Glu	Val	Pro	Val	Pro	Arg	Gln	Phe	Ala	Leu	Val	Cys	Phe	435	440	445
Arg	Leu	Arg	Ser	Ala	Ala	Ala	Gly	Glu	Lys	Arg	Ala	Arg	Asp	Gly	Asp	450	455	460
Asp	Ala	Glu	Pro	Asn	Glu	Leu	Asn	Arg	Arg	Leu	Leu	Glu	Ala	Val	Asn			

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465	470	475	480
Ala Thr Gly Arg Ala Tyr Met Ser Ser Ala Val Val Gly Gly Ile Tyr			
	485	490	495
Val Leu Arg Cys Ala Ile Gly Asn Ser Leu Thr Glu Glu Arg His Val			
	500	505	510
Arg Glu Ala Trp Cys Val Val Gln Glu Gln Ala Thr Val Val Leu Ala			
	515	520	525
Ala Ala Ala Cys Thr Glu Glu Arg Ala Val His Ser Ala Arg Cys Ala			
	530	535	540
Asp Ala Pro Ala Ala Val Pro Pro Val Gln Asn Glu Gly Tyr Gly Glu			
545	550	555	560
Pro Thr Ser Ile Ala Ala Lys Ile Phe Gly Thr Ser Ile Ala Arg Cys			
	565	570	575
Ser Ile Lys Ser Glu Ala Ser Thr Tyr His Ser Trp Ser Thr Leu Trp			
	580	585	590
Arg Thr Leu Met Phe Lys Leu Leu Thr Trp Ile Ile Ser Arg Leu			
	595	600	605

<210> SEQ ID NO 75
 <211> LENGTH: 510
 <212> TYPE: PRT
 <213> ORGANISM: Prunus persica
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 6G202600.1

<400> SEQUENCE: 75

Met Thr Ser Ala Leu Asp Pro Val Glu Phe Arg Arg Gln Gly His Met			
1	5	10	15
Met Val Asp Phe Ile Ala Asp Tyr Tyr Gln Asn Ile Asp Lys Tyr Pro			
	20	25	30
Val Leu Ser Gln Val Asp Pro Gly Tyr Leu Arg Lys Arg Leu Pro Glu			
	35	40	45
Ser Ala Pro Asp Asn Pro Glu Pro Ile Glu Thr Ile Leu Gln Asp Val			
	50	55	60
Gln Glu His Ile Val Pro Gly Leu Thr His Trp Gln Ser Pro Ser Phe			
65	70	75	80
Phe Ala Tyr Phe Ala Ser Asn Val Ser Ile Ala Gly Phe Leu Gly Glu			
	85	90	95
Met Leu Ser Thr Gly Phe Asn Val Val Gly Phe Asn Trp Val Ser Ser			
	100	105	110
Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Met Asp Trp Leu Gly Asn			
	115	120	125
Leu Leu Ser Leu Pro Lys Ser Phe Leu Phe Ser Gly Asn Gly Gly Gly			
	130	135	140
Val Ile His Gly Ser Thr Cys Glu Ala Ile Val Cys Thr Met Ala Ala			
145	150	155	160
Ser Arg Asp Gln Met Leu Ser Arg Ile Gly Gly Asp Asn Ile Gly Lys			
	165	170	175
Leu Val Val Tyr Gly Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala			
	180	185	190
Ser Gln Ile Val Gly Ile Asn Pro Lys Asn Phe Arg Ala Ile Glu Ala			
	195	200	205
Thr Arg Ser Thr Thr Phe Ala Leu Ser Pro Glu Ser Leu Lys Leu Ala			
	210	215	220
Ile Ser Ser Asp Ile Glu Ala Gly Leu Val Pro Leu Phe Leu Cys Ala			

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225	230	235	240
Thr Val Gly Thr Thr Ala Thr Thr Ala Val Asp Pro Leu Gly Pro Leu	245	250	255
Cys Asp Val Ala Lys His His Gly Met Trp Val His Val Asp Ala Ala	260	265	270
Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Phe Ile Asp	275	280	285
Gly Ile Glu Gly Val Asp Ser Phe Ser Phe Asn Ala His Lys Trp Phe	290	295	300
Phe Thr Gly Leu Asp Cys Cys Cys Leu Trp Val Lys Asn Pro Gly Ala	305	310	315
Leu Ile Ser Ser Leu Ser Ala Asn Pro Glu Phe Leu Arg Asn Lys Pro	325	330	335
Thr Asp Ser Lys Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Ala Leu	340	345	350
Ser Arg Arg Phe Arg Ala Met Lys Leu Trp Leu Val Leu Arg Ser Tyr	355	360	365
Gly Val Val Asn Leu Arg Asn Phe Leu Arg Ser His Val Lys Met Ala	370	375	380
Lys Leu Phe Glu Gly Leu Val Ala Met Asp Gln Arg Phe Glu Ile Val	385	390	395
Val Pro Arg Asn Phe Ser Met Val Pro Pro Thr Thr Pro Thr Ser Asn	405	410	415
Ser Phe His Gln Asn Gly Ile Glu Ile Asn Val Glu Lys Cys Thr Asn	420	425	430
Glu Val Asn Cys Lys Leu Leu Glu Ala Ile Asn Ala Ser Gly Arg Val	435	440	445
Phe Met Thr His Ala Met Val Gly Gly Met Tyr Val Ile Arg Cys Ala	450	455	460
Val Gly Val Thr Gln Thr Glu Glu Lys His Ile Ala Met Ala Trp Lys	465	470	475
Val Val Gln Glu His Ala Asp Val Ile Leu Lys Asn Asn Gly Asp Asp	485	490	495
Gly Asp Ala Asn Leu Lys Leu Pro Leu Leu Asp Lys Ile Ala	500	505	510

<210> SEQ ID NO 76

<211> LENGTH: 510

<212> TYPE: PRT

<213> ORGANISM: Prunus persica

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 4G086700.1

<400> SEQUENCE: 76

Met Gly Ser Leu Asn Phe Asp His Pro Gln Glu Asn Asn Ser Ala His	1	5	10	15
Met Ser Gly Pro Leu Asp Leu Val Glu Leu Arg Arg Gln Gly His Met	20	25	30	
Ile Ile Asp Phe Ile Thr Asp Tyr Tyr Gln Asn Ile Glu Lys His Pro	35	40	45	
Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Lys Gln Arg Leu Pro Glu	50	55	60	
Ser Ala Pro Tyr Asn Pro Glu Pro Ile Glu Thr Ile Leu Arg Asp Val	65	70	75	80
Gln Asp His Ile Val Pro Gly Leu Thr His Trp Gln Ser Pro Asn His				

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85								90				95			
Phe	Ala	Tyr	Phe	Pro	Ala	Thr	Ile	Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu
			100					105					110		
Met	Leu	Thr	Thr	Cys	Phe	Asn	Val	Val	Gly	Phe	Asn	Trp	Met	Ala	Ser
		115					120					125			
Pro	Ala	Ala	Thr	Glu	Leu	Glu	Thr	Ile	Val	Met	Asp	Trp	Leu	Gly	Asp
						135					140				
Met	Leu	Lys	Leu	Pro	Asn	Ser	Phe	Leu	Phe	Ser	Gly	Thr	Gly	Gly	Gly
145					150					155					160
Val	Leu	His	Gly	Ser	Thr	His	Glu	Ser	Val	Val	Cys	Thr	Met	Ala	Ala
			165						170					175	
Ala	Arg	Asp	Gln	Ile	Leu	Ser	Arg	Ile	Gly	Glu	Glu	Asn	Ile	Gly	Lys
			180					185					190		
Leu	Val	Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Ser	Val	Ile	Gln	Lys	Val
		195					200					205			
Ser	Gln	Ile	Val	Gly	Ile	Pro	Ser	Lys	Asn	Phe	Arg	Ala	Ile	Glu	Thr
	210					215					220				
Thr	Ile	Ser	Ser	Ser	Phe	Thr	Leu	Ser	Pro	Glu	Thr	Leu	Arg	Leu	Thr
225					230					235					240
Val	Cys	Ser	Asp	Met	Glu	Ala	Gly	Leu	Val	Pro	Phe	Tyr	Leu	Cys	Ala
				245					250					255	
Thr	Val	Gly	Thr	Thr	Ala	Thr	Thr	Ala	Val	Asp	Pro	Leu	Gly	Pro	Leu
			260					265					270		
Cys	Asp	Val	Ala	Lys	Asp	Tyr	Gly	Met	Trp	Val	His	Val	Asp	Ala	Ala
		275					280					285			
Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	Gln	Tyr	Ile	Asp
	290					295					300				
Gly	Ile	Glu	Gly	Ala	Asn	Ser	Phe	Ser	Phe	Asn	Ala	Gln	Lys	Trp	Phe
305					310					315					320
Phe	Thr	Ala	Leu	Asp	Cys	Cys	Cys	Leu	Trp	Val	Lys	Asn	Pro	Ser	Ala
				325					330					335	
Leu	Thr	Lys	Ser	Met	Ser	Thr	Asp	Leu	Glu	Val	Leu	Arg	Asn	Lys	Ala
			340					345					350		
Ser	Glu	Ser	Lys	Arg	Val	Val	Asp	Phe	Lys	Asp	Trp	Gln	Ile	Ala	Leu
		355					360					365			
Thr	Arg	Arg	Phe	Arg	Ala	Ile	Lys	Leu	Trp	Leu	Val	Leu	Arg	Ser	Tyr
	370					375					380				
Gly	Val	Ala	Asn	Leu	Arg	Asn	Phe	Leu	Arg	Ser	His	Val	Lys	Met	Ala
385					390					395					400
Lys	Arg	Phe	Glu	Gly	Leu	Val	Arg	Thr	Asp	Glu	Arg	Phe	Glu	Val	Val
			405						410					415	
Val	Pro	Arg	Ile	Phe	Ala	Leu	Val	Cys	Phe	Arg	Ile	Ser	Pro	Ser	Ala
			420					425					430		
Ile	Ser	Lys	Ala	Asn	Pro	Thr	Pro	Ser	Asp	Glu	Lys	Cys	Val	Asn	Glu
		435					440					445			
Val	Asn	Cys	Lys	Leu	Leu	Glu	Ala	Ile	Asn	Gly	Ser	Gly	Trp	Val	Tyr
	450					455					460				
Met	Thr	His	Ala	Val	Val	Gly	Gly	Met	Tyr	Val	Leu	Arg	Cys	Ala	Ile
465					470					475					480
Gly	Ala	Ser	Leu	Thr	Lys	Glu	Lys	His	Val	Ala	Met	Ala	Trp	Lys	Val
			485						490					495	
Val	Gln	Glu	His	Val	Asp	Ala	Ile	Leu	Pro	Leu	Thr	Met	Tyr		
		500						505					510		

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<210> SEQ ID NO 77
<211> LENGTH: 533
<212> TYPE: PRT
<213> ORGANISM: Prunus persica
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 4G087100.1

<400> SEQUENCE: 77

Met Met Gly Ser Val Glu Phe Glu His Pro Gln Glu Asn Asn Ser Ala
1      5      10      15
His Met Thr Thr Ser Pro Leu Asp Pro Glu Glu Phe Arg Arg Gln Gly
20     25     30
His Met Val Ile Asp Phe Ile Ala Asp Tyr Tyr Lys Thr Ile Glu Lys
35     40     45
Tyr Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Lys Lys Arg Leu
50     55     60
Pro Glu Ser Ala Pro Tyr Asp Pro Glu Pro Ile Glu Thr Ile Leu Gln
65     70     75     80
Asp Val Gln Asp His Leu Val Pro Gly Leu Thr His Trp Leu Ser Pro
85     90     95
Asn His Phe Gly Tyr Phe Pro Ala Ala Ile Ser Thr Ala Ala Phe Leu
100    105    110
Gly Glu Met Leu Thr Thr Gly Phe Asn Val Val Gly Phe Asn Trp Met
115    120    125
Ala Ser Pro Ala Ala Thr Glu Leu Glu Asn Ile Val Met Asp Trp Leu
130    135    140
Gly Asp Met Leu Lys Leu Pro Lys Ser Phe Leu Phe Ser Gly Asn Gly
145    150    155    160
Gly Gly Val Leu Gln Gly Thr Thr Cys Glu Ala Ile Val Cys Thr Met
165    170    175
Ala Ala Ala Arg Asp Gln Met Leu Arg Gln Ile Gly Arg Glu Asn Ile
180    185    190
Gly Lys Leu Val Val Tyr Gly Ser Asp Gln Thr His Ser Ala Leu Gln
195    200    205
Lys Ala Ser Gln Ile Val Gly Ile His Pro Lys Asn Phe Arg Ala Ile
210    215    220
Glu Thr Thr Thr Ser Thr Ser Phe Ala Leu Ser Pro Glu Val Leu Lys
225    230    235    240
Ser Thr Ile Cys Ser Asp Ile Glu Ala Gly Leu Val Pro Leu Phe Leu
245    250    255
Cys Ala Thr Val Gly Thr Thr Ala Ile Thr Ala Val Asp Pro Leu Gly
260    265    270
Pro Leu Cys Glu Val Ala Lys Glu His Asp Met Trp Val His Val Asp
275    280    285
Ala Ala Tyr Ala Gly Ser Ala Phe Ile Cys Pro Glu Phe Gln Tyr Phe
290    295    300
Ile Asp Gly Val Glu Gly Ala Asp Ser Phe Ser Leu Asn Ala His Lys
305    310    315    320
Trp Phe Phe Thr Thr Leu Asp Cys Cys Cys Leu Trp Val Lys Asn Pro
325    330    335
Ser Ala Leu Val Ser Ser Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn
340    345    350
Lys Ala Thr Asp Ser Lys Gln Val Val Asp Tyr Lys Asp Trp Gln Ile
355    360    365

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Ala Leu Ser Arg Arg Phe Lys Ala Ile Lys Leu Trp Leu Val Leu Arg
 370 375 380

Ser Tyr Gly Val Gly Asn Leu Arg Asn Phe Leu Arg Ser His Val Lys
 385 390 395 400

Met Ala Lys Ile Phe Glu Gly Leu Val Gly Met Asp Lys Arg Phe Glu
 405 410 415

Ile Val Ala Pro Arg His Phe Ser Leu Val Cys Phe Arg Val Ser Pro
 420 425 430

Ser Ala Ile Ser Lys Ala Asn Pro Ser Leu Ser Asp His Asp Asn Gly
 435 440 445

Lys Leu Lys Ala His Asn Tyr Glu Leu Leu Asn Gly Val Lys Cys Val
 450 455 460

Val Asn Glu Val Asn Ser Lys Leu Leu Glu Ala Ile Asn Gly Ser Gly
 465 470 475 480

Leu Val Tyr Met Ser His Ala Val Val Gly Gly Met Tyr Val Leu Arg
 485 490 495

Cys Ala Ile Gly Ala Ser Leu Thr Glu Glu Lys His Val Ala Met Ala
 500 505 510

Trp Lys Val Val Gln Glu His Ala Asp Ala Ile Leu Gly Thr Lys Ile
 515 520 525

Ile Val Asp Gln Thr
 530

<210> SEQ ID NO 78
 <211> LENGTH: 502
 <212> TYPE: PRT
 <213> ORGANISM: Medicago truncatula
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 31073039

<400> SEQUENCE: 78

Met Asn Thr Ser Ser Ser Asn Pro Pro Gln Ser Asp Pro Gln Lys Thr
 1 5 10 15

Met Asn Pro Leu Asp Leu Glu Glu Phe Lys Arg Gln Gly Tyr Met Met
 20 25 30

Ile Asp Phe Leu Thr Asp Tyr Tyr Lys Asn Ile Glu Asn Tyr Pro Val
 35 40 45

Leu Ser Lys Val Glu Pro Gly Tyr Leu Ala Lys Ile Leu Pro Ser Ser
 50 55 60

Ala Pro Phe Gln Pro Glu Ser Ile Glu Ser Ile Leu Glu Asp Val Gln
 65 70 75 80

Gln His Ile Ile Pro Gly Ile Thr His Trp Met Ser Pro Asn Tyr Tyr
 85 90 95

Ala Tyr Phe Pro Ser Ser Gly Ser Ile Ala Gly Phe Ile Gly Glu Met
 100 105 110

Leu Ser Thr Gly Phe Asn Val Val Gly Phe Asn Trp Leu Ser Ser Pro
 115 120 125

Ala Ala Thr Glu Leu Glu Thr Ile Val Met Asn Trp Leu Gly Lys Leu
 130 135 140

Leu Asn Leu Pro Lys Ser Phe Ile Phe Ser Ser Asn Ile Lys Gly Gly
 145 150 155 160

Gly Glu Ile Lys Lys Leu Ser Gln Ile Gly Lys Asp Asn Ile Gly Lys
 165 170 175

Leu Val Val Tyr Cys Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala
 180 185 190

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Thr Gln Ile Val Gly Ile His Ser Glu Asn Phe Arg Val Ile Lys Thr
    195                      200                      205

Lys Gly Ser Asn Leu Phe Ala Leu Ser Pro Asp Ser Leu Leu Ser Thr
    210                      215                      220

Ile Leu Leu Asp Val Asp Asn Gly Leu Ile Pro Tyr Phe Leu Cys Ala
    225                      230                      235                      240

Thr Ile Gly Thr Thr Ser Thr Asn Ala Val Asp Pro Ile Lys Leu Leu
    245                      250                      255

Cys Asn Val Thr Lys Glu Tyr Asp Ile Trp Val His Val Asp Ala Ala
    260                      265                      270

Tyr Ala Gly Ser Val Cys Ile Cys Pro Glu Phe Arg His Cys Ile Asp
    275                      280                      285

Gly Ile Glu Glu Leu Asn Ser Phe Ser Phe Asn Ala His Lys Trp Phe
    290                      295                      300

Leu Thr Asn Leu Ala Cys Cys Cys Leu Trp Val Lys Asp His Asn Ala
    305                      310                      315                      320

Leu Thr Thr Ser Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys Lys
    325                      330                      335

Ser Asp Ser Lys Glu Val Ile Asp Tyr Lys Asp Trp Gln Ile Pro Leu
    340                      345                      350

Ser Arg Lys Phe Asn Ala Leu Lys Leu Trp Ile Val Leu Arg Ser Tyr
    355                      360                      365

Gly Val Glu Asn Leu Lys Asn Phe Leu Arg Asn His Val Glu Met Ala
    370                      375                      380

Lys Ile Phe Glu Gly Leu Val Arg Lys Asp Glu Arg Phe Glu Ile Val
    385                      390                      395                      400

Val Pro Ser Lys Phe Ser Leu Val Cys Phe Arg Ile Ser Pro Phe Ala
    405                      410                      415

Ile Ser Ile Ala Asn Asp Ser Glu Gly Tyr Tyr Val Gly Lys Met Met
    420                      425                      430

Asn Asp Ala Tyr Leu Val Asn Glu Met Asn His Lys Leu Leu Asp Leu
    435                      440                      445

Ile Asn Ser Ser Gly Lys Ala Tyr Met Ser His Gly Glu Val Glu Gly
    450                      455                      460

Ser Phe Val Ile Arg Cys Ala Ile Gly Ala Thr Leu Thr Glu Glu His
    465                      470                      475                      480

His Val Thr Met Thr Trp Lys Leu Val Gln Gln Ile Ala Ser Phe Leu
    485                      490                      495

Leu Gly Thr Pro Leu Asn
    500

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<210> SEQ ID NO 79

<211> LENGTH: 577

<212> TYPE: PRT

<213> ORGANISM: Zea mays

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: GRMZM2G009400

<400> SEQUENCE: 79

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Met Ala Ile Leu Asn Arg Ala Asp Thr Ser His Thr Thr Thr Ala Ser
1          5          10          15

Asn Gly Ser Ala Thr Pro Ala Ala Pro Val Ala Ile Ala Met Pro Ser
20          25          30

Leu Pro His Pro Pro Leu Asp Ala Asp Glu Phe Arg Arg Gln Gly Arg
35          40          45

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Leu 50	Val	Val	Asp	Phe	Ile	Ala 55	Asp	Tyr	Tyr	Ala 60	Arg	Ile	Asp	Gly	Tyr
Pro 65	Val	Arg	Pro	Ala	Val 70	Ala	Pro	Gly	Phe	Leu 75	Ile	Arg	Gln	Leu	Pro 80
Glu	Ala	Ala	Pro	Ala 85	Arg	Pro	Glu	Pro	Asp 90	Ala	Leu	Ala	Ala	Ala 95	Leu
Arg	Asp	Val	Arg	Asp 100	Leu	Ile	Leu	Pro	Gly 105	Val	Thr	His	Trp 110	Gln	Ser
Pro	Arg	His	Phe	Ala	His	Phe 120	Ala	Ala	Thr	Ala	Ser	Asn 125	Val	Gly	Ala
Leu 130	Gly	Glu	Ala	Leu	Ala	Ala 135	Gly	Leu	Asn	Val	Asn 140	Pro	Phe	Thr	Trp
Ala 145	Ala	Ser	Pro	Ala	Ala	Thr 150	Glu	Leu	Glu	Val 155	Val	Val	Thr	Asp	Trp 160
Leu	Gly	Lys	Ala	Leu 165	His	Leu	Pro	Glu	Ser 170	Leu	Leu	Phe	Ser	Gly 175	Gly
Gly	Gly	Gly	Thr 180	Leu	Leu	Gly	Thr	Ser	Cys 185	Glu	Ala	Met	Leu	Cys	Thr
Ile	Val	Ala 195	Ala	Arg	Asp	Arg 200	Lys	Leu	Ala	Glu	Val	Gly 205	Glu	Glu	Arg
Ile 210	Gly	Asp	Leu	Val	Val	Tyr 215	Cys	Ser	Asp	Gln	Thr 220	His	Phe	Ser	Phe
Gln 225	Lys	Ala	Ala	Arg	Ile 230	Ala	Gly	Ile	Arg	Arg 235	Gly	Asn	Cys	Arg	Glu 240
Ile	Pro	Thr	Ser	Arg 245	Glu	Ser	Gly	Phe	Thr 250	Leu	Ser	Pro	Lys	Ala 255	Leu
Ala	Ala	Ala	Val 260	Arg	Ala	Asp	Glu	Ala	Ala 265	Gly	Arg	Val	Pro 270	Leu	Phe
Leu	Cys	Ala 275	Thr	Val	Gly	Thr 280	Thr	Pro	Thr	Ala	Ala 285	Val	Asp	Pro	Leu
Arg 290	Glu	Leu	Cys	Ala	Ala	Val 295	Ala	Gly	His	Asp	Val 300	Trp	Val	His	Val
Asp 305	Ala	Ala	Tyr	Ala	Gly 310	Ala	Ala	Cys	Val	Cys 315	Pro	Glu	Phe	Ser	His 320
Val	Val	Ala	Gly 325	Val	Glu	Ala	Ala	Glu	Ser 330	Phe	Ser	Thr	Asn	Pro 335	His
Lys	Trp	Leu 340	Leu	Ala	Asn	Met	Asp	Cys	Cys 345	Ala	Leu	Trp	Val	Arg 350	Arg
Pro	Ala 355	Ala	Leu	Thr	Ala	Ala	Leu	Gly	Thr 360	Asp	His	Asp 365	Val	Ile	Leu
Lys 370	Asp	Pro	Ala	Ala	Ala	Gln 375	Ala	Gln	Ala	Gln	Gln 380	Gln	Gln	Cys	Ser
Asp 385	Gly	Gly	Val	Val	Asp 390	Tyr	Lys	Asp	Trp	Gln 395	Val	Ala	Leu	Ser	Arg 400
Arg	Phe	Arg	Ala 405	Leu	Lys	Leu	Trp	Leu	Val 410	Leu	Arg	Cys	His	Gly 415	Val
Glu	Gly	Leu 420	Arg	Gly	Leu	Val	Arg	Ala	His 425	Val	Arg	Met	Ala	Ala 430	Ala
Phe	Glu 435	Ala	Met	Val	Arg	Gly	Asp 440	Ala	Arg	Phe	Glu 445	Val	His	Val	Pro
Arg 450	Gln	Phe	Ala	Leu	Val	Cys 455	Phe	Arg	Leu	Arg 460	Ala	Val	Ala	Val	Ala

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Val	Ala	Gly	Glu	Lys	Arg	Ala	Gly	Asp	Tyr	Asp	Gly	Val	Ala	Ala	Gly
465					470					475					480
Asn	Glu	Leu	Asn	Arg	Arg	Leu	Leu	Glu	Ala	Val	Asn	Ala	Thr	Gly	Arg
			485						490					495	
Val	Tyr	Met	Ser	Ser	Ala	Val	Val	Gly	Gly	Ala	Tyr	Ile	Leu	Arg	Cys
			500					505					510		
Ala	Ile	Gly	Asn	Ser	Leu	Thr	Glu	Glu	Arg	His	Val	Arg	Glu	Ala	Trp
		515					520					525			
Ser	Val	Val	Gln	Glu	Gln	Ala	Thr	Ala	Ile	Leu	Ser	Ala	Ala	Thr	Ala
	530					535					540				
Thr	Ala	Arg	Thr	Asn	Gly	Leu	Thr	Val	Arg	Arg	Ala	Arg	Cys	Asp	Ala
545				550					555						560
Glu	Ala	Asp	Val	Ser	Asp	Val	Pro	Thr	Pro	Gln	Gln	Pro	Leu	Pro	Leu
			565						570					575	

Gly

<210> SEQ ID NO 80
 <211> LENGTH: 519
 <212> TYPE: PRT
 <213> ORGANISM: Glycine max
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 07G059000.1

<400> SEQUENCE: 80

Met	Glu	Met	Lys	Asn	Thr	Met	Asn	Arg	Asn	Pro	Gln	Ser	Asp	Ala	Pro
1			5						10					15	
Ile	Ile	Lys	Pro	Leu	Asp	Pro	Glu	Glu	Phe	Lys	Arg	Gln	Gly	Tyr	Met
		20					25						30		
Met	Val	Asp	Phe	Leu	Ala	Asp	Tyr	Ile	Arg	Asn	Val	Ser	His	Tyr	Pro
		35					40					45			
Val	Leu	Ser	Lys	Val	Glu	Pro	Gly	Tyr	Leu	Lys	Gln	Arg	Leu	Pro	Thr
	50				55						60				
Ser	Ala	Pro	Cys	Gly	Pro	Glu	Pro	Ile	Glu	Ser	Ile	Leu	Lys	Asp	Val
65				70					75					80	
Gln	Asp	His	Ile	Ile	Pro	Gly	Leu	Thr	His	Trp	Gln	Ser	Pro	Asn	Phe
			85					90						95	
Tyr	Gly	Tyr	Phe	Pro	Ser	Ser	Gly	Ser	Ile	Ala	Gly	Phe	Met	Gly	Glu
		100					105						110		
Met	Leu	Ser	Ala	Gly	Leu	Asn	Val	Val	Gly	Phe	Asn	Trp	Val	Ser	Ser
		115				120						125			
Pro	Ser	Ala	Thr	Glu	Leu	Glu	Ser	Ile	Val	Met	Asp	Trp	Leu	Gly	Gln
		130			135						140				
Val	Leu	Asn	Leu	Pro	Lys	Ser	Phe	Leu	Phe	Cys	Gly	Asp	His	Gly	Gly
145				150					155					160	
Gly	Val	Val	Leu	Gly	Thr	Thr	Cys	Glu	Ala	Ile	Leu	Cys	Thr	Leu	Val
			165					170						175	
Ala	Ala	Arg	Glu	Lys	Lys	Leu	Ser	Gln	Val	Gly	Lys	Glu	Asn	Ile	Gly
		180						185					190		
Lys	Leu	Val	Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu	Gln	Lys
		195				200						205			
Ala	Ala	Gln	Ile	Ala	Gly	Ile	His	Pro	Ala	Asn	Phe	Arg	Val	Ile	Lys
	210				215					220					
Thr	Lys	Arg	Ser	Asn	Ser	Phe	Ala	Leu	Ser	Pro	Asp	Ser	Leu	Leu	Ser
225				230					235					240	
Thr	Ile	Leu	Leu	Asp	Val	Glu	Arg	Gly	Leu	Ile	Pro	Cys	Phe	Leu	Cys

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245					250					255					
Ala	Thr	Val	Gly	Thr	Thr	Ala	Ile	Ala	Thr	Ile	Asp	Pro	Ile	Gly	Pro
			260					265					270		
Leu	Cys	Asn	Val	Ala	Lys	Asp	Tyr	Gly	Ile	Trp	Val	His	Val	Asp	Ala
		275					280					285			
Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	His	Cys	Ile
		290				295					300				
Asp	Gly	Val	Glu	Glu	Val	Asn	Ser	Phe	Ser	Leu	Asn	Ala	His	Lys	Trp
		305				310					315				320
Phe	Leu	Thr	Asn	Leu	Thr	Cys	Cys	Cys	Leu	Trp	Val	Lys	Asp	His	Ile
			325						330					335	
Ala	Leu	Thr	Lys	Ser	Leu	Thr	Val	Asn	Pro	Gln	Phe	Leu	Arg	Asn	Lys
			340					345					350		
Ala	Ser	Glu	Ser	Lys	Arg	Val	Ile	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Pro
		355					360				365				
Leu	Ser	Arg	Lys	Phe	Asn	Ala	Leu	Lys	Leu	Trp	Leu	Val	Leu	Arg	Ser
		370				375					380				
Tyr	Gly	Val	Glu	Asn	Ile	Arg	Asn	Phe	Leu	Arg	Asn	His	Val	Gln	Met
		385				390					395				400
Ala	Lys	Thr	Phe	Glu	Gly	Leu	Val	Arg	Leu	Asp	Lys	Arg	Phe	Glu	Ile
			405						410					415	
Val	Val	Pro	Pro	Lys	Phe	Ser	Leu	Val	Cys	Phe	Arg	Ile	Ala	Pro	Ser
			420				425						430		
Ala	Ile	Ile	Ala	Asn	Gly	Leu	Ser	Lys	Gly	Val	Glu	Ala	Cys	Tyr	Asn
		435				440					445				
Gly	Lys	Leu	Val	Asn	Asp	Glu	Tyr	Met	Val	Asn	Glu	Val	Asn	Arg	Lys
		450				455					460				
Leu	Leu	Asp	Ser	Val	Asn	Ser	Ser	Gly	Asp	Ala	Phe	Met	Thr	His	Gly
		465				470					475				480
Glu	Val	Glu	Gly	Ala	Phe	Met	Ile	Arg	Cys	Ala	Ile	Gly	Gly	Thr	Leu
			485						490					495	
Thr	Glu	Glu	His	His	Val	Ile	Met	Ala	Trp	Lys	Leu	Val	Gln	Glu	His
			500				505						510		
Ala	Asn	Ser	Leu	Leu	Gly	Leu									
			515												

<210> SEQ ID NO 81

<211> LENGTH: 531

<212> TYPE: PRT

<213> ORGANISM: Panicum virgatum

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: Ca01381.1

<400> SEQUENCE: 81

Met	Ala	Ile	Leu	Asn	His	Gly	Asp	Thr	Thr	Ala	Ala	Ser	Gly	Thr	Ser
1				5					10					15	
Pro	Ala	Ala	Ala	Ala	Val	Asn	Val	Ala	Pro	Pro	Met	His	Ser	Leu	Val
			20					25					30		
Gln	Pro	Val	Leu	Asp	Ala	Asp	Glu	Phe	Arg	Arg	Gln	Gly	Arg	Leu	Val
		35					40					45			
Val	Asp	Phe	Ile	Ala	Asp	Tyr	Thr	Arg	Ile	Asp	Glu	Tyr	Pro	Val	
		50				55			60						
Arg	Pro	Ala	Val	Ala	Pro	Gly	Phe	Leu	Ala	Arg	Gln	Leu	Pro	Glu	Ala
		65			70				75					80	
Ala	Pro	Ala	Arg	Pro	Glu	Pro	Gly	Gly	Asp	Ala	Leu	Ala	Ala	Ala	Leu

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85								90					95				
Arg	Asp	Val	Arg	Asp	Leu	Ile	Leu	Pro	Gly	Val	Thr	His	Trp	Gln	Ser		
			100					105					110				
Pro	Arg	His	Phe	Ala	His	Phe	Ala	Thr	Thr	Gly	Ser	Asn	Val	Gly	Ala		
			115				120					125					
Leu	Gly	Glu	Ala	Leu	Ala	Ala	Gly	Leu	Asn	Ile	Asn	Pro	Phe	Thr	Trp		
			130				135				140						
Ala	Ala	Ser	Pro	Ala	Ala	Thr	Glu	Leu	Glu	Val	Val	Val	Thr	Asp	Trp		
145					150					155					160		
Leu	Gly	Lys	Ala	Leu	His	Leu	Pro	Glu	Arg	Leu	Leu	Phe	Ser	Gly	Gly		
				165					170					175			
Gly	Gly	Gly	Thr	Leu	Leu	Gly	Thr	Ser	Cys	Glu	Ala	Met	Leu	Cys	Thr		
			180					185					190				
Leu	Val	Ala	Ala	Arg	Asp	Arg	Lys	Leu	Ala	Glu	Ile	Gly	Glu	Glu	Arg		
		195					200					205					
Met	Gly	Asp	Leu	Val	Val	Tyr	Cys	Ser	Asp	Gln	Thr	His	Phe	Ser	Phe		
	210					215					220						
Arg	Lys	Ala	Ala	Arg	Ile	Ala	Gly	Ile	Arg	Arg	Gly	Asn	Cys	Arg	Glu		
225					230					235					240		
Ile	Pro	Thr	Ser	Arg	Glu	Ser	Gly	Phe	Ala	Leu	Gln	Pro	Arg	Thr	Leu		
				245					250					255			
Leu	Ala	Ala	Val	Arg	Ala	Asp	Glu	Ala	Ala	Gly	Arg	Val	Pro	Met	Phe		
			260					265					270				
Leu	Cys	Ala	Thr	Val	Gly	Thr	Thr	Pro	Thr	Ala	Ala	Val	Asp	Pro	Leu		
		275					280						285				
Arg	Glu	Leu	Cys	Ala	Ala	Val	Ala	Gly	Arg	Gly	Val	Trp	Val	His	Val		
	290					295					300						
Asp	Ala	Ala	Tyr	Ala	Gly	Ala	Ala	Cys	Val	Cys	Pro	Glu	Phe	Arg	Gly		
305					310					315					320		
Ala	Thr	Ala	Gly	Ala	Glu	Ala	Val	Asp	Ser	Phe	Ser	Thr	Asn	Pro	His		
				325					330					335			
Lys	Trp	Leu	Leu	Ala	Asn	Met	Asp	Cys	Cys	Ala	Leu	Trp	Val	Arg	Arg		
			340					345					350				
Pro	Glu	Ala	Leu	Thr	Ala	Ala	Leu	Gly	Thr	Asp	His	Asp	Val	Ile	Leu		
		355					360					365					
Lys	Asp	Pro	Ser	Ser	Glu	Arg	Gly	Gly	Gly	Val	Val	Asp	Tyr	Lys	Asp		
	370					375						380					
Trp	Gln	Val	Ala	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp	Leu		
385					390					395					400		
Val	Leu	Arg	Cys	His	Gly	Val	Glu	Gly	Leu	Arg	Gly	Leu	Val	Arg	Ala		
				405					410					415			
Asp	Ala	Arg	Phe	Glu	Val	Pro	Val	Pro	Arg	Gln	Phe	Ala	Leu	Val	Cys		
			420					425					430				
Phe	Arg	Leu	Arg	Ala	Ala	Ala	Ala	Ala	Ala	Val	Gly	Glu	Lys	Arg	Gly		
		435					440					445					
Arg	Asp	Arg	Asp	Asn	Asp	Ala	Glu	Pro	Asn	Glu	Leu	Asn	Arg	Arg	Leu		
	450					455					460						
Leu	Glu	Ala	Val	Asn	Ala	Thr	Gly	Arg	Ala	Tyr	Met	Ser	Ser	Ala	Val		
465					470					475					480		
Val	Gly	Gly	Ile	Tyr	Val	Leu	Arg	Cys	Ala	Ile	Gly	Asn	Ser	Leu	Thr		
				485					490					495			
Glu	Glu	Arg	His	Val	Arg	Glu	Ala	Trp	Arg	Val	Val	Gln	Glu	Gln	Ala		
			500					505					510				

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Thr Ala Val Leu Ala Ala Ala Ala Cys Thr Glu Glu Arg Ala Val Arg
 515 520 525

Ser Ala Arg
 530

<210> SEQ ID NO 82
 <211> LENGTH: 489
 <212> TYPE: PRT
 <213> ORGANISM: Theobroma cacao
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 27425420

<400> SEQUENCE: 82

Met Ser Ser Ala Ser Arg Lys Thr Phe Leu Pro Leu Glu Pro Thr Ser
 1 5 10 15

Phe Thr Asn Glu Ser Lys Ala Val Ile Asp Phe Ile Ala Asp Tyr Tyr
 20 25 30

Lys Asn Ile Glu Glu Tyr Pro Val Gln Ser Gly Val Glu Pro Gly Tyr
 35 40 45

Leu Ser Ala Lys Leu Pro Asp Ser Ala Pro Tyr Cys Pro Glu Ser Leu
 50 55 60

Glu Asp Ile Leu Lys Asp Val Asn Asp Cys Ile Ile Pro Gly Leu Thr
 65 70 75 80

His Trp Gln Ser Pro Asn Phe Phe Ala Tyr Phe Gln Ala Asn Ala Ser
 85 90 95

Thr Ala Gly Phe Leu Gly Glu Met Leu Cys Ser Gly Phe Asn Val Val
 100 105 110

Gly Phe Asn Trp Ile Ser Ser Pro Ala Ala Thr Glu Leu Glu Ser Ile
 115 120 125

Val Leu Asp Trp Met Gly Lys Leu Leu Lys Leu Pro Ser Ser Phe Leu
 130 135 140

Phe Ser Gly Thr Gly Gly Gly Val Leu His Gly Ser Thr Cys Glu Ala
 145 150 155 160

Ala Val Cys Thr Leu Ala Ala Ala Arg Asp Lys Ala Leu Lys Glu Leu
 165 170 175

Gly Gly Trp Glu Asn Ile Thr Lys Leu Met Val Tyr Ala Ser Asp Gln
 180 185 190

Thr His Phe Thr Phe Gln Lys Ala Ala Lys Leu Val Gly Ile Pro Pro
 195 200 205

Ser Asn Phe Arg Phe Ile Glu Thr Ser Leu Ser Thr Gly Phe Ser Met
 210 215 220

Ser Ser Asp Gln Val Arg Leu Ala Ile Glu His Asp Ile Lys Ser Gly
 225 230 235 240

Leu Val Pro Leu Phe Leu Cys Ala Thr Ile Gly Thr Thr Ala Cys Gly
 245 250 255

Ala Ile Asp Pro Ile Ala Glu Leu Gly Gln Val Ala Arg Glu Tyr Lys
 260 265 270

Leu Trp Leu His Ile Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys
 275 280 285

Pro Glu Leu Arg His Phe Leu Asp Gly Val Glu Leu Ala Asn Ser Val
 290 295 300

Ser Met Asn Pro His Lys Trp Phe Leu Thr Asn Met Asp Cys Cys Cys
 305 310 315 320

Leu Trp Ile Thr Glu Pro Arg Leu Leu Val Asp Ser Leu Ser Thr Asp
 325 330 335

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Pro Glu Ile Leu Arg Asn Lys Ala Ser Glu Phe Lys Ala Val Leu Asp
 340 345 350

Tyr Lys Asp Trp Gln Val Ala Leu Ser Arg Arg Phe Arg Ala Leu Lys
 355 360 365

Leu Trp Ile Val Ile Arg Arg His Gly Leu Ala Asn Leu Val Tyr His
 370 375 380

Ile Arg Ser Asp Ile Ser Met Ala Glu Arg Phe Glu Ala Phe Val Ala
 385 390 395 400

Lys Asp Asp Arg Phe Asp Ile Val Val Pro Arg Lys Phe Ala Leu Val
 405 410 415

Cys Phe Arg Leu Lys Pro Lys Gln Glu Leu Glu Gly Leu Glu Leu Asn
 420 425 430

Ser Arg Leu Leu Glu Ala Ile Asn Ser Ser Gly Arg Ala Phe Met Thr
 435 440 445

His Ala Val Val Gly Gly Ile Tyr Val Ile Arg Cys Ala Ile Gly Thr
 450 455 460

Thr Met Thr Glu Glu Arg His Val Asp Ala Leu Trp Lys Leu Ile Gln
 465 470 475 480

Glu Lys Ala Gln Gly Leu Leu Met Glu
 485

<210> SEQ ID NO 83
 <211> LENGTH: 625
 <212> TYPE: PRT
 <213> ORGANISM: Fragaria vesca
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 27274768

<400> SEQUENCE: 83

Met Gly Ser Leu Asp Phe His His Val Pro Glu Lys Thr Asn Ser Asp
 1 5 10 15

Pro Pro Met Ala Asn Pro Met Asp Pro Glu Glu Phe Arg Arg Gln Gly
 20 25 30

His Ile Met Ile Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu Lys
 35 40 45

Tyr Pro Val Leu Ser Gln Val Gln Pro Gly Tyr Leu Lys Lys Leu Leu
 50 55 60

Pro Glu Ser Ala Pro Tyr Asn Pro Glu Pro Ile Glu Thr Ile Leu Gln
 65 70 75 80

Asp Val Gln Asp His Ile Val Pro Gly Ile Thr His Trp Gln Ser Pro
 85 90 95

Ser Tyr Phe Ala Tyr Phe Pro Ser Ser Gly Ser Ile Ala Gly Phe Leu
 100 105 110

Gly Glu Met Leu Ser Thr Gly Phe Asn Val Val Gly Phe Asn Trp Met
 115 120 125

Ser Ser Pro Ala Ala Thr Glu Leu Glu Arg Thr Thr Cys Glu Ala Ile
 130 135 140

Val Cys Thr Met Ala Ala Ala Arg Asp Gln Met Leu Ser Arg Ile Gly
 145 150 155 160

Lys Asp Asn Ile Gly Lys Leu Val Val Tyr Gly Ser Asp Gln Thr His
 165 170 175

Ser Ala Leu Lys Lys Ala Ser Gln Ile Val Gly Ile His Pro Asn Asn
 180 185 190

Phe Arg Ala Ile Lys Thr Thr Lys Ser Thr Glu Phe Ala Leu Ser Pro
 195 200 205

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Glu	Leu	Leu	Arg	Ser	Thr	Ile	Cys	Ser	Asp	Ile	Asp	Lys	Gly	Leu	Val
210						215					220				
Pro	Leu	Phe	Leu	Cys	Ala	Thr	Met	Gly	Thr	Thr	Ala	Thr	Thr	Ser	Val
225					230					235					240
Asp	Pro	Leu	Arg	Gly	Leu	Cys	Asp	Val	Ala	Lys	Asp	Tyr	Asp	Leu	Trp
				245					250					255	
Val	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ile	Cys	Ile	Cys	Pro	Glu
			260					265					270		
Phe	Arg	His	Phe	Ile	Glu	Gly	Val	Asp	Gly	Ala	Asn	Ser	Phe	Ser	Phe
		275					280					285			
Asn	Ala	His	Lys	Trp	Phe	Phe	Thr	Thr	Leu	Asp	Cys	Cys	Cys	Leu	Trp
	290					295					300				
Val	Lys	Asn	Pro	Thr	Ala	Leu	Ile	Asn	Ser	Leu	Ser	Thr	Asn	Pro	Glu
305					310					315					320
Phe	Leu	Arg	Asn	Lys	Ala	Ser	Asp	Ser	Lys	Gln	Val	Val	Asp	Tyr	Lys
				325					330					335	
Asp	Trp	Gln	Val	Ala	Leu	Ser	Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp
			340					345					350		
Leu	Val	Leu	Arg	Ser	Tyr	Gly	Val	Ala	Asn	Leu	Arg	Ser	Phe	Leu	Arg
	355						360					365			
Ser	His	Val	Lys	Met	Ala	Glu	Val	Phe	Glu	Lys	Leu	Val	Arg	Glu	Asn
	370					375					380				
Lys	Trp	Phe	Glu	Val	Val	Val	Pro	Arg	Asn	Phe	Ala	Met	Val	Cys	Phe
385					390					395					400
Arg	Ile	Ser	Pro	Ser	Ala	Ile	Arg	Lys	Ala	Pro	Thr	Asp	Asp	Asp	Gly
				405					410					415	
Ile	Asp	Val	Val	Ile	Asn	Glu	Val	Asn	Ser	Lys	Leu	Leu	Glu	Ala	Met
		420						425					430		
Asn	Thr	Ser	Gly	Ser	Val	Tyr	Met	Thr	His	Ala	Val	Val	Gly	Gly	Met
	435						440					445			
Tyr	Val	Leu	Arg	Cys	Ala	Ile	Gly	Ala	Thr	Met	Thr	Glu	Glu	Lys	His
	450					455					460				
Val	Leu	Met	Ala	Trp	Lys	Cys	Gly	Ser	Ala	Leu	Glu	Arg	Lys	Asp	Val
465					470					475					480
Ala	Ala	Asn	Glu	Thr	Leu	Ser	Phe	Asn	Phe	Gln	Arg	Arg	Phe	Asp	Arg
				485					490					495	
Arg	Ala	Arg	Gln	Arg	Arg	Gly	His	Val	Gly	Phe	Arg	Leu	Ala	Ile	Thr
				500				505					510		
Met	Leu	Asp	Leu	Lys	Thr	Ser	Glu	Arg	Asp	Gly	Ala	Arg	Arg	Trp	Ser
	515						520				525				
Ile	Gly	Ala	Tyr	Ala	Asn	Gln	Ile	Thr	Thr	Ile	Ser	Gln	Ala	Asn	Ser
	530					535					540				
Ser	Val	Ala	Trp	Thr	Met	Glu	Phe	His	Ser	Cys	Phe	Ile	Phe	Phe	Cys
545					550					555					560
Gly	Ser	Ile	Lys	Leu	Asp	Thr	Gln	Val	Pro	Asn	Asp	Asp	Phe	Val	Leu
				565					570					575	
Ser	Ala	Arg	Trp	Pro	Pro	Ser	Phe	Pro	Val	Ser	Gly	Trp	Ser	Thr	Ile
				580				585					590		
Asn	Phe	His	Glu	Thr	Ile	Lys	Ile	Tyr	Val	Gly	Ser	Leu	Asp	Ser	Leu
	595						600					605			
Asp	Ser	Trp	Thr	Met	Glu	Phe	His	Ser	Cys	Phe	Thr	Phe	Phe	Cys	Gly
	610					615					620				

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Ser
625

<210> SEQ ID NO 84
 <211> LENGTH: 486
 <212> TYPE: PRT
 <213> ORGANISM: *Gossypium raimondii*
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 26786642

<400> SEQUENCE: 84

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Met Val Ser Ala Ser Arg Lys Thr Phe Leu Pro Leu Asp Pro Val Thr
 1           5           10          15

Phe Ser Asn Glu Ser Lys Ala Val Ile Asp Phe Ile Ala Asp Tyr Tyr
 20          25          30

Glu Asn Val Glu Lys Tyr Pro Val Gln Ser Thr Val Glu Pro Gly Tyr
 35          40          45

Leu Ser Ala Met Leu Pro Glu Ser Ala Pro Tyr Cys Pro Glu Pro Leu
 50          55          60

Gln Asp Ile Leu Glu Asp Val Ser Asn Cys Ile Ile Pro Gly Leu Thr
 65          70          75          80

His Trp Gln Ser Pro Asn Phe Phe Ala Tyr Phe His Ala Asn Ala Ser
 85          90          95

Thr Ala Gly Phe Phe Gly Glu Met Leu Cys Ser Gly Phe Asn Val Val
 100         105         110

Gly Phe Asn Trp Ile Ser Ser Pro Ala Ala Thr Glu Leu Glu Ser Ile
 115         120         125

Val Leu Asp Trp Met Gly Lys Met Leu Lys Leu Pro Ser Ser Phe Leu
 130         135         140

Phe Ser Gly Thr Gly Gly Gly Val Leu His Gly Ser Ser Cys Glu Ala
 145         150         155         160

Ala Val Cys Val Leu Ala Ala Ala Arg Asp Lys Ala Leu Lys Glu Leu
 165         170         175

Gly Gly Trp Glu Asn Ile Thr Lys Leu Val Val Tyr Ala Ser Asp Gln
 180         185         190

Ala His Phe Thr Phe Gln Lys Ala Ala Lys Leu Val Gly Ile Pro Pro
 195         200         205

Ser Asn Phe Arg Leu Ile Glu Thr Ser Phe Ser Thr Gly Phe Ser Leu
 210         215         220

Ser Pro Glu Asn Leu Arg Phe Val Ile Glu Asp Asn Ile Arg Ser Gly
 225         230         235         240

Leu Val Pro Leu Phe Leu Cys Ala Thr Ile Gly Thr Thr Pro Ser Gly
 245         250         255

Ala Val Asp Pro Ile Ala Glu Leu Gly Lys Val Ala Met Glu Phe Lys
 260         265         270

Leu Trp Leu His Ile Asp Ala Ala Tyr Ala Gly Ser Gly Cys Ile Cys
 275         280         285

Pro Glu Leu Arg His Tyr Leu Asp Gly Val Glu Leu Ala Asn Ser Ile
 290         295         300

Ser Met Asn Pro His Lys Trp Phe Leu Thr Asn Met Asp Cys Cys Cys
 305         310         315         320

Leu Trp Ile Lys Glu Pro Lys Leu Leu Val Asp Ser Leu Ser Thr Asp
 325         330         335

Pro Glu Ile Leu Arg Asn Asn Ala Ser Lys Ser Lys Ala Val Val Asp
 340         345         350

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Cys Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala Leu Lys
 355 360 365
 Leu Trp Val Val Ile Arg Arg His Gly Leu Ala Asn Leu Met Cys His
 370 375 380
 Ile Arg Ser Asp Ile Ala Met Ala Lys Arg Phe Glu Ala Leu Val Gly
 385 390 395 400
 Glu Asp Glu Arg Phe Glu Ile Val Val Pro Arg Lys Phe Ala Leu Val
 405 410 415
 Cys Phe Arg Leu Lys Pro Lys Val Glu Glu Glu Asp Leu Asn Cys Lys
 420 425 430
 Leu Val Glu Ala Ile Asn Ser Ser Gly Arg Ala Phe Met Ser His Ala
 435 440 445
 Val Leu Ser Gly Ile Tyr Val Ile Arg Cys Ala Ile Gly Thr Thr Leu
 450 455 460
 Thr Gln Gln His His Val Asp Ala Leu Trp Lys Leu Ile Gln Asp Lys
 465 470 475 480
 Ala Gln Ser Leu Leu Met
 485

<210> SEQ ID NO 85
 <211> LENGTH: 495
 <212> TYPE: PRT
 <213> ORGANISM: Populus trichocarpa
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 26994989

<400> SEQUENCE: 85

Met Gly Ser Leu Ser Thr Asn Thr Phe Ser Pro Leu Asp Pro Asn Gly
 1 5 10 15
 Phe Thr Asn Asp Ser Lys Met Val Ile Asp Phe Ile Ala Asp Tyr Tyr
 20 25 30
 Lys Asn Ile Glu Asn Asn Pro Val Gln Ser Gln Val Lys Pro Gly Tyr
 35 40 45
 Leu Leu Thr Gln Leu Pro Asp Thr Ala Pro Tyr Cys Glu Glu Ser Leu
 50 55 60
 Glu Asp Val Leu Lys Asp Val Thr Asp Ser Ile Ile Pro Gly Leu Thr
 65 70 75 80
 His Trp Gln Ser Pro Asn Phe Phe Ala Tyr Phe Gln Ala Asn Ala Ser
 85 90 95
 Thr Ala Gly Phe Val Gly Glu Met Leu Cys Thr Gly Leu Asn Val Val
 100 105 110
 Gly Phe Asn Trp Ile Ala Ser Pro Ala Ala Thr Glu Leu Glu Ser Ile
 115 120 125
 Val Met Asp Trp Met Gly Lys Met Leu Lys Leu Pro Ser Thr Phe Leu
 130 135 140
 Phe Ser Gly Asn Gly Gly Gly Val Leu His Gly Ser Thr Cys Glu Ala
 145 150 155 160
 Ile Val Cys Thr Leu Val Ala Ala Arg Asp Glu Thr Leu Arg Met Ile
 165 170 175
 Gly Ala Glu Asn Ile Thr Lys Leu Val Val Tyr Ala Ser Asp Gln Thr
 180 185 190
 His Ser Thr Leu Leu Lys Gly Val Lys Leu Val Gly Ile Pro Ser Ser
 195 200 205
 Asn Phe Arg Cys Leu Ser Thr Ser Phe Ser Ser Glu Phe Ser Leu Ser
 210 215 220

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Pro Gln Ala Leu Glu Asp Ala Ile Glu Asn Asp Ile Lys Ala Gly Leu
225                230                235                240

Val Pro Leu Phe Leu Cys Ala Thr Val Gly Thr Thr Ala Cys Gly Ala
                245                250                255

Val Asp Pro Val Met Asp Leu Gly Glu Ile Ala Arg Lys Tyr Asn Leu
                260                265                270

Trp Phe His Ile Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro
                275                280                285

Glu Phe Arg His Tyr Leu Asp Gly Val Glu Leu Ala Asp Ser Leu Ser
290                295                300

Met Asn Pro His Lys Trp Leu Leu Thr Asn Met Asp Cys Cys Cys Leu
305                310                315                320

Trp Val Lys Gln Pro Arg Leu Leu Ile Glu Ser Leu Ser Ser Asp Ala
                325                330                335

Glu Phe Leu Arg Asn Asn Ala Ser Glu Ser Ser Asp Val Val Asp Tyr
340                345                350

Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala Leu Lys Leu
355                360                365

Trp Ile Val Ile Arg Arg His Gly Leu Ala Asn Leu Met Cys His Ile
370                375                380

Arg Ser Asp Val Asn Leu Ala Lys Arg Phe Glu Ser Leu Val Ala Lys
385                390                395                400

Asp Ser Arg Phe Glu Val Val Val Arg Arg Arg Phe Ser Leu Val Cys
405                410                415

Phe Arg Leu Lys His Asn Asp Glu Cys Gln Gly Leu Glu Leu Asn Arg
420                425                430

Lys Leu Leu Ala Ala Val Asn Glu Ser Gly Arg Ala Phe Met Thr His
435                440                445

Ala Val Val Gly Gly Leu Phe Ile Ile Arg Cys Ala Ile Gly Ser Thr
450                455                460

Leu Thr Glu Glu Arg His Val Asp Asp Leu Trp Lys Leu Ile Gln Glu
465                470                475                480

Lys Ala Ala Asp Leu Leu Ser Lys Lys Gln Val Leu Leu Asp Asn
485                490                495

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<210> SEQ ID NO 86
<211> LENGTH: 565
<212> TYPE: PRT
<213> ORGANISM: Malus domestica
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 22679008

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<400> SEQUENCE: 86

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Met Ser Leu Leu Ala Phe Tyr Ser Asn Ser Gly Glu Arg Ser Lys Arg
1                5                10                15

Val His Leu Ser Ala Ser Thr Tyr Gly Asn Ser Thr Pro Asn Ser Tyr
20                25                30

Ile Ser Leu Pro Tyr Ala Leu Phe Ser Ser Ala Thr Gln Leu Ile Asn
35                40                45

Ile His Ser Asn Ser Ser Asn Phe Gln Met Gly Ser Leu Ile Ser Gln
50                55                60

Glu Asn Asn Ser Pro Asn Val Pro Thr Asn Pro Leu Asp Pro Glu Glu
65                70                75                80

Phe Arg Arg Gln Gly His Leu Val Ile Asp Phe Ile Ala Asp Tyr Tyr
85                90                95

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Lys 100	Ser 101	Ile 102	Glu 103	Lys 104	His 105	Pro 106	Val 107	Leu 108	Ser 109	Gln 110	Val 111	Gln 112	Pro 113	Gly 114	Tyr 115
Leu 116	Lys 117	Lys 118	Arg 119	Leu 120	Pro 121	Asp 122	Thr 123	Ala 124	Pro 125	Tyr 126	Asn 127	Pro 128	Glu 129	Pro 130	Leu 131
Glu 132	Thr 133	Ile 134	Leu 135	Gln 136	Asp 137	Val 138	Gln 139	Asp 140	His 141	Ile 142	Val 143	Pro 144	Gly 145	Ile 146	Thr 147
His 148	Trp 149	Gln 150	Ser 151	Pro 152	Asn 153	Tyr 154	Phe 155	Ala 156	Tyr 157	Phe 158	Pro 159	Ser 160	Ser 161	Gly 162	Ser 163
Val 164	Ala 165	Gly 166	Phe 167	Leu 168	Gly 169	Glu 170	Met 171	Leu 172	Ser 173	Ser 174	Gly 175	Phe 176	Asn 177	Val 178	Val 179
Gly 180	Phe 181	Asn 182	Trp 183	Met 184	Ser 185	Ser 186	Pro 187	Ala 188	Ala 189	Thr 190	Glu 191	Leu 192	Glu 193	Ser 194	Thr 195
Val 196	Arg 197	Asp 198	Trp 199	Phe 200	Gly 201	Asn 202	Met 203	Leu 204	Lys 205	Leu 206	Pro 207	Lys 208	Ser 209	Phe 210	Leu 211
Phe 212	Ser 213	Gly 214	Asn 215	Gly 216	Gly 217	Asp 218	Val 219	Ile 220	Gln 221	Gly 222	Thr 223	Thr 224	Cys 225	Glu 226	Ala 227
Leu 228	Val 229	Cys 230	Ala 231	Met 232	Val 233	Ala 234	Ala 235	Arg 236	Asp 237	Gln 238	Lys 239	Leu 240	Ser 241	Lys 242	Phe 243
Gly 244	Arg 245	His 246	Asn 247	Ile 248	Gly 249	Lys 250	Leu 251	Val 252	Val 253	Tyr 254	Gly 255	Ser 256	Asp 257	Gln 258	Thr 259
His 260	Ser 261	Ala 262	Leu 263	Gln 264	Lys 265	Ala 266	Ser 267	Gln 268	Ile 269	Val 270	Gly 271	Ile 272	His 273	Pro 274	Glu 275
Asn 276	Phe 277	Arg 278	Ser 279	Ile 280	Glu 281	Thr 282	Thr 283	Arg 284	Ser 285	Thr 286	Ser 287	Phe 288	Ala 289	Leu 290	Ser 291
Pro 292	Glu 293	Ser 294	Leu 295	Lys 296	Val 297	Ile 298	Ile 299	Tyr 300	Ser 301	Asp 302	Ile 303	Glu 304	Ala 305	Gly 306	Leu 307
Val 308	Pro 309	Leu 310	Phe 311	Leu 312	Cys 313	Ala 314	Thr 315	Val 316	Gly 317	Thr 318	Thr 319	Ala 320	Ile 321	Ala 322	Thr 323
Val 324	Asp 325	Pro 326	Leu 327	Gly 328	Pro 329	Leu 330	Cys 331	Gly 332	Val 333	Ala 334	Gly 335	Asp 336	Tyr 337	Gly 338	Met 339
Trp 340	Val 341	His 342	Val 343	Asp 344	Ala 345	Ala 346	Tyr 347	Ala 348	Gly 349	Ser 350	Ala 351	Cys 352	Ile 353	Cys 354	Pro 355
Ser 356	Phe 357	Asp 358	Ile 359	Ser 360	Leu 361	Met 362	Ala 363	Ser 364	Arg 365	Val 366	Gln 367	Ile 368	His 369	Ser 370	Val 371
Ser 372	Thr 373	Arg 374	Thr 375	Asn 376	Gly 377	Ser 378	Ser 379	Pro 380	Leu 381	Ser 382	Thr 383	Val 384	Val 385	Ala 386	Phe 387
Gly 388	Leu 389	Arg 390	Ile 391	Pro 392	Thr 393	Arg 394	Trp 395	Asn 396	Lys 397	Ala 398	Thr 399	Glu 400	Leu 401	Lys 402	Gln 403
Val 404	Val 405	Asp 406	Tyr 407	Lys 408	Asp 409	Trp 410	Gln 411	Ile 412	Ala 413	Leu 414	Ser 415	Arg 416	Arg 417	Phe 418	Arg 419
Ser 420	Met 421	Lys 422	Leu 423	Trp 424	Leu 425	Val 426	Leu 427	Arg 428	Ser 429	Tyr 430	Gly 431	Val 432	Ala 433	Asn 434	Leu 435
Arg 436	Asn 437	Phe 438	Leu 439	Arg 440	Ser 441	His 442	Val 443	Lys 444	Met 445	Ala 446	Lys 447	Ile 448	Phe 449	Glu 450	Gly 451
Leu 452	Val 453	Ala 454	Met 455	Asp 456	Lys 457	Arg 458	Phe 459	Glu 460	Ile 461	Val 462	Ala 463	Pro 464	Arg 465	Asn 466	Phe 467
Ser 468	Leu 469	Val 470	Cys 471	Phe 472	Arg 473	Val 474	Ser 475	Pro 476	Ser 477	Ser 478	Ile 479	Ser 480	Asn 481	Lys 482	Ala 483
Ser 484	Ser 485	Asp 486	Gln 487	Asn 488	Gly										

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515	520	525
Gly Leu Tyr Met Leu Arg	Phe Ala Val Gly Ala	Thr Leu Thr Glu Glu
530	535	540
His His Ile Ala Leu Ala Trp	Lys Val Val Gln Glu His Ala Asp Gln	
545	550	555 560
Ile Leu Thr Lys Tyr		
565		
<210> SEQ ID NO 87		
<211> LENGTH: 523		
<212> TYPE: PRT		
<213> ORGANISM: Citrus clementina		
<300> PUBLICATION INFORMATION:		
<308> DATABASE ACCESSION NUMBER: 20801973		
<400> SEQUENCE: 87		
Met Arg Ala Gly Glu Ala Ser Ile Ile Lys Met Gly Ser Phe Gly Leu		
1	5 10	15
Ser Ala Asn Asn Ile Thr His Gly Ser Ser Phe Ser Ala Asp Leu Glu		
	20 25	30
Pro Lys Ser Phe Ser Asp Glu Ser Lys Ala Val Ile Asp Phe Ile Ala		
	35 40	45
Asp Tyr Tyr Lys Asn Ile Glu Lys Tyr Pro Val Gln Ser Lys Val Glu		
	50 55	60
Pro Gly Tyr Leu Ser Ala Arg Leu Pro Asp Thr Ala Pro His Ser Pro		
65	70 75	80
Glu Ser Leu Asp Asp Ile Leu Lys Asp Val Thr Asp Cys Ile Leu Pro		
	85 90	95
Gly Leu Thr His Trp Gln Ser Pro Asn Phe Phe Gly Tyr Phe Gln Ala		
	100 105	110
Asn Ala Ser Thr Ala Gly Phe Leu Gly Glu Met Leu Cys Ser Gly Phe		
	115 120	125
Asn Val Val Gly Phe Asn Trp Leu Ala Ser Pro Val Ala Thr Glu Leu		
	130 135	140
Glu Ser Ile Val Met Asp Trp Met Gly Lys Met Leu Lys Leu Pro Ser		
145	150 155	160
Ser Phe Leu Phe Ser Gly Thr Gly Gly Gly Val Leu His Gly Ser Thr		
	165 170	175
Cys Glu Ser Leu Val Cys Thr Leu Ala Ala Ala Arg Asp Lys Ala Leu		
	180 185	190
Glu Lys Leu Gly Gly Gly Phe Asp Asn Ile Thr Lys Leu Ala Val Tyr		
	195 200	205
Ala Ser Asp Gln Thr His Phe Ala Leu Gln Lys Ser Ala Lys Leu Ile		
	210 215	220
Gly Ile Pro Pro Ala Asn Phe Arg Pro Leu Arg Thr Ser Phe Ser Thr		
225	230 235	240
Glu Phe Ser Leu Ser Pro Asp Thr Val Arg Ala Ala Ile Glu Asp Asp		
	245 250	255
Ile Lys Ser Gly His Val Pro Leu Tyr Leu Cys Ala Thr Val Gly Thr		
	260 265	270
Thr Gly Ala Gly Ala Val Asp Pro Ile Glu Glu Leu Gly Lys Ile Ala		
	275 280	285
Asn Glu Tyr Lys Leu Trp Leu His Ile Asp Ala Ala Tyr Ala Gly Ser		
	290 295	300
Ala Cys Ile Cys Pro Glu Tyr Arg His Tyr Leu Asn Gly Val Glu Leu		

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305	310	315	320
Ala Asp Ser Ile Ser Leu Asn Pro His Lys Trp Phe Leu Thr Asn Met			
	325	330	335
Asp Cys Cys Cys Leu Trp Val Lys His Pro Ser Phe Leu Val Asp Ser			
	340	345	350
Leu Ser Thr Glu Ser Asp Ile Met Arg Asn Arg Ser Pro Ala Ser Asn			
	355	360	365
Thr Ser Thr Asn Ala Ala Pro Val Ile Asp Tyr Lys Asp Trp Gln Ile			
	370	375	380
Ala Leu Ser Arg Arg Phe Lys Ala Leu Lys Leu Trp Thr Val Ile Arg			
	385	390	395
Lys His Gly Tyr Ser Gly Leu Met Tyr His Ile Arg Ser Asp Val Ser			
	405	410	415
Met Ala Lys Arg Phe Ala Ala Met Val Ala Lys Asp Glu Arg Phe Glu			
	420	425	430
Ile Val Val Pro Arg Lys Phe Ala Leu Val Cys Phe Arg Leu Lys Pro			
	435	440	445
Lys Arg Glu Ser Glu Gly Ser Glu Leu Asn Arg Glu Leu Val Asp Ala			
	450	455	460
Leu Asn Gly Ser Gly Arg Ala Phe Leu Thr Gln Ala Met Leu Gly Gly			
	465	470	475
Val Tyr Val Ile Arg Cys Ser Ile Gly Thr Thr Leu Thr Gln Asp Arg			
	485	490	495
His Val Asp Asp Leu Trp Lys Leu Ile Gln Glu Lys Ala Asp Arg Leu			
	500	505	510
Leu Ser Leu Gln Glu Pro Glu His Ala Ser Arg			
	515	520	

<210> SEQ ID NO 88

<211> LENGTH: 506

<212> TYPE: PRT

<213> ORGANISM: Citrus clementina

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 20818150

<400> SEQUENCE: 88

Met Gly Ser Leu Asn Ser Asp His Glu Leu Lys Thr Asn Ser Ala Ser			
1	5	10	15
Phe Asn Asn Pro Met Asp Ser Glu Glu Phe Arg Arg Gln Gly His Met			
	20	25	30
Ile Ile Asp Phe Ile Ala Asp Tyr Tyr Arg Asp Val Glu Lys Tyr Pro			
	35	40	45
Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Gln Lys Arg Leu Pro Glu			
	50	55	60
Ser Ala Pro Tyr Asn Pro Glu Pro Ile Glu Thr Ile Leu Gln Asp Val			
	65	70	75
Gln Gln His Ile Val Pro Gly Ile Thr His Trp Gln Ser Pro Tyr Tyr			
	85	90	95
Phe Ala Tyr Phe Pro Ser Ser Gly Ser Ile Ala Gly Phe Leu Gly Glu			
	100	105	110
Met Leu Ser Ser Gly Phe Asn Val Val Gly Phe Asn Trp Met Ser Ser			
	115	120	125
Pro Ala Ala Thr Glu Leu Glu Asn Ile Val Met Asp Trp Leu Gly Glu			
	130	135	140
Met Leu Lys Leu Pro Lys Ser Phe Leu Phe Ser Gly Thr Gly Gly Gly			

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145	150	155	160
Val Ile Gln Gly Thr Thr Cys Glu Ala Ile Leu Cys Thr Leu Ala Ala			
	165	170	175
Ala Arg Asp Gln Ile Leu Asn Glu Ile Gly Arg Glu Asn Ile Ser Arg			
	180	185	190
Leu Val Val Tyr Gly Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala			
	195	200	205
Ala Gln Ile Ala Gly Ile Asp Pro Lys Asn Phe Arg Ala Ile Lys Thr			
	210	215	220
Thr Lys Ser Ser Ser Phe Thr Leu Thr Pro Glu Ser Leu Gln Ala Ala			
	225	230	235
Ile Asp Leu Asp Ile Gln Ser Gly Leu Ile Pro Leu Phe Leu Cys Ala			
	245	250	255
Thr Val Gly Thr Thr Ala Ile Thr Thr Val Asp Pro Leu Gly Pro Leu			
	260	265	270
Cys Asp Ile Ala Lys Arg Tyr Ser Ile Trp Ile His Val Asp Ala Ala			
	275	280	285
Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Phe Ile Asp			
	290	295	300
Gly Ile Glu Ser Ala Asp Ser Phe Ser Leu Asn Ala His Lys Trp Phe			
	305	310	315
Phe Thr Thr Leu Asp Cys Cys Cys Met Trp Val Lys Asn Pro Asn Ala			
	325	330	335
Leu Ile Lys Ala Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys Ala			
	340	345	350
Ser Asp Ser Lys Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Ser Leu			
	355	360	365
Ser Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Leu Arg Ser Phe			
	370	375	380
Gly Val Ala Asn Leu Arg Asn Phe Leu Arg Ser His Val Gly Met Ala			
	385	390	395
Gln Leu Phe Gln Glu Leu Val Gly Gly Asp Asn Arg Phe Glu Ile Val			
	405	410	415
Ala Pro Arg Asn Phe Ala Val Val Cys Phe Arg Val Leu Pro Ser Ala			
	420	425	430
Ser Gly Leu Gly Asn Gly Lys Ala Asn Glu Gly Ala Asn Glu Leu Asn			
	435	440	445
Arg Lys Leu Leu Glu Ser Ile Asn Ala Ser Gly Gln Leu Tyr Val Ser			
	450	455	460
His Gly Met Val Ala Gly Ile Tyr Phe Ile Arg Phe Ala Val Gly Ala			
	465	470	475
Thr Leu Thr Glu Asp Arg His Val Ile Ala Ala Trp Lys Val Val Gln			
	485	490	495
Glu Lys Leu Asp Gly Ile Leu Ala Thr Ser			
	500	505	

<210> SEQ ID NO 89

<211> LENGTH: 467

<212> TYPE: PRT

<213> ORGANISM: Vitis vinifera

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: 17834108

<400> SEQUENCE: 89

Met Gly Ser Leu Ser Phe Asn Thr Phe Ser Pro Leu Asp Pro Gln Ser

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1	5	10	15
Phe Ser Glu Glu Ser Lys Met Val Val Asp Phe Ile Ala Asp Tyr Tyr	20	25	30
Lys Asn Val Glu Lys Tyr Pro Val Gln Ser Gln Val Asp Pro Gly Tyr	35	40	45
Leu Met His His Cys Pro Asp Thr Ala Pro Tyr Cys Pro Glu Pro Leu	50	55	60
Glu Thr Ile Leu Lys Asp Val Ser Asp Gly Ile Ile Pro Gly Leu Thr	65	70	75
His Trp Gln Ser Pro Asn Phe Phe Gly Tyr Phe Gln Ala Asn Ala Ser	85	90	95
Thr Ala Gly Phe Leu Gly Glu Met Leu Cys Thr Gly Leu Asn Val Val	100	105	110
Gly Phe Asn Trp Ile Ala Ser Pro Ala Ala Thr Glu Leu Glu Ser Ile	115	120	125
Ala Ile Ile Cys Ser Leu Ala Ala Ala Arg Asp Lys Val Leu Lys Lys	130	135	140
Leu Gly His His Lys Ile Thr Lys Leu Val Val Tyr Gly Ser Asp Gln	145	150	155
Thr His Ser Thr Leu Gln Lys Ala Ser Lys Leu Val Gly Ile Pro Ala	165	170	175
Ser Asn Phe Arg Ser Leu Pro Thr Ser Phe Ser Asn Tyr Phe Ala Leu	180	185	190
Cys Pro Asp Asp Val Arg Thr Ala Met Glu Glu Asp Ile Gly Ala Gly	195	200	205
Leu Val Pro Leu Phe Leu Cys Ala Thr Val Gly Thr Thr Ser Ser Gly	210	215	220
Ala Val Asp Pro Leu Glu Ala Leu Gly His Val Ala Lys Asp Phe Lys	225	230	235
Val His His Leu Asn Gly Val Glu Leu Ala His Ser Ile Ser Met Asn	245	250	255
Pro His Lys Trp Leu Leu Thr Asn Met Asp Cys Cys Cys Leu Trp Ile	260	265	270
Lys Glu Pro Lys Leu Phe Val Asp Ser Leu Ser Thr Ala Pro Glu Phe	275	280	285
Leu Arg Asn Asn Ala Ser Glu Ser Lys Lys Val Ile Asp Tyr Lys Asp	290	295	300
Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala Ile Lys Val Trp Ala	305	310	315
Val Val Pro Arg Arg Phe Ala Leu Val Cys Phe Arg Leu Arg Pro Arg	325	330	335
Glu Glu Gly Glu Ser Thr Glu Leu Asn Ser Arg Leu Leu Met Ala Val	340	345	350
Asn Gly Ser Gly Ala Ala Phe Met Thr His Ala Val Val Gly Gly Ile	355	360	365
Tyr Ile Ile Arg Cys Ala Ile Gly Ser Thr Leu Thr Glu Thr Arg His	370	375	380
Val Asp Ser Leu Trp Lys Leu Ile Gln Glu Lys Ala Gln Leu Val Leu	385	390	395
Gln Glu Pro Gly Leu Ala Leu Glu Glu Asp Tyr Ile Asp Pro Cys Ile	405	410	415
Gly Val Ser Ala Thr Ser Leu His Ala Val Val Arg Trp Tyr Cys Asn	420	425	430

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Tyr Ser Ser Glu Ile Asn Ala His Leu Val Phe Ile Ala Phe Phe Val
 435 440 445

Val Val Cys Lys Glu Asn Arg Glu Asn Tyr Val Leu Gly Val Asn Gly
 450 455 460

Pro Pro Asn
 465

<210> SEQ ID NO 90
 <211> LENGTH: 506
 <212> TYPE: PRT
 <213> ORGANISM: Petunia hybrida
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: ABB72475.1

<400> SEQUENCE: 90

Met Asp Thr Ile Lys Ile Asn Pro Glu Phe Asp Gly Gln Phe Cys Lys
 1 5 10 15

Thr Thr Ser Leu Leu Asp Pro Glu Glu Phe Arg Arg Asn Gly His Met
 20 25 30

Met Val Asp Phe Leu Ala Asp Tyr Phe His Asn Ile Glu Lys Tyr Pro
 35 40 45

Val Arg Ser Gln Val Glu Pro Gly Tyr Leu Glu Arg Leu Leu Pro Asp
 50 55 60

Ser Ala Pro Ile Gln Pro Glu Pro Ile Glu Lys Ile Leu Lys Asp Val
 65 70 75 80

Arg Ser Asp Ile Phe Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe
 85 90 95

Phe Ala Tyr Phe Pro Cys Ser Ser Ser Thr Ala Gly Ile Leu Gly Glu
 100 105 110

Met Leu Ser Ala Gly Leu Asn Val Val Gly Phe Ser Trp Ile Ala Ser
 115 120 125

Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Met Asp Trp Leu Gly Lys
 130 135 140

Leu Ile Asn Leu Pro Lys Thr Tyr Leu Phe Ser Gly Gly Gly Gly
 145 150 155 160

Val Met Gln Gly Thr Thr Cys Glu Val Met Leu Cys Thr Ile Val Ala
 165 170 175

Ala Arg Asp Lys Met Leu Glu Lys Phe Gly Arg Glu Asn Ile Asp Lys
 180 185 190

Leu Val Val Tyr Ala Ser Asp Gln Thr His Phe Ser Phe Gln Lys Ala
 195 200 205

Val Lys Ile Ser Gly Ile Lys Pro Glu Asn Phe Arg Ala Ile Pro Thr
 210 215 220

Thr Lys Ala Thr Glu Phe Ser Leu Asn Pro Glu Ser Leu Arg Arg Ala
 225 230 235 240

Ile Gln Glu Asp Lys Lys Ala Gly Leu Ile Pro Leu Phe Leu Cys Thr
 245 250 255

Ser Ile Gly Thr Thr Ser Thr Thr Ala Val Asp Pro Leu Lys Pro Leu
 260 265 270

Cys Glu Ile Ala Glu Glu Tyr Gly Ile Trp Val His Val Asp Ala Ala
 275 280 285

Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Gln His Phe Leu Asp
 290 295 300

Gly Val Glu His Ala Asn Ser Phe Ser Phe Asn Ala His Lys Trp Leu
 305 310 315 320

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Phe Thr Thr Leu Asp Cys Cys Cys Leu Trp Leu Lys Asp Pro Ser Ser
      325                      330                      335

Leu Thr Lys Ala Leu Ser Thr Asn Pro Glu Val Leu Arg Asn Asp Ala
      340                      345                      350

Thr Asp Ser Glu Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Thr Leu
      355                      360                      365

Ser Arg Arg Phe Arg Ser Leu Lys Leu Trp Leu Val Leu Lys Ser Tyr
      370                      375                      380

Gly Val Ala Asn Leu Arg Asn Phe Ile Arg Ser His Ile Glu Met Ala
      385                      390                      395                      400

Lys His Phe Glu Glu Leu Val Ala Met Asp Glu Arg Phe Glu Ile Met
      405                      410                      415

Ala Pro Arg Asn Phe Ser Leu Val Cys Phe Arg Val Ser Leu Leu Ala
      420                      425                      430

Leu Glu Lys Lys Phe Asn Phe Val Asp Glu Thr Gln Val Asn Glu Phe
      435                      440                      445

Asn Ala Lys Leu Leu Glu Ser Ile Ile Ser Ser Gly Asn Val Tyr Met
      450                      455                      460

Thr His Thr Val Val Glu Gly Val Tyr Met Ile Arg Phe Ala Val Gly
      465                      470                      475                      480

Ala Pro Leu Thr Asp Tyr Pro His Ile Asp Met Ala Trp Asn Val Val
      485                      490                      495

Arg Asn His Ala Thr Met Met Leu Asn Ala
      500                      505

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<210> SEQ ID NO 91
<211> LENGTH: 494
<212> TYPE: PRT
<213> ORGANISM: Carica papaya
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: 16421889

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<400> SEQUENCE: 91

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Met Ser Ser Leu Ser Arg Asp Leu Asn Ala Ser Pro Leu Glu Pro Glu
 1          5          10          15

Asn Phe Arg Val Glu Ser Lys Arg Val Ile Asp Phe Ile Ala Asp Tyr
 20          25          30

Tyr Lys Asn Ile Glu Thr Tyr Pro Val Gln Ser Arg Val Lys Pro Gly
 35          40          45

Tyr Leu Ala Gly Arg Leu Pro Ser Ser Ala Pro Phe Ser Pro Glu Ser
 50          55          60

Leu Glu Thr Ile Leu Gln Asp Ile Ala Glu Asn Ile Ser Pro Gly Leu
 65          70          75          80

Thr His Trp Gln Ser Pro Asn Phe Phe Gly Tyr Phe Gln Ala Asn Ala
 85          90          95

Ser Thr Ala Gly Phe His Gly Glu Met Leu Cys Ser Gly Leu Asn Val
100          105          110

Val Gly Phe Asn Trp Ile Ser Ser Pro Ala Ala Thr Glu Leu Glu Ser
115          120          125

Leu Val Met Asp Trp Met Gly Asn Met Leu Lys Leu Pro Ser Ser Phe
130          135          140

Leu Phe Ser Gly Ser Gly Gly Gly Val Leu His Gly Ser Thr Cys Glu
145          150          155          160

Ala Val Val Cys Thr Leu Ala Ala Ala Arg Asp Lys Thr Leu Asn Gln
165          170          175

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Leu Gly Gly Asn Tyr Gln Asn Ile Thr Lys Phe Val Val Tyr Ala Ser
 180 185 190
 Asp Gln Thr His Phe Thr Leu Gln Lys Ala Ala Lys Leu Ile Gly Ile
 195 200 205
 Pro Pro Ser Asn Phe Arg Ser Leu Thr Thr Ser Phe Pro Ser Gly Phe
 210 215 220
 Ser Leu Ser Pro Glu Lys Leu Gln Ser Ala Ile Lys Asp Asp Ile Lys
 225 230 235 240
 Ser Gly Tyr Val Pro Leu Tyr Val Cys Ala Thr Val Gly Thr Thr Ala
 245 250 255
 Ala Gly Ala Val Asp Pro Ile Leu Glu Leu Gly Lys Val Ala Gln Glu
 260 265 270
 Tyr Asn Leu Trp Phe His Ile Asp Ala Ala Tyr Ala Gly Ser Ala Cys
 275 280 285
 Ile Cys Thr Glu Phe Arg His Tyr Leu Asn Gly Val Glu Leu Ala Asp
 290 295 300
 Ser Ile Ser Thr Asn Pro His Lys Trp Leu Leu Thr Asn Met Glu Cys
 305 310 315 320
 Ser Cys Leu Trp Val Lys Ser Pro Ser Ser Leu Val Asp Ser Leu Ser
 325 330 335
 Thr Lys Ser Glu Ile Met Arg Asn Ala Ala Thr Asp Ser Asn Gln Val
 340 345 350
 Ile Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala
 355 360 365
 Leu Lys Leu Trp Ile Val Ile Arg Arg His Gly Leu Ser Gly Leu Thr
 370 375 380
 Ser His Ile His Lys Asp Ile Lys Met Ala Glu Leu Phe Glu Ser Leu
 385 390 395 400
 Val Ala Lys Asp Lys Arg Phe Glu Ile Val Val Pro Arg Lys Phe Ala
 405 410 415
 Leu Val Cys Phe Arg Phe Lys Pro Glu Lys Glu Asn Gln Asp Leu Ser
 420 425 430
 Glu Leu Asn Ser Lys Leu Leu Asn Ala Val Asn Ser Ser Gly Cys Ala
 435 440 445
 Phe Met Thr His Ala Val Leu Glu Gly Val Tyr Thr Ile Arg Cys Ala
 450 455 460
 Ile Gly Thr Thr Leu Thr Glu Glu His His Val Val Asn Leu Trp Lys
 465 470 475 480
 Leu Ile Gln Glu Lys Ala Gln Ser Leu Ile Ile Asn Glu Tyr
 485 490

<210> SEQ ID NO 92
 <211> LENGTH: 452
 <212> TYPE: PRT
 <213> ORGANISM: Sphagnum fallax
 <300> PUBLICATION INFORMATION:
 <308> DATABASE ACCESSION NUMBER: 0042s0024.1

<400> SEQUENCE: 92

Met Ser Ser Lys Val Ala Pro Trp Ser Arg Leu Ser Lys Pro Leu Asp
 1 5 10 15
 Val Glu Glu Phe Arg Thr His Ala His Arg Met Val Asp Phe Ile Ala
 20 25 30
 Asp Tyr His His Asn Ile Gln Ser Phe Pro Val His Ser Gln Leu Lys
 35 40 45

[illegible]

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<210> SEQ ID NO 93
<211> LENGTH: 486
<212> TYPE: PRT
<213> ORGANISM: Eucalyptus grandis
<300> PUBLICATION INFORMATION:
<308> DATABASE ACCESSION NUMBER: E01788.1

<400> SEQUENCE: 93

Met Asn Pro Leu Asp Pro Gly Glu Phe Arg Arg Gln Gly His Met Val
 1             5             10             15

Val Asp Phe Leu Ala Lys Tyr Tyr Glu Asn Ile Glu Lys Tyr Pro Val
      20             25             30

Leu Ser Gln Val Glu Pro Gly Tyr Leu Ser Lys Arg Leu Pro Ser Ser
      35             40             45

Ala Pro Gln Asp Glu Glu Pro Met Glu Ala Ile Leu Asp Asp Val His
      50             55             60

Gln His Ile Phe Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe Phe
      65             70             75             80

Ala Tyr Tyr Gln Thr Asn Thr Ser Thr Ala Ala Ile Leu Gly Glu Met
      85             90             95

Leu Cys Ala Gly Phe Asn Val Ala Gly Phe Asn Trp Val Ser Ser Pro
      100            105            110

Ala Ala Thr Glu Leu Glu Ser Leu Val Met Asp Trp Leu Gly Lys Met
      115            120            125

Leu Asp Leu Pro Arg Pro Phe Leu Pro Phe Gly Asn Gly Gly Gly Val
      130            135            140

Ile Glu Gly Asn Thr Ser Glu Ala Ile Ile Cys Thr Leu Thr Ala Ala
      145            150            155            160

Arg Asp Arg Val Leu Arg Lys Leu Gly His Asn Ser Ile Ala Lys Leu
      165            170            175

Val Val Tyr Gly Ser Asp Gln Thr Asn Cys Ser Phe Gln Lys Ala Ala
      180            185            190

Arg Val Val Gly Ile Asp Pro Arg Asn Phe Arg Ala Leu Lys Met Thr
      195            200            205

Arg Ser Thr Leu Phe Gly Leu Ser Pro Asp Ser Leu Glu Lys Ala Ile
      210            215            220

Arg Leu Asp Ile Asn Ala Gly Leu Ile Pro Leu Tyr Leu Cys Ala Thr
      225            230            235            240

Val Gly Thr Thr Ser Cys Ala Ala Val Asp Pro Leu Glu Pro Leu Cys
      245            250            255

Lys Val Ala Ser Lys Phe Ser Met Trp Ile His Val Asp Ala Ala Tyr
      260            265            270

Ala Gly Ala Ser Cys Ile Cys Pro Glu Tyr Arg Lys Phe Ile Asn Gly
      275            280            285

Val Glu Phe Ala Asp Ser Phe Ser Phe Asn Ala His Lys Trp Leu Leu
      290            295            300

Thr Pro Leu Asp Cys Cys Cys Leu Trp Val Lys Asp Pro Asn Ala Leu
      305            310            315            320

Val Lys Ser Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Glu Ala Thr
      325            330            335

Glu Ser Lys Gln Val Ile Asp Tyr Ala Asp Trp Gln Leu Ser Leu Ser
      340            345            350

Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Leu Arg Ser His Gly
      355            360            365

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Val Gln Asn Leu Arg Ser His Ile Lys Asn His Cys Arg Leu Ala Lys
 370 375 380

Leu Phe Glu Glu Leu Val Glu Glu Asp Pro Gln Phe Glu Val Val Phe
 385 390 395 400

Pro Arg Asn Phe Ala Leu Val Cys Phe Arg Ile His Pro Ser Gly Val
 405 410 415

Ala Gly Met Leu Asn Ala Gln Leu Leu His Ala Ile Asn Ala Ser Gly
 420 425 430

Arg Val Phe Met Ser His Thr Thr Val Gly Gly Val Tyr Val Leu Arg
 435 440 445

Phe Ala Val Gly Ala Thr Leu Val Thr Glu Lys His Val Ile Met Ala
 450 455 460

Trp Lys Val Val Gln Glu His Ala Asn Ser Leu Leu Ser Met Pro Ala
 465 470 475 480

Ser Glu Gln His Ser Ala
 485

<210> SEQ ID NO 94
 <211> LENGTH: 5336
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: pHis8-4 vector

<400> SEQUENCE: 94

tggcgaatgg gacgcgcct gtagcggcgc attaagcgcg gcgggtgtgg tggttacgcg	60
cagcgtgacc gctacacttg ccagcgcct agcgcgcct cctttcgctt tcttcccttc	120
ctttctcgcc acgttcgcgc gctttccccg tcaagctcta aatcgggggc tcccttagg	180
gttcgcgattt agtgctttac ggcacctcga ccccaaaaaa cttgattagg gtgatggttc	240
acgtagtggg ccatcgccct gatagacggt ttttcgcct ttgacgttgg agtccacgtt	300
ctttaatagt ggactcttgt tccaaactgg aacaacactc aaccctatct cggctctattc	360
ttttgattta taagggattt tgccgatttc ggcctattgg ttaaaaaatg agctgattta	420
acaaaaattt aacgcgaatt ttaacaaaat attaacgttt acaatttcag gtggcacttt	480
tcggggaaat gtgcgcggaa ccctatttg tttatttttc taaatacatt caaatatgta	540
tccgctcatg aattaattct tagaaaaact catcgagcat caaatgaaac tgcaatttat	600
tcatatcagg attatcaata ccatattttt gaaaaagccg tttctgtaat gaaggagaaa	660
actcaccgag gcagttccat aggatggcaa gatcctggtg tcggtctgcg attccgactc	720
gtccaacatc aatacaacct attaatcttc cctcgtcaaa aataaggtta tcaagtgaga	780
aatcaccatg agtgacgact gaatccggtg agaatggcaa aagtttatgc atttctttcc	840
agacttggtc aacaggccag ccattacgct cgtcatcaaa atcactcgca tcaaccaaac	900
cgttattcat tcgtgattgc gctgagcga gacgaaatac gcgatcgctg ttaaaaggac	960
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<210> SEQ ID NO 96
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<210> SEQ ID NO 97
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<400> SEQUENCE: 97

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<210> SEQ ID NO 98
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<210> SEQ ID NO 99
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<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: pEAQ-HT Forward primer for Rr4HPAAS
<400> SEQUENCE: 99

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<220> FEATURE:
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<400> SEQUENCE: 100

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 <213> ORGANISM: Artificial Sequence
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<210> SEQ ID NO 102
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: p423TEF Reverse primer for Rr4HPAAS

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<210> SEQ ID NO 103
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 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: pHis8-4 Forward primer for RrAAS

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<210> SEQ ID NO 104
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 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
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<210> SEQ ID NO 107
 <211> LENGTH: 56

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<212> TYPE: DNA
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<210> SEQ ID NO 108
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 108

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<210> SEQ ID NO 109
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 109

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<210> SEQ ID NO 110
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Reverse primer for RrPAR2

<400> SEQUENCE: 110

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<210> SEQ ID NO 111
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT1

<400> SEQUENCE: 111

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<210> SEQ ID NO 112
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT1

<400> SEQUENCE: 112

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<210> SEQ ID NO 113
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<220> FEATURE:
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 115

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<210> SEQ ID NO 116
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Reverse primer for RrUGT2

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<210> SEQ ID NO 117
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Forward primer for RrUGT2

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: pEAQ-HT Reverse primer for RrUGT2

<400> SEQUENCE: 118

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<210> SEQ ID NO 119
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT3

<400> SEQUENCE: 119

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<210> SEQ ID NO 120
<211> LENGTH: 44

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT3

<400> SEQUENCE: 120

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<210> SEQ ID NO 121
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<212> TYPE: DNA
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<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Forward primer for RrUGT3

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<210> SEQ ID NO 122
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<210> SEQ ID NO 123
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Forward primer for RrUGT3

<400> SEQUENCE: 123

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<210> SEQ ID NO 124
<211> LENGTH: 58
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Reverse primer for RrUGT3

<400> SEQUENCE: 124

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<210> SEQ ID NO 125
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT4

<400> SEQUENCE: 125

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<210> SEQ ID NO 126
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 126

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<400> SEQUENCE: 127

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<400> SEQUENCE: 128

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<210> SEQ ID NO 129
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<400> SEQUENCE: 129

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<210> SEQ ID NO 130
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 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT6

<400> SEQUENCE: 130

cagcccgagg gatccactag ttcagtttga attcctcgac aggagcac 48

<210> SEQ ID NO 131
 <211> LENGTH: 55
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: p426TEF Forward primer for RrUGT7

<400> SEQUENCE: 131

caatctaadc taagttttct agaactagta tggctgaaaa cactcatgct catgc 55

<210> SEQ ID NO 132
 <211> LENGTH: 52
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT7

<400> SEQUENCE: 132

cagcccgagg gatccactag ttcatttctt gaagatttgt aggtcgtgga tg 52

<210> SEQ ID NO 133
 <211> LENGTH: 58
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT8

<400> SEQUENCE: 133

caatctaatac taagttttct agaactagta tggcttctc ctcttttagct tgtgattc 58

<210> SEQ ID NO 134
<211> LENGTH: 61
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT8

<400> SEQUENCE: 134

cagcccgagg gatccactag tttatttaac tgtttctgt tttgcagga cagaatgaat 60
g 61

<210> SEQ ID NO 135
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT9

<400> SEQUENCE: 135

caatctaatac taagttttct agaactagta tggggtctga gccactagtc c 51

<210> SEQ ID NO 136
<211> LENGTH: 51
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT9

<400> SEQUENCE: 136

cagcccgagg gatccactag tttatgctga aattgcatcc ttagcaactg g 51

<210> SEQ ID NO 137
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT10

<400> SEQUENCE: 137

caatctaatac taagttttct agaactagta tgacgaggcg ccaccac 47

<210> SEQ ID NO 138
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT10

<400> SEQUENCE: 138

cagcccgagg gatccactag ttcattcaag gccattgaca aaacgac 47

<210> SEQ ID NO 139
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT11

<400> SEQUENCE: 139

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caatctaatac taagtttttct agaactagta tggcaggcga gattctaata ctcccg 56

<210> SEQ ID NO 140
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT11

<400> SEQUENCE: 140

cagccccggg gatccactag ttcacttggt ggagataatg aagtccttg 49

<210> SEQ ID NO 141
<211> LENGTH: 46
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT12

<400> SEQUENCE: 141

caatctaatac taagtttttct agaactagta tggaggaggc ggccag 46

<210> SEQ ID NO 142
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT12

<400> SEQUENCE: 142

cagccccggg gatccactag ttaacacag agtccaaatg tccagcaac 49

<210> SEQ ID NO 143
<211> LENGTH: 58
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT13

<400> SEQUENCE: 143

caatctaatac taagtttttct agaactagta tgctacctct cttacatggt acactaac 58

<210> SEQ ID NO 144
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT13

<400> SEQUENCE: 144

cagccccggg gatccactag ttacaagcc aatgttggtc ctgagatcac 50

<210> SEQ ID NO 145
<211> LENGTH: 46
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT14

<400> SEQUENCE: 145

caatctaatac taagtttttct agaactagta tggacaccac cgccgc 46

<210> SEQ ID NO 146
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT14

<400> SEQUENCE: 146

cagcccgagg gatccactag tttatcccct tccaagttga gtcaacgac 49

<210> SEQ ID NO 147
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT15

<400> SEQUENCE: 147

caatctaadc taagttttct agaactagta tggctgatgc tgctcaacat gtc 53

<210> SEQ ID NO 148
<211> LENGTH: 58
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT15

<400> SEQUENCE: 148

cagcccgagg gatccactag tttattgaac tttgtgaaat tgaagatgac tcaaaagg 58

<210> SEQ ID NO 149
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT16

<400> SEQUENCE: 149

caatctaadc taagttttct agaactagta tggcagagga aaacagaacc agc 53

<210> SEQ ID NO 150
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT16

<400> SEQUENCE: 150

cagcccgagg gatccactag ttcatacagc tgaagatatt ttggatatga attggtc 57

<210> SEQ ID NO 151
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT17

<400> SEQUENCE: 151

caatctaadc taagttttct agaactagta tgggctcact tccttcac 49

<210> SEQ ID NO 152
<211> LENGTH: 49
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT17

<400> SEQUENCE: 152

cagcccgagg gatccactag ttcagacgct aaactggacc acctttttcc 49

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<210> SEQ ID NO 153
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT18

<400> SEQUENCE: 153

caatctaatac taagttttct agaactagta tgggctcccg aggaaagcca catg 54

<210> SEQ ID NO 154
<211> LENGTH: 46
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT18

<400> SEQUENCE: 154

cagcccgagg gatccactag ttcattttgg ggaattagac agcagg 46

<210> SEQ ID NO 155
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT19

<400> SEQUENCE: 155

caatctaatac taagttttct agaactagta tgacgtcatc aacacctctc cctc 54

<210> SEQ ID NO 156
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT19

<400> SEQUENCE: 156

cagcccgagg gatccactag tctaaaaaaa tgctttaaca tagctagcgt ccg 53

<210> SEQ ID NO 157
<211> LENGTH: 50
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT20

<400> SEQUENCE: 157

caatctaatac taagttttct agaactagta tgggttcact cgacgtcgtc 50

<210> SEQ ID NO 158
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT20

<400> SEQUENCE: 158

cagcccgagg gatccactag ttcatttcat aatagcttca tcaatcaact cg 53

<210> SEQ ID NO 159
<211> LENGTH: 56
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT21

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<400> SEQUENCE: 159

caatctaadc taagttttct agaactagta tgaagtccaa cactcatcta ttcctc 56

<210> SEQ ID NO 160

<211> LENGTH: 44

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT21

<400> SEQUENCE: 160

cagcccgagg gatccactag ttcatacaac cggctccagt tgac 44

<210> SEQ ID NO 161

<211> LENGTH: 57

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT22

<400> SEQUENCE: 161

caatctaadc taagttttct agaactagta tgaaaactcc tcaaaatcca cacgtag 57

<210> SEQ ID NO 162

<211> LENGTH: 55

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT22

<400> SEQUENCE: 162

cagcccgagg gatccactag ttcaatcctg ataaatcttt gaactcatct tgctc 55

<210> SEQ ID NO 163

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT23

<400> SEQUENCE: 163

caatctaadc taagttttct agaactagta tggaaaggca gagtgatcac caag 54

<210> SEQ ID NO 164

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT23

<400> SEQUENCE: 164

cagcccgagg gatccactag ttcatctggt ggatatacaca tctctaacaa actg 54

<210> SEQ ID NO 165

<211> LENGTH: 45

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT24

<400> SEQUENCE: 165

caatctaadc taagttttct agaactagta tgagcaacgc cgccg 45

<210> SEQ ID NO 166

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<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT24

<400> SEQUENCE: 166

cagcccgagg gatccactag tttagtttat gacttcattc acttgctcca acaac 55

<210> SEQ ID NO 167
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT25

<400> SEQUENCE: 167

caatctaato taagttttct agaactagta tggcgcgcca ccactttg 48

<210> SEQ ID NO 168
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT25

<400> SEQUENCE: 168

cagcccgagg gatccactag tttagcaggt aacaaggcta ttaaccaa ccttgag 57

<210> SEQ ID NO 169
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT26

<400> SEQUENCE: 169

caatctaato taagttttct agaactagta tgatcatcaga ttccggccac attatcc 57

<210> SEQ ID NO 170
<211> LENGTH: 58
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT26

<400> SEQUENCE: 170

cagcccgagg gatccactag tctatattat ttttctaat gccatgactt gtcggacc 58

<210> SEQ ID NO 171
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT27

<400> SEQUENCE: 171

caatctaato taagttttct agaactagta tgagttcagt caatgctcaa aagcc 55

<210> SEQ ID NO 172
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT27

<400> SEQUENCE: 172

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cagcccgagg gatccactag ttcaaaagtg cattagtagt ccttcacaa atc 53

<210> SEQ ID NO 173
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT28

<400> SEQUENCE: 173

caatctaate taagttttct agaactagta tggactcggg tgatctgaac aag 53

<210> SEQ ID NO 174
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT28

<400> SEQUENCE: 174

cagcccgagg gatccactag tctagttggc acttggcaac acaatcg 47

<210> SEQ ID NO 175
<211> LENGTH: 57
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT29

<400> SEQUENCE: 175

caatctaate taagttttct agaactagta tgggatctct aggaagaag attcaac 57

<210> SEQ ID NO 176
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT29

<400> SEQUENCE: 176

cagcccgagg gatccactag tttaggttgt aactacaatt ttttttttgg ac 52

<210> SEQ ID NO 177
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Forward primer for RrUGT29

<400> SEQUENCE: 177

gaaaacttgt acttcaggc ccatggcatg ggatctctag gaaagaagat tcaac 55

<210> SEQ ID NO 178
<211> LENGTH: 53
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Reverse primer for RrUGT29

<400> SEQUENCE: 178

ctcgaattcg gatccgcat ggtaggttg taactacaat ttttttttg gac 53

<210> SEQ ID NO 179
<211> LENGTH: 56
<212> TYPE: DNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Forward primer for RrUGT29

<400> SEQUENCE: 179

gtatattctg cccaaattcg cgaccggtat gggatctcta ggaaagaaga ttcaac 56

<210> SEQ ID NO 180
<211> LENGTH: 66
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Reverse primer for RrUGT29

<400> SEQUENCE: 180

gaaaatttaa tgaaaccaga gttaaaggcc tcgagttagg ttgtaactac aatttttttt 60
ttggac 66

<210> SEQ ID NO 181
<211> LENGTH: 54
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT30

<400> SEQUENCE: 181

caatctaadc taagttttct agaactagta tgggctcccg aggaagcca catg 54

<210> SEQ ID NO 182
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT30

<400> SEQUENCE: 182

cagcccgagg gatccactag ttcattttgg ggaattagac agcagg 46

<210> SEQ ID NO 183
<211> LENGTH: 59
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT31

<400> SEQUENCE: 183

caatctaadc taagttttct agaactagta tggaatctgt acaaggtgtt caagaaaag 59

<210> SEQ ID NO 184
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT31

<400> SEQUENCE: 184

cagcccgagg gatccactag ttcagtttga attcctcgac aggagcac 48

<210> SEQ ID NO 185
<211> LENGTH: 58
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT32

<400> SEQUENCE: 185

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caatctaadc taagttttct agaactagta tggactcggg tgatctgaac aagaaacc 58

<210> SEQ ID NO 186
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT32

<400> SEQUENCE: 186

cagcccgagg gatccactag tctacaattt tttttggac agaagtacgt catttataag 60

tc 62

<210> SEQ ID NO 187
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT33

<400> SEQUENCE: 187

caatctaadc taagttttct agaactagta tgagcttaat tgaaaaacca ctcacg 56

<210> SEQ ID NO 188
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<400> SEQUENCE: 188

cagcccgagg gatccactag tctaaccgat atgttttgtt tttgagagca ggac 54

<210> SEQ ID NO 189
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Forward primer for RrUGT33

<400> SEQUENCE: 189

gaaaacttgt acttcaggc ccatggcatg agcttaattg aaaaaccact cacg 54

<210> SEQ ID NO 190
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Reverse primer for RrUGT33

<400> SEQUENCE: 190

ctcgaattcg gatccgcat ggctaaccga tatgttttgt tttgagagc aggac 55

<210> SEQ ID NO 191
<211> LENGTH: 55
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Forward primer for RrUGT33

<400> SEQUENCE: 191

gtatatattcg cccaaattcg cgaccggtat gagcttaatt gaaaaaccac tcacg 55

<210> SEQ ID NO 192

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<211> LENGTH: 68
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Reverse primer for RrUGT33

<400> SEQUENCE: 192

gaaaatttaa tgaaccaga gttaaaggcc tcgagctaac ggatatgttt tgtttttgag 60
agcaggac 68

<210> SEQ ID NO 193
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Forward primer for RrUGT34

<400> SEQUENCE: 193

gcatagcaat ctaatctaag ttttctagaa ctagtggac cctgacgaca gcgttttg 58

<210> SEQ ID NO 194
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: p426TEF Reverse primer for RrUGT34

<400> SEQUENCE: 194

cagcccgagg gatccactag tttagttttt gttctcgtac aaataatgca caaactcatc 60

<210> SEQ ID NO 195
<211> LENGTH: 46
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Forward primer for Pc4HPAAS

<400> SEQUENCE: 195

gaaaacttgt acttocaggc ccattggcatg ggctccatcg ataatac 46

<210> SEQ ID NO 196
<211> LENGTH: 48
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pHis8-4 Reverse primer for Pc4HPAAS

<400> SEQUENCE: 196

ctcgaattcg gatccgccat ggtaggata aaatattcac gatcttct 48

<210> SEQ ID NO 197
<211> LENGTH: 47
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Forward primer for Pc4HPAAS

<400> SEQUENCE: 197

gtatattctg cccaaattcg cgaccggat gggtccatc gataatc 47

<210> SEQ ID NO 198
<211> LENGTH: 60
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: pEAQ-HT Reverse primer for Pc4HPAAS

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<400> SEQUENCE: 198

gaaaatttaa tgaaaccaga gttaaaggcc tcgagttagg ataaaatatt cacgatcttc 60

<210> SEQ ID NO 199

<211> LENGTH: 55

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: pHis8-4 Forward primer for PsTyDC

<400> SEQUENCE: 199

gaaaacttgt acttcacaggc ccatggcatg ggaagccttc cgactaataa ccttg 55

<210> SEQ ID NO 200

<211> LENGTH: 49

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: pHis8-4 Reverse primer for PsTyDC

<400> SEQUENCE: 200

ctcgaattcg gatccgccat ggctaggcac caagtatggc atctgtatg 49

<210> SEQ ID NO 201

<211> LENGTH: 56

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: pEAQ-HT Forward primer for PsTyDC

<400> SEQUENCE: 201

gtatatctcg cccaaattcg cgaccggtat ggaagcctt ccgactaata accttg 56

<210> SEQ ID NO 202

<211> LENGTH: 62

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: pEAQ-HT Reverse primer for PsTyDC

<400> SEQUENCE: 202

gaaaatttaa tgaaaccaga gttaaaggcc tcgagctagg caccaagtat ggcatctgta 60

tg 62

<210> SEQ ID NO 203

<211> LENGTH: 48

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Forward primer for AAS55083

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: GenBank AAS55083

<400> SEQUENCE: 203

caatctaadc taagttttct agaactagta tggcaggcag tgggactg 48

<210> SEQ ID NO 204

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Reverse primer for AAS55083

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: GenBank AAS55083

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<400> SEQUENCE: 204

cagcccgagg gatccactag ttcagtgttt aactgaggat ctccactttt tagc 54

<210> SEQ ID NO 205

<211> LENGTH: 59

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Forward primer for EU567325

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: GenBank EU567325

<400> SEQUENCE: 205

gcatagcaat ctaatctaag ttttctagaa ctagtatggg ttctgaaact cggcctttg 59

<210> SEQ ID NO 206

<211> LENGTH: 54

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: p426TEF Reverse primer for EU567325

<300> PUBLICATION INFORMATION:

<308> DATABASE ACCESSION NUMBER: GenBank EU567325

<400> SEQUENCE: 206

cagcccgagg gatccactag tctagacttt ctttaacttg agttcctgaa gcag 54

<210> SEQ ID NO 207

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: C. roseus

<400> SEQUENCE: 207

Val Asp Phe Lys Asn Trp Gln Ile Ala Thr Gly Arg Lys Phe Arg Ser
1 5 10 15

Leu Lys Leu Trp Leu
20

<210> SEQ ID NO 208

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: C. annum

<400> SEQUENCE: 208

Val Asp Tyr Lys Asp Trp Gln Ile Gly Thr Gly Arg Lys Phe Lys Ser
1 5 10 15

Leu Arg Leu Trp Leu
20

<210> SEQ ID NO 209

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: O. pumila

<400> SEQUENCE: 209

Val Asp Phe Lys Asp Trp Gln Ile Gly Thr Gly Arg Arg Phe Lys Ala
1 5 10 15

Leu Arg Leu Trp Leu
20

<210> SEQ ID NO 210

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: C. acuminata

-continued

<400> SEQUENCE: 210

Val Asp Tyr Lys Asp Trp Gln Val Gly Thr Gly Arg Arg Phe Lys Ala
1 5 10 15

Leu Arg Leu Trp Phe
20

<210> SEQ ID NO 211

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: *C. acuminata*

<400> SEQUENCE: 211

Val Asp Phe Lys Asp Trp Gln Val Gly Thr Gly Arg Arg Phe Lys Ala
1 5 10 15

Leu Arg Leu Trp Phe
20

<210> SEQ ID NO 212

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: *O. sativa*

<400> SEQUENCE: 212

Thr Asp Leu Lys Asp Met Gln Val Gly Val Gly Arg Arg Phe Arg Gly
1 5 10 15

Leu Lys Leu Trp Met
20

<210> SEQ ID NO 213

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: *P. hybrid*

<400> SEQUENCE: 213

Val Asp Tyr Lys Asp Trp Gln Ile Thr Leu Ser Arg Arg Phe Arg Ser
1 5 10 15

Leu Lys Leu Trp Leu
20

<210> SEQ ID NO 214

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: *P. crispum*

<400> SEQUENCE: 214

Val Asp Tyr Lys Asp Trp Gln Ile Met Leu Ser Arg Arg Phe Arg Ala
1 5 10 15

Leu Lys Leu Trp Phe
20

<210> SEQ ID NO 215

<211> LENGTH: 21

<212> TYPE: PRT

<213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 215

Ile Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ser
1 5 10 15

Met Lys Leu Trp Leu
20

-continued

<210> SEQ ID NO 216
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 216

Val Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ser
1 5 10 15
Leu Lys Leu Trp Met
20

<210> SEQ ID NO 217
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: *T. flavum*

<400> SEQUENCE: 217

Val Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala
1 5 10 15
Met Lys Leu Trp Leu
20

<210> SEQ ID NO 218
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: *R. hybrid*

<400> SEQUENCE: 218

Val Asp Tyr Lys Asp Trp Gln Ile Ala Leu Ser Arg Arg Phe Arg Ala
1 5 10 15
Leu Lys Leu Trp Leu
20

<210> SEQ ID NO 219
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: *R. crenulata*

<400> SEQUENCE: 219

Val Asp Tyr Lys Asp Trp Gln Ile Ser Leu Ser Arg Arg Phe Arg Ala
1 5 10 15
Ile Lys Met Trp Met
20

<210> SEQ ID NO 220
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 220

Leu Val Lys Ala Leu Ser Thr Ser Ala Glu Tyr Leu Lys Asn Lys Ala
1 5 10 15
Thr Glu Ser Lys
20

<210> SEQ ID NO 221
<211> LENGTH: 21
<212> TYPE: PRT
<213> ORGANISM: *B. distachyon*

<400> SEQUENCE: 221

Leu Ile Ala Ala Leu Gly Thr Glu Gln Glu Tyr Ile Leu Lys Asp Ser

-continued

1	5	10	15
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Ala Ser Glu Gly His
20

<210> SEQ ID NO 222
 <211> LENGTH: 21
 <212> TYPE: PRT
 <213> ORGANISM: *O. sativa*

<400> SEQUENCE: 222

Leu Val Ala Ala Leu Gly Thr Glu Gln Glu Tyr Ile Leu Arg Asp Ala
1 5 10 15

Ala Ala Glu Gly His
20

<210> SEQ ID NO 223
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: *T. flavum*

<400> SEQUENCE: 223

Leu Ile Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys Ala
1 5 10 15

Thr Glu Ser His
20

<210> SEQ ID NO 224
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 224

Leu Val Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys Ala
1 5 10 15

Thr Glu Ser Arg
20

<210> SEQ ID NO 225
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: *C. acuminata*

<400> SEQUENCE: 225

Leu Val Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Gln Pro
1 5 10 15

Ser Glu Ser Lys
20

<210> SEQ ID NO 226
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: *C. acuminata*

<400> SEQUENCE: 226

Leu Val Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Lys Pro
1 5 10 15

Ser Glu Ser Asn
20

<210> SEQ ID NO 227
 <211> LENGTH: 20
 <212> TYPE: PRT

-continued

<213> ORGANISM: *O. pumila*

<400> SEQUENCE: 227

Met Val Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys Arg
1 5 10 15

Ser Glu Phe Asp
20

<210> SEQ ID NO 228

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: *O. sativa*

<400> SEQUENCE: 228

Leu Thr Gly Ser Leu Glu Thr Asn Pro Glu Tyr Leu Lys Asn His Ala
1 5 10 15

Ser Asp Ser Gly
20

<210> SEQ ID NO 229

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: *C. annuum*

<400> SEQUENCE: 229

Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Tyr Leu Lys Asn Lys Ala
1 5 10 15

Ser Gln Gly Asn
20

<210> SEQ ID NO 230

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: *A. thaliana*

<400> SEQUENCE: 230

Leu Ile Asp Ala Leu Lys Thr Asn Pro Glu Tyr Leu Glu Phe Lys Val
1 5 10 15

Ser Lys Lys Asp
20

<210> SEQ ID NO 231

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: *C. annuum*

<400> SEQUENCE: 231

Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Tyr Leu Lys Asn Lys Ala
1 5 10 15

Ser Gln Gly Asn
20

<210> SEQ ID NO 232

<211> LENGTH: 20

<212> TYPE: PRT

<213> ORGANISM: *C. roseus*

<400> SEQUENCE: 232

Leu Leu Arg Ala Leu Thr Thr Asn Pro Glu Tyr Leu Lys Asn Lys Gln
1 5 10 15

Ser Asp Leu Asp
20

-continued

<210> SEQ ID NO 233
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *R. rosea*

<400> SEQUENCE: 233

Leu Thr Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Gln Gln
1 5 10 15

Ser Glu Leu Asn
20

<210> SEQ ID NO 234
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *R. hybrid*

<400> SEQUENCE: 234

Leu Ala Ser Ser Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys Ala
1 5 10 15

Ser Asp Ser Lys
20

<210> SEQ ID NO 235
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *P. crispum*

<400> SEQUENCE: 235

Leu Ile Lys Ser Leu Ser Thr Tyr Pro Glu Phe Leu Lys Asn Asn Ala
1 5 10 15

Ser Glu Thr Asn
20

<210> SEQ ID NO 236
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *A. thaliana*

<400> SEQUENCE: 236

Leu Thr Leu Ala Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala
1 5 10 15

Ser Gln Ala Asn
20

<210> SEQ ID NO 237
<211> LENGTH: 20
<212> TYPE: PRT
<213> ORGANISM: *O. europaea*

<400> SEQUENCE: 237

Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala
1 5 10 15

Ser Glu Gly Asn
20

<210> SEQ ID NO 238
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: *R. crenulata*

<400> SEQUENCE: 238

-continued

Leu Ile Glu Ser Leu Ala Ala Glu Ala Asn Phe Leu Lys Gly Asn Ser
 1 5 10 15

<210> SEQ ID NO 239
 <211> LENGTH: 20
 <212> TYPE: PRT
 <213> ORGANISM: R. rosea

<400> SEQUENCE: 239

Leu Ile Gln Ser Leu Ser Thr Tyr Pro Glu Phe Leu Lys Asn Lys Ala
 1 5 10 15

Ser Gln Ser Asn
 20

<210> SEQ ID NO 240
 <211> LENGTH: 16
 <212> TYPE: PRT
 <213> ORGANISM: R. rosea

<400> SEQUENCE: 240

Leu Ile Glu Ser Leu Ala Ala Glu Ala Asn Phe Leu Lys Gly Gly Ser
 1 5 10 15

<210> SEQ ID NO 241
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: R. rosea

<400> SEQUENCE: 241

Val Leu Tyr Cys Ser Asp Gln Thr His Phe Thr Ile His Lys Gly Ala
 1 5 10 15

Lys

<210> SEQ ID NO 242
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: A. thaliana

<400> SEQUENCE: 242

Val Val Tyr Ser Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys
 1 5 10 15

Gln

<210> SEQ ID NO 243
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: A. thaliana

<400> SEQUENCE: 243

Val Val Tyr Gly Ser Asp Gln Thr His Ser Ser Phe Arg Lys Ala Cys
 1 5 10 15

Leu

<210> SEQ ID NO 244
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: C. annuum

<400> SEQUENCE: 244

Val Val Tyr Cys Ser Asp Gln Thr His Ser Ser Leu Gln Lys Ala Cys
 1 5 10 15

Gln

-continued

<210> SEQ ID NO 245
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: O. sativa

<400> SEQUENCE: 245

Val Val Tyr Ala Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Cys
1 5 10 15

Gln

<210> SEQ ID NO 246
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: O. sativa

<400> SEQUENCE: 246

Ala Val Tyr Ala Ala Asp Gln Thr His Ser Thr Phe Phe Lys Ala Cys
1 5 10 15

Arg

<210> SEQ ID NO 247
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: C. acuminata

<400> SEQUENCE: 247

Val Val Tyr Gly Ser Asp Gln Thr His Ser Thr Tyr Ala Lys Ala Cys
1 5 10 15

Lys

<210> SEQ ID NO 248
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: C. acuminata

<400> SEQUENCE: 248

Val Val Tyr Gly Ser Asp Gln Thr His Ser Thr Tyr Ala Lys Ala Cys
1 5 10 15

Asn

<210> SEQ ID NO 249
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: C. annum

<400> SEQUENCE: 249

Val Val Tyr Gly Ser Asp Gln Thr His Ser Met Tyr Ala Lys Ala Cys
1 5 10 15

Lys

<210> SEQ ID NO 250
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: O. pumila

<400> SEQUENCE: 250

Val Val Tyr Gly Ser Asp Gln Thr His Ser Phe Phe Gln Lys Thr Cys
1 5 10 15

Lys

-continued

<210> SEQ ID NO 251
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *C. roseus*

<400> SEQUENCE: 251

Val Cys Tyr Gly Ser Asp Gln Thr His Thr Met Phe Pro Lys Thr Cys
1 5 10 15

Lys

<210> SEQ ID NO 252
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *P. crispum*

<400> SEQUENCE: 252

Val Val Tyr Cys Ser Asp Gln Thr His Ser Ala Leu Gln Lys Ala Ala
1 5 10 15

Lys

<210> SEQ ID NO 253
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *R. hybrid*

<400> SEQUENCE: 253

Val Val Tyr Gly Ser Asp Gln Thr His Ser Thr Leu Gln Lys Ala Thr
1 5 10 15

Gln

<210> SEQ ID NO 254
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *P. hybrida*

<400> SEQUENCE: 254

Val Val Tyr Ala Ser Asp Gln Thr His Phe Ser Phe Gln Lys Ala Val
1 5 10 15

Lys

<210> SEQ ID NO 255
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *T. flavum*

<400> SEQUENCE: 255

Val Val Tyr Gly Ser Asp Gln Thr His Cys Ala Leu Gln Lys Ala Ala
1 5 10 15

Gln

<210> SEQ ID NO 256
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: *B. distachyon*

<400> SEQUENCE: 256

Val Val Tyr Cys Ser Asp Gln Thr His Phe Ala Phe Arg Lys Ala Ala
1 5 10 15

Arg

-continued

<210> SEQ ID NO 257

<211> LENGTH: 17

<212> TYPE: PRT

<213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 257

Val	Val	Tyr	Ala	Ser	Asn	Gln	Thr	His	Cys	Ala	Leu	Gln	Lys	Ala	Ala
1				5					10					15	

Gln

<210> SEQ ID NO 258

<211> LENGTH: 490

<212> TYPE: PRT

<213> ORGANISM: *R. rosea*

<400> SEQUENCE: 258

Met	Gly	Ser	Leu	Pro	Ser	Pro	Asn	Asp	Pro	Ser	Asn	Thr	Phe	Asn	Pro
1				5					10					15	

Met	Asp	Leu	Thr	Glu	Leu	Ser	Thr	Glu	Ser	Lys	Leu	Val	Val	Asp	Phe
			20					25					30		

Ile	Thr	Gln	Tyr	Tyr	Gln	Thr	Leu	Glu	Thr	Arg	Pro	Val	Gln	Pro	Arg
		35				40						45			

Val	Lys	Pro	Gly	Phe	Leu	Thr	Gly	Gln	Leu	Pro	Asp	Lys	Ala	Pro	Phe
	50					55					60				

His	Gly	Glu	Ser	Met	Glu	Val	Ile	Leu	Ser	Asp	Val	Asn	Glu	Lys	Ile
65					70					75				80	

Val	Pro	Gly	Leu	Thr	His	Trp	Gln	Ser	Pro	Asn	Phe	His	Ala	Tyr	Phe
			85						90					95	

Pro	Ala	Ser	Ser	Ser	Asn	Ala	Gly	Leu	Leu	Gly	Glu	Leu	Leu	Cys	Ser
			100					105					110		

Gly	Leu	Ser	Val	Ile	Gly	Phe	Thr	Trp	Ser	Ser	Ser	Pro	Ala	Ala	Thr
	115						120					125			

Glu	Leu	Glu	Asn	Val	Val	Val	Asp	Trp	Met	Ala	Lys	Met	Leu	Asn	Leu
	130					135					140				

Pro	Ser	Ser	Phe	Cys	Phe	Ser	Gly	Gly	Gly	Gly	Gly	Val	Leu	Gln	Ala
145				150					155					160	

Asn	Thr	Cys	Glu	Ala	Val	Leu	Cys	Thr	Leu	Ala	Ala	Ala	Arg	Asp	Lys
			165					170						175	

Ala	Leu	Asn	Arg	Val	Gly	Asp	Asp	Gln	Ile	Asn	Lys	Leu	Val	Leu	Tyr
		180						185					190		

Cys	Ser	Asp	Gln	Thr	His	Phe	Thr	Ile	His	Lys	Gly	Ala	Lys	Leu	Ile
	195						200					205			

Gly	Ile	Arg	Ser	Lys	Asn	Ile	Lys	Ser	Ile	Thr	Thr	Lys	Lys	Glu	Asn
	210				215					220					

Glu	Phe	Lys	Leu	Cys	Pro	Asn	Asp	Leu	Arg	Asp	Ala	Ile	Arg	Ser	Asp
225				230					235					240	

Leu	Glu	Ala	Gly	Leu	Val	Pro	Phe	Tyr	Val	Cys	Gly	Thr	Ile	Gly	Thr
			245						250				255		

Thr	Ala	Leu	Gly	Val	Val	Asp	Pro	Ile	Lys	Glu	Leu	Gly	Lys	Val	Ala
		260						265					270		

Arg	Glu	Phe	Asp	Leu	Trp	Leu	His	Val	Asp	Gly	Ala	Tyr	Gly	Gly	Ser
		275					280					285			

Ala	Cys	Ile	Cys	Pro	Glu	Phe	Gln	His	Tyr	Leu	Asp	Gly	Val	Asp	Leu
	290					295					300				

Val	Asp	Ser	Ile	Ser	Met	Asn	Ala	His	Lys	Trp	Leu	Leu	Ser	Asn	Leu
305					310					315				320	

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Asp Cys Cys Phe Leu Trp Leu Gln Ser Pro Asn Ala Leu Ile Glu Ser
 325 330 335
 Leu Ala Ala Glu Ala Asn Phe Leu Lys Gly Gly Ser Glu Met Val Asp
 340 345 350
 Tyr Lys Asp Trp Gln Ile Ser Leu Ser Arg Arg Phe Arg Ala Ile Lys
 355 360 365
 Met Trp Met Val Ile Arg Arg Tyr Gly Val Ser Asn Leu Ile Glu His
 370 375 380
 Ile Arg Ser Asp Val Ser Met Ala Val Arg Phe Glu Glu Met Val Ala
 385 390 395 400
 Ala Asp Asp Arg Phe Glu Ile Val Phe Pro Arg Lys Phe Ala Leu Val
 405 410 415
 Cys Phe Lys Leu Ser Ser Glu Lys Thr Pro Pro Gly Arg Asp Ser Glu
 420 425 430
 Leu Thr Arg Glu Leu Met Glu Arg Val Asn Ser Ser Gly Lys Ala Tyr
 435 440 445
 Leu Ser Gly Val Gln Met Gly Arg Ile Phe Phe Ile Arg Cys Val Ile
 450 455 460
 Gly Ser Ser Leu Thr Glu Glu Arg His Val Asp Asn Leu Trp Arg Leu
 465 470 475 480
 Ile Gln Glu Thr Ala Gln Ser Ile Val Ser
 485 490

<210> SEQ ID NO 259

<211> LENGTH: 490

<212> TYPE: PRT

<213> ORGANISM: A. thaliana

<400> SEQUENCE: 259

Met Glu Asn Gly Ser Gly Lys Val Leu Lys Pro Met Asp Ser Glu Gln
 1 5 10 15
 Leu Arg Glu Tyr Gly His Leu Met Val Asp Phe Ile Ala Asp Tyr Tyr
 20 25 30
 Lys Thr Ile Glu Asp Phe Pro Val Leu Ser Gln Val Gln Pro Gly Tyr
 35 40 45
 Leu His Lys Leu Leu Pro Asp Ser Ala Pro Asp His Pro Glu Thr Leu
 50 55 60
 Asp Gln Val Leu Asp Asp Val Arg Ala Lys Ile Leu Pro Gly Val Thr
 65 70 75 80
 His Trp Gln Ser Pro Ser Phe Phe Ala Tyr Tyr Pro Ser Asn Ser Ser
 85 90 95
 Val Ala Gly Phe Leu Gly Glu Met Leu Ser Ala Gly Leu Gly Ile Val
 100 105 110
 Gly Phe Ser Trp Val Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile
 115 120 125
 Val Leu Asp Trp Val Ala Lys Leu Leu Asn Leu Pro Glu Gln Phe Met
 130 135 140
 Ser Lys Gly Asn Gly Gly Gly Val Ile Gln Gly Ser Ala Ser Glu Ala
 145 150 155 160
 Val Leu Val Val Leu Ile Ala Ala Arg Asp Lys Val Leu Arg Ser Val
 165 170 175
 Gly Lys Asn Ala Leu Glu Lys Leu Val Tyr Ser Ser Asp Gln Thr
 180 185 190
 His Ser Ala Leu Gln Lys Ala Cys Gln Ile Ala Gly Ile His Pro Glu

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195					200					205					
Asn	Cys	Arg	Val	Leu	Thr	Thr	Asp	Ser	Ser	Thr	Asn	Tyr	Ala	Leu	Arg
210						215					220				
Pro	Glu	Ser	Leu	Gln	Glu	Ala	Val	Ser	Arg	Asp	Leu	Glu	Ala	Gly	Leu
225					230					235					240
Ile	Pro	Phe	Phe	Leu	Cys	Ala	Asn	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala
				245					250					255	
Val	Asp	Pro	Leu	Ala	Ala	Leu	Gly	Lys	Ile	Ala	Asn	Ser	Asn	Gly	Ile
			260					265					270		
Trp	Phe	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro
			275				280					285			
Glu	Tyr	Arg	Gln	Tyr	Ile	Asp	Gly	Val	Glu	Thr	Ala	Asp	Ser	Phe	Asn
	290					295					300				
Met	Asn	Ala	His	Lys	Trp	Phe	Leu	Thr	Asn	Phe	Asp	Cys	Ser	Leu	Leu
305					310					315					320
Trp	Val	Lys	Asp	Gln	Asp	Ser	Leu	Thr	Leu	Ala	Leu	Ser	Thr	Asn	Pro
			325						330					335	
Glu	Phe	Leu	Lys	Asn	Lys	Ala	Ser	Gln	Ala	Asn	Leu	Val	Val	Asp	Tyr
			340					345					350		
Lys	Asp	Trp	Gln	Ile	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu
	355						360					365			
Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Ser	Glu	Thr	Leu	Lys	Ser	Tyr	Ile
	370					375					380				
Arg	Asn	His	Ile	Lys	Leu	Ala	Lys	Glu	Phe	Glu	Gln	Leu	Val	Ser	Gln
385					390					395					400
Asp	Pro	Asn	Phe	Glu	Ile	Val	Thr	Pro	Arg	Ile	Phe	Ala	Leu	Val	Cys
			405						410					415	
Phe	Arg	Leu	Val	Pro	Val	Lys	Asp	Glu	Glu	Lys	Lys	Cys	Asn	Asn	Arg
			420					425					430		
Asn	Arg	Glu	Leu	Leu	Asp	Ala	Val	Asn	Ser	Ser	Gly	Lys	Leu	Phe	Met
	435					440						445			
Ser	His	Thr	Ala	Leu	Ser	Gly	Lys	Ile	Val	Leu	Arg	Cys	Ala	Ile	Gly
	450					455					460				
Ala	Pro	Leu	Thr	Glu	Glu	Lys	His	Val	Lys	Glu	Ala	Trp	Lys	Ile	Ile
465					470					475					480
Gln	Glu	Glu	Ala	Ser	Tyr	Leu	Leu	His	Lys						
			485					490							

<210> SEQ ID NO 260

<211> LENGTH: 547

<212> TYPE: PRT

<213> ORGANISM: A. thaliana

<400> SEQUENCE: 260

Met	Phe	Lys	Pro	Gln	His	Met	Tyr	Asp	Arg	Glu	Phe	Gly	Thr	Gly	Asn
1				5					10					15	
Gly	Tyr	Ser	Asn	Gly	Asn	Gly	Tyr	Thr	Asn	Gly	Asn	Gly	His	Thr	Asn
		20						25					30		
Gly	Asn	Gly	Asn	Tyr	Asn	Gly	Asn	Gly	His	Val	Asn	Gly	Asn	Gly	Lys
		35					40					45			
Ala	Asn	Gly	Ala	Lys	Val	Val	Lys	Met	Lys	Pro	Met	Asp	Ser	Glu	Leu
		50					55				60				
Leu	Arg	Glu	Gln	Gly	His	Ile	Met	Val	Asp	Phe	Ile	Ala	Asp	Tyr	Tyr
65					70					75					80

Lys	Asn	Leu	Gln	Asp	Ser	Pro	Gln	Asp	Phe	Pro	Val	Leu	Ser	Gln	Val	
				85					90					95		
Gln	Pro	Gly	Tyr	Leu	Arg	Asp	Met	Leu	Pro	Asp	Ser	Ala	Pro	Glu	Arg	
			100				105				110					
Pro	Glu	Ser	Leu	Lys	Glu	Leu	Leu	Asp	Asp	Val	Ser	Lys	Lys	Ile	Met	
			115				120				125					
Pro	Gly	Ile	Thr	His	Trp	Gln	Ser	Pro	Ser	Tyr	Phe	Ala	Tyr	Tyr	Ala	
			130				135				140					
Ser	Ser	Thr	Ser	Val	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu	Asn	Ala	Gly	
			145				150				155				160	
Leu	Ser	Val	Val	Gly	Phe	Thr	Trp	Leu	Thr	Ser	Pro	Ala	Ala	Thr	Glu	
			165				170				175					
Leu	Glu	Ile	Ile	Val	Leu	Asp	Trp	Leu	Ala	Lys	Leu	Leu	Gln	Leu	Pro	
			180				185				190					
Asp	His	Phe	Leu	Ser	Thr	Gly	Asn	Gly	Gly	Gly	Val	Ile	Gln	Gly	Thr	
			195				200				205					
Gly	Cys	Glu	Ala	Val	Leu	Val	Val	Val	Leu	Ala	Ala	Arg	Asp	Arg	Ile	
			210				215				220					
Leu	Lys	Lys	Val	Gly	Lys	Thr	Leu	Leu	Pro	Gln	Leu	Val	Val	Tyr	Gly	
			225				230				235				240	
Ser	Asp	Gln	Thr	His	Ser	Ser	Phe	Arg	Lys	Ala	Cys	Leu	Ile	Gly	Gly	
			245				250				255					
Ile	His	Glu	Glu	Asn	Ile	Arg	Leu	Leu	Lys	Thr	Asp	Ser	Ser	Thr	Asn	
			260				265				270					
Tyr	Gly	Met	Pro	Pro	Glu	Ser	Leu	Glu	Glu	Ala	Ile	Ser	His	Asp	Leu	
			275				280				285					
Ala	Lys	Gly	Phe	Ile	Pro	Phe	Phe	Ile	Cys	Ala	Thr	Val	Gly	Thr	Thr	
			290				295				300					
Ser	Ser	Ala	Ala	Val	Asp	Pro	Leu	Val	Pro	Leu	Gly	Asn	Ile	Ala	Lys	
			305				310				315				320	
Lys	Tyr	Gly	Ile	Trp	Leu	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Asn	Ala	
			325				330				335					
Cys	Ile	Cys	Pro	Glu	Tyr	Arg	Lys	Phe	Ile	Asp	Gly	Ile	Glu	Asn	Ala	
			340				345				350					
Asp	Ser	Phe	Asn	Met	Asn	Ala	His	Lys	Trp	Leu	Phe	Ala	Asn	Gln	Thr	
			355				360				365					
Cys	Ser	Pro	Leu	Trp	Val	Lys	Asp	Arg	Tyr	Ser	Leu	Ile	Asp	Ala	Leu	
			370				375				380					
Lys	Thr	Asn	Pro	Glu	Tyr	Leu	Glu	Phe	Lys	Val	Lys	Val	Ser	Lys	Lys	
			385				390				395				400	
Asp	Thr	Val	Val	Asn	Tyr	Lys	Asp	Trp	Gln	Ile	Ser	Leu	Ser	Arg	Arg	
			405				410				415					
Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu	Arg	Leu	Tyr	Gly	Ser	Glu	
			420				425				430					
Asn	Leu	Arg	Asn	Phe	Ile	Arg	Asp	His	Val	Asn	Leu	Ala	Lys	His	Phe	
			435				440				445					
Glu	Asp	Tyr	Val	Ala	Gln	Asp	Pro	Ser	Phe	Glu	Val	Val	Thr	Thr	Arg	
			450				455				460					
Tyr	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Ala	Pro	Val	Asp	Gly	Asp	Glu	
			465				470				475				480	
Asp	Gln	Cys	Asn	Glu												

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500	505	510
Leu Arg Phe Ala Val Gly Ala	Pro Leu Thr Glu Glu Lys His Val Thr	
515	520	525
Glu Ala Trp Gln Ile Ile Gln Lys His Ala Ser Lys Phe Thr Arg Asn		
530	535	540
Asp His Tyr		
545		
<210> SEQ ID NO 261		
<211> LENGTH: 487		
<212> TYPE: PRT		
<213> ORGANISM: C. annum		
<400> SEQUENCE: 261		
Met Glu Gly Glu Leu Lys Pro Met Asp Ala Glu Gln Leu Arg Glu Tyr		
1	5	10
Gly His Lys Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu		
20	25	30
Thr Leu Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Lys Leu		
35	40	45
Leu Pro Glu Thr Ala Pro Ala His Ser Glu Thr Leu Gln Asn Val Leu		
50	55	60
Glu Asp Val Gln Thr Lys Ile Leu Pro Gly Val Thr His Trp Gln Ser		
65	70	75
Pro Asp Tyr Phe Ala Tyr Phe Pro Ser Asn Ser Ser Val Ala Gly Phe		
85	90	95
Leu Gly Glu Met Leu Ser Ala Gly Ile Asn Met Val Gly Phe Ser Trp		
100	105	110
Ile Thr Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Leu Asp Trp		
115	120	125
Leu Ala Lys Ala Leu Lys Leu Pro Asp Glu Phe Leu Ser Thr Gly Gln		
130	135	140
Gly Gly Gly Val Ile Gln Gly Thr Ala Ser Glu Ala Val Leu Val Val		
145	150	155
Leu Leu Ala Ala Arg Asp Lys Val Leu Arg Arg Val Gly Lys Asp Ala		
165	170	175
Ile Ser Lys Leu Val Val Tyr Cys Ser Asp Gln Thr His Ser Ser Leu		
180	185	190
Gln Lys Ala Cys Gln Ile Gly Gly Ile His Pro Glu Asn Phe Arg Val		
195	200	205
Leu Lys Thr Asp Pro Ser Arg Asp Tyr Ala Leu Ser Pro Asp Thr Leu		
210	215	220
Ser Glu Ala Val Ser His Asp Met Ala Thr Asp Leu Ile Pro Phe Phe		
225	230	235
Phe Cys Ala Thr Ile Gly Thr Thr Ser Ser Thr Ala Val Asp Pro Leu		
245	250	255
Leu Asp Leu Gly Lys Ile Ala Gln Ser Asn Ser Ile Trp Phe His Val		
260	265	270
Asp Ala Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Tyr Arg Gly		
275	280	285
Tyr Ile Asn Gly Val Glu Glu Ala His Ser Phe Asn Met Asn Ala His		
290	295	300
Lys Trp Phe Leu Thr Asn Phe Asp Cys Ser Ala Leu Trp Val Lys Asp		
305	310	315
		320

Arg	Ser	Ala	Leu	Ile	Gln	Ser	Leu	Ser	Thr	Asn	Pro	Glu	Tyr	Leu	Lys		
				325					330							335	
Asn	Lys	Ala	Ser	Gln	Gly	Asn	Leu	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln		
				340					345							350	
Val	Pro	Leu	Gly	Arg	Arg	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Met	Val	Leu		
				355					360							365	
Arg	Leu	Tyr	Gly	Leu	Glu	Lys	Leu	Gln	Ala	Tyr	Ile	Arg	Asn	His	Ile		
				370					375							380	
Gln	Leu	Ala	Lys	Leu	Phe	Glu	Lys	Leu	Val	Ala	Gln	Asp	Gln	Arg	Phe		
				385					390							400	
Glu	Ile	Val	Thr	Pro	Arg	Lys	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Leu		
				405					410							415	
Pro	Pro	Pro	Ser	Asn	Glu	Asp	Tyr	Ala	Asn	Lys	Leu	Asn	His	Asn	Leu		
				420					425							430	
Leu	Asp	Ser	Val	Asn	Ser	Thr	Gly	Lys	Leu	Phe	Ile	Ser	His	Thr	Leu		
				435					440							445	
Leu	Ser	Asp	Lys	Tyr	Ile	Leu	Arg	Phe	Ala	Val	Gly	Ala	Pro	Leu	Thr		
				450					455							460	
Glu	Glu	Arg	His	Ile	Val	Gly	Ala	Trp	Lys	Val	Leu	Gln	Asp	Glu	Ala		
				465					470							475	480
Ala	Thr	Leu	Leu	Ser	Lys	Cys											
				485													

<400> SEQUENCE: 262

Met 1	Glu	Gly	Val 5	Gly	Gly	Gly	Gly	Gly	Gly 10	Glu	Glu	Trp	Leu	Arg 15	Pro
Met	Asp	Ala	Glu 20	Gln	Leu	Arg	Glu	Cys 25	Gly	His	Arg	Met	Val 30	Asp	Phe
Val	Ala	Asp 35	Tyr	Tyr	Lys	Ser	Ile 40	Glu	Ala	Phe	Pro	Val 45	Leu	Ser	Gln
Val	Gln 50	Pro	Gly	Tyr	Leu	Lys 55	Glu	Val	Leu	Pro	Asp 60	Ser	Ala	Pro	Arg
Gln 65	Pro	Asp	Thr	Leu	Asp 70	Ser	Leu	Phe	Asp	Asp 75	Ile	Gln	Gln	Lys	Ile 80
Ile	Pro	Gly	Val	Thr 85	His	Trp	Gln	Ser	Pro 90	Asn	Tyr	Phe	Ala	Tyr 95	Tyr
Pro	Ser	Asn 100	Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu	Met	Leu 110	Ser	Ala	
Ala	Phe 115	Asn	Ile	Val	Gly	Phe	Ser 120	Trp	Ile	Thr	Ser	Pro 125	Ala	Ala	Thr
Glu 130	Leu	Glu	Val	Ile	Val	Leu 135	Asp	Trp	Phe	Ala	Lys 140	Met	Leu	Gln	Leu
Pro 145	Ser	Gln	Phe	Leu	Ser 150	Thr	Ala	Leu	Gly	Gly 155	Gly	Val	Ile	Gln	Gly 160
Thr	Ala	Ser	Glu 165	Ala	Val	Leu	Val	Ala	Leu 170	Leu	Ala	Ala	Arg 175	Asp	Arg
Ala	Leu	Lys 180	Lys	His	Gly	Lys	His 185	Ser	Leu	Glu	Lys	Leu 190	Val	Val	Tyr
Ala	Ser	Asp 195	Gln	Thr	His	Ser	Ala 200	Leu	Gln	Lys	Ala	Cys 205	Gln	Ile	Ala

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Gly Ile Phe Ser Glu Asn Val Arg Val Val Ile Ala Asp Cys Asn Lys
 210 215 220
 Asn Tyr Ala Val Ala Pro Glu Ala Val Ser Glu Ala Leu Ser Ile Asp
 225 230 235 240
 Leu Ser Ser Gly Leu Ile Pro Phe Phe Ile Cys Ala Thr Val Gly Thr
 245 250 255
 Thr Ser Ser Ser Ala Val Asp Pro Leu Pro Glu Leu Gly Gln Ile Ala
 260 265 270
 Lys Ser Asn Asp Met Trp Phe His Ile Asp Ala Ala Tyr Ala Gly Ser
 275 280 285
 Ala Cys Ile Cys Pro Glu Tyr Arg His His Leu Asn Gly Val Glu Glu
 290 295 300
 Ala Asp Ser Phe Asn Met Asn Ala His Lys Trp Phe Leu Thr Asn Phe
 305 310 315 320
 Asp Cys Ser Leu Leu Trp Val Lys Asp Arg Ser Phe Leu Ile Gln Ser
 325 330 335
 Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys Ala Ser Gln Ala Asn
 340 345 350
 Ser Val Val Asp Phe Lys Asp Trp Gln Ile Pro Leu Gly Arg Arg Phe
 355 360 365
 Arg Ser Leu Lys Leu Trp Met Val Leu Arg Leu Tyr Gly Val Asp Asn
 370 375 380
 Leu Gln Ser Tyr Ile Arg Lys His Ile His Leu Ala Glu His Phe Glu
 385 390 395 400
 Gln Leu Leu Leu Ser Asp Ser Arg Phe Glu Val Val Thr Pro Arg Thr
 405 410 415
 Phe Ser Leu Val Cys Phe Arg Leu Val Pro Pro Thr Ser Asp His Glu
 420 425 430
 Asn Gly Arg Lys Leu Asn Tyr Asp Met Met Asp Gly Val Asn Ser Ser
 435 440 445
 Gly Lys Ile Phe Leu Ser His Thr Val Leu Ser Gly Lys Phe Val Leu
 450 455 460
 Arg Phe Ala Val Gly Ala Pro Leu Thr Glu Glu Arg His Val Asp Ala
 465 470 475 480
 Ala Trp Lys Leu Leu Arg Asp Glu Ala Thr Lys Val Leu Gly Lys Met
 485 490 495
 Val

<210> SEQ ID NO 263

<211> LENGTH: 514

<212> TYPE: PRT

<213> ORGANISM: O. sativa

<400> SEQUENCE: 263

Met Gly Ser Leu Asp Thr Asn Pro Thr Ala Phe Ser Ala Phe Pro Ala
 1 5 10 15
 Gly Glu Gly Glu Thr Phe Gln Pro Leu Asn Ala Asp Asp Val Arg Ser
 20 25 30
 Tyr Leu His Lys Ala Val Asp Phe Ile Ser Asp Tyr Tyr Lys Ser Val
 35 40 45
 Glu Ser Met Pro Val Leu Pro Asn Val Lys Pro Gly Tyr Leu Gln Asp
 50 55 60
 Glu Leu Arg Ala Ser Pro Pro Thr Tyr Ser Ala Pro Phe Asp Val Thr
 65 70 75 80

Met	Lys	Glu	Leu	Arg	85	Ser	Ser	Val	Val	Pro	Gly	Met	Thr	His	Trp	Ala
Ser	Pro	Asn	Phe	Phe	Ala	Phe	Phe	Pro	Ser	Thr	Asn	Ser	Ala	Ala	Ala	Ala
Ile	Ala	Gly	Asp	Leu	Ile	Ala	Ser	Ala	Met	Asn	Thr	Val	Gly	Phe	Thr	
Trp	Gln	Ala	Ser	Pro	Ala	Ala	Thr	Glu	Met	Glu	Val	Leu	Ala	Leu	Asp	
Trp	Leu	Ala	Gln	Met	Leu	Asn	Leu	Pro	Thr	Ser	Phe	Met	Asn	Arg	Thr	
Gly	Glu	Gly	Arg	Gly	Thr	Gly	Gly	Gly	Val	Ile	Leu	Gly	Thr	Thr	Ser	
Glu	Ala	Met	Leu	Val	Thr	Leu	Val	Ala	Ala	Arg	Asp	Ala	Ala	Leu	Arg	
Arg	Ser	Gly	Ser	Asp	Gly	Val	Ala	Gly	Leu	His	Arg	Leu	Ala	Val	Tyr	
Ala	Ala	Asp	Gln	Thr	His	Ser	Thr	Phe	Phe	Lys	Ala	Cys	Arg	Leu	Ala	
Gly	Phe	Asp	Pro	Ala	Asn	Ile	Arg	Ser	Ile	Pro	Thr	Gly	Ala	Glu	Thr	
Asp	Tyr	Gly	Leu	Asp	Pro	Ala	Arg	Leu	Leu	Glu	Ala	Met	Gln	Ala	Asp	
Ala	Asp	Ala	Gly	Leu	Val	Pro	Thr	Tyr	Val	Cys	Ala	Thr	Val	Gly	Thr	
Thr	Ser	Ser	Asn	Ala	Val	Asp	Pro	Val	Gly	Ala	Val	Ala	Asp	Val	Ala	
Ala	Arg	Phe	Ala	Ala	Trp	Val	His	Val	Asp	Ala	Ala	Tyr	Ala	Gly	Ser	
Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	His	His	Leu	Asp	Gly	Val	Glu	Arg	
Val	Asp	Ser	Ile	Ser	Met	Ser	Pro	His	Lys	Trp	Leu	Met	Thr	Cys	Leu	
Asp	Cys	Thr	Cys	Leu	Tyr	Val	Arg	Asp	Thr	His	Arg	Leu	Thr	Gly	Ser	
Leu	Glu	Thr	Asn	Pro	Glu	Tyr	Leu	Lys	Asn	His	Ala	Ser	Asp	Ser	Gly	
Glu	Val	Thr	Asp	Leu	Lys	Asp	Met	Gln	Val	Gly	Val	Gly	Arg	Arg	Phe	
Arg	Gly	Leu	Lys	Leu	Trp	Met	Val	Met	Arg	Thr	Tyr	Gly	Val	Ala	Lys	
Leu	Gln	Glu	His	Ile	Arg	Ser	Asp	Val	Ala	Met	Ala	Lys	Val	Phe	Glu	
Asp	Leu	Val	Arg	Gly	Asp	Asp	Arg	Phe	Glu	Val	Val	Val	Pro	Arg	Asn	
Phe	Ala	Leu	Val	Cys	Phe	Arg	Ile	Arg	Ala	Gly	Ala	Gly	Ala	Ala	Ala	
Ala	Thr	Glu	Glu	Asp	Ala	Asp	Glu	Ala	Asn	Arg	Glu	Leu	Met	Glu	Arg	
Leu	Asn	Lys	Thr	Gly	Lys	Ala	Tyr	Val	Ala	His	Thr	Val	Val	Gly	Gly	
Arg	Phe	Val	Leu	Arg	Phe	Ala	Val	Gly	Ser	Ser	Leu	Gln	Glu	Glu	His	

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His Val Arg Ser Ala Trp Glu Leu Ile Lys Lys Thr Thr Thr Glu Met
 500 505 510

Met Asn

<210> SEQ ID NO 264

<211> LENGTH: 502

<212> TYPE: PRT

<213> ORGANISM: C. acuminata

<400> SEQUENCE: 264

Met Gly Ser Leu Asp Ser Asn Tyr Asp Thr Glu Ser Pro Ala Ser Val
 1 5 10 15

Gly Gln Phe Asn Pro Leu Asp Pro Glu Glu Phe Arg Lys Gln Ala His
 20 25 30

Cys Ile Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu Ser Tyr
 35 40 45

Pro Val Leu Ser Gln Val Asp Pro Gly Tyr Arg His Ser Arg Leu Gly
 50 55 60

Lys Asn Ala Pro Tyr Arg Ser Glu Pro Phe Glu Ser Ile Leu Lys Asp
 65 70 75 80

Val Gln Lys Asp Ile Ile Pro Gly Met Thr His Trp Met Ser Pro Asn
 85 90 95

Phe Phe Ala His Phe Pro Ala Thr Val Ser Ser Ala Ala Phe Val Gly
 100 105 110

Glu Met Leu Cys Thr Cys Phe Asn Ser Val Gly Phe Asn Trp Leu Ala
 115 120 125

Ser Pro Ala Ala Thr Glu Leu Glu Met Val Val Ile Asp Trp Leu Ala
 130 135 140

Asn Met Leu Lys Leu Pro Lys Ser Phe Met Phe Ser Gly Thr Gly Gly
 145 150 155 160

Gly Val Leu Gln Gly Thr Thr Ser Glu Ala Ile Leu Cys Thr Leu Ile
 165 170 175

Ala Ala Ser Pro Met His Phe Glu Ile Val Gly Val Lys Thr Ser Thr
 180 185 190

Ser Phe Val Val Tyr Gly Ser Asp Gln Thr His Ser Thr Tyr Ala Lys
 195 200 205

Ala Cys Lys Leu Ala Gly Ile Leu Pro Cys Asn Ile Arg Ser Ile Pro
 210 215 220

Thr Thr Ala Asp Ser Asn Phe Ser Val Ser Pro Leu Leu Leu Arg Arg
 225 230 235 240

Ala Ile Glu Ala Asp Lys Ala Ala Gly Met Val Pro Leu Tyr Ile Cys
 245 250 255

Ala Thr Val Gly Thr Thr Ser Thr Thr Ala Ile Asp Pro Leu Ser Ser
 260 265 270

Leu Ala Asp Val Ala Asn Asp Tyr Gly Val Trp Phe His Val Asp Ala
 275 280 285

Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Tyr Leu
 290 295 300

Asp Gly Ile Glu Arg Ala Asp Ser Leu Ser Leu Ser Pro His Lys Trp
 305 310 315 320

Leu Leu Ser Tyr Leu Asp Cys Cys Cys Leu Trp Val Lys Ser Pro Ser
 325 330 335

Leu Leu Val Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Gln
 340 345 350

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Pro	Ser	Glu	Ser	Lys	Ser	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Val	Gly
		355					360					365			
Thr	Gly	Arg	Arg	Phe	Lys	Ala	Leu	Arg	Leu	Trp	Phe	Val	Met	Arg	Ser
	370					375					380				
Tyr	Gly	Val	Ala	Asn	Leu	Gln	Ser	His	Ile	Arg	Thr	Asp	Val	Gln	Met
	385				390					395					400
Ala	Lys	Met	Phe	Glu	Gly	Phe	Val	Lys	Ser	Asp	Pro	Arg	Phe	Glu	Ile
			405						410					415	
Leu	Val	Pro	Arg	Val	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Asn	Pro	Ile
			420					425					430		
Ser	Gly	Ser	Asp	Pro	Thr	Gly	Thr	Glu	Ala	Leu	Asn	Arg	Lys	Leu	Leu
		435					440					445			
Asp	Trp	Val	Asn	Ser	Thr	Gly	Arg	Val	Tyr	Met	Thr	His	Thr	Lys	Val
	450					455					460				
Gly	Gly	Ile	Tyr	Met	Leu	Arg	Phe	Ala	Val	Gly	Ala	Thr	Leu	Thr	Glu
	465				470					475					480
Lys	Arg	His	Val	Ser	Ser	Ala	Trp	Lys	Leu	Ile	Lys	Glu	Gly	Ala	Asp
			485					490						495	
Val	Leu	Leu	Lys	Glu	Asp										
			500												

<210> SEQ ID NO 265

<211> LENGTH: 498

<212> TYPE: PRT

<213> ORGANISM: C. acuminata

<400> SEQUENCE: 265

Met	Gly	Ser	Ile	Asp	Ser	Asn	Tyr	Asp	Thr	Glu	Ser	Ala	Gly	Gln	Cys
1				5					10					15	
Arg	Pro	Leu	Glu	Pro	Glu	Glu	Phe	Arg	Lys	Gln	Ala	His	Gln	Met	Val
		20					25						30		
Asp	Phe	Ile	Ala	Asp	Tyr	Tyr	Lys	Asn	Ile	Glu	Ser	Tyr	Pro	Val	Leu
	35						40					45			
Ser	Gln	Val	Glu	Pro	Gly	Tyr	Leu	Gln	Ser	Arg	Leu	Pro	Glu	Thr	Ala
	50				55						60				
Pro	Tyr	Arg	Pro	Glu	Pro	Phe	Glu	Ser	Ile	Leu	Lys	Asp	Val	His	Lys
	65			70					75						80
Asp	Ile	Ile	Pro	Gly	Val	Thr	His	Trp	Leu	Ser	Pro	Asn	Phe	Phe	Ala
			85					90						95	
Tyr	Phe	Pro	Ala	Thr	Val	Ser	Ser	Ala	Ala	Phe	Val	Gly	Glu	Met	Leu
		100						105					110		
Cys	Thr	Cys	Phe	Asn	Ala	Val	Gly	Phe	Asn	Trp	Leu	Ala	Ser	Pro	Ala
	115					120					125				
Glu	Leu	Glu	Leu	Glu	Met	Val	Val	Met	Asp	Trp	Leu	Ala	Ser	Met	Leu
	130				135					140					
Lys	Leu	Pro	Asn	Ser	Phe	Thr	Phe	Leu	Gly	Thr	Gly	Gly	Gly	Val	Ile
	145				150					155					160
Gln	Gly	Thr	Thr	Ser	Glu	Ala	Ile	Leu	Cys	Thr	Leu	Ile	Ala	Ala	Arg
			165					170						175	
Asp	Arg	Ala	Leu	Glu	Ser	Ile	Gly	Val	Asp	Ser	Ile	His	Lys	Leu	Val
		180						185					190		
Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Ser	Thr	Tyr	Ala	Lys	Ala	Cys	Asn
		195				200						205			
Leu	Ala	Gly	Ile	Leu	Pro	Cys	Asn	Ile	Arg	Ser	Ile	Arg	Thr	Glu	Ala
	210					215					220				

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Val Ala Asn Phe Ser Leu Ser Pro Asp Ser Leu His Arg Glu Ile Glu
 225 230 235 240
 Ala Asp Val Ala Ala Gly Met Val Pro Leu Tyr Leu Cys Ala Thr Val
 245 250 255
 Gly Thr Thr Ser Thr Thr Ala Ile Asp Ser Leu Ser Pro Leu Ala Asp
 260 265 270
 Val Ala Asn Asp Tyr Gly Leu Trp Phe His Val Asp Ala Ala Tyr Ala
 275 280 285
 Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Tyr Leu Asp Gly Ile
 290 295 300
 Glu Arg Ala Asp Ser Leu Ser Leu Ser Pro His Lys Trp Leu Leu Ser
 305 310 315 320
 Tyr Leu Asp Cys Cys Cys Leu Trp Val Lys Arg Pro Ser Val Leu Val
 325 330 335
 Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Lys Pro Ser Glu
 340 345 350
 Ser Asn Ser Val Val Asp Phe Lys Asp Trp Gln Val Gly Thr Gly Arg
 355 360 365
 Arg Phe Lys Ala Leu Arg Leu Trp Phe Val Met Arg Ser Tyr Gly Val
 370 375 380
 Ala Asn Leu Gln Ser His Ile Arg Ser Asp Ile Gln Met Ala Lys Met
 385 390 395 400
 Phe Glu Glu Phe Val Asn Ser Asp Pro Arg Phe Glu Ile Val Val Pro
 405 410 415
 Arg Val Phe Ser Leu Val Cys Phe Arg Leu Asn Pro Phe Ser Lys Ser
 420 425 430
 Asp Pro Cys Asn Thr Glu Leu Leu Asn Arg Lys Leu Leu Glu Trp Val
 435 440 445
 Asn Ser Thr Gly Gln Val Tyr Ile Thr His Thr Lys Val Gly Gly Val
 450 455 460
 Tyr Met Leu Arg Phe Ala Val Gly Ala Thr Leu Thr Glu Glu His His
 465 470 475 480
 Val Ser Ala Ala Trp Lys Leu Ile Arg Glu Gly Ala Asp Ala Leu Leu
 485 490 495

Cys Ser

<210> SEQ ID NO 266

<211> LENGTH: 503

<212> TYPE: PRT

<213> ORGANISM: C. annuum

<400> SEQUENCE: 266

Met Gly Ser Leu Asp Ser Asn Asn Ser Thr Gln Thr Gln Ser Asn Val
 1 5 10 15
 Thr Lys Phe Asn Pro Leu Asp Pro Glu Glu Phe Arg Thr Gln Ala His
 20 25 30
 Gln Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu Ser Tyr
 35 40 45
 Pro Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Arg Asn His Leu Pro
 50 55 60
 Glu Asn Ala Pro Tyr Leu Pro Glu Ser Leu Asp Thr Ile Met Lys Asp
 65 70 75 80
 Val Glu Lys His Ile Ile Pro Gly Met Thr His Trp Leu Ser Pro Asn
 85 90 95

Phe	Phe	Ala	Phe			Pro	Ala	Thr	Val	Ser	Ser	Ala	Ala	Phe	Leu	Gly	
			100				105			110							
Glu	Met	Leu	Cys	Asn	Cys	Phe	Asn	Ser	Val	Gly	Phe	Asn	Trp	Leu	Ala		
			115				120			125							
Ser	Pro	Ala	Met	Thr	Glu	Leu	Glu	Met	Ile	Ile	Met	Asp	Trp	Leu	Ala		
			130				135			140							
Asn	Met	Leu	Lys	Leu	Pro	Glu	Cys	Phe	Met	Phe	Ser	Gly	Thr	Gly	Gly		
			145				150			155							
Gly	Val	Ile	Gln	Gly	Thr	Thr	Ser	Glu	Ala	Ile	Leu	Cys	Thr	Leu	Ile		
			165				170			175							
Ala	Ala	Arg	Asp	Arg	Lys	Leu	Glu	Asn	Ile	Gly	Val	Asp	Asn	Ile	Gly		
			180				185			190							
Lys	Leu	Val	Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Ser	Met	Tyr	Ala	Lys		
			195				200			205							
Ala	Cys	Lys	Ala	Ala	Gly	Ile	Phe	Pro	Cys	Asn	Ile	Arg	Ala	Ile	Ser		
			210				215			220							
Thr	Cys	Val	Glu	Asn	Asp	Phe	Ser	Leu	Ser	Pro	Ala	Val	Leu	Arg	Gly		
			225				230			235							
Ile	Val	Glu	Val	Asp	Val	Ala	Ala	Gly	Leu	Val	Pro	Leu	Phe	Leu	Cys		
			245				250			255							
Ala	Thr	Val	Gly	Thr	Thr	Ser	Thr	Thr	Ala	Ile	Asp	Pro	Ile	Ser	Glu		
			260				265			270							
Leu	Gly	Glu	Leu	Ala	Asn	Glu	Phe	Asp	Ile	Trp	Leu	His	Val	Asp	Ala		
			275				280			285							
Ala	Tyr	Gly	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	Gln	Tyr	Leu		
			290				295			300							
Asp	Gly	Ile	Glu	Arg	Ala	Asn	Ser	Phe	Ser	Leu	Ser	Pro	His	Lys	Trp		
			305				310			315							
Leu	Leu	Ser	Tyr	Leu	Asp	Cys	Cys	Cys	Met	Trp	Val	Lys	Glu	Pro	Ser		
			325				330			335							
Val	Leu	Val	Lys	Ala	Leu	Ser	Thr	Asn	Pro	Glu	Tyr	Leu	Arg	Asn	Lys		
			340				345			350							
Arg	Ser	Glu	His	Gly	Ser	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Gly		
			355				360			365							
Thr	Gly	Arg	Lys	Phe	Lys	Ser	Leu	Arg	Leu	Trp	Leu	Ile	Met	Arg	Ser		
			370				375			380							
Tyr	Gly	Val	Ala	Asn	Leu	Gln	Ser	His	Ile	Arg	Ser	Asp	Val	Arg	Met		
			385				390			395							
Ala	Lys	Met	Phe	Glu	Gly	Leu	Val	Arg	Ser	Asp	Pro	Tyr	Phe	Glu	Val		
			405				410			415							
Ile	Val	Pro	Arg	Arg	Phe	Ser	Leu	Val	Cys	Phe	Arg	Phe	Asn	Pro	Asp		
			420				425			430							
Lys	Glu	Tyr	Glu	Pro	Ala	Tyr	Thr	Glu	Leu	Leu	Asn	Lys	Arg	Leu	Leu		
			435				440			445							
Asp	Asn	Val	Asn	Ser	Thr	Gly	Arg	Val	Tyr	Met	Thr	His	Thr	Val	Ala		
			450				455			460							
Gly	Gly	Ile	Tyr	Met	Leu	Arg	Phe	Ala	Val	Gly	Ala	Thr	Phe	Thr	Glu		
			465				470			475							
Asp	Arg	His	Leu	Ile	Cys	Ala	Trp	Lys	Leu	Ile	Lys	Asp	Cys	Ala	Asp		
			485				490			495							
Ala	Leu	Leu	Arg	Asn	Cys	Gln											
			500														

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<210> SEQ ID NO 267

<211> LENGTH: 506

<212> TYPE: PRT

<213> ORGANISM: *O. pumila*

<400> SEQUENCE: 267

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Met Gly Ser Ile Ser Glu Asn Cys Asp Asp Ser Ile Ser Leu Ala Ala
1          5          10          15
Pro Phe Arg Pro Leu Glu Pro Glu Glu Phe Arg Lys Gln Ala His Val
20        25        30
Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Ile Glu Asn Tyr Pro
35        40        45
Val Leu Ser Gln Val Glu Pro Gly Tyr Leu Lys Asn Arg Leu Pro Glu
50        55        60
Thr Ala Pro His Leu Pro Glu Ser Phe Glu Thr Ile Leu Lys Asp Ile
65        70        75        80
Lys Lys Asp Ile Val Pro Gly Met Thr Asn Trp Leu Ser Pro Asn Phe
85        90        95
Phe Ala Tyr Phe Pro Ala Thr Val Ser Ser Ala Ala Phe Val Gly Glu
100       105       110
Met Leu Cys Thr Gly Phe Asn Ser Val Gly Phe Asn Trp Leu Ala Ser
115       120       125
Pro Ala Ser Thr Glu Leu Glu Met Val Val Ile Asp Trp Leu Ala Asn
130       135       140
Met Leu Lys Leu Pro Lys Ser Phe Met Phe His Gly Thr Gly Gly Gly
145       150       155       160
Val Ile Gln Gly Thr Thr Ser Glu Ala Ile Leu Cys Thr Leu Ile Ala
165       170       175
Ala Arg Asp Gly Ala Leu Glu Lys Ile Gly Met Glu Asn Val Gly Lys
180       185       190
Leu Val Val Tyr Gly Ser Asp Gln Thr His Ser Phe Phe Gln Lys Thr
195       200       205
Cys Lys Val Ala Gly Ile Phe Pro Cys Asn Ile Lys Leu Ile Pro Thr
210       215       220
Thr Arg Glu Asp Asn Phe Ser Met Ser Pro Ile Ala Leu Arg Glu Gln
225       230       235       240
Ile Glu Ala Asp Val Ala Asp Gly Leu Val Pro Ile Phe Leu Cys Thr
245       250       255
Thr Val Gly Thr Thr Ser Thr Ala Ala Ile Asp Pro Val Ser Glu Val
260       265       270
Ala Lys Val Ala Asn Asp Phe Asn Ile Trp Val His Val Asp Ala Ala
275       280       285
Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg Gln Tyr Leu Asp
290       295       300
Gly Ile Glu Leu Val Asp Ser Phe Ser Leu Ser Pro His Lys Trp Leu
305       310       315       320
Leu Cys Phe Leu Asp Cys Cys Cys Leu Trp Leu Lys Lys Pro His Leu
325       330       335
Met Val Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys Arg
340       345       350
Ser Glu Phe Asp Gly Val Val Asp Phe Lys Asp Trp Gln Ile Gly Thr
355       360       365
Gly Arg Arg Phe Lys Ala Leu Arg Leu Trp Leu Val Met Arg Ser Tyr
370       375       380

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Gly Val Glu Asn Leu Lys Arg His Ile Leu Ser Asp Val Gln Met Ala
 385 390 395 400
 Lys Met Phe Glu Gly Leu Val Lys Ser Asp Pro Arg Phe Glu Ile Ile
 405 410 415
 Val Pro Arg Ala Phe Ala Leu Val Cys Phe Arg Leu Asn Pro Gly Lys
 420 425 430
 Gly Tyr Asp Asp Glu Ile Asp Lys Glu Ile Leu Asn Lys Glu Leu Leu
 435 440 445
 Asp Leu Ile Asn Ser Thr Gly Arg Ala Tyr Met Thr His Thr Lys Ala
 450 455 460
 Gly Gly Ile Tyr Met Leu Arg Phe Ala Val Gly Thr Thr Leu Thr Glu
 465 470 475 480
 Glu His His Val Tyr Ala Ala Trp Glu Leu Ile Lys Glu Cys Thr Asp
 485 490 495
 Ala Ser Leu Thr Lys Thr Asn Ile Ile Glu
 500 505

<210> SEQ ID NO 268

<211> LENGTH: 500

<212> TYPE: PRT

<213> ORGANISM: *C. roseus*

<400> SEQUENCE: 268

Met Gly Ser Ile Asp Ser Thr Asn Val Ala Met Ser Asn Ser Pro Val
 1 5 10 15
 Gly Glu Phe Lys Pro Leu Glu Ala Glu Glu Phe Arg Lys Gln Ala His
 20 25 30
 Arg Met Val Asp Phe Ile Ala Asp Tyr Tyr Lys Asn Val Glu Thr Tyr
 35 40 45
 Pro Val Leu Ser Glu Val Glu Pro Gly Tyr Leu Arg Lys Arg Ile Pro
 50 55 60
 Glu Thr Ala Pro Tyr Leu Pro Glu Pro Leu Asp Asp Ile Met Lys Asp
 65 70 75 80
 Ile Gln Lys Asp Ile Ile Pro Gly Met Thr Asn Trp Met Ser Pro Asn
 85 90 95
 Phe Tyr Ala Phe Phe Pro Ala Thr Val Ser Ser Ala Ala Phe Leu Gly
 100 105 110
 Glu Met Leu Ser Thr Ala Leu Asn Ser Val Gly Phe Thr Trp Val Ser
 115 120 125
 Ser Pro Ala Ala Thr Glu Leu Glu Met Ile Val Met Asp Trp Leu Ala
 130 135 140
 Gln Ile Leu Lys Leu Pro Lys Ser Phe Met Phe Ser Gly Thr Gly Gly
 145 150 155 160
 Gly Val Ile Gln Asn Thr Thr Ser Glu Ser Ile Leu Cys Thr Ile Ile
 165 170 175
 Ala Ala Arg Glu Arg Ala Leu Glu Lys Leu Gly Pro Asp Ser Ile Gly
 180 185 190
 Lys Leu Val Cys Tyr Gly Ser Asp Gln Thr His Thr Met Phe Pro Lys
 195 200 205
 Thr Cys Lys Leu Ala Gly Ile Tyr Pro Asn Asn Ile Arg Leu Ile Pro
 210 215 220
 Thr Thr Val Glu Thr Asp Phe Gly Ile Ser Pro Gln Val Leu Arg Lys
 225 230 235 240
 Met Val Glu Asp Asp Val Ala Ala Gly Tyr Val Pro Leu Phe Leu Cys

245										250					255				
Ala	Thr	Leu	Gly	Thr	Thr	Ser	Thr	Thr	Ala	Thr	Asp	Pro	Val	Asp	Ser				
260										265					270				
Leu	Ser	Glu	Ile	Ala	Asn	Glu	Phe	Gly	Ile	Trp	Ile	His	Val	Asp	Ala				
275										280					285				
Ala	Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	His	Tyr	Leu				
290										295					300				
Asp	Gly	Ile	Glu	Arg	Val	Asp	Ser	Leu	Ser	Leu	Ser	Pro	His	Lys	Trp				
305										310					315				
Leu	Leu	Ala	Tyr	Leu	Asp	Cys	Thr	Cys	Leu	Trp	Val	Lys	Gln	Pro	His				
325										330					335				
Leu	Leu	Leu	Arg	Ala	Leu	Thr	Thr	Asn	Pro	Glu	Tyr	Leu	Lys	Asn	Lys				
340										345					350				
Gln	Ser	Asp	Leu	Asp	Lys	Val	Val	Asp	Phe	Lys	Asn	Trp	Gln	Ile	Ala				
355										360					365				
Thr	Gly	Arg	Lys	Phe	Arg	Ser	Leu	Lys	Leu	Trp	Leu	Ile	Leu	Arg	Ser				
370										375					380				
Tyr	Gly	Val	Val	Asn	Leu	Gln	Ser	His	Ile	Arg	Ser	Asp	Val	Ala	Met				
385										390					395				
Gly	Lys	Met	Phe	Glu	Glu	Trp	Val	Arg	Ser	Asp	Ser	Arg	Phe	Glu	Ile				
405										410					415				
Val	Val	Pro	Arg	Asn	Phe	Ser	Leu	Val	Cys	Phe	Arg	Leu	Lys	Pro	Asp				
420										425					430				
Val	Ser	Ser	Leu	His	Val	Glu	Glu	Val	Asn	Lys	Lys	Leu	Leu	Asp	Met				
435										440					445				
Leu	Asn	Ser	Thr	Gly	Arg	Val	Tyr	Met	Thr	His	Thr	Ile	Val	Gly	Gly				
450										455					460				
Ile	Tyr	Met	Leu	Arg	Leu	Ala	Val	Gly	Ser	Ser	Leu	Thr	Glu	Glu	His				
465										470					475				
His	Val	Arg	Arg	Val	Trp	Asp	Leu	Ile	Gln	Lys	Leu	Thr	Asp	Asp	Leu				
485										490					495				
Leu	Lys	Glu	Ala																
500																			

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<210> SEQ ID NO 269
<211> LENGTH: 514
<212> TYPE: PRT
<213> ORGANISM: P. crispum
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<400> SEQUENCE: 269

Met	Gly	Ser	Ile	Asp	Asn	Leu	Thr	Glu	Lys	Leu	Ala	Ser	Gln	Phe	Pro
1				5					10					15	
Met	Asn	Thr	Leu	Glu	Pro	Glu	Glu	Phe	Arg	Arg	Gln	Gly	His	Met	Met
			20					25					30		
Ile	Asp	Phe	Leu	Ala	Asp	Tyr	Tyr	Arg	Lys	Val	Glu	Asn	Tyr	Pro	Val
		35					40					45			
Arg	Ser	Gln	Val	Ser	Pro	Gly	Tyr	Leu	Arg	Glu	Ile	Leu	Pro	Glu	Ser
	50					55					60				
Ala	Pro	Tyr	Asn	Pro	Glu	Ser	Leu	Glu	Thr	Ile	Leu	Gln	Asp	Val	Gln
65					70					75					80
Thr	Lys	Ile	Ile	Pro	Gly	Ile	Thr	His	Trp	Gln	Ser	Pro	Asn	Phe	Phe
			85						90					95	
Ala	Tyr	Phe	Pro	Ser	Ser	Gly	Ser	Thr	Ala	Gly	Phe	Leu	Gly	Glu	Met
			100						105				110		

Leu	Ser	Thr	Gly	Phe	Asn	Val	Val	Gly	Phe	Asn	Trp	Met	Val	Ser	Pro
115															
Ala	Ala	Thr	Glu	Leu	Glu	Asn	Val	Val	Thr	Asp	Trp	Phe	Gly	Lys	Met
130															
Leu	Gln	Leu	Pro	Lys	Ser	Phe	Leu	Phe	Ser	Gly	Gly	Gly	Gly	Gly	Val
145															
Leu	Gln	Gly	Thr	Thr	Cys	Glu	Ala	Ile	Leu	Cys	Thr	Leu	Val	Ala	Ala
			165												
Arg	Asp	Lys	Asn	Leu	Arg	Gln	His	Gly	Met	Asp	Asn	Ile	Gly	Lys	Leu
			180												
Val	Val	Tyr	Cys	Ser	Asp	Gln	Thr	His	Ser	Ala	Leu	Gln	Lys	Ala	Ala
			195												
Lys	Ile	Ala	Gly	Ile	Asp	Pro	Lys	Asn	Phe	Arg	Ala	Ile	Glu	Thr	Thr
			210												
Lys	Ser	Ser	Asn	Phe	Gln	Leu	Cys	Pro	Lys	Arg	Leu	Glu	Ser	Ala	Ile
			225												
Leu	His	Asp	Leu	Gln	Asn	Gly	Leu	Ile	Pro	Leu	Tyr	Leu	Cys	Ala	Thr
			245												
Val	Gly	Thr	Thr	Ser	Ser	Thr	Thr	Val	Asp	Pro	Leu	Pro	Ala	Leu	Thr
			260												
Glu	Val	Ala	Lys	Lys	Tyr	Asp	Leu	Trp	Val	His	Val	Asp	Ala	Ala	Tyr
			275												
Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	Gln	Tyr	Leu	Asp	Gly
			290												
Val	Glu	Asn	Ala	Asp	Ser	Phe	Ser	Leu	Asn	Ala	His	Lys	Trp	Phe	Leu
			305												
Thr	Thr	Leu	Asp	Cys	Cys	Cys	Leu	Trp	Val	Arg	Asn	Pro	Ser	Ala	Leu
			325												
Ile	Lys	Ser	Leu	Ser	Thr	Tyr	Pro	Glu	Phe	Leu	Lys	Asn	Asn	Ala	Ser
			340												
Glu	Thr	Asn	Lys	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Met	Leu	Ser
			355												
Arg	Arg	Phe	Arg	Ala	Leu	Lys	Leu	Trp	Phe	Val	Leu	Arg	Ser	Tyr	Gly
			370												
Val	Gly	Gln	Leu	Arg	Glu	Phe	Ile	Arg	Gly	His	Val	Gly	Met	Ala	Lys
			385												
Tyr	Phe	Glu	Gly	Leu	Val	Asn	Met	Asp	Lys	Arg	Phe	Glu	Val	Val	Ala
			405												
Pro	Arg	Leu	Phe	Ser	Met	Val	Cys	Phe	Arg	Ile	Lys	Pro	Ser	Ala	Met
			420												
Ile	Gly	Lys	Asn	Asp	Glu	Asp	Glu	Val	Asn	Glu	Ile	Asn	Arg	Lys	Leu
			435												
Leu	Glu	Ser	Val	Asn	Asp	Ser	Gly	Arg	Ile	Tyr	Val	Ser	His	Thr	Val
			450												
Leu	Gly	Gly	Ile	Tyr	Val	Ile	Arg	Phe	Ala	Ile	Gly	Gly	Thr	Leu	Thr
			465												
Asp	Ile	Asn	His	Val	Ser	Ala	Ala	Trp	Lys	Val	Leu	Gln	Asp	His	Ala
			485												
Gly	Ala	Leu	Leu	Asp	Asp	Thr	Phe	Thr	Ser	Asn	Lys	Leu	Val	Glu	Val
			500												
Leu	Ser														

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<211> LENGTH: 508

<212> TYPE: PRT

<213> ORGANISM: R. hybrid

<400> SEQUENCE: 270

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Met Gly Ser Phe Pro Phe His Arg Asp Leu Gln Glu Ile Ala Ser Ser
 1          5          10          15

Gln Leu Thr Lys Ala Leu Asp Pro Glu Glu Phe Arg Lys Gln Gly His
 20          25          30

Met Val Ile Asn Phe Ile Ala Asp Tyr Tyr Gln Asn Ile Glu Lys Tyr
 35          40          45

Pro Val Leu Ser Arg Val Glu Pro Gly Tyr Leu Lys Lys Cys Leu Pro
 50          55          60

Val Ser Ala Pro Tyr Asp Pro Glu Pro Ile Ser Thr Ile Leu Arg Asp
 65          70          75          80

Val Gln Asn His Ile Val Pro Gly Leu Thr His Trp Gln Ser Pro Asn
 85          90          95

Phe Phe Ala Tyr Phe Ser Ser Thr Ala Ser Thr Ala Gly Phe Leu Gly
100          105          110

Glu Ile Leu Thr Thr Gly Phe Asn Val Val Gly Phe Asn Trp Val Ser
115          120          125

Ser Pro Ala Ala Thr Glu Leu Glu Asn Ile Val Met Asp Trp Leu Gly
130          135          140

Asp Met Leu Gln Leu Pro Lys Ser Phe His Phe Ser Gly Asn Gly Gly
145          150          155          160

Gly Val Leu His Gly Ser Thr Cys Glu Ala Ile Val Cys Thr Met Val
165          170          175

Ala Ala Arg Asp Gln Met Leu Arg Arg Ile Gly Ser Glu Asn Leu Gly
180          185          190

Lys Leu Val Val Tyr Gly Ser Asp Gln Thr His Ser Thr Leu Gln Lys
195          200          205

Ala Thr Gln Ile Val Gly Ile Asn Thr Glu Asn Phe Arg Ala Ile Lys
210          215          220

Thr Thr Lys Ser Thr Gly Phe Ala Leu Ser Pro Glu Met Leu Arg Leu
225          230          235          240

Thr Ile Ser Ser Asp Leu Glu Lys Gly Leu Val Pro Leu Phe Leu Cys
245          250          255

Ala Thr Ile Gly Thr Thr Ala Thr Thr Ala Ile Asp Pro Leu Glu Ala
260          265          270

Leu Cys His Val Ala Lys Glu Tyr Gly Val Trp Val His Val Asp Ala
275          280          285

Ala Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Arg His Phe Ile
290          295          300

Asn Gly Val Glu Gly Ala Asn Ser Phe Ser Phe Asn Pro His Lys Trp
305          310          315          320

Leu Phe Thr Gly Met Asp Cys Cys Cys Leu Trp Val Lys Asn Pro Ser
325          330          335

Val Leu Ala Ser Ser Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys
340          345          350

Ala Ser Asp Ser Lys Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Ala
355          360          365

Leu Ser Arg Arg Phe Arg Ala Leu Lys Leu Trp Leu Val Leu Arg Ser
370          375          380

Tyr Gly Val Ala Asn Leu Arg Asn Phe Ile Arg Ile His Val Lys Met

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385	390	395	400
Ala Lys Thr Phe Glu Gly Leu Val Arg Met Asp Lys Arg Phe Glu Ile	405	410	415
Leu Val Pro Arg Asn Phe Ser Leu Val Cys Phe Arg Ile Ser Pro Ser	420	425	430
Ala Leu Ile Ser Ser Asn Glu Asp Asp Glu Ile Gly Met Val Asn Glu	435	440	445
Val Asn Cys Lys Leu Leu Glu Ala Ile Asn Ala Ser Gly Lys Ala Tyr	450	455	460
Met Thr His Ala Val Val Gly Gly Leu Tyr Val Leu Arg Cys Ala Val	465	470	475
Gly Ala Thr Leu Thr Glu Glu Lys His Ile Val Glu Ala Trp Asn Val	485	490	495
Val Gln Asp His Ala Gln Ala Ile Leu Ser Thr Tyr	500	505	

<210> SEQ ID NO 271
 <211> LENGTH: 506
 <212> TYPE: PRT
 <213> ORGANISM: P. hybrida

<400> SEQUENCE: 271

Met Asp Thr Ile Lys Ile Asn Pro Glu Phe Asp Gly Gln Phe Cys Lys	1	5	10	15
Thr Thr Ser Leu Leu Asp Pro Glu Glu Phe Arg Arg Asn Gly His Met	20	25	30	
Met Val Asp Phe Leu Ala Asp Tyr Phe His Asn Ile Glu Lys Tyr Pro	35	40	45	
Val Arg Ser Gln Val Glu Pro Gly Tyr Leu Glu Arg Leu Leu Pro Asp	50	55	60	
Ser Ala Pro Ile Gln Pro Glu Pro Ile Glu Lys Ile Leu Lys Asp Val	65	70	75	80
Arg Ser Asp Ile Phe Pro Gly Leu Thr His Trp Gln Ser Pro Asn Phe	85	90	95	
Phe Ala Tyr Phe Pro Cys Ser Ser Ser Thr Ala Gly Ile Leu Gly Glu	100	105	110	
Met Leu Ser Ala Gly Leu Asn Val Val Gly Phe Ser Trp Ile Ala Ser	115	120	125	
Pro Ala Ala Thr Glu Leu Glu Ser Ile Val Met Asp Trp Leu Gly Lys	130	135	140	
Leu Ile Asn Leu Pro Lys Thr Tyr Leu Phe Ser Gly Gly Gly Gly Gly	145	150	155	160
Val Met Gln Gly Thr Thr Cys Glu Val Met Leu Cys Thr Ile Val Ala	165	170	175	
Ala Arg Asp Lys Met Leu Glu Lys Phe Gly Arg Glu Asn Ile Asp Lys	180	185	190	
Leu Val Val Tyr Ala Ser Asp Gln Thr His Phe Ser Phe Gln Lys Ala	195	200	205	
Val Lys Ile Ser Gly Ile Lys Pro Glu Asn Phe Arg Ala Ile Pro Thr	210	215	220	
Thr Lys Ala Thr Glu Phe Ser Leu Asn Pro Glu Ser Leu Arg Arg Ala	225	230	235	240
Ile Gln Glu Asp Lys Lys Ala Gly Leu Ile Pro Leu Phe Leu Cys Thr	245	250	255	

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Ser Ile Gly Thr Thr Ser Thr Thr Ala Val Asp Pro Leu Lys Pro Leu
      260      265      270

Cys Glu Ile Ala Glu Glu Tyr Gly Ile Trp Val His Val Asp Ala Ala
      275      280      285

Tyr Ala Gly Ser Ala Cys Ile Cys Pro Glu Phe Gln His Phe Leu Asp
      290      295      300

Gly Val Glu His Ala Asn Ser Phe Ser Phe Asn Ala His Lys Trp Leu
      305      310      315      320

Phe Thr Thr Leu Asp Cys Cys Cys Leu Trp Leu Lys Asp Pro Ser Ser
      325      330      335

Leu Thr Lys Ala Leu Ser Thr Asn Pro Glu Val Leu Arg Asn Asp Ala
      340      345      350

Thr Asp Ser Glu Gln Val Val Asp Tyr Lys Asp Trp Gln Ile Thr Leu
      355      360      365

Ser Arg Arg Phe Arg Ser Leu Lys Leu Trp Leu Val Leu Lys Ser Tyr
      370      375      380

Gly Val Ala Asn Leu Arg Asn Phe Ile Arg Ser His Ile Glu Met Ala
      385      390      395      400

Lys His Phe Glu Glu Leu Val Ala Met Asp Glu Arg Phe Glu Ile Met
      405      410      415

Ala Pro Arg Asn Phe Ser Leu Val Cys Phe Arg Val Ser Leu Leu Ala
      420      425      430

Leu Glu Lys Lys Phe Asn Phe Val Asp Glu Thr Gln Val Asn Glu Phe
      435      440      445

Asn Ala Lys Leu Leu Glu Ser Ile Ile Ser Ser Gly Asn Val Tyr Met
      450      455      460

Thr His Thr Val Val Glu Gly Val Tyr Met Ile Arg Phe Ala Val Gly
      465      470      475      480

Ala Pro Leu Thr Asp Tyr Pro His Ile Asp Met Ala Trp Asn Val Val
      485      490      495

Arg Asn His Ala Thr Met Met Leu Asn Ala
      500      505

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<210> SEQ ID NO 272

<211> LENGTH: 518

<212> TYPE: PRT

<213> ORGANISM: T. flavum

<400> SEQUENCE: 272

```

Met Gly Ser Leu His Val Glu Asp Leu Asp Asn Ile Ser Lys Cys Thr
1      5      10      15

Val Glu Asn Pro Leu Asp Pro Glu Glu Phe Arg Arg Gln Gly His Met
20     25     30

Met Ile Asp Phe Leu Ala Asp Tyr Tyr Arg Asp Ile Glu Lys Tyr Pro
35     40     45

Val Arg Ser Gln Val Glu Pro Gly Tyr Leu Arg Lys Glu Ile Pro Asp
50     55     60

Ser Ala Pro Tyr Asn Pro Glu Ser Ile Glu Thr Ile Leu Glu Asp Val
65     70     75     80

His Lys Gln Ile Ile Pro Gly Ile Thr His Trp Gln Ser Pro Asn Tyr
85     90     95

Phe Ala Tyr Phe Pro Ser Ser Gly Ser Val Ala Gly Phe Leu Gly Glu
100    105    110

Met Leu Ser Thr Gly Phe Asn Val Val Gly Phe Asn Trp Met Ser Ser
115    120    125

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Pro	Ala	Ala	Thr	Glu		Leu	Glu	Ser	Ile	Val	Met	Asp	Trp	Leu	Gly	Lys
130							135					140				
Met	Leu	Lys	Leu	Pro	Lys	Ser	Phe	Leu	Phe	Ser	Gly	Asn	Gly	Gly	Gly	
145					150					155					160	
Val	Leu	Gln	Gly	Thr	Thr	Cys	Glu	Ala	Ile	Leu	Cys	Thr	Leu	Thr	Ala	
				165					170					175		
Ala	Arg	Asp	Arg	Met	Leu	Asn	Lys	Ile	Gly	Arg	Glu	Asn	Ile	Cys	Lys	
			180					185					190			
Leu	Val	Val	Tyr	Gly	Ser	Asp	Gln	Thr	His	Cys	Ala	Leu	Gln	Lys	Ala	
	195						200					205				
Ala	Gln	Ile	Ala	Gly	Ile	His	Pro	Asn	Asn	Phe	Arg	Ala	Val	Pro	Thr	
	210					215					220					
Thr	Lys	Ala	Asn	Asp	Tyr	Gly	Leu	Ser	Ala	Ser	Ala	Leu	Arg	Ser	Thr	
225					230					235					240	
Ile	Leu	Glu	Asp	Ile	Glu	Ala	Gly	Leu	Val	Pro	Leu	Phe	Leu	Cys	Ala	
			245						250					255		
Thr	Val	Gly	Thr	Thr	Ser	Ser	Thr	Ala	Val	Asp	Pro	Ile	Gly	Pro	Leu	
			260					265					270			
Cys	Lys	Val	Ala	Ser	Asp	Tyr	Ser	Ile	Trp	Val	His	Val	Asp	Ala	Ala	
	275						280					285				
Tyr	Ala	Gly	Ser	Ala	Cys	Ile	Cys	Pro	Glu	Phe	Arg	His	Phe	Ile	Asp	
	290					295					300					
Gly	Val	Glu	Asn	Ala	Asp	Ser	Phe	Ser	Leu	Asn	Ala	His	Lys	Trp	Phe	
305					310					315					320	
Phe	Thr	Thr	Leu	Asp	Cys	Cys	Cys	Leu	Trp	Val	Lys	Glu	Pro	Ser	Ala	
			325						330					335		
Leu	Ile	Lys	Ala	Leu	Ser	Thr	Asn	Pro	Glu	Tyr	Leu	Arg	Asn	Lys	Ala	
			340					345					350			
Thr	Glu	Ser	His	Gln	Val	Val	Asp	Tyr	Lys	Asp	Trp	Gln	Ile	Ala	Leu	
	355						360					365				
Ser	Arg	Arg	Phe	Arg	Ala	Met	Lys	Leu	Trp	Leu	Val	Leu	Arg	Ser	Tyr	
	370					375					380					
Gly	Val	Ala	Asn	Leu	Arg	Asn	Phe	Leu	Arg	Ser	His	Val	Lys	Met	Ala	
385					390				395						400	
Lys	Asn	Phe	Glu	Gly	Phe	Ile	Ala	Leu	Asp	Lys	Arg	Phe	Glu	Ile	Val	
			405						410					415		
Val	Pro	Arg	Thr	Phe	Ala	Met	Val	Cys	Phe	Arg	Leu	Leu	Pro	Pro	Arg	
			420					425					430			
Ser	Pro	Leu	Ile	Ile	Lys	Thr	Asn	Gly	Tyr	Gln	Asn	Gly	Asn	Gly	Val	
	435						440					445				
Tyr	His	Lys	Asp	Glu	Ser	Arg	Ala	Asn	Glu	Leu	Asn	Arg	Arg	Leu	Leu	
	450					455					460					
Glu	Ser	Ile														

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<210> SEQ ID NO 273
<211> LENGTH: 533
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<212> TYPE: PRT

<213> ORGANISM: B. distachyon

<400> SEQUENCE: 273

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Met Ala Pro Thr Ser Met Cys Phe Asp Ala Ile Asn Gly Ala Ala Ala
1      5      10      15
Ala Gln Asn Gly Thr Ala Pro Val Leu Ala Thr Lys Pro Ala Ala Gln
20      25      30
Ala Leu Gln Cys Pro Asn Ala Leu Asn Ala Asp Asp Phe Arg Arg Gln
35      40      45
Gly His Gln Val Ile Asp Phe Ile Ala Glu Tyr Tyr Gly Gly Met Ala
50      55      60
Asp Tyr Pro Val His Pro Ser Val Thr Pro Gly Phe Leu Arg Asn Leu
65      70      75      80
Leu Pro Ala Ser Ala Pro Ser Arg Ala Glu Pro Asp Ala Phe Ser Ser
85      90      95
Ala Leu Lys Asp Ile Arg Asp His Ile Leu Pro Gly Met Thr His Trp
100     105     110
Gln Ser Pro Arg His Phe Ala His Phe Pro Ala Ser Ser Ser Thr Val
115     120     125
Gly Ala Leu Gly Glu Ala Leu Thr Ala Gly Ile Asn Val Val Pro Phe
130     135     140
Thr Trp Ala Ala Ser Pro Ala Ala Thr Glu Leu Glu Met Val Val Val
145     150     155     160
Asp Trp Leu Gly Lys Ala Leu His Leu Pro Glu Thr Leu Leu Phe Ala
165     170     175
Gly Gly Gly Gly Gly Thr Leu Leu Gly Thr Ser Cys Glu Ala Ile Leu
180     185     190
Cys Ala Leu Val Ala Ala Arg Asp Arg Lys Leu Ala Glu Ile Gly Gly
195     200     205
Arg Arg Ile Gly Asp Leu Val Val Tyr Cys Ser Asp Gln Thr His Phe
210     215     220
Ala Phe Arg Lys Ala Ala Arg Ile Ala Gly Ile Leu Arg Glu His Ile
225     230     235     240
Arg Glu Ile Gln Thr Cys His Ala Asn Met Phe Ala Leu Ser Ala Thr
245     250     255
Ala Leu Glu Ala Ala Met Gln Ala Asp Val Glu Ala Gly Leu Val Pro
260     265     270
Leu Phe Val Cys Ala Thr Val Gly Thr Thr Gln Thr Thr Ala Val Asp
275     280     285
Pro Ile Gly Glu Leu Cys Thr Val Thr Ala Pro His Gly Val Trp Val
290     295     300
His Val Asp Ala Ala Tyr Ala Gly Ser Ala Leu Val Cys Pro Glu Phe
305     310     315     320
Arg His Val Ile Asn Gly Val Glu Ser Val Asp Ser Phe Ser Met Asn
325     330     335
Ala His Lys Trp Leu Leu Thr Asn Asn Asp Cys Cys Ala Met Trp Val
340     345     350
Lys Lys Pro Ser Glu Leu Ile Ala Ala Leu Gly Thr Glu Gln Glu Tyr
355     360     365
Ile Leu Lys Asp Ser Ala Ser Glu Gly His Asp Ile Val Asp Tyr Lys
370     375     380
Asp Trp Thr Met Thr Leu Thr Arg Arg Phe Arg Ala Leu Lys Met Trp
385     390     395     400

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Leu Val Leu Arg Cys Tyr Gly Ile Asp Gly Leu Arg Glu His Ile Arg
 405 410 415
 Ser His Val Arg Met Ala Glu Ala Phe Glu Asn Leu Val Arg Ala Asp
 420 425 430
 Glu Arg Phe Glu Val Val Thr Asp Arg Gln Phe Ala Leu Val Cys Phe
 435 440 445
 Arg Leu Arg Ser Pro Glu Lys Tyr Gly Gly Glu Lys Thr Ala Asn Glu
 450 455 460
 Leu Asn Arg Ser Leu Leu Glu Glu Val Asn Ala Val Thr Leu Gly Pro
 465 470 475 480
 Tyr Met Ser Ser Ala Asn Val Gly Gly Met Tyr Met Leu Arg Cys Ala
 485 490 495
 Val Gly Ser Thr Leu Thr Glu Asp Cys His Val Thr Asp Gly Trp Lys
 500 505 510
 Val Val Gln Asp Arg Ala Thr Ser Ile Leu Arg Lys Met Glu Ile Ile
 515 520 525
 Tyr Ser Val Leu Gly
 530

<210> SEQ ID NO 274

<211> LENGTH: 512

<212> TYPE: PRT

<213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 274

Met Gly Ser Leu Pro Thr Asn Asn Leu Glu Ser Ile Ser Leu Cys Ser
 1 5 10 15
 Gln Asn Pro Leu Asp Pro Asp Glu Phe Arg Arg Gln Gly His Met Ile
 20 25 30
 Ile Asp Phe Leu Ala Asp Tyr Tyr Lys Asn Val Glu Asn Tyr Pro Val
 35 40 45
 Arg Ser Gln Val Glu Pro Gly Tyr Leu Lys Lys Arg Leu Pro Glu Ser
 50 55 60
 Ala Pro Tyr Asn Pro Glu Ser Ile Glu Thr Ile Leu Glu Asp Val Thr
 65 70 75 80
 Asn Asp Ile Ile Pro Gly Leu Thr His Trp Gln Ser Pro Asn Tyr Phe
 85 90 95
 Ala Tyr Phe Pro Ser Ser Gly Ser Ile Ala Gly Phe Leu Gly Glu Met
 100 105 110
 Leu Ser Thr Gly Phe Asn Val Val Gly Phe Asn Trp Met Ser Ser Pro
 115 120 125
 Ala Ala Thr Glu Leu Glu Ser Ile Val Met Asn Trp Leu Gly Gln Met
 130 135 140
 Leu Thr Leu Pro Lys Ser Phe Leu Phe Ser Ser Asp Gly Ser Ser Gly
 145 150 155 160
 Gly Gly Gly Val Leu Gln Gly Thr Thr Cys Glu Ala Ile Leu Cys Thr
 165 170 175
 Leu Thr Ala Ala Arg Asp Lys Met Leu Asn Lys Ile Gly Arg Glu Asn
 180 185 190
 Ile Asn Lys Leu Val Val Tyr Ala Ser Asn Gln Thr His Cys Ala Leu
 195 200 205
 Gln Lys Ala Ala Gln Ile Ala Gly Ile Asn Pro Lys Asn Val Arg Ala
 210 215 220
 Ile Lys Thr Ser Lys Ala Thr Asn Phe Gly Leu Ser Pro Asn Ser Leu

225	230								235						240		
Gln Ser Ala Ile	Leu Ala Asp Ile	Glu Ser	Gly Leu Val	Pro Leu Phe	245	250	255										
Leu Cys Ala Thr	Val Gly Thr Thr	Ser Ser	Thr Ala Val	Asp Pro Ile	260	265	270										
Gly Pro Leu Cys	Ala Val Ala Lys	Leu Tyr	Gly Ile Trp	Val His Ile	275	280	285										
Asp Ala Ala Tyr	Ala Gly Ser Ala	Cys Ile	Cys Pro Glu	Phe Arg His	290	295	300										
Phe Ile Asp Gly	Val Glu Asp Ala	Asp Ser	Phe Ser Leu	Asn Ala His	305	310	315										
Lys Trp Phe Phe	Thr Thr Leu Asp	Cys Cys	Cys Leu Trp	Val Lys Asp	325	330	335										
Ser Asp Ser Leu	Val Lys Ala Leu	Ser Thr	Ser Ala Glu	Tyr Leu Lys	340	345	350										
Asn Lys Ala Thr	Glu Ser Lys Gln	Val Ile	Asp Tyr Lys	Asp Trp Gln	355	360	365										
Ile Ala Leu Ser	Arg Arg Phe Arg	Ser Met	Lys Leu Trp	Leu Val Leu	370	375	380										
Arg Ser Tyr Gly	Val Ala Asn Leu	Arg Thr	Phe Leu Arg	Ser His Val	385	390	395										
Lys Met Ala Lys	His Phe Gln Gly	Leu Met	Gly Met Asp	Asn Arg Phe	405	410	415										
Glu Ile Val Val	Pro Arg Thr Phe	Ala Met	Val Cys Phe	Arg Leu Lys	420	425	430										
Pro Ala Ala Ile	Phe Lys Gln Lys	Ile Val	Asp Asn Asp	Tyr Ile Glu	435	440	445										
Asp Gln Thr Asn	Glu Val Asn Ala	Lys Leu	Glu Ser Val	Asn Ala	450	455	460										
Ser Gly Lys Ile	Tyr Met Thr His	Ala Val	Val Gly Gly	Val Tyr Met	465	470	475										
Ile Arg Phe Ala	Val Gly Ala Thr	Leu Thr	Glu Glu Arg	His Val Thr	485	490	495										
Gly Ala Trp Lys	Val Val Gln Glu	His Thr	Asp Ala Ile	Leu Gly Ala	500	505	510										

Ala Ser Gln Ala Asn Leu Val Val

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<210> SEQ ID NO 277
<211> LENGTH: 26
<212> TYPE: PRT
<213> ORGANISM: A. thaliana

<400> SEQUENCE: 277

Ser Leu Ile Asp Ala Leu Lys Thr Asn Pro Glu Tyr Leu Glu Phe Lys
1 5 10 15

Val Lys Val Ser Lys Lys Asp Thr Val Val
20 25

<210> SEQ ID NO 278
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: C. annuum

<400> SEQUENCE: 278

Ala Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Tyr Leu Lys Asn Lys
1 5 10 15

Ala Ser Gln Gly Asn Leu Val Val
20

<210> SEQ ID NO 279
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: O. sativa

<400> SEQUENCE: 279

Phe Leu Ile Gln Ser Leu Ser Thr Asn Pro Glu Phe Leu Lys Asn Lys
1 5 10 15

Ala Ser Gln Ala Asn Ser Val Val
20

<210> SEQ ID NO 280
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: O. sativa

<400> SEQUENCE: 280

Arg Leu Thr Gly Ser Leu Glu Thr Asn Pro Glu Tyr Leu Lys Asn His
1 5 10 15

Ala Ser Asp Ser Gly Glu Val Thr
20

<210> SEQ ID NO 281
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: C. acuminata

<400> SEQUENCE: 281

Leu Leu Val Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Gln
1 5 10 15

Pro Ser Glu Ser Lys Ser Val Val
20

<210> SEQ ID NO 282
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: C. acuminata

<400> SEQUENCE: 282

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Val Leu Val Lys Ala Leu Ser Thr Asp Pro Glu Tyr Leu Lys Asn Lys
1 5 10 15

Pro Ser Glu Ser Asn Ser Val Val
20

<210> SEQ ID NO 283
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: C. annum

<400> SEQUENCE: 283

Val Leu Val Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys
1 5 10 15

Arg Ser Glu His Gly Ser Val Val
20

<210> SEQ ID NO 284
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: O. pumila

<400> SEQUENCE: 284

Leu Met Val Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys
1 5 10 15

Arg Ser Glu Phe Asp Gly Val Val
20

<210> SEQ ID NO 285
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: C. roseus

<400> SEQUENCE: 285

Leu Leu Leu Arg Ala Leu Thr Thr Asn Pro Glu Tyr Leu Lys Asn Lys
1 5 10 15

Gln Ser Asp Leu Asp Lys Val Val
20

<210> SEQ ID NO 286
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: P. crispum

<400> SEQUENCE: 286

Ala Leu Ile Lys Ser Leu Ser Thr Tyr Pro Glu Phe Leu Lys Asn Asn
1 5 10 15

Ala Ser Glu Thr Asn Lys Val Val
20

<210> SEQ ID NO 287
<211> LENGTH: 24
<212> TYPE: PRT
<213> ORGANISM: R. hybrid

<400> SEQUENCE: 287

Val Leu Ala Ser Ser Leu Ser Thr Asn Pro Glu Phe Leu Arg Asn Lys
1 5 10 15

Ala Ser Asp Ser Lys Gln Val Val
20

<210> SEQ ID NO 288

-continued

<211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: *P. hybrida*

<400> SEQUENCE: 288

Ser Leu Thr Lys Ala Leu Ser Thr Asn Pro Glu Val Leu Arg Asn Asp
 1 5 10 15
 Ala Thr Asp Ser Glu Gln Val Val
 20

<210> SEQ ID NO 289
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: *T. flavum*

<400> SEQUENCE: 289

Ala Leu Ile Lys Ala Leu Ser Thr Asn Pro Glu Tyr Leu Arg Asn Lys
 1 5 10 15
 Ala Thr Glu Ser His Gln Val Val
 20

<210> SEQ ID NO 290
 <211> LENGTH: 25
 <212> TYPE: PRT
 <213> ORGANISM: *B. distachyon*

<400> SEQUENCE: 290

Glu Leu Ile Ala Ala Leu Gly Thr Glu Gln Glu Tyr Ile Leu Lys Asp
 1 5 10 15
 Ser Ala Ser Glu Gly His Asp Ile Val
 20 25

<210> SEQ ID NO 291
 <211> LENGTH: 24
 <212> TYPE: PRT
 <213> ORGANISM: *P. somniferum*

<400> SEQUENCE: 291

Ser Leu Val Lys Ala Leu Ser Thr Ser Ala Glu Tyr Leu Lys Asn Lys
 1 5 10 15
 Ala Thr Glu Ser Lys Gln Val Ile
 20

What is claimed is:

1. A host cell comprising a transgene encoding a heterologous tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT) operably linked to a promoter, wherein the T8GT comprises an amino acid sequence having at least 95% identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.

2. The host cell of claim 1, wherein the host cell is a fungi cell, a bacterial cell, or a plant cell.

3. The host cell of claim 2, wherein the fungi cell is a yeast cell.

4. The host cell of claim 3, wherein the yeast cell is a *Saccharomyces cerevisiae* cell.

5. The host cell of claim 2, wherein the bacterial cell is a cell from a genus selected from *Agrobacterium*, *Escherichia*, *Bacillus*, *Pseudomonas* and *Streptomyces*.

6. The host cell of claim 5, wherein the bacterial cell is an *Agrobacterium tumefaciens* cell, an *Escherichia coli* cell, a *Bacillus subtilis* cell, a *Pseudomonas aeruginosa* cell, or a *Streptomyces griseus* cell.

7. The host cell of claim 2, wherein the plant cell is a *Nicotiana benthamiana* cell, or a cell from a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.

8. The host cell of claim 1, wherein the host cell further comprises a transgene encoding a 4-hydroxyphenylacetaldehyde reductase (4HPAR), wherein the 4HPAR comprises SEQ ID NO: 4.

9. The host cell of claim 8, wherein the host cell further comprises a transgene encoding a 4-hydroxyphenylacetaldehyde synthase (4HPAAS), wherein the 4HPAAS comprises SEQ ID NO: 2.

10. The host cell of claim 1, wherein the amino acid sequence of the T8GT has at least 99% identity to the one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.

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11. The host cell of claim 1, wherein the amino acid sequence of the T8GT comprises the amino acid sequence of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, or SEQ ID NO: 20.

12. A method of producing the host cell of claim 1, wherein the method comprises introducing into the host cell a vector comprising the transgene.

13. A method of producing tyrosol 8-O-glucoside (salidroside), wherein the method comprises inserting into a host cell a transgene that encodes a tyrosol:UDP-glucose 8-O-glucosyltransferase (T8GT), wherein the transgene is operably linked to a promoter, and wherein the T8GT comprises an amino acid sequence having at least 95% identity to one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20; and culturing the host cell in a culture media comprising tyrosol.

14. The method of claim 13, wherein the host cell is a fungi cell, a bacterial cell, or a plant cell.

15. The method of claim 14, wherein the host cell is the fungi cell, and wherein the fungi cell is a yeast cell.

16. The method of claim 15, wherein the yeast cell is a *Saccharomyces cerevisiae* cell.

17. The method of claim 14, wherein the host cell is the bacterial cell, and wherein the bacterial cell is a cell from a

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genus selected from the group consisting of *Agrobacterium*, *Escherichia*, *Bacillus*, *Pseudomonas* and *Streptomyces*.

18. The method of claim 17, wherein the bacterial cell is an *Agrobacterium tumefaciens* cell, an *Escherichia coli* cell, a *Bacillus subtilis* cell, a *Pseudomonas aeruginosa* cell, or a *Streptomyces griseus* cell.

19. The method of claim 14, wherein the host cell is the plant yeast cell, and wherein the plant cell is a *Nicotiana benthamiana* cell, or a cell from a genus selected from the group consisting of *Arabidopsis*, *Beta*, *Glycine*, *Helianthus*, *Solanum*, *Triticum*, *Oryza*, *Brassica*, *Medicago*, *Prunus*, *Malus*, *Hordeum*, *Musa*, *Phaseolus*, *Citrus*, *Piper*, *Sorghum*, *Daucus*, *Manihot*, *Capsicum*, and *Zea*.

20. The method of claim 13, wherein the amino acid sequence of the T8GT has at least 99% identity to the one or more of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, and SEQ ID NO: 20.

21. The method of claim 13, wherein the amino acid sequence of the T8GT comprises the amino acid sequence of SEQ ID NO: 14, SEQ ID NO: 16, SEQ ID NO: 18, or SEQ ID NO: 20.

* * * * *